SP 7 : 2026

राष्ट्रीय भवन निर्माण मानक 2026
NATIONAL BUILDING CONSTRUCTION STANDARDS 2026

खण्ड 2 / VOLUME 2

भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS

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VOLUME 2

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भारतीय मानक ब्यूरो

BUREAU OF INDIAN STANDARDS

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BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Section 1) covers the lighting and natural ventilation requirements in buildings. Thermal comfort conditions for the building occupants and illumination levels for different tasks are recommended to be achieved either by daylighting or artificial lighting or a combination of both. This section, read together with Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS, adequately covers the thermal comfort conditions for naturally ventilated buildings and mixed mode operation buildings, and the illumination levels required and methods of achieving the same.

Ventilation requirements to maintain air quality, control body odours and pollutants in terms of air changes per hour and to ensure thermal comfort and heat balance of body are laid for different occupancies and the methods of achieving the same by natural means or mixed mode operations means are covered in this Section. The provisions on mechanical ventilation are covered in Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS.

Climatic factors which normally help in deciding the orientation of the buildings to get desirable benefits of lighting and natural ventilation inside the buildings are also covered in this section.

This section was first published in 1970 as a part of National Building Code of India. The first revision of the section was brought out in 1983. In the second revision of 2005, some provisions were updated based on the information given in the SP 41 : 1987 'Handbook on functional requirements of buildings (other than industrial buildings)'; other major changes in the 2005 revision included rationalization of definitions and inclusion of definitions for some more terms; inclusion of climatic classification map of India based on a new criteria; updating of data on total solar radiations incident on various surfaces of buildings for summer and winter seasons; inclusion of design guidelines for natural ventilation; reference to Part 8 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of the Code for guidelines on mechanical ventilation, made, where these provisions were covered exhaustively; inclusion of rationalized method for estimation of desired capacity of ceiling fans and their optimum height above the floor for rooms of different sizes; incorporation of design sky illuminance values for different climatic zones of India, etc. Energy efficiency was another important aspect which was considered. Accordingly, the relevant requirements for energy efficient system for lighting and natural ventilation were duly included in the concerned provisions under the section.

In third revision of the Code in 2016, the changes that were incorporated include; elaboration of the calculation for solar load, and inclusion of 'Method of calculating solar load on vertical surfaces in different orientation' in Annex A; inclusion of provisions on sky component calculation procedure along with examples in Annex B; inclusion of reference to SP 41 : 1987 for obtaining coefficient utilization for determination of luminous flux; updation of provisions relating to efficient artificial light source and luminaires; inclusion of modern lighting techniques such as LED and induction light in comparison with their energy consumption; updation of provisions relating to photocontrols for artificial lights; inclusion of definitions and enabling provision for lighting shelves and light pipes; elaboration of the provisions related to thermal comfort including indices such as effective temperature, adaptive thermal comfort along with elaborations on tropical summer index; elaboration of design guidelines for natural ventilation with illustrations; elaboration of provisions related to determination of rate of ventilation particularly on combined effect of wind and thermal actions; and inclusion of provision on colour rendering in line with that in SP 72 : 2010 'National Lighting Code 2010'.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

As a result of experience gained since the implementation of the 2016 version of this NBCS and feedback received as well as revisions of Indian Standards on which this Section was based, a need was felt to revise this Section. The significant changes incorporated in this revision include the following:

a) Terminology clause has been updated with new terms and definitions;

b) Orientation of building clause has been renamed to site planning and building placement; and

c) The clause on aspects of daylighting has been updated;

d) Provisions for spatial and environmental consideration in building design; and operable windows and UV exposure have been added;

e) The aims of good lighting clause has been updated;

f) Table 6 for illuminance and lighting performance parameters have been updated;

g) Provisions for planning and design guidelines for circadian lighting including supplementary artificial circadian lighting have been introduced;

h) Planning and design guidelines for outdoor lighting and its energy efficiency aspects have been introduced;

j) Detailed provisions on design charts for natural ventilation and acceptable air temperature with the increase in the air speed have been added;

k) Provisions of the adaptive thermal comfort mode of ventilation have been reviewed and updated;

The provisions of this Section are without prejudice to the various acts, Rules and Regulations including the *Factories Act*, 1948 and Rules and Regulations framed thereunder.

The information contained in this Section is largely based on the following Indian Standards/Special Publications:

IS 2440 : 1975	Guide for daylighting of buildings (*second revision*)
IS 3103 : 1975	Code of practice for industrial ventilation (*first revision*)
IS 3362 : 1977	Code of practice for natural ventilation of residential buildings (*first revision*)
IS 3646 (Part 1) : 2025	Interior illumination — Code of practice: Part 1 General requirements and recommendations for working interiors (*second revision*)
IS 7662 (Part 1) : 1974	Recommendations for orientation of buildings: Part 1 Nonindustrial buildings
IS 11907 : 1986	Recommendations for calculation of solar radiation on buildings
SP 32 : 1986	Handbook on functional requirements of industrial buildings (lighting and ventilation)
SP 41 : 1987	Handbook on functional requirements of buildings other than industrial buildings

Provisions given in National Lighting Code of India, 'SP 72 : 2025' (*first revision*) may also be referred.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

Assistance has been taken from the following documents in the revision of this Section:

'Report on Energy Conservation in Buildings', submitted to Department of Power, Ministry of Energy by CSIR-Central Building Research Institute, Roorkee.

'Adaptive thermal comfort model based on fields studies in five climate zones across India', Building and Environment, Volume 219, 1 July 2022 by Rawal, R., Shukla, Y., Vardhan, V., et al. (2022)

'Field studies of thermal comfort across multiple climate zones for the subcontinent: India Model for Adaptive Comfort (IMAC)', *Building and Environment*, 98 (March 2106), (pp. 55–70) by Manu, S., Shukla, Y., Rawal, R., Thomas, L. E., and De Dear, R. (2016)

'Low Energy Cooling and Ventilation for Indian Residences: Design Guide', Loughborough University, UK and CEPT Research and Development Foundation (CRDF), India by Cook, M., Shukla, Y., Rawal, R., Loveday, D., Faria, L. C., and Angelopoulos, C. (2020).

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **'good practice'** in relation to design, constructional procedures or other related information, and where reference is made to **"accepted standard"** in relation to material specification, testing, or other related information, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/ Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [D-1(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Part D/Section 1, that is, IS 7662 (Part 1) : 1974 'Recommendations for orientation of buildings: Part 1 Non-industrial buildings'

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 1 LIGHTING AND NATURAL VENTILATION

1 SCOPE

1.1 This NBCS (Part D/Section 1) covers requirements and methods for lighting and natural ventilation of buildings.

1.2 For all buildings and facilities open to and used by the public, including all forms of public housing by the government/civic bodies and private developers, adequate lighting and natural ventilation for barrier-free access and movement within and around buildings by elderly and persons with disabilities and of different age groups shall be ensured. The ventilation through the building envelope elements such as windows, skylights, ventilators for thermal comfort and better air quality requirements are included in this section.

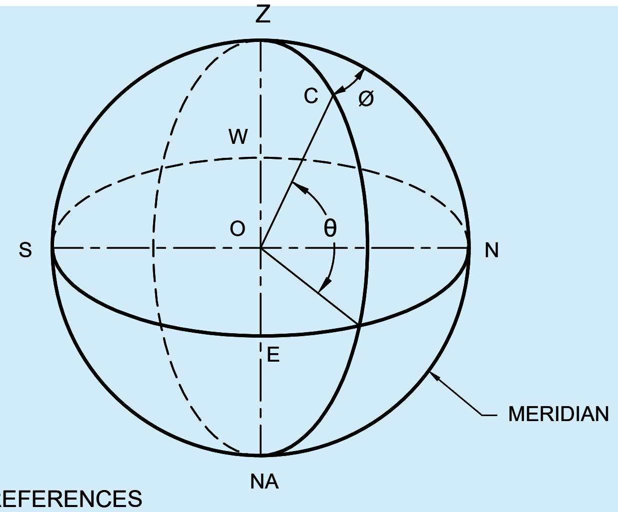
2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply.

2.1 Lighting

2.1.1 *Altitude* (θ) — The angular distance of any point of celestial sphere, measured from the horizon, on the great circle passing through the body and the zenith (*see* Fig. 1).

2.1.2 *Azimuth* (Φ) — The angle measured between meridians passing through the north point and the point in question (point C in Fig. 1).



REFERENCES

O - Observer's station	*S* - Geographical south
C - Celestial body	*E* - Geographical east
Z - Zenith	*W* - Geographical west
NA - Nadir	*N* - Geographical north

FIG. 1 ALTITUDE AND AZIMUTH OF A CELESTIAL BODY

2.1.3 *Brightness Ratio or Contrast* — The variations or contrast in brightness of the details of a visual task, such as white print on blackboard.

2.1.4 *Candela* (*cd*) — The SI unit of luminous intensity.

Candela = 1 lumen per steradian

2.1.5 *Central Field* — The area of circle around the point of fixation and its diameter, subtending an angle of about 2 degrees at the eye. Objects within this area are most critically seen in both their details and colour.

2.1.6 *Clear Design Sky* — The distribution of luminance of such a sky is non-uniform; the horizon can be brighter than the zenith depending on the sun position, and when L_z is the brightness at zenith, the brightness at an altitude (θ) in the region away from the sun, is given by the expression:

$$L_\theta = L_z \operatorname{cosec} \theta \ (\text{for } 15^\circ < \theta \leq 90^\circ)$$

$$L_\theta = L_z \operatorname{cosec} 15^\circ \ (\text{for } 0^\circ \leq \theta \leq 15^\circ) = 3.863\,7\ L_z$$

2.1.7 *Circadian Lighting* — A lighting means that uses electric lights in a way that supports natural sleep and wake cycle of humans. This is done by changing how bright the light is and what colour it is throughout the day, so it matches what the body expects from natural daylight and night time.

2.1.8 *Colour Rendering Index* (*CRI*) — Measure of the degree to which the psychophysical colour of an object illuminated by the test illuminant conforms to that of the same object illuminated by the reference illuminant, suitable allowance having been made for the state of chromatic adaptation.

2.1.9 *Correlated Colour Temperature* (*CCT*) (*K*) — The temperature of the Planckian radiator whose perceived colour most closely resembles that of a given stimulus at the same brightness and under specified viewing conditions.

2.1.10 *Daylighting* — The strategic use of natural optical radiation from the sun and sky to illuminate interior spaces, enhancing visual comfort and supporting human circadian regulation. In this context, daylight encompasses the portion of solar radiation within the visible spectrum, typically between 380 nm and 780 nm. Effective daylighting design aims to maximize natural light penetration, minimize reliance on electric lighting, and promote occupant health and well-being.

2.1.11 *Daylight Area* — The portion of an interior space where the working plane receives natural light at or above a specified daylight factor or meets a defined minimum illuminance threshold. This area is typically delineated by a daylight factor contour, representing zones with adequate daylight levels for visual tasks.

2.1.12 *Daylight Factor* — The ratio, expressed as a percentage, of the illuminance at a specific point on a given plane indoors to the simultaneous illuminance on a horizontal plane outdoors under an overcast sky. This factor measures the amount of daylight reaching the indoor point relative to outdoor conditions, providing an indication of natural light availability in interior spaces.

2.1.13 *Daylight Penetration* — The maximum distance to which a given daylight factor contour penetrates into a room.

2.1.14 *Direct Solar Illuminance* — The illuminance from the sun without taking into account the light from the sky.

2.1.15 *External Reflected Component* (*ERC*) — Daylight illuminance at a point which is received by direct reflection from external surfaces.

2.1.16 *Glare* — A condition of vision in which there is discomfort or a reduction in the ability to see significant objects or both due to an unsuitable distribution or range of luminance or due to extreme contrasts in space.

2.1.17 *Illuminance* — At a point on a surface, the ratio of the luminous flux incident on an infinitesimal element of the surface containing the point under consideration to the area of the element.

NOTE — The unit of illuminance (the measurement of illumination) is lux which is 1 lumen per m^2.

2.1.18 *Indoor Artificial Lighting* — Artificial lighting powered by electricity, designed to enhance visual comfort, task performance, and occupant well-being in indoor environments. Its effectiveness is assessed using several key parameters:

a) *Spectral power density* (*SPD*) — Describes the relative power of a light source at each wavelength across the visible spectrum. SPD influences colour rendering, visual perception and biological effects, including circadian response;

b) *Colour tuning* — The capability to dynamically adjust the correlated colour temperature (*CCT*) and spectral content of the light source, allowing adaptation to user preferences, visual tasks, or time-of-day considerations to support circadian health;

c) *Light flicker* — Rapid fluctuations in luminous intensity that may be visible or invisible to the human eye. Flicker can cause visual fatigue, headaches, or reduced task performance. The P_{st}^{LM} value (short-term flicker severity measured by the light flickermeter method) is used to quantify flicker in lighting systems; and

d) *Unified glare rating* (*UGR*) — A standardized metric that quantifies discomfort glare in indoor spaces. Lower UGR values indicate better visual comfort and help prevent visual strain without reducing task visibility.

2.1.19 *Internal Reflected Component* (*IRC*) — Daylight illuminance at a point in a given plane which is received by direct reflection or inter-reflection from the internal surfaces.

2.1.20 *Light Output Ratio* (*LOR*) *or Efficiency* (η) — The ratio of the luminous flux emitted from the luminaire to that emitted from the lamp(s) (nominal luminous flux). It is expressed in percent.

2.1.21 *Light Pipe* — A conduit made of a highly reflective material, which is capable of channelling light from one end to the other through successive internal reflections. Such a pipe may be flexible or rigid.

2.1.22 *Light Shelf* — A daylighting system based on sun path geometry used to bounce the light off a ceiling, project it deeper into a space, distribute it from above and diffuse it to produce a uniform light level below.

2.1.23 *Lumen* (*lm*) — SI unit of luminous flux. The luminous flux emitted within unit solid angle (one steradian) by a point source having a uniform intensity of one candela.

2.1.24 *Luminance* (*at a Point of a Surface in a Given Direction*) (*Brightness*) — The quotient of the luminous intensity in the given direction of an infinitesimal element of the surface containing the point under consideration by the orthogonally projected area of the element on a plane perpendicular to the given direction. The unit is candela per square metre (cd/m^2).

2.1.25 *Luminous Flux* (ϕ) — The quantity characteristic of radiant flux which expresses its capacity to produce visual sensation evaluated according to the values of relative luminous efficiency for the light adapted eye:

a) *Effective luminous flux* (ϕ_n) — Total luminous flux which reaches the working plane; and

b) *Nominal luminous flux* (ϕ_o) — Total luminous flux of the light sources in the interior.

2.1.26 *Maintenance Factor* (*d*) — The ratio of the average illuminance on the working plane after a certain period of use of a lighting installation to the average illuminance obtained under the same conditions for a new installation.

2.1.27 *Meridian* — It is the great circle passing through the zenith and nadir for a given point of observation.

2.1.28 *North and South Points* — The point in the respective directions where the meridian cuts the horizon.

2.1.29 *Orientation of Buildings* — In the case of non-square buildings, orientation refers to the direction of the normal to the long axis. For example, if the length of the building is east-west, its orientation is north-south.

2.1.30 *Outdoor Artificial Lighting* — Lighting systems specifically designed for exterior environments, intended to enhance visibility, safety and security during nighttime hours while minimizing adverse environmental and ecological impacts. Key considerations include:

a) *Fully shielded* — Luminaire designs that emit light only in the downward direction, preventing upward light emissions. This helps control glare and limits contributions to light pollution;

b) *Full cut-off* — A luminaire classification where zero light is emitted at or above the horizontal plane of the fixture. This design minimizes sky glow and enhances lighting efficiency by directing light only where it is needed;

c) *Light trespass* — The unintended or intrusive illumination of areas beyond the target zone, such as neighbouring properties, streets, or natural habitats. Light trespass can lead to discomfort, sleep disruption and privacy issues; and

d) *Sky glow* — The diffuse illumination of the night sky caused by artificial lighting, particularly from urban areas. Sky glow reduces the visibility of stars, interferes with astronomical observations and disrupts nocturnal ecosystems.

2.1.31 *Peripheral Field* — It is the rest of the visual field which enables the observer to be aware of the spatial framework surrounding the object seen.

NOTE — A central part of the peripheral field, subtending an angle of about 30º on either side of the point of fixation, is chiefly involved in the perception of glare.

2.1.32 *Reflected Glare* — The adverse impact on visual comfort and efficiency caused by unwanted reflections in the task area, which can hinder the ability to see clearly or cause discomfort, regardless of the source of the glare.

2.1.33 *Surface Reflectance* (*Reflectance*) — The ratio of the luminous flux reflected by an opaque surface (with or without diffusion) to the flux it receives. Here the term is used for lighting. Some symbols used for surface reflectance of common indoor surfaces are:

r_c = reflection factor of ceiling.

r_w = reflection factor of parts of the wall between the working surface and the luminaires.

r_f = reflection factor of floor.

2.1.34 *Reveal* — The side of an opening for a window.

2.1.35 *Room Index* (k_r) — An index relating to the shape of a rectangular interior, according to the formula:

$$k_r = \frac{L \times W}{(L + W)\,H_m}$$

where

L and W = length and width respectively of the interior; and

H_m = mounting height, that is, height of the fittings above the working plane.

NOTES

1 For rooms where the length exceeds 5 times the width, L shall be taken as $L = 5W$.

2 If the surface reflectance of the upper stretch of the walls is less than half the surface reflectance of the ceiling, for indirect or for the greater part of indirect lighting, the value H_m is measured between the ceiling and the working plane.

2.1.36 *Sky Component* (*SC*) — Daylight illuminance at a point on a given plane which is received directly from the sky.

2.1.37 *Solar Load* — The amount of heat received into a building due to solar radiation which is affected by orientation, materials of construction and reflection of external finishes and colour.

2.1.38 *Utilization Factor* (*Coefficient of Utilization*) (μ) — The ratio of the total luminous flux which reaches the working plane (effective luminous flux, ϕ_n) to the total luminous flux of the light sources in the interior (nominal luminous flux, ϕ_o).

2.1.39 *UV-A* — Ultraviolet radiation with wavelengths between 315 and 400 nm. UV-A penetrates the skin more deeply than UV-B and contributes to photoaging, oxidative stress, and potential DNA damage through indirect mechanisms. While it plays a role in skin ageing and long-term health effects, UV-A is significantly less effective for germicidal and disinfection purposes due to its lower energy.

2.1.40 *UV-B* — Ultraviolet radiation with wavelengths between 280 nm and 315 nm. UV-B plays a critical role in the synthesis of pre-vitamin D3 in the human skin upon exposure. It has higher biological activity than UV-A and can also cause sunburn and DNA damage with excessive exposure.

2.1.41 *Visual Field* — The visual field in the binocular which includes an area approximately 120º vertically and 160º horizontally centering on the point to which the eyes are directed. The line joining the point of fixation and the centre of the pupil of each eye is called its primary line of sight.

2.1.42 *Working Plane* — A horizontal plane at a level at which work will normally be done (*see* **4.1.4.4** and **4.1.4.5**).

2.2 Ventilation

2.2.1 *Air Change per Hour* — The amount of air leakage into or out of a building or room in terms of the number of building volume or room volume exchanged.

2.2.2 *Axial Flow Fan* — A fan having a casing in which the air enters and leaves the impeller in a direction substantially parallel to its axis.

2.2.3 *Centrifugal Fan* — A fan in which the air leaves the impeller in a direction substantially at right angles to its axis.

2.2.4 *Contaminants* — Dusts, fumes, gases, mists, vapours and such other substances present in air that are likely to be injurious or offensive to the occupants.

2.2.5 *Dilution Ventilation* — Supply of outside air to reduce the airborne concentration of contaminants in the building.

2.2.6 *Dry Bulb Temperature* — The temperature of the air, read on a thermometer, taken in such a way so as to avoid errors due to radiation.

2.2.7 *Effective Temperature* (*ET*) — An arbitrary index which combines into a single value the effect of temperature, humidity and air movement on the sensation of warmth or cold felt by the human body and its numerical value is that of the temperature of still saturated air which would induce an identical sensation.

2.2.8 *Exhaust of Air* — Removal of air from a building or a room and its disposal outside by means of a mechanical device, such as a fan.

2.2.9 *Fresh Air or Outside Air* — Air of that quality, which meets the criteria of Table 1 and in addition shall be such that the concentration of any contaminant in the air is limited to within one-tenth the threshold limit value (TLV) of that contaminant.

NOTES

1 Where it is reasonably believed that the air of quality is not expected as indicated above, sampling and analysis shall be carried out by a competent authority having jurisdiction and if the outside air of the specified quality is not available, filtration and other treatment devices shall be used to bring its quality to or above the levels mentioned in Table 1. Odour is to be essentially unobjectionable.

2 The above list of contaminants is not exhaustive and available special literature may be referred for data on other contaminants.

2.2.10 *General Ventilation* — Ventilation, either natural or mechanical or both, so as to improve the general environment of the building, as opposed to local exhaust ventilation for contamination control.

2.2.11 *Globe Temperature* — The temperature measured by a thermometer whose bulb is enclosed in a matt black painted thin copper globe of 150 mm diameter. It combines the influence of air temperature and thermal radiations received or emitted by the bounding surfaces.

2.2.12 *Humidification* — The process whereby the absolute humidity of the air in a building is maintained at a higher level than that of outside air or at a level higher than that which would prevail naturally.

Table 1 Maximum Allowable Contaminant Concentrations for Ventilation Air

(*Clause* 2.2.9)

Sl No.	Contaminants	Annual Average (Arithmetic Mean)	Short Term Level (Not to Exceed More than Once a Year)	Averaging Period
		µg/m^3	µg/m^3	h
(1)	(2)	(3)	(4)	(5)
i)	Suspended particulates	60	150	24
ii)	Sulphur oxides	80	400	24
iii)	Carbon monoxide	20 000	30 000	8
iv)	Photochemical oxidant	100	500	1
v)	Hydrocarbons (not including methanes)	1 800	4 000	3
vi)	Nitrogen oxide	200	500	24

2.2.13 *Humidity*, *Absolute* — The mass of water vapour per unit volume.

2.2.14 *Humidity*, *Relative* — The ratio of the partial pressure or density of the water vapour in the air to the saturated pressure or density respectively of water vapour at the same temperature.

2.2.15 *Local Exhaust Ventilation* — Ventilation effected by exhaust of air through an exhaust appliance, such as a hood with or without fan located as closely as possible to the point at which contaminants are released, so as to capture effectively the contaminants and convey them through ducts to a safe point of discharge.

2.2.16 *Make-up Air* — Outside air supplied into a building to replace the indoor air.

2.2.17 *Mechanical Ventilation* — Supply of outside air either by positive ventilation or by infiltration by reduction of pressure inside due to exhaust of air, or by a combination of positive ventilation and exhaust of air.

2.2.18 *Natural Ventilation* — Supply of outside air into a building through window or other openings due to wind outside and convection effects arising from temperature or vapour pressure differences (or both) between inside and outside of the building.

2.2.19 *Positive Ventilation* — The supply of outside air by means of a mechanical device, such as a fan.

2.2.20 *Propeller Fan* — A fan in which the air leaves the impeller in a direction substantially parallel to its axis designed to operate normally under free inlet and outlet conditions.

2.2.21 *Spray-Head System* — A system of atomizing water so as to introduce free moisture directly into a building.

2.2.22 *Stack Effect* — Convection effect arising from temperature or vapour pressure difference (or both) between outside and inside of the room and the difference of height between the outlet and inlet openings.

2.2.23 *Tropical Summer Index* (*TSI*) — The temperature of calm air at 50 percent relative humidity which imparts the same thermal sensation as the given environment.

2.2.24 *Threshold Limit Value* (*TLV*) — Refers to air-borne concentration of contaminants currently accepted by the American Conference of Governmental Industrial Hygienists and represents conditions under which it is believed that nearly all occupants may be repeatedly exposed, day after day, without adverse effect.

2.2.25 *Velocity*, *Capture* — Air velocity at any point in front of the exhaust hood necessary to overcome opposing air currents and to capture the contaminants in air at that point by causing the air to flow into the exhaust hood.

2.2.26 *Ventilation* — Supply of outside air into, or the removal of inside air from an enclosed space.

2.2.27 *Wet Bulb Temperature* — The steady temperature finally given by a thermometer having its bulb covered with gauze or muslin moistened with distilled water and placed in an air stream of not less than 4.5 m/s.

3 SITE PLANNING AND BUILDING PLACEMENT

3.1 The aim of site planning and building placement for optimum building orientation is to provide physically and psychologically comfortable living inside the building by creating conditions which suitably and successfully ward off the undesirable effects of severe weather to a considerable extent by judicious use of the recommendations and knowledge of climatic factors.

3.2 Climate Zones

3.2.1 For the purpose of design of buildings, the country may be divided into the major climatic zones as given in Table 2, which also gives the basis of this classification.

Table 2 Classification of Climate

(*Clause* 3.2.1)

Sl No.	Climatic Zone	Mean Monthly Maximum Temperature °C	Mean Monthly Relative Humidity Percent
(1)	(2)	(3)	(4)
i)	Hot-dry	Above 30	Below 55
ii)	Warm-humid	Above 30	Above 55
		Above 25	Above 75
iii)	Temperate	25 to 30	Below 75
iv)	Cold	Below 25	All values
v)	Composite	*See* **3.2.2**	*See* **3.2.2**

The climatic classification map of India is shown in Fig. 2.

3.2.2 Each climatic zone does not have same climate for the whole year; it has a particular season for more than six months and may experience other seasons for the remaining period. A climatic zone that does not have any season for more than six months may be called as composite zone.

3.3 Climatic Factors

From the point of view of lighting and natural ventilation, the following climatic factors influence the optimum orientation of the building:

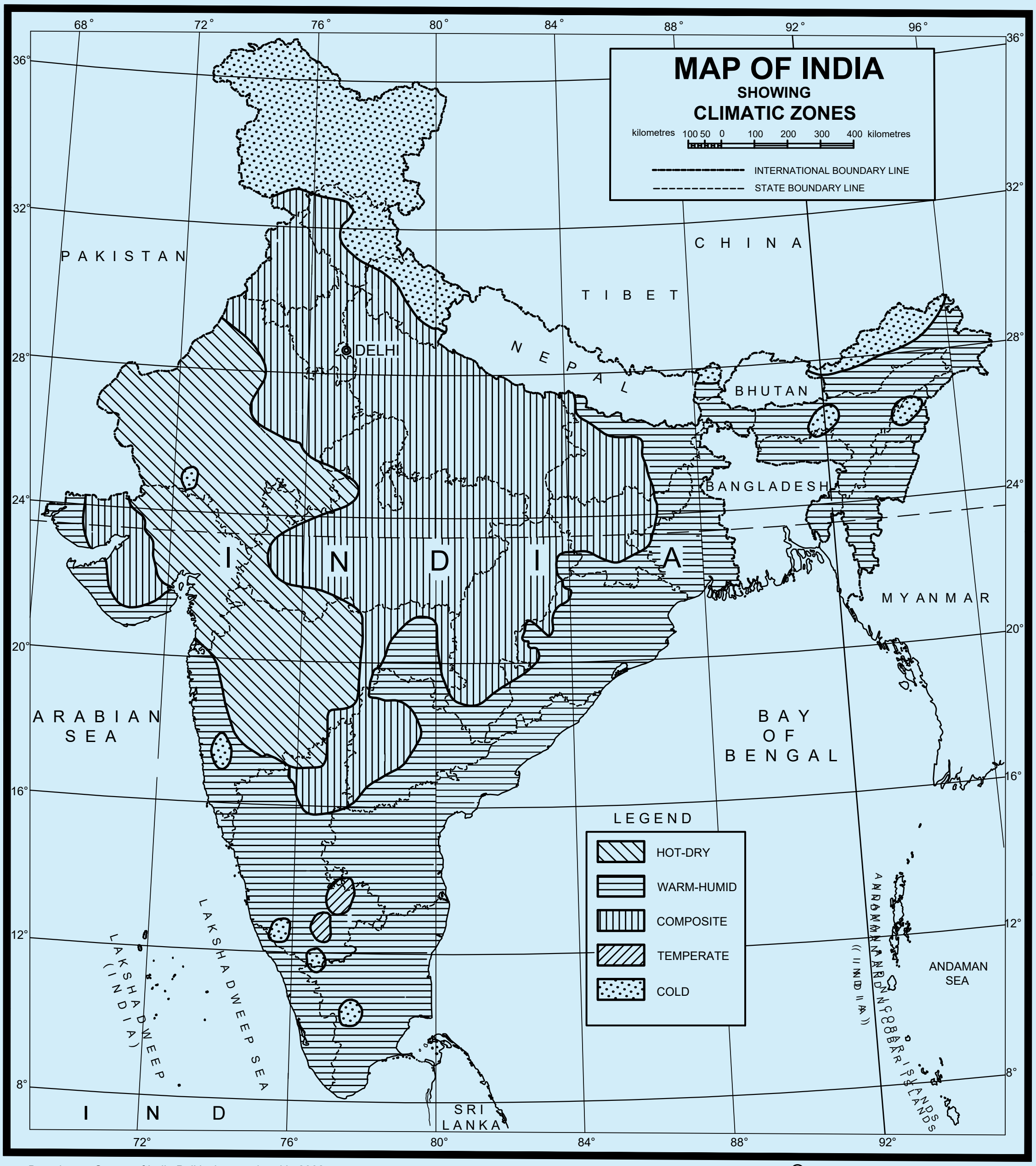
a) Solar radiation and temperature;

b) Relative humidity; and

c) Prevailing winds.

3.4 Solar Radiation

3.4.1 The best orientation from a solar point of view requires that the building as a whole should receive the maximum solar radiation in winter and the minimum in summer. For practical evaluation, it is necessary to know the duration of sunshine and hourly solar intensity on the various external surfaces on representative days of the seasons. The total direct plus diffused diurnal solar loads per unit area on the vertical surface facing different directions are given in Table 3 for two days in the year, that is, 22 June and 22 December, representative of summer and winter, for latitudes corresponding to some important cities all over India. From Table 3, the total heat intake can be calculated for all possible orientations of the building for these extreme days of summer and winter. Solar load on vertical surfaces of different orientations can be calculated as per the method given in Annex A.



Based upon Survey of India Political map printed in 2002.

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The territorial waters of India extend into the sea to a distance of twelve nautical miles measured from the appropriate baseline.

The interstate boundaries between Arunachal Pradesh, Assam and Meghalaya shown on this map are as interpreted from the North-Eastern Areas (Reorganization) Act, 1971, but have yet to be verified.

The external boundaries and coastlines of India agree with the Record/Master Copy certified by Survey of India.

The responsibility for the correctness of internal details rest with the publisher.

FIG. 2 MAP OF INDIA SHOWING CLIMATIC ZONES

3.4.1.1 Except in cold climatic zone, suitable sun-breakers have to be provided to cut off the incursion of direct sunlight to prevent heat radiation and to avoid glare. A location (latitude) and weather specific approach to sizing of shading devices should consider the annual start and end date of the summer season for the building. Table 3 shows irradiation (W/m^2) received per unit of the facade area in each orientation. Irradiation values maybe used for prioritizing shading devices and other measures of solar protection (for example, planting of vegetation).

Table 3 Total Solar Radiation (Direct Plus Diffused) Incident on Various Surfaces of Buildings, in W/m^2/day, for Summer and for Winter Seasons

(*Clause* 3.4.1, 3.4.1.1 *and* A-2.1.3)

Sl No.	Orientation	Season	Latitude					
			9° N	13° N	17° N	21° N	25° N	29° N
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	North	Summer	1 494	1 251	2 102	1 775	2 173	1 927
		Winter	873	859	840	825	802	765
ii)	North-East	Summer	2 836	2 717	3 144	3 092	3 294	3 189
		Winter	1 240	1 158	1 068	1 001	912	835
iii)	East	Summer	3 344	3 361	3 475	3 598	3 703	3 794
		Winter	2 800	2 673	2 525	2 409	2 211	2 055
iv)	South-East	Summer	2 492	2 660	2 393	2 629	2 586	2 735
		Winter	3 936	3 980	3 980	3 995	3 892	3 818
v)	South	Summer	1 009	1 185	1 035	1 117	1 112	1 350
		Winter	4 674	4 847	4 958	5 059	4 942	4 981
vi)	South-West	Summer	2 492	2 660	2 393	2 629	2 586	2 735
		Winter	3 936	3 980	3 980	3 995	3 892	3 818
vii)	West	Summer	3 341	3 361	3 475	3 598	3 703	3 794
		Winter	2 800	2 673	2 525	2 409	2 211	2 055
viii)	North-West	Summer	2 836	2 717	3 144	3 092	3 294	3 189
		Winter	1 240	1 158	1 068	1 001	912	835
ix)	Horizontal	Summer	8 107	8 139	8 379	8 553	8 817	8 863
		Winter	6 409	6 040	5 615	5 231	4 748	4 281

3.5 Relative Humidity and Prevailing Winds

3.5.1 The discomfort due to high relative humidity in air when temperatures are also high can be counteracted, to a great extent, by circulation of air with electric fans or by ventilation. In the past, simultaneously with heavy construction and surrounding *Verandahs* to counter the effect of sun's radiation, there was also an over emphasis on prevailing winds to minimise the adverse effects of high humidity with high temperatures. With the introduction of electric fan to effectively circulate air and owing to taking into account the rise in cost of construction of buildings, emphasis should be on protection from solar radiation where temperatures are very high. When, however, there is less diurnal variation between morning and mean maximum temperatures along with high humidity, as in coastal areas, the emphasis should be on prevailing winds.

3.5.1.1 For the purpose of orientation, it is necessary to study the velocity and direction of the wind at each hour and in each month instead of relying on generalizations of a month or a period or for the year as a whole. This helps to spot the right winds for a particular period of day or night.

3.5.1.2 It is generally found that variation up to 30° with respect to the prevalent wind direction does not materially affect indoor ventilation (average indoor air speed) inside the building.

3.5.2 In hot-dry climate, advantage can be taken of evaporative cooling in summer to cool the air before introducing it into the building. But in warm humid climate, it is desirable either to regulate the rate of air movement with the aid of electric fans or to take advantage of prevailing winds.

3.6 Aspects of Daylighting

The clear design sky concept accounts for the worst-case scenario, making building orientation less critical, except for the need to avoid direct sunlight and glare. However, mutual shading effects from opposite facades and significant vegetation elements shall be considered to maintain effective daylighting. For glare control reference may be made from relevant guidelines/best practices ('Handbook on Sustainability in Buildings'). Daylighting design is supported by three key pillars: daylight availability, sun control, and glare control, all of which are essential to ensure optimal visual comfort and energy efficiency within buildings.

3.7 Spatial and Environmental Considerations in Building Design

Sustainable building design focuses on optimizing the use of natural resources, improving energy efficiency and ensuring occupant well-being. Integrating natural elements such as vegetation and considering the spatial relationships between buildings play a crucial role in enhancing daylight access, ventilation, and thermal comfort, while promoting ecological balance.

The following considerations should guide the integration of these elements into building design:

a) It is suggested to plant tall trees or plants at least as tall as the building around new buildings. This can make the area look aesthetic and help the environment by giving shade, cleaning the air and making the place cooler. It is also better to use plants that naturally grow in the area;

b) Adequate spacing between a building and neighbouring structures is essential for ensuring access to natural light, ventilation, and privacy. The distance between buildings should be planned to minimize overshadowing, reduce light trespass and promote natural airflow, enhancing the overall livability of the area;

c) The orientation of building facades relative to cardinal directions should be planned to optimize solar gain, energy efficiency, and occupant comfort. Facades should maximize natural light while minimizing glare and overheating, contributing to overall energy efficiency and improving thermal performance; and

d) Tree planting in open spaces and streets should be carefully planned to provide shade during the summer season, allow intermittent or seasonal sunlight access, without obstructing natural winds or compromising daylight access. The selected trees should reduce glare, create cool microclimates, and improve air quality. Deciduous trees that shed leaves in winter and retain thick foliage in summer are especially advantageous, particularly for southern and western exposures, allowing maximum sunlight in winter and providing effective shading in summer. Trees with high-density foliage and low gap percentages (for example, *Ashoka*, *Asopalav*) should be placed at a distance proportional to their mature height to avoid obstructing light and airflow. Less dense or deciduous trees (for example, *Gulmohar*, *Amaltas*, *Neem*) may be planted with considerations beyond daylight access, including aesthetics and ecological benefits. In addition, it is recommended that native species of trees be preferred, as identified by the respective state environmental departments, to support region-specific environmental planning.

3.8 Operable Windows and UV Exposure

Operable windows should be provided in habitable spaces to allow for natural ventilation and daylighting. Most glazing types (single pane or double pane) block the UV part of the solar radiation to avoid over-exposure of interiors and occupants to damaging effects of UV radiation. However, some UV exposure has health benefits (UV-B for vitamin D3 production) that occupants can only access if the glass windows are operable. Operable windows should be designed for maximum accessibility to building occupants and direct sun access on facades that receive direct sun, operable windows should be preferred from healthy daylight access point-of-view.

3.9 For detailed information regarding orientation of buildings and recommendations for various climatic zones of country, reference may be made to good practice [D-1(1)].

4 LIGHTING

4.1 Principles of Lighting

4.1.1 *Aims of Good Lighting*

Good lighting is an essential element of building design, serving both visual and non-visual functions to promote occupant comfort, safety and well-being. It should ensure a visually comfortable and functional environment that enhances task performance, supports safe movement and complements the architectural and interior design of spaces. Beyond meeting visual requirements, lighting should also address non-visual impacts by supporting healthy circadian rhythms and aiding in proper eye development, particularly in children. Effective control of glare is critical, as poorly managed glare can lead to visual discomfort, eye strain and headaches. Proper glare management not only improves visual comfort and productivity but also contributes significantly to occupant satisfaction and overall quality of life in buildings.

4.1.1.1 Realization of these aims involves the following:

a) Careful planning of the brightness and colour pattern within both the working areas and the surroundings so that attention is drawn naturally to the important areas, detail is seen quickly and accurately and the room is free from any sense of gloom or monotony (*see* **4.1.4**);

b) Using directional lighting where appropriate to assist perception of task detail and to give good modelling;

c) Controlling direct and reflected glare from light sources to eliminate visual discomfort;

d) In artificial lighting installations, minimizing flicker from certain types of lamps and paying attention to the colour rendering properties of the light;

e) Correlating lighting throughout the building to prevent excessive differences between adjacent areas so as to reduce the risk of accidents; and

f) Installing emergency lighting systems, where necessary.

4.1.2 *Planning the Brightness Pattern*

The brightness pattern seen within an interior may be considered as composed of three main parts – the task itself, immediate background of the task and the general surroundings of walls, ceiling, floor, equipment and furnishings.

The illuminance requirements given in good practice [D-1(2)] for elders and persons with disabilities, shall be complied with in addition to the requirements specified in this Section.

4.1.2.1 In occupations where the visual demands are small, the levels of illumination derived from a criterion of visual performance alone may be too low to satisfy the other requirements. For such situations, therefore, illuminance recommendations are based on standards of welfare, safety and amenity judged appropriate to the occupations; they are also sufficient to give these tasks brightness which ensures that the visual performance exceeds the specified minimum.

4.1.2.2 Where work takes place over the whole utilizable area of room, the illumination over that area should be reasonably uniform and it is recommended that the uniformity ratio (minimum illuminance divided by average illuminance levels) should be not less than 0.7 for the working area.

4.1.2.3 To ensure visual comfort and minimize strain, the illuminance levels within a space should be carefully balanced. Large spatial variations in illuminance around the task or activity area shall be avoided, as they can lead to visual stress and discomfort. The illuminance of the immediate surrounding area shall be related to that of the task area or activity area, ensuring a well-balanced luminance distribution within the visual field. This immediate surrounding area shall be defined as a band with a minimum width of 0.5 m around the task area. While the illuminance of this surrounding area may be lower than that of the task area, it shall not fall below the minimum values specified in Table 4. Additionally, the size and position of the immediate surrounding area shall be clearly stated and documented in the lighting design specifications. Proper planning of these illuminance levels contributes to a visually comfortable and efficient built environment.

Table 4 Relationship of Average Illuminances on Immediate Surrounding to the Average Illuminance on the Task Area or Activity Area

(*Clause* 4.1.2.3)

Sl No.	Average Illuminance on the Task Area or Activity Area lx	Average Illuminance on Immediate Surrounding Areas lx
(1)	(2)	(3)
i)	≥750	500
ii)	500	300
iii)	300	200
iv)	200	150
v)	≤150	equal to task area

4.1.2.4 In indoor workplaces, particularly those devoid of daylight, ensuring adequate illumination beyond the immediate task surroundings is essential for visual comfort and spatial perception. A large area outside the immediate surrounding area, referred to as the background area, shall be illuminated to maintain a well-balanced visual environment. The background area is a horizontal surface at floor level, adjacent to the immediate surrounding area within the limits of the space. It shall have a maintained average illuminance of at least one-third of the illuminance value of the immediate surrounding area. In larger rooms, this background area shall have a minimum width of 3 m to ensure uniformity in lighting. Proper illumination of the background area may help in reducing visual fatigue and enhancing overall workplace efficiency.

4.1.2.5 In case of all buildings and facilities open to and used by the public, including all forms of public housing by the government/civic bodies and private developers, the requirements for visual contrast as given in good practice [D-1(2)] shall also be complied with for ensuring visual comfort for elders and persons with disabilities.

4.1.3 *Glare*

Excessive contrast or abrupt and large changes in brightness produce the effect of glare. When glare is present, the efficiency of vision is reduced and small details or subtle changes in scene cannot be perceived. It may be,

a) direct glare due to light sources within the field of vision;

b) reflected glare due to reflections from light sources or surfaces of excessive brightness; and

c) veiling glare where the peripheral field is comparatively very bright.

4.1.3.1 An example of glare sources in daylighting is the view of the bright sky through a window or skylight, especially when the surrounding wall or ceiling is comparatively dark or weakly illuminated. Glare can be minimized in this case either by shielding the open sky from direct sight by louvers, external hoods or deep reveals, curtains or other shading devices or by cross lighting the surroundings to a comparable level. A gradual transition of brightness from one portion to the other within the field of vision always avoids or minimizes the discomfort from glare.

For electric lamps the minimum shielding angles for lamp luminance shall not be less than the values given in the table below:

Sl No.	*Lamp Luminance* kcd/m^2	*Minimum Shielding Angle* Degrees
(1)	(2)	(3)
i)	1 to 20	10
ii)	20 to 50	15
iii)	50 to 500	20
iv)	≥500	30

The above mentioned shielding angle should not be applied to luminaries who do not appear in the field of view of a worker during usual work and/or do not give the worker any noticeable disability glare

Table 6 also gives recommended value of quality class of direct glare limitation for different tasks. These are numbers assigned to qualitative limits of direct glare; high, medium and low quality as 1, 2 and 3, respectively. For more details, reference may be made to good practice [D–1(3)].

4.1.4 *Recommended Values of Illuminance*

Table 6 gives the illuminance and lighting performance parameters commensurate with the general standards of lighting described in this Section and related to many occupations and buildings. These are valid under most of the conditions whether the illumination is by daylighting, artificial lighting or a combination of the two. The great variety of visual tasks makes it impossible to list them all and those given should be regarded as representing types of tasks.

The illuminance shall comply with the requirements given in good practice [D-1(2)] for persons with disabilities, in addition to the requirements specified in Table 6.

For comprehending and applying Table 6, reference may be made to **4.1.4.1** to **4.1.7**.

4.1.4.1 *Composition of Table* 6

a) Column (2) lists the reference number for each task area or activity area;

b) Column (3) lists those tasks areas or activities areas, for which specific requirements are given. If the particular task or activity is not listed, the values given for a similar, comparable situation should be adopted. Task areas or activity areas can also be a room, for example, a corridor or resting room;

c) Column (4) gives the required maintained average illuminance $\bar{E}_m$ on the reference surface (*see* **4.1.7**) for the interior (area) in which the task or activity from column (3) is performed;

d) Column (5) gives the minimum illuminance uniformity U_o on the reference surface for the maintained average illuminance $\bar{E}_m$ chosen according to **5**;

e) Column (6) gives the minimum colour rendering indices (R_a) (*see* **4.1.8**) for the situation listed in col (3);

f) Column (7) gives the unified glare rating limits (UGR limits, R_{UGL}) that are applicable to the situation listed in col (3) (*see* **4.1.3**);

g) Column (8) gives the maintained cylindrical illuminance $\bar{E}_{m,z}$ for the recognition of object and people;

h) Column (9) gives the maintained average illuminance on walls $\bar{E}_{m,wall}$; and

j) Column (10) gives the maintained average illuminance on ceilings $\bar{E}_{m,ceiling}$.

4.1.4.2 Indicative list detailing the different locations and tasks are grouped as per the following:

Sl No.	*Location*	*Tasks/Specific Location*
(1)	(2)	(3)
i)	Traffic zones inside buildings	–
ii)	General areas inside buildings	Rest, sanitation and first aid rooms Control rooms Store rooms, cold stores
iii)	Logistics and warehouses	–

Table (*Concluded*)

Sl No.	*Location*	*Tasks/Specific Location*
(1)	(2)	(3)
iv)	Industrial activities and crafts	Agriculture Bakeries Cement, cement goods, concrete, bricks Ceramics, tiles, glass, glassware Chemical, plastics and rubber industry Electrical and electronic industry Food stuffs and luxury food industry Foundries and metal casting Hair dressers Jewellery manufacturing Laundries and dry cleaning Leather and leather goods Metal working and processing Paper and paper goods Power stations Printers Rolling mills, iron and steel works Textile manufacture and processing Vehicle construction and repair Wood working and processing
v)	Offices	–
vi)	Retail premises	–
vii)	Places of public assembly	General areas Restaurants and hotels Theatres, concert halls, cinemas, places for entertainment Trade fairs, exhibition halls Museums Libraries Car parks (indoor)
viii)	Educational premises	Nursery school, play school Educational buildings
ix)	Health care premises	Rooms for general use Staff rooms Wards, maternity wards Examination rooms (general) Eye examination rooms Ear examination rooms Scanner rooms Delivery rooms Treatment rooms (general) Operating areas Intensive care unit Dental clinics/units Laboratories and pharmacies Decontamination rooms Autopsy rooms and mortuaries
x)	Transportation areas	Airports Railway installations

4.1.4.3 The illumination levels recommended in Table 6 are those to be maintained at all time on the task. As circumstances may be significantly different for different interiors used for the same application or for different conditions for the same kind of activity, a range of illuminances is recommended for each type of interior or activity instead of a single value of illuminance. Each range consists of three successive steps of the recommended scale of illuminances. They represent good practice and should be regarded as giving order of illumination commonly required rather than as having some absolute significance. For working interiors, the middle value of each range represents the recommended service illuminance that would be used unless one or more of the factors mentioned below apply.

4.1.4.3.1 The higher value of the range should be used when,

a) unusually low reflectances or contrasts are present in the task;

b) errors are costly to rectify;

c) visual work is critical;

d) accuracy or higher productivity is of great importance; and

e) visual capacity of the worker makes it necessary.

4.1.4.3.2 The lower value of the range may be used when,

a) reflectances or contrast are unusually high;

b) speed and accuracy is not important; and

c) the task is executed only occasionally.

4.1.4.4 Where a visual task is required to be carried out throughout an interior, general illumination level to the recommended value on the working plane is necessary; where the precise height and location of the task are not known or cannot be easily specified, the recommended value is that on horizontal plane 850 mm above floor level.

NOTE — For an industrial task, working plane for the purpose of general illumination levels is that on a work place which is generally 750 mm above the floor level. For certain purposes, such as viewing the objects of arts, the illumination levels recommended are for the vertical plane at which the art pieces are placed.

4.1.4.5 Where the task is localized, the recommended value is that for the task only; it need not, and sometimes should not, be the general level of illumination used throughout the interior. Some processes, such as industrial inspection process, call for lighting of specialized design, in which case the level of illumination is only one of the several factors to be taken into account.

4.1.4.6 In case of all buildings and facilities open to and used by the public, including all forms of public housing by the government/civic bodies and private developers, the minimum illuminance level as given in good practice [D-1(2)] shall also be complied with for ensuring sufficient lighting for accessibility by elders and persons with disabilities.

4.1.5 *Lighting for Movement about a Building*

Most buildings are complexes of working areas and other areas, such as passages, corridors, stairways, lobbies and entrances. The lighting of all these areas shall be properly correlated to give safe movement within the building at all times. In case of all buildings and facilities open to and used by the public, including all forms of public housing by the government/civic bodies and private developers, the illuminance in these areas shall comply with the requirements given in good practice [D-1(2)].

4.1.5.1 *Corridors*, *passages and stairways*

Transitions from brightly lit work areas to dimly lit corridors, passages, or stairways can pose safety risks, as the human eye requires time to adapt to significant changes in luminance. To mitigate these risks, such transitional spaces should be illuminated uniformly and adequately to ensure clear visibility and safe movement. In areas that are used intermittently, occupancy or motion-sensing lighting controls may be used to conserve energy while ensuring illumination is provided when the space is occupied. Stairways should be designed to prevent direct glare from luminaires and to eliminate harsh shadows that can obscure steps. Tread lighting or integrated step-strip lighting can be used to delineate step edges, especially in low-light environments or emergency egress paths. For outdoor applications, such as industrial catwalks and stairways, lighting should be robust, weather-resistant, and positioned to provide uniform coverage without creating glare. These strategies collectively enhance visual guidance, reduce fall hazards, and improve overall safety in circulation spaces.

4.1.5.2 *Entrances*

The problems of correctly grading the lighting within a building to allow adequate time for adaptation when passing from one area to another area are particularly acute at building entrances. These are given below:

a) By day, people entering a building will be adapted to the very high levels of brightness usually present outdoors and there is risk of accident if entrance areas, particularly any steps, are poorly lighted. This problem may often be overcome by arranging windows to give adequate natural lighting at the immediate entrance, grading to lower levels further inside the entrance area. Where this cannot be done, supplementary artificial lighting should be installed to raise the level of illumination to an appropriate value; and

b) At night it is desirable to light entrance halls and lobbies so that the illumination level reduces towards the exit and so that no bright fittings are in the line of sight of people leaving the building. Any entrance steps to the building should be well-lighted by correctly screened fittings.

4.1.6 *Brightness Appearance of Rooms*

To enhance the brightness appearance and visual comfort of rooms, it is important to define the task or activity areas within the space and assess the type of visual tasks performed. Appropriate lighting criteria, such as maintained average illuminance ($\bar{E}_m$), uniformity ratio (U_o), colour rendering index (R_a) and unified glare rating limit (R_{UGL}), should be selected based on the most demanding tasks. Key room surfaces, including walls and ceilings, should be illuminated to contribute to the perception of brightness. Higher illuminance levels in task areas may be considered to support better visual performance, while the illumination of objects and room surfaces, such as walls ($\bar{E}_{m,wall}$) and ceilings ($\bar{E}_{m,ceiling}$), should be optimized. In spaces with high ceilings or in industrial settings where ceiling lighting is less critical, a lower ceiling illuminance may be acceptable. Table 5 shows the assignment of columns to requirements, as given in **4.1.4.1**.

4.1.7 *Lighting Requirements for Task Areas*, *Activity Areas*, *Room and Space Brightness*

The requirements for task areas and activity areas are given in Table 5. The requirements for the specific tasks and activities are given by $\bar{E}_m$, U_o, R_a and R_{UGL}. The requirements for the space in which the task(s) or activities are carried out are given by $\bar{E}_{m,z}$ for the perception of objects and people within this space and $\bar{E}_{m,wall}$ and $\bar{E}_{m,ceiling}$ for room brightness. The latter are used for designing the room and the space including R_{UGL}. Glare (by R_{UGL}) is dedicated to the space in which a task is carried out.

The first four columns are used for of task area or activity area design and more than one of these areas can occur within one space.

Table 5 Assignment of Columns in Table 6 to Requirements

(*Clauses* 4.1.6 *and* 4.1.7)

Sl No.	Task Area or Activity Area Design				Room or Space Design Requirements		
(1)	(2)				(3)		
i)	Task or activity related requirements				Brightness appearance of rooms (**4.1.6**)		
ii)	$\bar{E}_m$ lux	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$ lux	$\bar{E}_{m,wall}$ lux	$\bar{E}_{m,ceiling}$ lux

where

$\bar{E}_m$ = minimum average illuminance level at defined area;
U_o = uniformity ratio (minimum illuminance/average illuminance);
R_a = colour rendering properties of a light source;
R_{UGL} = unified glare rating limit value;
$\bar{E}_{m,z}$ = maintained average cylindrical illuminance;
$\bar{E}_{m,wall}$ = average illuminance level at wall(s); and
$\bar{E}_{m,ceiling}$ = average illuminance level at ceiling.

In all enclosed places the maintained illuminances on the major surfaces shall have the values specified in Table 6.

Table 6 Illuminance and Lighting Performance Parameters

(*Foreword*, *Clauses* 4.1.3.1, 4.1.4, 4.1.4.1, 4.1.4.3, 4.1.7, 4.3.2, 4.3.2.1 *and* 4.4.8)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	**1**	**Areas Inside Buildings**							
	1.1	Corridors and circulation areas	75-100-150	0.40	70	28	–	–	–
	1.2	Stairs, escalators, moving walks	75-100-150	0.40	70	25	–	–	–
	1.3	Lifts	75-100-150	0.40	70	25	–	–	–
	1.4	Area in front of lifts and escalators	150-200-300	0.40	70	25	–	–	–
	1.5	Loading ramps/bays	100-150-200	0.40	70	25	–	–	–
	1.6	Building entrance with canopy	20-30-50	0.40	–	–	–	–	–
	1.7	Gangways: manned	100-150-200	0.40	70	25	–	–	–
ii)	**2**	**General Areas Inside Buildings-Rest, Sanitation and First Aid Rooms**							
	2.1	Canteens and break areas	150-200-300	0.40	80	22	–	–	–
	2.2	Resting rooms	75-100-150	0.40	80	22	–	–	–
	2.3	Rooms for physical exercise	200-300-500	0.40	80	22	–	–	–
	2.4	Cloakroom (area), washrooms, bathrooms, dressing, lockers, shower, sink and toilet areas	150-200-300	0.40	80	25	–	–	–
	2.5	Facial lighting in front of mirrors	150-200-300	0.40	80	–	–	–	–
	2.6	Sick bay	300-500-750	0.60	80	19	–	–	–
	2.7	Rooms for medical attention	300-500-750	0.60	90	19	–	–	–
	2.8	General cleaning	75-100-150	0.40	–	–	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$ lx	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$ lx	$\bar{E}_{m,wall}$ lx	$\bar{E}_{m,ceiling}$ lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
iii)	**3**	**General Areas Inside Buildings-Control Rooms**							
	3.1	Plant rooms, switch gear rooms	150-200-300	0.40	80	25	50	50	–
	3.2	Post sorting, switchboard	300-500-750	0.60	80	19	150	150	–
	3.3	Surveillance station	200-300-500	0.60	80	19	100	100	–
iv)	**4**	**General Areas Inside Buildings-Store Rooms, Cold Stores**							
	4.1	Store and stockrooms	75-100-150	0.40	80	25	–	–	–
	4.2	Dispatch packing handling areas	200-300-500	0.60	80	25	–	–	–
	4.3	Larder (large storage space for food)	150-200-300	0.40	80	25	–	–	–
v)	**5**	**Logistics and Warehouses**							
	5.1	Unloading/loading area	150-200-300	0.40	80	25	50	–	–
	5.2	Packing/grouping area	200-300-500	0.50	80	25	100	–	–
	5.3	Configuration and rehandling	500-750-1 000	0.60	80	22	150	–	–
	5.4	Open goods storage	150-200-300	0.40	80	25	50	–	–
	5.5	Rack storage floor	100-150-200	0.50	80	25	–	–	–
	5.6	Rack storage rack face	50-75-100	0.40	80	–	–	–	–
	5.7	Central logistics corridor (heavy traffic)	200-300-500	0.60	80	25	100	–	–
	5.8	Automated zones (unmanned)	50-75-100	0.40	80	25	–	–	–
vi)	**6**	**Industrial Activities and Crafts-Agriculture**							
	6.1	Loading and operating of goods, handling equipment and machinery	150-200-300	0.40	80	25	–	–	–
	6.2	Buildings for livestock	30-50-75	0.40	70	-	–	–	–
	6.3	Sick animal pens; calving stalls	150-200-300	0.60	80	25	–	–	–
	6.4	Feed preparation; dairy; utensil washing	150-200-300	0.60	80	25	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
vii)	**7**	**Industrial Activities and Crafts–Bakeries**							
	7.1	Preparation and baking	200-300-500	0.60	80	22	75	–	–
	7.2	Finishing, glazing, decorating	300-500-750	0.70	80	22	100	–	–
viii)	**8**	**Industrial Activities and Crafts-Cement, Cement Goods, Concrete, Bricks**							
	8.1	Drying	30-50-75	0.40	70	28	–	–	–
	8.2	Preparation of materials; work on kilns and mixers	150-200-300	0.40	70	28	50	–	–
	8.3	General machine work	200-300-500	0.60	80	25	100	–	–
	8.4	Rough forms	200-300-500	0.60	80	25	100	–	–
ix)	**9**	**Industrial Activities and Crafts-Ceramics, Tiles, Glass, Glassware**							
	9.1	Drying	30-50-75	0.40	70	28	–	–	–
	9.2	Preparation, general machine work	200-300-500	0.60	80	25	100	–	–
	9.3	Enamelling, rolling, pressing, shaping simple parts, glazing, glass blowing	200-300-500	0.60	80	25	100	–	–
	9.4	Grinding, engraving, glass polishing, shaping precision parts, manufacture of glass instruments	500-750-1 000	0.70	80	19	150	75	50
	9.5	Grinding of optical glass, crystal, hand grinding and engraving	500-750-1 000	0.70	80	19	150	75	50
	9.6	Precision work, for example, decorative grinding, hand painting	750-1 000-1 000	0.70	90	19	150	75	50
	9.7	Manufacture of synthetic precious stones	1 000-1 500-2 000	0.70	90	19	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
x)	**10**	**Industrial Activities and Crafts-Chemical, Plastics and Rubber Industry**							
	10.1	Remote operated processing installations	30-50-75	0.40	70	–	–	–	–
	10.2	Processing installations with limited manual intervention	100-150-200	0.40	70	28	50	–	–
	10.3	Constantly manned work stations in processing installations	200-300-500	0.60	80	25	100	–	–
	10.4	Precision measuring rooms, laboratories	300-500-750	0.60	80	19	150	75	50
	10.5	Pharmaceutical production	300-500-750	0.60	80	22	150	–	–
	10.6	Tyre production	300-500-750	0.60	80	22	150	–	–
	10.7	Colour inspection	750-1 000-1 500	0.70	90	19	150	75	50
	10.8	Cutting, finishing, inspection	500-750-1 000	0.70	80	19	150	75	50
xi)	**11**	**Industrial Activities and Crafts-Electrical and Electronic Industry**							
	11.1	Cable and wire manufacture	200-300-500	0.60	80	25	100	–	–
	11.2	Winding-large coils	200-300-500	0.60	80	25	100	–	–
	11.3	Winding-medium-sized coils	300-500-750	0.60	80	22	150	–	–
	11.4	Winding-small coils	500-750-1000	0.70	80	19	150	75	50
	11.5	Coil impregnating	200-300-500	0.60	80	25	100	–	–
	11.6	Galvanizing	200-300-500	0.60	80	25	100	–	–
	11.7	Assembly work:- rough, for example, large transformers	200-300-500	0.60	80	25	100	–	–
	11.8	Assembly work: Medium, for example, switchboards	300-500-750	0.60	80	22	150	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	11.9	Assembly work: Fine, for example, telephones, radios, IT equipment (computers)	500-750-1 000	0.70	80	19	150	75	50
	11.10	Assembly work: Precision, for example, measuring equipment, printed circuit boards	750-1 000-1 500	0.70	80	19	150	75	50
	11.11	Electronic workshops, testing, adjusting	1 000-1 500-2 000	0.70	80	19	150	75	50
xii)	**12**	**Industrial Activities and Crafts-Food Stuffs and Luxury Food Industry**							
	12.1	Work stations	150-200-300	0.40	80	25	50	–	–
	12.2	Sorting and washing of products, milling, mixing, packing	200-300-500	0.60	80	25	100	–	–
	12.3	Work stations and critical zones in slaughter houses, butchers, dairies, mills, on filtering floor in sugar refineries	300-500-750	0.60	80	25	150	–	–
	12.4	Cutting and sorting of fruit and vegetables	200-300-500	0.60	80	25	100	–	–
	12.5	Manufacture of delicatessen foods,	300-500-750	0.60	80	22	150	–	–
	12.6	Inspection of glasses and bottles, product control, trimming, sorting, decoration	300-500-750	0.60	80	22	150	–	–
	12.7	Laboratories	300-500-750	0.60	80	19	150	75	50
	12.8	Colour inspection	750-1 000-1 500	0.70	90	19	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
xiii)	**13**	**Industrial Activities and Crafts-Foundries and Metal Casting**							
	13.1	Man-size underfloor tunnels, cellars, etc	30-50-75	0.40	70	–	–	–	–
	13.2	Platforms	75-100-150	0.40	70	25	50	–	–
	13.3	Sand preparation	150-200-300	0.40	80	25	50	–	–
	13.4	Dressing	150-200-300	0.40	80	25	50	–	–
	13.5	Work stations at cupola and mixer	150-200-300	0.40	80	25	50	–	–
	13.6	Casting bay	150-200-300	0.40	80	25	50	–	–
	13.7	Shake out areas	150-200-300	0.40	80	25	50	–	–
	13.8	Machine moulding	150-200-300	0.40	80	25	50	–	–
	13.9	Hand and core moulding	200-300-500	0.60	80	25	100	–	–
	13.10	Die casting	200- 300-500	0.60	80	25	100	–	–
	13.11	Model building	300-500-750	0.60	80	22	150	–	–
xiv)	**14**	**Industrial Activities and Crafts-Hairdressers**							
	14.1	Hairdressing	300-500-750	0.60	90	19	150	75	50
xv)	**15**	**Industrial Activities and Crafts-Jewellery Manufacturing**							
	15.1	Working with precious stones	1 000-1 500-2 000	0.70	90	19	150	75	50
	15.2	Manufacture of jewellery	750-1 000-1 500	0.70	90	19	150	75	50
	15.3	Watch making (manual)	1 000-1 500-2 000	0.70	80	19	150	75	50
	15.4	Watch making (automatic)	300-500-750	0.60	80	19	150	75	50
xvi)	**16**	**Industrial Activities and Crafts-Laundries and Dry Cleaning**							
	16.1	Goods in, marking and sorting	200-300-500	0.60	80	25	100	–	–
	16.2	Washing and dry cleaning	200-300-500	0.60	80	25	100	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	16.3	Ironing, pressing	200-300-500	0.60	80	25	100	–	–
	16.4	Inspection and repairs	500-750–1 000	0.70	80	19	150	75	50
xvii)	**17**	**Industrial Activities and Crafts-Leather and Leather Goods**							
	17.1	Work on vats, barrels, pits	150-200-300	0.40	80	25	75	–	–
	17.2	Fleshing, skiving, rubbing, tumbling of skins	200-300-500	0.40	80	25	100	–	–
	17.3	Saddlery work, shoe manufacture: stitching, sewing, polishing, shaping, cutting, punching	300-500-750	0.60	80	22	150	–	–
	17.4	Sorting	300-500-750	0.60	90	22	150	–	–
	17.5	Leather dyeing (machine)	300-500-750	0.60	80	22	150	–	–
	17.6	Quality control	750-1 000-1 500	0.70	80	19	150	75	50
	17.7	Colour inspection	750-1 000-1 500	0.70	90	19	150	75	50
	17.8	Shoe making	300-500-750	0.60	80	22	150	–	–
	17.9	Glove making	300-500-750	0.60	80	22	150	–	–
xviii)	**18**	**Industrial Activities and Crafts-Metal Working and Processing**							
	18.1	Open die forging	150-200-300	0.60	80	25	50	–	–
	18.2	Drop forging	200-300-500	0.60	80	25	75	–	–
	18.3	Welding	200-300-500	0.60	80	25	75	–	–
	18.4	Rough and average machining: Tolerances $\geq$ 0.1 mm	200-300-500	0.60	80	22	75	–	–
	18.5	Precision machining; grinding: Tolerances < 0.1 mm	300-500-750	0.70	80	19	150	75	50
	18.6	Scribing; inspection	500-750-1 000	0.70	80	19	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$ lx	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$ lx	$\bar{E}_{m,wall}$ lx	$\bar{E}_{m,ceiling}$ lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	18.7	Wire and pipe drawing shops; cold forming	200-300-500	0.60	80	25	75	–	–
	18.8	Plate machining: Thickness ≥ 5 mm	150-200-300	0.60	80	25	50	–	–
	18.9	Sheet metalwork: Thickness < 5 mm	200-300-500	0.60	80	22	75	–	–
	18.10	Tool making, cutting equipment manufacture	500-750-1 000	0.70	80	19	150	–	–
	18.11	Assembly:						–	–
		1) Rough	150-200-300	0.60	60	25	50	–	–
		2) Medium	200-300-500	0.60	80	25	75	–	–
		3) Fine	300-500-750	0.60	80	22	150	–	–
		4) Precision	500-750-1 000	0.70	80	19	150	75	50
	18.12	Galvanizing	200-300-500	0.60	80	25	75	–	–
	18.13	Surface preparation and painting	500-750-1 000	0.70	80	25	150	–	–
	18.14	Tool, template and jig making, precision mechanics, micro- mechanics	750-1 000-1 500	0.70	80	19	150	75	50
xix)	**19**	**Industrial Activities and Crafts-Paper and Paper Goods**							
	19.1	Edge runners, pulp mills	150-200-300	0.40	80	25	50	–	–
	19.2	Paper manufacture and processing, paper and corrugating machines, cardboard manufacture	200-300-500	0.60	80	25	75	–	–
	19.3	Standard bookbinding work, for example, folding, sorting, gluing, cutting, embossing, sewing	300-500-750	0.60	80	22	150	–	–
xx)	**20**	**Industrial Activities and Crafts-Power Stations**							
	a) 20.1	Fuel supply plant	30-50-75	0.40	70	–	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	20.2	Boiler house	75-100-150	0.40	70	28	50	–	–
	20.3	Machine halls	150-200-300	0.40	80	25	50	–	–
	20.4	Side rooms, for example pump rooms, condenser rooms, etc; switchboards (inside buildings)	150-200-300	0.40	80	25	50	–	–
	20.5	Control rooms	300-500-750	0.70	80	19	150	75	50
xxi)	**21**	**Industrial Activities and Crafts-Printers**							
	21.1	Cutting, gilding, embossing, block engraving, work on	300-500-750	0.60	80	19	150	75	50
	21.2	Paper sorting and hand printing	300-500-750	0.60	80	19	150	75	50
	21.3	Type setting, retouching, lithography	750-1 000-1 500	0.70	80	19	150	75	50
	21.4	Colour inspection in multicoloured printing	1 000-1 500-2 000	0.70	90	19	150	75	50
	21.5	Steel and copper engraving	1 500-2 000-3 000	0.70	80	19	150	75	50
xxii)	**22**	**Industrial Activities and Crafts-Rolling Mills, Iron and Steel Works**							
	22.1	Production plants without manual operation	30-50-75	0.40	70	–	–	–	–
	22.2	Production plants with occasional manual operation	100-150-200	0.40	70	28	50	–	–
	22.3	Production plants with continuous manual operation	150-200-300	0.60	80	25	50	–	–
	22.4	Slab store	30-50-75	0.40	70	–	–	–	–
	22.8	Furnaces	150-200-300	0.40	70	25	50	–	–
	22.9	Mill train; coiler; shear line	200-300-500	0.60	70	25	75	–	–
	22.10	Control platforms; control panels	200-300-500	0.60	80	22	75	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	22.11	Test, measurement and inspection	300-500-750	0.60	80	22	150	–	–
	22.12	Underfloor man-sized tunnels; belt sections, cellars, etc	30-50-75	0.40	70	–	–	–	–
xxiii)	**23**	**Industrial Activities and Crafts-Textile Manufacture and Processing**							
	23.1	Work stations and zones in baths, bale opening	150-200-300	0.60	70	25	50	–	–
	23.2	Carding, washing, ironing, devilling machine work, drawing, combing, sizing, card cutting, pre-spinning, jute and hemp spinning	200-300-500	0.60	70	22	100	–	–
	23.3	Spinning, plying, reeling, winding	300-500-750	0.60	70	22	150	–	–
	23.4	Warping, weaving, braiding, knitting	300-500-750	0.60	70	22	150	–	–
	23.5	Sewing, fine knitting, taking up stitches	500-750-1 000	0.70	80	22	150	–	–
	23.6	Manual design, drawing patterns	500-750-1 000	0.70	90	22	150	–	–
	23.7	Finishing, dyeing	300-500-750	0.60	80	22	150	–	–
	23.8	Drying room	75-100-150	0.40	60	28	50	–	–
	23.9	Automatic fabric printing	300-500-750	0.60	90	25	100	–	–
	23.10	Burling, picking, trimming	750-1 000-1 500	0.70	80	19	150	75	50
	23.11	Colour inspection; fabric control	750-1 000-1 500	0.70	90	19	150	75	50
	23.12	Invisible mending	1 250-1 500-2 000	0.70	90	19	150	75	50
	23.13	Hat manufacturing	300-500-750	0.60	80	22	150	–	–
xxiv)	**24**	**Industrial Activities and Crafts-Vehicle Construction and Repair**							
	24.1	Press shop - Large parts	200-300-500	0.60	80	25	100	–	–
	24.2	Press shop - Visual inspection	300-500-750	0.60	80	22	150	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	24.3	Body work and assembly-Automatic line	200-300-500	0.60	80	25	100	–	–
	24.4	Body work and assembly- Manual welding	300-500-750	0.60	80	22	150	–	–
	24.5	Painting, spraying chamber, polishing chamber	500-750-1 000	0.70	80	22	150	–	–
	24.6	Painting, inspection, touch- up and polishing	750-1 000-1 500	0.70	90	19	150	75	50
	24.7	Upholstery manufacture (manual)	750-1 000-1 500	0.70	80	19	150	75	50
	24.8	Detailing: Subparts assembly (doors, dashboard, upholstery), under chassis assembly, motor and mechanical assembly, final assembly conveyor line	500-750-1 000	0.70	80	22	150	–	–
	24.9	Detailing - Work with electronics	500-750-1 000	0.60	90	22	150	–	–
	24.10	Final inspection	750-1 000-1 500	0.70	90	19	150	75	50
	24.11	General vehicle services, repair and testing	300-500-750	0.60	80	22	100	–	–
xxv)	**25**	**Industrial Activities and Crafts-Wood Working and Processing**							
	25.1	Automatic processing, for example, drying, plywood manufacturing	30-50-75	0.40	70	28	–	–	–
	25.2	Steam pits	100-150-200	0,40	70	28	50	–	–
	25.3	Saw frame	200-300-500	0.60	70	25	100	–	–
	25.4	Work at joiner's bench, gluing, assembly	200-300-500	0.60	80	25	100	–	–
	25.5	Polishing, painting, fancy joinery	500-750-1 000	0.70	80	22	150	–	–
	25.6	Work on wood working machines, for example, turning, fluting, dressing,	300-500-750	0.60	80	19	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
		rebating, grooving, cutting, sawing, sinking							
	25.7	Selection of veneer woods	500-750-1 000	0.90	70	22	150	–	–
	25.8	Marquetry, inlay work	500-750-1 000	0.90	70	22	150	–	–
	25.9	Quality control, inspection	750-1 000-1 500	0.90	70	19	150	75	50
xxvi)	**26**	**Offices**							
	26.1	Filing, copying, etc	200-300-500	0.40	80	19	100	75	50
	26.2	General office area - Writing, typing, reading, data processing	300-500-750	0.60	80	19	150	75	50
	26.3	Private cabins	300-500-750	0.70	80	19	150	75	50
	26.4	CAD work stations	300-500-750	0.60	80	19	150	75	50
	26.5	Huddle/breakout area	200-300-500	0.60	80	19	150	75	50
	26.6	Conference and meeting rooms	300-500-750	0.60	80	19	150	75	50
	26.7	Light on presenter	150-200-300		80	19			
	26.8	Reception desk	200-300 -500	0.60	80	22	100	75	50
	26.9	Archiving	150-200-300	0.40	80	25	75	–	–
	26.10	Canteens and break out areas	150-200-300	0.60	80	22	–	–	–
	26.11	Resting rooms	75-100-200	0.40	80	22	–	–	–
	26.12	Rooms for physical exercise	200-300-500	0.40	80	22	–	–	–
	26.13	Cloakroom (area), wash room	150-200-300	0.40	80	25	100	75	50
	26.14	Facial lighting in front of mirrors	200-300-500	0.40	80	–	–	–	–
	26.15	Sick bay	300-500-750	0.60	80	19	100	75	50
	26.16	Rooms for medical attention	300-500-750	0.40	90	19	100	75	50
	26.17	General cleaning	75-100-150	0.40	80	–	–	–	–

Table 6 (*Continued*)

Sl No.	**Ref No.**	**Type of Task/Activity Area**	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	26.18	Corridors and circulation areas	75 -100-150	0.40	70	28	–	–	–
	26.19	Stairs, escalators, travelators	75-100-150	0.40	70	25	–	–	–
	26.20	Lifts	75-100-150	0.40	70	25	–	–	–
	26.21	Area in front of lifts and escalators	150-200-300	0.40	70	25	–	75	50
	26.22	Auditorium, training hall, town hall, amphitheater	300-500-750	0.60	80	19	75	50	–
xxvii)	**27**	**Retail Premises**							
	27.1	General sales area	200-300-500	0.40	80	22	75	–	–
	27.2	Till area	300-500 -750	0.60	80	19	100	75	50
	27.3	Wrapper table	300-500-750	0.60	80	22	100	–	–
	27.4	Storage area	200-300-500	0.40	80	25	50	–	–
	27.5	Dressing/fitting room	200-300-500	0.4	90	–	–	–	–
xxviii)	**28**	**Places of Public Assembly General Areas**							
	28.1	Entrance halls	75-100-150	0.40	80	22	50	–	–
	28.2	Cloakrooms	150-200-300	0.40	80	25	75	–	–
	28.3	Lounges	150-200-300	0.40	80	22	75	–	–
	28.4	Ticket offices	200-300-500	0.60	80	22	75	–	–
xxix)	**29**	**Places of Public Assembly- Restaurants and Hotels**							
	29.1	Reception/cashier desk, porters desk	200-300-500	0.60	80	22	100	–	–
	29.2	Kitchen	300-500-750	0.60	80	22	100	–	–
	29.3	Restaurant, dining room, function room	–	–	80	–	–	–	–
	29.4	Self-service restaurant	150-200-300	0.40	80	22	75	–	–
	29.5	Buffet	200-300-500	0.60	80	22	75	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	29.6	Conference rooms	300-500-750	0.60	80	19	150	75	50
	29.7	Corridors	75-100-150	0.40	80	25	50		
xxx)	**30**	**Places of Public Assembly-Theatres, Concert Halls, Cinemas, Places for Entertainment**							
	30.1	Practice rooms	200-300-500	0.60	80	22	100	–	–
	30.2	Dressing rooms	200-300-500	0.60	90	22	100	–	–
	30.3	Seating areas-maintenance, cleaning	150-200-300	0.40	80	22	50	–	–
	30.4	Stage area rigging	200-300-500	0.40	80	25	75	–	–
xxxi)	**31**	**Places of Public Assembly-Trade Fairs, Exhibition Halls**							
	31.1	General lighting	200-300-500	0.40	80	22	50	–	–
xxxii)	**32**	**Places of Public Assembly-Museums**							
	32.1	Exhibits, insensitive to light	–	–	80	–	–	–	–
	32.2	Exhibits sensitive to light	–	–	80	–	–	–	–
xxxiii)	**33**	**Places of Public Assembly-Libraries**							
	33.1	Bookshelves	150-200-300	0.40	80	19	–	–	–
	33.2	Reading area	300-500-750	0.60	80	19	100	75	50
	33.3	Counters	300-500-750	0.60	80	19	150	75	50
	33.4	General lighting	200-300-500	0.40	80	22	75	–	–
xxxiv)	**34**	**Places of Public Assembly-Car Parks (Indoor)**							
	34.1	Entry/exit ramps (during daylight hours)	200-300-500	0.40	70	25	75	–	–
	34.2	Entry/exit ramps (at night)	50-75-100	0.40	70	25	50	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	34.3	Traffic lanes, internal ramps and pedestrian paths	50-75-100	0.40	70	25	50	–	–
	34.4	Parking areas - Not open to public	50-75-100	0.25	70	–	50	–	–
	34.5	Parking areas - Open to public with a large number of users for example, shopping centres, arenas	100-150-200	0.40	70	–	50	–	–
	34.6	Ticket office	200-300-500	0.60	80	19	75	75	50
xxxv)	**35**	**Educational Premises-Nursery School, Play School**							
	35.1	Play room	200-300-500	0.40	80	22	100	75	50
	35.2	Nursery	200-300-500	0.40	80	22	100	75	50
	35.3	Handicraft room	200-300-500	0.60	80	19	100	75	50
xxxvi)	**36**	**Educational Premises-Educational Buildings**							
	36.1	Classroom-General activities	300-500-750	0.60	80	19	150	75	50
	36.2	Auditorium, lecture halls	300-500-750	0.60	80	19	150	75	50
	36.3	Attending lecture in seating areas in auditoriums and lecture halls	150-200-300	0.60	80	19	75	75	50
	36.4	Black, green and white boards	300-500-750	0.70	80	19	–	–	–
	36.5	Black, green and white boards in auditorium and lecture halls	300-500-750	0.60	80	19	–	–	–
	36.6	Projector and smartboard presentation	–	–	–	–	–	–	–
	36.7	Display board	150-200-300	0.60	80	19	–	–	–
	36.8	Demonstration table in auditoriums and lecture halls	500-750-1 000	0.70	80	19	–	–	–
	36.9	Light on teacher/presenter	–	–	80	–	150	–	–
	36.10	Light on podium area	200-300-500	0.70	80	–	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	36.11	Computer work only	200-300-500	0.60	80	19	100	75	50
	36.12	Art rooms in art schools	500-750-1 000	0.70	90	19	150	75	50
	36.13	Technical drawing rooms	500-750-1 000	0.60	80	19	150	75	50
	36.14	Practical rooms and laboratories	300-500-750	0.60	80	19	150	75	50
	36.15	Handcraft rooms	300-500-750	0.60	80	19	150	75	50
	36.16	Teaching workshop	300-500-750	0.60	80	19	150	75	50
	36.17	Preparation rooms and workshops	300-500-750	0.60	80	22	150	–	–
	36.18	Entrance halls	150-200-300	0.40	80	22	75	–	–
	36.19	Circulation areas, corridors	75-100-150	0.40	80	25	50	–	–
	36.20	Stairs	100-150-200	0.40	80	25	50	–	–
	36.21	Student common rooms and assembly halls	150-200-300	0.40	80	22	75	–	–
	36.22	Teachers rooms	200-300-500	0.60	80	19	100	75	50
	36.23	Library – Bookshelves	150-200-300	0.60	80	19	75	75	50
	36.24	Library - Reading areas	300-500-750	0.60	80	19	100	75	50
	36.25	Stock rooms for teaching materials	75–100-150	0.40	80	25	50	–	–
	36.26	Sports halls, gymnasiums, swimming pools	200–300–500	0.60	80	22	100	–	–
	36.27	School canteens	150-200-300	0.40	80	22	75	–	–
	36.28	Kitchen	300–500-750	0.60	80	22	100	–	–
xxxvii)	**37**	**Health Care Premises-Rooms for General Use**							
	37.1	Waiting rooms	150-200-300	0.40	80	22	75	75	30
	37.2	Corridors: During the day	75-100-150	0.40	80	22	50	50	30
	37.3	Corridors: Cleaning	75-100-150	0.40	80	22	50	50	30

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	37.4	Corridors: During the night	30-50-75	0.40	80	22	–	–	–
	37.5	Corridors with multi-purpose use (for example, pre-examination of patients)	150-200-300	0.60	80	22	75	75	50
	37.6	Day rooms	200-300-500	0.60	80	22	75	75	50
	37.7	Lifts for persons and visitors	75-100-150	0.60	80	22	50	50	30
	37.8	Service lifts	150-200-300	0.60	80	22	75	75	50
xxxviii)	**38**	**Health Care Premises-Staff Rooms**							
	38.1	Staff office	300-500-750	0.60	80	19	150	75	50
	38.2	Staff rooms	200-300-500	0.60	80	19	100	75	50
xxxix)	**39**	**Health Care Premises-Wards, Maternity Wards**							
	39.1	General lighting	75-100-150	0.70	80	19	50	50	30
	39.2	Reading lighting	200-300-500	0.70	80	19	100	75	50
	39.3	Wards-simple examinations	200-300-500	0.70	80	19	100	75	50
	39.4	Examination and treatment	750-1 000-1 500	0.70	90	19	150	75	50
	39.5	Night lighting, observation lighting	5	–	80	–	–	–	–
	39.6	Bathrooms and toilets for patients	150-200-300	0.70	90	22	75	75	50
xl)	**40**	**Health Care Premises-Examination Rooms (General)**							
	40.1	General lighting	300-500-750	0.60	90	19	150	75	50
	40.2	Examination and treatment	750-1 000-1 500	0.70	90	19	150	75	50
xli)	**41**	**Health Care Premises-Eye Examination Rooms**							
	41.1	General lighting	300-500-750	0.60	90	19	150	75	50
	41.2	Examination of the outer eye	750-1 000-1 500	–	90	–	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	41.3	Reading and colour vision tests with vision charts	300-500-750	0.70	90	19	150	75	50
xlii)	**42**	**Health Care Premises-Ear Examination Rooms**							
	42.1	General lighting	300-500-750	0.60	90	19	150	75	50
	42.2	Ear examination	750-1 000-1 500	–	90	–	150	75	50
xliii)	**43**	**Health Care Premises-Scanner Rooms**							
	43.1	General lighting	200-300-500	0.60	80	19	100	75	50
	43.2	Scanners with image enhancers and television systems	30-50-75	–	80	19	–	–	–
xliv)	**44**	**Health Care Premises-Delivery Rooms**							
	44.1	General lighting	200-300-500	0.60	90	19	100	75	50
	44.2	Examination and treatment	750-1 000-1 500	0.70	90	19	150	75	50
xlv)	**45**	**Health Care Premises-Treatment Rooms (General)**							
	45.1	Dialysis	300-500-750	0.60	80	19	150	75	50
	45.2	Dermatology	300-500-750	0.60	90	19	150	75	50
	45.3	Endoscopy	200-300-500	0.60	80	19	100	75	50
	45.4	Plastering	300-500-750	0.60	80	19	150	75	50
	45.5	Medical baths	200-300-500	0.60	80	19	100	75	50
	45.6	Massage and radiotherapy	200-300-500	0.60	80	19	100	75	50
xlvi)	**46**	**Health Care Premises-Operating Areas**							
	46.1	Pre-op and recovery rooms	500-750-1 000	0.60	90	19	150	75	50
	46.2	Operating cavity surround	750-1 000-1 500	0.60	90	19	150	75	50
	46.3	Operating theatre	750-100-1 500	0.60	90	19	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	46.4	Operating cavity	–	–	90	–	–	–	–
xlvii)	**47**	**Health Care Premises-Intensive Care Unit**							
	47.1	General lighting	200-300-500	0.60	90	19	50	50	30
	47.2	Simple examinations	300-500-750	0.60	90	19	100	75	50
	47.3	Examination and treatment	1 000-1 250 -1 500	0.70	90	19	150	75	50
	47.4	Night watch	20	–	90	19	–	–	–
xlviii)	**48**	**Health Care Premises-Dental Clinics/Units**							
	48.1	General lighting	300-500-750	0.60	90	19	150	75	50
	48.2	At the patient	750-1 000-1 500	0.70	90	–	150	75	50
	48.3	Operating cavity	–	–	–	–	–	–	–
	48.4	White teeth matching	–	–	–	–	–	–	–
xlix)	**49**	**Health Care Premises-Laboratories and Pharmacies**							
	49.1	General lighting	300-500-750	0.60	80	19	150	75	50
	49.2	Colour inspection	750-1 000- 1 500	0.70	90	19	150	75	50
l)	**50**	**Health Care Premises-Decontamination Rooms**							
	50.1	Sterilization	300-500-750	0.60	80	22	100	75	50
	50.2	Disinfection	300-500-750	0.60	80	22	100	75	50
li)	**51**	**Health Care Premises-Autopsy Rooms and Mortuaries**							
	51.1	General lighting	300-500-750	0.60	90	19	150	75	50
	51.2	Autopsy table and dissecting table	500-7500	0.70	90	–	150	75	50

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
lii)	**52**	**Transportation Areas-Airports**							
	52.1	Arrival and departure halls, baggage claim areas	150-200-300	0.40	80	22	75	–	–
	52.2	Connecting areas	100-150-200	0.40	80	22	50	–	–
	52.3	Information desks, check-in desks	300-500-750	0.70	80	19	150	–	–
	52.4	Customs and passport control desks	300-500-750	0.70	80	19	150	–	–
	52.5	Waiting areas	150-200-300	0.40	80	22	50	–	–
	52.6	Luggage storage rooms	150-200-300	0.40	80	25	50	–	–
	52.7	Security check areas	200-300-500	0.60	80	19	100	–	–
	52.8	Air traffic control tower	300-500-750	0.60	80	19	50	75	50
	52.9	Tasks in hangars: Testing and repair areas, engine test areas, measuring areas	300-500-750	0.60	80	22	50	–	–
liii)	**53**	**Transportation Areas-Railway Installations**							
	53.1	Fully enclosed platforms: Small number of passengers	30-50-75	0.30	80	–	–	–	–
	53.2	Fully enclosed platforms: Medium number of passengers	75-100-150	0.40	80	–	–	–	–
	53.3	Fully enclosed platforms: Large number of passengers	150-200-300	0.50	80	–	–	–	–
	53.4	Fully enclosed passenger subways (underpasses): Small number of passengers	30-50-75	0.30	80	–	–	–	–
	53.5	Fully enclosed passenger subways (underpasses): Medium number of passengers	75-100-150	0.40	80	–	–	–	–

Table 6 (*Continued*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_m$	U_o	R_a	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	53.6	Fully enclosed passenger subways (underpasses): Large number of passengers	150-200-300	0.50	80	–	–	–	–
	53.7	Stairs, escalators: Small number of passengers	30-50-75	0.30	80	–	–	–	–
	53.8	Stairs, escalators: Medium number of passengers	75-100-150	0.40	80	–	–	–	–
	53.9	Stairs, escalators: Large number of passengers	150-200-300	0.50	80	–	–	–	–
	53.10	Ticket hall and concourse	150-200-300	0.50	80	28	75	–	–
	53.11	Ticket counters and luggage offices	200-300-500	0.50	80	19	100	75	50
	53.12	Waiting rooms	150-200-300	0.40	80	22	75	–	–
	53.13	Entrance halls, station halls	150-200-300	0.40	80	–	75	–	–
	53.14	Switch and plant rooms	150-200-300	0.50	80	28	50	–	–
	53.15	Railway control centre (area of dispatcher)	150-200-300	0.50	80	19	–	–	–
	53.16	Access tunnels	30-50-75	0.40	20	–	–	–	–
	53.17	Assembly work in maintenance sheds - Rough	150-200-300	0.40	80	–	–	–	–
	53.18	Assembly work in maintenance sheds - Medium	200-300-500	0.50	80	–	–	–	–
	53.19	Assembly work in maintenance sheds - Fine	300-500 -750	0.60	80	–	–	–	–
	53.20	Assembly work in maintenance sheds precision	500-750-1 000	0.70	80	–	–	–	–

Table 6 (*Concluded*)

Sl No.	Ref No.	Type of Task/Activity Area	$\bar{E}_{m}$	U_{o}	R_{a}	R_{UGL}	$\bar{E}_{m,z}$	$\bar{E}_{m,wall}$	$\bar{E}_{m,ceiling}$
			lx				lx	lx	lx
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	53.21	Circulation areas for maintenance halls for railway vehicles (without additional vehicular traffic)	75-100-150	0.25	80	–	–	–	–
	53.22	Circulation areas for maintenance halls for railway vehicles (with additional vehicular traffic)	100-150-200	0.40	80	–	–	–	–

NOTES

1 For 33.1, illuminance level mentioned is vertical at bookshelves.

2 For 36.4, 36.7 and 36.23, illuminance level mentioned is vertical illuminance.

3 For 40.1 and 40.2, preferred colour temperature should be 4 000 K to 5 000 K.

4 For 41.1, preferred colour temperature should be 4 000 K to 5 000 K.

5 For 41.2 and 41.3, preferred colour temperature should be 4 000 K to 6 500 K.

6 For 46.1, 46.2, 46.3 and 46.4, additional local lighting to be provided to achieve 10 000 lux to 50 000 lux.

4.1.8 *Colour Rendering*

The colour appearance of light and its colour rendering capability are different aspects of the light sources. A faithful reproduction of an object colour depends on the colour rendering capability of the light source. In 1965, International Commission on Illumination (CIE) developed a quantitative method of assignment of colour rendering property and is denoted as Colour Rendering Index (CRI).

CRI is arrived at by a test by which a number of specified samples are tested under a standard or reference light source and the chromaticity coordinate are plotted on the IE triangle as given in Fig. 7 of Part 2 'Physics of Light, Section 3 Colour' of good practice [D-1(3)]. The same test is repeated under the source under test and corresponding chromaticity coordinate are plotted on the same plot. The difference between the position of each sample for test and standard source is measured to scale. The general colour rendering index (R_a) is obtained by the average value for eight samples {*see* Fig. 8 of Part 2 'Physics of Light, Section 3 Colour' of good practice [D-1(3)]}. For perfect agreement of colour, the R1 value should be 100. In general:

$$R_a = 1/(R1 + R2 + R3 + R4 + \ldots\ldots + R8)$$

The specific colour rendering index for an individual sample is given by:

$$R_i = 100 - 4.6\Delta E_i$$

where

ΔE_i = chromaticity shift on the CIE chromaticity diagram for each sample.

From the obtained value of R_a, as calculated above, the colour rendering shall be evaluated as mentioned in the following table:

Sl No.	*Colour Rendering Evaluated*	R_a (*General Colour Rendering Index*)
(1)	(2)	(3)
i)	Excellent	$90 \leq R_a < 100$
ii)	Good	$80 \leq R_a < 90$
iii)	Moderate	$70 \leq R_a < 80$

4.1.9 For detailed information regarding principles of good lighting, reference may be made to good practice [D-1(4)].

4.2 Daylighting

While the sun is the primary source of natural light, the daylight received at the Earth's surface comprises two components: direct solar illuminance and sky illuminance (diffused light from the sky dome). For effective daylighting design in buildings, direct solar illuminance is typically excluded due to the potential for glare, excessive contrast and thermal discomfort (overheating). Glare from direct sun exposure in occupied zones should be mitigated through appropriate window orientation, sizing and external/internal shading strategies. Accordingly, daylighting strategies should emphasize sky illuminance, which provides diffused uniform light conducive to visual comfort and energy efficiency. By relying on sky illuminance, daylighting design achieves balanced illumination, reduces glare and cooling loads, and supports both visual comfort and sustainable building performance.

4.2.1 The relative amount of sky illuminance at a given point-in-time depends on the position of the sun defined by its altitude, which in turn, varies with the latitude of the locality, the day of the year and the time of the day, as indicated in Table 7.

4.2.2 The external available horizontal sky illuminance (diffuse illuminance) values which are exceeded for about 90 percent of the daytime working hours may be taken as outdoor design illuminance values for ensuring adequacy of daylighting design. The outdoor design sky illuminance varies for different climatic regions of the country. The recommended design sky illuminance values are 6 800 lux for cold climate, 8 000 lux for composite climate, 9 000 lux for warm humid climate, 9 000 lux for temperate climate and 10 500 lux for hot-dry climate. For integration with the artificial lighting during daytime working hours an increase of 500 lux in the recommended sky design illuminance for daylighting is suggested.

4.2.3 The daylight factor is dependent on the sky luminance distribution, which varies with atmospheric conditions. A clear design sky with its non-uniform distribution of luminance is adopted for the purposes of design in this section.

4.2.4 *Components of Daylight Factor*

Daylight factor is the sum of all the daylight reaching on an indoor reference point from the following sources:

a) Direct sky visible from the point;

b) External surfaces reflecting light directly (*see* Note 1) to the point; and

c) Internal surfaces reflecting and inter-reflecting light to the point.

NOTES

1 External surface reflection may be computed approximately only for points at the centre of the room, and for detailed analysis the procedures are complicated and these may be ignored for actual calculations.

2 Each of the three components, when expressed as a ratio or percent of the simultaneous external illuminance on the horizontal plane, defines respectively the sky component (SC), the external reflected component (ERC) and the internal reflected component (IRC) of the daylight factor.

4.2.4.1 The daylight factors on the horizontal plane only are usually taken, as the working plane in a room is generally horizontal; however, the factors in vertical planes should also be considered when specifying daylighting values for special cases, such as daylighting on classrooms, blackboards, pictures and paintings hung on walls.

4.2.5 *Sky Component* (*SC*)

Sky component for a window of any size is computed by the use of the appropriate Table 20, Table 21 and Table 22 of Annex B.

The recommended sky component level should be ensured generally on the working plane at the following positions:

a) At a distance of 3 m to 3.75 m from the window along the central line perpendicular to the window;

b) At the centre of the room if more appropriate; and

c) At fixed locations, such as school desks, black-boards and office tables.

The daylight area of the prescribed sky component should not normally be less than half the total area of the room.

4.2.5.1 The values obtainable from the tables are for rectangular, open unglazed windows, with no external obstructions. The values shall be corrected for the presence of window bars, glazing and external obstructions, if any. This assumes the maintenance of a regular cleaning schedule.

4.2.5.2 *Corrections for window bars*

The corrections for window bars shall be made by multiplying the values read from tables in Annex B by a factor equal to the ratio of the clear opening to the overall opening.

Table 7 Solar Altitudes (to the Nearest Degree) for Indian Latitudes

(*Clause* 4.2.1)

Sl No.	Period of Year	22 June						21 March and 23 September						22 December					
	Hours of Day (Sun or Solar) → Latitude ↓	07 00 17 00	08 00 16 00	09 00 15 00	10 00 14 00	11 00 13 00	12 00 -	07 00 17 00	08 00 16 00	09 00 15 00	10 00 14 00	11 00 13 00	12 00 -	07 00 17 00	08 00 16 00	09 00 15 00	10 00 14 00	11 00 13 00	12 00 -
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
i)	10ÁN	18	31	45	58	70	77	15	30	44	59	72	80	9	23	35	46	53	57
ii)	13ÁN	19	32	46	60	72	80	15	29	44	58	70	77	8	21	33	43	51	54
iii)	16ÁN	20	33	47	61	74	83	14	29	43	56	68	74	7	19	31	41	48	51
iv)	19ÁN	21	34	48	62	75	86	14	28	42	55	66	71	5	18	29	48	45	48
v)	22ÁN	22	35	49	62	75	89	14	28	41	53	64	68	4	16	27	36	42	45
vi)	25ÁN	23	36	49	63	76	88	13	27	40	52	61	65	3	14	25	34	39	42
vii)	28ÁN	23	36	49	63	76	86	13	26	39	50	59	62	1	13	23	31	37	39
viii)	31ÁN	24	37	50	62	75	82	13	25	37	48	56	56	–	11	21	28	34	36
ix)	34ÁN	25	37	49	62	73	79	12	25	36	46	53	56	–	9	18	26	31	33

4.2.5.3 *Correction for glazing*

Where windows are glazed, the sky components obtained from Annex B shall be reduced by 10 percent to 20 percent, provided the panes are of clear and clean glass. Where glass is of the frosted (ground) type, the sky components read from Annex A may be reduced by 15 percent to 30 percent. In case of tinted or reflective glass the reduction is about 50 percent. Higher indicated correction corresponds to larger windows and/or near reference points. In the case of openings and glazings which are not vertical, suitable correction shall be taken into account.

4.2.5.4 *Correction for external obstructions*

There is no separate correction, except that the values from tables in Annex B shall be read only for the unobstructed portions of the window.

4.2.6 *External Reflected Component* (*ERC*)

The value of the sky component corresponding to the portion of the window obstructed by the external obstructions may be found by the use of methods described in Annex C of good practice [D-1(5)].

These values when multiplied by the correction factors, corresponding to the mean elevation of obstruction from the point in question as given in Table 8, can be taken as the external reflected components for that point.

For method of calculating ERC, reference may be made to accepted standard {*see* examples given in Annex B of good practice [D-1(5)]}.

Table 8 Correction Factor for ERC

(*Clause* 4.2.6)

Sl No.	Mean Angle of Elevation	Correction Factor
(1)	(2)	(3)
i)	5°	0.086
ii)	15°	0.086
iii)	25°	0.142
iv)	35°	0.192
v)	45°	0.226
vi)	55°	0.274
vii)	65°	0.304
viii)	75°	0.324
ix)	85°	0.334

4.2.7 *Internal Reflected Component* (*IRC*)

The component of daylight factor contributed by reflection from the inside surfaces varies directly as the window area and inversely as the total area of internal surfaces and depends on the reflection factor of the floor, wall and roof surfaces inside and of the ground outside. For rooms white-washed on walls and ceiling and windows of normal sizes, the IRC will have sizeable value even at points far away from the window. External obstructions, when present, will proportionately reduce IRC. Where accurate values of IRC are desired, the same may be done in accordance with the good practice [D-1(5)].

4.2.8 *General Principles of Openings to Afford Good Lighting*

4.2.8.1 Generally, while taller openings give greater penetrations, broader openings give better distribution of light. It is preferable that some area of the sky at an altitude of 20º to 25º should light up the working plane.

4.2.8.2 Broader openings may also be equally or more efficient, provided their sills are raised by 300 mm to 600 mm above the working plane.

NOTE — It is to be noted that while placing window with a high sill level might help natural lighting, this is likely to reduce ventilation at work levels. While designing the opening for ventilation also, a compromise may be made by providing the sill level about 150 mm below the head level of workers.

4.2.8.3 For a given penetration, a number of small openings properly positioned along the same, adjacent or opposite walls will give better distribution of illumination than a single large opening. The sky component at any point, due to a number of openings may be easily determined from the corresponding sky component contour charts appropriately superposed. The sum of the individual sky component for each opening at the point gives the overall component due to all the openings. The same charts may also facilitate easy drawing of sky component contours due to multiple openings.

4.2.8.4 Unilateral lighting from side openings will, in general, be unsatisfactory if the effective width of the room is more than 2 to 2.5 times the distance from the floor to the top of the opening. In such cases provision of light shelves is always advantageous.

4.2.8.5 Openings on two opposite sides will give greater uniformity of internal daylight illumination, especially when the room is 7 m or more across. They also minimize glare by illuminating the wall surrounding each of the opposing openings. Side openings on one side and clerestory openings on the opposite side may be provided where the situation so requires.

4.2.8.6 Cross-lighting with openings on adjacent walls tends to increase the diffused lighting within a room.

4.2.8.7 Openings in deep reveals tend to minimize glare effects.

4.2.8.8 Openings shall be provided with *Chajjahs*, louvers, baffles or other shading devices to exclude, as far as possible, direct sunlight entering the room. *Chajjahs*, louvers, etc, reduce the effective height of the opening for which due allowance shall be made. Broad and low openings are, in general, much easier to shade against sunlight entry. Direct sunlight, when it enters, increases the inside illuminance very considerably. Glare will result if it falls on walls at low angles, more so than when it falls on floors, especially when the floors are dark coloured or less reflective.

4.2.8.9 Light control media, such as translucent glass panes (opal or matt) surfaced by grinding, etching or sandblasting, configurated or corrugated glass, certain types of prismatic glass, tinted glass and glass blasts are often used. They should be provided, either fixed or movable outside or inside, especially in the upper portions of the openings. The lower portions are usually left clear to afford desirable view. The chief purpose of such fixtures is to reflect part of the light on to the roof and thereby increase the diffuse lighting within, light up the farther areas in the room and thereby produce a more uniform illumination throughout. They will also prevent the opening causing serious glare discomfort to the occupants but will provide some glare when illuminated by direct sunlight.

4.2.9 *Availability of Daylight in Multistoreyed Block*

Proper planning and layout of building can add appreciably to daylighting illumination inside. Certain dispositions of building masses offer much less mutual obstruction to daylight than others and have a significant relevance, especially when intensive site planning is undertaken. As guidance, the relative availability of daylight in multi-storeyed blocks (up to 4 storeys) of different relative orientations may be taken as given in Table 9.

4.2.10 For specified requirements for daylighting of special occupancies and areas, reference may be made to good practice [D-1(6)]. For using annual climate-based daylight assessment of design, a reference may be made to specialist literature (*see* Note). It also presents a computer simulation-based approach for assessment of adequacy of daylight in indoor spaces.

NOTE — The specialist literature may be Chapter 5 of Energy Conservation and Sustainable Building Code 2024.

Table 9 Relative Availability of Daylight on the Window Plane at Ground Level in Four-Storeyed Building Blocks (Clear Design-Sky as Basis, Daylight Availability Taken as Unity on an Unobstructed Facade, Values are for the Centre of the Blocks)

(*Clause* 4.2.9)

Sl No.	Distance of Separation Between Blocks	Infinitely Long Parallel Blocks	Parallel Blocks Facing Each Other (Length = 2 × Height)	Parallel Blocks facing Gaps Between Opposite Blocks (Length = 2 × Height)
(1)	(2)	(3)	(4)	(5)
i)	0.5 Ht	0.15	0.15	0.25
ii)	1.0 Ht	0.30	0.32	0.38
iii)	1.5 Ht	0.40	0.50	0.55
iv)	2.0 Ht	0.50	0.60	0.68

NOTE — Ht = Height of the building.

4.3 Artificial Lighting

4.3.1 Artificial lighting may have to be provided,

a) Where the recommended illumination levels have to be obtained by artificial lighting only;

b) To supplement daylighting when the level of illumination falls below the recommended value; and

c) Where visual task may demand a higher level of illumination.

4.3.2 *Artificial Lighting Design for Interiors*

For general lighting purposes, the recommended practice is to design for a level of illumination on the working plane on the basis of the recommended levels for visual tasks given in Table 6 by a method called 'Lumen method'. In order to make the necessary detailed calculations concerning the type and quantity of lighting equipment necessary, advance information on the surface reflectances of walls, ceilings and floors is required. Similarly, calculations concerning the brightness ratio in the interior call for details of the interior décor and furnishing. Stepwise guidance regarding designing the interior lighting systems for a building using the 'Lumen method' is given in **4.3.2.1** to **4.3.2.4**.

4.3.2.1 *Determination of the illumination level*

Recommended value of illumination shall be taken from Table 6, depending upon the type of work to be carried out in the location in question and the visual tasks involved.

4.3.2.2 *Selection of the light sources and luminaires*

The selection of light sources and luminaires depends on the choice of lighting system, namely, general lighting, directional lighting and localized or local lighting.

4.3.2.3 *Determination of the luminous flux*

a) The luminous flux (ϕ) reaching the working plane depends upon the following:

1) Lumen output of the lamps;

2) Type of luminaire;

3) Proportion of the room (room index) (k_r);

4) Reflectance of internal surfaces of the room;

5) Depreciation in the lumen output of the lamps after burning their rated life; and

6) Depreciation due to dirt collection on luminaires and room surface.

b) Coefficient of utilization or utilization factor;

1) The compilation of tables for the utilization factor requires a considerable amount of calculations, especially if these tables have to cover a wide range of lighting practices. For every luminaire, the exact light distribution has to be measured in the laboratory and their efficiencies have to be calculated and measured exactly. These measurements comprise,

i) the luminous flux radiated by the luminaires directly to the measuring surface;

ii) the luminous flux reflected and re-reflected by the ceiling and the walls to the measuring surface; and

iii) the inter-reflections between the ceiling and wall which result in the measuring surface receiving additional luminous flux.

All these measurements have to be made for different reflection factors of the ceiling and the walls for all necessary room indices. These tables have also to indicate the maintenance factor to be taken for the luminous flux depreciation throughout the life of an installation due to ageing of the lamp and owing to the deposition of dirt on the lamps and luminaires and room surfaces.

2) The values of the reflection factor of the ceiling and of the wall are as follows:

White and very light colours	0.7
Light colours	0.5
Middle tints	0.3
Dark colours	0.1

For the walls this should be estimated, taking into account the influence of the windows without curtains, shelves, *Almirahs* and doors with different colours, etc.

c) Calculation for determining the luminous flux {see Table 22 of accepted standard [D-1(7)]}

$$E_{\mathrm{av}} = \frac{\mu . \phi}{A}$$

or, $\phi = \frac{E_a A}{\mu}$, for new condition, and

$\phi = \frac{E\mathrm{av}.A}{\mu . d}$, for working condition.

where

E_{av} = average illumination level required on the working plane, in lux;

μ = utilization factor in new conditions;

ϕ = total luminous flux of the light sources installed in the room, in lumens;

A = area of the working plane, in m^2; and

d = maintenance factor.

In practice, it is easier to calculate straightaway the number of lamps or luminaires from:

$$N_{lamp} = \frac{E_{av} . A}{\mu . d . \phi_{lamp}}$$

$$N_{luminaries} = \frac{E_{av} . A}{\mu . d . \phi_{luminaries}}$$

where

N_{lamp} = total number of lamps;

ϕ_{lamp} = luminous flux of each lamp, in lumens;

$N_{luminaires}$ = total number of luminaires; and

$\phi_{luminaries}$ = luminous flux of each luminaire, in lumens.

4.3.2.4 *Arrangement of the luminaires*

This is done to achieve better uniformly distributed illumination. The location of the luminaires has an important effect on the utilization factor:

a) In general, luminaires are spaced 'x' metre apart in either direction, while the distance of the end luminaire from the wall is '0.5 x' metre. The distance 'x' is more or less equal to the mounting height 'H_m' between the luminaire and the working plane. The utilization factor tables are calculated for this arrangement of luminaires {*see* Table 22 of the accepted standard [D-1(7)]}.

b) For small rooms where the room index (k_r) is less than 1, the distance 'x' should always be less than H_m, since otherwise luminaires cannot be properly located. In most cases of such rooms, four or two luminaires are placed for good general lighting. If, however, in such rooms only one luminaire is installed in the middle, higher utilization factors are obtained, but the uniformity of distribution is poor. For such cases, references should be made to the additional tables for k_r = 0.6 to 1.25 for luminaires located centrally.

4.3.3 *Artificial Lighting to Supplement Daylighting*

4.3.3.1 The need for general supplementary artificial lighting arises due to diminution of daylighting beyond design hours, that is, for solar altitude below 15° or when dark cloudy conditions occur.

4.3.3.2 The need may also arise for providing artificial lighting during the day in the innermost parts of the building which cannot be adequately provided with daylighting, or when the outside windows are not of adequate size or when there are unavoidable external obstructions to the incoming daylighting.

4.3.3.3 The need for supplementary lighting during the day arises, particularly when the daylighting on the working plane falls below 100 lux and the surrounding luminance drops below 19 cd/m^2.

4.3.3.4 The requirement of supplementary artificial lighting increases with the decrease in daylighting availability. Therefore, conditions near sunset or sunrise or equivalent conditions due to clouds or obstructions, etc, represent the worst conditions when the supplementary lighting is most needed.

4.3.3.5 The requirement of supplementary artificial lighting when daylighting availability becomes poor may be determined from Fig. 3 for an assumed ceiling height of 3.0 m, depending upon floor area, fenestration percentage and room surface reflectance. Cool daylight fluorescent tubes are recommended with semi-direct luminaires. To ensure a good distribution of illumination, the mounting height should be between 1.5 m and 2.0 m above the work plane for a separation of 2.0 m to 3.0 m between the luminaires. Also, the number of lamps should preferably be more in the rear half of the room than in the vicinity of windows. The following steps may be followed for using Fig. 3 for determining the number of fluorescent tubes required for supplementary daylighting:

a) Determine fenestration percentage of the floor area, that is:

$$\frac{Window\ area}{Floor\ area} \times 100$$

b) In Fig. 3, refer to the curve corresponding to the percent fenestration determined above and the set of reflectances of ceiling, walls and floor actually provided; and

c) For the referred curve of Fig. 3 read, along the ordinate, the number of 40 W fluorescent tubes required, corresponding to the given floor area on the abscissa.

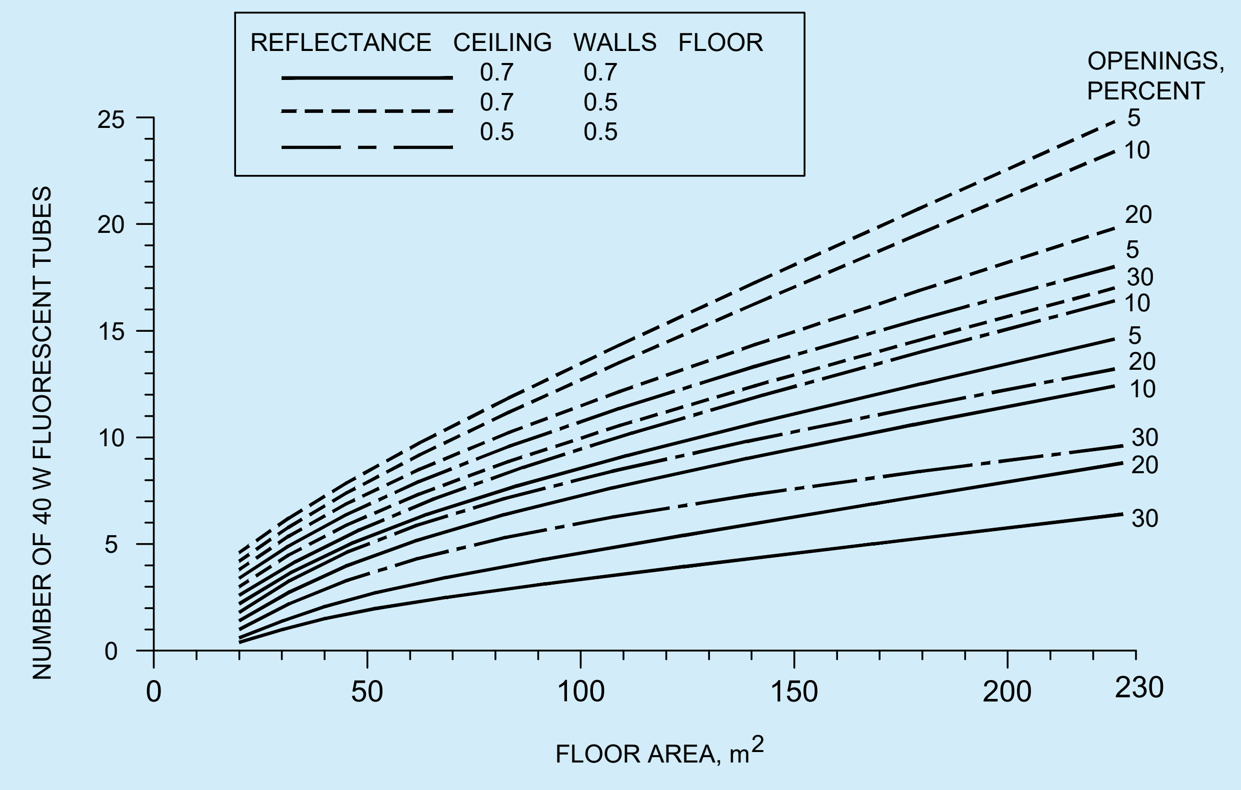


FIG. 3 SUPPLEMENTARY ARTIFICIAL LIGHTING FOR 40 W FLUORESCENT TUBES

4.3.4 For detailed information on the design aspects and principles of artificial lighting, reference may be made to good practice [D-1(4)].

4.3.5 For specific requirements for lighting of special occupancies and areas, reference may be made to good practice [D-1(8)].

4.3.6 Electrical installation aspect for artificial lighting should be in accordance with Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS.

4.4 Energy Conservation in Lighting

4.4.1 A substantial portion of the energy consumed on lighting may be saved by utilization of daylight and rational design of supplementary artificial lights.

4.4.2 Daytime use of artificial lights may be minimized by proper design of windows for adequate daylight indoors. Daylighting design should be according to **4.2**.

4.4.3 Fenestration expressed as percentage of floor area required for satisfactory visual performance of a few tasks for different separation to height (*S*/*H*) ratio of external obstructions such as opposite buildings may be obtained from the design nomogram (*see* Fig. 4). The obstructions at a distance of three times their height or more (*S*/*H* > 3) from a window facade are not significant and a window facing such an obstruction may be regarded as a case of unobstructed window.

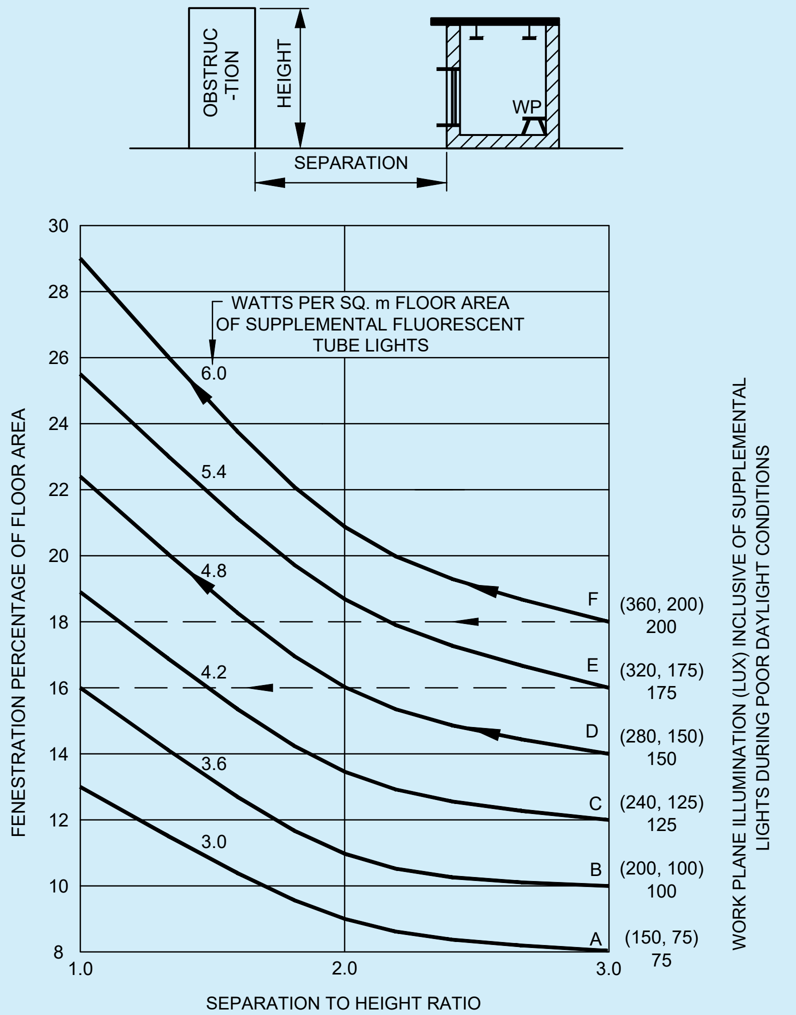


FIG. 4 NOMOGRAPH FOR DAYLIGHTING AND SUPPLEMENTAL LIGHTING DESIGN OF BUILDINGS

4.4.3.1 The nomogram consists of horizontal lines indicating fenestration percentage of floor area and vertical lines indicating the separation to height ratio of external obstructions such as opposite buildings. Any vertical line for separation to height ratio other than already shown in the nomogram (1.0, 2.0 and 3.0) may be drawn by designer, if required. For cases where there is no obstruction, the ordinate corresponding to the value 3.0 may be used. The value of percentage fenestration and separation to height ratio are marked on left hand ordinate and abscissa respectively. The illumination levels are marked on the right hand ordinate. The values given within brackets are the illumination levels on the work plane at centre and rear of the room. The wattage of fluorescent tubes required per square metre of the floor area for different illumination levels is shown on each curve.

4.4.3.2 Following assumptions have been made in the construction of the nomogram:

a) An average interior finish with ceiling white, walls off white and floor grey has been assumed;

b) Ceiling height of 3 m, room depths up to 10 m and floor area between 30 m^2 and 50 m^2 have been assumed. For floor area beyond 50 m^2 and less than 30 m^2, the values of percent fenestration as well as wattage/m^2 should be multiplied by a factor of 0.85 and 1.15, respectively;

c) It is assumed that windows are of metallic sashes with louvers of width up to 600 mm or a *Chhajja* (balcony projection) at ceiling level of width up to 2.0 m. For wooden sashes, the window area should be increased by a factor of about 1.1; and

d) Luminaires emanating more light in the downward direction than upward direction (such as reflectors with or without diffusing plastics) and mounted at a height of 1.5 m to 2.0 m above the work plane have been considered.

4.4.3.3 *Method of use*

The following steps shall be followed for the use of nomogram:

a) *Step* 1 — Decide the desired illumination level depending upon the task illumination requirement in the proposed room and read the value of watt per square metre on the curve corresponding to the required illumination level;

b) *Step* 2 — Fix the vertical line corresponding to the given separation to height ratio of opposite buildings on the abscissa. From the point of intersection of this vertical line and the above curve move along horizontal and read the value of fenestration percent on the left-hand ordinate; and

c) *Step* 3 — If the floor area is greater than 50 m^2 or if it is less than 30 m^2, the value of watt/m^2 as well as fenestration percent may be easily determined for adequate daylighting and supplemental artificial lighting for design purposes. However, if the fenestration provided is less than the required value, the wattage of supplementary artificial lights should be increased proportionately to make up for the deficiency of natural illumination.

4.4.4 For good distribution of day light on the working plane in a room, window height, window width and height of sill should be chosen in accordance with the following recommendations:

a) In office buildings windows of height 1.2 m or more in the centre of a bay with sill level at 1.0 to 1.2 m above floor and in residential buildings windows of height 1.0 m to 1.1 m with sill height as 0.9 m to 0.7 m above floor are recommended for good distribution of daylight indoors. Window width can accordingly be adjusted depending upon the required fenestration percentage of the floor area;

b) If the room depth is more than 10 m, windows should be provided on opposite sides for bilateral lighting; and

c) It is desirable to have a white finish for ceiling and off white (light colour) to white for walls. There is about 7 percent improvement in lighting levels in changing the finish of walls from moderate to white.

4.4.5 For good distribution and integration of daylight with artificial lights the following guidelines are recommended:

a) Employ cool daylight fluorescent tubes for supplementary artificial lighting;

b) Distribute luminaries with a separation of 2 m to 3 m in each bay of 3 m to 4 m width; and

c) Provide more supplementary lights such as twin tube luminaries in work areas where daylight is expected to be poor for example in the rear region of a room having single window and in the central region of a room having windows on opposite walls. In the vicinity of windows only single tube luminaries should be provided.

4.4.6 *Artificial Lighting*

Energy conservation in lighting is affected by reducing wastage and using energy effective lamps and luminaires without sacrificing lighting quality. Measures to be followed comprise utilization of daylight, energy effective artificial lighting design by providing required illumination where needed, turning off artificial lights when not needed, maintaining lighter finishes of ceiling, walls and furnishings, and implementing periodic schedule for cleaning of luminaires and group replacement of lamps at suitable intervals. Choice of light sources with higher luminous efficacy and luminaires with appropriate light distribution is the most effective means of energy saving in lighting. However, choice of light sources also depends on the other lighting quality parameters like colour rendering index and colour temperature or appearance. The choice of light sources type also depends on the desired mounting height in the interiors.

4.4.6.1 *Efficient artificial light sources and luminaires*

Luminous efficacies of some of the lamps used in lighting of buildings are given in Table 10 along with average life in burning hours, colour rendering index and colour temperature. Given the high luminous efficacy that can be achieved with most LED lighting, energy efficiency in lighting can be achieved via space appropriate zoning, identifying the application type (task versus ambient) and careful consideration of lighting quality requirements.

Table 10 Luminous Efficacy, Life, Lumen Maintenance and Colour Rendition of Light Sources

(*Clause* 4.4.6.1)

Sl No.	Light Source	Wattage Range W	Efficacy Lm/W	Average Life h	Lumen Maintenance	Colour Rendition
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Incandescent lamps	15 to 200	12 to 20	500 to 1 000	Fair to good	Very good
ii)	Tungsten halogen	300 to 1 500	20 to 27	200 to 2 000	Good to very good	Very good
iii)	Slim line fluorescent	18 to 58	57 to 67	5 000	Fair to good	Good
iv)	High pressure mercury vapour lamps	60 to 1 000	50 to 65	5 000	Very low to fair	Federate
v)	Blended-light lamps	160 to 250	20 to 30	5 000	Low to fair	Federate
vi)	High pressure sodium vapour lamps	50 to 1 000	90 to 125	10 000 to 15 000	Fair to good	Low to good
vii)	Metal halide lamps	35 to 2 000	80 to 95	4 000 to 10 000	Very low	Very good
viii)	Low pressure sodium	10 to 180	100 to 200	10 000 to 20 000	Good to very good	Poor
ix)	LED	0.5 to 2.0	60 to 100	10 000	Very good	Good for white LED

NOTES

1 This includes lamps and wattages currently in use in buildings in India.

2 Luminous efficacy varies with the wattage of the lamp.

3 Average life values are from available Indian Standards. Where Indian Standard is not available, values given are only indicative.

4 For exact values, it is advisable to contact manufacturers.

4.4.6.2 For the same lumen output, it is possible to save 50 percent to 70 percent energy if CFL lamps are replaced with induction lighting, and 40 percent to 60 percent if replaced with LED lamps. Similar energy effective solutions are to be chosen for every application area.

Similarly, with white fluorescent tubes recommended for corridors and staircases, the electrical consumption reduces to 1/4.5 of the energy consumption with incandescent lamps.

4.4.6.3 Efficient luminaire also plays an important role in energy conservation in lighting. The choice of a luminaire should be such that it is efficient not only initially but also throughout its life. Following luminaries are recommended for different locations:

a) For offices semi-direct type of luminaries are recommended so that both the work plane illumination and surround luminance can be effectively enhanced;

b) For corridors and stair cases direct type of luminaries with wide spread of light distributions are recommended; and

c) In residential buildings, glare free wide spread of light distributions are recommended.

4.4.7 *Cleaning Schedule for Window Panes and Luminaires*

Adequate schedule for cleaning of window panes and luminaries will result in significant advantage of enhanced daylight and lumen output from luminaries. Frequent cleaning of windows will tend to reduce the duration over which artificial lights will be used and minimize the wastage of energy. Depending upon the location of the building a cleaning interval of three to six months is recommended for cleaning of luminaires.

4.4.8 *Photocontrols for Artificial Lights*

There is a considerable wastage of electrical energy in lighting of buildings due to carelessness in switching off lights even when sufficient daylight is available indoors. In offices and commercial buildings, occupants may switch on lights in the morning and keep them on throughout the day. When sufficient daylight is available inside, suitable photocontrols can be employed to switch off the artificial lights and thus prevent the wastage of energy.

The photocontrol should have the following features:

a) An integrated photocontrol system continually measures the amount of visible light under the lighting fixture and maintains the lux levels as referred in Table 6;

b) An integrated photocontrol system should maintain six daylighting scenarios that can be adjusted by the user namely; daytime occupied, daytime unoccupied, sunset occupied, sunset unoccupied, night time occupied and night time unoccupied;

c) The photocontrol sensor should have a 60° cone of reference to measure the amount of light on the work surface;

d) Photocontrol, along with supporting lighting based on influx of natural light, should have controls for intelligent 'occupancy based shut-off' system to control which areas need lighting at any moment; and

e) The controls should maintain a decent standard of 'ease of use' for occupants as well as building staff.

4.4.9 *Solar Photovoltaic Systems* (*SPV*)

Solar photovoltaic system enables direct conversion of sunlight into electricity and is viable option for lighting purpose in remote non grid areas. The common SPV lighting systems are:

a) Solar lantern;

b) Fixed type solar home lighting system; and

c) Street lighting system.

4.4.9.1 SPV lighting system should preferably be provided with CFL for energy efficiency.

4.4.9.2 Invertors used in buildings for supplying electricity during the power cut period should be charged through SPV system.

4.4.9.3 Regular maintenance of SPV system is necessary for its satisfactory functioning.

4.4.10 Lighting shelves and light pipes may be explored for utilization and integration in the lighting design.

4.5 Circadian Lighting

Light is fundamental to human health and well-being, extending beyond its role in vision. It is the principal regulator of our circadian rhythm, profoundly influencing visual experience, physiological processes and emotional state. Thus, human-centric lighting, designed for human health and well-being, is essential.

Human physiology is evolved under natural sunlight. Ideally, we would experience abundant daylight exposure, gradually decreasing in the evening, with minimal light at night. Modern indoor lifestyles disrupt this natural rhythm. Optimizing indoor environments to emulate this 24 h light-dark cycle can significantly improve sleep quality and promote rejuvenation, with the added benefit of naturally sustained energy.

The influence of light on human health stems from its role in regulating the internal clock of the body. The circadian rhythm governs diverse physiological functions, from sleep-wake cycles and mood to overall health. This system is orchestrated by the suprachiasmatic nucleus (SCN) in the brain, which relies on consistent light-dark patterns, reflecting our attunement to the diurnal and seasonal rhythms of daylight availability.

The melanopic daylight efficacy ratio (melanopic DER) and melanopic equivalent daylight illuminance (melanopic EDI) provide metrics for assessing light's efficacy in supporting our biological rhythms.

In practice, morning light acts as a critical entraining cue, signalling the body to transition to a state of wakefulness and activity. Exposure to bright light during the day further strengthens the circadian rhythm, which plays a vital role in regulating sleep-wake cycles, daytime engagement, and mood. Conversely, darkness at night-time is essential for promoting restorative sleep, a process crucial for overall health and well-being.

NOTE — The early 2 000 s saw the discovery of specialized retinal light-sensitive cells, peaking at around 480 nm (cyan), leading to the concepts of 'biologically active' or 'circadian light'.

4.5.1 *Need for Supplementary Artificial Circadian Lighting*

Artificial circadian lighting is advised in spaces where permanent day-active occupants spend more than 4 h between sunrise and sunset and do not have a direct line of sight to any external windows from their general working area.

In spaces where artificial circadian lighting cannot be incorporated due to functional considerations (for example, museums, movie theatres), break rooms for day-active permanent occupants may be equipped with circadian lighting.

4.5.2 *Characteristics of Supplementary Artificial Circadian Lighting*

In spaces where supplementary artificial circadian lighting is implemented, it should have the following features:

a) The lighting system shall include dynamic controls that enable gradual changes in light intensity and colour temperature to mimic natural daylight cycles; and

b) Lux levels (vertical illuminance at the occupant positions) shall be modulated based on time of day to ensure sufficient light exposure during the day and reduced exposure in the evening time following the best-practice for circadian lighting design.

4.6 Outdoor Lighting

Outdoor lighting plays a crucial role in ensuring safety, security and functionality in various environments, while contributing to energy conservation and minimizing environmental impacts. To achieve these objectives, outdoor lighting should be designed and managed in accordance with the good practice [D-1(3)], focussing on energy efficiency and minimizing light pollution.

4.6.1 *Lighting Design and Light Zones*

Outdoor lighting designs should adhere to energy efficiency measures and address the requirements of different light zones, as per the good practice [D-1(3)]. The following guidelines should be considered for the design in various zone:

a) *Urban zones* — Lighting intensity should be optimized for high activity areas, using energy-efficient technologies such as LEDs and lighting controls to ensure energy savings;

b) *Suburban zones* — Lighting should be designed for moderate activity areas with an emphasis on reducing unnecessary energy consumption, utilizing dimming controls or motion sensors; and

c) *Rural zones* — Lighting should be kept minimal, ensuring safety while limiting light pollution. Energy-efficient, low-intensity lighting should be implemented.

These zones should serve as guidelines and lighting levels should align with the specific needs of the location while minimizing energy consumption as required by specialist literature.

NOTE — The specialist literature may be Energy Conservation and Sustainable Building Code 2024.

4.6.2 *Energy Efficiency and Light Pollution Mitigation*

To comply with the good practice [D-1(3)] and reduce light pollution, outdoor lighting systems should incorporate the following:

a) *Energy-efficient light sources* — Use of LEDs or other high-efficiency light sources with appropriate lumen output to reduce energy consumption;

b) *Fully shielded and full cut-off fixtures* — These should be implemented to prevent upward light emission and sky glow, focusing light where needed to minimize spill over and improve energy use;

c) *Lighting controls* — The good practice [D-1(3)] compliant controls, such as timers, dimmers, and motion sensors, should be employed to reduce lighting during periods of low activity, contributing to energy savings; and

d) *Luminaire efficacy* — All outdoor lighting luminaires should meet the minimum efficacy requirements as specified in the good practice [D-1(3)] for energy conservation and reduced power consumption.

4.6.3 *Effects of Light Pollution on Life*

Excessive outdoor lighting, if not properly controlled, may have adverse effects on human, animal and plant life:

a) *Human health* — Exposure to artificial light at night can disrupt circadian rhythms, leading to potential health risks such as sleep disorders and increased stress. Lighting design should prioritize human well-being by controlling glare and unnecessary brightness;

b) *Animal life* — Wildlife, particularly nocturnal species, may experience disruptions in natural behaviours, such as feeding and mating, due to poorly managed artificial lighting. The good practice [D-1(3)] compliant lighting should minimize impact on ecosystems; and

c) *Plant life* — Artificial lighting may affect plant growth cycles, leading to ecological imbalance. Properly designed lighting should reduce this risk by controlling light spill and adhering to the lighting recommendations of good practice [D-1(3)].

5 VENTILATION

5.1 General

Natural ventilation of buildings is required to supply outdoor air for the respiration of occupants, to remove any products of combustion or other contaminants in the air and to provide such thermal environments as will assist in the maintenance of heat balance of the body in order to prevent discomfort and injury to the health of the occupants. The natural ventilation through building envelope elements such as operable windows, skylights and ventilators help maintain the desired quality of air and thermal comfort conditions when outdoor air and thermal environments are conducive to the cause.

5.2 Design Considerations

5.2.1 *Respiration*

Supply of fresh air to provide oxygen for the human body for elimination of waste products and to maintain carbon dioxide concentration in the air within safe limits rarely calls for special attention as enough outside air for this purpose normally enters the areas of occupancy through crevices and other openings.

In normal habitable rooms devoid of smoke generating source, the content of carbon dioxide in air rarely exceeds 0.5 percent to 1 percent and is, therefore, incapable of producing any ill effect. The amount of air required to keep the concentration down to 1 percent is very small. The change in oxygen content is also too small under normal conditions to have any ill effects; the oxygen content may vary quite appreciably without noticeable effect, if the carbon dioxide concentration is unchanged.

5.2.2 *Vitiation by Body Odours*

Where no products of combustion or other contaminants are to be removed from air, the amount of fresh air required for dilution of inside air to prevent vitiation of air by body odours, depends on the air space available per person and the degree of physical activity; the amount of air decreases as the air space available per person increases, and it may vary from 20 m^3 to 30 m^3 per person per hour. In rooms occupied by only a small number of persons such an air change will automatically be attained in cool weather by normal leakage around windows and other openings and this may easily be secured in warm weather by keeping the openings open.

No standards have been laid down under the *Factories Act*, 1948 as regards the amount of fresh air required per worker or the number of air changes per hour. Section 16 of the *Factories Act*, 1948 relating to over-crowding requires that at least 14 m^3 to 16 m^3 of space shall be provided for every worker and for the purpose of that section no account shall be taken of any space in a work room which is more than 4.25 m above the floor level.

NOTE — Vitiation of the atmosphere can also occur in factories by odours given off due to contaminants of the product itself, say for example, from tobacco processing in a '*Beedi'* factory. Here the ventilation will have to be augmented to keep odours within unobjectionable levels.

5.2.2.1 *Recommended values for air changes*

The standards of general ventilation are recommended/based on maintenance of required oxygen, carbon dioxide and other air quality levels and for the control of body odours when no products of combustion or other contaminants are present in the air; the values of air changes should be as follows:

Sl No.	*Application*	*Air Change per Hour*
(1)	(2)	(3)
i)	Assembly rooms	4 to 8
ii)	Bakeries	20 to 30
iii)	Banks/building societies	4 to 8
iv)	Bathrooms	6 to 10
v)	Bedrooms	2 to 4
vi)	Billiard rooms	6 to 8
vii)	Boiler rooms	*See* Note 2
viii)	Cafes and coffee bars	10 to 12
ix)	Canteens	8 to 12
x)	Cellars	3 to 10
xi)	Changing rooms	6 to 10
xii)	Churches	1 to 3
xiii)	Cinemas and theatres	10 to 15
xiv)	Club rooms	12, *Min*
xv)	Compressor rooms	10 to 12
xvi)	Conference rooms	8 to 12
xvii)	Corridors	5 to 10
xviii)	Dairies	8 to 12
xix)	Dance halls	12, *Min*
xx)	Dye works	20 to 30
xxi)	Electroplating shops	10 to 12
xxii)	Engine rooms/DG Rooms/GG Rooms	*See* Note 2
xxiii)	Entrance halls	3 to 5
xxiv)	Factories and work shops	8 to 10
xxv)	Foundries	15 to 30
xxvi)	Garages	6 to 8
xxvii)	Glass houses	25 to 60
xxviii)	Gymnasium	6, *Min*

Table (*Concluded*)

Sl No.	*Application*	*Air Change per Hour*
(1)	(2)	(3)
xxix)	Hair dressing saloon	10 to 15
xxx)	Hospitals sterilising	15 to 25
xxxi)	Hospital wards	6 to 8
xxxii)	Hospital domestic	15 to 20
xxxiii)	Laboratories	6 to 15
xxxiv)	Launderettes	10 to 15
xxxv)	Laundries	10 to 30
xxxvi)	Lavatories	6 to 15
xxxvii)	Lecture theatres	5 to 8
xxxviii)	Libraries	3 to 5
xxxix)	Lift cars	20, *Min*
xl)	Living rooms	3 to 6
xli)	Mushroom houses	6 to 10
xlii)	Offices	6 to 10
xliii)	Paint shops (not cellulose)	10 to 20
xliv)	Photo and X-ray dark room	10 to 15
xlv)	Public house bars	12, *Min*
xlvi)	Recording control rooms	15 to 25
xlvii)	Recording studios	10 to 12
xlviii)	Restaurants	8 to 12
xlix)	Schoolrooms	5 to 7
l)	Shops and supermarkets	8 to 15
li)	Shower baths	15 to 20
lii)	Stores and warehouses	3 to 6
liii)	STP rooms	30, *Min*
liv)	Squash courts	4, *Min*
lv)	Swimming baths	10 to 15
lvi)	Toilets	6 to 10
lvii)	Underground vehicle parking	6, *Min*
lviii)	Utility rooms	15 to 30
lix)	Welding shops	15 to 30

NOTES

1 The ventilation rates may be increased by 50 percent where heavy smoking occurs or if the room is below the ground.

2 The ventilation rate shall be as per **14.2.1** of Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS.

5.2.3 *Heat Balance of Body*

Especially in hot weather, when the thermal environment inside the room is worsened by the heat given off by machinery, occupants and other sources, the prime need for ventilation is to provide such a thermal environment as will assist in the maintenance of heat balance of the body in order to prevent discomfort and injury to health. Excess of heat either from increased metabolism due to physical activity of persons or gains from a hot environment has to be offset to maintain normal body temperature (approximately 37 °C). Heat exchange of the human body with respect to the surroundings is determined by the temperature and humidity gradient between the skin and the surroundings and other factors, such as the age of person, clothing, etc and the latter depends on air temperature (dry bulb temperature), relative humidity, radiation from the solid surroundings and rate of air movement. The volume of outside air to be circulated through the room is, therefore, governed by the physical considerations of controlling the temperature, air distribution or air movement. Air movement and air distribution may, however, be achieved by recirculation of the inside air rather than bringing in all outside air. However, fresh air supply or the circulated air will reduce heat stress by dissipating heat from the body by evaporation of the sweat, particularly when the relative humidity is high and the air temperature is near body temperature.

5.2.3.1 *Indices of thermal comfort*

Thermal comfort is that condition of a thermal environment under which a person can maintain a body heat balance at normal body temperature and without perceptible sweating. Limits of comfort vary considerably according to studies carried out in India and abroad.

The thermal indices which find applications for Indian climate are as follows:

a) Effective temperature (ET);
b) Tropical summer index (TSI); and
c) Operative temperature based adaptive thermal comfort.

5.2.3.1.1 *Effective temperature* (*ET*)

Effective temperature is defined as the temperature of still, saturated air which has the same general effect upon comfort as the atmosphere under investigation. Combinations of temperature, humidity and wind velocity producing the same thermal sensation in an individual are taken to have the same effective temperature.

Initially two scales were developed, one of which referred to men stripped to the waist and called the basic scale. The other applies to men fully clad in indoor clothing and called the normal scale of effective temperature. Bedfort (1946) proposed the use of globe temperature reading instead of the air temperature reading to make allowance for the radiant heat. This scale is known as the corrected effective temperature (CET) scale. No allowance, however, was made for the different rates of energy expenditure. The scale was compiled only for men either seated or engaged in light activity.

Figure 5 represents the corrected effective temperature nomogram. The CET can be obtained by connecting the appropriate points representing the dry bulb (or globe) and wet bulb temperatures and reading the CET value at the intersection of this line with the relevant air velocity curve from the family of curves for various air velocities running diagonally upwards from left to right.

The effective temperature scale may be considered to be reasonably accurate in warm climates where the heat stress is not high but it may be misleading at high levels of heat stress. There appears to be an inherent error in this scale if used as an index of physiological strain, the error increasing with the severity of the environmental conditions. For low and moderate degrees of heat stress, the effective temperature scales appear to assess climatic heat stress with an accuracy which is acceptable for most practical purposes.

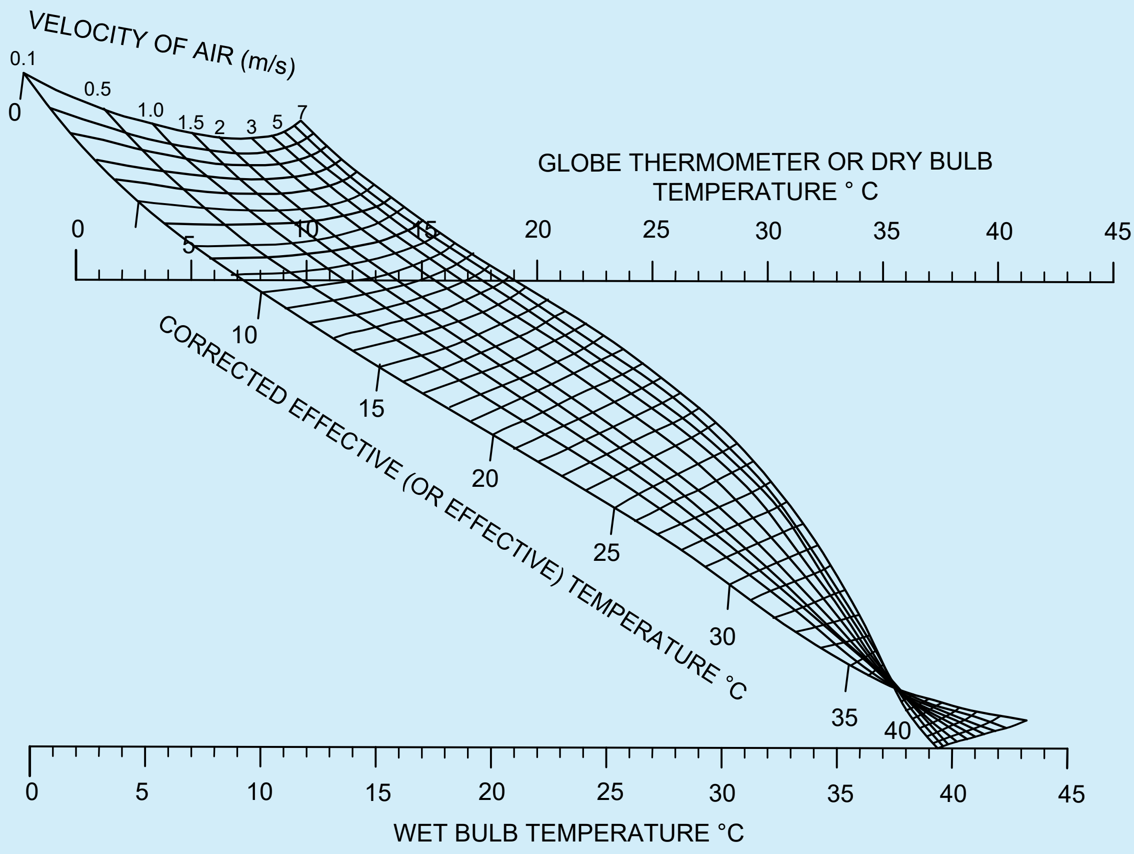


FIG. 5 CORRECTED EFFECTIVE TEMPERATURE NOMOGRAM

5.2.3.1.2 *Tropical Summer Index* (*TSI*)

The TSI is defined as the temperature of calm air, at 50 percent relative humidity which imparts the same thermal sensation as the given environment. The 50 percent level of relative humidity is chosen for this index as it is a reasonable intermediate value for the prevailing humidity conditions. Mathematically, TSI (°C) is expressed as:

$$\mathrm{TSI} = 0.745\mathrm{t_g} + 0.308\mathrm{t_w}\text{-}2.06\sqrt{v + 0.841}$$

where

t_w = wet bulb temperature, in °C;
t_g = globe temperature, in °C; and
v = air speed, in m/s.

The thermal comfort of a person lies between TSI values of 25 °C and 30 °C with optimum condition at 27.5 °C. Air movement is necessary in hot and humid weather for body cooling. A certain minimum desirable wind speed is needed for achieving thermal comfort at different temperatures and relative humidity. Such wind speeds are given in Table 11. These are applicable to sedentary work in offices and other places having no noticeable sources of heat gain. Where somewhat warmer conditions are prevalent, such as in godowns and machine shops and work is of lighter intensity, and higher temperatures can be tolerated without much discomfort, minimum wind speeds for just acceptable warm conditions are given in Table 12. For obtaining values of indoor wind speed above 2.0 m/s, mechanical means of ventilation may have to be adopted (*see also* Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS).

The warmth of the environment was found tolerable between 30 °C and 34 °C (TSI), and too hot above this limit. On the lower side, the coolness of the environment was found tolerable between 19 °C and 25 °C (TSI) and below 19 °C (TSI), it was found too cold.

Table 11 Desirable Wind Speeds (m/s) for Thermal Comfort Conditions

(*Clause* 5.2.3.1.2)

Sl No.	Dry Bulb Temperature, °C	Relative Humidity, percent						
		30	40	50	60	70	80	90
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	28	[1]	[1]	[1]	[1]	[1]	[1]	[1]
ii)	29	[1]	[1]	[1]	[1]	[1]	0.06	0.19
iii)	30	[1]	[1]	[1]	0.06	0.24	0.53	0.85
iv)	31	[1]	0.06	0.24	0.53	1.04	1.47	2.10
v)	32	0.20	0.46	0.94	1.59	2.26	3.04	[2]
vi)	33	0.77	1.36	2.12	3.00	[2]	[2]	[2]
vii)	34	1.85	2.72	[2]	[2]	[2]	[2]	[2]
viii)	35	3.20	[2]	[2]	[2]	[2]	[2]	[2]

Table 12 Minimum Wind Speeds (m/s) for Just Acceptable Warm Conditions

(*Clause* 5.2.3.1.2)

Sl No.	Dry Bulb Temperature, °C	Relative Humidity, percent						
		30	40	50	60	70	80	90
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	28	[1]	[1]	[1]	[1]	[1]	[1]	[1]
ii)	29	[1]	[1]	[1]	[1]	[1]	[1]	[1]
iii)	30	[1]	[1]	[1]	[1]	[1]	[1]	[1]
iv)	31	[1]	[1]	[1]	[1]	[1]	0.06	0.23
v)	32	[1]	[1]	[1]	0.09	0.29	0.60	0.94
vi)	33	[1]	0.04	0.24	0.60	1.04	1.85	2.10
vii)	34	0.15	0.46	0.94	1.60	2.26	3.05	[2]
viii)	35	0.68	1.36	2.10	3.05	[2]	[2]	[2]
ix)	36	1.72	2.70	[2]	[2]	[2]	[2]	[2]

5.2.4 *Design of Indoor Conditions as per Adaptive Thermal Comfort Model*

Air conditioning systems for interior spaces intended for human occupancy should be sized for not more than 26 °C for cooling and for not less than 18 °C for heating at an occupied level. Design based on an adaptive thermal comfort model can play a major role in reducing energy use whilst maintaining the comfort, productivity and well-being of occupants. This approach recognizes that people's thermal comfort needs depend on their past and present context and that these needs vary with the outdoor environmental conditions of their location.

People living year-round in air-conditioned spaces are likely to develop high expectations for homogeneity and cool temperatures and may become quite critical if thermal conditions deviate from the centre of the comfort zone

[1] None.

[2] Higher than those acceptable in practice.

they have come to expect. In contrast, people who live or work in naturally ventilated buildings, can control their immediate interior spaces and get accustomed to variable indoor thermal conditions that reflect local patterns of daily and seasonal climate changes. Their thermal perceptions, preferences, as well as tolerances, are likely to extend over a wider range of temperatures. It allows buildings to operate within broader temperature bands. The specification of a broader comfort band suited to the Indian context has the potential to reduce the use of energy-intensive space cooling for buildings in India. The provisions here deal with the adaptive thermal comfort model, which differentiates the thermal response of occupants in air-conditioned and naturally ventilated buildings. It is imperative to specify indoor comfort conditions separately for buildings depending upon their operation, namely, air-conditioned buildings as well as naturally ventilated buildings:

a) Operative temperature is a more suitable index to measure thermal comfort in a building having low indoor air velocities since the index also accounts for the impact of building surface temperatures (radiant temperatures) on human comfort. Indoor operative temperature, for sedentary activities, can be approximated as an arithmetic mean of indoor air and radiant temperatures. For operating an air conditioning system based on operative temperature, the prevalent conditions along with the historical outdoor trend can be used to derive suitable operative temperature limits. Typical climatic conditions of various Indian cities are in the public domain; and

b) In case of buildings having higher indoor air velocity (more than 0.5 m/s), an effective temperature-based approach is recommended since, in addition to all factors considered in operative temperature, it also takes into account heat dissipation from the human body through convective heat transfer. High air velocity can allow keeping higher air temperature without compromising thermal comfort. However, it is also suggested to keep in consideration noise and other effects of high indoor air velocity. For annual indoor design conditions as per the adaptive model, for Indian cities, reference may be made to Annex A of Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS. These are indicative, simplified values. They have been derived using typical meteorological year (TMY) files. More accurate design conditions can be derived using the following equations.

5.2.4.1 *Design Conditions for Commercial Buildings*

It is recommended to use the below-mentioned set of adaptive thermal comfort equations for commercial buildings. Equations (a) to (b) are best suited for the 30 day outdoor running mean temperatures between 13 °C and 38.5 °C:

a) For naturally ventilated (NV) commercial buildings — The following equation should be used for the design and operation of naturally ventilated (NV) buildings. It indicates that occupants in NV buildings thermally adapt to the outdoor temperature of their location:

$$Indoor\ operative\ temperature = (0.54 \times Outdoor\ temperature) + 12.83$$

where, indoor operative temperature (in °C) is the neutral temperature, and outdoor temperature is the 30 day outdoor running mean air temperature (in °C). The 90 percent acceptability range for naturally ventilated buildings is ± 2.38 °C.

b) For mixed-mode (MM) operation commercial buildings — Mixed-mode buildings, where HVAC is operated only during extreme outdoor conditions, are becoming prevalent in India. The occupants in mixed-mode buildings are more adaptive when compared to those in air conditioned buildings and somewhat less adaptive compared to occupants in naturally ventilated buildings:

$$Indoor\ operative\ temperature = (0.28 \times Outdoor\ temperature) + 17.87$$

where, indoor operative temperature (in °C) is the neutral temperature, and outdoor temperature is the 30 day outdoor running mean air temperature (in °C). The 90 percent acceptability range for mixed-mode buildings is ± 3.46 °C.

5.2.4.2 *Design Conditions for Residential Buildings*

The following equation should be used for determining adaptive thermal comfort in residential buildings, applicable for outdoor running mean temperatures between 7 °C and 33 °C. It is recommended to use the below-mentioned adaptive thermal comfort equation for residential buildings, for all operation modes:

$$Indoor\ operative\ temperature = (0.42 * Outdoor\ temperature) + 17.45$$

where, indoor operative temperature (in °C) is neutral temperature, and outdoor temperature is the 30 day outdoor running mean air temperature (in °C). The 90 percent acceptability range for conditioned buildings is ± 2.15 °C.

The adaptive comfort values derived using equations given in **5.2.4.1** for commercial, and the equation given in **5.2.4.2** for residential buildings are best suited for indoor relative humidity conditions (RH in percentage) between 25 percent and 75 percent.

5.2.5 *Minimum Outside Fresh Air*

Minimum outside fresh air applicable to only fully air-conditioned buildings. Table 13 prescribes minimum supply rates of acceptable outdoor air required for acceptable indoor air quality. These requirements for ventilation of health care facilities (hospitals, nursing and convalescent homes) are separately covered in Table 14. These values have been prescribed to dilute human bio effluents and other contaminants with an adequate margin of safety and to account for health variations among people and varied activity levels. The values prescribed in the table should be used in conjunction with the accompanying notes. Naturally ventilated spaces and mixed mode spaces, while operating in naturally ventilated mode, will have to rely on outdoor air using fenestration systems. The quantity and distribution of introduced fresh air should take into account the natural infiltration of the building. The proportion of fresh air introduced into air conditioned building may be varied to achieve economical and efficient operation. When the fresh air can provide a useful cooling effect, the quantity should be controlled through an air-side economizer to balance the cooling demand. However, when the air is too warm or humid the quantity may be reduced to a minimum to reduce the cooling load.

5.2.5.1 *Air class characteristic*

The following are the characteristics of the four air classes:

a) Class 1 — Air with low contaminant concentration, low sensory-irritation intensity and inoffensive odour;

b) Class 2 — Air with moderate contaminant concentration, mild sensory-irritation intensity, or mildly offensive odours. This air also includes air that is not necessarily harmful or objectionable but that is inappropriate for transfer or recirculation to spaces used for different purposes;

c) Class 3 — Air with significant contaminant concentration, significant sensory-irritation intensity, or offensive odour; and

d) Class 4 — Air with highly objectionable fumes or gases or with potentially dangerous particles, bio aerosols, or gases, at concentrations high enough to be considered harmful.

Table 13 Minimum Ventilation Rates in Breathing Zone

(*Clauses* 5.2.5)

(This table is not valid in isolation; it shall be used in conjunction with the accompanying notes.)

Sl No.	Occupancy Category	People Outdoor Air Rate, R_p	Area Outdoor Air Rate, R_a	Notes	Default Values		Air Class
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	**Correctional facilities:**						
	a) Cell	2.5	0.6	–	25	4.9	2
	b) Dayroom	2.5	0.3	–	30	3.5	1
	c) Guard stations	2.5	0.3	–	15	4.5	1
	d) Booking/waiting	3.8	0.3	–	50	4.4	2
ii)	**Educational facilities:**						
	a) Daycare (through age 4)	5	0.9	–	25	8.6	2
	b) Daycare sickroom	5	0.9	–	25	8.6	3
	c) Classrooms (age 5 to 8)	5	0.6	–	25	7.4	1
	d) Classrooms (age 9 plus)	5	0.6	–	35	6.7	1
	e) Lecture classroom	3.8	0.3	–	65	4.3	1
	f) Lecture hall (fixed seats)	3.8	0.3	–	150	4.0	1
	g) Art classroom	5	0.9	–	20	9.5	2
	h) Science laboratories	5	0.9	–	25	8.6	2
	j) University/college laboratories	5	0.9	–	25	8.6	2

Table 13 (*Continued*)

Sl No.	Occupancy Category	People Outdoor Air Rate, R_p	Area Outdoor Air Rate, R_a	Notes	Default Values		Air Class
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	k) Wood/metal shop	5	0.9	–	20	9.5	2
	m) Computer lab	5	0.6	–	25	7.4	1
	n) Media centre	5	0.6	*See* Note 6	25	7.4	1
	p) Music/theatre/dance	5	0.3	–	35	5.9	1
	q) Multi-use assembly	3.8	0.3	–	100	4.1	1
iii)	**Food and beverage service:**						
	a) Restaurant dining rooms	3.8	0.9	–	70	5.1	2
	b) Cafeteria/fast-food dining	3.8	0.9	–	100	4.7	2
	c) Bars, cocktail lounges	3.8	0.9	–	100	4.7	2
iv)	**General:**						
	a) Break rooms	2.5	0.3	–	25	5.1	1
	b) Coffee stations	2.5	0.3	–	20	5.5	1
	c) Conference/meeting rooms	2.5	0.3	–	50	3.1	1
	d) Corridors	–	0.3	–	–	–	1
	e) Storage rooms	–	0.6	*See* Note 7	–	–	1

Table 13 (*Continued*)

Sl No.	Occupancy Category	People Outdoor Air Rate, R_p	Area Outdoor Air Rate, R_a	Notes	Default Values		Air Class
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
v)	**Hotels, motels, resorts, dormitories:**						
	a) Bedroom/living room	2.5	0.3	–	10	5.5	1
	b) Barracks sleeping areas	2.5	0.3	–	20	4.0	1
	c) Laundry rooms, central	2.5	0.6	–	10	8.5	2
	d) Laundry rooms within dwelling units	2.5	0.6	–	10	8.5	1
	e) Lobbies/pre-function	3.8	0.3	–	30	4.8	1
	f) Multipurpose assembly	2.5	0.3	–	120	2.8	1
vi)	**Office buildings:**						
	a) Office space	2.5	0.3	–	5	8.5	1
	b) Reception Areas	2.5	0.3	–	30	3.5	1
	c) Telephone/data entry	2.5	0.3	–	60	3.0	1
	d) Main entry lobbies	2.5	0.3	–	10	5.5	1
				–			
vii)	**Miscellaneous spaces**:						
	a) Bank vaults/safe deposit	2.5	0.3	–	5	8.5	2
	b) Computer (not printing)	2.5	0.3	–	4	10.0	1

Table 13 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	c) Electrical equipment rooms	–	0.3	*See* Note 7	–	–	1
	d) Elevator machine rooms	–	0.6	*See* Note 7	–	–	1
	e) Pharmacy (preparation area)	2.5	0.9	–	10	11.5	2
	f) Photo studios	2.5	0.6	–	10	8.5	1
	g) Shipping/receiving	–	0.6	*See* Note 7	–	–	1
	h) Telephone closets	–	0.0	–	–	–	1
	j) Transportation waiting	3.8	0.3	–	100	4.1	1
	k) Warehouses	–	0.3	*See* Note 7	–	–	2
viii)	**Public assembly spaces:**						
	a) Auditorium seating area	2.5	0.3	–	150	2.7	1
	b) Places of religious worship	2.5	0.3	–	120	2.8	1
	c) Courtrooms	2.5	0.3	–	70	2.9	1
	d) Legislative chambers	2.5	0.3	–	50	3.1	1
	e) Libraries	2.5	0.6	–	10	8.5	1
	f) Lobbies	2.5	0.3	–	150	2.7	1
	g) Museums (children's)	3.8	0.6	–	40	5.3	1

Table 13 (*Continued*)

Sl No.	Occupancy Category	People Outdoor Air Rate, R_p	Area Outdoor Air Rate, R_a	Notes	Default Values		Air Class
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	h) Museums/galleries	3.8	0.3	–	40	4.6	1
ix)	**Residential:**						
	a) Dwelling unit	2.5	0.3	*See* Notes 8 and 9	*See* Note 8	–	1
	b) Common corridors	–	0.3	–	–	–	1
x)	**Retail:**						
	a) Sales (except as below)	3.8	0.6	–	15	7.8	2
	b) Mall common areas	3.8	0.3	–	40	4.6	1
	c) Barber shop	3.8	0.3	–	25	5.0	2
	d) Beauty and nail salons	10	0.6	–	25	12.4	2
	e) Pet shops (animal areas)	3.8	0.9	–	10	12.8	2
	f) Supermarket	3.8	0.3	–	8	7.6	1
	g) Coin-operated laundries	3.8	0.3	–	20	5.3	2
xi)	**Sports and entertainment:**						
	a) Sports arena (play area)	–	1.5	*See* Note 10	–	–	1
	b) Gym, stadium (play area)	–	1.5	–	30	–	2
	c) Spectator areas	3.8	0.3	–	150	4.0	1

Table 13 (*Concluded*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s/person	l/s/m^2		persons per 100 m^2	l/s/person	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	d) Swimming (pool and deck)	-	2.4	*See* Note 11	–	–	2
	e) Disco/dance floors	10	0.3	–	100	10.3	1
	f) Health club/aerobics room	10	0.3	–	40	10.8	2
	g) Health club/weight rooms	10	0.3	–	10	13.0	2
	h) Bowling alley (seating)	5	0.6	–	40	6.5	1
	j) Casinos	3.8	0.9	–	120	4.6	1
	k) Game arcades	3.8	0.9	–	20	8.3	1
	m) Stages, studios	5	0.3	*See* Note 12	70	5.4	1

NOTES

1 The rates in this table are based on all other applicable requirements being met.

2 This table applies to no-smoking areas only. Rates for smoking-permitted spaces shall be determined using other methods.

3 Volumetric airflow rates are based on an air density of 1.2 kg of dry air/m^3, which corresponds to dry air at a barometric pressure of 1 atm (101.3 kPa) and an air temperature of 21 °C. Rates may be adjusted for actual density but such adjustment is not required for compliance with this standard.

4 Actual occupant density should be considered, the default occupant density shall be used only when actual occupant density is not known. Default combined outdoor air (per person) rate is based on the default occupant density.

5 If the occupancy category for a proposed space or zone is not listed, the requirements for the listed occupancy category that is most similar in terms of occupant density, activities and building construction shall be used.

6 For high school and college libraries, use values shown for public assembly spaces-libraries.

7 The prescribed value may not be sufficient when stored materials include those having potentially harmful emissions.

8 Default occupancy for dwelling units shall be two persons for studio and one-bedroom units, with one additional person for each additional bedroom.

9 Air from one residential dwelling shall not be recirculated or transferred to any other space outside of that dwelling.

10 When combustion equipment is intended to be used on the playing surface, additional dilution ventilation and/or source control shall be provided.

11 The prescribed value does not allow for humidity control. Additional ventilation or dehumidification may be required to remove moisture.

12 The prescribed value does not include special exhaust for stage effects, for example, dry ice vapours, smoke.

Table 14 Outdoor Air Requirements for Ventilation of Health Care Facilities

(*Clause* 5.2.5)

Sl No.	Application	Estimated Maximum Occupancy[1)]	Outdoor Air Requirements		Remarks
		persons/100 m²	l/s per person	l/s per m²	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Patient rooms	10	13	–	Special requirements and pressure relationships may determine minimum ventilation rates and filter efficiency. Procedures generating contaminants may require higher rates.
ii)	Medical procedure	20	8	–	
iii)	Operating rooms	20	15	–	
iv)	Recovery and ICU	20	8	–	
v)	Autopsy rooms	20	–	2.5	Air shall not be recirculated into other spaces.
vi)	Physical therapy	20	8	–	

[1)] Net occupiable space.

5.2.6 There will be a limit of heat tolerance when air temperatures are excessive and the degree of physical activity is high. This limit is determined when the bodily heat balance is upset, that is, when the bodily heat gain due to conduction, convection and the radiation from the surroundings exceeds the bodily heat loss, which is mostly by evaporation of sweat from the surface of the body. The limits of heat tolerance for Indian workers are based on the study conducted by the Chief Adviser Factories, Government of India, Ministry of Labour and are given in his report on Thermal Stress in Textile Industry (Report No. 17) issued in 1956. According to this report, where workers in industrial buildings wearing light clothing are expected to do work of moderate severity with the energy expenditure in the range 273 W to 284 W, the maximum wet bulb temperature shall not exceed 29 °C and adequate air movement subject to a minimum air velocity of 30 m/min shall be provided, and in relation to the dry bulb temperature, the wet bulb temperature of air in the work room, as far as practicable, shall not exceed that given in Table 15.

Table 15 Maximum Permissible Wet Bulb Temperatures for Given Dry Bulb Temperatures

(*Clause* 5.2.6)

Sl No.	Dry Bulb Temperature °C	Maximum Wet-Bulb Temperature °C
(1)	(2)	(3)
i)	30	29.0
ii)	35	28.5
iii)	40	28.0
iv)	45	27.5
v)	50	27.0

NOTES

1 These are limits beyond which the industry should not allow the thermal conditions to go for more than 1 h continuously. The limits are based on a series of studies conducted on Indian subjects in psychrometric chamber and on other data on heat casualties in earlier studies conducted in Kolar Gold Fields and elsewhere.

2 Figures given in this table are not intended to convey that human efficiency at 50 °C will remain the same as at 30 °C, provided appropriate wet bulb temperatures are maintained. Efficiency decreases with a rise in the dry bulb temperature as well, as much as possible. Long exposures to temperatures of 50 °C dry bulb/27 °C wet bulb may prove dangerous.

3 Refrigeration or some other method of cooling is recommended in all cases where conditions would be worse than those shown in this table.

5.3 Methods of Ventilation

General ventilation involves providing a building with relatively large quantities of outside air in order to improve general environment of the building. This may be achieved in one of the following ways:

a) Natural supply and natural exhaust of air;

b) Natural supply and mechanical exhaust of air;

c) Mechanical supply and natural exhaust of air; and

d) Mechanical supply and mechanical exhaust of air.

5.3.1 *Control of Heat*

Although it is recognized that general ventilation is one of the most effective methods of improving thermal environmental conditions in factories, in many situations, the application of ventilation should be preceded by and considered along with some of the following other methods of control. This would facilitate better design of buildings for general ventilation, either natural or mechanical or both and also reduce their cost.

5.3.1.1 *Isolation*

Sometimes it is possible to locate heat producing equipment, such as furnaces in such a position as would expose only a small number of workers to hot environment. As far as practicable, such sources of heat in factories should be isolated.

In situations where relatively few people are exposed to severe heat stress and their activities are confined to limited areas as in the case of rolling mill operators and crane operators, it may be possible to enclose the work areas and provide spot cooling or supply conditioned air to such enclosures.

5.3.1.2 *Insulation*

A considerable portion of heat in many factories is due to the solar radiation falling on the roof surfaces, which, in turn, radiate heat inside the building. In such situations, insulations of the roof or providing a false ceiling or double roofing would be very effective in controlling heat. Some reduction can also be achieved by painting the roof in heat reflective shades.

Hot surfaces of equipment, such as pipes, vessels, etc, in the building should also be insulated to reduce their surface temperature.

5.3.1.3 *Substitution*

Sometimes, it is possible to substitute a hot process by a method that involves application of localized or more efficiently controlled method of heating, examples include induction hardening instead of conventional heat treatment, cold riveting or spot welding instead of hot riveting, etc.

5.3.1.4 *Radiant shielding*

Hot surfaces, such as layers of molten metal emanate radiant heat, which can best be controlled by placing a shield having a highly reflecting surface between the source of heat and the worker, so that a major portion of the heat falling on the shield is reflected back to the source. Surfaces such as of tin and aluminium have been used as materials for shields. The efficiency of the shield does not depend on its thickness, but on the reflectivity and emissivity of its surface. Care should be taken to see that the shield is not heated up by conduction and for this purpose adequate provision should be made for the free flow upwards of the heated air between the hot surface and the shield by leaving the necessary air space and providing opening at the top and the bottom of the sides.

5.3.2 *Volume of Air Required*

The volume of air required shall be calculated by using both the sensible heat and latent heat gain as the basis. The larger of the two values obtained should be used in actual practice.

In places without sufficient wind speeds and/or in buildings where effective cross ventilation is not possible due to the design of the interior, the indoor air may be exhausted by a fan, with outdoor air entering the building through the open windows.

5.3.2.1 *Volume of air required for removing sensible heat*

When the amount of sensible heat given off by different sources, namely, the sun, the manufacturing processes, machinery, occupants and other sources, is known and a suitable value for the allowable temperature rise is assumed, the volume of outside air to be provided for removing the sensible heat may be calculated from:

$$Q_1 = \frac{2.976\,8\,K_s}{t}$$

where

Q_1 = quantity of air, in m^3/h;

K_s = sensible heat gained, in W; and

t = allowable temperature rise, in °C.

5.3.2.2 Temperature rise refers mainly to the difference between the air temperatures at the outlet (roof exit) and at the inlet openings for outside air. As very little data exist on allowable temperature rise values for supply of outside air in summer months, the values given in Table 16 related to industrial buildings may be used for general guidance.

Table 16 Allowable Temperature Rise Values

(*Clause* 5.3.2.2)

Sl No.	Height of Outlet Opening m	Temperature Rise °C
(1)	(2)	(3)
i)	6	3 to 4.5
ii)	9	4.5 to 6.5
iii)	12	6.5 to 11

NOTES

1 The conditions are limited to light or medium heavy manufacturing processes, freedom from radiant heat and inlet openings not more than 3 m to 4.5 m above floor level.

2 At the working zone between floor level and 1.5 m above floor level, the recommended maximum allowable temperature rise for air is 2 °C to 3 °C above the air temperature at the inlet openings.

5.3.2.3 *Volume of air required for removing latent heat*

If the latent heat gained from the manufacturing processes and occupants is also known and a suitable value for the allowable rise in the vapour pressure is assumed, the volume of air to be provided to remove the latent heat may be calculated from:

$$Q_2 = \frac{4\ 127.26\,K_1}{h}$$

where

Q_2 = quantity of air, in m^3/h;

K_1 = latent heat gained, in W; and

h = allowable vapour pressure difference, in mm of mercury.

NOTE — In majority of the cases, the sensible heat gain will far exceed the latent heat gain, so that the amount of outside air to be drawn by ventilating equipment can be calculated on the basis of the equation given in **5.3.2.1**.

5.3.2.4 Ventilation is also expressed as m^3/h/m^2 of floor area. This relationship fails to evaluate the actual heat relief provided by a ventilation system, but it does give a relationship which is independent of building height. This is a more rational approach, because, with the same internal load, the same amount of ventilation air, properly applied to the work zone with adequate velocity, will provide the desired heat relief quite independently of the ceiling height of the space, with few exceptions. Ventilation rates of 30 m^3/h/m^2 to 60 m^3/h/m^2 have been found to give good results in many plants.

5.4 Natural Ventilation

The rate of ventilation by natural means through windows or other openings depends on,

a) direction and velocity of wind outside and sizes and disposition of openings (wind action); and

b) convection effects arising from temperature of vapour pressure difference (or both) between inside and outside the room and the difference of height between the outlet and inlet openings (stack effect).

5.4.1 *Ventilation of Non-Industrial Buildings*

Ventilation in non-industrial buildings due to stack effect, unless there is a significant internal load, could be neglected, except in cold regions and wind action may be assumed to be predominant.

5.4.1.1 In hot dry regions, the main problem in summer is to provide protection from sun's heat so as to keep the indoor temperature lower than those outside under the sun. For this purpose, windows and other openings are generally kept closed during day time and only minimum ventilation is provided for the control of odours or for removal of products of combustion.

5.4.1.2 In warm humid regions, the problem in the design of non-industrial buildings is to provide free passage of air to keep the indoor temperature as near to those outside in the shade as possible, and for this purpose the buildings are oriented to face the direction of prevailing winds and windows and other openings are kept open on both windward and leeward sides.

5.4.1.3 In winter months in cold regions, the windows and other openings are generally kept shut, particularly during night; and ventilation necessary for the control of odours and for the removal of products of combustion can be achieved either by stack action or by some infiltration of outside air due to wind action.

5.4.2 *Ventilation of Industrial Buildings*

In providing natural ventilation of all industrial buildings having significant internal heat loads due to manufacturing process, proper consideration should be given to the size and distribution of windows and other inlet openings in relation to outlet openings so as to give, with due regard to orientation, prevailing winds, size and configuration of the building and manufacturing processes carried on, maximum possible control of thermal environment.

5.4.2.1 In the case of industrial buildings wider than 30 m, the ventilation through windows may be augmented by roof ventilation.

5.4.3 *Design Guidelines for Natural Ventilation*

5.4.3.1 *By wind action*

a) A building need not necessarily be oriented perpendicular to the prevailing outdoor wind; it may be oriented at any convenient angle between 0° and 30° without losing any beneficial aspect of the breeze. If the prevailing wind is from east or west, building may be oriented at 45° to the incident wind so as to diminish the solar heat without much reduction in air motion indoors;

b) Inlet openings in the buildings should be well distributed and should be located on the windward side at a low level, and outlet openings should be located on the leeward side. Inlet and outlet openings at high levels may only clear the top air at that level without producing air movement at the level of occupancy;

c) Maximum air movement at a particular plane is achieved by keeping the sill height of the opening at 85 percent of the critical height (such as head level) for the following recommended levels of occupancy:

    1) For sitting on chair 0.75 m;

    2) For sitting on bed 0.60 m; and

    3) For sitting on floor 0.40 m.

d) Inlet openings should not as far as possible be obstructed by adjoining buildings, trees, sign boards or other obstructions or by partitions inside in the path of air flow;

e) In rooms of normal size having identical windows on opposite walls the average indoor air speed increases rapidly by increasing the width of window up to two-third of the wall width; beyond that the increase is in much smaller proportion than the increase of the window width. The air motion in the working zone is maximum when window height is 1.1 m. Further increase in window height promotes air motion at higher level of window, but does not contribute additional benefits as regards air motion in the occupancy zones in buildings;

f) Greatest flow per unit area of openings is obtained by using inlet and outlet openings of nearby equal areas at the same level;

g) For a total area of openings (inlet and outlet) of 20 percent to 30 percent of floor area, the average indoor wind velocity is around 30 percent of outdoor velocity. Further increase in window size increases the available velocity but not in the same proportion as shown in Fig. 6. In fact, even under most favourable conditions the maximum average indoor wind speed does not exceed 40 percent of outdoor velocity;

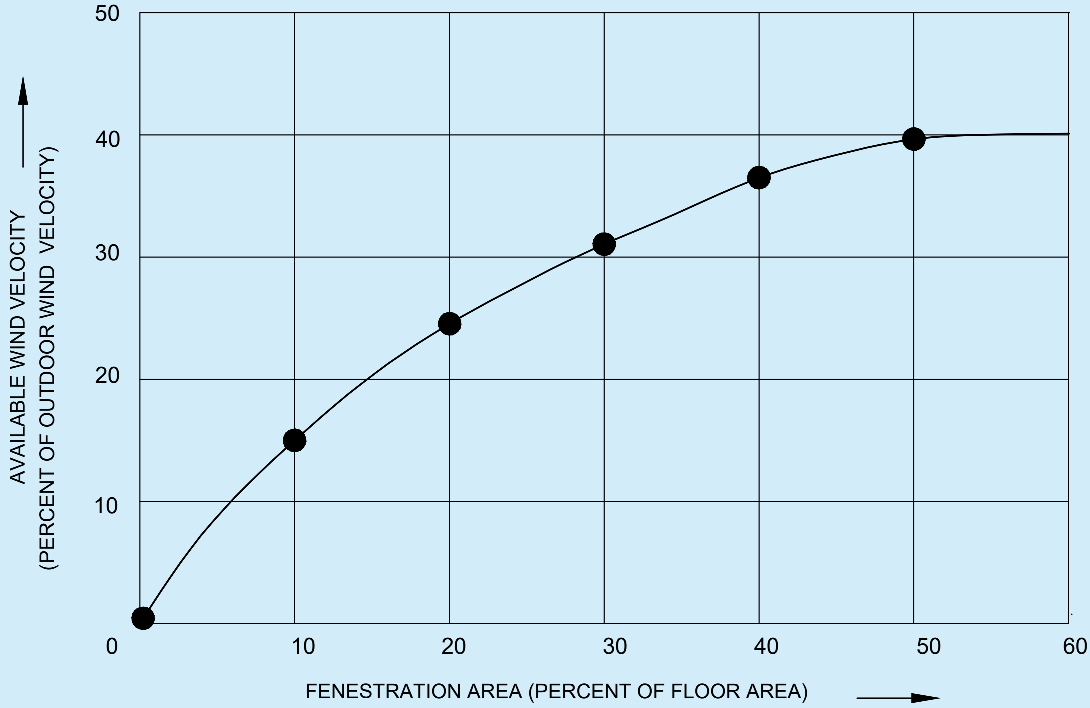


FIG. 6 EFFECT OF AREA OF OPENING ON AVERAGE INDOOR WIND VELOCITY

h) Where the direction of wind is quite constant and dependable, the size of the inlet should be kept within 30 percent to 50 percent of the total area of openings and the building should be oriented perpendicular to the incident wind. Where direction of the wind is quite variable the openings may be arranged so that as far as possible there is approximately equal area on all sides. Thus, no matter what the wind direction be, there would be some openings directly exposed to wind pressure and others to air suction and effective air movement through the building would be assured;

j) Windows of living rooms should open directly to an open space. In places where building sites are restricted, open space may have to be created in the buildings by providing adequate courtyards;

k) In the case of rooms with only one wall exposed to outside, provision of two windows on that wall is preferred to that of a single window;

m) Windows located diagonally opposite to each other with the windward window near the upstream corner give better performance than other window arrangements for most of the building orientations;

n) Horizontal louvers, that is, sunshades atop windows deflect the incident wind upward and reduce air motion in the zone of occupancy. A horizontal slot between the wall and horizontal louver prevents upward deflection of air in the interior of rooms. Provision of inverted L type (Γ) louver increases the room air motion provided that the vertical projection does not obstruct the incident wind (*see* Fig. 7);

p) Provision of horizontal sashes inclined at an angle of 45° in appropriate direction helps to promote the indoor air motion. Sashes projecting outward are more effective than projecting inward;

q) Air motion at working plane 0.4 m above the floor can be enhanced by 30 percent using a pelmet type wind deflector (*see* Fig. 8);

r) Roof overhangs help promoting air motion in the working zone inside buildings;

s) In case of room with windows on one wall, with single window, the room wind velocity inside the room on the windward side is 10 percent of outdoor velocity at points up to a distance one-sixth of room width from the window and then decreases rapidly and hardly any air movement is produced in the leeward half portion of the room. The average indoor wind velocity is generally less than 10 percent of outdoor velocity. When two windows are provided and wind impinges obliquely on them, the inside velocity increases up to 15 percent of the outdoor velocity;

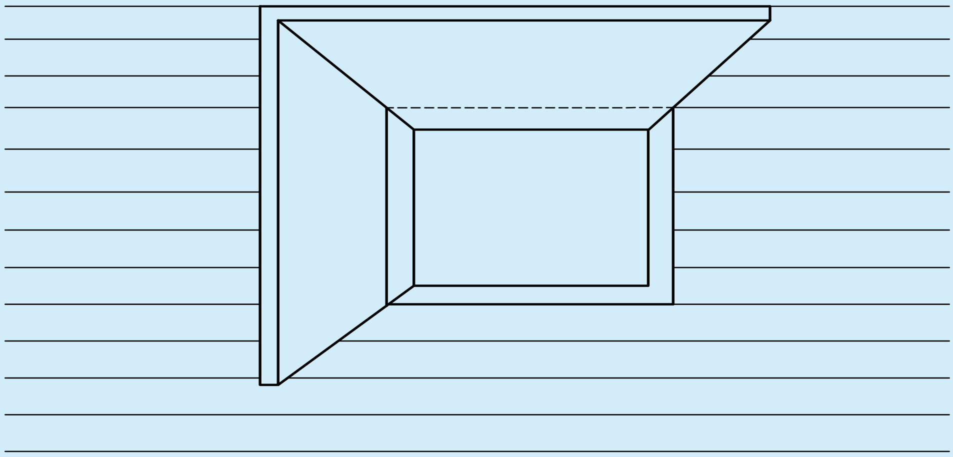


FIG. 7 L-TYPE LOUVER

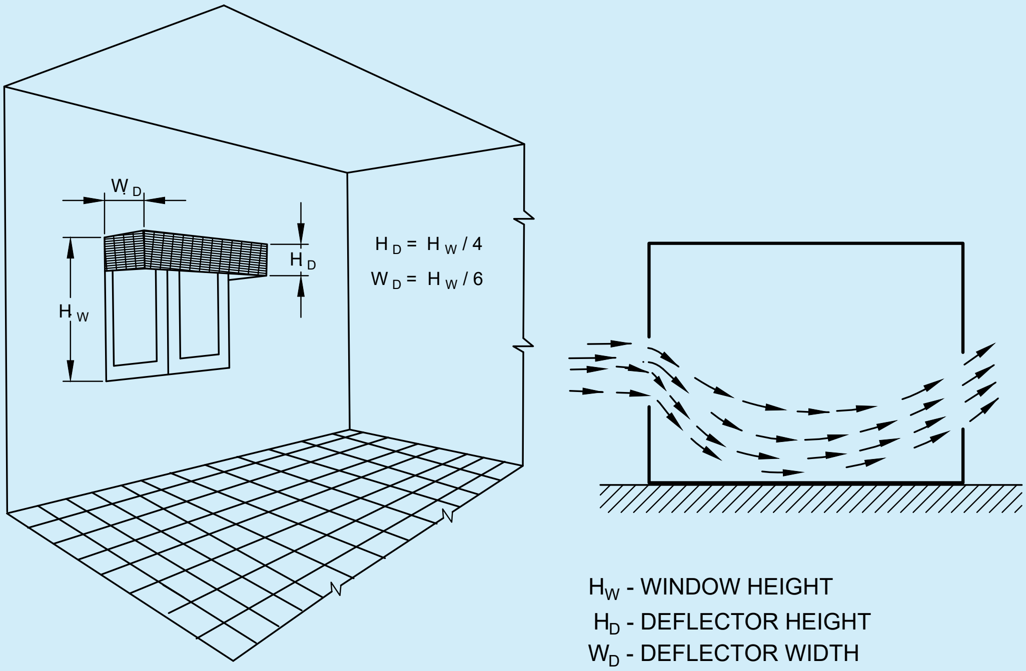


FIG. 8 SKETCH OF A PELMET TYPE WIND DEFLECTOR

t) Cross ventilation can be obtained through one side of the building to the other, in case of narrow buildings with the width common in the multistoreyed type by the provision of large and suitably placed windows or combination of windows and wall ventilators for the inflow and outflow of air;

u) *Verandah* open on three sides is to be preferred since it causes an increase in the room air motion for most of the orientations of the building with respect to the outdoor wind;

v) A partition placed parallel to the incident wind has little influence on the pattern of the air flow, but when located perpendicular to the main flow, the same partition creates a wind shadow. Provision of a partition with spacing of 0.3 m underneath helps augmenting air motion near floor level in the leeward compartment of wide span buildings;

w) Air motion in a building unit having windows tangential to the incident wind is accelerated when another unit is located at end-on position on downstream side (*see* Fig. 9);

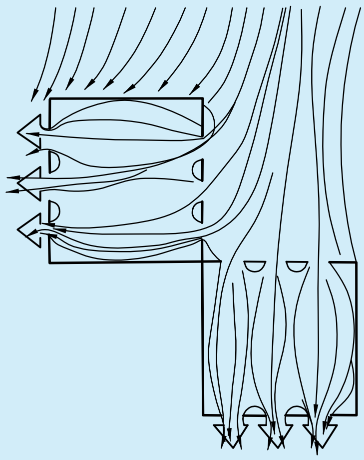


FIG. 9 TWO UNITS LOCATED AT THE END-ON POSITION

y) Air motion in two wings oriented parallel to the prevailing breeze is promoted by connecting them with a block on downstream side;

z) Air motion in a building is not affected by constructing another building of equal or smaller height on the leeward side; but it is slightly reduced if the leeward building is taller than the windward block;

aa) Air motion in a shielded building is less than that in an unobstructed building. To minimize shielding effect, the distances between two rows should be 8 H for semi-detached houses and 10 H for long rows houses. However, for smaller spacing the shielding effect is also diminished by raising the height of the shielded building;

bb) Hedges and shrubs deflect the air away from the inlet openings and cause a reduction in indoor air motion. These elements should not be planted at a distance of about 8 m from the building because the induced air motion is reduced to minimum in that case. However, air motion in the leeward part of the building can be enhanced by planting a low hedge at a distance of 2 m from the building;

cc) Trees with large foliage mass having trunk bare of branches up to the top level of window, deflect the outdoor wind downwards and promotes air motion in the leeward portion of buildings;

dd) Ventilation conditions indoors can be ameliorated by constructing buildings on earth mound having a slant surface with a slope of 10° on upstream side;

ee) In case of industrial buildings, the window height should be about 1.6 m and width about two-third of wall width. These should be located at a height of 1.1 m above the floor. In addition to this, openings around 0.9 m high should be provided over two-third length of the glazed area in the roof lights;

ff) Height of industrial buildings, although determined by the requirements of industrial processes involved, are generally kept large enough to protect the workers against hot stagnant air below the ceiling as also to dilute the concentration of contaminant inside. However, if high level openings in roof or walls are provided, building height can be reduced to 4 m without in any way impairing the ventilation performance; and

gg) The maximum width up to which buildings of height usually found in factories, being effectively ventilated by natural means by wind action, is 30 m, beyond which sufficient reliance cannot be placed on prevailing winds. Approximately half the ventilating area of openings should be between floor level and a height of 2.25 m from the floor.

NOTE — For data on outdoor wind speeds at a place, reference may be made to 'The Climatic Data Handbook prepared by Central Building Research Institute, Roorkee, 1999'.

5.4.3.2 *By stack effect*

Natural ventilation by stack effect occurs when air inside a building is at a different temperature than air outside. Thus, in heated buildings or in buildings wherein hot processes are carried on and in ordinary buildings during summer nights and during pre-monsoon periods, the inside temperature is higher than that of outside, cool outside air will tend to enter through openings at low level and warm air will tend to leave through openings at high level. It would, therefore, be advantageous to provide ventilators as close to ceilings as possible. Ventilators can also be provided in roofs as, for example, cowl, vent pipe, covered roof and ridge vent.

5.4.4 *Design Charts for Natural Ventilation*

Openings for natural ventilation can be sized without the need for complex and expensive computer simulation methods using analytical techniques. The design charts (DCs) provided hereunder can be used to assist in the sizing of ventilation openings to deliver natural ventilation.

The design charts are derived from analytical techniques. They provide either the geometrical free area (A_f) of openings to deliver a recommended air flow rate, or the achieved flow rate for a given free area.

Based on information for inside-outside air temperature difference (ΔT), wind speed (U) and wind pressure co-efficient (ΔC_p), the DCs are applicable for a wide range of weather conditions and parameters appropriate for residential buildings located in the selected Indian climatic zones.

These are considered to be the most common configurations encountered in practice in Indian residences. The corresponding DCs are:

a) DC-01 — Buoyancy-driven flow; single-sided ventilation with one opening (Fig. 10A and Fig. 11);

b) DC-02 — Buoyancy-driven flow; cross-ventilation with multiple openings (Fig. 10B and Fig. 12);

c) DC-03 — Wind-driven flow; single-sided ventilation with one opening (Fig. 10C and Fig. 13); and

d) DC-04 — Wind-driven flow; cross-ventilation with multiple openings (Fig. 10D and Fig. 14).

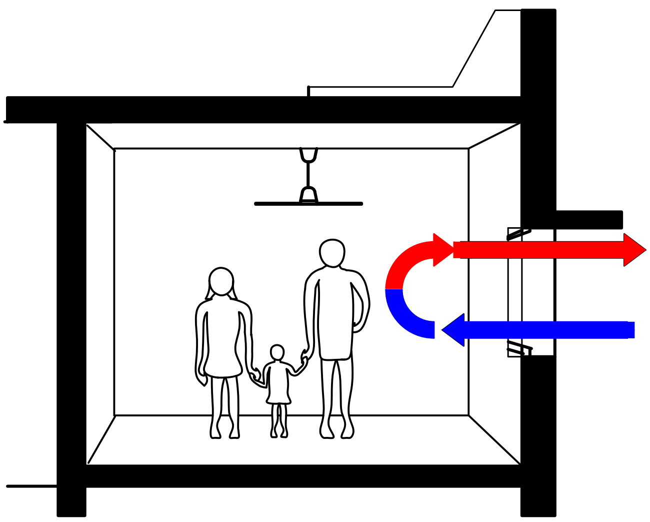


FIG. 10A DC-01

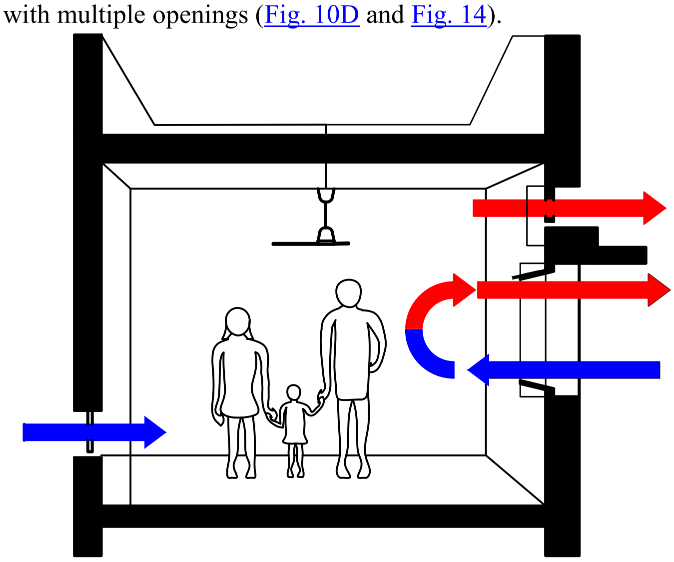


FIG. 10B DC-02

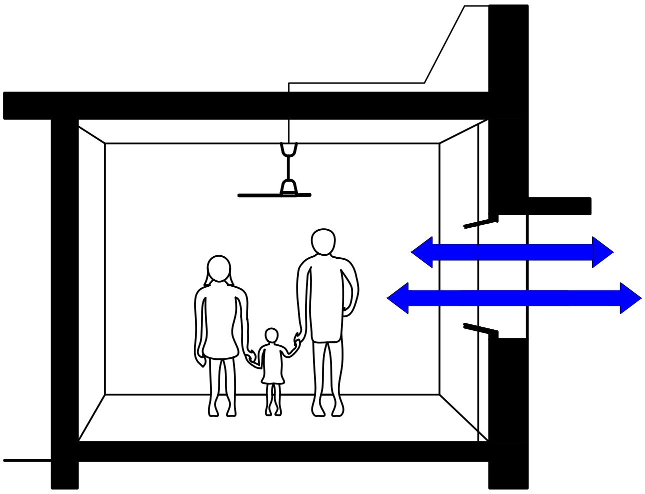


FIG. 10C DC-03

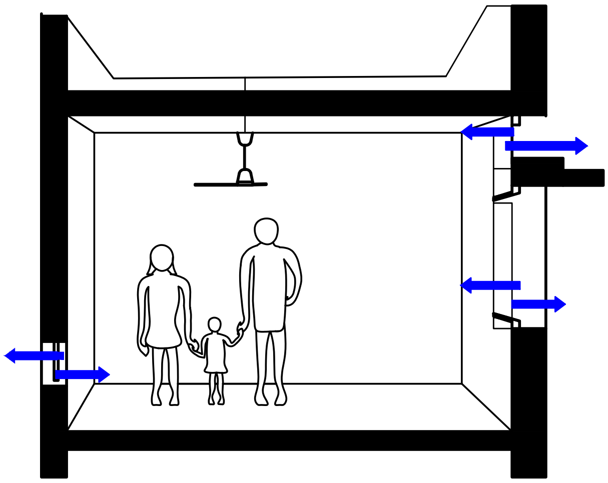


FIG. 10D DC-04

FIG. 10 CROSS-SECTION SKETCHES OF THE DRIVING FORCES FOR THE NATURAL VENTILATION SYSTEMS PRESENTED IN THE FOUR DESIGN CHARTS

5.4.4.1 *Design charts for natural ventilation — by stack effect*

a) DC-01 — Buoyancy-driven flow; single-sided ventilation with one opening;

The DC-01 is derived from the equation below (*see also* Fig. 11). For single opening, the value for C_d adopted is 0.25:

$$A_f = \frac{q_{rec}}{C_d}\sqrt{\frac{(T_{inside}+273)}{g\, h_a \Delta T_{inside\text{-}outside}}} \quad (\textit{see also } \textbf{C-1.2.1})$$

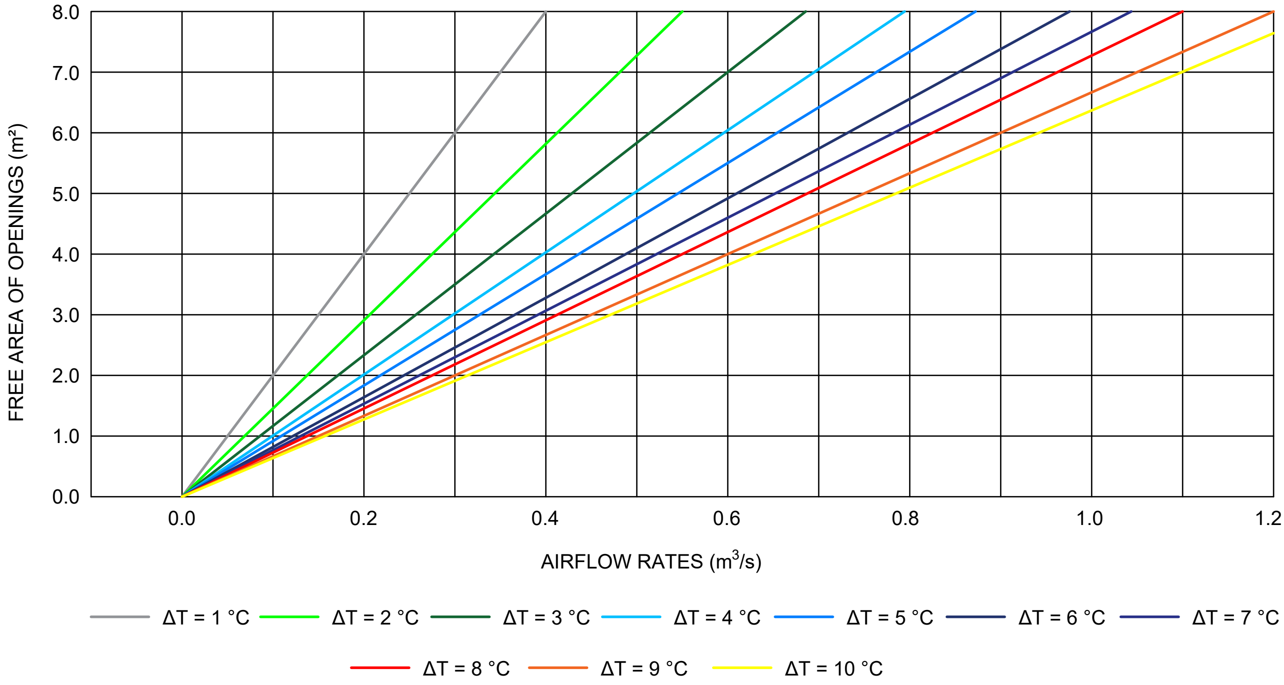


FIG. 11 DESIGN CHART DC-01 FOR SINGLE-SIDED SINGLE OPENING BUOYANCY DRIVEN FLOW

b) DC-02 — buoyancy-driven flow; cross-ventilation;

The DC-02 is also derived from the equation below (*see also* Fig. 12). Conversely, for multiple openings, the value for C_d adopted is 0.61:

$$V_z = V_{ref}\, k\, z^{\alpha} (m/s) \quad (\textit{see also } \textbf{C-1.2.3.1})$$

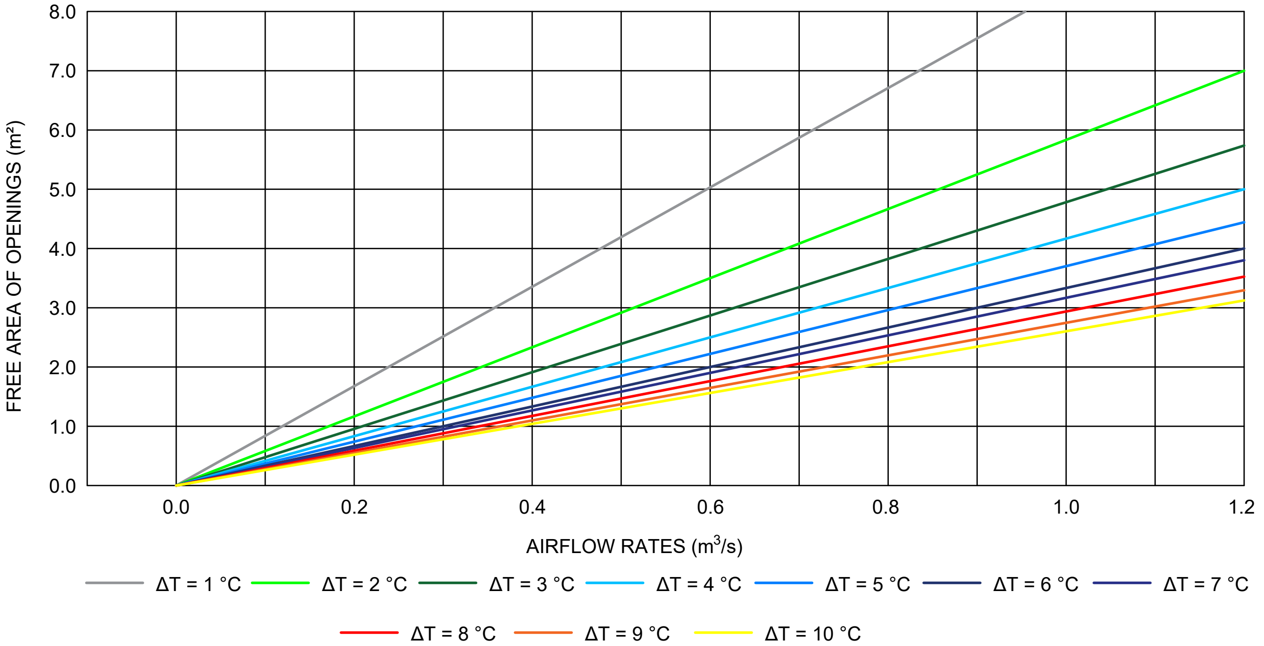


FIG. 12 DESIGN CHART DC-02 FOR MULTIPLE OPENINGS AND BUOYANCY-DRIVEN FLOW

5.4.4.2 *Design charts for natural ventilation — by wind effect*

a) DC-03 — Wind-driven flow; single-sided ventilation with one opening;

The DC-03 is derived from the following equation (*see also* Fig. 13):

$$A_f = {q_{rec}}/{C\,V_z} \text{ (see also C-1.2.3)}$$

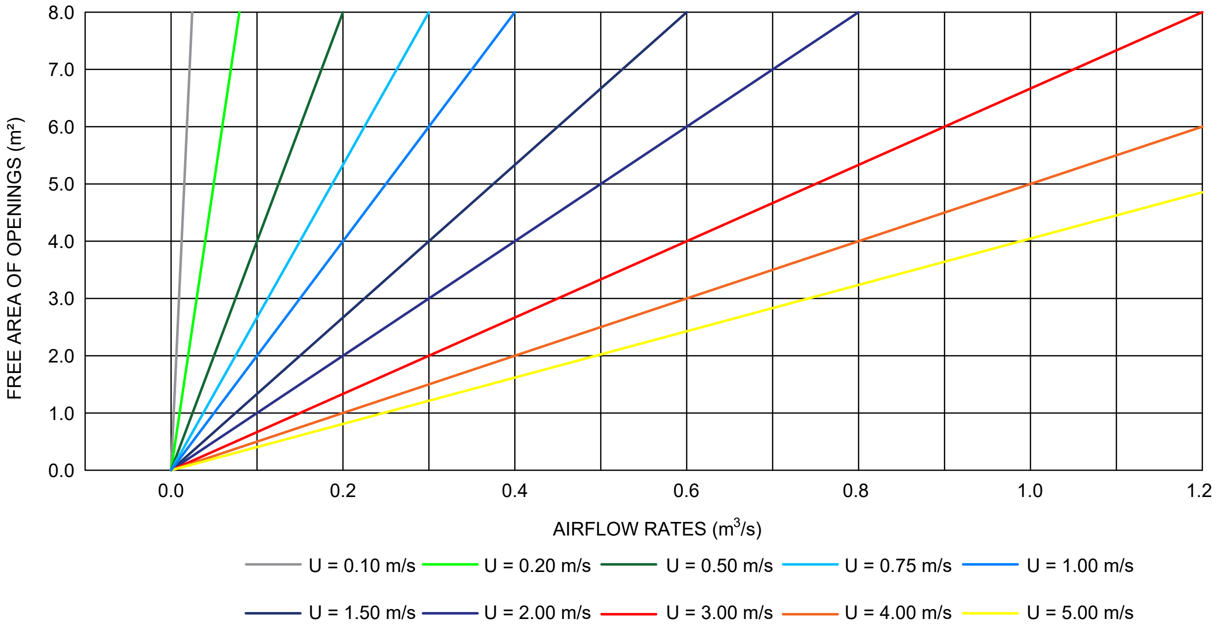


FIG. 13 DESIGN CHART DC-03 FOR SINGLE-SIDED SINGLE OPENINGS AND WIND-DRIVEN FLOW

b) DC-04 — Wind-driven flow; cross-ventilation with multiple openings;

The DC-04 is derived from the following equation (*see also* Fig. 14):

$$A_f = q_{ref}\left(C_d V_z \sqrt{\frac{\Delta C_p}{2}}\right)^{-1} \text{ (see also C-1.2.4)}$$

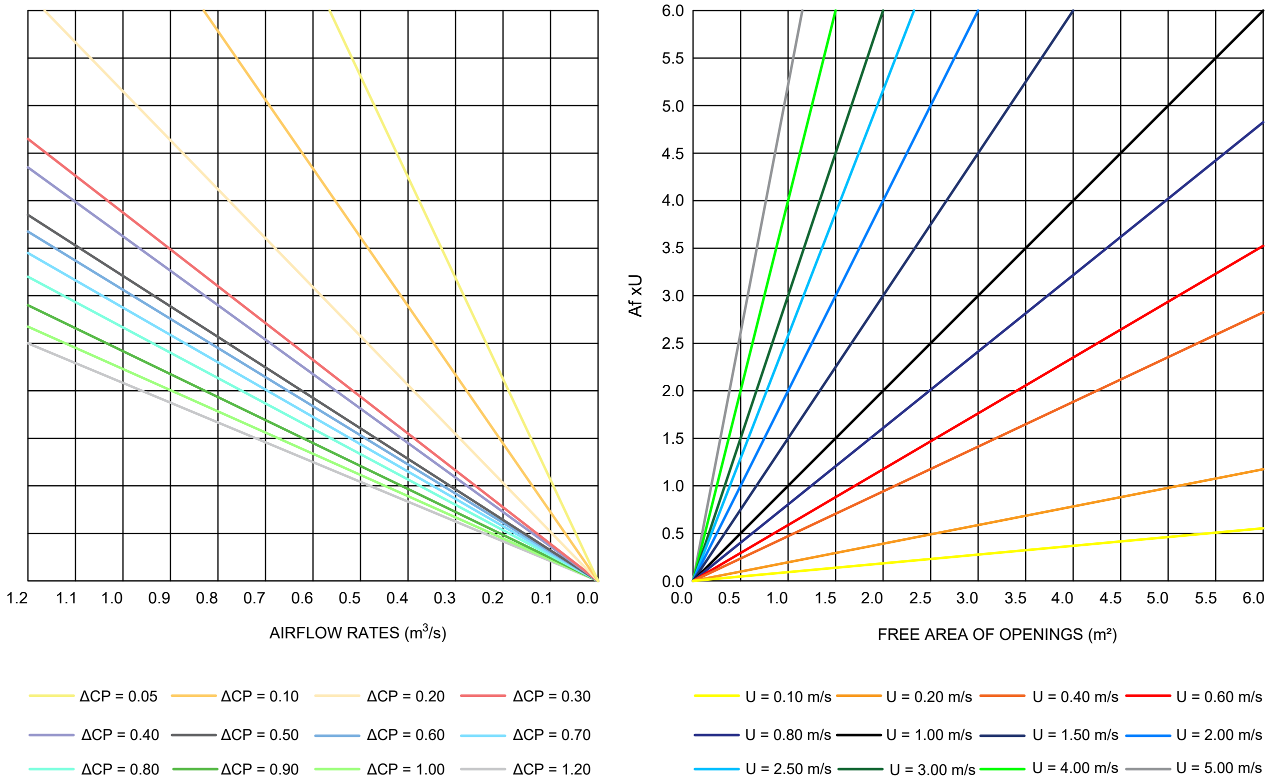


FIG. 14 DESIGN CHART DC-04 FOR MULTIPLE OPENINGS AND WIND-DRIVEN FLOW

5.4.4.3 For detailed application and use of the design charts for natural ventilation, reference may be made to Annex C.

5.5 Mechanical Ventilation

The requirements of mechanical ventilation should be in accordance with Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS.

5.6 Determining Rate of Ventilation

5.6.1 *Natural Ventilation*

This is difficult to measure as it varies from time to time. The amount of outside air through windows and other openings depends on the direction and velocity of wind outside (wind action) and/or convection effects arising from temperature or vapour pressure differences (or both) between inside and outside of the building (stack effect).

5.6.1.1 *Wind action*

For determining the rate of ventilation based on wind action the wind may be assumed to come from any direction within 45° of the direction of prevailing wind. Ventilation due to external wind is given by the following formula:

$$Q_w = K \times A \times V$$

where

Q_w = rate of air flow, in m^3/h;

K = coefficient of effectiveness, which may be taken as 0.6 for wind perpendicular to openings and 0.3 for wind at an angle less than 45° to the openings;

A = free area of inlet openings, in m^2; and

V = wind speed, in m/h.

NOTE — For wind data at a place, the local Meteorological Department may be consulted.

5.6.1.2 *Stack effect* (*thermal action*)

Ventilation due to convection effects arising from temperature difference between inside and outside is given by:

$$Q_T = 7.0\,A\sqrt{h\,(t_r - t_o)}$$

where

Q_T = rate of air flow, in m^3/h;

A = free area of inlet openings, in m^2;

h = vertical distance between inlets and outlets, in m;

t_r = average temperature of indoor air at height h, in °C; and

t_o = temperature of outdoor air, in °C.

NOTE — The equation is based on 0.65 times the effectiveness of openings. This should be reduced to 0.50 if conditions are not favourable.

5.6.1.3 When areas of inlet and outlet openings are unequal, the value of *A* may be calculated using the equation:

$$\frac{2}{A^2} = \frac{1}{A^2_{\text{inlet}}} + \frac{1}{A^2_{\text{outlet}}}$$

5.6.1.4 *Combined effect of wind and thermal action*

When both forces (wind and thermal) act together in the same direction, even without interference, the resulting air flow is not equal to the two flows estimated separately.

When acting simultaneously, the rate of air flow through the building may be computed by the following equation:

$$Q^2 = Q_w{}^2 + Q_t{}^2$$

where

Q = resultant volume of air flow, in m^3/min;

Q_w = volume of air flow due to wind force, in m^3/min; and

Q_t = volume of air flow due to thermal force, in m^3/min.

Wind velocity and direction, outdoor temperature, and indoor distribution cannot be predicted with certainty, and refinement in calculation is not justified. A simple method is to calculate the sum of the flows produced by each force separately. Then using the ratio of the flow produced by thermal forces to the aforementioned sum, the actual flow due to the combined forces can be approximated from Fig. 15. When the two flows are equal, the actual flow is about 30 percent greater than the flow caused by either force acting independently (*see* Fig. 15).

Judgment is necessary for proper location of openings in a building specially in the roof, where heat, smoke and fumes are to be removed. Usually, windward monitor openings should be closed, but if wind is so slight that temperature head can overcome it, all openings may be opened.

5.6.1.5 For method for determining the rate of ventilation based on probable indoor wind speed with typical illustrative example for residential building, reference may be made to **A-4** of good practice [D-1(9)].

5.6.2 *Mechanical Ventilation*

The determination of rate of ventilation in case of mechanical ventilation shall be done in accordance with Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS.

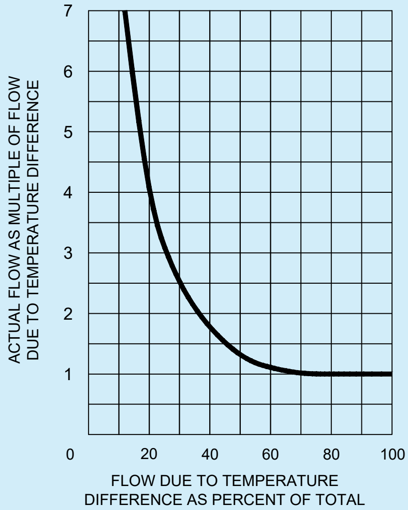


FIG. 15 DETERMINATION OF FLOW CAUSED BY COMBINED FORCES OF WIND AND TEMPERATURE DIFFERENCE

5.6.3 *Combined effect of Different Methods of Ventilation*

When combination of two or more methods of general ventilation is used, the total rate of ventilation shall be reckoned as the highest of the following three, and this rule shall be followed until an exact formula is established by research:

a) 1.25 times the rate of natural ventilation;

b) Rate of positive ventilation; and

c) Rate of exhaust of air.

5.6.4 *Measurement of Air Movement*

The rate of air movement of turbulent type at the working zone shall be measured either with a Kata thermometer (dry silvered type) or heated thermometer or properly calibrated thermocouple anemometer. Whereas anemometer gives the air velocity directly, the Kata thermometer and heated thermometer give cooling power of air and the rate of air movement is found by reference to a suitable nomogram using the ambient temperature.

5.7 Energy Conservation in Ventilation System

5.7.1 Maximum possible use should be made of wind induced natural ventilation. This may be accomplished by following the design guidelines given in 5.7.1.1.

5.7.1.1 Adequate number of circulating fans should be installed to serve all interior working areas during summer months in the hot dry and warm humid regions to provide necessary air movement at times when ventilation due to wind action alone does not afford sufficient relief.

5.7.1.1.1 The capacity of a ceiling fan to meet the requirement of a room with the longer dimension *D* metre should be about 55 D m^3/min.

5.7.1.1.2 The height of fan blades above the floor should be (3 H + W)/4, where H is the height of the room and W is the height of work plane.

5.7.1.1.3 The minimum distance between fan blades and the ceiling should be about 0.3 m.

5.7.2 Electronic regulators should be used instead of resistance type regulators for controlling the speed of fans.

5.7.3 When actual ventilated zone does not cover the entire room area, then optimum size of ceiling fan should be chosen based on the actual usable area of room, rather than the total floor area of the room. Thus, smaller size of fan can be employed and energy saving could be achieved.

5.7.4 Power consumption by larger fans is obviously higher, but their power consumption per square metre of floor area is less and service value higher. Evidently, improper use of fans irrespective of the rooms' dimensions is likely to result in higher power consumption. From the point of view of energy consumption, the number of fans and the optimum sizes for rooms of different dimensions are given in Table 17.

Table 17 Optimum Size and Number of Fans for Rooms of Different Sizes

(*Clause* 5.7.4)

Sl No.	Room Width m	Optimum Size (mm)/Number of Fans, For Room Length										
		4 m	5 m	6 m	7 m	8 m	9 m	10 m	11 m	12 m	14 m	16 m
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
i)	3	1 200/1	1 400/1	1 500/1	1 050/2	1 200/2	1 400/2	1 400/2	1 400/2	1 200/3	1 400/3	1 400/3
ii)	4	1 200/1	1 400/1	1 200/2	1 200/2	1 200/2	1 400/2	1 400/2	1 500/2	1 200/3	1 400/3	1 500/3
iii)	5	1 400/1	1 400/1	1 400/2	1 400/2	1 400/2	1 400/2	1 400/2	1 500/2	1 400/3	1 400/3	1 500/3
iv)	6	1 200/2	1 400/2	900/4	1 050/4	1 200/4	1 400/4	1 400/4	1 500/4	1 200/6	1 400/6	1 500/6
v)	7	1 200/2	1 400/2	1 050/4	1 050/4	1 200/4	1 400/4	1 400/4	1 500/4	1 200/6	1 400/6	1 500/6
vi)	8	1 200/2	1 400/2	1 200/4	1 200/4	1 200/4	1 400/4	1 400/4	1 500/4	1 200/6	1 400/6	1 500/6
vii)	9	1 400/2	1 400/2	1 400/4	1 400/4	1 400/4	1 400/4	1 400/4	1 500/4	1 400/6	1 400/6	1 500/6
viii)	10	1 400/2	1 400/2	1 400/4	1 400/4	1 400/4	1 400/4	1 400/4	1 500/4	1 400/6	1 400/6	1 500/6
ix)	11	1 500/2	1 500/2	1 500/4	1 500/4	1 500/4	1 500/4	1 500/4	1 500/4	1 500/6	1 500/6	1 500/6
x)	12	1 200/3	1 400/3	1 200/6	1 200/6	1 200/6	1 400/6	1 400/6	1 500/6	1 200/7	1 400/9	1 400/9
xi)	13	1 400/3	1 400/3	1 200/6	1 200/6	1 200/6	1 400/6	1 400/6	1 500/6	1 400/9	1 400/9	1 500/9
xii)	14	1 400/3	1 400/3	1 400/6	1 400/6	1 400/6	1 400/6	1 400/6	1 500/6	1 400/9	1 400/9	1 500/9

5.7.5 *Acceptable Air Temperature with the Increase in the Air Speed*

The natural ventilation (NV) – mixed mode (MM) solutions combine designed openings for natural ventilation with the simultaneous use of mechanical ceiling fans to deliver thermal comfort with minimal use of energy. The air movement induced by the fan allows a rise in the acceptable operative temperature which is proportional to the increase in the air speed. This combination will be referred to as NV-MM solutions for 'enhanced natural ventilation (NV+).

The total number of hours of the year for which the outdoor conditions are feasible for natural ventilation, calculated as per the adaptive thermal comfort model for India-mixed mode building, and estimated with the NV-MM solutions for NV+, are shown in Fig. 16 for each of the eight Indian cities. This estimation utilizes the values for the corresponding rise in the acceptable air temperature with the increase in the air speed shown in Table 18 for the scenario with $T_{radiant}$ = T_{air} (as in specialist literature) and applies to clothing insulation values (in clo) ranging from 0.50 clo to 0.70 clo and to metabolic rates (in met) ranging from 1.00 met ranging from 0.50 clo to 0.70 clo and to metabolic rates (in met) ranging from 1.00 met to 1.30 met.

NOTE — Specialist literature may be ASHRAE 2010.

This scenario is considered likely to occur in a typical residential environment during daytime and in summer season, with NV operating and without any mechanical cooling source (a radiant cooling panel). This scenario yields more conservative values than the scenario when occupants have some control over the local environment (as per specialist literature).

NOTE — Specialist literature may be ASHRAE 2017.

Table 18 Corresponding Rise in the Acceptable Air Temperature with the Increase in the Air Speed

(*Clause* 5.7.5)

Sl No.	Fan Speed Mode	Air Speed m/s	Corresponding Rise in Temperature °C		
			Applying: 0.50 clo to 0.70 clo; 1.00 met to 1.30 met		Applying: 0.50 clo to 1.00 clo; 1.10 met
			$T_{radiant} - T_{air}$ = -10	$T_{radiant} - T_{air}$ = 0	
(1)	(2)	(3)	(4)	(5)	(6)
i)	OFF	0.1	–	–	–
ii)	I	0.6	1.20	2.00	2.80
iii)	II	0.9	1.80	2.70	3.40
iv)	III	1.2	2.20	3.30	3.75
v)	IV	1.5	2.50	3.60	4.00

5.7.6 *Mixed-Mode Principles*

Buildings that are designed with a mixed-mode ventilation strategy rely on natural ventilation for much of the occupied hours but also have integrated mechanical ventilation and cooling systems that are used in certain climatic or internal heat gain conditions. Mixed-Mode (MM) systems include naturally ventilated buildings that are designed using one of the following approaches:

a) Concurrent mixed-mode (*see* Fig. 17A) — Where both systems operate at the same space at the same time. The mechanical system operates as supplementary ventilation or cooling while the occupants are free to operate the windows based on their personal preference. This strategy is the most common form of mixed mode ventilation strategy typically employed for open-plan offices and residential buildings;

b) Change-over mixed-mode (*see* Fig. 17B and Fig. 17C) — Where the mechanical systems and natural ventilation operate in the same space at different times of the day or year. This approach requires an automated control system to determine which mode should be used at a particular point in time. The selection is based on a variety of input parameters such as outdoor conditions and occupancy. The system operating as change-over MM in a residence can either run automatically or inform the occupants to open or close the windows and switch on/off HVAC systems; and

c) Zoned mixed-mode (*see* Fig. 17D) — Where the mechanical cooling and natural ventilation operate at the same time but in different spaces of the building. This ventilation strategy is suitable for large office buildings where some offices could use natural ventilation by operating the windows whereas other rooms such as conference rooms may require a mechanical system for fresh air distribution.

Mixed-mode ventilation presents an energy efficient alternative to full air conditioning by capitalizing on the potential for natural ventilation when conditions permit.

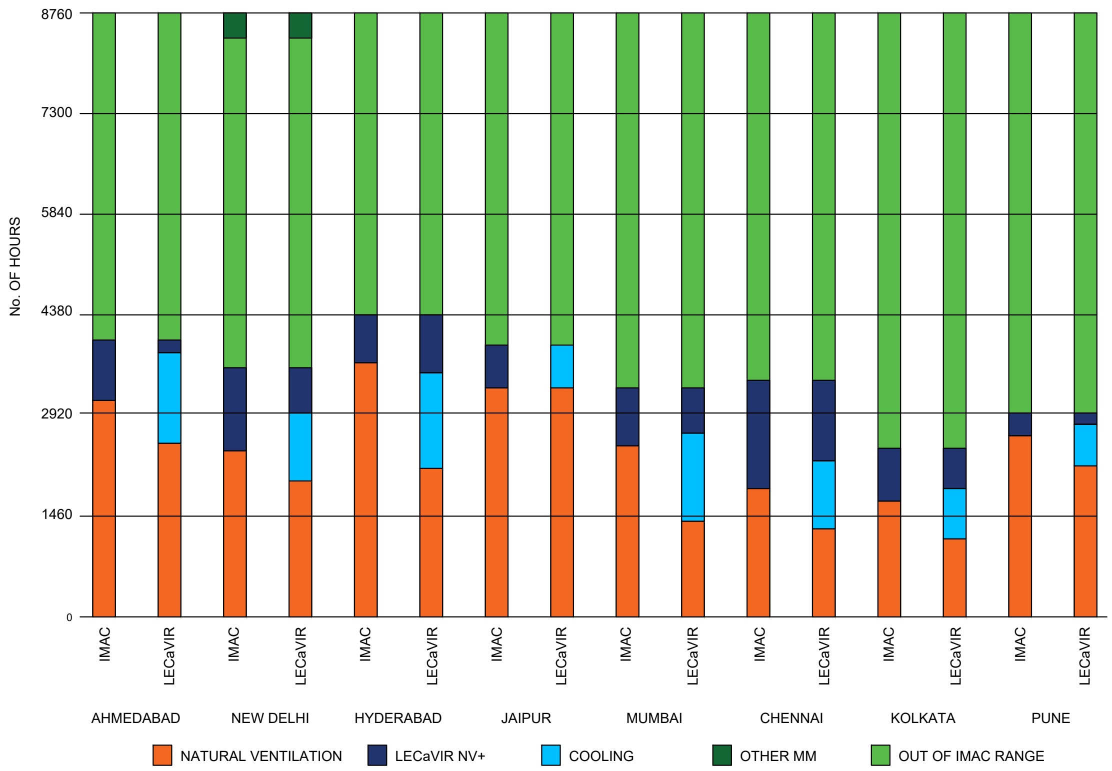


FIG. 16 NUMBER OF HOURS IN A YEAR USING 'NATURAL VENTILATION, MECHANICAL COOLING AND OTHER MM', FROM THE ADAPTIVE THERMAL COMFORT MODEL FOR INDIA-MIXED MODE BUILDINGS, AND USING 'NATURAL VENTILATION, ENHANCED NATURAL VENTILATION (NV+), MECHANICAL COOLING AND OTHER MM', FROM NV-MM SOLUTIONS FOR NV+ FOR EIGHT INDIAN CITIES

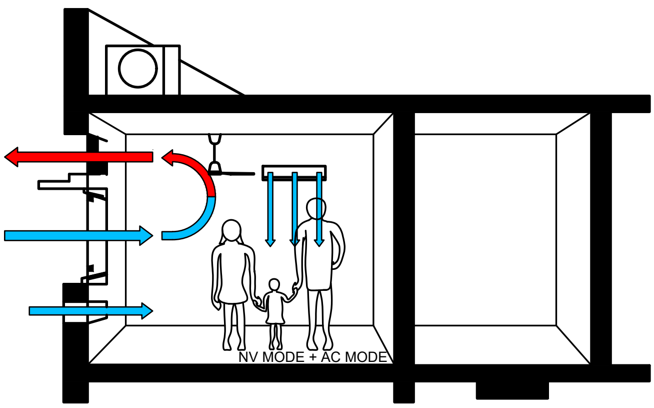


FIG. 17A CONCURRENT MIXED-MODE SYSTEM

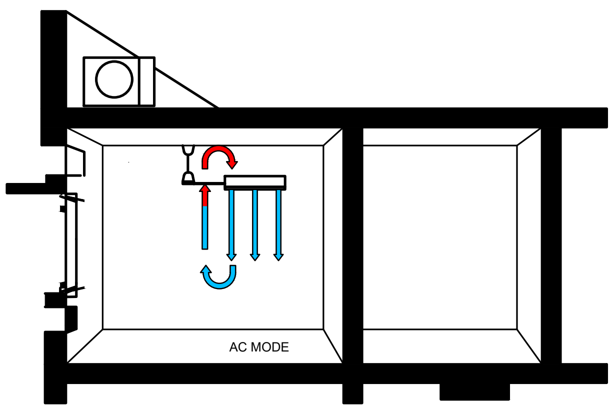


FIG. 17B CHANGE-OVER MIXED-MODE SYSTEM (AC MODE)

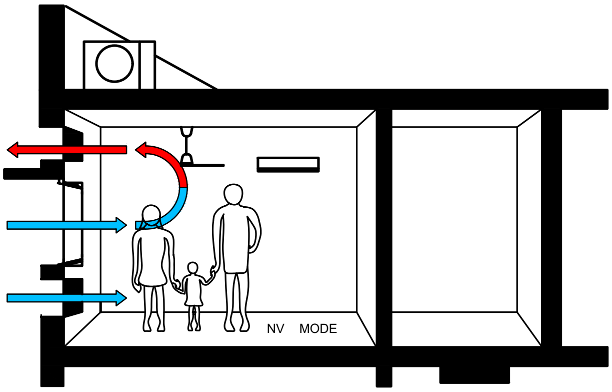


FIG. 17C CHANGE-OVER MIXED-MODE SYSTEM (NV MODE)

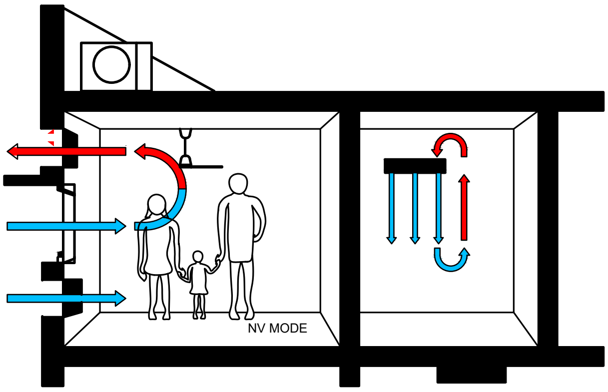


FIG. 17D ZONED MIXED-MODE SYSTEM FOR HOT DAY

FIG. 17 CLASSIFICATION OF MIXED-MODE SYSTEMS

ANNEX A

(*Clauses* 3.4.1 *and* 4.2.5.3)

METHOD OF CALCULATING SOLAR LOAD ON VERTICAL SURFACES OF DIFFERENT ORIENTATION

A-1 DETAILS OF CALCULATION

A-1.1 The solar energy above the earth's atmosphere is constant and the amount incident on unit area normal to sun's rays is called solar constant (1.360 8 kWm^{-2} or 2 cal/cm^2/min). This energy, in reaching the earth's surface, is depleted in, the atmosphere due to scattering by air molecules, water vapour, dust particles, and absorption by water vapour and ozone. The depletion varies with varying atmospheric conditions. Another important cause of depletion is the length of path traversed by sun's rays through the atmosphere. This path is the shortest when sun is at the zenith and, as the altitude of the sun decreases, the length of path in the atmosphere increases. Fig. 18 gives the computed incident solar energy/hour on unit surface area normal to the rays under standard atmospheric conditions (*see* Note below) for different altitudes of the sun.

NOTE — The standard atmospheric conditions assumed for this computation are: cloud-free, 300 dust particles per cm^3, 15 mm of precipitable water, 2.5 mm of ozone, at sea level.

A-1.2 In order to calculate the solar energy on any surface other than normal to the rays, the altitude of the sun at that time should be known. The corresponding value of direct solar radiation (I_N) should then be found with the help of Fig. 19. The solar radiation incident on any surface (I_S) is given by:

$$I_S = I_N\left(\sin\beta\sin\phi + \cos\beta\cos\alpha\cos\phi\right)$$

where

β = solar altitude;

ϕ = angle tilt of the surface from the vertical (*see* Fig. 19); and

α = wall solar azimuth angle.

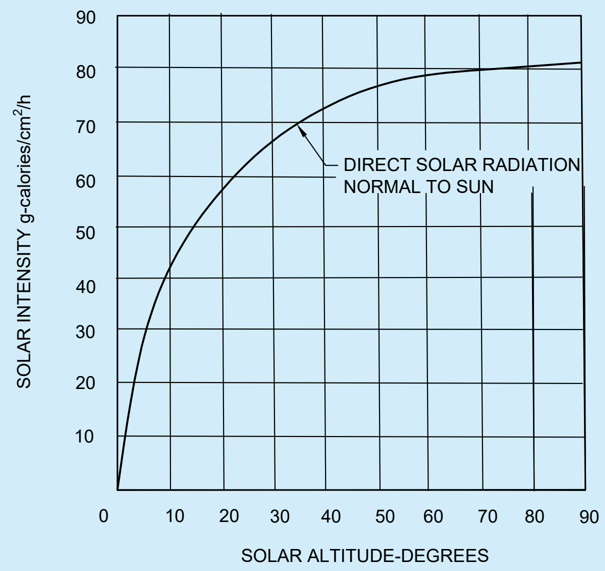


FIG. 18 DIRECT SOLAR INTENSITIES NORMAL TO SUN AT SEA LEVEL FOR STANDARD CONDITION (COMPUTED)

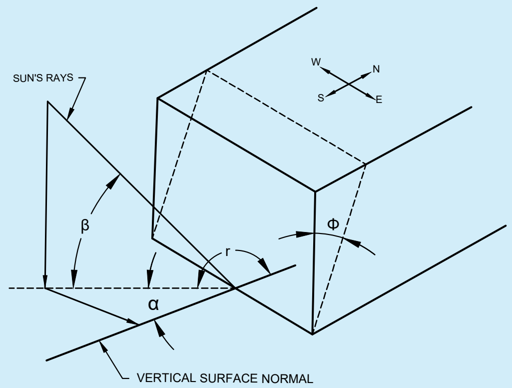


FIG. 19 DEFINITION OF SOLAR ANGLES

A-2 EXAMPLE TO FIND OUT ORIENTATION ON THE BASIS OF SOLAR LOAD

A-2.1 Example

A-2.1.1 As an example, a simple building with flat roof, 10 m × 20 m and 4 m high is dealt with below. For the sake of generalization, no shading device or verandah is taken.

A-2.1.2 As the roof is horizontal, it will receive the same solar heat in any orientation.

A-2.1.3 The area of the vertical surfaces are 4 m × 10 m = *A* (say) and 4 m × 20 m = 2*A*. Since, the external wall surfaces are not in shade except when the sun is not shining on them, the total solar load in a day on a surface can be obtained by multiplying the total load per unit area per day (*see* Table 3) by the area of the surface. For four principal orientations of the building, the total solar load on the building is worked out in Table 19.

A-2.1.4 From Table 19, it can be seen that for the above type of building, orientation 3 (longer surface facing North and South) is appropriate as it affords maximum solar heat gain in winter and in summer. This is true for all places of India from the point of solar heat gain. By further increasing the length to breadth ratio, the advantage of this orientation will be more pronounced. It may also be noted that in higher altitudes, the relative merit of this orientation is more.

A-2.1.5 It is also seen that the total solar heat on the building is the same for orientation 2 and 4. But if the site considerations require a choice between these two, orientation 2 should be preferred at places north of latitude 23° N and orientation 4 at southern places. This is so because the total solar load per unit area in summer on the north western wall decreases with the increase in latitude and that on the south western wall increases. It would, therefore, be advantageous to face only smaller surface of the building to greater solar load in the summer afternoons, when the air temperature also is higher.

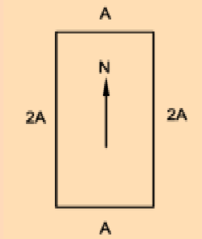
A-2.1.6 At hill stations, winter season cause more discomfort and so sole criterion for optimum orientation should be based on receiving maximum solar energy on building in winter.

Table 19 Solar Heat Gained Due to Orientation of Buildings

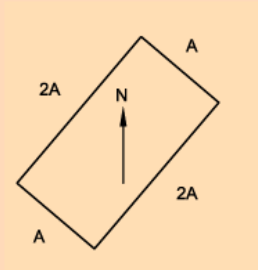
(*Clauses* A-2.1.3 *and* A-2.1.4)

Sl No.	Orientation	Solar Heat Gained							
		8º N THIRUVANANTHAPURAM		13º N CHENNAI		19º N MUMBAI		23º N KOLKATA	
		May 16	Dec 22	May 16	Dec 22	May 16	Dec 22	May 16	Dec 22
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	North	2 177 × A = 2 177A	–	1 625 × A = 1 625A	–	962 × A = 962A	–	741 × A = 741A	–
	East	2 618 × 2A = 5 236A	2 177 × 2A = 4 354A	2 697 × 2A = 5 394A	2 019 × 2A = 4 038A	2 795 × 2A = 5 590A	1 830 × 2A = 3 660A	2 871 × 2A = 5 742A	1 703 × 2A = 3 406A
	South	–	4 164 × A = 4 164A	–	4 385 × A = 4 385A	–	4 574 × A = 4 574A	205 × A = 205A	4 637 × A = 4 637A
	West	2 618 × 2A = 5 236A	2 177 × 2A = 4 354A	2 697 × 2A = 5 394A	2 019 × 2A = 4 038A	2 795 × 2A = 5 590A	1 830 × 2A = 3 660A	2 871 × 2A = 5 742A	1 703 × 2A = 3 406A
	Total	12 649A	12 872A	12 413A	12 461A	12 142A	11 894A	12 430A	11 449A
ii)	NE	2 650 × A = 2 650A	410 × A = 410A	2 492 × A = 2 492A	315 × A = 315A	2 255 × A = 2 255A	237 × A = 237A	2 192 × A = 2 192A	173 × A = 173A
	SE	1 167 × 2A = 2 334A	3 391 × 2A = 6 782A	1 341 × 2A = 2 682A	3 423 × 2A = 6 846A	1 640 × 2A = 3 280A	3 438 × 2A = 6 876A	1 845 × 2A = 3 690A	3 454 × 2A = 6 908A
	SW	1 167 × 2A = 2 334A	3 391 × A = 3 391A	1 341 × A = 1 341A	3 423 × A = 3 423A	1 640 × A = 1 640A	3 438 × A = 3 438A	1 845 × A = 1 845A	3 454 × A = 3 454A
	NW	2 650 × 2A = 5 300A	410 × 2A = 820A	2 492 × A = 4 984A	315 × 2A = 630A	2 255 × 2A = 4 510A	237 × 2A = 474A	2 192 × 2A = 4 384A	173 × 2A = 346A
	Total	12 618A	11 403A	11 499A	11 214A	11 685A	11 025A	12 111A	10 881A
iii)	North	2 177 × 2A = 4 354A	–	1 625 × 2A = 3 250A	–	962 × 2A = 1 924A	–	741 × 2A = 1 482A	–
	East	2 618 × A = 2 618A	2 177 × A = 2 177A	2 697 × A = 2 697A	2 019 × A = 2 019A	2 795 × A = 2 795A	1 830 × A = 1 830A	2 871 × A = 2 871A	1 703 × A = 1 703A
	South	–	4 164 × 2A = 8 328A	–	4 385 × 2A = 8 770A	–	4 574 × 2A = 9 148A	205 × 2A = 410A	4 637 × 2A = 9 274A
	West	2 618 × A = 2 618A	2 177 × A = 2 177A	2 697 × A = 2 697A	2 019 × A = 2 019A	2 795 × A = 2 795A	1 830 × A = 1 830A	2 871 × A = 2 871A	1 703 × A = 1 703A
	Total	9 590A	12 602A	8 644A	12 808A	7 514A	12 808A	7 634A	12 680A
iv)	NE	2 650 × 2A = 5 300A	–	2 492 × A = 4 984A	315 × 2A = 630A	2 255 × 2A = 4 510A	237 × 2A = 474A	2 192 × 2A = 4 384A	173 × 2A = 346A
	SE	1 167 × A = 1 167A	2 177 × A = 2 177A	1 341 × A = 1 341A	3 423 × A = 3 423A	1 640 × A = 1 640A	3 438 × A = 3 438A	1 845 × A = 1 845A	3 454 × A = 3 454A
	SW	1 167 × 2A = 2 334A	4 164 × 2A = 8 328A	1 341 × 2A = 2 682A	3 423 × 2A = 6 846A	1 640 × 2A = 3 280A	3 438 × 2A = 6 876A	1 845 × 2A = 3 690A	3 454 × 2A = 6 908A
	NW	2 650 × A = 2 650A	2 177 × A = 2 177A	2 492 × A = 2 492A	315 × A = 315A	2 255 × A = 2 255A	237 × A = 237A	2 192 × A = 2 192A	173 × A = 173A
	Total	11 451A	12 682A	11 499A	11 214A	11 685A	11 025A	12 111A	10 881A

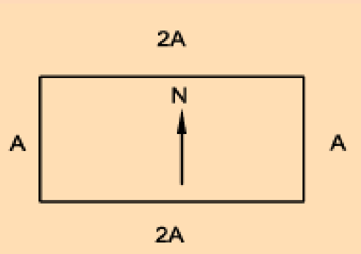
		29° N DELHI	
		May 16	Dec 22
i)	North	536 × A = 536A	–
	East	2 950 × 2A = 5 900A	1 467 × 2A = 2 934A
	South	741 × A = 741A	4 543 × A = 4 543A
	West	2 950 × 2A = 5 900A	1 467 × 2A = 2 934A
	Total	13 077A	10 411A
ii)	NE	2 098 × A = 2 098A	110 × A = 110A
	SE	2 192 × 2A = 4 384A	3 265 × 2A = 6 530A
	SW	2 192 × A = 2 192A	3 265 × A = 3 265A
	NW	2 098 × 2A = 4 196A	110 × 2A = 220A
	Total	12 870A	10 125A
iii)	North	536 × 2A = 1 072A	–
	East	2 950 × A = 2 950A	1 467 × A = 1 467A
	South	741 × 2A = 1 482A	4 543 × 2A = 9 086A
	West	2 950 × A = 2 950A	1 467 × A = 1 467A
	Total	8 454A	12 020A
iv)	NE	2 098 × 2A = 4 196A	110 × 2A = 220A
	SE	2 192 × A = 2 192A	3 265 × A = 3 265A
	SW	2 192 × 2A = 4 384A	3 265 × 2A = 6 530A
	NW	2 098 × A = 4 196A	110 × A = 110A
	Total	12 870A	10 125A



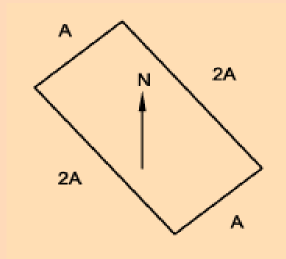
ORIENTATION 1



ORIENTATION 2



ORIENTATION 3



ORIENTATION 4

ANNEX B

(*Clauses* 4.2.5, 4.2.5.2, 4.2.5.3, 4.2.5.4 *and* 4.2.6)

SKY COMPONENT TABLES

B-1 DESCRIPTION OF TABLES

B-1.1 The three sky component tables are as given below:

a) Table 20 — Percentage sky components on the horizontal plane due to a vertical rectangular opening for the clear design sky;

b) Table 21 — Percentage sky components on the vertical plane perpendicular to a vertical rectangular opening for the clear design sky; and

c) Table 22 — Percentage sky components on the vertical plane parallel to a vertical rectangular opening for the clear design sky.

B-1.2 All the Table 20, Table 21 and Table 22 are for an unglazed opening illuminated by the clear design sky.

B-1.3 The values tabulated are the components at a point *P* distant from the opening on a line perpendicular to the plane of the opening through one of its lower corners, and *l* and *h* are the width and height respectively of the rectangular opening (*see* Fig. 20).

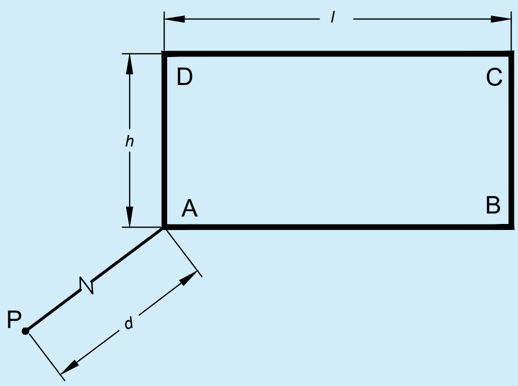


FIG. 20 DEPICTION OF OBSERVATIN POINT ' P '

B-1.4 Sky component for different *h*/*d* and *l*/*d* values are tabulated, that is, for windows of different size and for different distances of the point *P* from the window.

B-1.5 By suitable combination of the values obtained from the Table 20, Table 21 and Table 22, for a given point for a given window, the sky component in any plane passing through the point may be obtained.

B-1.6 Method of Using the Tables

B-1.6.1 Method of using the Table 20, Table 21 and Table 22 to get the sky component at given point is explained with help of the following example.

B-1.6.2 *Example*

It is desired to calculate the sky component (F) due to a vertical window ABCD with width 1.8 m and height 1.5 m at a point P on a horizontal plane 3.0 m from the window wall located as shown in the Fig. 21. Foot of the perpendicular N is 0.6 m below the sill and 0.9 m to the left of AD.

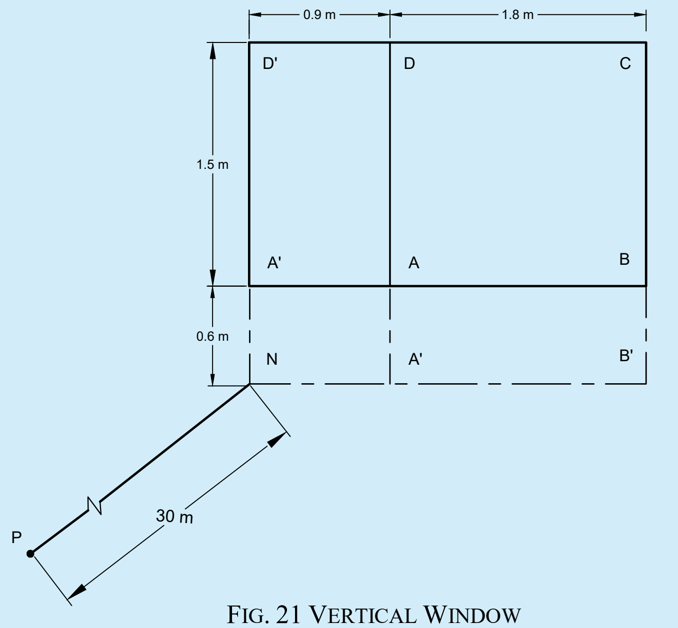


FIG. 21 VERTICAL WINDOW

Consider ABCD extended to NB'CD'

a) For NB'CD'

$l/d = (1.8 + 0.9)/3 = 0.9$

$h/d = (1.5 + 0.6)/3 = 0.7$

$F_1 = 5.708$ percent (from Table 20)

b) For NA'DD'

$l/d = 0.9/3 = 0.3$

$h/d = (1.5 + 0.6)/3 = 0.7$

$F_2 = 2.441$ percent (from Table 20)

c) For NB'BA'

$l/d = (1.8 + 0.9)/3 = 0.9$

$h/d = 0.6/3 = 0.2$

$F_3 = 0.878$ percent (from Table 20)

d) For NA'AA'

$l/d = 0.9/3 = 0.3$

$h/d = 0.6/3 = 0.2$

F_4 = 0.40 3 percent (from Table 20)

Since ABCD = NB'CD' – NA'DD' – NB'BA' + NA'AA'

Sky component, F = $F_1 - F_2 - F_3 + F_4$
= 5.708 – 2.441 – 0.878 + 0.403
= 2.792

B-2 GENERAL INSTRUCTIONS

B-2.1 For irregular obstructions like row of trees parallel to the plane of the window, equivalent straight boundaries horizontal and vertical, maybe drawn.

B-2.2 Waldram diagrams are an effective early-stage design tool for planning daylight and direct solar illuminance for extremely irregular obstructions or obstructions not in a plane parallel to the window. Fig. 22 and Fig. 23 show two possible massing schemes for a site located at 17° N latitude, while Fig. 24 and Fig. 25 illustrate the corresponding Waldram diagrams drawn from the central open area of the site. The white area above the shadow silhouettes in Fig. 24 and Fig. 25 indicate unobstructed sky (48 percent sky view in case of Fig. 24 and 65 percent sky view in case of Fig. 25). The yellow lines in Fig. 24 and Fig. 25 indicate the annual sun path as seen from the middle of the central open area of the site, aiding in evaluating sky visibility and optimizing site layouts for improved daylight performance.

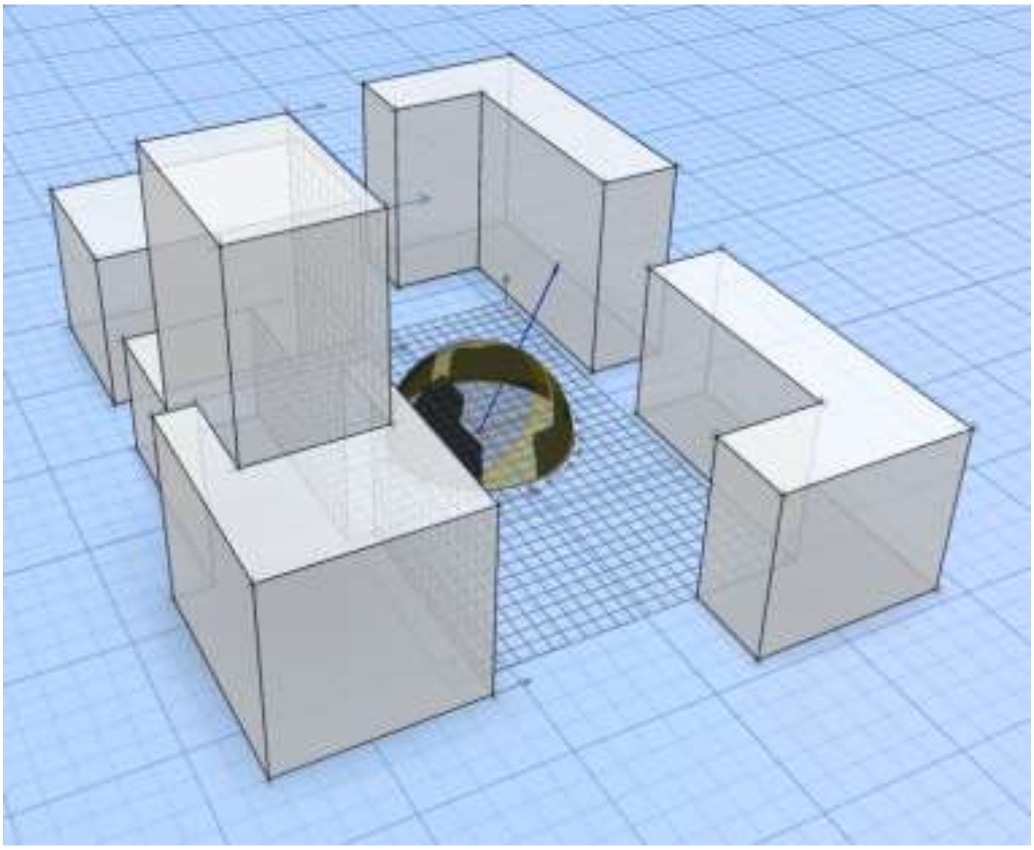


FIG. 22 MASSING SCHEME A

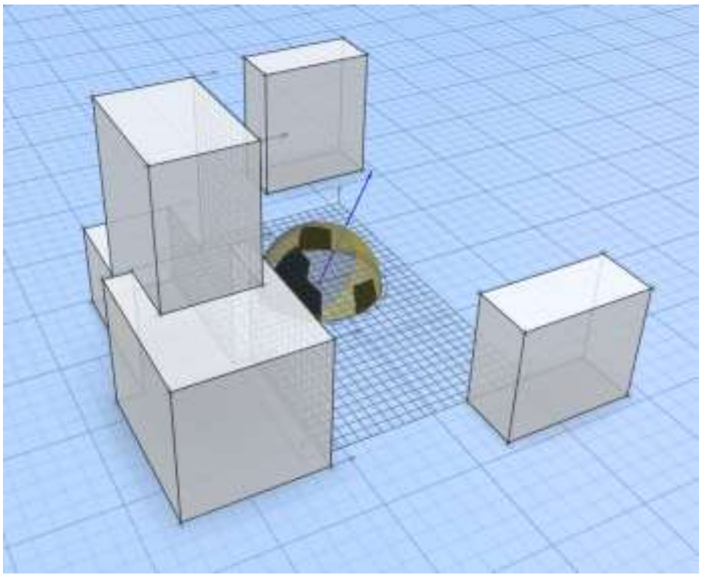


FIG. 23 MASSING SCHEME B

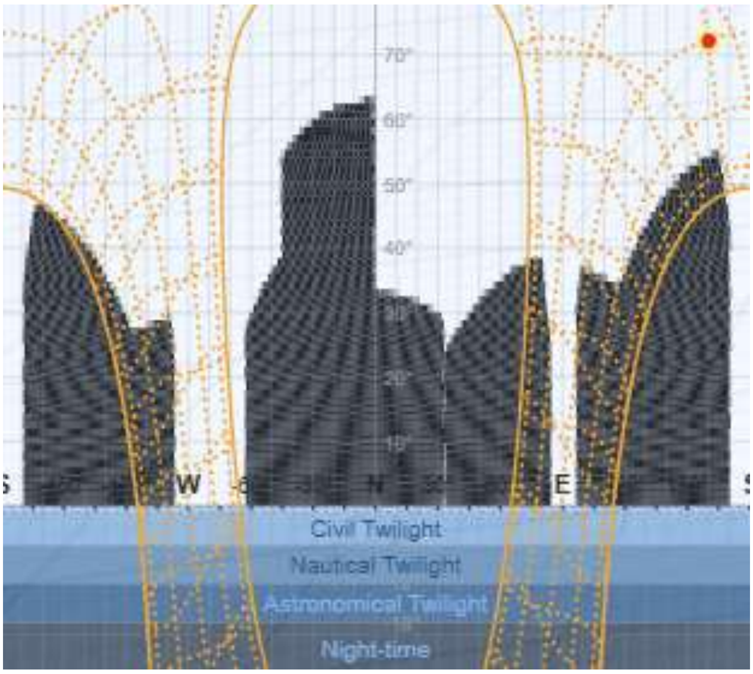


FIG. 24 WALDRAM DIAGRAM A

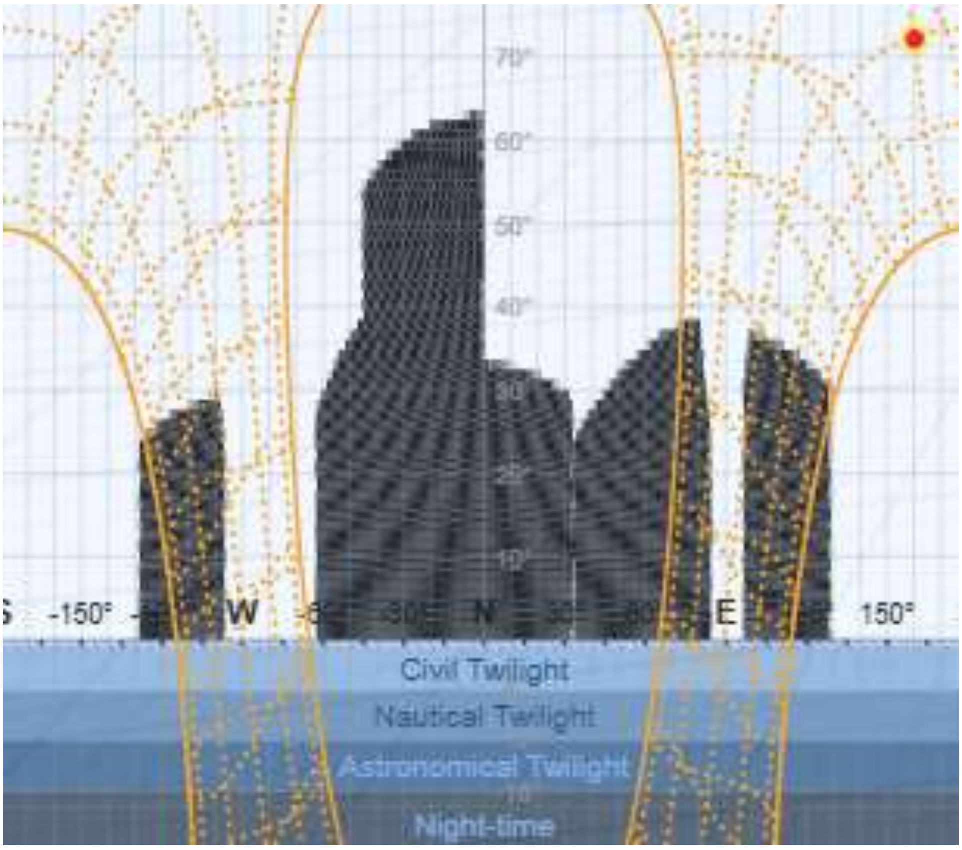


FIG. 25 WALDRAM DIAGRAM B

B-2.3 For bay windows, dormer windows or corner windows the effective dimensions of window opening computed should be taken when using the Table 20, Table 21 and Table 22 to find the sky components.

B-3 CALCULATION OF INTERNAL REFLECTED COMPONENT

B-3.1 The internal reflected component (IRC) is a variable quantity which varies from point to point in a room depending upon the interior finish. IRC value is maximum at the centre of the room and decreases elsewhere in all directions. For processing calculations of IRC at any given point of the room, special techniques have to be made out. The internal reflected component may be calculated by using the formula:

$$IRC = \frac{0.85W}{A(1\text{-R})}\ (\mathrm{C}R_{\mathrm{fw}} + 10R_{\mathrm{cw}})$$

where

W = window area;

C = constant of value 78 when there is no external obstruction but it has different values as shown in the following table when there are obstructions;

R_{fw} = average reflection factor of the floor and those parts of the wall below the plane of the mid-height of the window (excluding the window wall);

R_{cw} = average reflection factor of the ceiling and those parts of the wall above the plane of the mid-height of the window (excluding the window wall);

A = area of all the surfaces in the room (ceiling walls, floor and windows); and

R = average reflection factor of all surfaces in the room (ceiling, walls, floor and windows) expressed as a decimal part of unity.

Sl No. (1)	*Angle of Obstruction* Degree (2)	*Sky + External Obstruction* C (3)
i)	5	68.9
ii)	15	50.6
iii)	25	36.2
iv)	35	26.7
v)	45	20.1
vi)	55	15.8
vii)	65	12.9
viii)	75	11.1
ix)	85	10.36

B-3.2 Example

Consider two rooms of dimensions:

Room X = 6 m (l) × 5 m (w) × 3 m (ht)

Room Y = 3.7 m × 3 m × 3 m

Let the window area be 15 percent of the floor area and be glazed.

Window size in room X = 2.5 m × 1.8 m

Window size in room Y = 3.7 m × 3 m

The windows are on the 6 m × 3 m side in room X and 3.7 m × 3 m side in room Y, and the sill heights are 0.9 m from floor level.

Reflection coefficients of:

walls and ceiling = 70 percent
floor = 20 percent
glazing = 15 percent

a) <u>Value of IRC in room X:</u>

1) Total interior area, $A = 2\ (30 + 18 + 15) = 126\ \text{m}^2$;

2) Average reflection factor of interior:

$$R = \frac{(61.5 \times 0.7) + (30 \times 0.7) + (30 \times 0.2) + (4.5 \times 0.15)}{(61.5 + 30 + 30 + 4.5)} = 0.56$$

3) $1 - R = 0.44$;

4) Mid-height of window is 1.83 m from floor, average reflection factor of room below 1.83 m level excluding the wall containing the window:

$$R_{fw} = \frac{(29.28 \times 0.7) + (30 \times 0.2)}{(29.28 + 30)} = 0.45$$

5) Average reflection factor of room above 1.83 m level excluding the wall containing the window:

$$R_{\mathrm{cw}} = \frac{(18.72 \times 0.7) + (30 \times 0.7)}{(18.72 + 30)} = 0.7$$

6) IRC = $\frac{0.85 \times 4.5}{126 \times 0.44}$ $[(78 \times 0.45) + (10 \times 0.7)] = 2.904$.

b) Value of IRC in room Y:

1) Total interior area:

$$A = 2[(3.7 \times 3) + (3.7 \times 3) + (3 \times 3)] = 62.4 \text{ m}^2$$

2) Average reflection factor:

$$R = \frac{(38 \times 55 \times 0.7 \times 3 \times 0.7) + (3.7\times 3 \times 0.2) + (1.5 \times 1.1 \times 0.15)}{(38.55 + 11.1 + 11.1 + 1.65)} = 0.596$$

3) Mid-height of window from floor = 1.46 m;

4) Average reflection factor below 1.46 m level:

$$R_{\mathrm{fw}} = \frac{(3.7 \times 3 \times 0.7) + (1.54 \times 9.7 \times 0.7)}{(11.1 + 14.94)} = 0.48$$

5) Average reflection factor above 1.46 m level:

$$R_{\mathrm{cw}} = \frac{(3.7 \times 3 \times 0.7) + (1.54 \times 9.7 \times 0.7)}{(11.1 + 14.94)} = 0.7$$

6) IRC = $\frac{0.85 \times 1.65}{62.4 \times 0.404}$ $[(78 \times 0.48) + (10 \times 0.7)] = 2.472$.

B-4 GENERAL NOTE ON DAYLIGHTING OF BUILDING

B-4.1 The main aim of day lighting design is how to admit enough light for good visibility without setting up uncomfortable glare. No simple solution may be given as the sky varies so much in its brightness from hour to hour, and from season to season.

B-4.2 Different visual tasks need differing amounts of lights for the same visual efficiency. The correct amount of light for any task is determined by the following:

a) *Characteristics of the tasks* — Size of significant detail, contrast of detail with background and how close it is to the eyes;

b) *Sight of the worker* — For example, old people need more light;

c) Speed and accuracy necessary in the performance of work. If no errors are permissible, much more light is needed; and

d) *Ease and comfort of working* — Long and sustained tasks should be done easily whereas workers can make a special effort for tasks of very short duration.

These factors have been made the subject of careful analysis as a result of which tables of necessary levels of illumination have been draw up.

B-4.3 Levels of lighting determined analytically shall be translated into levels of daylight and then into size of window opening or vice versa for checking the size of window assumed for required levels of daylight.

B-4.4 One of the many important factors involved in the translation is the lightness of the room surface. The illumination levels in a given room with a finite window will be higher when the walls are light coloured than when these are dark coloured. It is necessary, therefore, at an early stage to consider the colouring of the rooms of the building and not to leave this until later. Lighting is not merely a matter of window openings and quite half the eventual level of lighting may be dependent on the decoration in the room. Whatever may be the colour the occupant wants to use, it is most desirable to maintain proper values of reflectance factors for ceiling, wall and floors so that the level of daylight illumination is maintained.

Table 20 Percentage Sky Components on the Horizontal Plane Due to a Vertical Rectangular Opening for the Clear Design Sky

(*Clauses* 4.2.5, B-1.1, B-1.2, B-1.5, B-1.6.1, B-1.6.2, *and* B-2.3)

Sl No.	***l/d*** → ***h/d*** ↓	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**	**1.0**	**1.1**	**1.2**
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	0.1	0.036	0.071	0.104	0.133	0.158	0.179	0.198	0.213	0.225	0.235	0.243	0.250
ii)	0.2	0.141	0.277	0.403	0.516	0.614	0.699	0.770	0.829	0.878	0.918	0.950	0.977
iii)	0.3	0.300	0.589	0.859	1.102	1.315	1.499	1.653	1.782	1.888	1.976	2.048	2.108
iv)	0.4	0.460	0.905	1.322	1.702	2.041	2.337	2.590	2.804	2.984	3.134	3.258	3.361
v)	0.5	0.604	1.189	1.741	2.247	2.700	3.099	3.444	3.740	3.992	4.204	3.383	4.553
vi)	0.6	0.732	1.443	2.114	2.732	3.289	3.781	4.211	4.582	4.900	5.171	5.401	5.596
vii)	0.7	0.844	1.665	2.441	3.159	3.808	4.385	4.891	5.330	5.708	6.034	6.311	6.548
viii)	0.8	0.942	1.858	2.727	3.532	4.262	4.914	5.488	5.989	6.423	6.798	7.119	7.395
ix)	0.9	1.026	2.025	2.974	3.855	4.657	5.375	6.011	6.567	7.051	7.470	7.832	8.144
x)	1.0	1.099	2.169	3.188	4.135	5.000	5.776	6.465	7.071	7.600	8.060	8.458	8.803
xi)	1.1	1.161	2.294	3.372	4.377	5.296	6.124	6.861	7.510	8.079	8.576	9.008	9.383
xii)	1.2	1.215	2.401	3.531	4.586	5.553	6.425	7.204	7.893	8.498	9.027	9.489	9.892
xiii)	1.3	1.262	2.493	3.668	4.767	5.775	6.687	7.503	8.226	8.863	9.422	9.912	10.339
xiv)	1.4	1.302	2.573	3.787	4.924	5.968	6.915	7.764	8.517	9.183	9.769	10.283	10.733
xv)	1.5	1.337	2.643	3.891	5.060	6.136	7.114	7.991	8.772	9.664	10.073	10.609	11.080
xvi)	1.6	1.367	2.703	3.981	5.179	6.283	7.287	8.190	8.996	9.710	10.341	10.897	11.386
xvii)	1.7	1.394	2.756	4.060	5.283	6.412	7.440	8.366	9.192	9.927	10.577	11.151	11.657
xviii)	1.8	1.417	2.803	4.129	5.375	6.526	7.574	8.520	9.366	10.119	10.786	11.376	11.898
xix)	1.9	1.438	2.844	4.190	5.456	6.626	7.693	8.656	9.520	10.289	10.972	11.577	12.112
xx)	2.0	1.456	2.880	4.244	5.527	6.714	7.798	8.778	9.656	10.440	11.137	11.755	12.303
xxi)	3.0	1.559	3.087	4.553	5.937	7.223	8.403	9.478	10.448	11.321	12.103	12.804	13.431
xxii)	4.0	1.600	3.168	4.676	6.100	7.426	8.646	9.759	10.768	11.678	12.498	13.235	13.897
xxiii)	5.0	1.620	3.208	4.735	6.179	7.525	8.765	9.897	10.925	11.854	12.693	13.448	14.128
xxiv)	10.0	1.648	3.263	4.818	6.289	7.662	8.930	10.089	11.144	12.100	12.965	13.747	14.454
xxv)	INF	1.657	3.282	4.846	6.327	7.710	8.986	10.155	11.220	12.186	13.060	13.851	14.567

Table 20 (*Concluded*)

Sl No.	***l/d*** / ***h/d***	**1.3**	**1.4**	**1.5**	**1.6**	**1.7**	**1.8**	**1.9**	**2.0**	**3.0**	**4.0**	**5.0**	**10.0**	**INF**
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)
i)	0.1	0.256	0.261	0.264	0.268	0.270	0.272	0.274	0.276	0.284	0.286	0.287	0.288	0.288
ii)	0.2	0.999	1.018	1.033	1.046	1.056	1.065	1.072	1.079	1.110	1.118	1.122	1.125	1.125
iii)	0.3	2.157	2.197	2.231	2.259	2.282	2.302	2.318	2.333	2.401	2.421	2.429	2.436	2.437
iv)	0.4	3.446	3.516	3.574	3.623	3.664	3.699	3.728	3.753	3.873	3.909	3.922	3.935	3.937
v)	0.5	4.659	4.765	4.853	4.928	4.990	5.043	5.088	5.126	5.312	5.366	5.387	5.408	5.410
vi)	0.6	5.761	5.901	6.020	6.121	6.208	6.281	6.344	6.397	6.661	6.739	6.769	6.798	6.802
vii)	0.7	6.751	6.924	7.071	7.198	7.307	7.400	7.481	7.551	7.902	8.006	8.047	8.087	8.092
viii)	0.8	7.632	7.836	8.011	8.162	8.292	8.405	8.502	8.587	9.029	9.164	9.217	9.268	9.276
ix)	0.9	8.413	8.645	8.846	9.019	9.170	9.301	9.415	9.515	10.045	10.214	10.280	10.345	10.355
x)	1.0	9.102	9.361	9.585	9.780	9.950	10.098	10.228	10.343	10.957	11.162	11.243	11.323	11.335
xi)	1.1	9.709	9.992	10.239	10.454	10.642	10.806	10.951	11.078	11.776	12.017	12.114	12.209	12.224
xii)	1.2	10.243	10.549	10.816	11.050	11.254	11.434	11.593	11.732	12.509	12.786	12.900	13.013	13.030
xiii)	1.3	10.713	11.040	11.326	11.577	11.797	11.992	12.163	12.314	13.167	13.478	13.609	13.742	13.762
xiv)	1.4	11.127	11.473	11.777	12.044	12.279	12.487	12.670	12.833	13.758	14.102	14.251	14.404	14.427
xv)	1.5	11.493	11.857	12.176	12.458	12.707	12.927	13.122	13.295	14.289	14.666	14.832	15.006	15.033
xvi)	1.6	11.817	12.196	12.531	12.826	13.088	13.319	13.525	13.708	14.768	15.176	15.359	15.555	15.585
xvii)	1.7	12.104	12.498	12.846	13.154	13.427	13.669	13.885	14.078	15.199	15.638	15.838	16.056	16.091
xviii)	1.8	12.359	12.766	13.127	13.446	13.730	13.983	14.208	14.409	15.590	16.058	16.274	16.516	16.554
xix)	1.9	12.587	13.006	13.378	13.708	14.002	14.264	14.498	14.707	15.944	16.441	16.673	16.937	16.980
xx)	2.0	12.789	13.220	13.603	13.943	14.246	14.516	14.758	14.975	16.265	16.790	17.037	17.325	17.372
xxi)	3.0	13.993	14.496	14.947	15.353	15.718	16.048	16.346	16.676	18.301	19.051	19.432	19.943	20.046
xxii)	4.0	14.493	15.030	15.514	15.951	16.347	16.706	17.033	17.330	19.241	20.142	20.623	21.322	21.495
xxiii)	5.0	14.742	15.296	15.798	16.252	16.664	17.040	17.382	17.695	19.740	20.740	21.293	22.148	22.393
xxiv)	10.0	15.094	15.674	16.201	16.681	17.118	17.518	17.885	18.222	20.491	21.681	22.390	23.676	24.238
xxv)	INF	15.217	15.806	16.342	16.831	17.278	17.688	18.064	18.410	20.770	22.046	22.838	24.463	26.111

Table 21 Percentage Sky Components on the Vertical Plane Perpendicular to a Vertical Rectangular Opening for the Clear Design Sky

(*Clause* 4.2.5, B-1.1, B-1.2, B-1.5, B-1.6.1 *and* B-2.3)

Sl No.	***l/d*** ***h/d***	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**	**1.0**	**1.1**	**1.2**
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	0.1	0.036	0.141	0.303	0.506	0.734	0.971	1.207	1.432	1.643	2.836	1.011	2.168
ii)	0.2	0.071	0.277	0.594	0.993	1.442	1.910	2.374	2.820	3.236	3.618	3.964	4.276
iii)	0.3	0.103	0.401	0.863	1.445	2.100	2.793	3.475	4.180	4.743	5.306	5.818	6.278
iv)	0.4	0.126	0.491	1.059	1.779	2.597	3.460	4.326	5.166	5.958	6.691	7.359	7.967
v)	0.5	0.142	0.554	1.197	2.015	2.947	3.937	4.938	5.914	6.842	7.707	8.503	9.228
vi)	0.6	0.154	0.600	1.298	2.187	3.204	4.288	5.389	6.468	7.498	8.464	9.358	10.177
vii)	0.7	0.162	0.634	1.372	2.316	3.397	4.552	5.729	6.887	7.997	9.042	10.013	10.907
viii)	0.8	0.169	0.660	1.429	2.413	3.543	4.754	5.990	7.209	8.382	9.490	10.523	11.476
ix)	0.9	0.174	0.680	1.472	2.487	3.655	4.909	6.192	7.460	8.683	9.841	10.924	11.926
x)	1.0	0.178	0.695	1.505	2.545	3.743	5.030	6.350	7.657	8.921	10.120	11.243	12.284
xi)	1.1	0.181	0.707	1.532	2.591	3.812	5.126	6.475	7.814	9.110	10.342	11.498	12.573
xii)	1.2	0.183	0.716	1.552	2.626	3.866	5.202	6.575	7.939	9.261	10.521	11.705	12.807
xiii)	1.3	0.185	0.723	1.568	2.655	3.910	5.263	6.655	8.040	9.384	10.666	11.873	12.998
xiv)	1.4	0.186	0.729	1.582	2.678	3.945	5.312	6.720	8.122	9.484	10.785	12.011	13.155
xv)	1.5	0.188	0.734	1.592	2.697	3.973	5.352	6.773	8.189	9.566	10.883	12.124	13.285
xvi)	1.6	0.189	0.738	1.601	2.712	3.996	5.385	6.816	8.244	9.634	10.963	12.219	13.394
xvii)	1.7	0.189	0.741	1.608	2.724	4.016	5.412	6.852	8.290	9.690	11.031	12.298	13.484
xviii)	1.8	0.190	0.744	1.614	2.735	4.032	5.434	6.882	8.328	9.737	11.087	12.364	13.561
xix)	1.9	0.191	0.746	1.619	2.743	4.045	5.453	6.908	8.360	9.777	11.135	12.420	13.625
xx)	2.0	0.191	0.748	1.623	2.751	4.056	5.469	6.929	8.387	9.811	11.175	12.468	13.680
xxi)	3.0	0.193	0.756	1.642	2.785	4.109	5.544	7.030	8.517	9.972	11.371	12.699	13.950
xxii)	4.0	0.194	0.759	1.648	2.794	4.124	5.566	7.058	8.540	10.018	11.427	12.767	14.029
xxiii)	5.0	0.194	0.760	1.650	2.798	4.129	5.574	7.069	8.568	10.036	11.449	12.793	14.060
xxiv)	10.0	0.194	0.761	1.652	2.801	4.135	5.581	7.080	8.582	10.053	11.470	12.818	14.095
xxv)	INF	0.194	0.761	1.652	2.802	4.136	5.582	7.081	8.584	10.056	11.473	12.822	14.095

Table 21 (*Concluded*)

Sl No.	***l/d*** ***h/d***	**1.3**	**1.4**	**1.5**	**1.6**	**1.7**	**1.8**	**1.9**	**2.0**	**3.0**	**4.0**	**5.0**	**10.0**	**INF**
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)
i)	0.1	2.308	2.433	2.544	2.642	2.730	2.808	2.878	2.940	3.309	3.461	3.536	3.641	3.678
ii)	0.2	4.554	4.802	5.022	5.219	5.393	5.549	5.688	5.812	6.547	6.850	7.000	7.211	7.284
iii)	0.3	6.690	7.058	7.385	7.677	7.936	8.168	8.375	8.560	9.657	10.110	10.335	10.651	10.760
iv)	0.4	8.507	8.900	9.420	9.804	10.146	10.451	10.724	10.968	12.421	13.024	13.323	13.743	13.889
v)	0.5	9.883	10.472	10.999	11.476	11.897	12.273	12.610	12.912	14.712	15.462	15.835	16.360	16.542
vi)	0.6	10.922	11.596	12.204	12.752	13.244	13.686	14.084	14.441	16.583	17.478	17.924	18.552	18.771
vii)	0.7	11.723	12.465	13.138	13.746	14.296	14.793	15.241	15.646	18.111	19.148	19.665	20.397	20.653
viii)	0.8	12.350	13.147	13.873	14.531	15.129	15.670	16.161	16.606	19.361	20.538	21.127	21.961	22.253
ix)	0.9	12.847	13.690	14.459	15.159	15.796	16.375	16.902	17.381	20.387	21.701	22.360	23.397	23.625
x)	1.0	13.245	14.126	14.931	15.666	16.337	16.948	17.504	18.012	21.237	22.680	23.408	24.446	24.810
xi)	1.1	13.356	14.478	15.314	16.079	16.778	17.416	17.999	18.531	21.946	23.508	24.303	25.441	25.841
xii)	1.2	13.827	14.766	15.628	16.418	17.141	17.802	18.407	18.961	22.543	24.208	25.072	26.309	26.745
xiii)	1.3	14.041	15.003	15.887	16.698	17.442	18.123	18.747	19.320	23.049	24.809	25.735	27.070	27.542
xiv)	1.4	14.217	15.198	16.101	16.931	17.692	18.391	19.032	19.621	23.480	25.326	26.308	27.441	28.249
xv)	1.5	14.364	15.361	16.280	17.125	17.902	18.616	19.272	19.875	23.850	25.772	26.808	28.336	28.880
xvi)	1.6	14.486	15.497	16.430	17.289	18.079	18.806	19.475	20.090	24.169	26.161	27.245	28.866	29.445
xvii)	1.7	14.589	15.511	16.556	17.427	18.229	18.968	19.648	20.274	24.444	26.501	27.629	29.340	29.955
xviii)	1.8	14.675	15.708	16.663	17.545	18.357	19.105	19.795	20.431	24.684	26.799	27.969	29.765	30.416
xix)	1.9	14.749	15.791	16.755	17.645	18.466	19.224	19.922	20.567	24.893	27.062	28.270	30.149	30.835
xx)	2.0	14.811	15.861	16.833	17.731	18.560	19.325	20.031	20.684	25.077	27.294	28.537	30.496	31.217
xxi)	3.0	15.120	16.211	17.224	18.164	19.036	19.844	20.594	21.289	26.082	28.619	30.108	32.676	32.742
xxii)	4.0	15.212	16.316	17.343	18.298	19.185	20.008	20.772	21.483	26.439	29.128	30.745	33.687	35.064
xxiii)	5.0	15.248	16.357	17.390	18.351	19.243	20.073	20.844	21.562	26.592	29.359	31.049	34.232	35.872
xxiv)	10.0	15.283	16.398	17.436	18.403	19.302	20.138	20.917	21.641	26.758	29.624	31.419	35.049	37.513
xxv)	INF	15.288	16.404	17.443	18.411	19.311	20.148	20.928	21.654	26.785	29.672	31.490	35.274	39.172

Table 22 Percentage Sky Components on the Vertical Plane Parallel to a Vertical Rectangular Opening for the Clear Design Sky

(*Clauses* 4.2.5, B-1.1, B-1.2, B-1.5, B-1.6.1 *and* B-2.3)

Sl No.	*l/d* *h/d*	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
i)	0.1	0.728	1.429	2.078	2.600	3.167	3.660	3.964	4.265	4.513	4.717	4.883	5.020	5.132
ii)	0.2	1.429	2.803	4.007	5.221	6.220	7.073	7.790	8.385	8.876	9.278	9.609	9.880	10.103
iii)	0.3	2.068	4.061	5.913	7.580	9.040	10.285	11.337	12.212	12.934	13.528	14.016	14.417	14.747
iv)	0.4	2.529	4.970	7.249	9.312	11.133	12.707	14.042	15.164	16.097	16.870	17.507	18.025	18.458
v)	0.5	2.852	5.608	8.186	10.529	12.606	14.401	15.952	17.256	18.350	19.262	20.021	20.652	21.177
vi)	0.6	3.086	6.070	8.867	11.415	13.681	15.656	17.353	18.793	20.008	21.027	21.879	22.592	23.189
vii)	0.7	3.259	6.413	9.373	12.074	14.482	16.588	18.402	19.949	21.257	22.359	23.285	24.063	24.716
viii)	0.8	3.389	6.672	9.755	12.573	15.090	17.296	19.201	20.830	22.212	23.380	24.365	25.195	25.895
ix)	0.9	3.489	6.869	10.046	12.955	15.556	17.840	19.817	21.511	22.952	24.173	25.206	26.078	26.816
x)	1.0	3.565	7.024	10.272	13.250	15.917	18.263	20.297	22.043	23.531	24.795	25.866	26.773	27.542
xi)	1.1	3.625	7.139	10.447	13.481	16.200	18.594	20.674	22.462	23.989	25.288	26.391	27.326	28.121
xii)	1.2	3.672	7.233	10.586	13.663	16.423	18.857	20.973	22.795	24.353	25.681	26.810	27.770	28.587
xiii)	1.3	3.709	7.307	10.696	13.807	16.602	19.067	21.213	23.062	24.646	25.998	27.148	28.128	28.963
xiv)	1.4	3.739	7.366	10.784	13.924	16.745	19.236	21.406	23.278	24.884	26.255	27.424	28.420	29.271
xv)	1.5	3.763	7.414	10.856	14.018	16.861	19.373	21.563	23.454	25.077	26.465	27.649	28.660	29.523
xvi)	1.6	3.783	7.453	10.914	14.095	16.956	19.485	21.692	23.599	25.236	26.638	27.835	28.857	29.732
xvii)	1.7	3.799	7.485	10.962	14.158	17.034	19.578	21.798	23.718	25.368	26.781	27.989	29.022	29.906
xviii)	1.8	3.812	7.512	11.002	14.211	17.099	19.655	21.886	23.817	25.478	26.900	28.118	29.160	30.052
xix)	1.9	3.824	7.534	11.035	14.254	17.153	19.719	21.960	23.900	25.570	27.001	28.226	26.276	30.175
xx)	2.0	3.833	7.553	11.062	14.291	17.199	19.773	22.022	23.970	25.647	27.086	28.318	29.374	31.279
xxi)	3.0	3.876	7.639	11.192	14.463	17.412	20.027	22.316	24.302	26.016	27.491	28.757	29.846	30.783
xxii)	4.0	3.888	7.663	11.228	14.511	17.471	20.098	22.398	24.398	26.121	27.606	28.884	29.983	30.930
xxiii)	5.0	3.893	7.672	11.241	14.529	17.494	20.125	22.430	24.432	26.161	27.650	28.932	30.035	30.986
xxiv)	10.0	3.897	7681	11.254	14.546	17.515	20.150	22.459	24.466	26.199	27.693	28.978	30.085	31.041
xxv)	INF	3.898	7.682	11.256	44.548	17.518	20.154	22.464	24.471	26.205	27.699	28.985	30.093	31.049

Table 22 (*Concluded*)

Sl No.	*l/d* *h/d*	**1.4**	**1.5**	**1.6**	**1.7**	**1.8**	**1.9**	**2.0**	**3.0**	**4.0**	**5.0**	**10.0**	**INF**
(1)	(2)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)
i)	0.1	5.225	5.301	5.365	5.418	5.463	5.501	5.533	5.687	5.733	5.749	5.765	5.766
ii)	0.2	10.286	10.439	10.565	10.671	10.760	10.835	10.899	11.207	11.296	11.330	11.362	11.365
iii)	0.3	15.020	15.246	15.434	15.591	15.724	15.836	15.931	16.390	16.523	16.574	16.623	16.627
iv)	0.4	18.816	19.113	19.360	19.568	19.742	19.890	20.015	20.624	20.801	20.868	20.933	20.939
v)	0.5	21.613	21.978	22.275	22.530	22.746	22.923	23.082	23.836	24.056	24.140	24.222	24.229
vi)	0.6	23.689	24.109	24.462	24.761	25.014	25.229	25.412	26.229	26.561	26.662	26.759	26.768
vii)	0.7	25.267	25.731	26.124	26.458	26.742	26.984	27.192	28.214	28.517	28.634	28.748	28.758
viii)	0.8	26.486	26.987	27.412	27.775	28.084	28.350	28.578	29.720	30.065	30.198	30.327	30.339
ix)	0.9	27.441	27.972	28.424	28.810	29.141	29.426	29.672	30.927	31.303	31.451	31.596	31.610
x)	1.0	28.196	28.572	29.226	29.633	29.982	30.283	30.544	31.889	32.302	32.467	32.627	32.643
xi)	1.1	28.798	29.375	29.869	30.293	30.658	30.973	31.246	32.670	33.117	33.297	33.473	33.491
xii)	1.2	29.283	29.878	30.388	30.826	31.204	31.532	31.816	33.309	33.796	33.981	34.173	34.193
xiii)	1.3	29.676	30.286	30.810	31.261	31.651	31.989	32.283	33.836	34.350	34.550	34.756	34.779
xiv)	1.4	29.998	30.621	31.157	31.618	32.018	32.365	32.667	34.374	34.813	35.035	35.247	35.271
xv)	1.5	30.262	30.897	31.443	31.914	32.322	32.677	32.986	34.641	35.202	35.436	35.663	35.689
xvi)	1.6	30.482	31.226	31.680	32.160	32.575	32.937	33.253	34.950	35.532	35.776	36.017	36.046
xvii)	1.7	30.665	31.317	31.879	32.366	32.888	33.156	33.477	35.211	35.812	36.067	36.321	36.352
xviii)	1.8	30.818	31.477	32.046	32.539	32.967	33.340	33.666	35.435	36.052	36.316	36.584	36.617
xix)	1.9	30.948	31.613	32.188	32.686	33.119	33.497	33.828	35.626	35.259	36.532	36.812	36.847
xx)	2.0	31.058	31.728	32.308	32.811	33.249	33.631	33.965	35.791	36.438	36.719	37.011	37.048
xxi)	3.0	31.592	32.291	32.898	33.427	33.889	34.294	34.551	36.640	37.380	37.715	38.107	38.157
xxii)	4.0	31.748	32.457	33.074	33.611	34.082	34.496	34.860	36.915	37.699	38.063	38.510	38.579
xxiii)	5.0	31.808	32.521	33.142	33.683	34.157	34.574	34.943	37.028	37.834	38.214	38.696	38.781
xxiv)	10.0	31.867	32.584	33.208	33.753	34.231	34.652	35.024	37.144	37.978	38.382	38.927	39.057
xxv)	INF	31.876	32.593	33.218	33.764	34.243	34.664	35.037	37.162	38.003	38.411	38.978	39.172

ANNEX C

(*Clauses* 4.2.6 *and* 5.4.4.3)

USE OF DESIGN CHARTS FOR NATURAL VENTILATION

C-1 APPLICATION OF THE DESIGN CHARTS

C-1.1 The application of the design charts for natural ventilation (NV) is demonstrated here. The openings for the master-bedroom of the apartment two BHK case in Fig. 26 are sized to deliver the recommended airflow rates for two scenarios of input parameters.

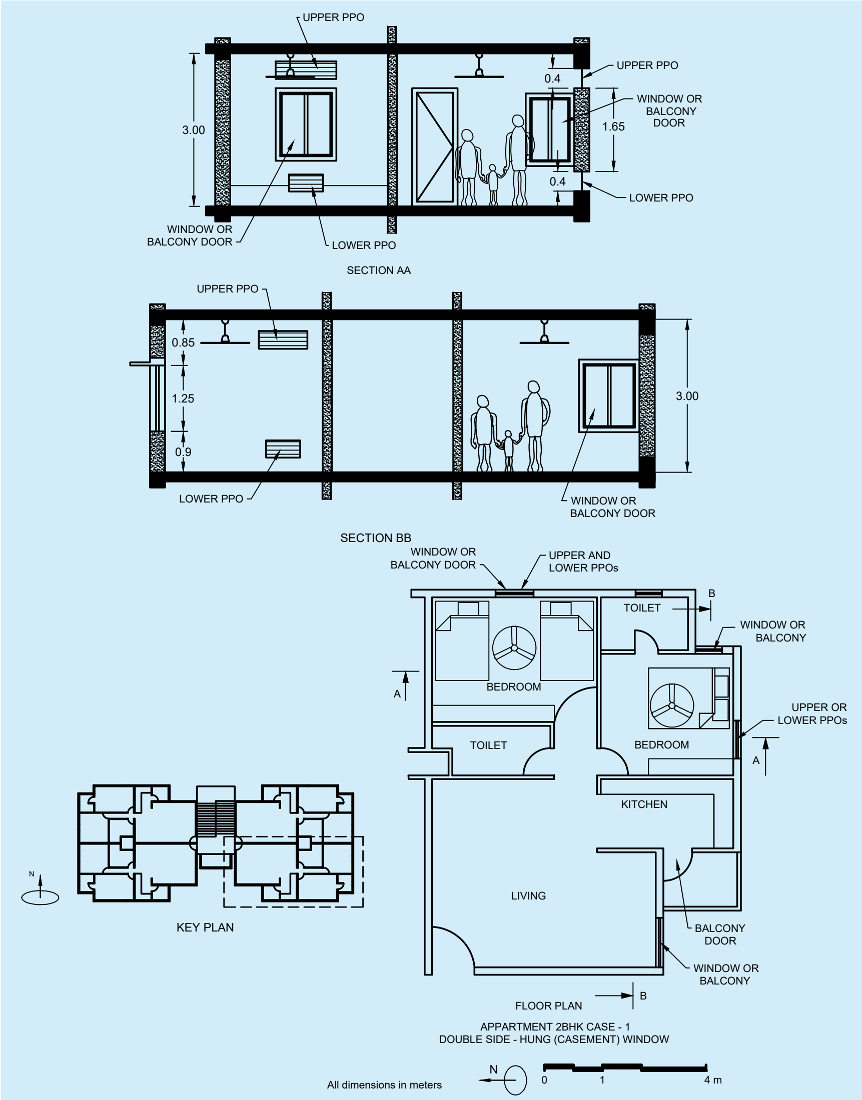


FIG. 26 CASE STUDY APARTMENT TWO BHK CASE: THE FLOOR PLAN WITH SUGGESTED LAYOUT, CROSS - SECTION AA SHOWING THE HORIZONTAL SLIDING (SASH) WINDOWS AND LOUVERS OF THE AUXILIARY PPOs FOR VENTILATION

C-1.2 The input parameters used for the demonstration of the design charts are shown in Table 23. The demonstration is shown for DC-01 in Fig. 26, for DC-02 in Fig. 27, for DC-03 in Fig. 28 and for DC-04 in Fig. 29. The free area for the openings A_f calculated with the DCs are shown in Table 25.

C-1.2.1 *DC*-01 — *Buoyancy-Driven Flow*; *Single-Sided Ventilation with One Opening*

$$A_f = \frac{q_{rec}}{C_d}\sqrt{\frac{(T_{inside} + 273)}{g\, h_a \Delta T_{inside\text{-}outside}}}$$

where

A_f	=	free area of the openings for NV, m^2;
C_d	=	discharge coefficient [-];
q_{rec}	=	ventilation rate achieved, m^3/s;
T_{inside}	=	temperature inside, °C;
g	=	acceleration due to gravity, m/s^2;
h_a	=	vertical distance between the centres of the openings, m; and
$\Delta T_{inside\text{-}outside}$	=	[temperature inside (°C) - temperature outside (°C)].

DC-01 is derived from the equation given in **C-1.2.1**. For single openings, the value for C_d adopted is 0.25. The demonstration of the application of this DC for two scenarios (a set of values) and for the input parameters as per Table 23 is shown in Fig. 27.

C-1.2.2 *DC-02 — Buoyancy-Driven Flow; Cross-Ventilation with Multiple Openings Shown in Fig. 28*

The DC-02 is also derived from the equation given in **C-1.2.3.1**. Conversely, for multiple openings, the value for C_d adopted is 0.61. The demonstration of the application of this DC for two scenarios (a set of values) and for the input parameters shown in Table 23 is shown in Fig. 28.

C-1.2.3 *DC-03 — Wind-Driven Flow; Single-Sided Ventilation with One Opening Shown in Fig. 29*

DC-03 is derived from the following equation:

$$A_f = q_{rec} / CV_z$$

where

A_f	=	free area of the openings for NV, m^2;
q_{rec}	=	ventilation rate achieved, m^3/s;
C	=	opening shape coefficient, [-]; and
V_z	=	wind speed at height 'Z', m/s.

C-1.2.3.1 The wind speed is calculated based on the height of the opening and the wind profile adjusted according to the terrain roughness. The equation below defines the wind speed (V_z) at height 'Z' based on the mean wind velocity at 10 m height (V_{ref}) and the characteristics of the roughness of the terrain, employed as an exponent of this power law. For this demonstration 'city terrain' is adopted in the calculations of V_z (input parameters and calculated values are shown in Table 23 and Table 24).

$$V_z = V_{ref}\, k\, z^{\alpha} \text{(m/s)}$$

where

V_z	=	wind speed at height 'Z', m/s;
V_{ref}	=	mean wind velocity, m/s;
k	=	terrain-roughness coefficient;
z	=	height above ground level, m; and
α	=	terrain roughness exponent.

Table 23 Input Parameters Used for the Demonstration of the Design Charts

(*Clauses* C-1.2, C-1.2.1, C-1.2.2, C-1.2.3.1, C-1.2.3.2, C-1.2.4.1 *and* C-1.2.5)

Sl No.	Input Parameters	Scenario 1	Scenario 2
(1)	(2)	(3)	(4)
i)	q_{rec} (m/s)	0.18	0.35
ii)	ΔT (°C)	3.4	2.6
iii)	T_{inside} (°C)	31.3	31
iv)	$T_{outside}$ (°C)	27.9	28.4
v)	V_{ref} (m/s)	1.8	2.03
vi)	V_r (m/s)	0.98	1.10
vii)	ΔC_p	0.36	0.36
viii)	C_d for single openings	0.25	0.25
ix)	C_d for multiple openings	0.61	0.61
x)	C for single openings	0.05	0.05
xi)	h_a (m)	1.20	1.20
xii)	z (m)	18.00	18.00
xiii)	α	0.33	0.33
xiv)	k	0.21	0.21

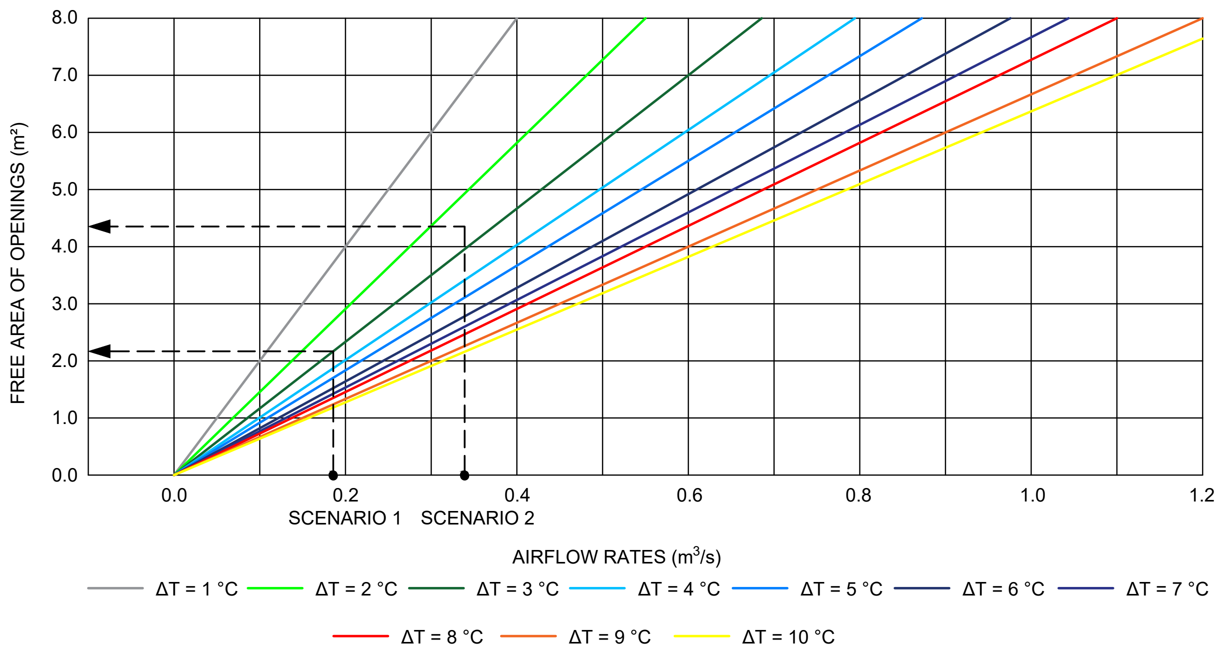


FIG. 27 DEMONSTRATION OF THE APPLICATION OF THE DC-01 (BUOYAN CY-DRIVEN FLOW; SINGLE-SIDED VENTILATION WITH ONE OPENING) FOR BOTH SCENARIOS OF INDIAN CITIES

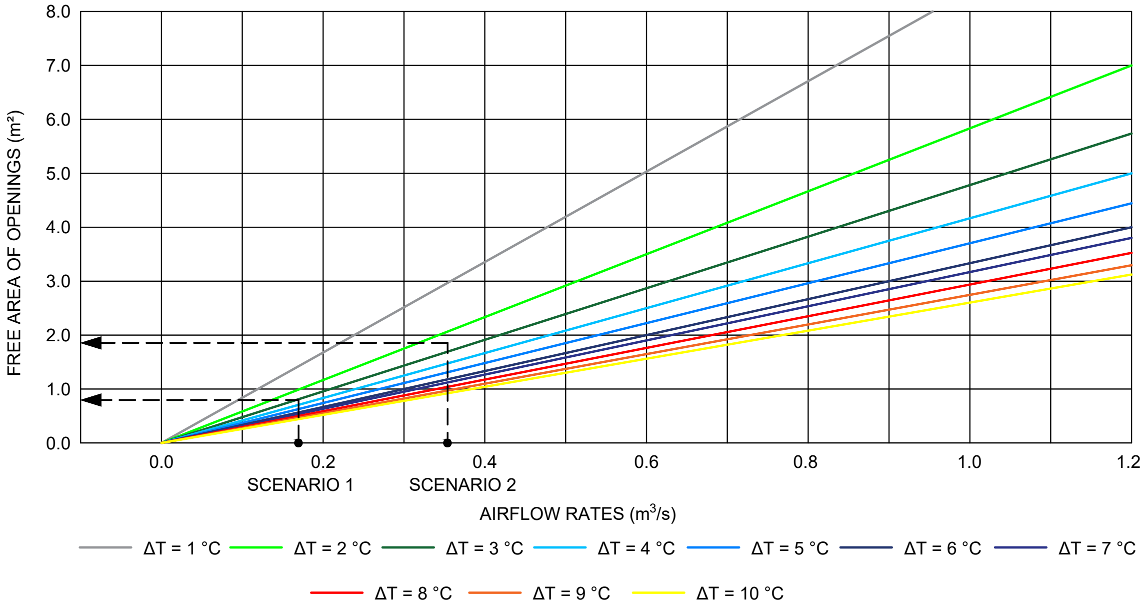


FIG. 28 DEMONSTRATION OF THE APPLICATION OF THE DC-02 (BUOYANCY-DRIVEN FLOW; CROSS-VENTILATION WITH MULTIPLE OPENINGS) FOR BOTH SCENARIOS OF INDIAN CITIES

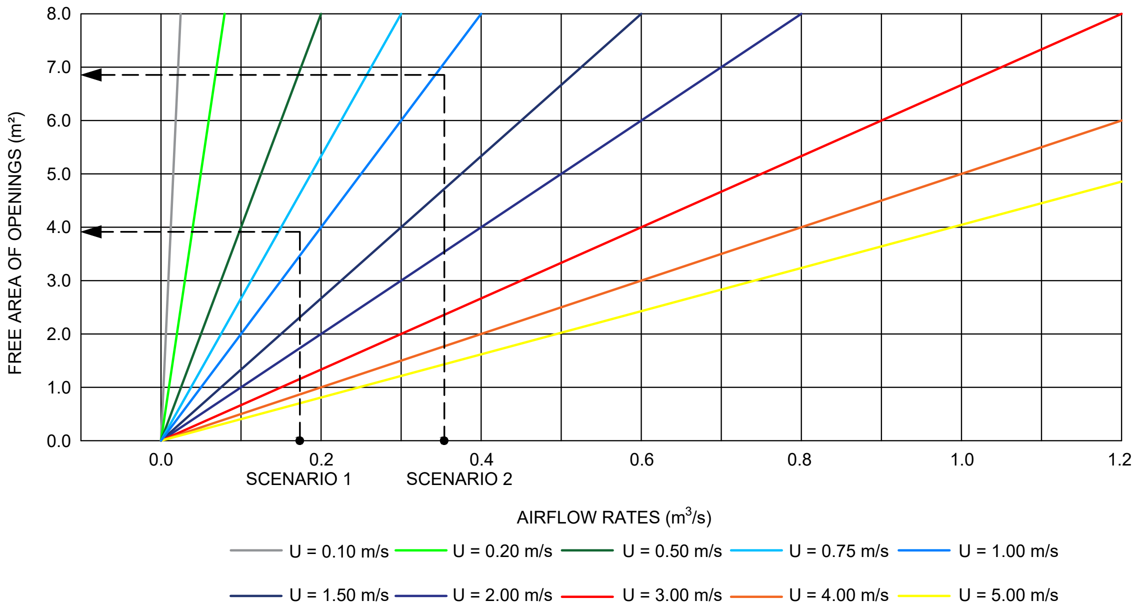


FIG. 29 DEMONSTRATION OF THE APPLICATION OF THE DC-03 (WIND-DRIVEN FLOW; SINGLE-SIDED VENTILATION WITH ONE OPENING) FOR BOTH SCENARIOS OF INDIAN CITIES

Table 24 Exponent Values Based on the Terrain Roughness

(*Clause* C-1.2.3.1)

Sl No.	Terrain Roughness	α	k
(1)	(2)	(3)	(4)
i)	Open country	0.17	0.68
ii)	Few blockages	0.20	0.52
iii)	Urban	0.25	0.35
iv)	City	0.33	0.21

C-1.2.3.2 The demonstration of the application of this DC for both scenarios of Indian cities and for the input parameters shown in Table 23 is shown in Fig. 29.

C-1.2.4 *DC-04 — Wind-Driven Flow; Cross-Ventilation with Multiple Openings is Derived from the Equation:*

$$A_f = q_{rec}\left(C_d V_z \sqrt{\frac{\Delta C_p}{2}}\right)^{-1}$$

C-1.2.4.1 For orthogonal wind and for the floor plan presented in Fig. 26 for the two bedroom, kitchen and hall, apartment block a pressure coefficient difference (ΔC_p) between inlets and outlets of 0.36 is adopted for this application demonstration (*see* Table 23).

C-1.2.5 The free areas of the openings calculated utilizing the four design charts provided in this design guide for the recommended airflow rates presented in Table 23 and for two scenarios of Indian cities, are given in Table 25.

Table 25 Free Areas of the Openings Calculated Utilizing the Design Charts for Both Scenarios of Indian Cities

(*Clauses* C-1.2 *and* C-1.2.5)

Sl No.	Design Chart	Scenario 1 m^2	Scenario 2 m^2
(1)	(2)	(3)	(4)
i)	DC-01	2.03	4.36
ii)	DC-02	0.83	1.79
iii)	DC-03	4.02	6.90
iv)	DC-04	0.78	1.33

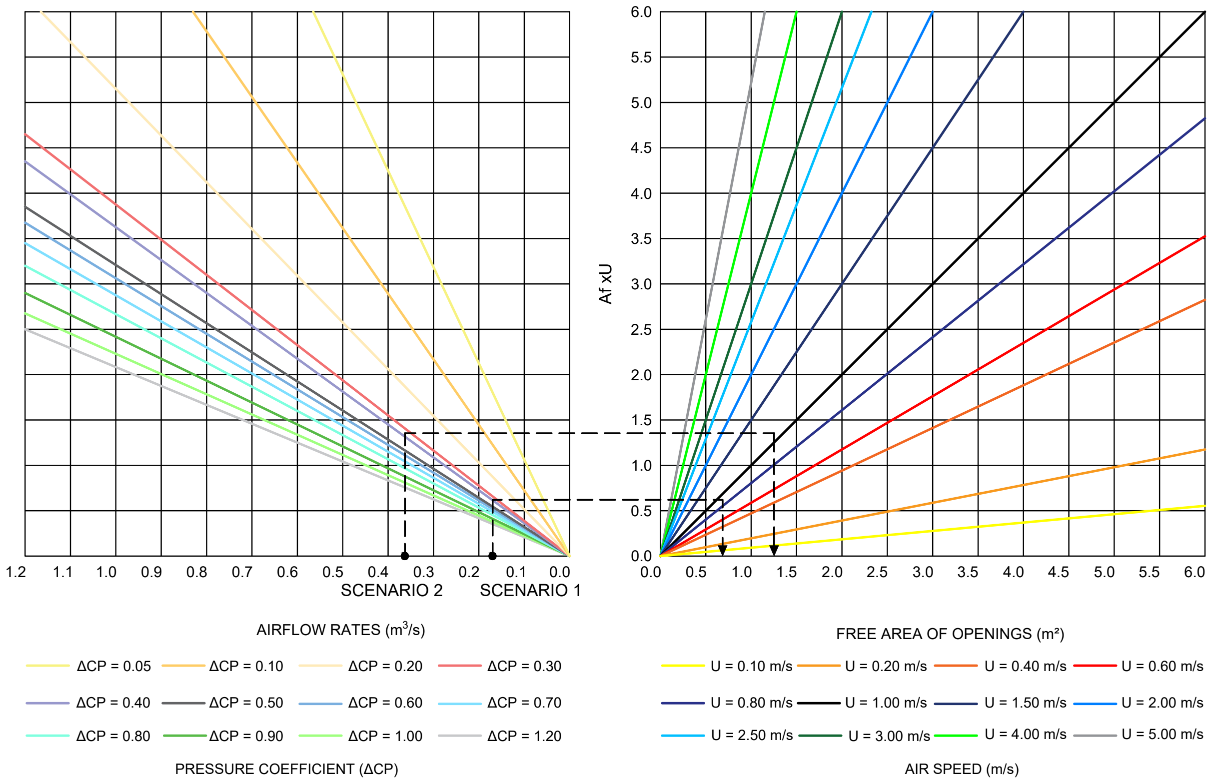


FIG. 30 DEMONSTRATION OF THE APPLICATION OF THE DC-04 (WIND-DRIVEN FLOW; CROSS-VENTILATION WITH MULTIPLE OPENINGS) FOR BOTH SCENARIOS OF INDIAN CITIES

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

	IS No.	*Title*
(1)	IS 7662 (Part 1) : 1974	Recommendations for orientation of buildings: Part 1 Nonindustrial buildings
(2)	IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities — Requirements (*second revision*)
(3)	SP 72 : 2025	National Lighting Code of India 2025 (*first revision*)
(4)	IS 3646 (Part 1) : 2025	Interior illumination — Code of practice: Part 1 General requirements and recommendations for working interiors (*second revision*)
(5)	IS 2440 : 1975	Guide for daylighting of buildings (*second revision*)
(6)	IS 6060 : 1971	Code of practice for daylighting of factory buildings
	IS 7942 : 1976	Code of practice for daylighting of educational buildings
(7)	SP 41 (S&T) : 1987	Handbook on functional requirements of buildings (other than industrial buildings)
(8)	IS 1944	Code of practice for lighting of public thoroughfares:
	(Parts 1 and 2) : 1970	Part 1 General Principles and Part 2 Lighting for traffic routes (Groups A1, A2, B1 and B2) (*first revision*)
	(Part 6) : 1981	Lighting for town and city centres and areas of civic importance (Group E)
	IS 2672 : 1966	Code of practice for library lighting
	IS 4347 : 1967	Code of practice for hospital lighting
	IS 6074 : 1971	Code of practice for functional requirements of hotels, restaurants and other food service establishments
	IS 6665 : 1972	Code of practice for industrial lighting
	IS 10894 : 1984	Code of practice for lighting of educational institutions
	IS 10947 : 1984	Code of practice for lighting for ports and harbours
(9)	IS 3362 : 1977	Code of practice for natural ventilation of residential buildings (*first revision*)

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 2 ELECTRICAL AND ALLIED INSTALLATIONS

BUREAU OF INDIAN STANDARDS

CONTENTS

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Section 2) covers the general requirements for electrical and allied installations in buildings. For more details, it is essential to refer to the relevant Indian Standards on the subject.

This Section was first published in 1970 as part of the National Building Code of India and was subsequently revised in 1983, 2005 and 2016. In the first revision in 1983, general guidance for electrical wiring installation in industrial location where voltage supply normally exceeds 650 V was included. This Section was also updated based on the existing version of the Indian Standards. The importance of pre-planning and exchange of information among all concerned agencies from the earlier stages of building work was emphasized.

In the second revision of 2005, the title of this Section was modified from the erstwhile 'Electrical Installations' to 'Electrical and Allied Installations' to reflect the provisions included on certain allied installations. The significant changes incorporated in this revision included, thorough change in the risk assessment procedure for lightning including some other changes in the provision of lightning protection of building; alignment of some of the provisions of wiring with the practices prevalent at that time; modification of definitions in line with terminologies used at national and international levels and addition of some new definitions; incorporation of provisions on installation of distribution transformer inside the multistoreyed buildings; introduction of concept of energy conservation in lighting and introduction of concept of various types of earthing in building installations.

In the third revision in 2016, various updates were made including: location and other requirements relating to layout, environmental and safety aspects for different substation apparatus/equipment and generating sets were reviewed and updated; location of compact substations was included; requirements for electrical supply system for life and safety services were included; provisions relating to reception and distribution of supply and wiring installations were updated; installation best practices of energy meters were updated; discrimination, cascading and limitation concepts for the coordination of protective devices in electrical circuits were introduced; socket outlets with suitable circuit breakers, conforming to Indian Standards were recommended for industrial and commercial applications, either indoors or outdoors; provisions relating to earthing/grounding were substantially revised and updated; provisions relating to lightning protection of buildings were revamped based on the national and international developments then; provisions relating to renewable energy sources for building, such as solar PV system, aviation obstacle lights, electrical supply for electric vehicle charging and car park management, etc, were included; provisions relating to electrical installations for construction sites and demolition sites were included; provisions relating to protection of human beings from electrical hazards and protection against fire in the building due to leakage current were included; typical formats for checklists for handing over and commissioning of substation equipment and earthing pit were included; and the terminologies updated.

In this revision, the publication has been renamed as National Building Construction Standards (NBCS).

All electrical installations in India come under the purview of *The Indian Electricity Act*, 2003 and the rules and regulations framed thereunder. In the context of the buildings, both buildings (the structure itself) and the building services (not just the electrical services, but all other services that use electricity or have an interface with the electrical system) are required to follow these. The erstwhile *Indian Electricity Rules*, 1956 were modified by various Central Electricity Authority Regulations. While revising the provisions of this Section of the NBCS, care has been taken to align the same with the provisions of the relevant regulations, particularly, *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023. In this revision, in addition to above, the following major modifications have been incorporated:

a) Various terms and their definitions have been reviewed based on current developments at national and international level. In addition, few more definitions have been added for more clarity and better understanding;

b) Voltage categorization for building has been made considering revision in voltage norms by CEA, Indian Standards and various State Electricity Boards;

c) Provisions relating to location and other requirements relating to layout, environmental and safety aspects for different substation apparatus/equipment and generating sets have been reviewed and updated;

d) Requirements for electrical supply system for life and safety services have been reviewed;

e) Provisions relating to reception and distribution of supply and wiring installations have been updated with due cognizance to Indian Standards formulated for various wiring systems;

f) Provisions relating to installation of energy meters have been updated;

g) Discrimination, cascading and limitation concepts for the coordination of protective devices in electrical circuits have been updated;

h) Socket-outlets with suitable circuit breakers, conforming to IS 1293 : 2019 'Plugs and socket-outlets for household and similar purposes of rated voltage up to and including 250 V and rated current up to and including 16 A — Specification (*fourth revision*)' have been recommended for industrial and commercial applications, either indoors or outdoors.

j) Provisions relating to earthing/grounding have been substantially revised and updated;

k) Provisions relating to lightning protection of buildings have been reviewed and updated based on the current national and international developments;

m) Provisions relating to renewable energy sources for building, such as rooftop solar photovoltaic (PV) system has been updated and building integrated photovoltaic (BIPV) system and solar water heating system have been included;

n) Provisions relating to electrical installations for construction sites and demolition sites have been reviewed and updated;

p) Provisions related to electric vehicles (EV) charging have been updated as per present norms;

q) Provisions relating to protection of human beings from electrical hazards and protection against fire in the building due to leakage current have been reviewed and updated;

r) Provisions on building management systems (BMS) have been updated in **10.4**; and

s) Provisions on power fence (electrical fence) system has been included in **10.11**.

This Section has to be read together with Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS for making provision for the desired levels of illumination as well as ventilation for different locations in different occupancies; Part D 'Building Services, Section 6 Information and Communication Enabled Installations'; Part F 'Fire and Life Safety' of this NBCS for list of emergency fire and life safety services and other sections of Part D 'Building Services' and Part E 'Plumbing Services' for electricity related requirements and integration thereof. Utmost importance should be given in the installation of electrical wiring to prevent short circuiting and the hazards associated therewith.

Notwithstanding the provisions given in this Section and the National Electrical Code 2023, the provisions of the *Central Electrical Authority* regulations framed under the *Indian Electricity Act*, 2003 as amended up to date have to be necessarily complied with.

The information contained in this section is largely based on the following Indian Standards/Special Publications:

IS 732 : 2019	Code of practice for electrical wiring installations (*fourth revision*)
IS 3043 : 2018	Code of practice for earthing (*second revision*)
IS 4648 : 1968	Guide for electrical layout in residential buildings
IS 19127 : 2025	Graphical symbols for diagrams in the field of electrotechnology
IS/IEC 62305-1 : 2010	Protection against lightning: Part 1 General principles
IS/IEC 62305-2 : 2010	Protection against lightning: Part 2 Risk management
IS/IEC 62305-3 : 2010	Protection against lightning: Part 3 Physical damage to structures and life hazard

IS/IEC 62305-4 : 2010	Protection against lightning: Part 4 Electrical and electronic systems within structures
IS 17017-1 : 2018	Electric vehicle conductive charging system: Part 1 General requirement
IS 17017-2 : 2020	Electric vehicle conductive charging system: Part 2 Plugs, socket-outlets, vehicle connectors, and vehicle inlets (*all sections*)
IS 18925 : 2025 (Parts 1 to 8)	Lightning protection system components
SP 30 : 2023	National Electrical Code of India, 2023 (*second revision*)

It may be noted that some of the above standards are currently under revision. The revised version when available should be referred.

Considerable assistance has also been drawn from following International Standards while formulating this Section:

IEC 62305 : 2024	Protection against lightning:
(Part 1) : 2024	General principles
(Part 2) : 2024	Risk management
(Part 3) : 2024	Physical damage to structures and life hazard
(Part 4) : 2024	Electrical and electronic systems within structures
IEC 60335-2-76 : 2018	Household and similar electrical appliances — Safety — Part 2-76: Particular requirements for electric fence energizers
IEC 60364-4-41 : 2005	Low-voltage electrical installations — Part 4-41: Protection for safety — Protection against electric shock
IEC 60364-4-43 : 2023	Low-voltage electrical installations — Part 4-43: Protection for safety — Protection against overcurrent
IEC 60364-4-44 : 2007	Low-voltage electrical installations — Part 4-44: Protection for safety — Protection against voltage disturbances and electromagnetic disturbances
IEC 60364-5-51 : 2005	Electrical installations of buildings — Part 5-51: Selection and erection of electrical equipment – Common rules
IEC 60364-5-54 : 2011	Low-voltage electrical installations — Part 5-54: Selection and erection of electrical equipment — Earthing arrangements and protective conductors
IEC 60364-7 series	Low-voltage electrical installations — Part 7: Requirements for special installations or locations
IEC 61439-1 : 2020	Low-voltage switchgear and controlgear assemblies — Part 1: General rules
IEC 61439-2 : 2020	Low-voltage switchgear and controlgear assemblies — Part 2: Power switchgear and controlgear assemblies
IEC 61439-6 : 2012	Low-voltage switchgear and controlgear assemblies — Part 6: Busbar trunking systems (busways)

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section of the NBCS.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **'good practice'** in relation to design, constructional procedures or other related information, and where reference is made to **'accepted standard'** in relation to material specification, testing, or other related information, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [D-2(1)] refers to the Indian Standard/Special Publication given at serial number (1) of the List of Standards given at the end of this Part D/Section 2, that is, SP 30 : 2023, 'National Electrical Code of India, 2023 (*second revision*)'

NATIONAL BUILDING COSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 2 ELECTRICAL AND ALLIED INSTALLATIONS

1 SCOPE

This NBCS (Part D/Section 2) covers the essential requirements for electrical installations in buildings to ensure efficient use of electricity including safety from fire and shock, reliability and sustainability. This Section also includes general requirements relating to renewable energy sources, EV charging, lightning protection of buildings and brief provisions on allied installations.

2 TERMINOLOGY AND CONVENTIONAL SYMBOLS

2.1 For the purpose of this Section, the following definitions shall apply. For definition of other terms, reference may be made to good practice [D-2(1)] and accepted standard [D-2(2)].

2.1.1 *Accessory* — A device, other than current using equipment, associated with such equipment or with the wiring of an installation.

2.1.2 *Apparatus* — Electrical apparatus including all machines, appliances and fittings in which conductors are used or of which they form a part.

2.1.3 *Appliance* — An item of current using equipment other than a luminaire or an independent motor.

2.1.4 *Back-up Protection* — Protection which is intended to operate when a system fault is not cleared or abnormal condition not detected in the required time, because of failure or inability of other protection to operate or failure of appropriate circuit-breaker to trip.

2.1.5 *Barrier* — A part providing a defined degree of protection against contact with live parts, from any usual direction of access.

2.1.6 *Basic Protection* — Protection against electric shock under fault-free condition.

NOTE — For low voltage installations, systems and equipment, basic protection generally corresponds to protection against direct contact that is 'contact of persons or live parts'.

2.1.7 *Battery Energy Storage System* (*BESS*) — Electrical energy storage system with an accumulation subsystem based on batteries fitted with secondary cells.

2.1.8 *Bonding Conductor* — A protective conductor providing equipotential bonding.

2.1.9 *Bonding Ring Conductor* (*BRC*) — A bus earthing conductor in the form of a closed ring.

NOTE — Normally the bonding ring conductor, as part of the bonding network, has multiple connections to the common bonding network (CBN) that improves its performance.

2.1.10 *Branch Switch* — A miniature circuit breaker that protects a specific branch circuit within a larger power distribution circuit, such as a house or building. It acts as a safety device, automatically interrupting the flow of electricity in the event of an overload or short circuit, preventing damage to wiring and appliances.

2.1.11 *Bunched* — Cables are said to be 'bunched' when two or more are contained within a single conduit, duct, ducting, or trunking, or, if not enclosed, are not separated from each other.

2.1.12 *Buried Direct* — A cable laid in the ground in intimate contact with the soil.

2.1.13 *Busbar Trunking System* — A type-tested assembly, in the form of an enclosed conductor system of comprising solid conductors separated by insulating materials. The assembly may consist of units such as:

a) Busbar trunking units, with or without tap-off facilities;

b) Tap-off units where applicable; and

c) Phase-transposition, expansion, building-movement, flexible, end-feeder and adaptor units.

2.1.14 *Bypass Equipotential Bonding Conductor* — Bonding conductor connected in parallel with the screens of cables.

2.1.15 *Cable* — A length of single-insulated conductor (solid or stranded), or two or more such conductors, each provided with its own insulation, which are laid up together. The insulated conductor or conductors may or may not be provided with an overall mechanical protective covering.

2.1.16 *Cable*, *Fire Survival* — A cable which continues to function and maintain the circuit integrity under circumstances of fire (against a specified temperature and period of the test) {*see* accepted standards [D-2(3)]}.

2.1.17 *Cable, Flame Retardant* (*FR*) — A cable which is flame retardant as per the accepted standard [D-2(4)].

2.1.18 *Cable*, *Flame Retardant Low Smoke and Halogen* (FR-LSH) — A cable which is flame retardant and emits low smoke and halogen as per the accepted standard [D-2(5)].

2.1.19 *Cable*, *Flexible* — A cable containing one or more cores, each formed of a group of wires, the diameters of the cores and of the wires being sufficiently small to afford flexibility.

2.1.20 *Cable*, *Metal-Sheathed* — An insulated cable with a metal sheath.

2.1.21 *Cable*, *PVC Sheathed-Insulated* — A cable in which the insulation of the conductor is a polyvinylchloride (PVC) compound; with PVC sheath also providing mechanical protection to the conductor core or cores in the cable.

2.1.22 *Cable*, *Weatherproof* — A cable so constructed that when installed in uncovered locations, it will withstand all kinds of weather variations (*see* **2.1.195** for definition of weatherproof).

2.1.23 *Cable*, *XLPE* — A cable in which the insulation of the conductor is cross-linked polythene and the mechanical protection is provided for the core or cores by a sheath of a polyvinyl chloride compound.

2.1.24 *Cable Armoured* — A cable provided with a wrapping of metal (usually in the form of tape or wire) serving as a mechanical protection.

2.1.25 *Cable Bracket* — A cable support consisting of single devices fixed to elements of building or plant construction.

2.1.26 *Cable Channel* — An enclosure situated above or in the ground, open or ventilated or closed, and having dimensions which do not permit the access of persons but allow access to the conductor and/or cables throughout their length during and after installation.

NOTE — A cable channel may or may not form part of the building construction.

2.1.27 *Cable Cleat* — A component of a support system which consists of elements spread at intervals along the length of the cable or conduits and which mechanically retains the cable or conduit.

2.1.28 *Cable Coupler* — A means enabling the connection, at will, of two flexible cables. It consists of a connector and a plug.

2.1.29 *Cable Ducting* — A manufactured enclosure of metal or insulating material, other than conduit or cable trunking, intended for the protection of cables which are drawn-in after erection of the ducting, but which is not specifically intended to form part of a building structure.

2.1.30 *Cable Ladder* — A cable support occupying less than 10 percent of the plan area and consisting of a series of supporting elements rigidly fixed to each other or to a main supporting member or members.

2.1.31 *Cable Raceways* — An enclosed channel of metal or non-metallic materials designed expressly for holding wires, cables or busbars, with openable/maintainable construction having provision of ventilation. These include electrical non-metallic tubing, electrical metallic tubing, underfloor raceways, cellular concrete floor raceways, cellular metal floor raceways, surface raceways and wireways.

2.1.32 *Cable Support and Fixing Materials* (*Fire Survival*) — Supports and fixing materials for supporting fire survival cable (*see* **2.1.16**), which continues in service after exposure to fire for a specified duration.

2.1.33 *Cable Tray* — A cable support consisting of a continuous base with raised edges and no covering. The classification and perforation of cable tray shall be as per accepted standard [D-2(6)].

2.1.34 *Cable Trunking* — A factory made closed support and protection system into which conductors and/or cables are laid after removal of the cover.

2.1.35 *Cable Tunnel* — An enclosure (corridor) containing supporting structures for conductors and/or cables and joints and whose dimensions allow free access to persons throughout the entire length.

2.1.36 *Cartridge Fuse Link* — A device comprising a fuse element or several fuse elements connected in parallel enclosed in a cartridge usually filled with an arc-extinguishing medium and connected to terminations. The fuse link is the part of a fuse which requires replacing after the fuse has operated.

2.1.37 *Ceiling Rose* — A fitting (usually used to attach to the ceiling) designed for the connection between the electrical installation wiring and a flexible cord (which is in turn connected to a lamp holder).

2.1.38 *Circuit* — An assembly of electrical equipment supplied from the same origin and protected against overcurrent by the same protective device(s). Circuits are categorized as follows:

a) *Category* 1 *circuit* — A circuit (other than a fire alarm annunciation or emergency lighting circuit and other circuits required to work during fire in a building) operating at low voltage and supplied directly from a mains supply system;

b) *Category* 2 *circuit* — With the exception of Category 3 circuits, any circuit for extra low-voltage (ELV)/telecommunication [for example, radio, telephone, sound distribution, building management system (BMS), public address system (PAS), intruder alarm, bell and call and data transmission circuits] which is supplied from a safety source; and

c) *Category* 3 *circuit* — A fire alarm circuit or an emergency lighting circuit and other circuits required to work during fire in a building.

2.1.39 *Circuit Breaker* — A mechanical switching device, capable of making, carrying and breaking currents under normal circuit conditions and also of making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions such as those of short circuit.

NOTE — A circuit breaker is usually intended to operate infrequently, although some types are suitable for frequent operation.

2.1.39.1 *Miniature circuit breaker* (*MCB*) — A compact mechanical switching device capable of making, carrying and breaking currents under normal circuit conditions and also making and carrying currents for specified times and automatically breaking currents under specified abnormal circuit conditions, such as those of overload and short circuits.

2.1.39.2 *Circuit breaker*, *linked* — A circuit breaker, the contacts of which are so arranged as to make or break all poles simultaneously or in a definite sequence.

2.1.39.3 *Moulded case circuit breaker* (*MCCB*) — A circuit breaker having a supporting housing of moulded insulating material forming an integral part of the circuit breaker.

2.1.39.4 *Air circuit breaker* (*ACB*) — A circuit breaker in which the contacts open and close in air at atmospheric pressure.

2.1.40 *Circuit*, *Final Sub* — An outgoing circuit connected to one-way distribution board and intended to supply electrical energy at one or more points to current, using appliances without the intervention of a further distribution board other than a one-way board. It includes all branches and extensions derived from that particular way in the board.

2.1.41 *Compact Substation or Prefabricated Substation* — Prefabricated and type-tested assembly which can to be operated from inside (walk-in type) or outside (non-walk-in type) comprising components such as power transformer, high-voltage switchgear and controlgear, low-voltage switchgear and controlgear, corresponding interconnections (cable, busbar or other) and auxiliary equipment and circuits located next to each other, maintaining segregation and integrity of each compartment in which they are located along with external interconnecting cables, earthing, protections, etc. The components shall be enclosed by either a common enclosure or by an assembly of enclosures.

NOTE — *See* accepted standard [D-2(7)] for requirements of prefabricated substation.

2.1.42 *Conductor, Aerial* — Any conductor which is supported by insulators above the ground and is directly exposed to the weather.

NOTE — Following four classes of aerial conductors are recognized:

a) Bare aerial conductors;

b) Covered aerial conductors;

c) Insulated aerial conductors; and

d) Weatherproof neutral-screened cable.

2.1.43 *Conductor*, *Bare* — A conductor not covered with insulating material.

2.1.44 *Conductor, Earthed* — A conductor with no provision for its insulation from earth.

2.1.45 *Conductor*, *Insulated* — A conductor adequately covered with insulating material of such quality and thickness as to prevent danger.

2.1.46 *Conductor of a Cable or Core* — The conducting portion consisting of a single wire or group of wires, assembled together and in contact with each other or connected in parallel.

2.1.47 *Conduit* — A part of a closed wiring system, a circular or non-circular cross section for conductors and/or cables in electrical installations, allowing them to be drawn in and/or replaced. Conduits should be sufficiently closed-jointed so that the conductors can only be drawn in and not inserted laterally.

2.1.48 *Connector* — The part of a cable coupler or of an appliance coupler which is provided with female contact and is intended to be attached to the flexible cable connected to the supply.

2.1.49 *Connector Box or Joint Box* — A box forming a part of wiring installation, provided to contain joints in the conductors of cables of the installations.

2.1.50 *Connector for Portable Appliances* — A combination of a plug and socket arranged for attachment to a portable electrical appliance or to a flexible cord.

2.1.51 *Consumer's Terminals* — The ends of the electrical conductors situated upon any consumer's premises and belonging to him, at which the supply of energy is delivered from the service line.

2.1.52 *Continuous Operating Voltage* (U_c) — Maximum rms voltage which may be continuously applied to a surge protection device’s mode of protection. This is equal to rated voltage.

2.1.53 *Cord, Flexible* — A flexible cable having a large number of strands of conductors of small cross-sectional area. Two flexible cords twisted together are known as twin 'flexible cord'.

NOTE — Large number of fine strands of wires for each conductor makes the conductor capable of withstanding frequent bends thereby improving their flexibility.

2.1.54 *Core of a Cable* — A single conductor of a cable with its insulation but not including any mechanical protective covering.

2.1.55 *Current Carrying Capacity of a Conductor* — The maximum current which can be carried by a conductor under specified conditions without its steady state temperature exceeding a specified value.

2.1.56 *Current Using Equipment* — Equipment which converts electrical energy in to another form of energy, such as light, heat, or motive power.

2.1.57 *Cut-out* — Any appliance for automatically interrupting the transmission of energy through any conductor when the current rises above a pre-determined amount.

2.1.58 *Damp Situation* — A situation in which moisture is either permanently present or intermittently present to such an extent as likely to impair the effectiveness of an installation conforming to the requirements for ordinary situations.

2.1.59 *Danger* — Danger to health or danger to life or limb from shock, burn or injury from mechanical movement to persons (and livestock where present), or from fire attendant upon the use of electrical energy.

2.1.60 *Datacentre* — A structure, or group of structures, dedicated to the centralized accommodation, interconnection and operation of information technology and network telecommunications equipment providing data storage and processing with the necessary levels of resilience and security required to provide the desired service availability.

2.1.61 *Dead* — A portion of an electrical circuit (normally expected to carry a voltage) at or near earth potential or apparently disconnected from any live system. A circuit apparently disconnected from all sources is expected to be at earth potential; but capacitive storage of charge in cables, capacitors, etc, can keep the electric circuit at a significant voltage (and often dangerous voltages from aspects of shock). Such circuits with storage components will be dead only on connection to earth.

2.1.62 *Design Current* (*of a Circuit*) — The magnitude of the current intended to be carried by the circuit in normal service.

2.1.63 *Direct Contact* — Contact of persons or live stock with live parts which may result in electric shock.

2.1.64 *Discrimination* (*Selectivity-Over-Current Discrimination*) — Coordination of the operating characteristics of two or more over-current protective devices should be such that, on the incidence of over-currents within stated limits, the device intended to operate would be the device closest to the point of fault or abnormality, and if proper discrimination is achieved within these limits, only that device closest should operate, while the other circuit breakers upstream do not operate, thereby ensuring that there is minimum area of power supply which is interrupted.

NOTES

1 Protective devices should have discrimination so that only the affected part (minimum section) of the circuit is isolated, even though a number of protective devices may be in the path of the overcurrent.

2 The electrical network requires the discrimination for all the fault circuits, including overload, short-circuit, etc. The downstream device should take care of the fault up to the level of ultimate short-circuit breaking capacity (*I*cu) of the downstream breaker which should be equal to the bus which is connected.

3 Distinction is made between series discrimination involving different over-current protective devices passing substantially the same over-current and network discrimination involving identical protective devices passing different proportions of the over-current.

4 Different types of protective devices may have to be used to ensure effective discrimination in circuits where proper and effective discrimination is necessary. Apart from the built-in sensors and actuators in circuit breakers, external relays operating on different parameters, and comparison of parameters between two or more points will have to be used for complex installations.

5 *See also* relevant parts of the accepted standard [D-2(8)].

2.1.65 *Disconnector* — A mechanical switching device which, in the open position, complies with the requirements specified for the isolation function.

NOTES

1 A disconnector is otherwise known as isolator.

2 A disconnector is capable of opening and closing a circuit when either a negligible current is broken or made, or when no significant change in the voltage across the terminals of each pole of the disconnector occurs. It is also capable of carrying currents under normal circuit conditions and carrying for a specified time, current under abnormal conditions, such as those of short-circuit.

2.1.66 *Distance Area or Resistance Area* (*for an Earth Electrode Only*) — The surface area of ground (around an earth electrode) on which a significant voltage gradient may exist.

2.1.67 *Diversity Factor* — A measure of the probability that a particular piece of equipment will turn on coincidentally to another piece of equipment. For aggregate systems it is defined as the ratio of the sum of the individual non-coincident maximum loads of various subdivisions of the system to the maximum demand of the complete system.

2.1.68 *Duct* — A closed passage way formed underground or in a structure and intended to receive one or more cables which may be drawn in.

2.1.69 *Ducting — See* **2.1.29**.

2.1.70 *Earth* — The conductive mass of the earth, whose electric potential at any point is conventionally taken as zero.

2.1.71 *Earth Continuity Conductor* — The conductor, including any clamp, connecting to the earthing lead or to each other, those parts of an installation which are required to be earthed. It may be in whole or in part, the metal conduit or the metal sheath or armour of the cables, or the special continuity conductor of a cable or flexible cord incorporating such a conductor.

2.1.72 *Earthed Concentric Wiring* — A wiring system in which one or more insulated conductors are completely surrounded throughout their length by a conductor, for example a sheath, which acts as a PEN conductor.

2.1.73 *Earth Electrode* — A conductor or group of conductors in intimate contact with the ground to provide a low resistance path for flow of current to earth.

2.1.74 *Earth Electrode Network* — Part of an earthing arrangement comprising of only the earth electrodes and their interconnections.

2.1.75 *Earth Electrode Resistance* — The resistance of an earth electrode to earth.

2.1.76 *Earth Fault* — An unintended and undesirable connection of phase/neutral conductor to earth. When the impedance is negligible, the connection is called a dead earth fault.

2.1.77 *Earth Fault Current* — A current resulting from a fault of negligible impedance between a line conductor and an exposed conductive part or a protective conductor.

2.1.78 *Earth Fault Loop Impedance* (Z_s) — The total impedance of the path that a fault current would take during an earth fault, starting at the point of the fault and returning to the supply's neutral point.

2.1.79 *Earthing* — Connection of the exposed conductive parts of an installation to the main earthing terminal of that installation.

2.1.80 *Earthing Conductor* — A protective conductor connecting the main earth terminal (or equipotential bonding conductor of an installation when there is no earth bus) to an earth electrode or to other means of earthing.

2.1.81 *Earthing Lead* — The final conductor by which the connection to the earth electrode is made.

2.1.82 *Earth Leakage Current* — A current which flows to earth, or to extraneous conductive parts, in a circuit which is electrically sound.

NOTE — This current may have a capacitive component including that resulting from the deliberate use of capacitors.

2.1.83 *Earthing Resistance*, *Total* — The resistance between the main earthing terminal and the earth.

2.1.84 *Electrical Equipment* (*abb: Equipment*) — Any item for such purposes as generation, conversion, transmission, distribution or utilization of electrical energy, such as machines, transformers, apparatus, measuring instruments, protective devices, wiring materials, accessories, and appliances.

2.1.85 *Electrically Independent Earth Electrodes* — Earth electrodes located at such a distance from one another that the maximum current likely to flow through one of them does not significantly affect the potential of the other(s).

2.1.86 *Electrical Supply System for Life and Safety Services* — A supply system intended to maintain the operation of essential parts of an electrical installation and equipment:

a) for health and safety of persons and livestock; and
b) to avoid damage to the environment and to other equipment.

NOTES

1 The supply system includes the source and the circuit(s) up to the terminals of the electrical equipment.

2 *See also* Part F 'Fire and Life Safety' of this NBCS regarding emergency fire and life safety services.

2.1.87 *Electric Shock* — A dangerous patho-physiological effect resulting from the passing of an electric current through a human body or an animal.

2.1.88 *Electric Vehicle* (*EV*) — A vehicle that is propelled by one or more electric motors, using electrical energy stored in rechargeable batteries offering various benefits, including reduced emissions, lower running costs and potentially quieter operation compared to traditional gasoline-powered vehicles.

2.1.89 *EV Connector* — Different types of EV charging connectors, also called plugs, that works with different vehicles, charging speeds and locations.

2.1.90 *Electric Vehicle Supply Equipment* (*EVSE*) — Equipment or a combination of equipment, providing dedicated functions to supply electric energy from a fixed electrical installation or supply network to an EV for the purpose of charging.

2.1.91 *Emergency Switching* — Rapid cutting off of electrical energy to remove any hazard to persons, livestock, or property which may occur unexpectedly.

2.1.92 *Enclosure* — A part providing protection of equipment against certain external influences and, in any direction, protection against direct contact.

2.1.93 *Enclosed Distribution Board* — An enclosure containing busbars with one or more control and protected devices for the purpose of protecting, controlling or connecting more than one outgoing circuits fed from one or more incoming circuits.

2.1.94 *Equipotential Bonding* — Electrical connection putting various exposed conductive parts and extraneous conductive parts at a substantially equal potential.

NOTE — In a building installation equipotential bonding conductors shall interconnect the following conductive parts:

a) Protective conductor;
b) Earth continuity conductor; and
c) Risers of air conditioning system and heating systems (if any).

2.1.95 *Exothermic Welding* — A welding process that uses a chemical reaction to generate intense heat, typically reaching temperatures of around 1 400 °C. This heat is used to melt metal and create a permanent, high-quality weld without the need for an external heat or power source. The process is particularly useful for joining conductors, such as in earthing and lightning protection systems, and can be used to weld dissimilar metals.

2.1.96 *Exposed Conductive Part* — A conductive part of electrical equipment, which can be touched and which is not normally live, but which may become live under fault conditions.

2.1.97 *Exposed Metal* — All metal parts of an installation which are easily accessible other than:

a) parts separated from live parts by double insulation;

b) metal name-plates, screw heads, covers, or plates, which are supported on, or attached, or connected to substantial non-conducting material only in such a manner that they do not become alive in the event of failure of insulation of live parts and whose means of fixing do not come in contact with any internal metal; and

c) parts which are separated from live parts by other metal parts which are themselves earthed or have double insulation.

2.1.98 *External Influence* — Any influence external to an electrical installation which affects the design and safe operation of that installation.

2.1.99 *Extraneous Conductive Part* — A conductive part not forming part of the electrical installation and liable to introduce a potential, generally the earth potential.

2.1.100 *Fault* — A circuit condition in which current flows through an abnormal or unintended path. This may result from an insulation failure or a bridging of insulation. Conventionally the impedance between live conductors or between live conductors and exposed or extraneous conductive parts at the fault position is considered negligible.

2.1.101 *Fault Current* — A current resulting from a fault.

2.1.102 *Fault Protection* — Protection against electric shock under single fault conditions.

NOTE — For low voltage installation, system's and equipment's fault protection generally corresponds to protection against indirect contact, mainly with regards to failure of basic insulation. Indirect contact is 'contact of persons or livestock with exposed-conductive parts which have become live under fault conditions'.

2.1.103 *Final Circuit* — A circuit connected directly to current using equipment, or to socket-outlets or other outlet points for the connection of such equipment.

2.1.104 *Fire Survival Distribution Board* — A distribution board which continues in service after exposure to fire to the required system rating.

2.1.105 *Fish Wire* — A draw wire to pull electrical wires or other cables through concealed spaces like conduits in ceilings/walls/floors.

2.1.106 *Fitting, Lighting* — A device for supporting or containing a lamp or lamps [for example, fluorescent or incandescent or halogen or compact fluorescent lamp (CFL) or light-emitting diode (LED)] together with any holder, shade, or reflector, for example, a bracket, a pendant with ceiling rose, an electrolier, or a portable unit.

2.1.107 *Fixed Equipment* — Equipment fastened to a support or otherwise secured.

2.1.108 *Flameproof Enclosure* — An enclosure which will withstand without injury any explosion of inflammable gas that may occur within it under practical conditions of operation within the rating of the apparatus (and recognized overloads, if any, associated therewith) and will prevent the transmission of flame which may ignite any inflammable gas that may be present in the surrounding atmosphere.

NOTES

1 Hazardous areas are classified into different zones, depending upon the extent to which an explosive atmosphere may exist at that place. In such areas, flame proof switchgear, fittings, accessories have to be used/installed in flameproof enclosure.

2 An electrical apparatus is not considered as flameproof unless it complies with the appropriate statutory regulations.

3 Other types of fittings are also in vogue in wiring installations, for example, 'increased safety'.

2.1.109 *Functional Earthing* — Connection to earth necessary for proper functioning of electrical equipment.

2.1.110 *Hand-Held Equipment* — Portable equipment intended to be held in the hand during normal use, in which the motor, if any, forms an integral part of the equipment.

NOTE — A hand held equipment is an item of equipment, the functioning of which requires constant manual support or guidance.

2.1.111 *Harmonics* (*Current and Voltage*) — All alternating current which is not absolutely sinusoidal is made up of a fundamental and a certain number of current and voltage harmonics [multiples of 50 Hz (basic frequency)] which are the cause of its deformation (distortion) when compared to the theoretical sine-wave.

2.1.112 *Hazardous Live-Part* — A live part which can give, under certain condition of external influence, an electric shock.

2.1.113 *Impulse Current* — A parameter used for the classification test for SPDs (*see* **2.1.177**); it is defined by three elements, a current peak value, a charge Q and a specific energy W/R.

2.1.114 *Impulse Withstand Voltage* — The highest peak value of impulse voltage of prescribed form and polarity which does not cause breakdown of insulation under specified condition.

2.1.115 *Indirect Contact* — Contact of persons or livestock with exposed conductive parts made live by a fault and which may result in electric shock.

2.1.116 *Industrial Plugs and Sockets* — Plugs and socket-outlets, cable couplers and appliance couplers, primarily intended for industrial use, either indoors or outdoors.

NOTE — For the purpose of this NBCS, industrial plugs and sockets conforming to accepted standards [D-2(9)] shall be used for industrial purpose.

2.1.117 *Inflammable Material* — A material capable of being easily ignited.

2.1.118 *Installation* (*Electrical*) — An assembly of associated electrical equipment to fulfil a specific purpose or purposes and having coordinated characteristics.

2.1.119 *Insulated* — Insulated shall mean separated from adjacent conducting material or protected from personal contact by a non-conducting substance or an air space, in either case offering permanently sufficient resistance to the passage of current or to disruptive discharges through or over the surface of the substance or space, to obviate danger or shock or injurious leakage of current.

2.1.120 *Insulation* — Suitable non-conducting material, enclosing, surrounding or supporting a conductor.

2.1.120.1 *Insulation*, *basic* — Insulation applied to live parts to provide basic protection against electric shock and which does not necessarily include insulation used exclusively for functional purposes.

2.1.120.2 *Insulation*, *double* — Insulation comprising both basic and supplementary insulation.

NOTE — Double insulation for small hand held equipment allows them to be used without a safety earth connection, without shock risk of such hand held equipment.

2.1.120.3 *Insulation*, *reinforced* — Single insulation applied to live parts, which provides a degree of protection against electric shock equivalent to double insulation under the conditions specified in the relevant standard.

NOTE — The term 'single insulation' does not imply that the insulation is a homogeneous piece. It may comprise several layers which cannot be tested singly as supplementary or basic insulation.

2.1.120.4 *Insulation*, *supplementary* — Independent insulation applied in addition to basic insulation in order to provide protection against electric shock in the event of a failure of basic insulation.

2.1.121 *Inverter* — Electric energy converter that changes direct electric current to single-phase or polyphase alternating currents.

2.1.122 *Isolation* — Cutting off an electrical installation, a circuit, or an item of equipment from every source of electrical energy.

2.1.123 *Isolator* — A mechanical switching device which, in the open position, complies with the requirements specified for the isolating function. An isolator is otherwise known as a disconnector.

2.1.124 *Junction Box* — A box forming a part of wiring installation, intended to conceal electrical connections and joints of conductors/cables in order to protect the connection from external influences such as direct contact, dust, water, moisture, UV radiation, etc depending upon the protection requirement of the space or utility.

2.1.125 *Leakage Current* — Electric current in an unwanted conductive path under normal operating conditions.

2.1.126 *LEMP Protection Measures* (*SPM*) — Measures taken to protect internal systems against the effects of LEMP.

2.1.127 *Lightning Electromagnetic Impulse* (*LEMP*) — All electromagnetic effects of lightning current *via* resistive, inductive and capacitive coupling that create surges and radiated electromagnetic fields.

2.1.128 *Lightning Protection* — Complete system for protection of structures against lightning, including their internal systems and contents, as well as persons, in general consisting of an LPS and SPM.

2.1.129 *Lightning Protection Level* (*LPL*) — A number related to a set of lightning current parameters values relevant to the probability that the associated maximum and minimum design values will not be exceeded in naturally occurring lightning.

NOTE — Lightning protection level is used to design protection measures according to the relevant set of lightning current parameters.

2.1.130 *Lightning Protection System* (*LPS*) — Complete system used to reduce physical damage due to lightning flashes to a structure.

2.1.130.1 *Lightning protection system*, *External* — Part of the LPS consisting of an air-termination system, a down-conductor system and an earth-termination system.

2.1.130.2 *Lightning protection system, Internal* — Part of the LPS consisting of lightning equipotential bonding and/or electrical insulation of external LPS.

2.1.131 *Lightning Protection Zone* — Zone where the lightning electromagnetic environment is defined.

2.1.132 *Line Conductor* — A conductor of an a.c. system for the transmission of electrical energy other than a neutral conductor or a PEN conductor. This also means the equivalent conductor of a d.c. system unless otherwise specified in this NBCS.

2.1.133 *Live Part* — A conductor or conductive part intended to be energised in normal use including a neutral conductor but, by convention, not a PEN conductor.

2.1.134 *Live or Alive* — Electrically charged so as to have a potential different from that of earth.

2.1.135 *Locations*, *Industrial* — Locations where tools and machinery requiring electrical wiring are installed for manufacture or repair.

2.1.136 *Locations*, *Non-Industrial* — Locations other than industrial locations, and shall include residences, offices, shops, showrooms, stores and similar premises requiring electrical wiring for lighting, or similar purposes.

2.1.137 *Low-Voltage Switchgear and Controlgear Assembly* — A combination of one or more low voltage switching devices together with associated control, measuring, signalling, protective, regulating equipment, etc, completely assembled under the responsibility of the manufacturer with all the internal electrical and mechanical interconnections and structural parts. The components of the assembly may be electromechanical or electronic.

2.1.138 *Luminaire* — Equipment which distributes, filters or transforms the light from one or more lamps and which includes any parts necessary for supporting, fixing and protecting the lamps, but not the lamps themselves, and, where necessary, circuit auxiliaries together with the means for connecting them to the supply.

NOTE — For the purposes of this NBCS a batten lamp holder, or a lamp holder suspended by flexible cord, is a luminaire.

2.1.139 *Main Earthing Terminal* — The terminal or bar which is the equipotential bonding conductor of protective conductors and conductors for functional earthing, if any, to the means of earthing.

2.1.140 *Meshed Bonding Network* (*MESH-BN*) — Bonding network in which all associated equipment frames, racks and cabinets and usually the d.c. power return conductor are bonded together as well as at multiple points to the CBN and may have the form of a mesh.

2.1.141 *Mobile Equipment* — Electrical equipment which is moved while in operation or which can be easily moved from one place to another while connected to the supply.

2.1.142 *Monitoring* — Observation of the operation of a system or part of a system to verify correct functioning or detect incorrect functioning by measuring system variables and comparing the measured value with the specified value.

2.1.143 *Multiple Earthed Neutral System* — A system of earthing in which the parts of an installation specified to be earthed are connected to the general mass of earth and, in addition, are connected within the installation to the neutral conductor of the supply system.

2.1.144 *Neutral Conductor* — Includes the conductor of a three-phase four-wire system; the conductor of a single-phase or d.c. installation, which is earthed by the supply undertaking (or otherwise at the source of the supply), and the middle wire or common return conductor of a three-wire d.c. or single-phase a.c. system.

2.1.145 *Origin of an Electrical Installation* — The point at which electrical energy is delivered to an installation.

NOTE — An electrical installation may have more than one origin.

2.1.146 *Overcurrent* — A current exceeding the rated value. For conductors the rated value is the current carrying capacity.

2.1.147 *Overload Current* (*of a Circuit*) — An overcurrent occurring in a circuit in the absence of an electrical fault.

2.1.148 *PEN Conductor* — A conductor combining the functions of both protective conductor and neutral conductor.

2.1.149 *Phase Conductor* — *See* **2.1.132**.

2.1.150 *Plug* — A device, provided with contact pins, which is intended to be attached to a flexible cable and which can be engaged with a socket-outlet or with a connector.

2.1.151 *Point* (*in Wiring*) — A termination of the fixed wiring intended for the connection of current using equipment.

2.1.152 *Portable Equipment* — Equipment which is moved while in operation or which can easily be moved from one place to another while connected to the supply.

2.1.153 *Protection*, *Ingress* — The degree of protection against intrusions (body parts such as hands and fingers), dust, accidental contact and water.

NOTE — The classification of degrees of ingress protection provided by enclosures for electrical equipment shall be as per the accepted standard [D-2(10)].

2.1.154 *Protection*, *Mechanical Impact* — The degrees of protection provided by enclosures for electrical equipment against external mechanical impacts.

NOTE — The classification of degrees of protection against mechanical impact provided by enclosures for electrical equipment shall be as per accepted standard [D-2(10)]

2.1.155 *Prospective Fault Current* (I_{pf}) — The value of overcurrent at a given point in a circuit resulting from a fault of negligible impedance between live conductor having a difference of potential under normal operating conditions, or between a live conductor and an exposed-conductive part.

2.1.156 *Protective Conductor* — A conductor used for some measures of protection against electric shock and intended for connecting together any of the following parts:

a) Exposed conductive parts;

b) Extraneous conductive parts;

c) Main earthing terminal; and

d) Earthed point of the source, or an artificial neutral.

2.1.157 *Protective Conductor Current* — Electric current appearing in a protective conductor, such as leakage current or electric current resulting from an insulation fault.

2.1.158 *Protective Earthing* — Earthing of a point or points in a system or in equivalent, for the purpose of safety.

2.1.159 *Protective Separation* — Separation of one electric circuit from another by means of:

a) double insulation;

b) basic insulation and electrically protective screening (shielding); or

c) reinforced insulation.

2.1.160 *Rated Current* — Value of current used for specification purposes, established for a specified set of operating conditions of a component, device, equipment or system.

2.1.161 *Rated Impulse Withstand Voltage Level* (U_w) — The level of impulse withstand voltage assigned by the manufacturer to the equipment, or to part of it, characterizing the specified withstand capability of its insulation against overvoltage.

2.1.162 *Residual Current* — The algebraic sum of the instantaneous values of current flowing through all live conductors of a circuit at a point of the electrical installation.

2.1.163 *Residual Current Device* (*RCD*) — Mechanical switching device or association of devices designed to make, carry and break currents under normal service conditions and to cause the opening of the contacts when the residual current attains a given value under specified conditions.

2.1.164 *Residual Current Operated Circuit Breaker with Integral Overcurrent Protection* (*RCBO*) — A residual current operated circuit breaker designed to perform the functions of protection against overload and/or short-circuit.

2.1.165 *Residual Current Operated Circuit-Breaker without Integral Overcurrent Protection* (*RCCB*) — A residual current operated circuit breaker not designed to perform the functions of protection against overload and/or short-circuits.

2.1.166 *Residual Operating Current* — Residual current which causes the residual current device to operate under specified conditions.

2.1.167 *Service* — The conductors and equipment required for delivering energy from the electric supply system to the wiring system of the premises served.

2.1.168 *Shock Current* — A current passing through the body of a person or an animal and having characteristics likely to cause dangerous patho-physiological effects.

2.1.169 *Short-Circuit Current* — An overcurrent resulting from a fault of negligible impedance between live conductors having a difference in potential under normal operating conditions.

2.1.170 *Space Factor* — The ratio (expressed as a percentage) of the sum of the overall cross-sectional areas of cables (including insulation and sheath) to the internal cross-sectional area of the conduit or other cable enclosure in which they are installed. The effective overall cross-sectional area of a non-circular cable is taken as that of a circle of diameter equal to the major axis of the cable.

2.1.171 *Standby Supply System* — A system intended to maintain supply to the installation or part thereof, in case of interruption of the normal supply, for reasons other than safety of persons.

NOTE — Standby supplies are necessary, for example, to avoid interruption of continuous industrial processes or data processing.

2.1.172 *Stationary Equipment* — Either fixed equipment or equipment not provided with a carrying handle and having such a mass that it cannot easily be moved.

2.1.173 *Step Voltage* — The potential difference between two points on the earth's surface, separated by distance of one pace, that will be assumed to be one metre in the direction of maximum potential gradient.

2.1.174 *Socket-Outlet* — A device, provided with female contacts, which is intended to be installed with the fixed wiring and intended to receive a plug.

NOTE — A luminaire track system is not regarded as a socket-outlet system.

2.1.175 *Substation* — A facility containing power/distribution transformers interconnecting two or more networks of different voltages.

2.1.176 *Surge* — A transient created by LEMP (*see* **2.1.127**) that appears as an overvoltage and/or an overcurrent.

2.1.177 *Surge Protective Devices* (*SPD*) — A device intended to limit transient overvoltages and divert surge currents. It contains at least one non-linear component. Types of SPDs are as follows:

a) *Type* 1 *SPDs* (*Surge Arresters*) — For protection of high-energy surges, caused by direct lightning strikes or switching surges from utility grid, installed at the service entrance near the main distribution board;

b) *Type* 2 *SPDs* (*Transient Voltage Surge Suppressors or TVSS*) — For protection against secondary surges caused by utility grid switching or internal electrical faults. Installed in electrical distribution panel/sub-distribution board; and

c) *Type* 3 *SPDs* (*Plug-in Surge Protectors*) — For providing localized protection for sensitive electronic equipment like computers and TVs installed near the point of use.

2.1.178 *Sustainability* — Under electrical this refers to the generation of electricity using renewable sources such as solar, wind, hydro, geothermal and bioenergy that are naturally replenished and have minimal negative impact on the environment and to meet present energy needs without compromising the ability of future generations to meet their own, promoting long-term energy security and environmental protection.

2.1.179 *Switch* — A mechanical switching device capable of making, carrying and breaking current under normal circuit conditions, which may include specified operating overload conditions; and also of carrying for a specified time currents under specified abnormal circuit conditions, such as those of short circuit.

NOTE — A switch may also be capable of making, but not breaking, short-circuit currents.

2.1.180 *Switchboard* — An assembly of switchgear with or without instruments, but the term does not apply to a group of local switches in a final circuit.

NOTE — The term 'switchboard' includes a distribution board.

2.1.181 *Switch Disconnector* — A switch which, in the open position, satisfies the isolating requirements specified for a disconnector.

NOTE — A switch disconnector is otherwise known as an isolating switch.

2.1.182 *Switch Disconnector Fuse* — A composite unit, comprising a switch with the fuse contained in or mounted on the moving member of the switch.

2.1.183 *Switch*, *Linked* — A switch, the contacts of which are so arranged as to make or break all poles simultaneously or in a definite sequence.

2.1.184 *Switchgear* — An assembly of main and auxiliary switching apparatus for operation, regulation, protection or other control of electrical installations.

2.1.185 *Systems*, *Earthing* — An electrical system consisting of a single source or multiple sources running in parallel of electrical energy and an installation. Types of system are identified as follows, depending upon the relationship of the source and of exposed-conductive parts of the installation, to earth:

a) *TN system* — A system having one or more points of the source of energy directly earthed, the exposed conductive-parts of the installation being connected to that point by protective conductors;

b) *TN-C system* — A system in which neutral and protective conductor functions are combined in a single conductor throughout the system;

c) *TN-S system* — A system having separate protective conductor throughout the system;

d) *TN-C-S system* — A system in which neutral and protective conductor functions are combined in a single conductor in part of the system;

e) *TT system* — A system having one point of the source of energy directly earthed, the exposed-conductive-parts of the installation being connected to the earth electrodes electrically independent of the earth electrodes of the source. This is also known as direct earthing system; and

f) *IT system* — A system in which all live parts are isolated from earth or one point connected to earth through an impedance. The exposed-conductive-parts of the electrical installation are earthed independently or collectively or to the earthing of the system.

NOTE — The types of systems depending upon the relationship to the source and of the exposed conductive parts of the insulations to earth are defined in good practice [D-2(11)].

2.1.186 *Thermal Runaway* — An unstable condition arising during constant voltage charging in which the rate of heat generation exceeds the heat dissipation capability, causing a continuous temperature increase and a further rise in charging current, which can lead to battery destruction.

2.1.187 *Touch Voltage* — The potential difference between the ground potential rise (GPR) of a grounded metallic structure and the surface potential at the point where a person could be standing while at the same time having a hand in contact with the grounded metallic structure. Touch voltage measurements can be 'open circuit' (without the equivalent body resistance included in the measurement circuit) or 'closed circuit' (with the equivalent body resistance included in the measurement circuit) voltage by which an installation or part of an installation is designated.

2.1.188 *Usable Wall Space* — All portions of a wall, except that occupied by a door in its normal open position, or occupied by a fire place opening, but excluding wall spaces which are less than 1 m in extent measured along the wall at the floor line.

2.1.189 *Utility Building* — A standalone separate single or two storied service building structure outside the main building structure meant for only accommodating services' spaces, such as electric substation, diesel generator plant room, a.c. plant room, plumbing plant room, sewerage treatment plant, medical gases, electrical and mechanical maintenance rooms. Such buildings do not have any permanent occupancy other than by personnel on duty.

2.1.190 *Voltage Nominal* (*of an Installation*) — Voltage by which an installation or part of an installation is designated.

2.1.191 *Voltage*, *Extra Low* (*ELV*) — The voltage which does not normally exceed 50 V.

2.1.192 *Voltage*, *Low* (*LV*) — The voltage which normally exceeds 50 V but does not normally exceed 1 000 V in a.c. (alternating current) and 1 500 V in d.c. (direct current).

2.1.193 *Voltage*, *High* (*HV*) — The voltage which normally exceeds 1 000 V a.c. but less than or equal to 33 kV.

2.1.194 *Voltage*, *Extra High* (*EHV*) — The voltage, which normally exceeds 33 kV.

NOTE – The Central Electricity Authority and the Bureau of Indian Standards introduced norms for different voltages during 2017-2018, which have been adopted in good practice [D-2(1)]. These norms may differ from those adopted by various State Governments in India. In buildings across India over the past 75 years, 11 kV and 33 kV have been termed as high tension/high voltage. Electrical control switchgears available for 11 kV/33 kV are known as high voltage switchgears. Accordingly, the voltage norms specified in **2.1.191** to **2.1.194** are adopted for buildings in relation to this NBCS.

2.1.195 *Weatherproof* — Accessories, lighting fittings, current-using appliances and cables are said to be of the weatherproof type with ingress protection according to the application, if they are so constructed that when installed in open situation they will withstand the effects of rain, snow, dust and temperature variations.

2.2 Conventional Symbols

The architectural symbols that are to be used in all drawings, wiring plans, etc, for electrical installations in buildings shall be as given in Annex A.

For other graphical symbols used in electrotechnology, reference may be made to good practice [D-2(1)].

3 GENERAL REQUIREMENTS

3.1 Statutory Requirements

All electrical installations that are in use/being executed shall conform to *The Indian Electricity Act*, 2003 and *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 as amended up to date; and rules if any made there under. This NBCS covers brief information about the statutory requirements.

NOTE — After India's independence, India formulated *The IE Rules*, 1956 based on *The IE Act*, 1910. Thereafter in 2003, Government of India adopted *The Electricity Act*, 2003 and nominated CEA to adopt necessary regulations for electrical supply, safety, generation and distribution. Based on these directions, CEA issued *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations* in 2010 which is superseded with *CEA Regulations*, 2023. Consequent to this, *The IE Act*, 1910 and *IE Rules*, 1956 have been repealed. The norms for voltages under regulations were established by CEA during 2017-18 and based on the same good practice [D-2(12)] was issued in 2018 which has been adopted by good practice [D-2(1)]. Under this, good practice [D-2(1)] has categorized up to 33 kV as medium voltage. However, all State Electricity Boards have fixed voltage norms which are different from CEA/NEC norms. It has been the practice for over 75 years to refer to 11 kV and 33 kV as high tension or high voltage in buildings. Hence, it has been decided to give separate voltage norms for buildings in this NBCS.

3.1.1 *The Electricity Act*, 2003

Section 53 of *The Electricity Act*, 2003, states requirements relating to 'Provisions relating to safety and electrical supply'. Some of the important points included in this NBCS are protecting the public from the dangers arising out of electricity, reducing the risk of personal injury, and damage to property. In accordance, Central Electricity

Authority under Section 70 of the Act, has been empowered to specify suitable measures. Such measures are updated and notified from time to time in the form of regulations. It may be noted that any failure to comply with any order, or any attempt or abetment that contravenes the rules and regulations framed under the act, may attract appropriate action as per the provisions therein.

NOTE — It may be noted that any failure to comply with any order, or any attempt or abetment that contravenes the rules and regulations framed under the Act, may attract appropriate action as per Section 146 of the Act. Section 149 decides responsibility and Section 150 stipulates that any act of abetment in relation to an offence may invite appropriate consequences. It means that the persons who is not directly involved in construction of electrical installation but indirectly relate to integrated part of electrical installation, such as architectural or civil construction, such as keeping provision of spaces, reserving routes, maintaining clearances, civil construction with required fire rating, etc come under the ambit of these sections. Accepted standard [D-2(13)] provides list of such stakeholders involved in electrical installation.

3.1.2 *Central Electricity Authority*

The Central Electricity Authority, formed under Section 70, is empowered to make regulations as directed under Section 53 and Section 177 of the *Indian Electricity Act*, 2003. In accordance, the following regulations have been notified and are in force. However, reference shall always be made to the latest versions/amendments.

3.1.2.1 *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023

These regulations state various requirements to be observed in the following activities:

a) Design;

b) Construction;

c) Verification and testing;

d) Maintenance;

e) Use;

f) Electric supply; and

g) Various activities integrated with the work of electrical installation.

NOTE — Regulation No. 14 (2) requires electrical installations to follow relevant standards, including this NBCS, the National Electrical Code of India, or the International Standards, where Indian Standards are unavailable. So, though the BIS Standards and Codes of practice are guidelines, their observance is no longer optional but mandatory when read with Section 146 of the Act.

3.1.2.2 *Central Electricity Authority* (*Technical Standards for the Construction of Electrical Plants and Electrical Lines*) *Regulations*, 2022

From the perspective of this NBCS, these regulations apply to transmission and distribution licensees. That is, whenever any electrical line, whether overhead or underground, passes through the plot area of the building/building-complex, observance of these provisions, which are in the scope of the respective owner of the line, may require verification.

NOTE — Important sections of the above regulations are given in Annex B.

3.2 Project Management of Electrical Works

3.2.1 From the perspective of project management, electrical work activities shall broadly be divided into the following stages:

a) Planning;

b) Designing;

c) Construction;

d) Verification and testing; and

e) Maintenance.

NOTE — For existing buildings, (a) to (c) may not apply unless there is a major change or complete renovation.

3.2.2 *Coordination*

At all work stages and the activities within these stages, different persons/agencies concerning planning, civil construction and building services are involved. Mutual coordination between the stakeholders involved, directly or indirectly, is very important to avoid conflicts.

3.3 Stakeholders and their Responsibility

The role of stakeholders is very important in streamlining electrical work. Accepted standard [D-2(13)] identifies stakeholders as below:

a) Owner/developer;

b) Architect;

c) Civil contractor;

d) Other service providers;

e) Electrical consultant;

f) Electrical contractor;

g) Electrical inspector; and

h) Electricity supplier.

The standard states the responsibility of each stakeholder as per the related statutory compliances, quoting the clauses from related regulations, standards and codes. While assuming responsibility, each stakeholder shall have the required qualifications and experience in the respective field as set out in rules, regulations, bylaws, or any government notification. Where such eligibility criteria are not mandated, it will be the responsibility of the employer to appoint a person with qualifications and experience comparable to similar/correlating tasks. In addition, the provisions given in accepted standard [D-2(13)] may also be complied with.

NOTE — Most of the time, the electrical contractor who executes the work follows the design received from the consultant. But, for an electrical consultant who plays a key role of electrical design, there are no set rules or regulations to determine eligibility or accreditation procedures. Under the circumstances, until such rules come into force, reference to **3.5** from accepted standard [D-2(13)] may be taken, or the qualification and experience of such a person may be considered on par with the Chartered Electrical Safety Engineer as stated in regulation 6 of the *CEA* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 (*see* Annex B).

It should be noted that appointing technical experts like consultants/contractors from respective fields and assigning them responsibility are inherent parts of project management and need to be observed at the planning stage.

3.4 Appointment of Technical Professionals Concerning Electrical Installation

In addition to the requirements stated in accepted standard [D-2(13)], an electrical engineer shall be appointed on site as a registered building professional to streamline, monitor and certify, jointly with the consultant and the electrical contractor, all works relating to electrical services.

3.5 Roles and Responsibilities of the Persons Appointed for Electrical Work

There shall be a written agreement between the developer and the appointee deciding the scope of work and the responsibilities to be undertaken. Such an agreement shall include a declaration concerning the work the appointee will undertake (*see* Annex G).

3.5.1 *Electrical Engineer on Site*

To monitor the coordination of work between all stakeholders, from planning to completion, the electrical engineer on site has an important role. In the team of technical persons often referred to as 'Registered building professional' or 'Licensed technical professional/person', with different nomenclature in various building rules/regulations/byelaws, the inclusion of an electrical engineer shall be treated as an important requirement for controlling and monitoring electrical work activities. His duties and functions shall be:

a) observe all regulatory requirements;

b) maintain coordination between all stakeholders;

c) check electrical requirements from the perspective of location and spaces in the architectural plans;

d) check the electrical design prepared by the design consultant;

e) check specifications related to material and method of construction;

f) participate in preparing the activities chart, considering the smooth and coordinated work progress;

g) monitor and maintain surveillance from planning to completion and handing over, with special attention to the work that will be concealed;

h) check and jointly sign certificates/declarations and checklists as recommended in accepted standard [D-2(13)];

j) check required NOCs/Permissions from respective authorities; and

k) maintain a record of all documents.

3.5.2 *Electrical Consultant*

To perform the desired role as an electrical consultant, he shall have the necessary qualifications and experience. Reference may be taken from the note under **3.3**. Such a person shall have an updated knowledge of all regulatory requirements, standards and codes of practice related to the design, construction, verification, testing, and maintenance of electrical work. He shall be responsible for preparing designs with drawings and specifications that adhere to quality standards. He shall consult the architect regarding necessary points to consider at the planning and construction stage. Table 1 from accepted standard [D-2(13)] provides a list of activities and responsibilities. Some important points in this regard are given below:

a) Identify the existing overhead and underground electrical lines on the proposed development plot and the mandatory requirements, compliances, and NOCs related to clearances;

b) Spaces, locations and routes of the distribution network required for electrical installation;

c) Specific requirements related to RCC structure (static/dynamic load of electrical equipment, requirement of recess within RCC structure to avoid core cutting, required height between finished floor level and soffit of the beam);

d) Specific requirement related to wall thickness to accommodate concealed electric boxes, raised floor (false flooring) to accommodate cable work below;

e) Specific requirements where an aluminium formwork system is used to cast concrete for building elements like walls, floors, columns, beams and slabs. Similar provisions have to be used in case of prefabricated concrete works;

f) Specific requirements related to ventilation and avoiding the accumulation of heat;

g) Assessment of electrical load including future expansion (while assessing load, other service providers shall be consulted for their requirements);

h) Space requirement for the electrical sub-station as per the assessment of the required load. *See* Annex D for suggested space provision as per the substation's capacity;

j) Consult with the electric supply company for load sanction and requirements, if any, such as space/room (*see* Note), etc;

NOTE — The supply company may state the requirement for space within the consumer's premises for substation/equipment. There may also be a case where there is no spare capacity or possibility to augment the existing electrical distribution. The consultant shall be well aware of the conditions of the Supply Code approved by the respective State Electricity Regulatory Commission and, in accordance, give the architect the requirements related to space. All related correspondence shall be documented.

k) Prepare drawings showing the electrical equipment locations and the electric supply distribution network on site plans. Supplier and consumer distribution networks shall be segregated;

m) Prepare drawings of electrical equipment rooms of substation, switch rooms and consult with the architect and civil engineer about the requirements related to technical specifications of civil construction like fire rating, ventilation facilities, doors/openings, ducts, floor level, clear ceiling height below beam soffit, access facility, dust-free areas, etc. Some of such requirements may be:

   1) *Transformer sub-station* — Oil draining arrangement, oil soak pit, curb to prevent the spread of spillage/leaked/drained oil, the required clearances around apparatus considering the type of transformer dry/oil-cooled (O class)/liquid-cooled (K class), adjacent structure whether combustible/non-combustible, provision of baffle wall, etc;

   2) *Standby power supply* — Genset, PUC norms, stack height and the building in the vicinity, safe location; and

   3) *Standby power supply* — UPS system, location and specific requirements related to battery storage.

n) Drawings showing proposed internal routes and layouts of cables (mains, sub-mains, circuit mains, light, fan power outlet points, distribution boxes, boards with electrical accessories, outlet points and the required clearances between other services, accessibility, etc;

   NOTE — Where aluminium formwork technology is used to cast concrete for building elements like walls, floors, columns, beams, and slabs, drawings showing precise locations of points and boxes are most important. Similar provisions have to be provided in case of prefabricated concrete works.

p) Design electrical installation with due consideration for general characteristics that adhere to protection for safety;

q) Consider the related requirements stated under the Energy Conservation Building Code to qualify buildings under a certain class;

r) Prepare specifications for the selection and erection of electrical equipment with due consideration to the factors of external influences;

s) State requirements relating to fire prevention, specific to electrical installation as given under CEA Regulations, for example, placement of extinguishers, PPE, automatic fire suppression system, etc;

t) State design requirements related EV charging stations and the clearances around;

u) State if there are any specific requirements relating to the access;

   NOTE — In the event, the electric control panel room is being located in the basement there shall be at least two independent access to the ground floor to avoid any one access getting blocked leading to a serious situation in accessing, operating and controlling the electric supply of critical services in case of fire.

w) Provide technical support and guidance to site engineers/contractors during project execution to ensure material and installation quality, and assist in resolving issues arising during installation, verification, and commissioning;

y) State requirements related to verifying and testing of electrical installations; and

z) Maintain respective checklists updated as required under the accepted standard [D-2(13)].

3.5.3 *Electrical Contractor**

Electrical contractor shall have a licence issued by the appropriate government. It shall be valid at the time of agreement and renewed when necessary. He shall be aware of the obligations mandated as per the rules and regulations. The supervisor(s) appointed by him shall have competency and updated technical knowledge of all regulatory requirements and the codes of practice. The contractor shall be responsible for:

a) Resolving differences regarding the design, drawings, and specifications provided by the consultant, if any, and carrying out work as per the final design. In case of unresolved points, maintain the record;

   *NOTE — Where the electrical contractor is also performing the job of electrical consultant, he shall also be responsible for observing requirements stated under **3.5.2**.

b) Before commencing work, get the drawing approvals from the respective authority, where applicable;

c) Employ electrical supervisor and the workmen having permits of the appropriate class or designated as per regulation 3 of the *CEA* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 to carry out work on/for the electrical installation (*see* Annex B);

d) Weatherproof lockable storage space for materials at site to be arranged by the electrical contractor;

e) Observe workplace safety in respect of:

   1) temporary electrical installation on site; and

   2) provision of adequate tools and personal protective equipment (PPE) to workmen.

f) Check that requirements related to civil construction concerning electrical installation are satisfied;

g) Observe the activity chart based on a time schedule, which is mutually agreed upon between all concerned service providers and plan the work in coordination;

   NOTE — Where an aluminium formwork is used to cast concrete for building elements like walls, floors, columns, beams and slabs, the coordination with civil work needs to be observed precisely. Similar provisions have to be ensured in case of prefabricated concrete works.

h) Carry out electrical installation according to the plans and finalised design by the consultant or, where applicable, the design approved by the authority (electrical inspector);

j) The execution shall be according to the approved methods of construction stated under the National or International codes of practice, where the NBCS has not set any parameters;

k) All work should be executed under the strict supervision of a competent person with the necessary permit issued by the government;

m) A record of hidden work shall be maintained with certificates of required compliance, which shall be jointly signed with the engineer on site before concealing;

n) Maintain electrical installation safety throughout the construction period and the defect liability period;

p) Get the installation verified and tested, preferably through a third party, at the following stages:

   1) During construction; and

   2) After completion, before certifying for its beneficial use.

q) Help the electrical engineer on site in maintaining updated checklists as required under the accepted standards [D-2(13)]; and

r) Prepare handing-over documents for the respective owner, stating details of the electrical installation as given the checklist.

3.6 Materials

All materials, fittings, appliances, etc used in electrical and allied installations, shall conform to Part B 'Building Materials' of this NBCS and other concerned Indian Standards.

3.7 Coordination with Local Supply Authority

a) In all cases, that is, whether the proposed electrical work is a new installation or extension of an existing one, or a modification involving major changes, the electricity supply undertaking shall be consulted about the feasibility, etc, at an early date. The wattage per square metre and permissible diversity consideration shall be defined as per the type of building (residential, commercial, mercantile, industrial, retail, convention, exhibition, hotel, hospital, institution, flatted factory, group housing, etc). The wattage per square metre (W/m^2) shall be defined considering probable loads as per city grading such that future loading into the development is accounted; and

b) *Addition to an installation* — An addition, temporary or permanent, shall not be made to the authorized load of an existing installation, until it has been definitely ascertained that the current carrying capacity and the condition of existing accessories, conductors, switches, etc, affected, including those of the supply authority are adequate for the increased load. The size of the cable/conductor shall be suitably selected on the basis of the ratings of the protective devices. Ratings of protective devices and their types shall be based on the installed load, switching characteristics and power factor.

Load assessment and application of suitable diversity factor to estimate the full load current shall be made as a first step. This should be done for every circuit, submain and feeder. Power factor, harmonics (*see* **5.3.6.6**) and efficiency of loads shall also be considered. Diversity factor assumed shall be based on one's own experience or as per table under **4.2.2.2**. Allowance should be made for about 15 percent to 20 percent for extension in near future. The wiring system should be adopted taking into account the environmental requirements and hazards, if any in the building. The sizes of wiring cables are decided not merely to carry the load currents, but also to withstand thermal effects of likely overcurrents, short circuit and ensure acceptance level of voltage drop.

3.8 Power Factor Improvement in Consumers' Installation

3.8.1 Conditions of supply of electricity boards or licensees stipulate the lower-limit of power factor which is generally 0.95 lag or better.

3.8.2 Principal causes of low power factor are many. For guidance to the consumers of electric energy who take supply at low and medium voltages for improvement of power factor, reference shall be made to good practice [D-2(1)] and accepted standard [D-2(14)].

3.9 Execution of Work

Unless otherwise exempted under the appropriate regulation of the *CEA* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023 as amended from time to time, the work of electrical installations shall be carried out by a licensed electrical contractor and under the direct supervision of a person holding a certificate of competency and by persons holding a valid permit issued and recognized by the State government.

3.10 Safety procedures and practices shall be kept in view during execution of the work in accordance with accepted standard [D-2(15)].

3.11 Safety provisions given in Part F 'Fire and Life Safety' of this NBCS should be followed.

4 PLANNING OF ELECTRICAL INSTALLATIONS

4.1 General

The design and planning of an electrical wiring installation involve consideration of all prevailing conditions and is usually influenced by the type and requirements of the consumer. Various utility services including EV charging, ELV systems namely intercom, data cabling (*see* Part D 'Building Services, Section 6 Information and Communication Enabled Installations' of this NBCS), CCTV, fire detection, etc shall also be taken into account with anticipated future requirements. A competent electrical design engineer should be involved at the planning stage with a view to providing for an installation that will prove adequate for its intended purpose and ensure safety, reliability and energy efficiency in its use. The information/requirements given in **3** shall also be kept into consideration while designing and planning an electrical wiring installation. With the proliferation of the use of electrical and electronic devices in buildings as well as the increase in the generation/distribution capacities of power systems, the hazards of energy feed to a fault or defect in the electrical installation has increased. Reliability of power supply and continued supply even under abnormal conditions are becoming very important not only for the operation of services and activities in a building, but also for the life safety of occupants. . Electricity is linked to all services and addition of standby and emergency power supply systems adds to the complexity requiring proper coordinated design. Generally, it is not difficult to provide proper pathways and equipment installation spaces, if an integrated approach is taken from the beginning. The designs should also have to keep the availability of optimum access to installations to ensure proper maintenance. Considering various utility services and to avoid conflict amongst them, it is most important to estimate space requirement for electrical work including LV systems, at planning stage and allocate it in consultation with architect/civil engineer.

4.1.1 The design and planning of an electrical wiring installation shall take into consideration the following:

a) Type of supply, building utility, occupancy, envisaged load and the earthing arrangement available;
b) Provisioning and designing of circuits (circuit length, voltage drop and other aspects), cabling, equipment loads, selection of switchgears, protective devices, fittings and accessories, earthing system, electrical panel rooms, etc complete;

c) Climatic condition, such as cooling air temperature, moisture or such other conditions which are likely to affect the installation adversely including provisioning of air conditioning systems for present and/or future expansion;

d) Provisions for lightning protection system (LPS) based on risk assessment including using building structure for the purpose;

e) Possible presence of inflammable or explosive dust, vapour or gas;

f) Degree of electrical and mechanical protection necessary;

g) Importance of continuity of service including the possible need for standby supply;

h) Probability of need for modification or future extension;

j) Probable operation and maintenance cost taking into account the electricity supply tariffs available;

k) Relative cost of various alternative methods;

m) Need for radio and telecommunication interference suppression;

n) Ease of maintenance;

p) Safety aspects;

q) Energy conservation;

r) Importance of proper discrimination between protective devices for continuity of supply and limited isolation of only the affected portion;

s) Reliability of power supply and redundancy (of sources and distribution paths) to cater to the needs for emergency power and standby power for continued operation of systems as well as integration of alternate sources of energy such as diesel generation, solar energy, wind power, etc; and

t) Integration of different types of energy sources such as (single or multi) grid power, DG set power backup, battery storage energy system, on grid solar PV power, on grid other renewable energy sources such as wind power, micro wind wheel, etc.

NOTE — The design and planning of electrical installations shall also comply with relevant provisions of good practice [D-2(1)].

4.1.2 All electrical apparatus shall be suitable for the services these are intended for and shall confirm to relevant Indian Standards.

4.1.3 *Coordination*

Proper coordination and collaboration among the architect, civil engineer, electrical engineer and mechanical engineer shall be ensured from the planning stage of the installation. The electrical engineer shall be conversant with the needs of the electrical supply provider for making electrical supply arrangement. Electrical supplier’s installation, as per regulations, needs to be segregated from consumer’s installation. Wherever required, prior approval of drawings shall be taken from concerned electrical supplier/electrical inspector. Further, depending on load and regulation provisions, consumer will need to submit to the electrical supplier the details regarding the accommodation of substation including transformers, switch-rooms, standby power, solar photovoltaic panels, lighting and ventilation scheme for the approval. Additional information sought by the Authority regarding details of electrical safety apparatus, fire safety apparatus, cable ducts, rising mains and distribution cables, sub-distribution boards, openings in floors and walls, etc for all electrical installations shall be provided. Provisions as-given in accepted standard [D-2(13)] shall also be complied.

4.1.4 While planning wiring and installation of fittings and accessories, information should be exchanged between the owner of the building/architect/consultant/electrical contractor and the local supply authority in respect of tariffs applicable, types of apparatus that may be connected under each tariff, requirement of space for installing meters, switches, etc, and for total load requirements of lights, fans and power. Consideration should also be taken for the anticipated increase in the use of electricity for lighting, general purpose socket-outlet, kitchen equipment, air conditioning, utility sockets, heating, etc.

4.1.5 It is essential that adequate provision should be made for all the services which may be required immediately and during the intended useful life of the building. If not provided, the user may be tempted to carry out extension of the installation himself or to rely upon use of multi-plug adaptors and long flexible cords, both of which are not recommended.

4.2 Substation and Switchrooms

4.2.1 *Location and Other Requirements*

The location and other requirements of a substation and switchrooms shall be as given below:

a) Availability of power lines nearby may be kept in view while deciding the location of the substation.

b) The substation should preferably be located near the load centre in a separate utility building adjacent to the generator room, if any. Location of substation in the basement should be avoided, as far as possible. If it is unavoidable, the floor level of substation shall be at least 0.6 m above the lowest point of the basement.

c) In case of more than 2 basements in a building/buildings, the substation/switchroom shall be provided in either first or second basement with floor of the substation not more than 10 m below the adjoining (site) ground level.

d) In places where flooding can occur and water level may go above 1 000 mm, the base substation may be located on one level above the ground level of a utility building or of main building. In such cases, only one feeder should feed ground level and levels below with automatic tripping of the feeder to avoid electrocution in case of live electricity coming in contact with water. Designers shall use their discretion in special cases depending on the degree of reliability, redundancy and the category of load and make suitable provisions.

NOTE — In cases, where the substation is located one level above ground level of utility building or main building, this should be after due evaluation of the other risks posed by such a location combined with the concurrence for such a decision from State Electricity Authority comprising the electrical inspectorate and the distribution licensee and the fire service. There should be direct access from outside the building for substation without entering building to substation.

e) Depending upon the topology of the area (in mountains/stepped terrain), location of substation and DG set should be such that flooding of the equipment area is avoided, with substation and DG set located at least 300 mm above maximum flood level with natural drainage of the substation/DG sets areas.

f) Ideal location for an electrical substation for a group of buildings will be at the electrical load centre. Generally, the load centre will be somewhere between the geometrical centre and the air conditioning plant room, as air conditioning plant room will normally be the largest load, if the building(s) are centrally air conditioned.

g) In order to prevent storm water entering the transformer and switch rooms, the floor level of the substation/switchroom shall be at least 300 mm above the highest flood water level that may be anticipated in the locality. Also, facility shall be provided for automatic removal of water.

h) Substation shall not be located immediately above or below plumbing water tanks or sewage treatment plant (STP) water tanks at the same location.

j) All door openings from substation, electrical rooms, etc, should open outwards. Vertical shutters (like fire rated rolling shutters) may also be acceptable provided they are combined with a single leaf door opening outwards for exit in case of emergency. For large substation room/electrical room having multiple equipment, two or more doors shall be provided which shall be remotely located from each other.

k) If substation is located at a height 1 000 m above MSL, then adequate derating of equipment shall be considered.

m) HV panel room and transformers should have direct access and located as near as possible. In case HV panel and transformers are located at different floors or at a distance of more than 20 m, HV isolator shall be provided nearer to the transformer.

n) In case transformer and main HV/LV panel room are located at different floors, the designer should also take care of the safety requirements caused by lack of direct visibility of the status of the controlling switch. To cater to the safety requirements under different conditions of operation as well as maintenance, it may be necessary to provide additional isolator or an emergency push button in the vicinity to trip the supply. Decision has to be taken based on the possible risks.

p) No services or ventilation shafts shall open into substation or switch room unless specific to substation or switch room.

q) *Oil-filled installation* — Substations with oil-filled equipment require great consideration for the fire detection, protection and suppression. Oil-filled transformers require a suitable soak pit with gravity flow to contain the oil in the event of the possibility of oil spillage from the transformer on its failure. Installation of oil-filled equipment shall meet the following requirements:

   1) Substations with oil-filled equipment/apparatus (transformers and high voltage panels) shall be either located in open or in a utility building. They shall not be located in any floor other than the ground floor or the first basement of a utility building. They shall not be located below first basement slab of utility building. They shall have direct access from outside the building for operation and maintenance of the equipment.

   2) Substations/utility buildings (where the substation or oil-filled transformer is located) shall be separated from the adjoining buildings including the main building by at least 6 m clear distance to allow passage of fire tender between the substation/utility building and adjoining building/main building.

   3) There shall be no interconnecting basement with the main building underneath the oil-filled transformers.

   4) Provisions for oil drainage to a tank (from which oil can be decanted and reused) at a lower level and separated by adequate fire barrier shall be provided. If there is a floor directly below the ground floor level or first basement where the oil-filled transformers and oil-filled circuit breakers are placed, then they should be separated by a fire barrier of appropriate fire rating as per Part F 'Fire and Life Safety' of this NBCS and proper oil drainage system shall be provided to avoid possible leakage of oil into the lower floor.

   5) Substation equipment having more than 2 000 of oil whether located indoors in the utility building or outdoors shall have baffle walls of 4 h fire rating between apparatus (*see also* Part F 'Fire and Life Safety' of this NBCS for fire safety related requirements).

   6) Provisions shall be made for suitable oil tank, and where use of more than 9 000 litre of oil in any one oil tank, receptacle or chamber is involved, provision shall be made for the draining away or removal of any oil which may leak or escape from the tank, receptacle or chamber containing the same. Special precautions shall be taken to prevent the spread of any fire resulting from the ignition of the oil from any cause and adequate provision shall be made for extinguishing any fire which may occur.

   7) Oil filled transformers/stabilizers/auto transformers (with oil capacity more than 50 litre) or any equivalent electrical equipment shall not be used within the buildings other than utility buildings. Utility building should not be used for any other activity other than services. In case of less than 50 litre equipment placed in building, it should be kept in naturally ventilated areas and fire segregated from occupied areas and other areas prone to accidental fires.

   8) Adequate fire barriers or deflectors shall be provided to avoid flames from the substation reaching or affecting the adjacent areas including upper floors (*see also* Part F 'Fire and Life Safety' of this NBCS).

   9) Transformer, HV panel, LV isolator shall not be kept one above the other in a way that in case of oil leakage or fire in an equipment the other becomes unapproachable. Stacking of electrical equipment should not be done.

   10) For transformers having large oil content (more than 2 000 litre), *Rule* 46(2) of the *Central Electricity Authority* (*MRSES*) *Regulations*, 2023 shall apply (*see* Annex B).

r) *Dry-type installation* — In case electric substation has to be located within the main multi-storeyed building itself for unavoidable reasons, it shall be a dry-type installation with very little combustible material, such as, a dry type transformer with vacuum breakers as HT switchgear and ACB or MCCB as low voltage (LV) switchgear. Such substations shall be located on the ground level or on first basement and shall have direct access from the outside of the building for operation and maintenance of the equipment.

Exceptionally, in case of functional buildings, such as air traffic control towers, high rise non – residential buildings, data centres and mixed use buildings of height more than 150 m having high electrical load requirement, dry-type installations/substations may also be provided at upper level without emergency backup systems requiring diesel or any inflammable material storage. This measure will decrease the current flow and short-circuit rating at various points, thereby reducing vulnerability to fire. In such cases, a base substation shall be located at ground floor/first basement to cater to the main HV/LV panel which feeds life and safety services loads as defined in **4.2.1**(h). The base substation shall be located in such a way to provide direct access to the firemen in case of any emergency. The power supply control to any substation or transformer located at upper floors shall be from the base substation so that in case of fire, the electrical supply can be easily disconnected to avoid additional losses.

s) The power supply HV cables voltage shall not be more than 36 kV and a separate dedicated and fire compartmented shaft should be provided for carrying such high voltage cables to upper floors in a building. These shall not be mixed with any other shaft and suitable fire detection and suppression measures shall be provided throughout the length of the cable on each floor.

t) The provision for installation and removal of substation equipment should be provided from inside or outside the building without disturbing the associate major equipment in the substation. In case of inside building path a designated elevator should be installed which is capable of handling weight and size of transformer parts being transported to designated floor for installation, maintenance and removal from the floor.

u) In case of compact substation {*see* accepted standard [D-2(7)]}, design and location of the substation shall ensure safety of the people around the compact substation installed along walkways, playgrounds, etc. Compact substation with incomer voltage of 12 kV or less, when located in open areas shall have fencing or barrier (of any metal based protection, such as wire mesh or chain link, which is duly earthed) against unauthorized contact possibility around it at a minimum distance of 750 mm around it with access for maintenance from all four sides. For incomer voltage more than 12 kV and less than 24 kV the fencing distance from substation may be 1 000 mm minimum. In case of more than 24 kV incomer, the distance may be further increased accordingly. The fencing design should take care of the servicing and maintenance requirements of the substation equipment.

w) In case of two transformers (dry type or transformers with cumulative oil quantity less than 2 000 litre) located next to each other without intermittent wall, the distance between the two shall be minimum 1 500 mm for 11 kV, minimum 2 000 mm for 22 kV and 33 kV. Beyond 33 kV, two transformers shall be separated by baffle wall of 4 h fire rating.

y) Horizontal routing of HT cable through functional/occupied areas should be avoided in view of safety.

z) If dry type transformer is used, it may be located adjacent to high voltage switchgear in the form of unit type substation. In such a case, no separate room or fire barrier for the transformer is required either between transformers or between transformer and the switchgear, thereby decreasing the room space requirement. however, minimum distances as specified in **4.2.1**(w) shall be maintained between the apparatus depending upon voltage ratings. Layout of equipment should take care of the need that any one piece of equipment or sub-assembly can be taken out of service and out of the installed location, while keeping the remaining system in service. Working space for access for maintenance of equipment, while keeping an adjoining section of the substation live to maintain power supply to essential loads, may require additional space between such sections of equipment.

aa) For acoustical enclosures/treatment, reference may be made to Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS and CPCB norms requirements.

bb) The minimum recommended spacing between the walls and the transformer periphery from the point of proper ventilation shall be in accordance with good practice [D-2(16)] (*see also* Fig. 1A). The actual spacing may be more than those given in the figure, depending on the circumstances, such as access to the accessories. Other requirements relating to installation of transformers shall also be in accordance with good practice [D-2(16)].

cc) *High voltage switch room/space* — The design should take care of HV equipment space and clearance required around for maintenance and personnel safety as given in **5.3.6.8**. This room may preferably have direct access from outside. In case of substation having one transformer and one source of supply, the owner shall provide one high voltage switch. In case of single point supply with two or more transformers, the number of switch required will be one for incoming supply and one for each transformer. Additional space may be provided keeping in mind future extension, if any. In case of dual supply, two switches shall be provided with mechanical/electrical inter locking arrangement. In case the number of incoming and outgoing switches exceed five, bus coupler of suitable capacity should invariably be provided.

dd) *Low voltage switch room/space* — The floor area required in respect of low voltage switchgear room may be determined keeping in view the number and type of incoming, outgoing, bus coupler switches including likely expansion in future, space requirement as given in **5.3.6.8**. The additional requirements of LV switchroom when located separate from the substation shall be as per **4.4**.

ee) Other requirements relating to installation of switchgears and controlgears as given in good practice [D-2(17)] shall also be complied with.

ff) The minimum height of substation room/HV switch room/LV switch room shall be arrived at considering 1 200 mm clearance requirement from top of the equipment to below the soffit of the beam (*see also* Annex C). In case cable entry/exit is from above the equipment (transformer, HV switchgear, LV switchgear), height of substation room/HV switch room/LV switch room shall also take into account requirement of space for turning radius of cable above the equipment height.

gg) All the rooms shall be provided with partitions up to the ceiling and shall have proper ventilation. Special care should be taken to dissipate transformer heat and where necessary fresh air louvers at lower level and exhaust fans at higher level shall be provided at suitable locations.

hh) In case of cable trench in substation/HV switch room/LV switch room, the same shall be adequately drained to ensure no water is stagnated at any time with live cables.

jj) *Power supply to emergency fire and life safety systems* — Emergency power supply of distribution system for critical requirement for functioning of fire and life safety system and equipment, shall be planned for efficient and reliable power and control supply to the following systems and equipment where provided:

1) Fire pumps;
2) Pressurization and smoke venting; including its ancillary systems such as dampers and actuators;
3) Fireman's lifts (including all lifts).
4) Exit signage lighting;
5) Emergency lighting;
6) Fire alarm system;
7) Public address (PA) system (relating to emergency voice evacuation and annunciation);
8) Magnetic door hold open devices; and
9) Lighting in fire command control and security room.

Power supply to these systems and equipment shall be from normal and emergency (standby generator) power sources with change over facility. It shall be ensured that in case the power supply is from HT source/HT generation, transformers should be planned in stand-by capacity to ensure continuity of power to such systems. Wherever transformers are installed at higher levels in buildings and backup DG sets are of higher voltage rating, then dual redundant cables shall be taken to all transformers. The generator shall be capable of taking starting current of all the fire and life safety systems and equipment as above. Where parallel HV/LV supply from a separate substation fed from different grid is provided with appropriate transformer for emergency, the provision of generator may be waived in consultation with the Authority.

NOTE — Standby generator shall not be kept at upper levels, *see* **4.3.1**.

The power supply to the panel/distribution board of these fire and life safety systems shall be through fire proof enclosures or fire survival cables or through alternate route in the adjoining fire compartment to ensure that supply of power is reliable to these systems and equipment. It is to be ensured that the cabling from the adjoining fire compartment is to be protected within the compartment of vulnerability.

The location of the panel/distribution board feeding the fire and life safety system shall be in fire safe zone ensuring supply of power to these systems. The cable used shall as far as possible be fire survival cable as per the accepted standard [D-2(3)].

Cables for fire alarm and PA system shall be laid in metal conduits or armoured to provide physical segregation from the power cables.

NOTE — Fire control room to be located at or near the main entrance of the building which shall be treated as fire safe zone for all purposes.

kk) Other requirements as given in *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023 shall also be complied with. The fire safety requirements for substation and electrical rooms, including fire rating requirements of substations enclosure, that is, walls, floor, ceiling, openings, doors, etc, as given in Part F 'Fire and Life Safety' of this NBCS should also be complied with.

4.2.2 *Layout of Substation*

4.2.2.1 In allocating the area of substation, it is to be noted that the flow of electric power is from supply company's meter room to HV room, then to transformer and finally to the LV switchgear room. The layout of the room and trenches of required depth shall be in accordance with this flow, so as to optimize the cables, bus-trunking, etc. Visibility of equipment controlled from the operating point of the controlling switchgear is also a desirable feature, though it may not be achievable in case of large substations. Substations shall not be located at or across expansion joints. The rooms/spaces required in a substation shall be provided as below (*see also* Fig. 1):

a) *Supply company's meter room*, generally at the periphery of the premise with direct access from the road/outside;

b) *HV isolation room*, required in case the substation is away from the meter room and is planned adjacent to meter room for disconnecting supply in case of any repair required between meter room and substation;

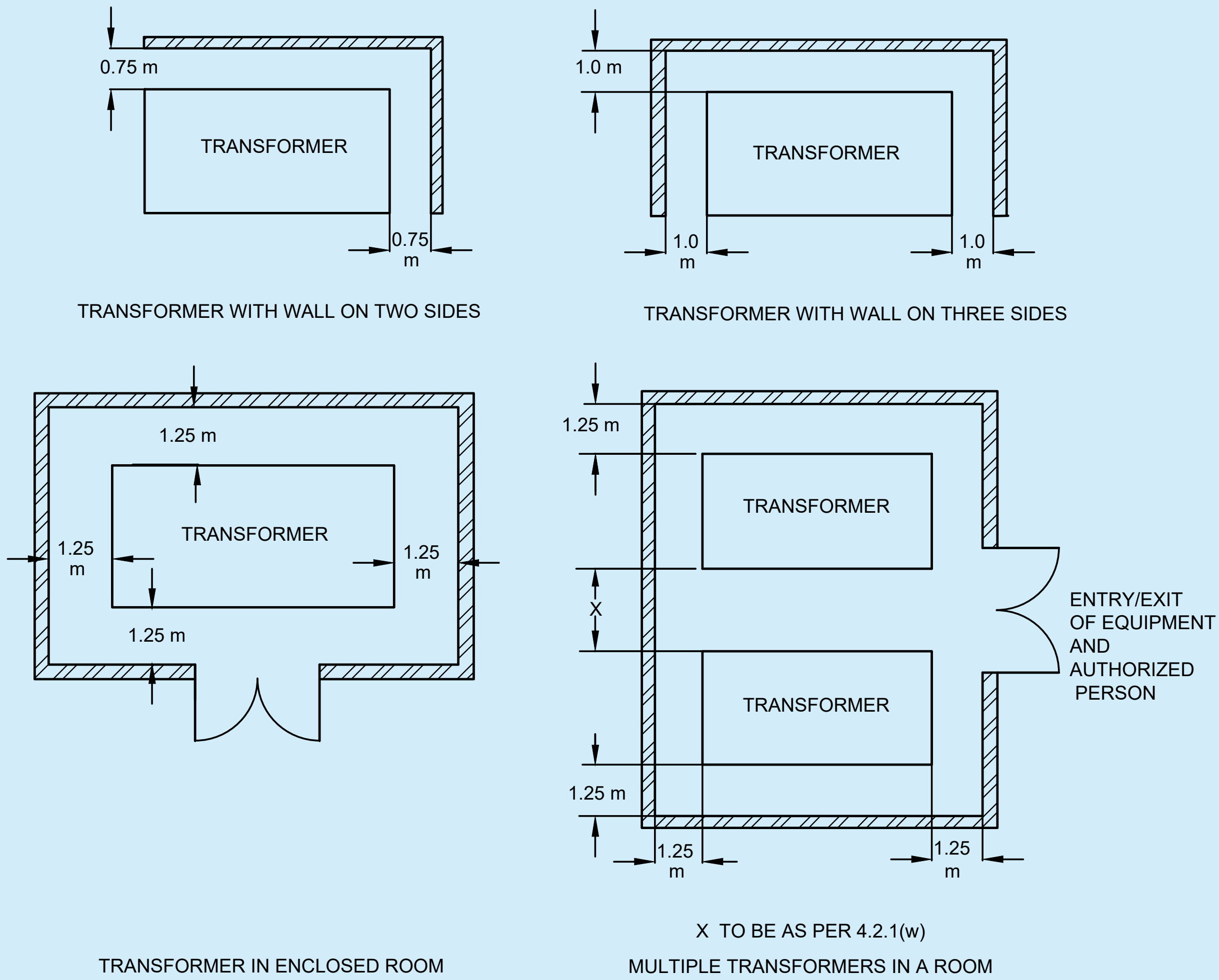
c) *HV panel room/space*, located adjacent to transformer;

d) *Transformer room/space*, separate space in case of oil-filled transformer and combined space in case of dry type transformer;

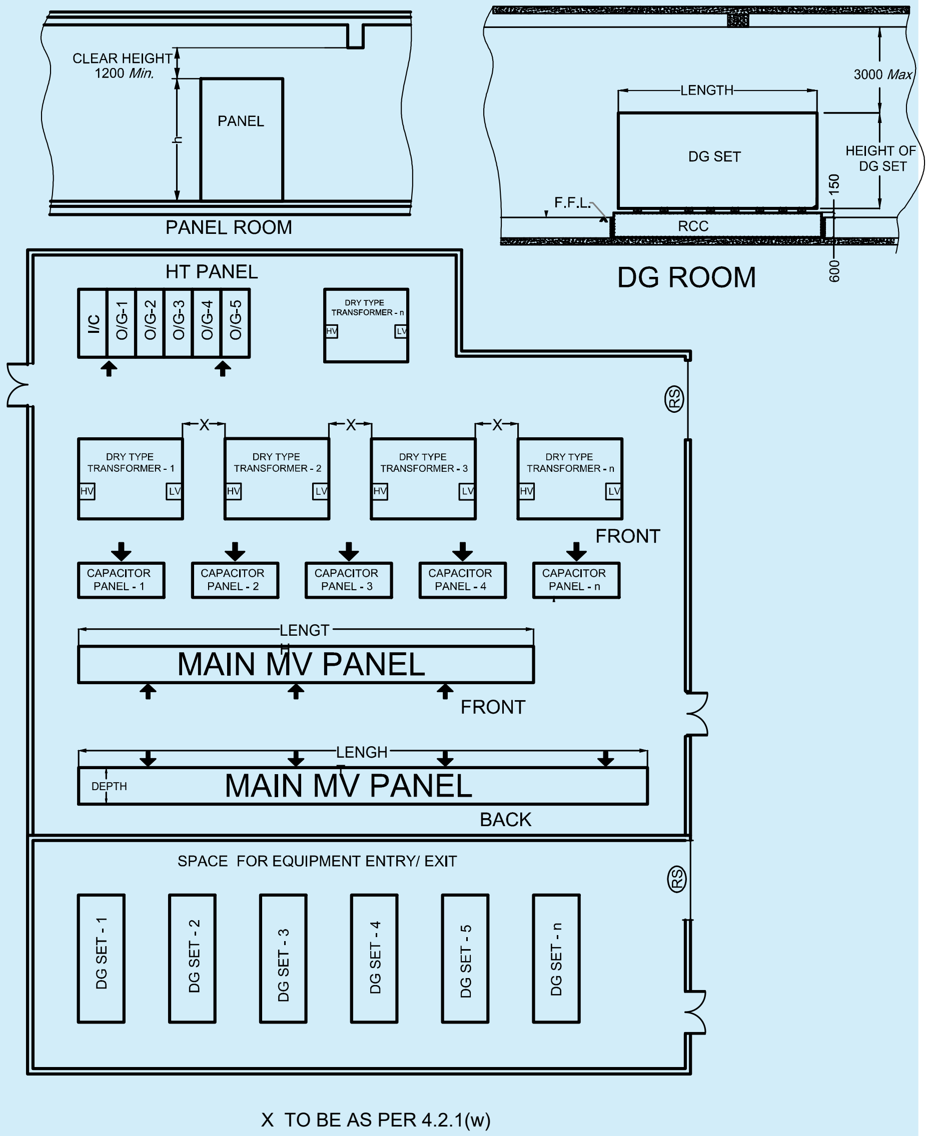
e) *LV isolation room/space* required, in case LV panel is away from transformer or on a different level for isolating supply in case of any repair required between transformer and LV switchgear; and

f) *Main LV panel room/space*, required for distribution to different facility/utility in a building.

A typical layout of a substation is shown in Fig. 1B.



1A MINIMUM RECOMMENDED SPACING BETWEEN THE TRANSFORMER PERIPHERY AND WALLS



ALL DIMENSIONS IN MILLIMETRES.

1B TYPICAL LAYOUT OF SUBSTATION WITH DRY TYPE EQUIPMENT IN A SINGLE ROOM

FIG. 1 TYPICAL LAYOUT OF SUBSTATION INCLUDING MINIMUM RECOMMENDED SPACING OF TRANSFORMER PERIPHERY FROM WALLS

4.2.2.2 *Capacity and size of substation*

The capacity of a substation depends upon the area of the building and its type. The capacity of substation may be determined based on the load requirements [*see also* **3.5.2** (g)]. The Annex F of **6.5.1** under Part 1 'General and Common Aspects, Section 9 Wiring Installations' of good practice [D-2(1)] gives broad values for different types of loads. Ratings of some electrical equipment are given in **6.1**. The Table given below provides broad values for different types of electrical loads. All these aspects shall be considered while calculating the load assessment for the building.

Sl No.	*Purpose of Final Circuit Fed from Conductors or Switchgear to which Diversity Applies*	*Typical Allowances for Diversity Based on*: *Type of Building*		
		Individual House-Hold Installations, Including Individual Dwelling of a Block	Small Shops, Stores, Offices and Business Premises	Small Hotels, Boarding Houses, etc
(1)	(2)	(3)	(4)	(5)
i)	Lighting and convenience power	66 percent of total current demand	90 percent of total current demand	75 percent of total current demand
ii)	Heating and power [*see also* Sl No. (iii) to (iv)]	100 percent of total current demand up to 10 A + 50 percent of any current demand in excess of 10 A	100 percent of full load of largest appliance + 75 percent of remaining appliances	100 percent of full load of largest appliance + 60 percent of second largest appliance + 40 percent of remaining appliances
iii)	Air-conditioning (non-centralized)	100 percent of connected AC load for spaces up to 100 m^2 and 60 percent to 75 percent of connected AC load beyond 100 m^2	100 percent of connected AC load for spaces up to 200 m^2 and 80 percent to 85 percent of connected AC load beyond 200 m^2	90 percent of full load for public spaces and FOTH/BOTH spaces + 60 percent – 75 percent of rooms
iv)	Air-conditioning (centralized)	70 percent of high side equipment and 80 percentage of low side equipment	80 percent of high side equipment and 90 percentage of low side equipment	80 percent of high side equipment and 90 percentage of low side equipment
v)	Cooking appliances	10 A + 30 percent full load of connected cooking appliances in excess of 10 A + 6 A if socket-outlet incorporated in the unit	100 percent of full load of largest appliance + 80 percent of full load of second largest appliance + 60 percent of full load of remaining appliances	70 percent to 60 percent of full load of kitchen equipment and its corresponding ventilation equipment
vi)	Laundry equipment's	100 percent of full load of largest motor + 50 percent of full load of remaining motors	100 percent of full load of largest motor + 50 percent of full load of remaining motors	100 percent of full load of largest motor + 50 percent of full load of remaining motors
vii)	Motors (other than lift, escalators and travellators motors which are subject to special consideration)	100 percent of full load of largest motor + 50 percent of full load of remaining motors	100 percent of full load of largest motor + 80 percent of full load of second largest motor + 60 percent of full load of remaining motors	100 percent of full load of largest motor + 50 percent of full load of remaining motors

Table (*Concluded*)

Sl No.	*Purpose of Final Circuit Fed from Conductors or Switchgear to which Diversity Applies*	*Typical Allowances for Diversity Based on:* *Type of Building*		
		Individual House-Hold Installations, Including Individual Dwelling of a Block	Small Shops, Stores, Offices and Business Premises	Small Hotels, Boarding Houses, etc
(1)	(2)	(3)	(4)	(5)
viii)	Lift, escalators and travellators	100 percentage of cumulative loads	100 percentage of cumulative loads	100 percentage of cumulative loads
ix)	Water heater/heat pump (instantaneous type[1])	100 percent of full load of largest appliance + 100 percent of full load of second largest appliance + 25 percent of full load of remaining appliances	100 percent of full load of largest appliance + 100 percent of full load of second largest appliance + 25 percent of full load of remaining appliances	100 percent of full load of largest appliance + 100 percent of full load of second largest appliance + 25 percent of full load of remaining appliances
x)	Water heater (thermostatically controlled)	No diversity allowable[2]	–	–
xi)	Floor warming installations	No diversity allowable[2]	–	–
xii)	Water heaters thermal storage space heating installations	No diversity allowable[2]	–	–
xiii)	Standard arrangements of final circuits in accordance with good practice [D-2(18)]	100 percent of the current demand of the largest circuit + 40 percent of the current demand of every other circuit	100 percent of the current demand of the largest circuit + 50 percent of the current demand of every other circuit	100 percent of the current demand of the largest circuit + 40 percent of the current demand of every other circuit
xiv)	Socket-outlets other than those included in Sl No. (ix) and stationary equipment other than those listed above	100 percent of the current demand of the largest point + 40 percent of the current demand of every other point	100 percent of the current demand of the largest point + 75 percent of the current demand of every other point	100 percent of the current demand of the largest point + 75 percent of the current demand of every point in main rooms (dining rooms, etc) + 40 percent of the current demand of every other point

NOTE — Diversity may be considered if multiple units of water heater are there in an individual house-hold installation, including individual dwelling of a block.

[1] For the purpose of the table, an instantaneous water heater is deemed to be a water heater of any loading which heats water only while the tap is turned on and therefore uses electricity intermittently.

[2] It is important to ensure that the distribution boards are of sufficient rating to take the total load connected to them without the application of any diversity.

After calculating the electrical load on the above basis, an overall load factor of 50 percent to 90 percent is to be applied to arrive at the minimum capacity of substation. A future load may also be considered for substation sizing [*see* **3.5.2** (g)]. The area required for substation and transformer room for different capacities is given in Annex C for general guidance. For reliability, it is recommended to split the load into more than one transformer and also provide for standby transformer as well as multiple sources, bus-section, etc.

4.3 Emergency Power Backup System (DG Sets/Gas Generators)

4.3.1 *Location*

The emergency power supply (such as generating sets) should not be allowed to be installed above ground floor or below the first basement level of the building in case of multiple basements. However, if necessary DG set can be installed in first floor with dedicated approach and in separate fire compartment. In case of DG set located in basement, facilities for forced ventilation shall be provided such that there is minimum derating of the equipment. The generating set should preferably be housed adjacent to LV switchgear in the substation building to enable transfer of electrical load efficiently and also to avoid transfer of vibration and noise to the main building.

4.3.2 *Room for Emergency Power Backup System*

The capacity of standby generating set shall be chosen on the basis of essential light load, essential air conditioning load, essential life and safety services load [*see* **4.2.1**(jj)], essential equipment load and essential services load, such as minimum one lift out of the bank of lifts and minimum one or all water pumps. In certain cases, more than one generating set is provided based on the type of essential and sensitive loads that are to be operational during power breakdowns. Hence depending on the change over time, the connected switchgears/automatic mains failure (AMF) panels are to be selected. Having chosen the capacity and number of generating sets, required space may be provided for their installation (*see* Annex D for general guidance). There shall be provision of separate direct escape and entry from outside so that in case of fire, electrical supplies can be disconnected to avoid additional losses which may be caused due to electrical supply, present at the time of fire. The height of diesel generating (DG) set rooms shall however be not more than 3 000 mm above the DG set height, unless required due to DG room ventilation requirements. Adequate space shall be provided for storing of fuel. Facilities including space at appropriate positions, relative to the location of the installed equipment has to be kept in the layout design for removal of equipment or sub-assemblies for repair or maintenance. When it is located at a place, other than the ground level with direct equipment access, a hatch or ramp shall be provided.

4.3.3 *Installation and Other Requirements*

Following installation and other requirements shall also be complied with:

a) Day-oil tanks for the DG sets shall be in compliance with *The Petroleum Act*, 1934. It is desirable to have day oil tanks of capacity less than 1 000 litre of diesel/high speed diesel (HSD) with the DG set and not exceeding 1 500 litre of HSD in separate rooms adjacent to DG sets/DG room. Additional quantities required should be kept in underground tanks as per provisions of *The Petroleum Act*, 1934 after taking necessary license;

b) The emergency installation shall comply with the norms laid down by the Central Pollution Control Board (CPCB) and shall also be in compliance with *The Petroleum Act*, 1934 and guidelines of Oil Industry Safety Directorate (OISD). Compartmentation for fire protection with detection and first-aid protection measures is essential;

   NOTE — Different type of fire safety requirements exists for the DG set, for the oil storage area and for the switchgear (*see also* Part F 'Fire and Life Safety' of this NBCS).

c) Acoustic enclosure for DG sets/acoustic lining of the DG room and ventilation system for DG room shall be in line with the requirements of CPCB. If DG set is located outdoor, it shall be housed in acoustics enclosure as per the requirements of CPCB norms. For acoustical enclosures/treatment, reference shall also be made to Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS;

d) Vertical stacking of DG Sets should not be done to ensure at least one DG set is in operation when the other set is on fire or under maintenance. In case stacking of DG sets is unavoidable on ground, a proper fire barrier of 120 min fire rating between the two DG sets should be ensured. Both DG sets should be kept on free standing civil structure with independent access for each generating set and fire barriers in between, such that fire in one does not jump/drip to other. Also, both should be available independently for repair and maintenance. In case of horizontal stacking of DG sets not less than 1.2 m space between the DG sets shall be maintained. Proper ventilation arrangement within the space shall be provided where stacking of DG sets is done;

e) The DG set room should have proper ventilation for engine combustion requirements, heat produced by the body of DG set and the heat removal from radiator or cooling tower, etc to maintain ambient temperature inside the room. The other requirements given in Part F 'Fire and Life Safety' of this NBCS for room for emergency power backup system including DG set room should also be complied with;

f) Other environmental requirements under the provisions of *The Petroleum Act/Rules*, 2002 and norms laid down by CPCB, as amended from time to time, shall be taken into account particularly from the aspect of engine emissions including the height of exhaust pipe and permitted noise levels/controls; and

g) Provisions contained in the *CEA* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023, as amended up to date, good practice [D-2(1)] and relevant Indian Standards shall also be complied with.

4.4 Battery Storage Energy Power Backup System (Li-ion or Similar)

Bulk storage of batteries for power backup supply shall be allowed to be installed in rooms which are properly ventilated to avoid fire accidents (18 air changes per hour). The system shall comply with provisions under relevant IEC standards for storage batteries. There shall be proper space around the container having storage batteries. Necessary gas monitoring system with automatic ventilation arrangement shall be provided for the space. No other material shall be stored in the battery storage area. These batteries should be properly maintained and connected to avoid any sparking/flashing leading to fire. Protection measures shall be taken to avoid thermal runaway, short circuiting, overheating of connecting cables, etc. Container should be air-conditioned/air cooled/liquid cooled and kept as per OEM guidelines. All protection and safety measures as given in the relevant Indian Standards shall be complied. Only a.c. wiring should be taken out of the container for synchronizing with other emergency sources or feeding solar power to premise as individual supply.

4.5 Multiple Energy Sources

When multiple energy sources such as grid, DG set power backup with AMF panel, battery storage energy system, on grid solar PV power, on grid other renewable energy sources such as wind power, micro wind wheel, etc are to be used in the same project simultaneously, synchronization of voltage and frequency of each source is required for energies to be used optimally including setting priorities of source utilization. All protection and safety measures as given in the relevant Indian Standards for each of the energy sources shall also be complied with.

4.6 Location of LV Switch Room Other Than in Substation

In large installations other than where a substation is provided, a separate switch room shall be provided. This shall be located as close to the electrical load centre as possible, on the ground floor or on the first basement level of the building, preferably on the same floor as Transformers. In case LT panel and transformer and DG sets are on different floors then LT isolators shall be proposed adjacent to transformer and DG sets.

Suitable cable trays/busducts shall be laid with minimum number of bends from the points of entry of the main supply cable to the position of the main switchgear. The switch room shall also be placed in such a position that riser shafts may readily be provided therefrom to the upper floors of the building in one straight vertical run. In larger buildings, more than one riser shaft may be required and then horizontal cable trays/bus ducts may also be required for running cables from the switch room to the foot of each rising main. The support system shall be sufficient to take into account the load of the cables to be carried with seismic considerations.

Such cable trays shall either be reserved for specific voltage grades or provided with a means of segregation for low and extra low voltage installations, such as nurses call-bell systems in hospitals, intercom telephone installations, fire detection and alarm system, security systems, data cables and announcement or public address system. Cables/wires for emergency fire and life safety services and their routing should be in accordance with **4.2.1**(jj) and Part F 'Fire and Life Safety' of this NBCS so that these services are maintained even in the event of fire.

Clear space around panels should be kept for access and maintenance depending on circuit breaker type as required. In case of ACB/MCCB front and back access should be planned with all sides movement clearance with 1 000 mm access on back and sides and 2 000 mm access in front of panel. However, space of less than or up to 200 mm is permitted on the back of the panel where there is no access or sufficient space.

4.7 Location and Requirements of Distribution Panels

All distribution panels including switchgears shall be installed in readily accessible position. The electrical controlgear distribution panels and other apparatus, which are required on each floor may conveniently be mounted adjacent to the rising mains, and adequate space considering clearances required as per **5.3.6.8** shall be provided at each floor for this purpose.

4.8 Substation Safety

The owner and the operator of any substation shall be collectively and severally be responsible for any lapse or neglect leading to an accident or an incidence of an avoidable abnormality and shall take care of the following safety requirements:

a) Enclose the substation or similar equipment where necessary to prevent, so far as is reasonably practicable, danger of electric shock or unauthorized access.

b) All equipment, entry and exit shall be always kept accessible.

c) Substation should be a separate fire compartment and its ventilation should be independent of any other fire compartment.

d) All substation rooms shall have two doors at opposite ends. Distance between all equipment such that exit/entry of equipment is feasible without removing of any equipment, light, cable or bus duct, etc.

e) Any part of the substation which is open to the air shall be enclosed, with a fence (earthed efficiently at both ends) or wall at least 600 mm above the height of tallest equipment; to prevent, so far as is reasonably practicable, danger of electric shock or unauthorized access.

f) Transformers (oil filled) when placed in open shall have fencing/wall at least 600 mm above the height of transformer. If trees are there around the substation, then a temporary cover should be provided to avoid leaves falling on the terminals of HT/LT terminations.

g) It shall be ensured that the following are displayed at all times in the substation:

   1) Sufficient safety signs of such size and placed in such positions as are necessary to give due warning of such danger as is reasonably foreseeable in the circumstances;

   2) A notice which is placed in a conspicuous position and which gives the location or identification of the substation, the name of each owner/operator who owns or operates the substation equipment making up the substation and the telephone number where a suitably qualified person appointed for this purpose by the owner/operator will be in constant attendance; and

   3) Such other signs, which are of such size and placed in such positions, as are necessary to give due warning of danger having regard to the siting of, the nature of, and the measures taken to ensure the physical security of, the substation equipment;

h) All reasonable precautions are taken to minimize the risk of fire associated with the equipment.

j) In addition to provisions mentioned in (c), name and emergency telephone number of the authorized personnel shall also be displayed at the substation and instructions covering schematic diagram; requirements of switchgear interlocking, if any; and permission requirements, if any, for load limitations on (incoming) feeders; be also prominently displayed.

4.9 Overhead Lines, Wires and Cables

All erections/alterations having relation to overhead lines, wires and cables shall comply with *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023 with special reference to regulation number 62 and 63 and the following points. However, in case of any conflict, the regulations shall prevail (*see* Annex B).

4.9.1 *Height Requirement*

4.9.1.1 While overhead lines may not be relevant within buildings, regulations related to overhead lines are of concern from different angles as follows:

a) Overhead lines may be required in building complexes, though use of underground cables is the preferred alternative; and

b) Overhead lines may be passing through the site of a building. In such a case the safety aspects are important for the construction activity in the vicinity of the overhead line as well as portions of low height buildings that may have to be constructed below the overhead lines. Overhead lines running adjacent to buildings pose hazard from the aspect of certain maintenance activity (such as use of a ladder on external face of a building) causing temporary compromise of the minimum safety clearance.

4.9.1.2 If at any time subsequent to the erection of an overhead line, whether covered with insulating material or not, or underground cable, any person who proposes to erect a new building or structure or to raise any road level or to carry out any other type of work whether permanent or temporary or to make in or upon any building, or structure or road, any permanent or temporary addition or alteration, in proximity to an overhead line or underground cable, such person and the contractor whom he employs to carry out the erection, addition or alteration, shall give intimation in writing of his intention to do so to the supplier or owner and to the Electrical Inspector and shall furnish therewith a scale drawing showing the proposed building, structure, road or any addition or alteration and scaffolding thereof required during the construction. In this connection, Regulation 63 of *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023, shall also be complied with (*see* Annex B).

4.9.1.3 Any person responsible for erecting an overhead line will keep informed the authority(s) responsible for services in that area for telecommunication, gas distribution, water and sewage network; roads and so as to have proper coordination to ensure safety. He shall also publish the testing, energizing programme for the line in the interest of safety.

4.9.1.4 For minimum distance (vertical and horizontal) of electric lines/wires/cables from buildings, Regulations 60, 62, 63, and 67 of *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023, shall be complied with (*see* Annex B).

4.9.1.5 Regulation 66 of *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023, which govern conditions related to the storage of material including storage of construction material at a construction site, or other materials in a building in vicinity of overhead lines and underground cables, shall also be complied with (*see* Annex B).

4.9.2 *Position, Insulation and Protection of Overhead Lines*

4.9.2.1 Any part of an overhead line which is not connected with earth and which is not ordinarily accessible shall be supported on insulators or surrounded by insulation. Any part of an overhead line which is not connected with earth and which is ordinarily accessible shall be:

a) made dead; or

b) so insulated that it is protected, so far it is reasonably practicable, against mechanical damage or interference; or

c) adequately protected to prevent danger.

4.9.2.2 Any person responsible for erecting a building or structure which will cause any part of an overhead line which is not connected with earth to become ordinarily accessible shall give reasonable notice to the licensee or distributor who owns or operates the overhead line of his intention to erect that building or structure.

The expression 'ordinarily accessible' means the overhead line might be reachable by hand if any scaffolding, ladder or other construction was erected or placed on/in, against or near to a building or structure.

4.9.2.3 Any bare conductor not connected with earth, which is part of a low voltage overhead line, shall be situated throughout its length directly above a bare conductor which is connected with earth.

4.9.3 *Precautions against Access and Warnings of Dangers*

4.9.3.1 Every support carrying a high voltage overhead line shall be fitted with anti-climbing devices to prevent any unauthorized person from reaching a position at which any such line will be a source of danger. In this connection, Regulation 76 of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023, shall also be complied with (*see* Annex B).

4.9.3.2 Every support carrying a high voltage overhead line, and every support carrying a low voltage overhead line incorporating bare phase conductors, shall have attached to it sufficient safety signs and placed in such positions as are necessary to give due warning of such danger as is reasonably foreseeable in the circumstances.

4.9.3.3 Poles supporting overhead lines near the road junctions and turnings shall be protected by a masonry or earth fill structure or metal barricade, to prevent a vehicle from directly hitting the pole, so that the vehicle, if out of control, is restrained from causing total damage to the live conductor system and lead to a hazardous condition on the road or footpath or building.

4.9.4 *Fitting of Insulators to Stay Wires*

Every stay wire which forms part of, or is attached to, any support carrying an overhead line incorporating bare phase conductors (except where the support is a lattice steel structure or other structure entirely of metal and connected to earth) shall be fitted with an insulator, no part of which shall be less than 3 m above ground or above the normal height of any such line attached to that support.

4.10 Maps of Underground Networks

4.10.1 Any person or organization or authority laying cables shall contact the local authority in charge of that area and find out the layout of:

a) water distribution pipe lines in the area;

b) sewage distribution network;

c) telecommunication network;

d) gas pipeline network; and

e) existing power cable network;

and plan the cable network in such a manner that the system is compatible, safe and non-interfering either during its installation or during its operation and maintenance. Plan of the proposed cable installation shall be brought to the notice of the other authorities referred above.

4.10.2 Suitable cable markers and danger sign as will be appropriate for the safety of the workmen of any of the systems shall be installed along with the cable installation. Cable route markers shall be provided at every 20 m and also at turnings and/or crossings.

4.10.3 Notification of testing and energization of the system shall also be suitably publicized for ensuring safety.

4.10.4 Any person or organization or authority associated with the operation and maintenance of services in a complex is required to have a complete integrated diagram or drawings of all services with particular emphasis on the hidden pipes, cables, etc, duly kept up-to-date by frequent interaction with all agencies associated with the maintenance work.

Organization or agency responsible for laying cables shall have and, so far it is reasonably practicable, keep up-to-date, a map or series of maps indicating the position and depth below surface level of all networks or parts thereof which he owns or operates. Where adequate mapping has not been done and the excavation for cable laying reveals lines pertaining to any of the other services, three dimensional location should be marked and recorded. Even where mapping exists, it may be examined if the records have become obsolete due to change such as, in road level. Any map prepared or kept shall be available for inspection by any authority, such as municipality, water supply, sewage, service providers and general public, provided they have a reasonable cause for requiring reference to any part of the map.

4.10.5 Any agency working on any one or more service (occupying the underground space for service pipes, cables, etc) should keep the other agencies informed of the work so that an inadvertent action will not cause a disruption of service. Each agency should be responsible for keeping the latest information with the central authority of such records and should be responsible to ensure that the modifications, if any, are duly updated and notified among the other agencies.

5 DISTRIBUTION OF SUPPLY AND CABLING

5.1 General

5.1.1 In the planning and design of an electrical wiring installation, due consideration shall be made of all the prevailing conditions. It is recommended that advice of a competent electrical engineer be sought at the initial stage itself with a view to providing an installation that will prove adequate for its intended purpose, be reliable, safe and efficient.

While planning of routing of cables and bus trunking it shall be ensured to avoid passing of cables and bus trunking from lift lobbies, staircase lobbies and fire exits. Proper ventilations or air-changes shall be provided for heat rejected from bus trunking and cables.

5.1.2 A certain redundancy in the electrical system is necessary and has to be built in from the initial design stage itself. The extent of redundancy will depend on the type of load, its criticality, normal hours of use, quality of power supply in that area, coordination with the standby power supply, capacity to meet the starting current requirements of large motors, etc.

5.1.3 In modern building technology, following high demands are made of the power distribution system and its individual components:

a) long life and good service quality;

b) safe protection in the event of fire;

c) low fire load;

d) flexibility in load location and connection, but critical in design;

e) low space requirement; and

f) minimum effort involved in carrying out retrofits.

5.1.4 The high load density in modern large buildings and high-rise buildings demands compact and safe solution for the supply of power. The use of busbar trunking system is ideal for such applications. Busbar trunking can be installed in vertical riser's shafts or horizontally in passages for transmission and distribution of power. They allow electrical installations to be planned in a simple and neat manner. In the building complexes, additional safety demands with respect to fire barriers and fire load can also be met with the use of busbar trunking. Busbar trunking system also reduces the combustible material near the area with high energy in comparison with other distribution systems such as cables and makes the building safe from the aspect of vulnerability to fire of electrical origin. In addition, unlike cable systems the reliability of a busbar trunking system is very high. These systems also require very little periodic maintenance. Choice of busbar trunking for distribution in buildings can be made on the basis of:

a) reduced fire load (drastically reduced in comparison to the cable system);

NOTE — Insulation materials of cables are required to be fire resistant and an essential performance requirement is that the insulation material may burn or melt and flow when directly exposed to a temperature (or fire) higher than what it is class designated, but should not continue to burn after the flame or the source of heat or fire is extinguished. Even if the above fire resistant property is exhibited by the cable insulation, a large collection of cables will make the cable insulation fail to exhibit this retardant property. While specific guidelines for limiting number of cable and bunching is not available and in such cases the switch over to a busbar trunking system is the proper alternative.

b) reduced maintenance over its entire lifetime;

c) longer service lifetime in comparison with a cable distribution; and

d) enhanced reliability due to rigid bolted joints and terminations and extremely low possibility of insulation failure.

NOTE — Provisions contained in electric supply and cabling shall also comply with the provisions relating to *the CEA* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023, good practice [D-2(1)] and relevant Indian Standards as amended up to date.

5.2 System of Supply

5.2.1 All electrical apparatus shall be suitable for the voltage and frequency of supply.

5.2.2 In case of connected load of 100 kVA and above, the relative advantage of high voltage three-phase supply may be considered. Though the use of high voltage supply entails the provisions of space and the capital cost of providing suitable transformer substation at the consumer's premises, the following advantages are gained:

a) advantage in tariff;

b) more effective earth fault protection for heavy current circuits;

c) elimination of interference with supplies to other consumers permitting the use of large size motors, welding plant, etc; and

d) better control of voltage regulation and more constant supply voltage.

NOTE — Additional safety precautions required to be observed in HV installations shall also be kept in view.

In many cases there may be no choice available to the consumer, as most of the licensees have formulated their policy of correlating the supply voltage with the connected load or the contract demand. Generally, the supply is at 240 V single phase up to 5 kVA, 415/240 V 3 phase from 5 kVA to 100 kVA, 11 kV (or 22 kV) for loads up to 5 MVA and 33 kV for consumers of connected load or contract demand more than 5 MVA.

In case of load exceeding 10 MVA at 33 kV for building complex, designer may explore option of having a combination of power transformers and distribution transformers to reduce the numbers and long lengths of power cables. 33 kV/11 kV power substation at site level/utility building for stepping down at 11 kV and then having 11 kV/415 Volt distribution substation at basement/ground/first floor/usage floors, near load centres to avoid long lengths of 415 volt cables.

5.2.3 In very large industrial buildings where heavy electric demands occur at scattered locations, the economics of electrical distribution at high voltage from the main substation to other subsidiary transformer substations or to certain items of plant, such as large motors and furnaces, should be considered. The relative economy attainable by use of high voltage distribution and high voltage equipment is a matter for expert judgment and individual assessment in the light of experience by a professionally qualified electrical engineer. *See also* good practice [D-2(1)] and accepted standard [D-2(19)].

5.2.4 In case of complex buildings/mixed use buildings where individual load of usage category (such as commercial, mercantile, residential, hotels, hospitals, storage, industry, etc has cumulative load of 1 000 kVA, then for EV charging requiring more than 100 kVA power should be proposed with independent distribution transformer(s) (200 kVA or more) with separate meter to avoid mixing of different type of loads, harmonics, etc.

5.2.5 As far as possible, EV charging arrangement shall be provided with independent transformer to ensure that harmonics created during EV charging do not interfere with power supply to other system. It shall be ensured that EV charging is not fed from diesel generator sets or gas generators or similar fuel based power generating systems (*see* **10.7** for more details).

5.3 Substation Equipment and Accessories

Substations require an approval from the Electrical Inspectorate. Such approval is mandatory before energizing the substation. Therefore, it is essential to get the approval for the general layout, schematic layout, protection plan, etc, before the start of the work from the Inspectorate. All substation equipment and accessories and materials, etc, shall conform to relevant Indian Standards, wherever they exist, otherwise the consumer (or his consultant) shall specify the standards to which the equipment to be supplied and that shall be approved by the authority. Manufacturers of equipment have to furnish certificate of conformity as well as type test certificates for record, in addition to specified test certificates for acceptance tests and installation related tests for earthing, earth continuity, load tests and tests for performance of protective gear. All works under this, shall comply with the requirements of relevant Indian Standards and provisions of **5.1** of Part 1 'General and Common Aspects, Section 9 Wiring Installations' of good practice [D-2(1)].

5.3.1 *Supply Company's High Voltage Meter Board*

In case of single point high voltage metering, energy meters shall be installed in building premise as per **4.2.2.1**, at such a place which is readily accessible/provides unobstructed access to the electric supply company. The supplier or owner of the installation shall provide at the point of commencement of supply a suitable isolating device fixed in a conspicuous position at not more than 1.7 m above the ground so as to completely isolate the supply to the building in case of emergency.

In case of multipoint metering, main energy meter and reference meters shall be installed in building premises as per **4.2.2.1**, at a place which is readily accessible to the owner/operator of the building and the authority. These should be accompanied with smart metering system (with dual source pre-paid meters).

In this connection, the *Central Electricity Authority* (*Installation and Operation of Meters*) *Regulations*, 2006, as amended from time-to-time shall also be complied with.

5.3.2 *High Voltage Switchgear*

5.3.2.1 The selection of the type of high voltage switchgear for any installation inter alia depends upon the following:

a) voltage of the supply system;
b) prospective short-circuit current at the point of supply;
c) size and layout of electrical installation;
d) availability of space; and
e) nature of installation.

Making and breaking capacity of switchgear shall be commensurate with short-circuit potentialities of the supply system and the supply authority shall be consulted on this subject. The HV switchgear and controlgear shall conform to the relevant Indian Standards. For details see accepted standard [D-2(7)].

5.3.2.2 Guidelines on various types of switchgear equipment and their choice for a particular application shall be in accordance with good practice [D-2(20)].

5.3.2.3 In extensive installations of switchgear (having more than four incoming supply or having more than 12 circuit breakers), banks of switchgears shall be segregated from each other in order to prevent spreading of the risk of damage by fire or explosion arising from switch failure. Where a bus-bar section switch is installed, it shall also be segregated from adjoining banks in the same way {*see* good practice [D-2(21)]}.

5.3.2.4 It should be possible to isolate any section from the rest of the switchboards such that work might be undertaken on this section without the necessity of making the switchboard dead. Isolating switches used for the interconnection of sections or for the purpose of isolating circuit-breakers of other apparatus, shall also be segregated within its compartment so that no live part is accessible when work in a neighbouring section is in progress.

5.3.2.5 In the case of double or ring main supply, switchgears with interlocking arrangement shall be provided to prevent simultaneous switching of two different supply sources. Electrical and/or mechanical interlocks may preferably be provided.

5.3.2.6 The selection and erection shall also comply with the provisions of accepted standard [D-2(22)].

5.3.3 *HV Cables*

5.3.3.1 The sizing of the cable shall depend upon the method of laying cable, current to be carried, permissible maximum temperature it shall withstand, voltage drop over the length of the cable, the prospective short-circuit current to which the cable may be subjected, the characteristics of the overload protection gear installed, load cycle, thermal resistivity of the soil and the operating voltage {*see also* good practice [D-2(23)]}.

5.3.3.2 All HV cables shall be installed in accordance with good practice [D-2(23)]. The HV cables shall either be laid on the cable rack/built-up concrete trenches/tunnel/basement or directly buried in the ground depending upon the specific requirement. When HV cable is hanging/running below the basement ceiling slab, the cable shall be laid in a fire rated enclosure/cable tray. The advice of the cable manufacturer with regard to installation, jointing and sealing should also be followed.

5.3.4 *High Voltage Busbar Trunking/Ducting*

HV busbar system is used for transporting power between HV generators, transformers and the infeed main switchgear of the main HV switchgear.

Generally, three types of bus ducts, namely non-segregated, segregated and isolated phase bus ducts are used. The non-segregated bus ducts consist of three phase busbars running in a common enclosure made of steel or aluminium. The enclosure shall provide safety for the operational personnel and shall reduce chances of faults. HV interconnecting busbar trunking for a.c. voltage above 1 kV up to and including 36 kV shall conform to accepted standard [D-2(24)]. The enclosures shall be effectively grounded.

Segregated phase bus ducts are similar to non-segregated phase duct except that metal or isolation barriers are provided between phase conductors to reduce chances of phase to phase faults. However, it is preferable to use metal barriers.

In the case of isolated bus ducts, each phase conductor shall be housed in a separate non-magnetic enclosure. The bus duct shall be made of sections which are assembled together at site to make complete assembly. The enclosure shall be of either round or square shape and welded construction. The enclosures of all phases in general should be supported on a common steel structure.

Seismic supports shall be provided for busbar trunking having continuous straight lengths of more than 24 m at a single stretch in horizontal formation and 12 m in vertical formation in accordance with relevant standards.

The bus duct system shall be coordinated with connecting switchgear so as to provide adequate protection.

When busbar trunking is crossing different fire compartments, they shall have fire barriers of same rating as that of the compartment (*see also* Part F 'Fire and Life Safety' of this NBCS).

5.3.5 *Transformers*

5.3.5.1 General design objective while selecting the transformer(s) for a substation should be to provide at least two or more transformers, so that a certain amount of redundancy is built in, even if a standby system is provided. With growing emphasis on energy conservation, the system design should be made for both extremes of loading. During the periods of lowest load in the system, it would be desirable to operate only one transformer and subsequently switching on the additional transformers as the load increases during the day. Total transformer capacity is generally selected on the basis of present load, possible future load, operation and maintenance cost and other system conditions. The selection of the maximum size (capacity) of the transformer is guided by the short-circuit making and breaking capacity of the switchgear used in the high voltage distribution system. Maximum size limitation is important from the aspect of feed to a downstream fault. The transformers shall conform to accepted standards [D-2(25)].

5.3.5.2 For reasons of reliability and redundancy it is normal practice to provide at least two transformers (especially where electrical load is beyond 1 500 kVA) for functional buildings, hospitals, hotels, institutes, etc. Providing distribution transformers beyond 1 600 kVA/2 000 kVA is not recommended. Interlinking by tie lines is an alternative to enhance reliability/redundancy in areas where there are a number of substations in close vicinity, such as a campus with three or four multi-storeyed blocks, each with a substation. Ring main type of distribution is preferred for complexes having a number of substations.

5.3.6 *Low Voltage Switchgear*, *Controlgear and their Assemblies*

5.3.6.1 The selection of the type of low voltage switchgear for any installation inter alia depends upon the following:

a) voltage of the distribution system;

b) prospective circuit current at the point at which the switchgear is proposed;

c) prospective short-circuit current at which the switchgear is proposed;

d) availability of space; and

e) nature of installation.

The switchgear, controlgear and their assemblies so selected, shall comply with the accepted standards [D-2(26)].

5.3.6.2 Switchgear (and its protective device) shall have breaking capacity not less than the anticipated fault level in the system at that point. System fault level at a point in distribution systems is predominantly dependent on the transformer size and its reactance. Parallel operation of transformers increases the fault level.

5.3.6.3 Where two or more transformers are to be installed in a substation to supply a low voltage distribution system, the distribution system shall be divided into separate sections each of which shall be normally fed from one transformer only unless the low voltage switchgear has the requisite short-circuit capacity. Provision may, however, be made to interconnect separate sections, through a bus coupler in the event of failure or disconnection of one transformer. *See* **4.2** for details of location and requirements of substation.

5.3.6.4 Isolation and controlling circuit breaker shall be interlocked so that the isolator cannot be operated unless the corresponding breaker is in open condition. The choice between alternative types of equipment may be influenced by the following considerations:

a) In certain installations supplied with electric power from remote transformer substations, it may be necessary to protect main circuits with circuit-breakers operated by earth fault, in order to ensure effective earth fault protection;

b) Where large electric motors, furnaces or other heavy electrical equipment is installed, the main circuits shall be protected from short-circuits by switch disconnector fuse or circuit breakers. For motor protection, the combination of contactor overload device and fuse or circuit breakers shall have total co-ordination at least for motor ratings up to 10 kW and for ratings above 10 kW, it shall be Type 2 co-ordination in accordance with relevant part of accepted standards [D-2(26)]. Wherever necessary, back up protection and earth fault protection shall be provided to the main circuit; and

c) Where means of isolating main circuits is separately required, switch disconnector fuse or switch disconnector may form part of main switchboards.

5.3.6.5 It shall be mandatory to provide power factor improvement at the substation bus with suitable capacity of inductor-capacitor combination and relative switchgear in consultation with the respective manufacturers depending upon the nature of electrical load anticipated on the system. Necessary switchgear/feeder circuit breaker shall be provided for controlling of inductor-capacitor bank.

Power factor of individual motor may be improved by connecting individual capacitor banks in parallel. For higher range of motors, which are running continuously without much variation in load, individual power factor correction at load end is advisable.

NOTE — Care should be taken in deciding the kVAr rating of the capacitor in relation to the magnetizing kVA of the motor. Over rating of the capacitor may cause injury to the motor and capacitor bank. The motor still rotating after disconnection from the supply, may act as generator by self-excitation and produce a voltage higher than supply voltage. If the motor is again switched on before the speed has fallen to about 80 percent of the normal running speed, the high voltage will be superimposed on the supply circuits and will damage both the motor and the capacitor.

As a general rule, the kVAr rating of the capacitor should not exceed the no-load magnetizing kVA of the motor.

Generally, it will be necessary to provide an automatic control for switching on the capacitors matching the load power factor and the bus voltage. Such a scheme will be necessary as capacitors permanently switched in the circuit may cause overvoltage at times of light load. Capacitor panel shall be provided with adequate ventilation facility.

With the wide spread use of thyristor and rectifier based loads, there is a necessity of providing a full size neutral; but this requirement is generally limited to the 3-phase 4-wire distribution generally in the 415 V.

5.3.6.6 Harmonics on the supply systems are becoming a greater problem due to the increasing use of electronic equipment, computer, fluorescent lamps, LEDs and CFLs (both types have control/driver circuits operating in switch mode), mercury vapour and sodium vapour lighting, TV, microwave ovens, latest air conditioners, refrigerators, controlled rectifier and inverters for variable speed drives, power electronics and other non-linear loads. Harmonics may lead to almost as much current in the neutral as in the phases. This current is almost third, fifth, seventh and ninth harmonic. In such cases, phase rectification devices may be considered at the planning stage itself for the limits of harmonic voltage distortion. Use of active filters in combination/independent use with capacitor banks should be proposed with either 3-phase 3-wire or 3-phase 4 wire depending upon the equipment for which electricity is utilized. The active filters along with capacitor panel should be proposed to be located in controlled environment.

5.3.6.7 Sufficient clearances as below shall be provided for isolating the switchboard to allow access for servicing, testing and maintenance (*see* Fig. 2):

a) A clear space of not less than 1 m in width shall be provided in front of the switchboard;

   NOTE — In case the board has a shutter in the front for aesthetic reasons, the opening of the shutter shall satisfy the requirement of working/safety space of 1 m in front of the switchgear.

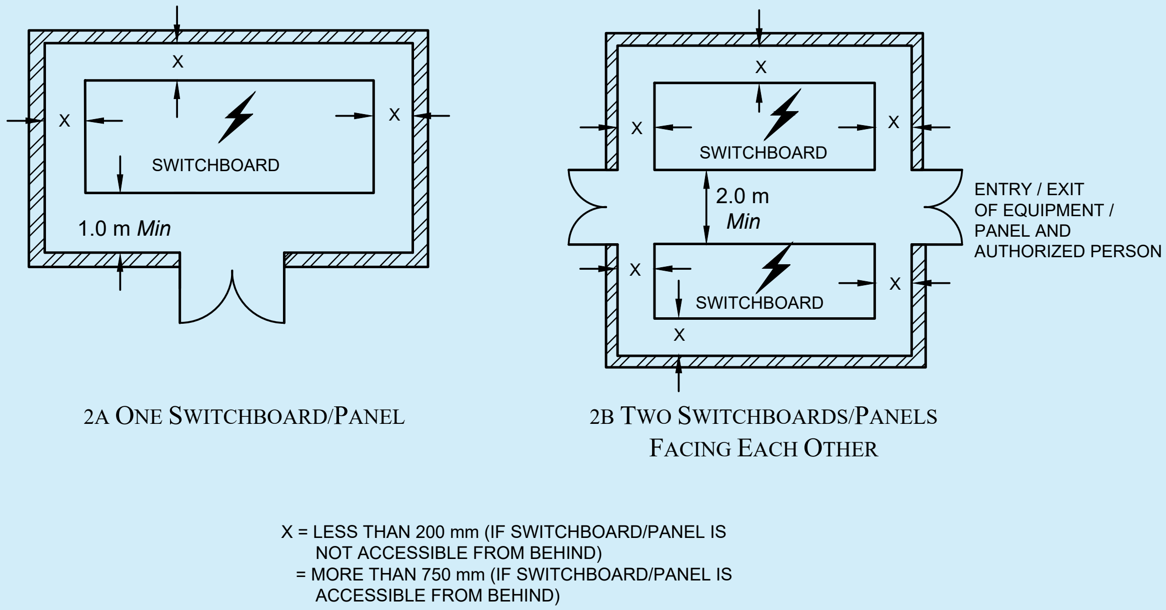
b) If access behind the switchboard is provided then it shall be either less than 200 mm or more than 750 mm in width, measured from the farthest protruding part of any attachment or conductor;

c) If the space behind the switchboard exceeds 750 mm in width, there shall be a passageway from either end of the switchboard, clear to a height of 1.8 m; and

d) If two switchboards are facing each other, a minimum distance of 2.0 m shall be maintained between them.

The connections between the switchgear mounting and the outgoing cable up to the wall shall be enclosed in a protection pipe.

There shall be a clear distance of not less than 250 mm between the board and the insulation cover, the distance being increased for larger boards in order that on closing of the cover, the insulation of the cables is not subjected to damage and no excessive twisting or bending in any case. The cable alley in the metal board should enable within prescribed limit, twisting or bending of cable such that insulation of the cables is not subjected to damage.



NOTE — X TO BE MEASURED FROM THE FARTHEST PROTRUDING PART OF ANY ATTACHMENT OR CONDUCTOR.

FIG. 2 CLEARANCES AROUND SWITCHBOARD IN ENCLOSED ROOM

In this connection, for installation of voltages exceeding 250 V single phase, Regulation 37 of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 shall also be complied with (*see* Annex B).

5.3.6.8 Sufficient additional space shall be allowed in substations and switch rooms to allow operation and maintenance. Sufficient additional space shall be allowed for temporary location and installation of standard servicing and testing equipment. Space should also be provided to allow for anticipated future extensions.

5.3.6.9 Panels in a room or cubicle, access to which is controlled by lock and key, shall be accessible to authorized persons only.

5.3.6.10 Except main LV panel, it will be preferable to locate the sub-panels/distribution boards/sub-meter boards near load centre. Further, it should be ensured that these panels are easily approachable. The panels should have clear access from common areas excluding staircase.

Where the switchboard is erected in a room of a building isolated from the source of supply or at a distance from it, adequate means of control and isolation shall be provided both near the boards and at the origin of supply. Sufficient clearances as per **5.3.6.8** shall be provided.

5.3.6.11 All switchboards shall be of metal clad totally enclosed type or any insulated enclosed pattern.

5.3.6.12 The selection and erection shall also comply with the provisions of accepted standard [D-2(22)].

5.3.7 *Medium/Low Voltage Cables*

5.3.7.1 The sizing of the cable shall depend upon the current to be carried, method of laying cable, permissible maximum temperature it shall withstand, voltage drop over the length of the cable, the prospective short-circuit current to which the cable may be subjected, the characteristics of the overload protection gear installed, load cycle, thermal resistivity of the soil and the operating voltage {*see also* good practice [D-2(18)]}.

It is desirable to use flame retardant/flame retardant low smoke and halogen free cables and wires in electrical distribution systems.

It is recommended to use four core cable in place of three and half core in respect of sizes below 50 mm^2 to minimize heating of neutral core due to harmonic content in the supply system and to also avoid overload failures. As far as possible, balancing of single phase system shall be done to ensure minimum unbalanced current in neutral of three phase system. Only four core cables shall be used for internal electrical installation. Also, all cable sizes up to 25 mm^2 shall be of copper conductor to prevent overheating and fire incidence in internal electrical installations and capacity/size considerations. All cables shall be installed in accordance with good practice [D-2(18)]. The advice of the cable manufacturer with regard to installation, jointing and sealing should also be followed.

5.3.7.2 *Colour identification of cores of non-flexible cables* {*see also* accepted standard [D-2(27)]}

The colour of cores of non-flexible cables shall be in accordance with the following:

Sl No.	*Function*	*Colour Identification of Core of Rubber of PVC Insulated Non-flexible Cable, or of Sleeve or Disc to be Applied to Conductor or Cable Code*
(1)	(2)	(3)
i)	Protective or earthing	Green and yellow (*see* Note 1)
ii)	Phase of a.c. single-phase circuit	Red [or yellow or blue (*see* Note 2)]
iii)	Neutral of a.c. single or three-phase circuit	Black
iv)	Phase R of 3 phase a.c. circuit	Red
v)	Phase Y of 3 phase a.c. circuit	Yellow
vi)	Phase B of 3 phase a.c. circuit	Blue
vii)	Positive of d.c. 2 wire circuit	Red
viii)	Negative of d.c. 2 wire circuit	Black
ix)	Outer (positive or negative) of d.c. 2 wire circuit derived from 3 wire system	Red
x)	Positive of 3 wire system (positive of 3 wire d.c. circuit)	Red
xi)	Middle wire of 3 wire d.c. circuit	Black
xii)	Negative of 3 wire d.c. circuit	Blue
xiii)	Functional earth-telecommunication	Cream/grey

NOTES

1 Bare conductors are also used for earthing and earth continuity conductors. But it is preferable to use insulated conductors with green coloured insulation with yellow stripes.

2 As alternative to the use of red, if desired in large installations, up to the final distribution board.

3 For armoured PVC-insulated cables and paper-insulated cables, *see* relevant Indian Standard.

5.3.8 *LV Busbar Trunking/Rising Mains*

5.3.8.1 Where heavy loads and/or multiple distribution feeders are required to be supplied, busbar/rising main systems are preferred. The busbars are available for continuous run from point to point or with tap offs at standard intervals and have to be chosen as per specific requirement. Seismic supports shall be provided for busbar trunking having continuous straight lengths of more than 24 m at a single stretch in horizontal formation and 12 m in vertical formation in accordance with relevant standards. There are following two types of LV bus duct system for power distribution:

a) air insulated (conventional) type; and

b) compact and sandwich type.

5.3.8.1.1 *Air insulated* (*conventional*) *type bus duct*

These are used for large power handling between transformer and switchgear or between switchgear and large power loads such as compressor drive motor, etc. This type is generally used in plant rooms, riser shafts, substations, etc. These are generally air insulated with intermediate ceramic support insulators enclosed in a metallic enclosure, which should be earthed. They have the least amount of combustible material. However, when these are crossing different fire compartments, they shall have fire barriers of same rating as that of the compartment (*see also* Part F 'Fire and Life Safety' of this NBCS).

Air insulated (conventional) type bus ducts with non-metallic enclosures are also available. However, such bus ducts shall be used only if essential and with appropriate additional care.

5.3.8.1.2 *Compact and sandwich type bus duct*

Compact and sandwich type bus ducts are used within areas of the building which have space restrictions, functional reasons and for aesthetic considerations. The busbars shall be of copper considering the safety and longevity aspects. They may be used in false ceiling spaces or even in corridors and shafts for distribution without any false ceiling as they provide an aesthetically acceptable finish to merge with other building elements such as beams, ducts or pipes in functional buildings.

The insulation material in such ducts are generally glass fibre tape or epoxy encapsulation in combination with ceramic supports/spacers. These bus ducts should be duly enclosed by a metallic enclosure, which should be earthed.

5.3.8.1.3 The selection and erection shall also comply with the provisions of accepted standard [D-2(28)].

5.3.8.2 The bus duct system shall be coordinated with connecting switchgear so as to provide adequate protection.

5.3.8.3 When busbar trunking is crossing different fire compartments, they shall have fire barriers of same rating as that of the compartment (*see also* Part F 'Fire and Life Safety' of this NBCS).

5.3.8.4 The low voltage busbar trunking shall conform to accepted standard [D-2(28)].

5.4 Reception and Distribution of Main Supply (LV Connections)

5.4.1 *Control at Point of Commencement of Supply*

5.4.1.1 The supplier shall provide a suitable metering switchboard in which each meter units shall have its individual compartment housing meter, wiring, circuit breaker, etc within a consumer's premises, in an accessible position and such metering switchboard shall be contained within adequately enclosed fireproof receptacle. Where more than one consumer is supplied through a common service line, such consumer shall be provided with an independent metering unit in a switchboard housed in common area with an incoming breaker. Such meter switchboards should be housed in an enclosed and lockable space. Cables/wires from such switchboards connecting to individual consumers should terminate directly to consumer's distribution board without any interim termination or change of conductor. Such switchboards maybe located at ground floor or first basement, if building height is not exceeding 15 m. All metering panel/connection arrangements will have interconnection between circuit breakers, fuses, individual meters, etc should be using solid busbar instead of cables (where length of cables is less than 10 m).

However, for buildings above height of 15 m [Regulation 38 of the *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023] shall be having meters on individual floors. In case of commercial buildings/shopping malls/multi-use spaces where, combining of units maybe envisaged, meter switchboards should be placed on the floor where consumer unit is located (*see* Annex B).

In case of high rise buildings, the supplier or owner of the installation shall provide at the point of commencement of supply a suitable isolating device with cut-out or breaker to operate on all phases except neutral in the 3 phase, 4 wire circuit and fixed in a conspicuous position at not more than 1.7 m above the ground so as to completely isolate the supply to the building in case of emergency. In this connection, Regulation 38 of *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023 shall also be complied with (*see* Annex B).

The supplier in accordance with Regulation 18 of *Central Electricity Authority* (*Measures relating to Safety and Electric Supply*) *Regulations*, 2023 shall provide and maintain on the consumer's premises for the consumer's use, a suitable earth terminal in an accessible position or near the point of commencement of supply as per relevant standards (*see* Annex B).

No cut-out, link or switch other than a linked switch arranged to operate simultaneously on the earthed or earthed neutral conductor and live conductor shall be inserted or remain inserted in any earthed or earthed neutral conductor of a two wire-system or in any earthed or earthed neutral conductor of a multi-wire system or in any conductor connected thereto. This requirement shall however not apply in case of:

a) a link for testing purposes; or

b) a switch for use in controlling a generator or transformer.

In this connection, Regulation 17 (ii) of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 shall also be complied with (*see* Annex B).

The neutral shall also be distinctly marked.

5.4.1.2 The main switch shall be easily accessible and situated as near as practicable to the termination of service line.

5.4.1.3 Where the conductors include an earthed conductor of a two-wire system or an earthed neutral conductor of a multi-wire system or a conductor which is to be connected thereto, an indication of a permanent nature shall be provided for identification in accordance with Regulation 17 (i) of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 (*see* Annex B).

5.4.1.4 *Energy meters*

5.4.1.4.1 Energy meters conforming to accepted standards [D-2(29)] shall be installed in all buildings at such a place which is readily accessible to the owner/operator/occupant of the building and the Authority. Meters should not be located at an elevated area or a depressed area that does not have access by means of a stairway of normal rise. The height of meter display shall be between 750 mm and 1 800 mm. In case the meter is provided with a secondary display unit, this requirement applies to the secondary display unit only. A minimum clearance of 50 mm should be maintained around the meter itself for better inspection. This includes the space between two meters, between meter and the mounting box and between two mounting boxes as the case may be. The energy meters should either be provided with a protecting covering, enclosing it completely except the glass window through which the readings are noted or should be mounted inside a completely enclosed panel provided with hinged arrangement for locking. Additionally, for outdoor installations, the meters and associated accessories shall be protected by appropriate enclosure of level of protection IP 55 and ensuring compliance with above conditions.

In large multi-storeyed buildings, installation of a large number of energy meters at the ground floor (or first basement) switch-room for the convenience of the meter-reader poses high fire hazard. More than 24 energy meters on one switchboard is undesirable. In such cases, where number of energy meters to be installed for feeding exceeds 24, energy meters shall be installed at each floor and therefore, the rising main (bus trunking) with tapping point at individual floor shall be provided for meters.

The energy meters shall be protected by suitable circuit breaker. The provisions of **5.3.6.7** shall apply in case of energy meters installed in boards.

5.4.1.4.2 Main sources of energy, as given below shall be metered, as required at entry into the premises/ control panel:

a) utility grid points (high voltage/low voltage);

b) captive generator sets; and

c) on-site renewable energy system (if installed/operational) including net metering for grid connected system.

5.4.1.4.3 Testing, evaluation, installation and maintenance of energy meters shall be in accordance with the good practice [D-2(30)].

5.4.1.4.4 *Centralized metering system*

Smart metering and energy monitoring in a centralized metering system are used to monitor, measure and control the demand of electrical loads in a building. These systems are designed specifically for the control and monitoring of those facilities in a building which have significant electrical consumption, such as heating, ventilation, air conditioning, lifts, pumps and lighting installations at multiple locations in a campus. The scope may span from a single building to a group of buildings, such as residential apartments under common ownership, large multi-storeyed buildings, malls, university campuses, office buildings, retail store networks, factories, or any building with multi-tenanted occupancy. These systems provide metering, submetering and monitoring functions to allow facility and building managers to gather data and insight that allows them to make more informed decisions about demand management and demand control across their sites.

For such buildings with centralized metering, several main meters and sub-meters with following requirements should be provided:

a) Main meters should be digital energy meters with high accuracy, high sampling rates and power quality parameters, that is, harmonics, etc, for meters installed at incomer level;

b) Separate sub-meters should be provided for all energy end uses and functional areas that individually account for reasonable energy consumption in the building. These may include, but are not limited to, sub-meters for HVAC system; common area lighting, raw power, UPS, other common utility; lifts and escalators, pumps, external and internal lighting, individual units/flats/shops/offices; etc;

c) The sub-meters should be able to communicate data for monitoring. At a minimum, the sub-metering infrastructure should facilitate the aggregation of total energy use. For individual residential or commercial meters, if required it shall be controlled by the local electrical authority to take data for grid metering and later data be shared for emergency metering; and

d) Adequate smart metering and energy monitoring infrastructure should be installed in order to help monitor operational energy use and costs and to enable continuous energy performance improvement.

5.4.1.4.5 The *Central Electricity Authority* (*Installation and Operation of Meters*) *Regulations*, 2006, as amended from time-to-time shall also be complied with.

5.4.2 *Main Switches and Switchboard*

5.4.2.1 All main switches shall be either of metal-clad enclosed pattern or of any insulated enclosed pattern which shall be fixed at close proximity to the point of entry of supply. Every switch shall have suitable ingress protection level rating (IP), so that its operation is satisfactory and safe in the environment of the installation. All works under **5.4** shall comply with the provisions of relevant Indian Standards and good practice [D-2(1)].

NOTE — Woodwork shall not be used for the construction or mounting of switches and switch boards installed in a building.

5.4.2.2 *Location*

The main switchboard shall comply with the following requirements relating to its location:

a) The location of the main board should be such that it is easily accessible to fireman and other personnel to quickly disconnect the supply in case of emergencies. If the room is locked for security reasons, means of emergency access, by schemes such as break glass cupboard, shall be incorporated;

b) Main switch board shall be installed in rooms or fire safe cupboards so as to safeguard against operation by unauthorized personnel. Otherwise, the main switch board shall have lock and key facility for small installations in residences or other occupancies having sanctioned loads less than 5 kW;

c) Switchboards shall be placed only in dry situations and in ventilated rooms and they shall not be placed in the vicinity of storage batteries or exposed to chemical fumes;

d) In damp situation or where inflammable or explosive dust, gas or vapour is likely to be present, the switchboard shall be totally enclosed and shall have adequate degree of ingress protection (IP). In some cases, flameproof enclosure may be necessitated by particular circumstances {*see* accepted standard [D-2(31)]};

e) Switchboards shall not be erected above gas stoves or sinks, or within 2.5 m of any washing unit in the washing rooms or laundries, or in bathrooms, lavatories or toilets, or kitchens;

f) In case of switchboards unavoidably fixed in places likely to be located outdoor, exposed to weather, drip, or to abnormal moist temperature, the outer casing shall be weatherproof and shall be provided with glands or bushings or adopted to receive screwed conduit, according to the manner in which the cables are run. The casing as well as cable entries shall have suitable IP ratings according to the installation;

g) Adequate illumination shall be provided for all working spaces around the switchboards; and

h) Easy access to the enclosure around switchgear is essential to enable easy and safe operation and maintenance. The provisions as given in **5.3.6.7** including requirements for sufficient clearances shall be complied with.

5.4.2.3 Metal-clad switchgear shall be mounted on any of the following types of boards:

a) *Fixed-type metal boards* — These shall consist of an angle or channel iron frame fixed on the wall or on floor and supported on the wall at the top, if necessary;

   NOTE — Such type of boards is suitable for both small and large switchboards. They are particularly suitable for large switchboards for mounting number of switchgears or high capacity metal-clad switchgear or both in an arrangement which do not require rear access.

b) *Protected-type switchboards* — A protected switchboard is one where all of the switchgear and conductors are protected by metal or halogen free plastic enclosures. They may consist of a metal/plastic cubicle panel, or an iron frame upon which metal-clad switchgears are mounted. They usually consist of a main switch, busbars and circuit breakers or fuses controlling outgoing circuits; and

c) *Outdoor-type switchboards* — An outdoor-type switchboard is one which is totally enclosed and UV ray protected and having high ingress protection against dust and moisture and vermin-proof and high impact resistant (IP 55 or higher and IK 10). Such switchboards are of cubicle type and also provide high impact resistance. Cubicle type boards shall be with hinged doors interlocked with switch-operating mechanisms. The doors of these switchboards shall have facility to ensure that it is always in closed conditions. All such switches shall bear labels indicating their functions.

   NOTE — Such switchboards shall be located away from areas likely to be crowded by the public.

Open type switchboards, wherever existing in old buildings, shall be phased out and replaced with protected-type switchboards with suitable circuit breakers. Switchboards should be composite including incoming and outgoing circuit breakers, meters, cable alleys, bus-bars, accessories, etc.

5.4.2.4 *Recessing of boards*

Where so specified, the switchboards shall be recessed in the wall. Ample room shall be provided at the back for connection and at the front between the switchgear mountings.

5.4.2.5 *Marking of apparatus* {*see also* accepted standards [D-2(32)]}

Where a board is connected to voltage higher than 250 V, all the apparatus mounted on it shall be marked with the following colours to indicate the different poles or phases to which the apparatus or its different terminals may have been connected:

a) *Alternating current* (*three-phase*) *system*:

   Phase 1 – red, Phase 2 – yellow and Phase 3 – blue; and
   1 Neutral – black

b) *Direct current* (*three-wire system*):

   2 outer wire, Positive – red and Negative – blue; and
   1 Mid wire (Neutral) – black

Where four-wire three-phase wiring is done, the neutral shall be in one colour and the other three phase conductors in another colour (preferably red) or shall be suitably tagged or sleeved to ensure fool proof identification.

NOTE — Generally red colour identification is adopted for the phase conductors and black for neutral with additional tags or sleeves or coloured tapes at terminations.

Earth continuity conductor shall be marked with green colour or green with yellow line.

Where a board has more than one switchgear, each such switchgear shall be marked to indicate the section of the installation it controls. The main switchgear shall be marked as such. Where there is more than one main switchboard in the building, each such switchboard shall be marked to indicate the section of the installation and/or building it controls.

All markings shall be clear and permanent.

5.4.2.6 *Drawings*

Before proceeding with the actual construction, a proper drawing showing the detailed dimensions and design including the disposition of the mountings of the boards, which shall be symmetrically and neatly arranged for arriving at the overall dimensions, shall be prepared along the building drawing. Such drawings will show the mandatory clearance spaces if any, and clear height below the soffit of the beam required to satisfy regulations and safety considerations, so that other designers or installers do not get into such areas or spaces for their equipment.

5.4.3 *Distribution Boards*

A distribution board comprises of one or more protective devices against overcurrent and ensuring the distribution of electrical energy to the various circuits. Distribution board shall provide plenty of wiring space, to allow working as well as to allow keeping the extra length of connecting cables, likely to be required for maintenance.

5.4.3.1 Main distribution board shall be provided with a circuit breaker on each pole of each circuit, on the phase or live conductor and a link on the neutral or earthed conductor of each circuit. The switches shall always be linked.

All incomers should be provided with surge protection devices depending upon the current carrying capacity and fault level (*see* **11**). Surge protecting devices should be provided with backup circuit breaker/fuses, wherever required.

5.4.4 *Branch Distribution Boards*

5.4.4.1 Branch MCB distribution boards shall be provided, along with earth leakage protective device (RCCB/RCD) in the incoming, with a miniature circuit breaker of adequate rating/setting chosen on the live conductor of each sub-circuit and the neutral conductor shall be connected to a common link and be capable of being disconnected individually for testing purposes. At least one spare circuit of the same capacity shall be provided on each branch distribution board. Further, the individual branching circuits (outgoing) shall be protected against over-current with miniature circuit breaker of adequate rating. In residential/industrial lighting installations, the various circuits shall be separated and each circuit shall be individually protected so that in the event of fault, only the particular circuit gets disconnected. In order to provide protection against electric shock due to leakage current for human being, a 30 mA RCCB/RCD shall be installed at distribution board incomer of buildings, such as residential, schools and hospitals. For all other buildings, a 100 mA RCCB/RCD will suffice for protection against leakage current.

In case of phase segregated distribution boards, earth leakage protective device shall be provided in the sub-incomer to provide phase-wise earth fault protection. The provision of sub-incomer in distribution board shall be as per consumer requirement.

5.4.4.2 Common circuit shall be provided for installations at higher level (those in the ceiling and at higher levels, above 1 m, on the walls) and for installations at lower level but with separate switch control (sockets for portable or stationery plug in equipment). For devices consuming high power and which are to be supplied through supply cord and plug, separate wiring shall be done. For plug-in equipment provisions shall be made for providing RCCB/RCD protection in the distribution board. They shall be selected as per the type of load. The details of RCD classes are available in Part 1 'General and common aspects, Section 10 Protection for safety — Protection against electric shock' of good practice [D-2(1)].

5.4.4.3 It is preferable to have additional circuit for kitchen and bathrooms. Such sub-circuit shall not have more than a total of ten points of light, fans and 6 A socket-outlets. The load of such circuit shall be restricted to 800 W and the wiring with minimum 1.5 mm^2 copper conductor cable is recommended. If a dedicated circuit is planned for light fixtures, the load of such circuit shall be restricted to 800 W or 20 points whichever is less and the wiring with 1.5 mm^2 copper conductor cable is recommended. If a dedicated circuit is planned for 6 A sockets the load of such circuit shall be restricted to 800 W or a maximum of 8 numbers whichever is less, controlling MCB should be sized accordingly. The wiring shall be with 1.5 mm^2 copper conductor cable. If a separate fan circuit is provided, the number of fans in the circuit shall not exceed ten. Power sub-circuit shall be designed according to the load but in no case shall there be more than two 16A outlets on each sub-circuit which can be wired with 4 mm^2 copper conductor cable for miscellaneous socket loads and shall be with 4 mm^2 copper conductor cable for equipment consuming more than 1 kW. Power sockets complying with the accepted standards [D-2(33)] with current rated according to their starting load, wiring, MCB, etc, shall be designed for special equipment space heaters, air conditioners, heat pumps, VRF, etc.

For feeding final single phase domestic type of loads or general office loads it is advisable to introduce additional cables if required to allow lowering of short circuit rating of the switchgear required at user end so that use of hand held equipment, etc fed through flexible cords is safe.

5.4.4.4 The circuits for lighting of common area shall be separate. For large halls 3-wire control with individual control and master control installed near the entrance shall be provided for effective conservation of energy. Occupancy sensors, movement sensors, lux level sensors, etc, may also be considered as switching options for lights, fans, TV, etc, for different closed spaces [*see* also relevant guidelines/best practices (Handbook on 'Sustainability in Buildings')].

5.4.4.5 Where daylight is abundantly available, particularly in large halls, lighting in the area near the windows likely to receive daylight shall have separate controls for lights, so that they can be switched off/automatically reduce intensity selectively when daylight is adequate, while keeping the lights in the areas remote from the windows on [*see* also relevant guidelines/best practices (Handbook on 'Sustainability in Buildings')].

5.4.4.6 Circuits for socket-outlets may be kept separate from circuits feeding fans and lights. Normally, fans and lights may be wired on a common circuit. In large spaces, circuits for fans and lights may also be segregated. Lights may have group control in large halls and industrial areas. While providing group control, consideration may be given for the nature of use of the area lit by a group. Consideration has to be given for the daylight utilization, while grouping, so that a group feeding area near windows receiving daylight can be selectively switched off during daylight period.

5.4.4.7 The load on any low voltage sub-circuit shall not exceed 3 000 W. In case of a new installation, all circuits and sub-circuits shall be designed with an initial load of about 2 500 W, so as to allow a provision of 20 percent increase in load due to any future modification. Power sub-circuits shall be designed according to the load, where the circuit is meant for a specific equipment. Good practice is to limit a circuit to a maximum of three sockets, where it is expected that there will be diversity due to use of very few sockets in large spaces (example sockets for use of vacuum cleaner). General practice is to limit it to two sockets in a circuit, in both residential and non-residential buildings and to provide a single socket on a circuit for a known heavy load appliance such as air conditioner, cooking range, etc.

5.4.4.8 In wiring installations at special places like construction sites, stadia, shipyards, open yards in industrial plants, etc, where a large number of high wattage lamps may be required, there shall be no restriction of load on any circuit but conductors used in such circuits shall be of adequate size for the load and proper circuit protection shall be provided. The distribution boards (DBs) used in these areas shall be of UV resistant, double insulated type with IP 66 or higher degree of protection. Power tools and other temporary equipment connected to these DBs shall be sufficiently protected against electrical faults. Insulated IP 66 sockets complying with the accepted standards [D-2(28)] used in these DBs shall have interlocking facility in addition to protection to ensure safe plugging and unplugging of these equipment.

5.4.5 *Location of Distribution Boards*

a) The distribution boards shall be located as near as possible to the centre of the load they are intended to control;

b) These shall be fixed on suitable stanchion or wall and shall be accessible for replacement/reset of protective devices and shall not be more than 1.8 m from floor level;

c) These shall be of either metal-clad type, or polycarbonate enclosure of minimum IP 42. But, if exposed to weather or damp situations, these shall be of the weatherproof type conforming to IP 55 and, if installed where exposed to explosive dust, vapour or gas, these shall be of flameproof type in accordance with accepted standards [D-2(34)]. In corrosive atmospheres, these shall be treated with anti-corrosive preservative or covered with suitable plastic compound;

d) Where two and/or more distribution boards feeding low voltage circuits are fed from a supply of low voltage, the metal case shall be marked ‘Danger 415 Volts’ and identified with proper phase marking and danger marks;

e) Each shall be provided with a circuit list giving diagram of each circuit which it controls and the current rating of the circuit and capacity of MCB;

f) In wiring branch distribution board, total load of connecting devices shall be divided as far as possible evenly between the number of ways in the board leaving spare circuits for future extension;

g) Distribution board shall not be located at structural expansion joints of the building;

h) Distribution board/other electrical outlets shall have a minimum calculated separation distance from lightning protection down-conductors to avoid flash over in case of lightning; and

j) Walls with flushed distribution boards shall have adequate support behind and surrounding so that there is no physical weight of the distribution board on the civil structure around. Electrical switch sockets, etc shall also be avoided to be mounted behind the distribution board to avoid touching the board from behind.

5.4.6 *Protection of Circuits*

a) Appropriate protection shall be provided at switchboards, distribution boards and at all levels of panels for all circuits and sub-circuits against short circuit, over-current and other parameters as required. The protective device shall be capable of interrupting maximum short circuit current and the prospective earth fault current that may occur, without danger. The ratings and settings of the protective devices such as MCB shall be coordinated so as to afford selectivity in operation and in accordance with accepted standards [D-2(35)];

b) Where circuit-breakers are used for protection of a main circuit and of the sub-circuits derived therefrom, discrimination in operation may be achieved by adjusting the protective devices of the sub-main circuit-breakers to operate at lower current settings and shorter time-lag than the main circuit-breaker;

c) Where HRC type fuses are used for back-up protection of circuit-breakers, or where HRC fuses are used for protection of main circuits and circuit-breakers for the protection of sub-circuits derived there from, in the event of short-circuits protection exceeding the capacity of the circuit-breakers, the HRC fuses shall operate earlier than the circuit-breakers; but for smaller overloads within the short-circuit capacity of the circuit-breakers, the circuit-breakers shall operate earlier than the HRC fuse blows;

d) The current rating of a circuit breaker shall not exceed the current rating of the smallest cable in the circuit protected by it; and

e) All distribution board or panel incomer may be protected by a surge protection device, if found necessary (*see* **11**). Separate CB with proper enclosure may be required in series with the surge protection device with main incomer. Back-up CB shall be of the capacity not lower than that recommended by the SPD manufacturer. Short circuit withstand capability of the SPDs should be coordinated with the CB and SPD should be selected to be matching the fault power expected/calculated at that point.

5.4.7 *Cascading, Discrimination and Limitation*

Cascading and discrimination in switchgear downstream and upstream shall be designed and maintained such that the continuity of power in case of any abnormal conditions such as overload, short circuit and earth faults, etc is maintained and only faulty circuit is isolated and power is made available to other loads.

Cascading technique allows the designer to select circuit breakers of lower breaking capacities. Utilizing the current limiting effect of the incoming breaker, outgoing breaker can sustain the higher faults than its capacity and even maintain the discrimination. The provisions of good practice [D-2(1)] and relevant Indian Standards shall be complied with while carrying out necessary works under **5.4**.

5.5 Protection Class of Equipment and Accessories

The class of ingress protection (IP) and protection against mechanical impact (IK) {*see also* accepted standard [D-2(36)]} shall be specific depending on the requirement at the place of installation.

5.6 Voltage and Frequency of Supply

It should be ensured that all equipment connected to the system including any appliances to be used on it are suitable for the voltage and frequency of supply of the system. The nominal values of low systems in India are 240 V single phase and 415 V a. c. three phase, respectively, and the frequency is 50 Hz.

NOTE — The design of the wiring system and the sizes of the cables should be decided taking into account following factors:

a) *Voltage drop* — This should be kept below 6 percent to ensure proper functioning of all electrical appliances and equipment including motors.

b) Thermal limit based current carrying capacity of the cable with appropriate derating factors applicable to the installation conditions.

c) Capacity to withstand the let through fault current based on the fault level and the controlling switchgear disconnection characteristics.

5.7 Rating of Cables and Equipment

The rating of the cables and the equipment shall be chosen based on the relevant Indian Standards, good practice [D-2(1)] and accepted standard [D-2(37)] read with availability of the material in the market.

5.8 Installation Circuits

5.8.1 The nominal cross-sectional area of copper phase conductors in a. c. circuits and of live conductors in d. c.

circuits shall be not less than the values specified below:

Sl No.	*Type of Circuit*	*Minimum Copper Wire Size*	*Remarks*
(1)	(2)	(3)	(4)
i)	Lighting	1.5 mm^2	Per circuit for 2 or more circuits
ii)	Socket-outlets, 6 A	2.5 mm^2	Earthing of the 3rd pin shall be ensured
iii)	Socket-outlets, 16 A	2.5 mm^2	1
iv)	Water heater < 3 kW	2.5 mm^2	1
v)	Heaters or electric equipment more than or equal to 3 kW	4.0 mm^2	1
vi)	Free standing electric range Separate oven and/or cook top	4.0 mm^2	1
vii)	Air conditioner up to 2 T	4.0 mm^2	1
viii)	Permanently connected appliances including dishwashers, heaters, etc	2.5 mm^2	1 above 10 A. Up to 10 A can be wired as part of a socket-outlet circuit
ix)	Appliance rated > 3 kW < 6 kW	6.0 mm^2	–
x)	Submains to garage or out-building	2.5 mm^2	1 for each
xi)	Mains cable		It should be based on demand load/peaking loads and future loading

NOTE — The minimum conductor size in point wiring shall not be less than 1.5 mm^2 copper conductor cable for fixed (permanent) wiring insulations.

5.8.2 For conductor sizes less than or equal to 25 mm^2, only copper conductor cables shall be used since aluminium conductor cables in sizes less than 25 mm^2 may cause termination problems leading to heating at the terminals and enhance the possibility of a fire.

In case of very important public buildings and multi-storey buildings, for connections to meter board and connections from tap-off box to floor panel and beyond, use of copper conductor cable up to 50 mm^2 be considered to avoid frequent failure and fire accidents.

5.8.3 Switch or isolator controlling a water heater or geyser should not be located within 1 m from the location of a shower or bath tub, to avoid a person in wet condition reaching the switch or isolator. It is preferable to provide the control switch outside the bath room near the entrance and provide an indication at the water heater. A socket or a connector block with suitable protection against water spray should be provided to connect the water heater. The above considerations apply to switches for outdoor lights and other appliances, with the object of avoidance of operation of a switch when a person is wet.

5.8.4 Sockets in kitchen, bathroom, toilet, garage, etc, should not be provided within a height of 1 m from the ground level. Similar care has to be taken for installations involving fountains, swimming pools, etc. Light fittings in such areas should be fed at low voltage, preferably through an isolating transformer with a proper earth leakage protection. Where possibility of a person in contact with a wet surface has to operate or touch an electrical switch or an appliance the circuit should be protected by a 30 mA RCCB/RCD, as applicable.

5.8.5 *Selecting and Installing Cables*

5.8.5.1 *Cable types*

For the purpose of this NBCS, cables above 1 mm^2 shall have stranded conductors. All cables when installed, shall be adequately protected against mechanical damage. This can be carried out by either having additional protection, such as being enclosed in PVC conduit or metal pipes, or placing the cables in a suitable location that requires no additional protection. The cables for wiring circuits in electrical installation shall have the appropriate wire size matching the requirement of the loads. The following table gives the recommendations for different types of loads:

For the mains cable	Cross-linked polyethylene (XLPE) cable
For internal wiring	Class II copper conductor flame retardant, low smoke halogen (FRLSH) cable
Fire alarm and other related services	Halogen free flame retardant (HFFR)/Fire survival copper conductor cable
For main earth or main equipotential wire	Poly vinyl chloride (PVC) insulated copper wire
Underground installation and installation in cable trench, feeders between buildings, etc	PVC insulated, PVC sheathed armoured cables or XLPE insulated, PVC sheathed cables armoured cables
Installation in plant rooms, switch rooms, etc, on cable tray or ladder or protected trench, where risk of mechanical damage to cable does not exist	PVC insulated, PVC sheathed or XLPE insulated, PVC sheathed unarmoured cables

5.8.5.2 *Circuit wire sizes*

Recommended minimum wire sizes for various circuits is given below:

Sl No.	*Circuits*	*Minimum Size of Copper Conductor Cable*	*Wire Colour*
(1)	(2)	(3)	(4)
i)	1-way lighting	2 + E cable of 1.5 mm^2	Red-black and green-yellow
ii)	2-way lighting control	3 wire cable of 1.5 mm^2	Red-white-blue
iii)	Storage water heaters up to 3 kW	2 + E cable of 2.5 mm^2 (stranded conductors)	Red-black and green-yellow
iv)	Storage water heaters between 3 kW and 6 kW	2 + E cable of 4 mm^2 (stranded conductors)	Red-black and green-yellow
v)	Socket-outlets and permanent connection units	2 + E cable of 2.5 mm^2 (stranded conductors)	Red-black and green-yellow

Table (*Concluded*)

Sl No.	*Circuits*	*Minimum Size of Copper Conductor Cable*	*Wire Colour*
(1)	(2)	(3)	(4)
vi)	Submains to garages or out buildings	2 + E cable 2.5 mm^2 (stranded conductors)	Red-black and green-yellow
vii)	Cooking hobs	2 + E cable 1.5 mm^2	Red-black and green-yellow
viii)	Separate ovens	2 + E cable 4 mm^2 (stranded conductors)	Red-black and green-yellow
ix)	Electric range	2 + E cable 6 mm^2 (stranded conductors)	Red-black and green-yellow
x)	Mains	2 wire cable Upto 25 mm^2 (stranded conductors)	Red-black
xi)	Main equipotential bonding wire	4 mm^2 (stranded conductors)	Green-yellow
xii)	Main earth wire	6 mm^2 (stranded conductors)	Green-yellow

NOTES

1 2 + E cable is also known as twin and earth wire cable (two wires and earth).

2 Earth wire can be as per the following:

a) Green-Yellow throughout their length with, in addition, light blue markings at the terminations; or

b) Light blue throughout their length with, in addition, green/yellow markings at the terminations.

3 The above sizes are recommendatory and shall be modified as per voltage drop, starting current, distance from DB, etc.

5.8.6 *Requirements for Physical Protection of Underground Cables*

Sl No.	*Protective Element*	*Specifications*
(1)	(2)	(3)
i)	Bricks	a) 100 mm minimum width b) 25 mm thick c) sand cushioning 100 mm and sand cover 100 mm
ii)	Concrete slabs	At least 50 mm thick
iii)	Plastic slabs (polymeric cover strips) fibre reinforced plastic	At least 10 mm thick, depending on properties and has to be matched with the protective cushioning and cover
iv)	PVC conduit or PVC pipe or stoneware pipe or hume pipe	The pipe diameter should be such so that the cable is able to easily slip down the pipe
v)	Galvanized pipe	The pipe diameter should be such so that the cable is able to easily slip down the pipe

The trench shall be backfilled to cover the cable initially by 200 mm of sand fill; and then a plastic marker strip shall be put over the full length of cable in the trench. The marker signs shall be provided where any cable enters or leaves a building. This will identify that there is a cable located underground near the building. The trench shall then be completely filled. If the cables rise above ground to enter a building or other structure, a mechanical protection such as a GI pipe or PVC pipe for the cable from the trench depth to a height of 2.0 m above ground shall be provided.

5.9 Lighting and Levels of Illumination

5.9.1 *General*

Lighting installation shall take into consideration the numerous factors on which the quality and quantity of artificial lighting depends. Recent practice in illumination is to provide the required illumination with a large number of light sources (not of higher illumination level) instead of fewer number of light sources of higher illumination level, to produce higher uniformity in illumination level.

With the increase in energy costs and awareness of the need to conserve energy for the protection of the environment, there has been a drastic change in the use of light sources during the last 10 years. Fluorescent lamps [tubular lamp (TL) and compact fluorescent lamp (CFL)] are no longer available in the market. Use of general lighting service (GLS) lamps, known as light bulbs, has almost become extinct. The space vacated by these light sources has been taken over by LED lamps in different configurations in the form of LED based bulbs, tube lights, ceiling flush light sources and floodlights with different wattages.

With this the lighting design has become more complex. With the developments in the types of light sources and their control systems now available, lighting design goes with the concept of better light with less energy and least impact on environment.

Automatic lighting control schemes may be considered to have efficient utilization of lights. Automatic controls can take care of the switching off when the space served has no activity or is illuminated by daylight.

While carrying out lighting design and layout reference shall also be made to good practice [D-2(38)].

5.9.2 *Electrical Installations for Lighting*

The concepts or needs of energy conservation today require more lights to be provided so that different sets of lights are used to light up the area of activity to the required higher level of lighting needed for the activity and provide a general minimum background level of lighting. Any space requires two or three different combination of lighting sets associated with the activity and this may require the wiring to be provided to accommodate the lighting groups in different circuits with group controls, automatic controls and remote controls.

Availability of LED lights with a wide range from 1 W to 100 W allows designers to provide spot task lighting of a high illumination level combined with a general space lighting of low illumination. As light follows the inverse square law, provision of the light source close to the task reduces the energy need.

Lighting demand for buildings should be considered as per type of building. Where nothing is specified, for lighting demand of any type of building a maximum of 13 W/m^2 of all built-up areas including balconies. Covered parking areas may be considered at 3.23 W/m^2 including balconies is to be considered. Service areas, corridors, etc, may be considered with very basic diversity of 80 percent to 100 percent. Power requirements shall be considered at least 55 W/m^2 with an overall diversity not exceeding 50 percent. These shall be excluding defined loads such as lifts, plumbing system, firefighting systems, ventilation requirement, etc.

Light emitting diode (LED) lamps have non-linear characteristics and require specifically made controlgear for each type of lamp for their proper operation. In some cases the controlgear is integral with the lamp, in some with light fitting and in some it is external. The electrical installation and wiring has to take this into account and provide appropriate space for such controlgear. There will be heat emission, introduction of harmonics, etc, and they also consume some energy. The electrical and lighting system design has to keep this aspect in the wiring design and installation.

5.9.3 *Principles of Lighting*

When considering the function of artificial lighting, attention shall be given to the following principle characteristics before designing an installation:

a) Illumination and its uniformity.

b) Special distribution of light. This includes a reference to the composition of diffused and directional light, direction of incidence, the distribution of luminances and the degree of glare.

c) Colour of the light and colour rendition.

d) Natural light sources, if possible such as light tubes.

e) System wattage of the luminaire proposed.

5.9.4 The variety of purposes which have to be kept in mind while planning the lighting installation may be broadly grouped as:

a) industrial buildings and processes;

b) offices, schools and public buildings;

c) surgeries and hospitals; and

d) hostels, restaurants, shops and residential buildings.

5.9.4.1 It is important that appropriate levels of illumination for these and the types and positions of fittings determined to suit the task and the disposition of the working planes are ensured.

5.9.5 For detailed requirements for lighting and lighting design and installations, reference shall be made to the good practice [D-2(38)] and provisions in Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS.

5.10 In locations where the system voltage exceeds 415 V three phase, as in the case of industrial locations, for details of design and construction of wiring installation, reference may be made to good practice [D-2(18)].

5.11 Guideline for Electrical Layout in Residential Buildings

For guidelines for electrical installation in residential buildings, reference may be made to accepted standard [D-2(39)] and good practice [D-2(1)].

A typical distribution scheme in a residential building with separate circuits for lights and fans and for power appliances is given in Fig. 3.

5.12 For detailed information regarding the installation of different electrical equipment, reference may be made to good practice [D-2(40)].

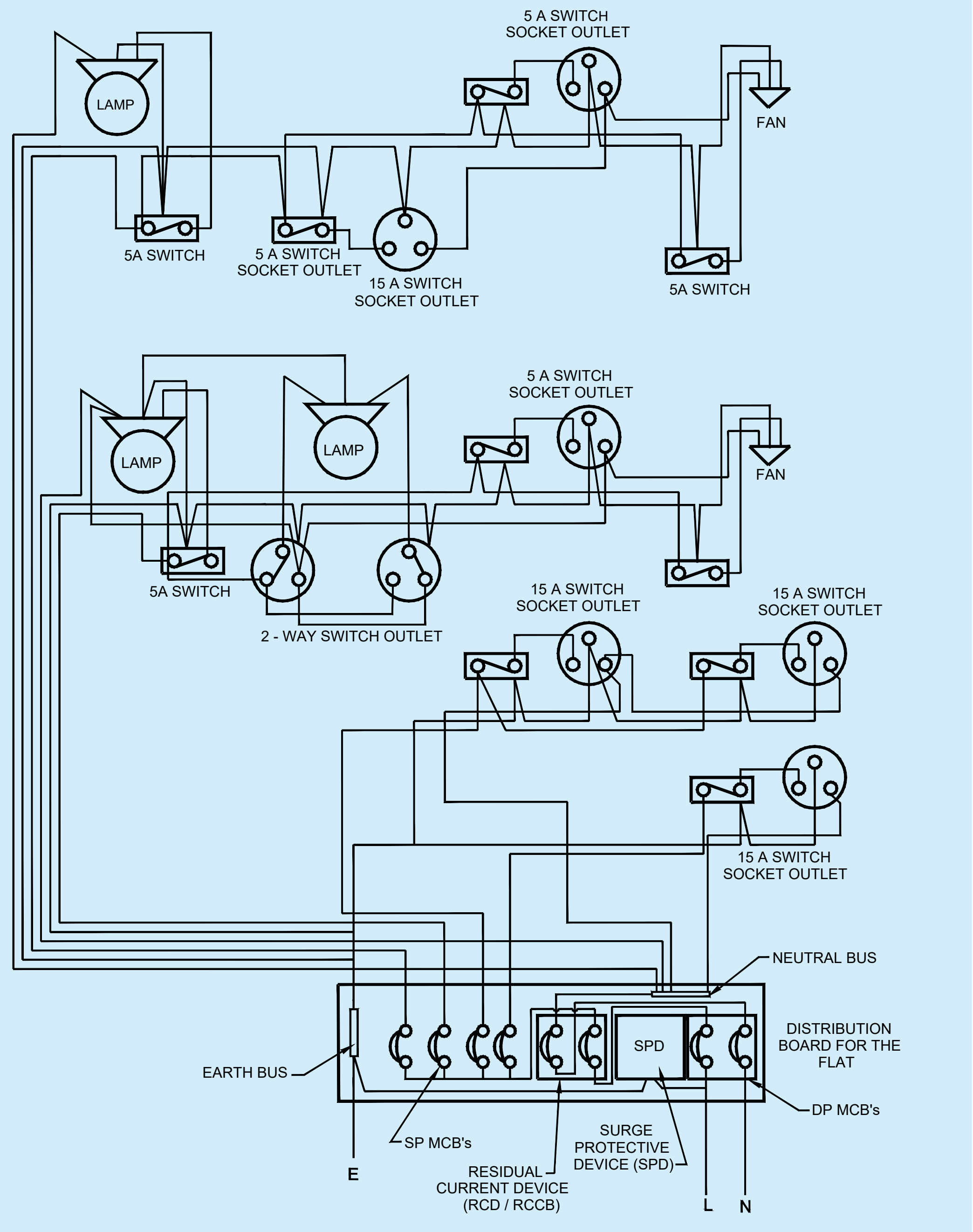


FIG. 3 WIRING DIAGRAM FOR A TYPICAL DISTRIBUTION SCHEME IN A RESIDENTIAL BUILDING FLAT

6 WIRING

6.1 General

The generic guidelines for wiring in buildings depending on the type of occupancy, local conditions, material selection and other relevant points are covered hereunder. Also, the construction requirements, safety aspects to be considered during construction and maintenance aspects which should be considered during construction are addressed. All works under this shall also comply with the relevant Indian Standards and provisions in good practice [D-2(1)].

6.1.1 *Estimation of Load Requirements*

All conductors, switches and accessories shall be of such size as to be capable of carrying, without their respective ratings being exceeded, the maximum current which will normally flow through them.

In estimating the current to be carried by any conductor the following ratings shall be taken, unless the actual values are known or specified for these elements:

Sl No. (1)	*Element* (2)	*Rating*, W (3)
i)	Light point	60
ii)	Fan point	75
iii)	6 A 3/5 pins socket-outlet	100
iv)	16 A 3/6 pins socket-outlet	2 000
v)	High pressure mercury vapour (HPMV) lamps, high pressure sodium vapour (HPSV) lamps	According to their capacity, controlgear losses shall be also considered as applicable
vi)	Geyser (storage type)	2 000
vii)	Geyser (instant)	3 000
viii)	Computer point	150
ix)	Printer, laser	1 500
x)	Air conditioner:	
	a) Up to 1.5 TR	2 000
	b) 1.5 to 2 TR	3 000

6.1.2 *Commencement of Electrical Work*

Electrical installation in a new building shall normally begin with conduit laying at the commencement of preparations for first roof concreting work. The necessary conduits and ducts shall be positioned firmly by tying the conduit to the reinforcement before concreting in an RCC building. Location of fan points should be properly marked and fan clamp boxes should be provided along with conduit laying in roof. Care should be taken to avoid use of damaged conduit or ducts, the conduits ends shall be given suitable anti-corrosive treatment and holes blocked off by putties or caps to protect conduits from getting blocked. All conduit openings and junction box openings, etc shall be properly protected against entry of mortar, concrete, etc during construction. Laying of conduits in walls shall start with the taking up of wall construction work. In case of surface wiring, the work will start after completion of interior plaster including curing. The wiring, fixing of switch boxes and fixing of switches will start only after ensuring necessary weatherproof area with locking arrangement is available to take up electrical work.

6.2 Selection of Size of Conductors

The size of conductors for circuits shall be selected after fixing the location of main sub-station, main switchboard for the building, location of distribution board and the electrical points under the planning of electrical services. The overall voltage drop from the transformer end to consumer's final distribution board shall not exceed six percent. The voltage drop of six percent will be divided between different points indicated above in such a way that from the distribution board to the last point shall be three percent. From main board to distribution board, from distribution board to sub distribution board if any, circuit wiring from DB/SDB to switch board and from switch board to light/fan points/plug points shall be with Class II category copper conductors. The size of the conductor cable shall be decided based on the maximum load on each point. In main/submain/circuit/sub-circuit, the circuit-breaker shall be selected to match the current rating of the circuit to ensure the desired protection.

6.3 Branch Switches

Where the supply is derived from a three-wire or four-wire source, and distribution is done on the two-wire system, all branch switches shall be placed in the outer or live conductor of the circuit and no single phase switch or protective device shall be inserted in the middle wire, earth or earthed neutral conductor of the circuit. Single-pole switches (other than for multiple control) carrying not more than 16 A may be of flush/modular type. For controlling points for air conditioners and geysers, suitable MCB type switches shall be provided.

6.4 Layout and Installation Drawing

6.4.1 The electrical layout should be drawn indicating properly the locations of all outlets, such as lamps, fans, appliances (both fixed and movable), and motors and best route for wiring.

6.4.2 All runs of wiring and the exact positions of all points of switch-boxes and other outlets shall be first marked on the plans of the building and approved by the engineer-in-charge or the owner before actual commencement of the work.

6.4.3 Industrial layout drawings should indicate the relative civil and mechanical details.

6.4.4 *Layout of Wiring*

The layout of wiring should be designed keeping in view disposition of the lighting system to meet the illumination levels. All wirings shall be done on the distribution system with main and branch distribution boards at convenient physical and electrical load centres. All types of wiring, whether concealed or unconcealed, should be as near the ceiling as possible. In all types of wirings due consideration shall be given for neatness and good appearance.

6.4.5 Balancing of circuits in three-wire or poly-phase installation shall be arranged beforehand. Proper balancing can be done only under actual load conditions. Conductors shall be so enclosed in earthed metal or incombustible insulating material that it is not possible to have ready access to them. Means of access shall be marked to indicate the voltage present.

Where terminals or other fixed live parts between which a voltage exceeding 250 V exists are housed in separate enclosures or items of apparatus which, although separated are within reach of each other, a notice shall be placed in such a position that anyone gaining access to live parts is warned of the magnitude of the voltage that exists between them.

Where loads are single phase, balancing should be for the peak load condition based on equipment usage. Facility for change should be built into the distribution design.

NOTE — The above requirements apply equally to three-phase circuits in which the voltage between lines or to earth exceeds 250 V and to groups of two or more single-phase circuits, between which medium voltage may be present, derived therefrom. They apply also to 3 wire d.c. or 3 wire single-phase a.c. circuits in which the voltage between lines or to earth exceeds 250 V and to groups of 2 wire circuits, between which medium voltage may be present, derived there from.

6.4.6 Low voltage wiring and associated apparatus shall comply, in all respects, with the requirements of Regulation 35, 36, 37, 39, 40, 41 and 42 of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023, as amended from time to time (*see* Annex B).

6.5 Conductors and Accessories

6.5.1 *Conductors*

Conductors for all the internal wiring shall be of copper. Conductors for power and lighting circuits shall be of adequate size to carry the designed circuit load without exceeding the permissible thermal limits for the insulation. For final selection of wiring to larger loads, the current carrying capacity will preside. The conductor size shall also be based on the voltage drop in the line so as to provide a terminal voltage not below the prescribed voltage requirement.

The conductor for final sub-circuit for fan and light wiring shall have a nominal cross-sectional area not less than 1.50 mm^2 copper. The cross-sectional area of conductor for power wiring shall be not less than 2.5 mm^2 copper. The minimum cross-sectional area of conductor of flexible cord shall be 1.50 mm^2 copper. All copper conductors used in wiring shall be of class II category for permanent wiring system.

In existing buildings where aluminum wiring has been used for internal electrification, changeover from aluminum conductor cables to copper conductor cables is recommended as it has been found that aluminum conductors pose a number of hazards.

NOTE — It is advisable to replace wiring which is more than 30 years old as the insulation also would have deteriorated, and will be in a state to cause failure on the slightest of mechanical or electrical disturbance.

6.5.2 *Flexible Cables and Flexible Cords*

Flexible cables and cords shall be of copper and stranded, and protected by flexible conduits or tough rubber or PVC sheath to prevent mechanical damage.

6.5.3 *Cable Ends*

When a stranded conductor having a nominal sectional area less than 6 mm^2 is not provided with cable sockets (thimbles/lugs) for end termination, all strands at the exposed ends of the cable shall be soldered together or crimped using suitable sleeve or ferrules.

6.5.4 *Special Risk*

Special forms of construction, such as flameproof enclosures, shall be adopted where there is risk of fire or explosion.

6.5.5 *Connection to Ancillary Buildings*

Unless otherwise specified, electric connections to ancillary buildings, such as out-houses, garages, etc adjacent to the main building and when no roadway intervenes shall be taken in an earthed GI pipe or heavy duty PVC or HDPE pipe of suitable size. This pipe can be taken either underground or over ground, however, in latter case, its height from the ground shall not be less than 5.8 m. This applies to both runs of mains or sub-mains or final sub-circuit wiring between the buildings.

6.5.6 *Expansion Joints*

Distribution boards shall be so located that the conduits shall not normally be required to cross expansion joints in a building. Where such crossing is found to be unavoidable, special care shall be taken to ensure that the conduit runs and wiring are not in any way put to strain or damaged due to expansion of building structure. Anyone of the standard methods of connection at a structural expansion joint shall be followed:

a) Flexible conduit shall be inserted at place of expansion joint;

b) Oversized conduit overlapping the conduit; and

c) Expansion box.

Supports and flexible joints shall be of same requirement as the rising main/bus duct in so far as resistance to seismic forces is concerned. This is further important when rising mains and bus-ducts cross expansion joints.

6.5.7 *Extra Low Voltage* (*Types of Wires/Cables*)

Extra low voltage services utilize various categories of cables/wires, such as fibre optic cable, co-axial cable, etc. These shall be laid at least at a distance of 300 mm from any power wire or cable. The distance may be reduced only by using completely closed earthed metal trunking with metal separations for various kind of cable. Special care shall be taken to ensure that the conduit runs and wiring are laid properly for low voltage signal to flow through it.

The power cable and the signal or data cable may run together under floor and near the equipment. However, separation may be required from the insulation aspect, if the signal cable is running close to a conductor carrying power at low voltage. All types of signal cables are required to have insulation level for withstanding 2 kV impulse voltage even if they are meant for service at low voltage.

6.6 Passing Through Walls, Floors and Expansion Joints

6.6.1 Where wires/cables are required to pass through walls, care shall be taken to see that wires/cables pass freely through protective pipe or box and that the wires pass through in a straight line without any twist or cross in wires. In case of wires crossing expansion joints in a recessed conduit system there shall be two loop wire boxes on two sides of the expansion joint which shall be connected with concentric pipes of different diameter in such a way that one box will hold the bigger diameter conduit while the other box will hold the smaller diameter conduit pipe. This way, the two sides can move freely as per structure without having any effect either on the structure or on the wiring system.

One of the following methods shall be employed for laying wires/cables:

a) *Conduit wiring system* (*see* **6.9**) — The conductor shall be drawn in either in a rigid steel conduit or a rigid non-metallic conduit conforming to relevant Indian Standards {*see* accepted standards [D-2(41)] and [D-2(42)]}. All conduits used for fire alarm and other fire related emergency relief services as well as security and UPS system shall only be of metallic conduit system.

   The conduits shall be colour coded as per the purpose of wire carried in the same. The recommended colour coding may be in form of bands of colour (100 mm thick, with centre to centre distance of 300 mm) or coloured throughout. The colour scheme may be as follows:

Sl No.	*Conduit Type*	*Colour Scheme*
(1)	(2)	(3)
i)	Power conduit	Black
ii)	Security conduit	Blue
iii)	Fire alarm conduit	Red
iv)	Extra low voltage conduit	Brown
v)	UPS conduit	Green

   Conduit wiring system shall comply with relevant Indian Standards. The number of insulated conductors that can be drawn into rigid conduit is given in Table 1 (*see* Table 1A for rigid steel conduits and Table 1B for rigid non-metallic conduits). As far as possible, cables of one circuit shall be housed in one conduit only.

b) *Cable trunking/cable ways/race ways* (*see* **6.10**) — Cable trunking/cable ways/race ways system should be used when number of wires/small cable sizes to be laid is more than the conduit capacity. Care should be taken to have space in the trunking system to minimize heating of wires and to provide identification of the different circuits. These can be laid in both floor and ceiling. Adequate junction boxes, access points shall be provided to pull in and out wire.

   Cable trunking or ducting system shall comply with accepted standards [D-2(43)].

c) *Tray and ladder rack* — As tray provides continuous support, unless mounted on edge or in vertical runs (when adequate strapping or clipping is essential), the mechanical strength of supported cable is not as important as with ladder-racking or structural support methods. Consequently, tray is eminently suitable for the smaller unarmoured cabling while ladder racks call for armoured cables or larger unarmoured cables as they provide the necessary strength to avoid sagging between supports. Both tray and ladder racks are provided with accessories to facilitate changes of route, and they provide no difficulty in this respect on vertical runs.

Cable tray/ladder racks and support systems shall be installed in such a way that the deflection between the spans shall be less than 1 percent of the span.

Power cables running in cable ladders both horizontal and vertical shall be fixed with proper clamps which can withstand the mechanical force created on the cable in case of short circuit current. The complete installation consisting of cables, ladders, clamps, ladder supports and fixtures shall also withstand the mechanical force of short circuit current. Only one layer of power cable shall be laid in a ladder. The minimum space between two cables shall be equal to the diameter of the biggest cable.

Cable tray and ladder system shall comply with accepted standard [D-2(44)].

6.6.2 Insulated conductors while passing through floors shall be protected from mechanical injury by means of rigid steel/non-metal conduit or by mechanical protection up to a height not less than 1.5 m above the floor and flush with the ceiling below. These steel conduits shall be earthed and securely bushed.

Power outlets and wiring in the floor shall be generally avoided. If not avoidable, false floor trunking or metal floor trunking should be used. Power sockets of adequate IP/IK rating shall be used.

False floor shall be provided where density of equipment and interconnection between different pieces of equipment is high. Examples are, mainframe computer station, telecommunication switch rooms, etc. Floor trunking shall be used in large halls, convention centres, open plan offices, laboratory, etc.

In case of floor trunking, drain points/weep holes shall be provided within the trunking for installation for convention centres, exhibition areas or any other place where water may spill into the trunking, as there might be possibility of water seepage in the case of wiring passing through the floors. Proper care should be taken for providing suitable means of draining of water. Possibility of water entry exists from floor washing, condensation in some particular weather and indoor temperature conditions. At the design stage, these aspects shall be assessed and an appropriate means of avoiding or reducing ingress of water and draining method shall be built in. Floor trunking outlets shall be suitably IP rated for protection against dust and water. Floor trunking shall be of adequate IK rating. Floor or ceiling trunking shall have a maximum of 40 percent fill.

Floor outlet boxes are generally provided for the use of appliances, which require a signal or communication connection. The floor box and trunking system should cater to serve both power distribution and the signal distribution, with appropriate safety and non-interference.

6.6.3 Where a wall tube passes outside a building and be exposed to weather, the outer end shall be bell-mouthed and turned downwards and properly bushed at the open end.

6.7 Wiring of Distribution Boards

6.7.1 All connections between parts of apparatus or between apparatus and terminals on a board shall be neatly arranged in a definite sequence, following the arrangements of the apparatus mounted thereon, avoiding unnecessary crossings.

6.7.2 Cables shall be connected to a terminal only by soldered or welded or crimped lugs using suitable sleeve, lugs or ferrules unless the terminal is of such a form that it is possible to securely clamp them without the cutting away of cables strands. Cables in each circuit shall be bunched together.

6.7.3 All bare conductors shall be rigidly fixed in such a manner that a clearance of at least 25 mm is maintained between conductors or opposite polarity or phase and between the conductors and any material other than insulation material.

6.7.4 If required, a pilot lamp shall be fixed and connected through an independent single pole switch and fuse to the busbars of the board. Leads connecting bus-bars to any instrument or indicating lamp or an outgoing connection switch or breaker face the same fault current that is applicable to the busbar and as such should be provided with a fuse capable of handling the prospective fault current.

6.7.5 In a hinged type board, the incoming and outgoing cables shall be fixed at one or more points according to the number of cables on the back of the board leaving suitable space in between cables and shall also, if possible, be fixed at the corresponding points on the switchboard panel. The cables between these points shall be of such length as to allow the switchboard panel to swing through an angle of not less than 90° and cables arranged and clamped in such a manner that the cables do not face bending, but only face a twist, when the hinged door is opened. The circuit breakers in such cases shall be accessible without opening the door of distribution board. Circuit breakers or any other equipment (having cable size more than 1.5 mm^2 multi-strand wire) shall also not be mounted on the door.

NOTE — Use of hinged type boards is discouraged, as these boards lead to deterioration of the cables in the hinged portion, leading to failures or even fire.

6.7.6 Wires terminating and originating from the protective devices shall be properly lugged and taped.

6.8 PVC-Sheathed Wiring System

6.8.1 *General*

Wiring with PVC-sheathed cables may be used for temporary installations for medium voltage installation and may be installed directly under exposed conditions of sun and rain or damp places.

6.8.2 *PVC Clamps/PVC Channel*

The clamps shall be used for temporary installations of 1 to 3 sheathed wires only. The clamps shall be fixed on wall at intervals of 100 mm in the case of horizontal runs and 150 mm in the case of vertical runs.

PVC channel shall be used for temporary installations in case of more than 3 wires or unsheathed wires. The channel shall be clamped on wall at intervals not exceeding 300 mm. PVC clamps/PVC channel shall conform to accepted standards.

6.8.3 *Protection of PVC-Sheathed Wiring from Mechanical Damage*

a) In cases where there are chances of any damage to the wirings, such wirings shall be covered with sheet metal protective covering, the base of which is made flush with the plaster or brickwork, as the case may be, or the wiring shall be drawn through a conduit complying with all requirements of conduit wiring system (*see* **6.9**); and

b) Such protective coverings shall in all cases be fitted on all down-drops within 1.5 m from the floor.

6.8.4 *Bends in Wiring*

The wiring shall not in any circumstances be bent so as to form a right angle but shall be rounded off at the corners to a radius not less than six times the overall diameter of the cable.

6.8.5 *Passing Through Floors*

All cables taken through floors shall be enclosed in an insulated heavy gauge steel conduit extending 1.5 m above the floor and flush with the ceiling below, or by means of any other approved type of metallic covering. The ends of all conduits or pipes shall be neatly bushed with porcelain, wood or other approved material.

6.8.6 *Passing Through Walls*

The method to be adopted shall be according to good practice. There shall be one or more conduits of adequate size to carry the conductors [*see* **6.9.1**(a)]. The conduits shall be neatly arranged so that the cables enter them straight without bending.

6.8.7 *Stripping of Outer Covering*

While cutting and stripping of the outer covering of the cables, care shall be taken that the sharp edge of the cutting instrument does not touch the rubber or PVC-sheathed insulation of conductors. The protective outer covering of the cables shall be stripped off near connecting terminals, and this protective covering shall be maintained up to the close proximity of connecting terminals as far as practicable. Care shall be taken to avoid hammering on link clips with any metal instruments after the cables are laid. Where junction boxes are provided, they shall be made moisture-proof with an approved plastic compound.

6.8.8 *Painting*

If so required, the tough rubber-sheathed wiring shall, after erection, be painted with one coat of oil-less paint or distemper of suitable colour over a coat of oil-less primer, and the PVC-sheathed wiring shall be painted with a synthetic enamel paint of quick drying type.

6.9 Conduit Wiring System

Conduit wiring system shall comply with accepted standards [D-2(45)]. Requirements relating to conduit wiring system with rigid steel and non-metallic conduits shall be as per **6.9.1** to **6.9.3**.

6.9.1 *Surface Conduit Wiring System with Rigid Steel Conduits*

a) *Type and size of conduit* — All conduit pipes shall conform to accepted standards [D-2(46)], finished with galvanized or stove enamelled surface. All conduit accessories shall be of threaded type and under no circumstance pin grip type or clamp type accessories shall be used. No steel conduit less than 16 mm in diameter shall be used. The number of insulated conductors that can be drawn into rigid steel conduit is given in Table 1A.

b) *Bunching of cables* — Unless otherwise specified, insulated conductors of a.c. supply and d.c. supply shall be bunched in separate conduits. For lighting and small power outlet circuits phase segregation in separate conduits is recommended.

c) *Conduit joints* — Conduit pipes shall be joined by means of screwed couplers and screwed accessories only {*see* accepted standard [D-2(47)]}. In long distance straight runs of conduit, inspection type couplers at reasonable intervals shall be provided or running threads with couplers and jam-nuts (in the latter case the bare threaded portion shall be treated with anti-corrosive preservative) shall be provided. Threaded portion on conduit pipes in all cases shall be between 11 mm and 27 mm long sufficient to accommodate pipes to full threaded portion of couplers or accessories. Cut ends of conduit pipes shall have no sharp edges or any burrs left to avoid damage to the insulation of conductors while pulling them through such pipes.

d) *Protection against dampness* — In order to minimize condensation or sweating inside the tube, all outlets of conduit system shall be properly drained and ventilated, but in such a manner as to prevent the entry of insects as far as possible.

e) *Protection of conduit against rust* — The outer surface of the conduit pipes, including all bends, unions, tees in the conduit system shall be adequately protected against rust particularly when such system is exposed to weather. In all cases, no bare threaded portion of conduit pipe shall be allowed unless such bare threaded portion is treated with anti-corrosive preservative or covered with suitable plastic compound.

f) *Fixing of conduit* — Conduit pipes shall be fixed by heavy gauge saddles, secured to suitable wood plugs or other plugs with screws in an approved manner at an interval of not more than 1 m, but on either side of couplers or bends or similar fittings, saddles shall be fixed at a distance of 300 mm from the centre of such fittings. Conduit fittings shall be avoided as far as possible on conduit system exposed to weather; where necessary, solid type fittings shall be used.

g) *Bends in conduit* — All necessary bends in the system including diversion shall be done by bending pipes; or by inserting suitable solid or inspection type normal bends, elbows or similar fittings; or fixing cast iron, thermoplastic or thermosetting plastic material inspection boxes; whichever is more suitable. Radius of such bends in conduit pipes shall be not less than 75 mm. No length of conduit shall have more than the equivalent of four quarter bends from outlet to outlet, the bends at the outlets not being counted.

h) *Outlets* — All outlets for fittings, switches, etc, shall be boxes of suitable metal or any other approved outlet boxes for either surface mounting or flush mounting system.

j) *Conductors* — All conductors used in conduit wiring shall preferably be stranded. No single-core cable of nominal cross-sectional area greater than 130 mm^2 shall be enclosed along in a conduit and used for alternating current.

k) *Erection and earthing of conduit* — The conduit of each circuit or section shall be completed before conductors are drawn in. The entire system of conduit after erection shall be tested for continuity using a fish wire and permanently connected to earth conforming to the requirements. Gas or water pipes shall not be used as earth medium. If conduit pipes are liable to mechanical damage they shall be adequately protected.

m) Inspection type conduit fittings, such as inspection boxes, draw boxes, bends, elbows and tees shall be so installed that they can remain accessible for such purposes withdrawal of existing cables or of new cables.

Table 1A Maximum Permissible Number of Single-Core Copper Cables up to and Including 1 100 V that can be Drawn into Rigid Steel Conduits

[*Clauses* 6.6.1(a) *and* 6.9.1(a)]

Sl No.	Size of Cable	Size of Conduit mm											
	Nominal Cross-Sectional Area mm^2	20		25		32		40		50		63	
		Number of Cables, *Max*											
		S	B	S	B	S	B	S	B	S	B	S	B
(1)	(2)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
i)	1.5	7	5	12	10	20	14	–	–	–		–	–
ii)	2.5	6	5	10	8	18	12	–	–	–	–	–	–
iii)	4	4	3	7	5	12	10	–	–	–	–	–	–
iv)	6	3	2	6	5	10	8	–	–	–	–	–	–
v)	10	2	–	4	3	6	5	8	6	–	–	–	–
vi)	16	–	–	2	–	4	3	7	6	–	–	–	–
vii)	25	–	–	–	–	3	2	5	4	8	6	9	7
viii)	35	–	–	–	–	2	–	4	3	7	5	8	6

NOTES

1 The table shows the maximum capacity of conduits for the simultaneously drawing of cables. The columns headed S apply to runs of conduit which have distance not exceeding 4.25 m between draw-in boxes and which do not deflect from the straight by an angle of more than 15. The columns headed B apply to runs of conduit which deflect from the straight by an angle of more than 15.

2 In case an inspection type draw-in box has been provided and if the cable is first drawn through one straight conduit, then through the draw-in box and then through the second straight conduit, such systems may be considered as that of a straight conduit even if the conduit deflects through the straight by more than 15.

3 Conductor sizes for cables and wires above and including 2.5 mm^2 core size shall be multi-stranded.

6.9.2 *Recessed Conduit Wiring System with Rigid Steel Conduit*

Recessed conduit wiring system shall be laid in concrete in the roof slab and in walls, and shall comply with all the requirements for surface conduit wiring system specified in **6.9.1(a)** to **6.9.1(k)** and also conform with the requirements specified under:

a) *Making of chase* — The chase in the wall shall be neatly made and be of ample dimensions to permit the conduit to be fixed in the manner desired. In the case of buildings under construction, chases shall be provided in the wall, ceiling, etc at the time of their construction and shall be filled up neatly after erection of conduit and brought to the original finish of the wall. In case of exposed brick/rubble masonry work, special care shall be taken to fix the conduit and accessories in position along with the building work.

b) *Fixing of conduit in chase* — The conduit pipe shall be fixed by means of staples or by means of saddles not more than 600 mm apart. Fixing of standard bends or elbows shall be avoided, as far as practicable and all curves maintained by bending the conduit pipe itself with a long radius which will permit easy drawing-in of conductors. All threaded joints of rigid steel conduit shall be treated with preservative compound to secure protection against rust.

c) *Inspection boxes* — Suitable inspection boxes shall be provided to permit periodical inspection and to facilitate removal of wires, if necessary. These shall be mounted flush with the wall. Suitable ventilating holes shall be provided in the inspection box covers. The minimum sizes of inspection boxes shall be 75 mm × 75 mm.

d) *Types of accessories to be used* — All outlet, such as switches and wall sockets, may be either of flush mounting type or of surface mounting type, as given below:

   1) *Flush mounting type* — All flush mounting outlets shall be of cast-iron or mild steel boxes with a cover of insulating material or shall be a box made of a suitable insulating material. The switches and other outlets shall be mounted on such boxes. The metal box shall be efficiently earthed with conduit by a suitable means of earth attachment.

   2) *Surface mounting type* — If surface mounting type outlet box is specified, it shall be of any suitable insulating material and outlets mounted in an approved manner.

The switches/socket-outlets shall have adequate IP rating for various utilizations.

6.9.3 *Conduit Wiring System with Rigid Non-Metallic Conduits*

Rigid non-metallic conduits are used for concealed conduit wiring. This type of wiring system is more preferred for coastal areas.

6.9.3.1 *Type and size*

All non-metallic conduits used shall conform to accepted standards [D-2(45)] and shall be used with the corresponding accessories {*see* accepted standards [D-2(48)]}. The conduits shall be circular or rectangular cross sections.

6.9.3.2 *Bunching of cables*

Conductors of a.c. supply and d.c. supply shall be bunched in separate conduits. For lighting and small power outlet circuits phase segregation in separate circuits is recommended. The number of insulated cables that may be drawn into the conduits are given in Table 1B. In Table 1B, the space factor does not exceed 40 percent of the available space.

Table 1B Maximum Permissible Number of 250 V Grade Single-Core Copper Cables that may be Drawn into Rigid Non-Metallic Conduits

[*Clauses* 6.6.1(a) *and* 6.9.3.2]

Sl No.	Sizes of Cable	Size of Conduit, mm				
	Nominal Cross-Sectional Area mm^2	20	25	32	40	50
		Number of Cables, *Max*				
(1)	(2)	(5)	(6)	(7)	(8)	(9)
i)	1.5	6	10	14	–	–
ii)	2.5	5	10	14	–	–
iii)	4	3	6	10	14	–
iv)	6	2	5	9	11	–
v)	10	–	4	7	9	–
vi)	16	–	2	4	5	12
vii)	25	–	–	2	2	6
viii)	35	–	–	–	2	5

6.9.3.3 *Conduit joints*

Conduits shall be joined by means of couplers. Where there are long runs of straight conduit, inspection type couplers shall be provided at intervals. For conduit fittings and accessories reference may be made to the accepted standard [D-2(48)].

6.9.3.4 *Fixing of conduit in chase*

The conduit pipe shall be fixed by means of stapples or by means of non-metallic saddles placed at not more than 600 mm apart or by any other approved means of fixing. Fixing of standard bends or elbows shall be avoided, as far as practicable and all curves shall be maintained by bending the conduit pipe itself with a long radius which will permit easy drawing in of conductors. At either side of bends, saddles/stapples shall be fixed at a distance of 150 mm from the centre of bends.

6.9.3.5 *Inspection boxes*

Suitable inspection boxes to the nearest minimum requirements shall be provided to permit periodical inspection and to facilitate replacement of wires, if necessary. The inspection/junction boxes shall be mounted flush with the wall or ceiling concrete. Where necessary deeper boxes of suitable dimensions shall be used. Suitable ventilating holes shall be provided in the inspection box covers, where required.

6.9.3.6 The outlet boxes such as switch boxes, regulator boxes and their phenolic laminated sheet covers shall be as per requirements of **6.9.1**(h). They shall be mounted flush with the wall.

6.9.3.7 *Types of accessories to be used*

All accessories such as switches, wall sockets, etc, may be either flush mounting type or of surface mounting type.

6.9.3.8 *Bends in conduits*

Wherever necessary, bends or diversions may be achieved by bending the conduits or by employing normal bends, inspection bends, inspection boxes, elbows or similar fittings. Heat may be used to soften the conduit for bending and forming joints in case of plain conduits.

6.9.3.9 *Outlets*

In order to minimize condensation or sweating inside the conduit, all outlets of conduit system shall be properly drained and ventilated, but in such a manner as to prevent the entry of insects.

6.10 Cable Trunking/Cable Ways

Cable trunking and ducting system of insulating material are used for laying of cables and wire both in floor and ceiling. The maximum number of cables with sizes that can be drawn in a cable trunking or ducting shall be limited to a maximum of 40 percent of space available.

7 FITTINGS AND ACCESSORIES

7.1 General

This covers the requirements of fittings and accessories which are normally used in fixed wiring installations of buildings. This includes type of light fixtures which are typically selected based on the requirement of light at a particular location and fans based on the requirement of air movement for the purpose of comfort of occupants, especially during hot summer months, and also of exhaust fans to maintain the enclosed area with fresh outside air so that a hygienic and comfortable atmosphere is created inside the enclosed area. For buildings other than small dwellings the requirement of lighting is normally selected based on the illumination level for different areas and that of the fans based on the type of occupancy and different comfort levels. The fittings and accessories shall comply with relevant Indian Standards and ISI marked (where mandated).

7.2 Ceiling Roses and Similar Attachments

A ceiling rose or any other similar attachments shall only be used as a final terminating point in single phase and voltage not exceeding 250 V.

Normally, only one flexible cord shall be attached to a ceiling rose. Specially designed ceiling roses shall be used for multiple pendants.

A ceiling rose shall not embody fuse terminal as an integral part of it.

7.3 Socket-Outlets and Plugs

There are two types of socket-outlets, one for the use of light loads not exceeding 100 W which are designated as 6 A and is normally provided with 3 pin socket-outlets in which both 3 pin plug tops and 2 pin appliances like mobile chargers can be used. This type of socket-outlet shall normally be kept on the switch board as well as individually at convenient heights not exceeding 1.2 m from the floor level based on requirement.

The second type of socket-outlets is normally of 16 A capacity and are provided with 6 pin socket-outlets in which 3 pin plug tops of 16 A capacity as well as smaller type 3 pin plug tops of about 10 A capacity can be used. These types of socket-outlets are recommended for use in respect of domestic appliances not exceeding 2 kW. These are normally placed independently at suitable heights and also above kitchen slab at appropriate level based on the requirement (*see* Fig. 4).

Each socket-outlet shall be controlled by a switch which shall preferably be located immediately adjacent thereto or combined therewith. The switch controlling the socket-outlet shall be on the live side of the line.

In respect of individual air conditioners, it is recommended to use MCB controlled socket-outlet so that proper overload protection can be maintained based on the capacity of air conditioner used which may not exceed 3 TR.

In case of public buildings, to facilitate operation of switches/socket-outlets by persons with disabilities and the elderly, these shall be installed at an accessible height for reaching and operating, between 800 mm and 1 100 mm above floor level and shall be located at a minimum of 600 mm with a preference of minimum 700 mm from any internal corner. They shall be so fixed so as to be away from danger of mechanical injury.

Suitable earth wire shall be drawn along with the power wiring for the respective socket-outlets so that the third pin is effectively earthed to ensure the equipment connected to the respective socket-outlet is properly earthed at the time of use.

NOTES

1 In situations where a socket-outlet is accessible to children, it is necessary to install an interlocked plug and socket or alternatively a socket-outlet which automatically gets screened by the withdrawal of plug. In industrial premises socket-outlet of rating 20 A and above shall preferably be provided with interlocked type switch.

2 As an exception, electrical wall socket-outlets, telephone points and TV sockets can be located at a minimum height of 400 mm above floor level.

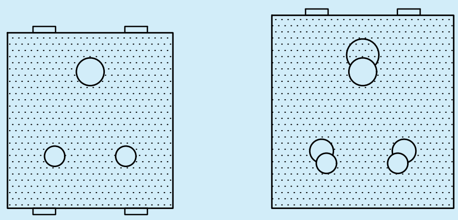


FIG. 4 6 A 3 PIN, 6/16 A COMBINED SOCKET-OUTLET

7.4 Light Fittings

7.4.1 *Lamp Holders*

There are different types of lamp holders namely pendant type, angle bracket type and batten mounted type lamp holders, each of which is further categorized as pin type and different sizes of screw type. These lamp holders are used in combination with different type of lamp shades to form ceiling suspended light fittings, wall mounted bracket type light fittings and chandelier type light fittings. These fittings can hold any of the GLS lamps of different wattage or CFL lamps and even the LED bulbs.

A switch shall be provided for control of every lighting fitting or a group of lighting fittings in the live wire. The neutral wire shall normally be looped in a light circuit. The earth wire of each point shall be used to earth the lamp holder or any such metallic point in the light fitting which does not carry electric current. Where control at more than one point is necessary, either two way or intermediate switches may be provided as required.

Shade made of flammable material shall not form a part of lighting fittings unless such shade is well protected against all risks of fire.

General and safety requirements for electrical lighting fittings shall be in accordance with provisions contained in good practice [D-2(1)] and relevant Indian Standards.

7.4.2 *Lamps*

All bare lamps including light fittings unless otherwise required and suitably protected, shall be hung at a height of not less than 2.4 m above the finished floor level. All electric lamps and accessories shall conform to relevant Indian Standards.

7.4.3 *Outdoor Lighting*

External street light fittings, bollards light fittings, bulk-head fittings, etc shall be weatherproof to prevent the ingress of moisture and dust effectively.

7.5 Fans, Regulators and Clamps

7.5.1 *Ceiling Fans*

Ceiling fans, including their suspension, shall conform to the relevant Indian Standards and good practice [D-2(1)]. There are two main types of ceiling fans available, the conventional type regular a.c. fans and the brushless direct current (BLDC) type fans which consume less power but may be expensive. Selection of the fans shall be on the basis of following technical parameters:

a) fan sweep, in mm (900/1 050/1 200/1 400/1 500);

b) air delivery, in m^3/min;

c) maximum power input, in w; and

d) service value, in m^3/min/W.

The bottom most point of the ceiling fan shall not be less than 2 400 mm above the finished floor level {*see* good practice [D-2(1)]}.

All ceiling fans shall meet the following requirements:

a) Control of a ceiling fan shall be through its own regulator as well as a switch in series.

b) All ceiling fans shall be wired with normal wiring to ceiling roses or to special connector boxes to which fan rod wires shall be connected and suspended from hooks or shackles with insulators between hooks and suspension rods. There shall be no joint in the suspension rod. The suspension rod shall be of adequate strength to withstand the dead and impact forces imposed on it. Suspension rods should preferably be procured along with the fan.

c) Fan hooks (clamp) shall be of suitable design according to the nature of construction of ceiling on which these clamps are to be fitted. In all cases fan clamps shall be fabricated from new metal of suitable sizes and they shall be as close fitting as possible. Fan clamps for reinforced concrete roofs shall be buried with the casting of roof and care shall be taken to tie the fan clamp with the reinforcement of the roof slab. Fan clamps for wooden beams shall be of suitable flat iron fixed on two sides of the beam, and according to the size and section of the beam one or two mild steel bolts passing through the beam shall hold both flat irons together. Fan clamps for steel joist shall be fabricated from flat iron to fit rigidly to the bottom flange of the beam. *See also* good practice [D-2(1)]. Care shall be taken during fabrication that the metal does not crack while hammer to shape. Other fan clamps shall be made to suit the position, but in all cases, care shall be taken to see that they are rigid and safe.

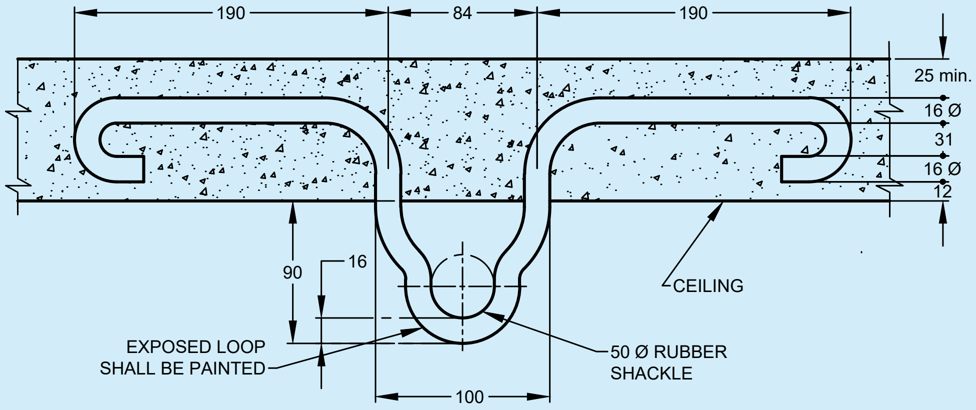
d) Canopies on top and bottom of suspension rods shall effectively conceal suspensions and connections to fan motors, respectively.

e) The lead-in-wire shall be of nominal cross-sectional area, not less than 1.5 mm^2 copper and shall be protected from abrasion.

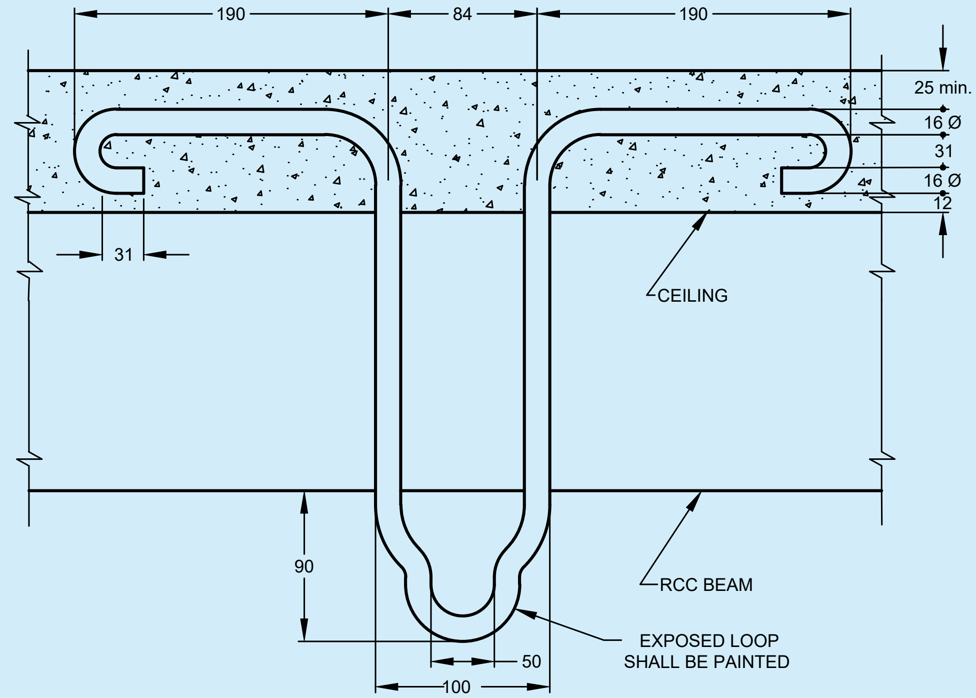
f) Unless otherwise specified, the clearance between the bottom most point of the ceiling fan and the floor shall be not less than 2.4 m. The minimum clearance between the ceiling and the plane of the blades shall be not less than 300 mm.

A typical arrangement of a fan clamp is given in Fig. 5.

NOTE — All fan clamps shall be so fabricated that fans revolve steadily.



5A SLAB



5B SLAB WITH BEAM

ALL DIMENSIONS IN MILLIMETRES.

NOTES

1 RCC SLAB STEEL REINFORCEMENT NOT SHOWN.

2 FAN CLAMP SHALL BE PLACED IN POSITION SUCH THAT ITS PROJECTING ARMS ARE IN THE LINE OF LENGTH OF BEAM.

FIG. 5 TYPICAL DESIGN OF FAN CLAMPS

7.5.2 *Exhaust Fans*

Exhaust fans are necessary for spaces, such as community toilets, kitchens and canteens and godowns to provide the required number of air changes (*see* Part D 'Building Services, Section 3 Air Conditioning, Heating and

Mechanical Ventilation' of this NBCS). Since the exhaust fans are located generally on the outer walls of a room, appropriate openings in such walls shall be provided for, in the planning stage.

NOTE — Exhaust fan requirement is based on the recommended air changes (*see* Part D 'Building Services, Section 3 Air Conditioning, Heating and Mechanical Ventilation' of this NBCS). Reference should also be made to Part F 'Fire and Life Safety' of this NBCS for exhaust fan requirements for smoke extraction.

Selection of exhaust fan can be made on the basis of:

a) blade sweep size in mm;

b) air changes in CFM or m^3/min; and

c) power input in W.

For fixing of an exhaust fan, a circular hole shall be provided in the wall to suit the size of the frame which shall be fixed by means of rag-bolts embedded in the wall. The hole shall be neatly plastered with cement and brought to the original finish of the wall. The exhaust fan shall be connected to exhaust fan point which shall be wired as near to the hole as possible by means of a flexible cord, care being taken that the blades rotate in the proper direction.

7.5.3 *Air Circulation* (*Ceiling Fans*)

7.5.3.1 Proper air circulation of rooms and halls of residential buildings/office can be achieved by suitable selection of ceiling fans based on their size and air displacement. In big halls, this can be done either by larger number of smaller fans or smaller number of larger fans. The economics of the system as a whole should be a guiding factor along with the bays of the building. For design guidelines in this regard, reference shall be made to Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS and good practice [D-2(1)].

7.5.3.2 Where ceiling fans are provided, the bay sizes of a building, which control fan point locations, play an important part. Fans of 1 200 mm/1 400 mm sweep normally cover an area of 9 m^2 to 10 m^2, for every part of a bay to be served by the ceiling fans, it is necessary that the bays shall be so designed that full number of fans can be suitably located for the bay, otherwise it will result in ill-ventilated pockets. In general, fans in long halls may be spaced at 3 m in both the directions. If building modules do not lend themselves for proper positioning of the required number of ceiling fans, other fans such as, air circulators or bracket fans will have to be employed for the areas uncovered by the ceiling fans. For this, suitable electrical outlets shall be provided although result may be disproportionate to cost on account of fans.

7.5.3.3 High volume low speed (HVLS) fans are generally ceiling fans having more number of blades which move slowly and distribute large amounts of air at low rotational speed. These are installed in very large areas like airports, hangers, warehouses and in commercial spaces like malls, etc, which have very big area with large movement of people. These are primarily recommended for bigger areas from the consideration of energy conservation.

7.5.3.4 Nowadays BLDC type fans having low wattage and higher air displacement are available with remote control operation. Choice of the same may be decided based on their operational efficiency in relation to cost.

7.6 Interchangeability

Similar parts of all switches, lamp holders, distribution fuse-boards, ceiling roses, brackets, pendants, fans and all other fittings shall be so chosen that they are of the same type and interchangeable in each installation.

7.7 Equipment

7.7.1 Electrical equipment which form integral part of wiring intended for switching or control or protection of wiring installations shall conform to the relevant Indian Standards, wherever they exist.

7.7.2 Positioning of fans and light fittings shall be chosen to make these effective without causing shadows and stroboscopic effect on the working planes.

8 EARTHING

8.1 General

8.1.1 Earthing is an important part of any electrical installation. It is essential to provide safety of human beings from electrical shock. It is also necessary to prevent fire from electrical fault and to operate the protective devices to disconnect power during an electrical fault and to ensure safe operation of the equipment. Earthing is therefore meant to divert the fault current to the protective earth conductor without causing any major damage to the installations. Thus earthing/earth conductor (also known as a ground conductor or earthing conductor) is a safety feature in electrical systems which provides a low-impedance path to the earth and helps to prevent hazardous voltages from appearing on equipment ensuring that faults are cleared quickly. In principle, it acts as a safeguard, protecting both people and equipment from electrical hazards. With the proliferation in electrical/electronic gadgets and greater dependency on voice and data communication systems, proper and effective earthing or grounding is very important. Earthing is also necessary for safe dissipation of the effects of lightning strikes from the buildings and its contents, including from sensitive equipment. All works under earthing shall comply with the provisions of good practice [D-2(1)] and [D-2(11)] apart from other Indian Standards, where applicable.

8.1.2 Different types of earth electrodes and different types of earthing systems are available. For low voltage and high voltage systems which apply to almost all electrical systems of buildings, the common earthing system followed is with the neutral solidly earthed at the source. This system requires that there is always a protective earth continuity conductor running all through the system and all metal parts of electrical appliances connected to an electrical system are connected to the earth continuity conductor. The exception to this system is the double insulated appliances which are connected to the line and neutral and operate on low voltage (single phase) and are also of low power consumption. All appliances (other than double insulated devices) use the earthing through the earth continuity conductor. Single phase appliances use a 3 wire connection with line (live wire), neutral (return path wire) and the earth-wire. Three phase appliances use a connection with 4 wires for a load which does not require a neutral connection or use a 5 wire connection if the appliance requires a neutral connection. Care should be taken to ensure that the earthing system, the protective earth conductor and the earthing pin of the socket-outlets are not disconnected.

8.1.3 Different earthing systems have features which are suitable for different applications. Earthing system adopted should be so selected so as to match with the type of load, protection device, application, degree of reliability, etc. For classification of electrical systems based on the relationship of the source and of exposed-conductive parts of the installation, to earth, *see* **2.1.185**.

8.1.4 *System Earthing*

Earthing of system is designed primarily to preserve the security of the system by ensuring that the potential on each conductor is restricted to a value consistent with the level of applied insulation. It is equally important that earthing should ensure efficient and fast operation of protective gear in the case of earth faults.

a) *TN system* — A system having one or more points of the source of energy directly earthed, the exposed conductive-parts of the installation being connected to that point by protective conductors. This is divided into TN-C system, TN-S system and TN-C-S system which are described below:

    1) *TN-C system* — A system in which neutral and protective conductors are combined in a single conductor throughout the system. This is normally provided at their source by the electrical supply company in single phase two wire or three phase four wire system.

    2) *TN-S system* — A system having separate neutral and protective conductor throughout the system. This is provided by the consumer at the point of receipt of electric supply.

    3) *TN-C-S system* — A system in which neutral and protective conductors are combined in a single conductor in part of the system. It involves a shared neutral and earth conductor from the source up to the consumer's intake, where it is then separated into individual neutral (N) and protective earth (PE) conductors.

b) *TT system* — A system having one point of the source of energy directly earthed, the exposed-conductive-parts of the installation being connected to the earth electrodes electrically independent of the earth electrodes of the source.

c) *IT system* — A system in which all live parts are isolated from earth or point connected to earth through high impedance. The exposed-conductive-parts of the electrical installation are earthed independently or collectively to the earthing of the system. This system is mostly used in hospitals, sensitive areas and places where stable power supply is essential for critical loads.

8.2 Selection and Design of Earthing System

8.2.1 Earthing shall generally be carried out in accordance with the requirements of Regulation 18, 43 and 50 of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 (*see* Annex B), relevant Indian Standards and the relevant regulations of the Electricity Supply Authority concerned. All materials, fittings, etc used in earthing shall conform to relevant Indian Standard specification.

8.2.2 Conductors and earth electrodes in an earthing system shall be so designed and constructed that in normal use their performance is reliable and without danger to persons and surrounding equipment. Earthing system shall be designed so as to have touch potential, step potential and transfer potential within the specified limits as indicated in good practice [D-2(18)] and [D-2(11)] as amended up to date. The choice of a material depends on its ability to match the particular application requirement. The requirements for earthing arrangements are intended to provide a connection to earth which:

a) is reliable and suitable for the protective requirements of the installation;

b) can carry earth fault currents and protective conductor currents to earth without danger from thermal, thermo-mechanical and electromechanical stresses and from electric shock arising from these currents;

c) if relevant, is also suitable for functional requirements; and

d) is suitable for the foreseeable external influences as provided in good practices [D-2(1)], [D-2(18)], [D-2(11)] and IEC 60364-5-51 : 2005 'Electrical installations of buildings — Part 5-51: Selection and erection of electrical equipment — Common rules', for example, mechanical stresses and corrosion.

8.2.3 The main earthing system of an electrical installation shall consist of:

a) an earth electrode (electrode can be vertical rod/pipe/buried plate or an earth mat with several vertical installations or a ring earthing strip/conductor);

b) a main earthing wire;

c) main earthing terminal (MET), an earth bar (located on the main switchboard for small installation and installed in the wall/room in case of large industrial electronic installations) for the connection of the main earthing wire, protective earthing wires and/or bonding wires within the installation; and

d) a removable link, which effectively disconnects the neutral bar from the earth bar.

NOTE — The requirements of (c) and (d) above shall be carried out by a licensed electrician as part of the switchboard installation.

8.2.4 The main earthing wire connection shall:

a) be mechanically and electrically sound;

b) be protected against damage, corrosion, and vibration;

c) not place any strain on various parts of the connection;

d) not damage the wire or fittings; and

e) be secured at the earth electrode.

8.2.4.1 The main earthing wire termination shall be readily accessible at the earth electrode except for **8.2.22**. As far as possible, all earth connections, except exothermically welded, shall be visible for inspection.

8.2.5 Consideration shall be given to the earthing arrangements where currents with high frequencies are expected to flow {*see* good practice [D-2(18)] and IEC 60364-4-44 : 2007 'Low-voltage electrical installations — Part 4-44: Protection for safety — Protection against voltage disturbances and electromagnetic disturbances'}.

8.2.6 Protection against electric shock {*see* good practice [D-2(18)] and IEC 60364-4-41: 2005 'Low-voltage electrical installations — Part 4-41: Protection for safety — Protection against electric shock'}, shall not be adversely affected by any foreseeable change of the earth electrode resistance (for example, due to corrosion, drying or freezing).

8.2.7 Where the supply to an installation is at high or extra high voltage, requirements concerning the earthing arrangements of the high or extra high voltage supply and of the low-voltage installation shall also comply with provisions of good practice [D-2(18)] and IEC 60364-4-44 : 2007 'Low-voltage electrical installations — Part 4-44: Protection for safety — Protection against voltage disturbances and electromagnetic disturbances'.

8.2.8 A permanent fitting (like a screwed-down label, or one that can be threaded onto the cable) shall be used at the connection point that is clearly marked with the words: 'EARTHING LEAD — DO NOT DISCONNECT' or 'EARTHING CONDUCTOR — DO NOT DISCONNECT'.

8.2.9 All low voltage, three phase equipment shall be earthed by two separate and distinct connections with earth. In case of single phase equipment, single distinct earth connection is sufficient. The contact area of earth conductor/plate shall be determined in accordance with good practice [D-2(11)].

8.2.9.1 The 415/240 V, 4 wire, 3 phase systems are normally operated with the neutral solidly earthed at source. At low voltage, the Central Electricity Authority regulations require that the neutral be earthed by two separate and distinct connections with earth. Neutral of transformers of 33 kV/415 V, 11 kV/415 V and neutral of low voltage DG sets and similar equipment shall be of the same size as the phase conductor or higher as per relevant Indian Standard.

NOTE — Neutral conductor of half the size of the phase conductor was permitted in earlier installations. But with the proliferation of equipment using non-linear devices and consequent increase in harmonics, the neutral will carry a current more than the notional out-of-balance current and therefore neutral conductor shall be of the same size as the phase conductor.

8.2.10 In the case of high and extra high voltages, the neutral points shall be earthed by not less than two separate and distinct connections to earth, each having its own electrode at the generating station or substation and may be earthed at any other point, provided no interference is caused by such earthing. The neutral may be earthed through suitable impedance. Neutral earthing conductor shall be sized as to have a current carrying capacity not less than the phase current.

8.2.11 For industrial/commercial installations having a transformer within the facility, soil resistivity of the place of installation shall be measured and recorded {*see* good practice [D-2(11)]}. For the adopted type of earth electrode configuration, earth fault impedance of the configuration shall be calculated and recorded {*see* good practice [D-2(11)]}.

8.2.12 It is recommended that a drawing showing the main earth connection and earth electrodes be prepared for each installation.

8.2.13 Earth system shall be so devised that the testing of individual earth electrode is possible. It is recommended that the value of any earth system resistance shall be such as to conform to the degree of shock protection desired. For measuring purposes, the joint shall be capable of being opened with the aid of a tool. In normal use it shall remain closed. Inclusion of online earth monitoring system with real time earthing grid resistance and integrity measurement at site level can be considered for buildings which house lot of sensitive systems and also number of electronic devices to ensure human safety and equipment protection.

8.2.14 Conductors, other than live conductors and any other parts intended to carry a fault current, shall be capable of carrying that current without attaining an excessive temperature.

8.2.15 No addition to the current carrying system, either temporary or permanent, shall be made which will increase the maximum earth fault current or its duration until it has been ascertained that the existing arrangement of earth electrodes, earth busbar, etc are capable of carrying the new value of earth fault current which may be obtained by this addition.

8.2.16 No cut-out, link or switch other than a linked switch arranged to operate simultaneously on the earthed conductor or earthed neutral conductor and the live conductors, shall be inserted on any supply system. This, however, does not include the case of a switch used in controlling a generator or a transformer or a link for test purposes.

8.2.17 The number of connections along the conductors shall be minimized to reduce potential failure points. The high energy lightning currents can induce significant thermal expansion, if bonding is inadequate and overlap is poor. Also, mechanical connections should be minimized. All conductors used in earthing and lightning protection system should be joined using methods that ensure long-term electrical, mechanical and environmental reliability. Critical connections, especially those below ground or exposed to moisture shall be made using exothermic welding which provides a permanent molecular bond with superior thermal and corrosion resistance. Exothermic welded joints shall conform to the relevant standards.

8.2.18 Earthing associated with current-carrying conductor is normally essential for the function of the system and is generally known as system earthing or functional earthing, while earthing of non-current carrying metal work and conductor is essential for the safety of human life, animals and of property and it is generally known as equipment earthing or equipotential bonding. The requirements for protective purposes shall always take precedence.

8.2.19 *Equipotential Bonding*

Equipotential bonding is through connecting exposed and extraneous conductive parts in a building to the main earthing terminal of the installation by protective bonding conductors. Where such conductive parts originate outside the building, they shall be bonded near their entry point within the building.

It shall be ensured that all exposed metal parts and extraneous conductive parts within a specific area or system are at the same potential. The same is achieved by creating a low resistance path between these conductive parts. Equipotential bonding allows fault currents to flow to earthed conductor without creating dangerous voltage difference that could harm individuals.

Equipotential bonding is used in various settings including buildings, industrial facilities, swimming pools and areas with lightning protection system. In the event of a fault equipotential bonding helps limit touch voltages and ensures that fault currents are safely directed to earth preventing electric shock (*see* Fig. 6).

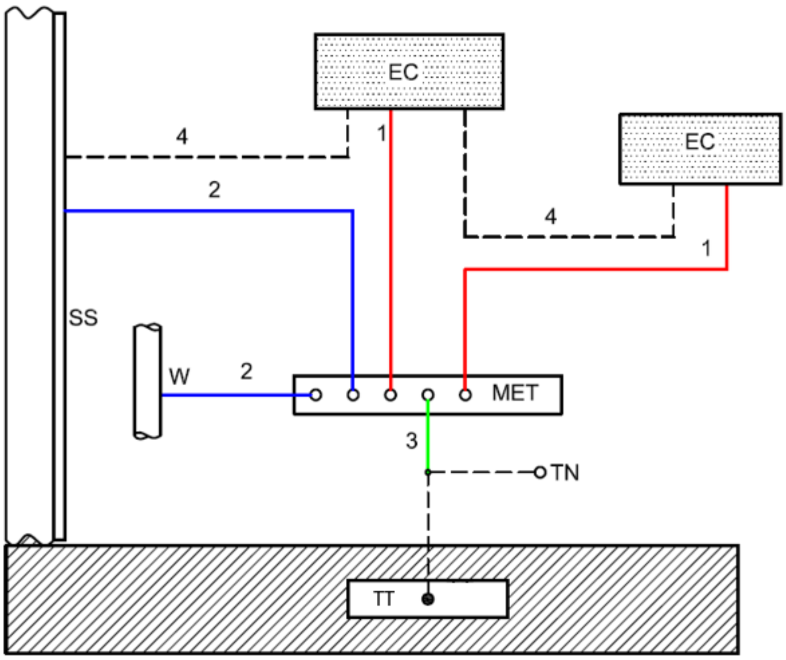
8.2.20 All earthing systems in a building shall be interconnected at the earth level (ground).

8.2.21 For selection of electrodes for use in corrosive environments, reference shall be made to good practice [D-2(11)].

8.2.22 Test joints are not required in the case of natural down-conductors combined with foundation earth electrodes (*see* Fig. 7).

8.2.23 For computer and other sensitive electronic equipment system in industrial and commercial application, special bonding techniques with isolation transformer can be employed (*see* Fig. 8).

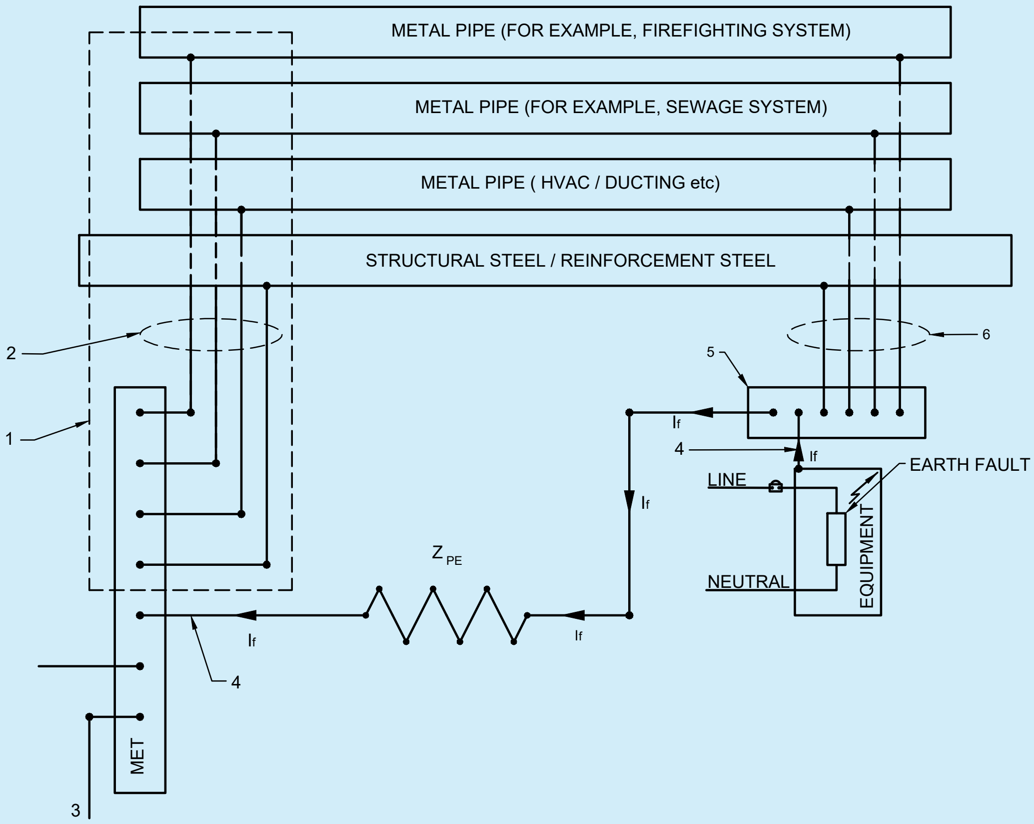
8.2.24 Isolated earthing is unsafe during a transient condition. In unavoidable conditions if isolated earthing is used to reduce potential difference between isolated earthing, earth couplers or isolating spark gaps shall be installed. This will reduce potential difference during a transient condition such as lightning.



Key

EC	Exposed conductive part (panel board body)
MET	Main earthing terminal
SS	Structural steel (extraneous conductive part)
W	Metallic water pipe (extraneous conductive part)
TT	Earth electrode in a TT or IT system
TN	Other means of earthing in a TN system, for example, connection to earthed point of the power system
1	Circuit protective conductor
2	Main protective bonding conductor
3	Earthing conductor
4	Supplementary protective bonding conductors (additional bonding, if required)
1,2,3,4	Protective conductors

6A ILLUSTRATION OF EARTHING AND PROTECTIVE CONDUCTORS



KEY

1 MAIN EQUIPOTENTIAL BONDING (CONNECTION BETWEEN MET AND EXTRANEOUS CONDUCTIVE PARTS).

2 MAIN EQUIPOTENTIAL BONDING CONDUCTOR (INDIVIDUAL CONNECTION, NO LOOPING). IN CASE OF LARGE NUMBER OF EXTRANEOUS CONDUCTIVE PARTS, AN EARTHING TERMINAL CAN BE LOCATED NEAR THE EXTRANEOUS CONDUCTIVE PARTS FOR INDIVIDUAL CONNECTION TO THESE PARTS AND A COMMON CONNECTION FROM THIS EARTHING TERMINAL TO MET. SUITABLE INDICATIONS SHALL BE MADE TO EVERY CONNECTION OF EQUIPOTENTIAL BONDING CONDUCTOR TO AVOID ACCIDENTAL DISCONNECTIONS.

3 EARTHING CONDUCTOR. (CONNECTS MET AND EARTHED POINT OF THE SOURCE IN TN SYSTEM OR MET AND EARTH ELECTRODE IN CASE OF TT/IT SYSTEM). ONE CONNECTION FROM MET IS ALSO NECESSARY TO LPS (IF INSTALLED).

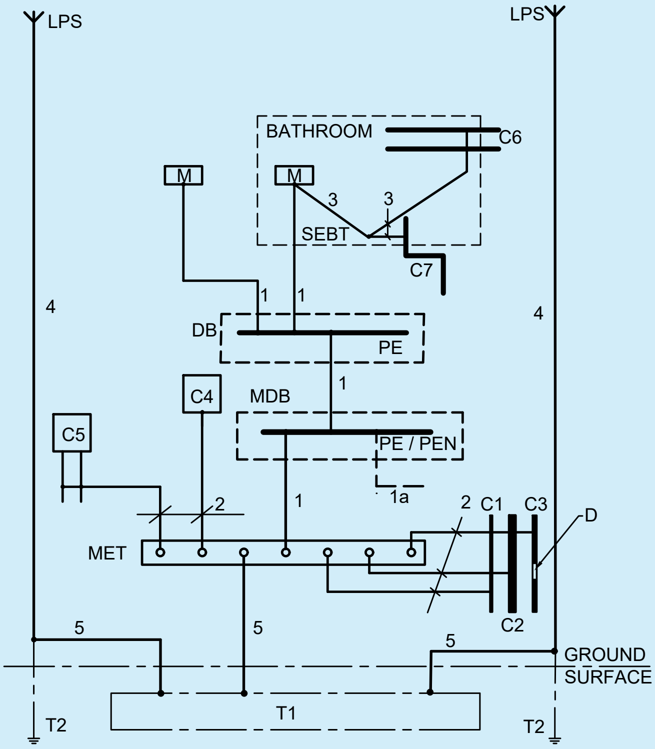
4 PROTECTIVE EARTHING BY PROTECTIVE EARTHING CONDUCTOR (PE).

5 SUPPLEMENTARY EQUIPOTENTIAL BONDING TERMINAL (SEBT) (TO RE DUCE TOUCH VOLTAGE < 50 V).

6 SUPPLEMENTARY EQUIPOTENTIAL BONDING CONDUCTOR.

6B MAIN AND ADDITIONAL PROTECTIVE EQUIPOTENTIAL BONDING

FIG. 6 PROTECTIVE EQUIPOTENTIAL BONDING



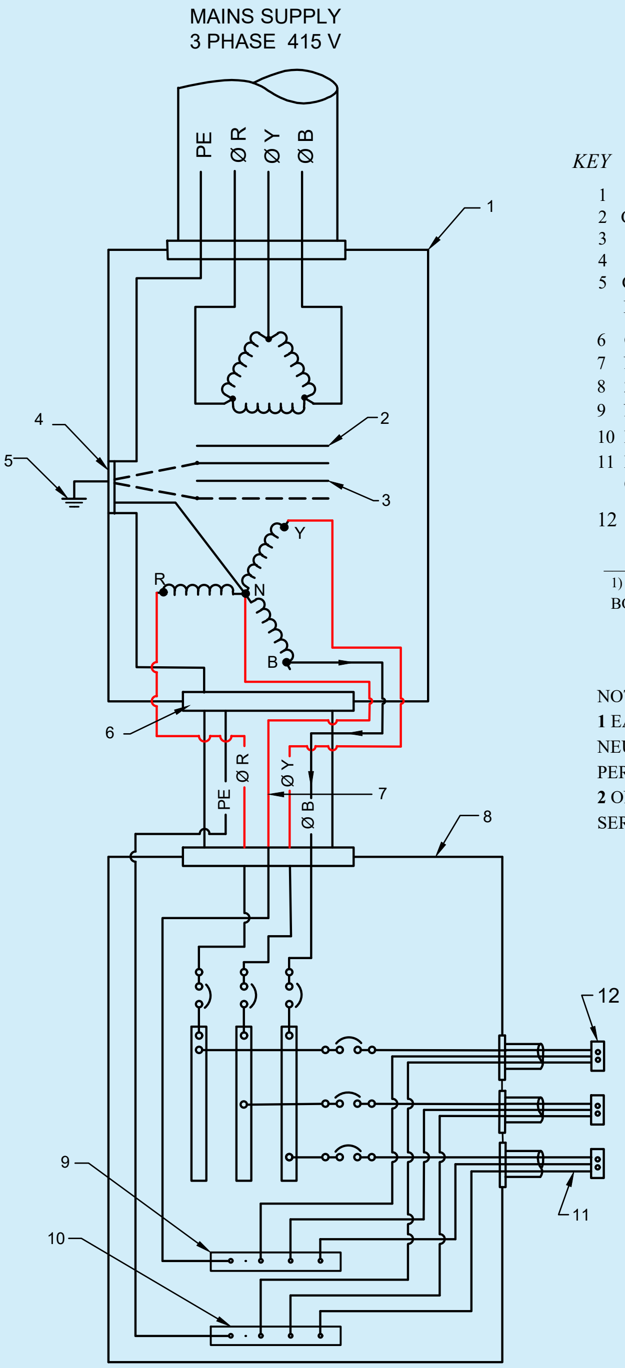
NOTES

1 FUNCTIONAL EARTHING CONDUCTORS ARE NOT SHOWN IN FIGURE.

2 WHERE A LIGHTNING PROTECTION SYSTEM IS INSTALLED, THE ADDITIONAL REQUIREMENTS ARE GIVEN IN 11.

KEY	
C	Extraneous-conductive-part
C1	Water pipe, metal from outside
C2	Waste water pipe, metal from outside
C3	Gas pipe with insulating insert, metal from outside
C4	Air conditioning
C5	Heating system
C6	Water pipe, metal for example in a bathroom
C7	Waste water pipe, metal for example in a bathroom
D	Insulating insert
MDB	Main distribution board
DB	Distribution board
MET	Main earthing terminal
SEBT	Supplementary equipotential bonding terminal
T1	Concrete-embedded foundation earth electrode or soil-embedded foundation earth electrode
T2	Earth electrode for LPS, if necessary
LPS	Lightning protection system (if any)
PE	PE terminal(s) in the distribution board
PE/PEN	PE/PEN terminal(s) in the main distribution board
M	Exposed-conductive-part
1	Protective earthing conductor (PE)
1A	Protective conductor, or PEN conductor, if any, from supplying network
2	Protective bonding conductor for connection to the main earthing terminal
3	Protective bonding conductor for supplementary bonding
4	Down conductor of a lightning protection system (LPS), if any
5	Earthing conductor

FIG. 7 EXAMPLE OF AN EARTHING ARRANGEMENT FOR FOUNDATION EARTH ELECTRODE, PROTECTIVE CONDUCTORS AND PROTECTIVE BONDING CONDUCTORS



KEY

1 ISOLATION TRANSFORMER
2 CORE
3 SHIELD
4 ALL EARTHING CONNECTIONS MADE AT A POINT
5 CONNECTIONS TO BUILDING STEELS/EARTH PITS/RING EARTHING
6 CONDUIT EARTHING
7 NEUTRAL
8 240/415 VOLT POWER PANEL
9 NEUTRAL BUS[1)]
10 EARTH BUS [1)]
11 EARTH CONNECTION FOR SOCKET/WORK STATION/ COMPUTERS
12 3 PIN SOCKETS WITH ISOLATED EARTH PIN

[1)] BOTH BUSBARS ISOLATED FROM PANEL ENCLOSURE. NO BONDING CONNECTION BETWEEN NEUTRAL AND EARTH.

NOTES

1 EARTH BRANCH CIRCUIT SHALL HAVE A SEPARATE NEUTRAL AND EARTH WIRE. NO DAISY CHAINING PERMITTED.

2 ONLY COMPUTER OR CONTROL SYSTEM SHOULD BE SERVED FROM THIS PANEL.

FIG. 8 RECOMMENDED POWER DISTRIBUTION FOR A COMPUTER AND CONTROL SYSTEM WITH A DELTA/STAR ISOLATION TRANSFORMER

8.3 Earth Electrodes

The efficacy of any earth electrode depends on its configuration and upon local soil conditions. Number of earth electrodes suitable for the soil conditions and the value of resistance to earth as well as earth fault impedance shall be considered. It is essential that earthing conductors of the earth system shall follow the provisions as per good practice [D-2(1)] read with good practice [D-2(11)] in view of provisions in CEA Regulation No. 43 (*see* Annex B) regarding earth fault loop impedance. Examples of earth electrodes which may be used are:

a) concrete-embedded foundation earth electrode;

b) soil-embedded foundation earth electrode;

c) metallic electrode embedded directly in soil vertically or horizontally (for example rods, wires, tapes, pipes or plates);

d) metal sheath and other metal coverings of cables according to local conditions or requirements;

e) other suitable underground metalwork (for example pipes) according to local conditions or requirements; and

f) welded metal reinforcement of concrete (except pre-stressed concrete) embedded in the earth.

The type, materials and dimensions of earth electrodes shall be selected to withstand corrosion and to have adequate mechanical strength for the intended lifetime. For materials commonly used for earth electrodes, the minimum sizes, from the point of view of corrosion and mechanical strength, when embedded in the soil or in concrete, shall be as specified in Table 2. In the case of lightning protection system, the provisions of **11.6.3** shall also be complied with.

NOTES

1 For corrosion, the parameters to be considered are: the soil *p*H at the site, soil resistivity, soil moisture, stray and leakage a.c. and d.c. current, chemical contamination, and proximity of dissimilar materials.

2 The minimum thickness of protective coating is greater for vertical earth electrodes than for horizontal earth electrodes because of their greater exposure to mechanical stresses while being embedded.

3 Interconnection of dissimilar metals should be avoided in earthing systems and earth conductors to prevent corrosion and loose connections.

Earth electrode either in the form of solid rod, pipe, plate or earth grid should be provided at all premises for providing an earth system. Details of typical pipe, rod and plate earth electrodes are given in Fig. 9 and Fig. 10. Other electrode configurations can be as in Fig. 11 {*see* good practice [D-2(11)]}.

Although electrode material does not affect initial earth resistance, care should be taken to select a material which is resistant to corrosion in the type of soil in which it is used. In case where soil condition leads to excessive corrosion of the electrode and the connections, it is recommended to use either copper/stainless steel or copper coated steel electrode and copper/stainless steel connections. Exothermic welding may also be adopted to have enhanced life and strength to the connection. It is recommended to use similar material for earth electrodes and earth conductors or otherwise precautions should be taken to avoid corrosion. As an alternate, 50 mm GI earth strip and 100 mm CI pipe can be provided as per good practice [D-2(11)].

Table 2 Minimum Size of Commonly Used Earth Electrodes, Embedded in Soil or Concrete, Used to Prevent Corrosion and Provide Mechanical Strength[1)]

[*Clauses* 8.3, 9.2.2.1(s) *and Fig. 9*]

Sl No.	Material and Surface	Shape	Diameter	Cross-Sectional Area	Thickness	Weight of Coating	Thickness of Coating/ Sheathing
			mm	mm^2	mm	g/m^2	µm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Steel embedded in concrete (bare, hot galvanized or stainless)	Round wire	10	–	–	–	–
		Solid tape or strip	–	75	3	–	–
ii)	Steel hot-dip galvanized[2)]	Strip[3)] or shaped strip/plate – solid plate – lattice plate	–	90	3	500	63
		Round rod installed vertically	16	–	–	350	45
		Round wire installed horizontally	10	–	–	350	45
		Pipe	25	–	2	350	45
		Stranded (embedded in concrete)	–	70	–	–	–
		Cross profile installed vertically	–	(290)	3	–	–
iii)	Steel copper sheathed	Round rod installed vertically	(15)	–	–	–	2 000
iv)	Steel with electro-deposited copper coating	Round rod installed vertically	14	–	–	–	250[6)]
		Round wire installed horizontally	(8)	–	–	–	70
		Strip installed horizontally	–	90	3	–	70
v)	Stainless steel [4)]	Strip[3)] or shaped strip/plate	–	90	3	–	–
		Round rod installed vertically	16	–	–	–	–
		Round wire installed horizontally	10	–	–	–	–

Table 2 (*Concluded*)

Table 2 Minimum Size of Commonly Used Earth Electrodes, Embedded in Soil or Concrete, Used to Prevent Corrosion and Provide Mechanical Strength[1)]

[*Clauses* 8.3, 9.2.2.1(s) *and Fig.* 9]

Sl No.	Material and Surface	Shape	Diameter	Cross-Sectional Area	Thickness	Weight of Coating	Thickness of Coating/ Sheathing
			mm	mm^2	mm	g/m^2	µm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Pipe	25	–	2	–	–
vi)	Copper	Strip		50	2	–	–
		Round wire installed horizontally	–	(25)[5)] 50	–	–	–
		Round rod installed vertically	(12) 15	–	–	–	–
		Stranded wire	1.7 for individual strands of wire	(25)[5)] 50	–	–	–
		Pipe[6)]	20	–	2	–	–
		Solid plate	–	–	(1.5) 2	–	–
		Lattice plate	–	–	2	–	–

[1)] As per Table B-1 under Part 1 'General and Common Aspects, Section 18: Earthing' of good practice [D-2(1)].

[2)] The coating shall be smooth, continuous and free from flux stains.

[3)] As rolled strip or slit strip with round edges.

[4)] Chromium ≥ 16 percent, nickel ≥ 5 percent, molybdenum ≥ 2 percent, carbon ≤ 0.08 percent.

[5)] Where experience shows that the risk of corrosion and mechanical damage is extremely low, 16 mm^2 can be used.

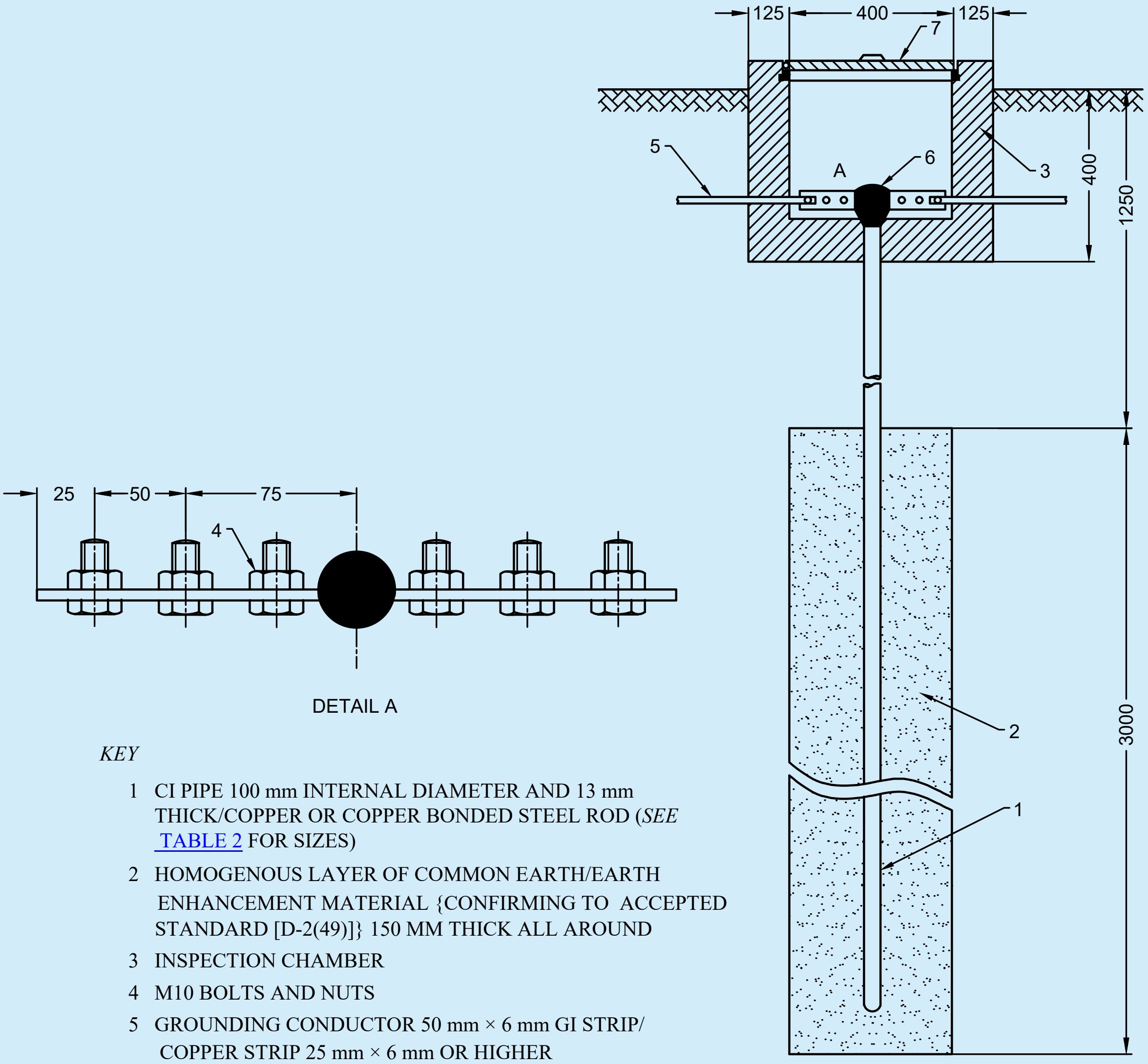
[6)] This thickness is provided to withstand mechanical damage of copper coating during the installation process. It may be reduced to not less than 100 µm where special precautions to avoid mechanical damage of copper during the installation process (for example, drilling holes or special protective tips) are taken according to the manufacturer's instruction.

NOTES

1 Values in bracket are applicable for protection against electric shock only, while values not in brackets are applicable for lightning protection and for protection against electric shock.

2 Metals inserted inside pipe will not influence in the final earth resistance value.

3 Unprotected ferrous materials are not recommended due to high corrosion (*see* IEC 60364-4-43 : 2008 'Low-voltage electrical installations — Part 4-43: Protection for safety — Protection against overcurrent').



KEY

1 CI PIPE 100 mm INTERNAL DIAMETER AND 13 mm THICK/COPPER OR COPPER BONDED STEEL ROD (*SEE* TABLE 2 FOR SIZES)

2 HOMOGENOUS LAYER OF COMMON EARTH/EARTH ENHANCEMENT MATERIAL {CONFIRMING TO ACCEPTED STANDARD [D-2(49)]} 150 MM THICK ALL AROUND

3 INSPECTION CHAMBER

4 M10 BOLTS AND NUTS

5 GROUNDING CONDUCTOR 50 mm × 6 mm GI STRIP/ COPPER STRIP 25 mm × 6 mm OR HIGHER

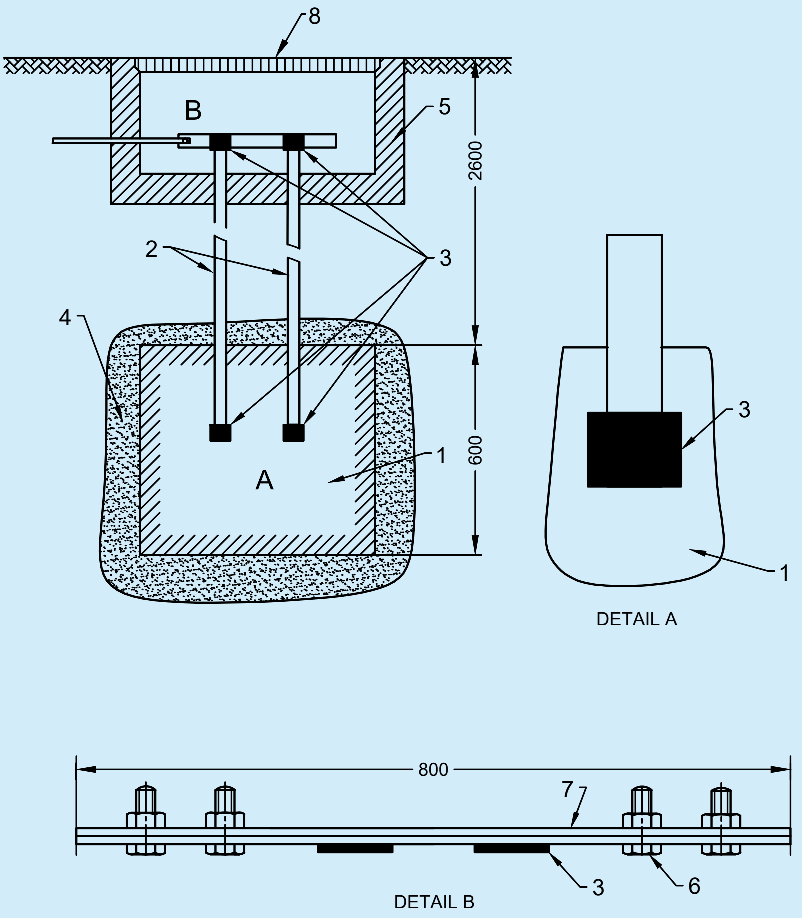
6 EXOTHERMIC WELDING

7 CI LID/COVER

ALL DIMENSIONS IN MILLIMETRES.

NOTE — INSPECTION HOUSING CAN ALSO BE OF FRP MATERIALS WITH CI COVER TESTED ACCORDING TO ACCEPTED STANDARD [D-2(50)].

FIG. 9 TYPICAL ARRANGEMENT OF EARTHING WITH COPPER/COPPER BONDED ELECTRODE WITH EXOTHERMIC WELDING (MAINTENANCE FREE ARRANGEMENT)



KEY

1 600 mm × 600 mm × 3 mm COPPER PLATE
2 30 mm × 6 mm COPPER STRIP
3 EXOTHERMIC WELDING
4 EARTH ENHANCEMENT MATERIAL {CONFORMING TO ACCEPTED STANDARD [D-2(49)]}
5 INSPECTION CHAMBER
6 M12 × 40 STAINLESS STEEL BOLTS AND NUTS
7 50 mm × 6 mm COPPER STRIP
8 CI LID/COVER

ALL DIMENSIONS IN MILLIMETRES.

NOTE — INSPECTION HOUSING CAN ALSO BE OF FRP MATERIALS WITH CI COVER TESTED ACCORDING TO ACCEPTED STANDARD [D-2(50)].

FIG. 10 TYPICAL ARRANGEMENT OF COPPER ELECTRODE PLATE EARTHING WITH EXOTHERMIC WELDING (MAINTENANCE FREE ARRANGEMENT)

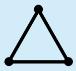
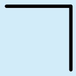
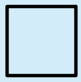
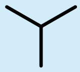
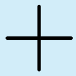
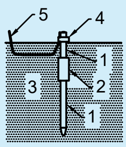
1		THREE RODS AT THE VERTICES OF AN EQUILATERAL TRIANGLE
2		TWO STRIPS SET AT RIGHT ANGLES TO EACH OTHER MEETING AT ONE CORNER
3		VERTICAL ELECTRODES INTERCONNECTED AS A SQUARE
4		THREE STRIPS SET AT 120°, MEETING AT THE STAR POINT ALL OF EQUAL LENGTH
5		FOUR STRIPS SET IN A CRUCIFORM
6		VERTICAL ROD 1 EXTENSIBLE EARTH ROD 2 ROD COUPLING 3 SOIL 4 CONDUCTOR TO ROD CLAMP 5 EARTHING CONDUCTOR

FIG. 11 MISCELLANEOUS ELECTRODE CONFIGURATIONS

8.4 Earth Enhancing Compound

Multiple rods, even in large numbers, may sometime fail to produce an adequately low resistance to earth. This condition arises in installations involving soils of high resistivity. The alternative is to reduce the resistivity of the soil immediately surrounding the earth electrode. To reduce the soil resistivity, artificial soil treatment shall be adopted.

8.4.1 Earthing enhancing compound is a conductive compound producing low resistance of an earth-termination system. These compounds used for artificial treatment of soil shall satisfy the requirements as per accepted standard [D-2(49)].

8.4.2 The material of the earthing enhancing compound shall be chemically inert to subsoil. It shall not pollute the environment. It shall provide a stable environment in terms of physical and chemical properties and exhibit low resistivity. The earthing enhancing compound shall not be corrosive to the earth electrodes being used. The usage of earth enhancing compound shall adhere to the conditions detailed in accepted standard [D-2(49)].

8.4.3 The materials used for artificial treatment should also fulfil toxicity characteristic leaching procedure (TCLP) requirements.

8.4.4 Use of salt [sodium chloride (NaCl)] for artificial treatment of soil should be avoided as it accelerates corrosion of ferrous materials.

8.4.5 The requirement and use of earth enhancing compound shall comply with the provisions as per accepted standard [D-2(49)].

8.5 Earth Electrode Inspection Housings and Earth Electrode Seals

8.5.1 *Earth Electrode Inspection Housing*

Earth electrode inspection housing is the metallic or non-metallic enclosure that houses the down-conductor/earth-termination connection for inspection and testing purposes and consists of a housing and a removable lid. The design of the earth electrode inspection housing shall be such that it carries out its function of enclosing the down-conductor/earth rod termination in an acceptable and safe manner, and has sufficient internal dimensions to permit the assembly/disassembly of the earth rod clamp. The material of the earth electrode inspection housing shall be compatible with its surrounding environment and shall comply with the tests given in accepted standard [D-2(50)]. The work shall also comply with the provisions of good practice [D-2(11)].

8.5.2 *Earth Electrode Seal*

Water pressure seal used in conjunction with an earth rod electrode shall be such that there will be no effect on the water proofing treatment for the foundation of the building. The design of the earth electrode seal shall be such that it carries out its function of preventing ground water bypassing the earth rod and entering the basement of a building, in an acceptable and safe manner and shall be compatible with its surrounding environment and comply with the tests given in accepted standard [D-2(50)].

8.6 Bonding and Inter Connection

All connections made in an earthing system above or below ground should meet electrical conductivity, corrosion resistance, current carrying capacity and mechanical strength of the conductor. These connections should be strong enough to maintain a temperature rise below that of the conductor and to withstand the effect of heating and the mechanical forces caused by fault currents. Consideration shall be given to electrolytic corrosion when using different materials in an earthing arrangement. The complete connections shall be able to resist corrosion for the intended life of the installation.

8.6.1 For external conductors (for example, earthing conductor) connected to a concrete-embedded foundation earth electrode, the connection made from hot-dip galvanized steel shall not be embedded in the soil.

8.6.2 Where an earth electrode consists of parts that should be connected together, the connection shall be by exothermic welding, pressure connectors, clamps or other suitable mechanical connectors.

8.6.3 All connection components shall meet the requirements according to accepted standard [D-2(51)] and also provisions of good practice [D-2(11)] and [D-2(1)] where applicable.

8.7 Equipment and Portions of Installations which shall be Earthed

8.7.1 *Equipment to be Earthed*

Except for equipment provided with double insulation, all the non-current carrying metal parts of electrical installations are to be earthed properly. All metal conduits, trunking, cable sheaths, switchgear, distribution fuse boards, lighting fittings and all other parts made of metal shall be bonded together and connected by means of two separate and distinct conductors to an efficient earth electrode.

8.7.2 *Structural Metal Work*

Earthing of metallic parts of the structure shall be done according to good practice [D-2(11)].

8.8 Neutral Earthing

To comply with relevant Central Electricity Authority regulations, no fuses or circuit breakers other than a linked circuit breaker shall be inserted in an earthed neutral conductor. A linked switch or linked circuit breaker shall be arranged to break the neutral either with or after breaking all the related phase conductors. This shall make (or close) the neutral before making (or closing) the phases.

If this neutral point of the supply system is connected permanently to earth, then the above rule applies throughout the installation including 2 wire final circuits. This means that no fuses may be inserted in the neutral or common return wire. And the neutral should consist of a bolted solid link or part of a linked switch, which completely disconnects the whole system from the supply. This linked switch shall be so arranged that the neutral makes before and break after the phases. For further details, reference should be made to good practice [D-2(11)].

8.9 System of Earthing

Equipment and portions of installations shall be deemed to be earthed only if the earthing is done as per TN-S system with equipotential bonding as required. In all cases, the relevant provisions of the Central Electricity Authority regulations shall be complied along with provisions under good practice [D-2(11)].

8.10 The earthing of electrical installations for non-industrial and industrial buildings shall be done in accordance with good practice [D-2(11)] and good practice [D-2(1)].

9 INSPECTION, TESTING, VERIFICATION AND COMMISSIONING OF INSTALLATION

9.1 General Requirements

9.1.1 Before the completed installation, or an addition to the existing installation, is put into service, inspection and testing shall be carried out in accordance with the *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023, with special reference to Regulation numbers 5, 8, 14, 17, 18, 20, 21, 29 to 39, 42, 43, 46 to 48 and 50 (*see* Annex B), and also as per provisions in Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations' of good practice [D-2(1)]. In the event of defects being found, these shall be rectified, as soon as practicable and the installation retested.

9.1.2 Initial and periodic inspection and testing shall be carried out in order to maintain the installation in a sound condition after putting into service. The periodicity shall be as per provisions in Annex E under Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations' of good practice [D-2(1)].

9.1.3 Where an addition is to be made to the fixed wiring of an existing installation, the latter shall be examined for compliance with the recommendations of this NBCS.

9.1.4 The individual equipment and materials which form part of the installation shall generally conform to the relevant Indian Standard, wherever applicable. If there is no relevant Indian Standard for any item, these shall be approved by the appropriate authority.

9.1.5 *Completion Drawings*

On completion of the electric work, a wiring diagram shall be prepared and submitted to the Engineer-in-Charge or the owner. All wiring diagrams shall indicate clearly the main switch board, the runs of various mains and submains cables and the position of all points and their controls. All circuits shall be clearly indicated and numbered in the wiring diagram and all points shall be given the same number as the circuit in which they are electrically connected. Also the location and number of earth points and the run of each loads should be clearly shown in the completion drawings.

9.1.6 *Inspection and Testing*

The substations and other electrical installations shall be checked and tested as detailed in **4.2** and **4.3** under Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations' of good practice [D-2(1)] and necessary test reports shall be prepared in the prescribed proforma therein. In addition, additional checks and tests as mentioned under the **9.2** and **9.3** shall also be carried out if not included in good practice [D-2(1)]. In case any defects are noticed, the same shall be rectified, retested and their test format shall be completed.

9.2 Inspection of the Installation

9.2.1 *General*

On completion of wiring a general inspection shall be carried out by competent personnel in order to verify that the provisions of this NBCS and that of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 have been complied with. This, among other things, shall include checking whether all equipment, fittings, accessories, wires/cables used in the installation are of adequate rating and quality to meet the requirement of the load. General workmanship of the electrical wiring with regard to the layout and finish shall be examined for neatness that would facilitate easy identification of circuits of the system, adequacy of clearances, soundness, contact pressure and contact area. A complete check shall also be made of all the protective devices, with respect to their ratings, range of settings and co-ordination between the various protective devices. It is preferable to have a third party inspection to maintain quality assurance.

9.2.2 *Items to be Inspected*

All equipment in a substation including HV panel, transformer, LV panel, emergency DG sets, battery bank, cables, cable terminations, etc need inspection. Healthiness of the main distribution boards, metering panels, distribution cables, rising mains, bus ducts, etc need to be verified along with earthing system. In case of provisions of rooftop solar PV system and other services, the equipment connected therein shall also be included in the items to be inspected.

9.2.2.1 *Substation installations*

In substation installation, it shall be checked whether:

a) the installation has been carried out in accordance with the approved drawings;

b) phase to phase and phase to earth clearances are provided as required;

c) all equipment are efficiently earthed and properly connected to the required number of earth electrodes;

d) HV and LV switchgears are all vermin and damp-proof and all unused openings or holes are blocked properly;

e) the required ground clearance to live-terminals is provided;

f) suitable fencing is provided (with gate, lockable arrangements) and suitably earthed as required;

g) there is no vegetation in outdoor substation;

h) the required number of caution boards, firefighting equipment, operating rods, insulation mats of appropriate rating, etc, are kept in the substation;

j) in case of indoor substation sufficient ventilation and draining arrangements are made;

k) all cable trenches are provided with non-inflammable covers;

m) free accessibility is provided for all equipment for normal operation;

n) all name plates are fixed and the equipment are fully painted;

p) all construction materials and temporary connections are removed;

q) oil-level, busbar tightness, transformer tap position, etc are in order;

r) earth pipe troughs and cover slabs are provided for earth electrodes/earth pits and the neutral earth pits and earth pits of lightning protection system are marked for easy identification;

s) earth electrodes are of GI pipes or CI pipes or copper plates or Cu bonded rods as per Table 2 and, for earth connections, brass bolts and nuts with lead washers are provided in the pipes/plates;

t) earth pipe troughs and oil sumps/pits are free from rubbish and dirt and stone jelly, and the earth connections are visible and easily accessible;

u) earthing system designed are periodically checked for permissible limits of step and touch potential;

w) the earth busbars have tight connections and corrosion-free joint surfaces;

y) operating handle of protective device are provided at an accessible height from ground, that is, 300 mm to 1 800 mm;

z) adequate headroom is available in the transformer room for easy topping-up of oil, maintenance, etc;

aa) safety devices, horizontal and vertical barriers, busbar covers/shrouds, automatic safety shutters/doors interlock and handle interlock are safe and in reliable operation in all panels and cubicles;

bb) clearances in the front, rear and sides of the main HV, LV and sub-switch boards are adequate;

cc) the switches operate freely, the 3 blades make contact at the same time, the arcing horns contact in advance, and the handles are provided with locking arrangements;

dd) insulators are free from cracks, and are clean;

ee) in oil cooled transformers, leakage of oil, if any should be checked;

ff) connections to bushing in transformers are tight and have good contact;

gg) bushings are free from cracks and are clean;

hh) accessories of transformers like breathers, vent pipe, buchholz relay, etc, are in order;

jj) connections to gas relay in transformers are in order;

kk) oil and winding temperature are set for specific requirements in transformers;

mm) in case of cable cellars, adequate arrangements made to pump out water that has entered due to seepage or other reasons;

nn) all incoming and outgoing circuits of HV and LV panels are clearly and indelibly labelled for identification;

pp) no cable is damaged;

qq) there is adequate clearance around the equipment installed; and

rr) cable terminations are proper.

9.2.2.2 *High/low voltage installation*

In high and low voltage installations, it shall be checked whether:

a) all blocking materials that are used for safe transportation in switchgears, contactors, relays, etc are removed;

b) all connections to the earthing system are feasible for periodical inspection;

c) sharp cable bends are avoided and cables are taken in a smooth manner in the trenches or alongside the walls and ceilings using suitable support clamps at regular intervals;

d) suitable circuit breaker or lockable push button is provided near the motors/apparatus for controlling supply to the motor/apparatus in an easily accessible location;

e) two separate and distinct earth connections are provided for the motor/apparatus;

f) control switch-fuse is provided at an accessible height from ground for controlling supply to overhead travelling crane, hoists, overhead busbar trunking;

g) the metal rails on which the crane travels are electrically continuous and earthed and bonding of rails and earthing at both ends are done;

h) four core cables are used for overhead travelling crane and portable equipment, the fourth core being used for earthing, and separate supply for lighting circuit is taken;

j) if flexible metallic hose is used for wiring to motors and other equipment, the wiring is enclosed to the full lengths, and the hose secured properly by approved means;

k) the cables are not taken through areas where they are likely to be damaged or chemically affected;

m) the screens and armours of the cables are earthed properly;

n) the belts of the belt driven equipment are properly guarded;

p) adequate precautions are taken to ensure that no live parts are so exposed as to cause danger;

q) ammeters and voltmeters are tested;

r) the relays are inspected visually by removing covers for deposits of dusts or other foreign matter;

s) wherever bus ducts/rising mains/overhead bus trucking are used, special care being taken for earthing the system, and all tap off points are provided with adequately rated protective device like MCB, MCCB, fuses, RCCB/RCD, SPD (*see* **11**), etc;

t) all equipment are weather, dust and vermin proof; and

u) any and all equipment having air insulation as media maintain proper distances between phases, phase to neutral, phase to earth and earth to neutral.

9.2.2.3 *Overhead lines*

For overhead lines, it shall be checked whether:

a) all conductors and apparatus including live parts thereof are inaccessible;

b) the types and size of supports are suitable for the overhead lines/conductors used and are in accordance with approved drawing and standards;

c) clearances from ground level to the lowest conductor of overhead lines, sag conditions, etc, are in accordance with the relevant Indian Standards and the CEA Regulations;

d) where overhead lines cross the roads or cross each other or are in proximity with one another, suitable guarding is provided at road crossings and to also protect against possibility of the lines coming in contact with one another;

e) every guard wire is properly earthed;

f) the type, size and suitability of the guarding arrangement provided is adequate;

g) stays are provided suitably on the overhead lines, as required and are efficiently earthed or provided with suitable stay insulators of appropriate voltages;

h) anti-climbing devices and danger board/caution board notices are provided on all HV supports;

j) clearances along the route are checked and all obstructions, such as, trees/branches and shrubs are cleared on the route to the required distance on either side;

k) clearance between the live conductor and the earthed metal parts are adequate;

m) for the service connections tapped-off from the overhead lines, cut-outs of adequate capacity are provided;

n) all insulators are properly and securely mounted, also they are not damaged;

p) all poles are properly grouted/insulated so as to avoid bending of pole towards tension; and

q) steel poles, if used is properly earthed.

9.2.2.4 *Lighting and convenience power circuits*

For the lighting and convenience power circuits, it shall be checked whether:

a) wooden boxes are avoided;

b) panels are provided for mounting the lighting boards and switch controls, etc;

c) neutral links are provided in double pole switch-fuses which are used for lighting control and no protective device (such as MCB, MCCB, fuses, RCCB/RCD, etc) is provided in the neutral;

d) the plug points (6 A) in the lighting circuit are all of 3/5-pin type, the third pin being properly earthed;

e) the plug points (16 A) in the lighting circuit are all of 3/6-pin type, the third pin being properly earthed;

f) tamper-proof interlocked switch socket and plug are used for locations easily accessible;

g) wiring for lighting is taken in surface or recessed conduit and conduit properly earthed, or alternatively, armoured cable wiring is used in factories;

h) a separate earth wire is run in the lighting installation to provide earthing for plug points, fixtures and equipment;

j) proper connectors and junction boxes are used wherever joints are to be made in conductors or cross-over of conductors takes place;

k) clear and permanent identification marks are painted in all distribution boards, switchboards, sub-main boards and switches as necessary;

m) the polarity has been checked and all protective devices (such as MCB, MCCB, fuses, RCCB/RCD, etc) and single pole switches are connected on the phase conductor only and wiring is correctly connected to socket-outlets;

n) spare knockouts are provided in distribution boards and switch fuses are blocked;

p) the ends of conduits enclosing the wiring leads are provided with ebonite or other suitable bushes;

q) the fittings and fixtures used for outdoor use are all of weather-proof construction and of suitable IP and IK rating and similarly, fixtures, fittings and switchgears used in the hazardous area, are of flame-proof application;

r) proper terminal connectors are used for termination of wires (conductors and earth leads) and all strands are inserted in the terminals;

s) flat ended screws are used for fixing conductor to the accessories;

t) as desirable, flat washers backed up by spring washers are used for making end connections; and

u) all metallic parts of installation, such as conduits, distribution boards, metal boxes, etc, have been properly earthed.

9.3 Testing of Installation

9.3.1 *General*

After inspection, the tests as indicated in **9.3.2** shall be carried out, before an installation or an addition to the existing installation is put into service. Any testing of the electrical installation in an already existing installation shall commence after obtaining permit to work from the Engineer-in-Charge and after ensuring the safety provisions.

Testing of the installations will cover the testing of equipment, connections, cables, switchgear, protective devices, circuit breakers and the associated relays, measuring instruments and earthing. Periodicity of testing should not exceed six months. More frequent testing may be prescribed for complex installations and installations feeding sensitive loads. Due date of next test cycle should be displayed on the equipment.

The tests as detailed in **4.3** under Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations' of good practice [D-2(1)] shall be carried out before energizing the installation and shall be recorded in the proforma attached under Annex E.

9.3.2 *Testing*

9.3.2.1 *Switchboards*

HV and LV switchboards shall be tested in the manner indicated below:

a) All high voltage switchboards shall be tested for insulation (phase to phase, phase to earth and body to earth);

b) In case of oil type switchboards, dielectric strength of oil shall also be tested;

c) All earth connections shall be checked for continuity;

d) The operation of the protective devices shall be tested by means of secondary or primary injection tests;

e) The operation of the breakers shall be tested from all control stations;

f) Indication/signalling lamps shall be checked for proper working;

g) The operation of the breakers shall be tested for all interlocks;

h) The closing and opening timings of the breakers shall be tested wherever required for auto-transfer schemes;

j) Contact resistance of main and isolator contacts shall be measured; and

k) The specific gravity and the voltage of the control battery shall be measured.

9.3.2.2 *Transformers*

Transformers shall be tested in the manner indicated below:

a) All commissioning tests shall be in accordance with good practice [D-2(16)]; and

b) Insulation resistance on HV and LV windings shall be measured at the end of 1 min, as also at the end of 10 min of measuring the polarization index. The absolute value of insulation resistance should not be the sole criterion for determining the state of dryness of the insulation. Polarization index values should form the basis for determining the state of dryness of insulation. For any class of insulation, the polarization index should be greater than 1.5. In case of oil type transformers, the dielectric strength shall be tested.

9.3.2.3 *Cables*

Cable installations shall be checked as given below:

a) It shall be ensured that the cables conform to the relevant Indian Standards. Tests shall also be done in accordance with good practice [D-2(23)]. The insulation resistance before and after the tests shall be checked;

b) The insulation resistance between each conductor and against earth shall be measured. The insulation resistance varies with the type of insulation used and with the length of cable;

c) Physical examination of cables shall be carried out;

d) Cable terminations shall be checked; and

e) Continuity test shall be performed before charging the cable with current.

9.3.2.4 *Motors and other equipment*

The following tests shall be made on motor and other equipment.

The insulation resistance of each phase winding against the frame and between the windings shall be measured. Megger of 500 V or 1 000 V rating shall be used. Star points should be disconnected. Minimum acceptable value of the insulation resistance varies with the rated power and the rated voltage of the motor.

The following relation may serve as a reasonable guide:

$$R_i = \frac{20 \times E_n}{1\,000 + 2P}$$

where

R_i = insulation resistance in MΩ at 25 °C;

E_n = rated phase to phase voltage; and

P = rated power, in kW.

If the resistance is measured at a temperature different from 25 °C, the value shall be corrected to 25 °C.

The insulation resistance as measured at ambient temperature does not always give a reliable value, since moisture may have been absorbed during shipment and storage. When the temperature of such a motor is raised, the insulation resistance will initially drop considerably, even below the acceptable minimum. If any suspicion exists on this score, motor winding shall be dried out.

9.3.2.5 *Wiring installation*

The following tests shall be done:

a) The insulation resistance shall be measured by applying between earth and the whole system of conductor or any section thereof with all fuses in place and all switches closed, and except in earthed concentric wiring, all lamps in position or both poles of installation otherwise electrically connected together, a d.c. voltage of not less than twice the working voltage, provided that it does not exceed 500 V for low voltage circuits. Where the supply is derived from three-wire (a.c. or d.c.) or a poly-phase system, the neutral pole of which is connected to earth either direct or through added resistance, the working voltage shall be deemed to be that which is maintained between the outer or phase conductor and the neutral;

b) The insulation resistance in mega-ohm (MΩ) of an installation measured as in (a) shall be not less than 50 divided by the number of points on the circuit, provided that the whole installation need not be required to have an insulation resistance greater than 1 MΩ;

c) Control rheostats, heating and power appliances and electric signs, may, if desired, be disconnected from the circuit during the test, but in that event the insulation resistance between the case of framework, and all live parts of each rheostat, appliance and sign shall be not less than that specified in the relevant Indian Standard or where there is no such specification, shall be not less than 0.5 MΩ; and

d) The insulation resistance shall also be measured between all conductors connected to one pole or phase conductor of the supply and all the conductors connected to the middle wire or to the neutral on to the other pole of phase conductors of the supply. Such a test shall be made after removing all metallic connections between the two poles of the installation and in these circumstances the insulation resistance between conductors of the installation shall be not less than that specified in (b).

9.3.2.6 *Earthing*

The following tests shall be done for the earth electrodes (earthing):

a) The earth resistance of each electrode shall be measured;

b) Earth resistance of earthing grid shall be measured;

c) All electrodes shall be connected to the grid and the earth resistance of the entire earthing system shall be measured; and

d) After receipt of electrical supply, the fault loop impedance shall be measured to ensure that the value obtained is minimal and to comply with the provisions under the CEA Regulations No. 43 (*see* Annex B) and good practice [D-2(11)].

These tests shall preferably be done during the summer months.

9.3.2.7 *Lightning protection system* (*LPS*)

Lightning protection system (LPS) verification shall be done as per accepted standards [D-2(52)] including the details of calculation of risk assessment and the selection of lightning protection level. The objective of inspection of LPS shall include whether all components of LPS are in good condition and capable of performing their designed functions and there is no corrosion. Inspections shall include checking of embedded electrodes, earth resistance values for the earth termination system, conditions of connections, equipotential bonding and fixings. The complete LPS shall be tested by opening one of the test joints to ensure that the total LPS system does not have more than 10 Ω to the earth. In case, if the total resistance of LPS exceeds 10 Ω then additional down conductors should be planned from air terminations and mesh conductor system, based on rolling sphere/mesh/protection angle method, to the earth so that the overall resistance is brought down to below 10 Ω by uniformly placing the additional down conductors. The testing shall also comply with all provisions included in **4.4** of Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations', of good practice [D-2(1)] and IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'.

9.4 Checklist for Installation

The testing and submission of reports of all electrical installations shall be done as detailed in the model form in Annex D of Part 1 'General and common aspects, Section 17 Erection and pre-commissioning — Testing of installations, of good practice [D-2(1)]. In addition, the checklist covering set of checks for installation, handing over and commissioning of typical equipment of a substation as given in Annex E is for general reference and use. For other allied installation, the proper checklist should be developed keeping in view the type of loads, quality of service, environmental conditions and operating requirements of redundancy and reliability.

9.5 Completion Certificate

On completion of an electrical installation (or an extension to an installation) a certificate shall be furnished by the contractor, counter-signed by the certified supervisor under whose direct supervision the installation was carried out. This certificate shall be in a prescribed form as required by the local electric supply authority. One such recommended form is given in Annex F. This is a general form giving the minimum basic requirements and the items in the form have to be augmented keeping in view the features of the particular equipment, system or installation and environmental conditions of its operation.

10 ALLIED/MISCELLANEOUS SERVICES

10.0 General

The allied and miscellaneous services covered under this are special type of services which may or may not be part of the main electrical installations of the building but are services which are used in the building and may sometimes use electrical supply to run the services. This also covers the photovoltaic and other rooftop generation system which will get integrated with the electrical supply system for the building. This clause also covers inverters, UPS and small portable generation system to meet certain emergency requirements. The chapter also covers broadly the provisions for EV charging, Data centres and multeystorey buildings beyond 24 storeys. This clause provides the requirements related to these services.

10.1 Telecommunication and Information and Communication Technology Services

10.1.1 *Telephone/Intercom Services*

10.1.1.1 House wiring of telephone subscribers in small buildings may be on the surface of walls or desirably, in a concealed manner through conduits. In large multi-storeyed buildings intended for commercial, business and office use as well as for residential purposes, wiring for telephone connections should be done in a concealed manner through conduits. The requirements of telecommunication facilities like telephone connections, private branch exchange and intercommunication facilities, should be planned well in advance so that suitable provisions are made in the building plan in such a way that the demand for telecommunication services in any part of the building at any floor are met at any time during the life of the building.

10.1.1.2 Layout arrangements, methods for internal block wiring and other requirements regarding provisions of space, etc may be decided depending on the number of phone outlets and other details in consultation with engineer/architect and user. *See also* Part D 'Building Services, Section 6 Information and Communication Enabled Installations' of this NBCS.

10.1.2 *Fibre to The Home* (*FTTH*) *Services*

Fibre to the home (FTTH) services provide high speed internet, television and phone directly to a home or business using fibre optic cables. These services provide significantly higher bandwidth, faster speeds and more reliable connections compared to traditional broadband with advanced features such as high-definition video streaming, online gaming, etc. Instead of copper wires, FTTH uses thin strands of glass or plastic fibre(s) which transmit data as light signals.

Installing FTTH network is straightforward fibre installation with some additional components like splitters, an optical network terminal (ONT) and associated hardware at the customer premises. For more details, reference may be made to Part D 'Building Services, Section 6 Information and Communication Enabled Installations' of this NBCS.

10.1.3 *Information and Communication Technology Services including Computer Networking*

See Part D 'Building Services, Section 6 Information and Communication Enabled Installations' of this NBCS.

10.2 Public Address System

See Part F 'Fire and Life Safety' of this NBCS.

10.3 Emergency and Standby Power Supply Systems

10.3.1 *General*

Use of electricity has grown tremendously and for various activities the dependency on electricity has increased to such an extent as to cause serious problems even with loss of electrical power for a very short time. As a result, a wide variety of alternate sources of electricity are being used in our built environment.

The different alternative sources of power are the uninterrupted power supply (UPS) system, inverter, diesel (CNG/LPG) generator sets, petrol/kerosene oil generator sets and bio-gas generator sets.

In addition to the above, there is a proliferation of power sources, such as solar photo-voltaic cells, wind generators, bio-mass and waste-based power plants, etc, primarily oriented towards reduction of the environmentally harmful CO_2 emissions.

These systems give electricity during the periods of the failure of the conventional grid based public energy system and keep our critical systems in continued operation. However, introduction of more than one source of electrical power results in safety related issues. For safety from electrical shock to human beings or livestock, the hazard is not just dependent on the main high-powered source such as the grid, but the hazard is the same from a low powered source also. Shock from a small 20 W inverter can be as dangerous as a shock from the grid with megawatts of power at the back. As such precautions from the angle of safety apply equally to all sources of power. Electric shock hazards are dependent on the system voltage and as such even a low capacity generator or an inverter (of capacity 100 VA) poses the same level of shock hazard as a multi-kilovolt ampere capacity generator and all protection provisions (such as earthing, earth leakage and overload breakers) shall be provided as done for a large capacity system.

Power devices contain fuel, batteries which are points of concentrated sources of energy constrained in a small place. Any unintended improper release of this bottled up energy can unleash devastating consequences, such as fire and as such care is required in location which houses any of these sources of electrical power and its associated components.

10.3.2 *Uninterrupted Power Supply* (*UPS*) *System*

UPS is an electrical device providing an interface between the mains power supply and sensitive loads (computer systems, instrumentation, etc). The UPS supplies sinusoidal a.c. power free of disturbances and within strict amplitude and frequency tolerances. It is generally made up of a rectifier/charger and an inverter together with a battery for backup power in the event of a mains failure with virtually no time lag.

In general UPS system shall be provided for sensitive electronic equipment like computers, printers, fire alarm panel, public address system equipment, access control panel, EPABX, etc, with the following provisions. In addition, provisions as given in good practice [D-2(1)] shall also be followed.

a) Isolation transformers may be provided for UPS systems to provide higher grade of power supply quality to the loads fed by the UPS.

b) UPS shall have dedicated neutral earth pits of IT system. This earth pit shall be interconnected with other earth pits below soil for equipotential bonding and comply with the provisions of good practice [D-2(11)].

c) Adequate rating of protective devices such as MCB, MCCB, fuses, RCD, etc shall be provided at both incoming and outgoing sides.

d) UPS room shall be provided with adequate ventilation and/or air conditioning as per requirement, to avoid thermal runaway.

e) For all 3 phase UPS, 4 pole circuit breaker (CB) shall be used and for all 1 phase UPS, double pole CB shall be used.

f) The system shall confirm to set of accepted standards [D-2(53)].

10.3.3 *Inverter*

In general inverter system shall be provided for house lighting, shop lighting, etc.

NOTE — While a UPS system is provided to maintain power supply without any break even in the event of a failure of the incoming power supply, an inverter system is provided where a short break is acceptable.

Inverter systems also have a battery bank to supply power during the failure of the main power supply and the battery charged through a rectifier. The following provisions shall apply to inverter systems:

a) Adequate rating of protective devices such as MCB, MCCB, fuses, RCD, etc, shall be provided at both incoming and outgoing sides;

b) Earthing shall be done in accordance with good practice [D-2(11)];

c) Adequate ventilation space shall be provided around the battery section of the inverter; and

d) Care shall be taken in circuit design to keep the connected load in such a manner that the demand at the time of mains failure is within the capability of the inverter.

NOTE — If the inverter fails to take over the load at the time of the mains failure, the purpose of providing the inverter and battery backup is defeated.

10.3.4 The following provisions shall apply to both inverter and UPS systems:

a) Circuits which are fed by the UPS or inverter systems should have suitable marking to ensure that a worker does not assume that the power is off, once the mains have been switched off from the DB for maintenance;

b) Electric shock hazards are dependent on the system voltage and as such even a low capacity generator or an inverter (of capacity 100 VA) poses the same level of shock hazard as a multi-kilovolt ampere capacity generator and all protection provisions (such as safety earthing, earth leakage and overload breakers) shall therefore be provided as done in case of a large capacity system;

c) UPS and inverter systems should be provided with protection to shut the output during abnormal conditions, such as overload or short circuit. Such systems may also have the choice of auto-restoration after a preset time delay and an ultimate lock-out after a number of restoration attempts. Warning should be displayed wherever the possibility of automatic restoration is possible;

d) Batteries that go with UPS and inverter systems are required to be placed in well-ventilated spaces as oxygen and hydrogen gasses are produced in the batteries, which unless ventilated, can cause explosive conditions; and

e) The flooring for the battery room should be with acid (or alkali as the case may be) resistant tiles or coating.

10.4 Building Management System (BMS)

10.4.1 *General*

Building management system (BMS) is a computer-based system for monitoring and managing the building's mechanical, electrical and electro-mechanical systems including electrical substation equipment, HVAC, lighting, energy management, lifts, escalators, fire safety, security systems car parking management, etc. By its usage, building operations are optimized, energy efficiency is enhanced besides ensuring performance of all systems, safety and improving occupant's comfort.

10.4.2 *Working*

BMS provides a central platform for effective monitoring and efficient management of various building services. It allows remote monitoring and control from a single desk/location.

Separate standalone monitoring and control system provided for each service like electrical sub-station equipment; solar power and renewable energy system; standby generation system; HVAC system for control of operation and monitoring of all equipment including comfort conditions like temperature, humidity and air quality; fire protection system combining fire detection and firefighting arrangements, public address system (evacuation system); security system consisting of access control, CCTV, security fencing, water management system; etc can all be further integrated with BMS for remote monitoring, control and management .

By using sensors, it collects data on various parameters of all systems including comfort conditions of individual rooms/areas such as temperature, humidity, air quality, energy consumption based on occupancy level and adjusts system settings automatically for optimization of performance and eliminating wastage of energy thereof. Additionally, it can also spot inconsistencies and can send alerts to maintenance engineers to address the problems promptly.

The software of BMS also includes various features of reporting and analytics for improving building performance and identification of improvement areas. By integrating it with other building systems a more comprehensive and efficient solution of building management can be attained.

10.4.3 *Compliance with Regulations and Standards*

Where mandated, the relevant energy efficiency and safety standards prescribed by the Bureau of Energy Efficiency (BEE), Part F ‘Fire and Life Safety’ of this NBCS and other relevant parts and sections of this NBCS should be complied with. It should also be capable of seamless integration and exchange of data with other systems and devices functioning within the building. Designed for optimal performance, this system should be protected against unauthorized access and cyber threats.

10.5 Security System

Security system may comprise an integrated closed circuit television system, access control system, perimeter protection systems, movement sensors, etc. These have a central control panel, which has a defined history storage capacity. This main control panel may be located where it can be manned suitably with provision for integration to the fire detection and alarm system when considered necessary.

These may be considered for high security areas or large crowded areas or complexes. High security areas may consider uncorded, high-resolution, black and white cameras in place of coloured cameras. These cameras may be accompanied or automatically controlled with movement sensors. Cameras may be linked to access controls so that proper recording of the movement at the points of access to high security areas is maintained.

Access control may be provided for entry to high security areas. The systems may have proximity card readers, magnetic readers, etc.

NOTE — *See* relevant guidelines/best practices (Handbook on ‘Asset and Facility Management in Buildings’) for provisions relating to security system.

10.6 Car Parking Area

10.6.1 *Car Parking Management System*

Wherever car park management system is provided in multi-level parking or other parking lots with features of boom barriers, pay and display machines (manned or unmanned type) and parking guidance system (for displaying number of car spaces vacant on various floors, direction of entry and exit, etc), the electrical provisions for the same shall be adequately backed with UPS for protection of vehicle and for efficient car parking management.

10.6.2 *Emergency Lighting in Basement Car Parking*

Lighting at suitable points should be provided on the passageway of the cars so that the public in the area can safely move away from the passage in the event of power failure.

10.7 Electrical Vehicle Charging

10.7.1 *General*

Increased use of electric vehicles (EVs) here resulted in the need for providing adequate EV charging infra structure in the built environment. It is recommended to provide for EV charging infrastructure for 20 percent of the parking area of buildings. Provisions of charging infra structure should be adequately increased such that slow charging of vehicles in parked condition for long duration can be undertaken in parking area of residential, office, institutional and in about 40 percent of commercial parking areas. Further, considering the rising status of EV adoption, the initial proposal of 20 percent parking space allocation for EVs may be increased up to 30 percent, say by 2030.

10.7.2 *Charging Arrangement*

Standards for charging of electric vehicles presently restrict the charging arrangements to open area and upto first basement with provisions of necessary sprinkler arrangement for firefighting services and necessary water curtain/compartment for fire protection. The charging infrastructure in podium/above ground parking do not provide significant risk considering the availability of cross ventilation in most such parking arrangements. However, with increased demand for charging infrastructure, the use of other basement may have to be considered. The charging arrangement in parking should be segregated individually for two wheelers and for four wheelers.

Charging arrangement for cars (four wheelers), can be provided both in ground level and first basement having adequate ventilation. However, two wheelers need to be segregated from four wheelers in parking and charging arrangement; and should be restricted to only ground floor level with adequate ventilation and firefighting arrangement using water mist fire system. The use of water mist fire system having up to 85 percent reduced water required for fighting the fire makes it a better option for both cooling effect and controlling electric fire in electric vehicles. The parking facility for the two wheelers shall be so arranged that not more than two vehicles should be parked side by side for charging and a space of not less than 1 m should be kept between the vehicle groups. The charging infrastructure should be provided in groups of four outlet points with three pin socket-outlets of 10 A capacity with 30 mA RCD control for each socket-outlet with metering arrangement. The spacing of the charging outlet shall be at a distance of 4 m/5 m on the basis of the parking arrangement provided.

In residential complexes, charging of vehicles would be preferred while parked for the day. Accordingly, it will not be advisable to keep parking of electrical vehicles contiguous to/vicinity of the petrol/diesel/CNG gas based vehicles. Hence separate segregated parking lot is required to be created for EVs (four wheelers) preferably in ground level/first basement with compartmentalized and water sprinkler fire barrier arrangement (for the basement). Suitable fire alarm/fire detection and fire ventilation system shall also be provided. Building professionals can consider developing suitable independent tower arrangement (podium) for car parking of electrical vehicles among the residential/commercial tower blocks and the building plan approval authorities can consider for suitable exemptions from the permitted area for construction.

It is necessary to provide for at least 20 percent of the vehicles in a residential complex with electric charging using a slow charging arrangement, as per Government guidelines. Hence each charging point shall be provided with two numbers 15 kW single phase two wire and two numbers 15 kW three phase four wire outlets controlled by suitable MCCB and RCD of 30 mA capacity with suitable metering arrangement. Each outlet shall be controlled by appropriate RCD/RCBO with rated residual operating current not exceeding 30 mA and necessary metering arrangement. In case of covered parking (first basement/podium), each location shall be provided with two sets of charging points. Each charging point shall be provided with access to portable firefighting arrangement within 15 m. The charging arrangement will be initially permitted to first basement only with compartment/sprinkler based water curtain and normal ventilation requirement of 15 to 18 air changes per hour with gas monitoring system. No charging infrastructure or EV parking shall be provided in the last basement. EV charging is a critical application from the point of fire safety, it is essential that all cables used for feeding the

EV charging stations of both two wheelers and four wheelers shall be of copper conductors for sizes upto 50 mm^2 4 core cables to prevent overheating with adequate earthing connections. All terminals shall be crimped to prevent loose connections.

It is also necessary to provide in the open area of the residential complex slow and fast charging points for 5 percent of the number of flats subject to minimum of two points with independent meter reading so that in case the number of electric vehicles increase in the complex, the users can take advantage of the fast charging arrangement. The arrangement shall be with two numbers 15 kW single phase two wire, 15 kW three phase four wire and 30 kW four wire outlets each controlled by MCCB of appropriate capacity, RCD of 30 mA capacity and metering arrangement.

10.7.3 As far as possible, EV charging arrangement shall be provided with independent transformer to ensure that harmonics created during EV charging do not interfere with power supply to other system and also to maintain other electrical parameters. A separate transformer is recommended for EV charging points where the total connected load exceeds 100 kW. It shall be ensured that EV charging is not fed from diesel generator sets or gas generators or similar fuel based power generating systems.

In case of hospitals, commercial complexes and big residential complexes, necessary public charging infrastructure shall be provided with 15 kW/30 kW/60 kW charging points duly controlled by appropriate capacity RCD/RCBO having rated residual current of 30 mA, necessary metering arrangement for each point with fast charger provisions. Spaces for EVs to be provided with charging facility for at least 20 percent of total vehicles. However, in the first basement, charging points shall be of capacity less 15 kW only.

All charging points, cabling, etc shall comply with accepted standards [D-2(54)] and good practice [D-2(1)] and the *CEA* (*Measures Relating to Electric Supply and Safety*) *Regulations*, 2023, with special reference to Regulation No. 123, 124, 125, 126, 127 and 128 (*see* Annex B).

10.7.4 *Battery Swapping System*

Battery swapping systems provide quick, safe and reliable swapping of the swappable battery system (SBS) of electric vehicles preferably two wheelers. The batteries will be loaded in a battery swap station (BSS). The SBS are stored in a battery rack in the BSS. The battery swapping is carried out by means of appropriate manipulators. EV battery swap system consists of:

a) battery swap station (BSS);

b) supporting systems;

c) swappable battery system (SBS); and

d) power supply system.

See accepted standard [D-2(55)].

Battery swap system shall be permitted only on ground floor (or on above ground floors) or outside the building line including battery storage zone and battery charging zone.

10.7.5 *Charging Infrastructure for Electric Buses*

Each electric bus depot shall be equipped with a minimum of two direct current (d.c.) fast chargers, each rated at not less than 240 kW, conforming to power level 3 or level 4. Chargers shall comply with the following Indian Standards:

a) d.c. charging protocol for heavy-duty vehicles — as per accepted standards [D-2(56)].

b) High-power d.c. charging systems — as per accepted standards [D-2(57)].

c) EV-EVSE communication protocol (power line communication) — as per accepted standards [D-2(58)].

d) Connector and plug interface for M3 category vehicles — as per accepted standards [D-2(59)].

All installations shall conform to applicable safety norms as per the CEA Regulations and relevant Indian Standards.

10.7.6 *Safety Requirements for Residential Charging of Electric Vehicles*

10.7.6.1 *Scope*

This clause specifies safety requirements for the installation and use of electric vehicle (EV) charging systems in residential premises.

10.7.6.2 *General safety provisions*

a) Only devices listed by a qualified testing laboratory shall be used.

b) Manufacturer's instructions shall be read and complied with.

c) Chargers and cords shall be inspected for damage prior to use; damaged components shall not be used.

d) Use of extension cords with EV chargers shall not be permitted.

e) Charging equipment shall be installed in locations that are:

   1) away from high-traffic areas; and

   2) free from proximity to combustible materials.

f) Charging accessories shall be stored out of reach of children and animals when not in use.

g) Charging station outlets shall be covered to prevent ingress of water.

h) Prior to initiating home charging, a qualified electrician shall:

   1) assess the capacity of the existing electrical system; and

   2) install a dedicated circuit for the charging device.

10.7.6.3 *Precautionary measures*

Warning signs shall be prominently displayed to ensure the safety of maintenance personnel. The signage shall include the following statements:

a) The vehicle is equipped with high-voltage batteries;

b) Battery maintenance shall be undertaken only by the manufacturer or authorized personnel; and

c) Orange-coloured high-voltage cables shall not be touched.

10.8 Solar Photovoltaic Power Generating System in Buildings

10.8.1 *General*

Solar energy, which is available in two forms, heat and light, is a renewable and inexhaustible natural resource and can supplement/augment the depleting fossil fuel resources. Greenhouse gases and pollutant emissions which result from fossil fuel generation can also be offset by solar photovoltaic power generation. Most parts of the country receive good solar radiation of 4 kWh/m^2 to 7 kWh/m^2 per day and almost 300 sunny days in a year making solar PV system one of the most preferred renewable energy sources in the country.

10.8.2 Solar PV power generating system consists of components and subsystems that are used to convert incident solar radiation directly into electrical energy. The energy converter (namely, solar photovoltaic cells which convert solar energy directly into d.c. electric power) does not have moving parts and has a comparatively long lifetime. Also, it can be used in decentralized/distributed mode.

10.8.3 *Rooftop Solar System*

The energy received from the sun is converted into electricity through photovoltaic cells and is called solar photovoltaic electricity. This is done by solar photo-voltaic system consisting of solar PV panels containing group of solar cells which generate d.c. power and using inverter and other equipment of appropriate capacity, the d.c. power is converted into a.c. power. The complete system comprising of inverter, cables both d.c. as well as a.c., protection system, interconnections, etc, which together is known as balance of system. Thus, solar PV system consists of solar PV panel and balance of system. The basic unit of solar panel, that is, solar cells are semiconductors made of silicon. The PV cells are made using different technologies namely monocrystalline silicon, polycrystalline silicon, thin film, etc. The choice among them is based on efficiency, cost and application among other factors. (*See* Fig. 12).

The installation of rooftop solar PV system shall be in accordance with the Regulation No. 119 and 121 of *Central Electricity Authority* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 (*see* Annex B).

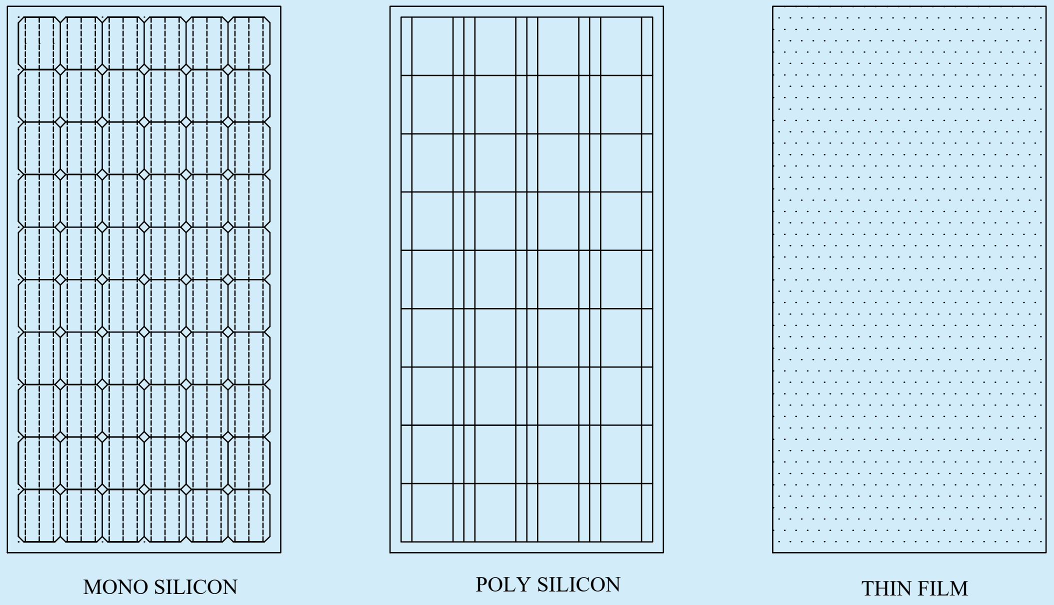


FIG. 12 MONO, POLY AND AMORPHOUS SILICON SOLAR CELL MODULE.

The solar PV systems used in buildings are of two types. Those installed on the rooftop are known as rooftop solar (RTS) system while solar PV system integrated with building facade is known as building integrated photovoltaic (BIPV) system.

10.8.3.1 *Solar PV components*

A brief description of all the major components are listed below:

Sl No.	*Component*	*Purpose*	*Technical Requirements and Compliance Standard*
(1)	(2)	(3)	(4)
i)	PV modules	To generate electricity utilizing the sunlight	Higher efficiency and lower cost meeting performance and safety standard as per accepted standards [D-2(60)] Modules to be connected in series and parallel to meet the inverter input requirement *See* accepted standard [D-2(61)].
ii)	Module mounting structure	To provide stability and rigidity to modules, to orient them towards the south direction at the appropriate latitude angle	Made of galvanized iron {*see* accepted standard [D-2(62)]} It should be capable of withstanding highest wind speed {*see* accepted standard [D-2(63)]}; and should have more than 25 years life.
iii)	d.c. and a.c. distribution boxes	To combine PV strings, to protect interconnections from corrosion, to house protective devices-fuses, surge protection devices, earth termination, breakers and isolators	GRP/FRP/powder coated aluminum enclosure. Should be weather and vermin proof, having required number of string entries and one cable exit having the required power rating and protection (IP 65) and as per relevant Indian Standard and good practice [D-2(1)]
iv)	Inverter	To convert d.c. power to a.c.	The inverter shall comply with all requirements as mandated in relevant Indian Standards, good practice [D-2(1)] and should produce grid quality power and be able to synchronize with it and having all mandated protection features of IP 54; accepted standards [D-2(64)]; and **5** of [D-2(65)] Interconnection and power quality shall be as per CEA/state regulation
v)	Cables (d.c. and a.c.)	To connect solar panels to inverter and enable safe flow of current	Copper cables with the required voltage rating and current carrying capacity shall be used for interconnection. The voltage drop should not exceed 2 percent. All d.c. cables shall be solar d.c. cable as per accepted standard [D-2(66)] and a.c. cables as per accepted standard [D-2(4)] and shall be copper conductor cable for sizes upto 50 mm^2.

Table (*Concluded*)

Sl No.	*Component*	*Purpose*	*Technical Requirements and Compliance Standard*
(1)	(2)	(3)	(4)
vi)	Earthing	To provide safe and alternate path for fault current and surge currents	Should be provided independently both on the d.c. and a.c. side of the circuit using copper/copper bonded steel down conductors {*see* good practice [D-2(11)]} and *CEA Safety Regulation*, 2023
vii)	Lightning protection system	To prevent damage caused by lightning surges	LPS should be provided as per accepted standard [D-2(52)]
viii)	Net meter	To measure net of energy exported to and imported from the grid	Installed by the DISCOM after calibration and sealing

10.8.3.2 *Solar PV systems classification*

There are mainly two types of PV systems namely grid tied system and the off-grid system.

In the grid tied system the building electrical loads are supported by both the solar power and the utility power. When solar power is insufficient to meet the load demand or is not available as during nights, power is drawn from the grid and when the generation is more, the excess is fed to the grid. A net meter is installed to measure the net of the energy drawn from and injected into the grid. The tariff for feeding excess power is typically fixed every year by the respective state regulatory commissions. A typical grid tie system is shown in Fig. 13.

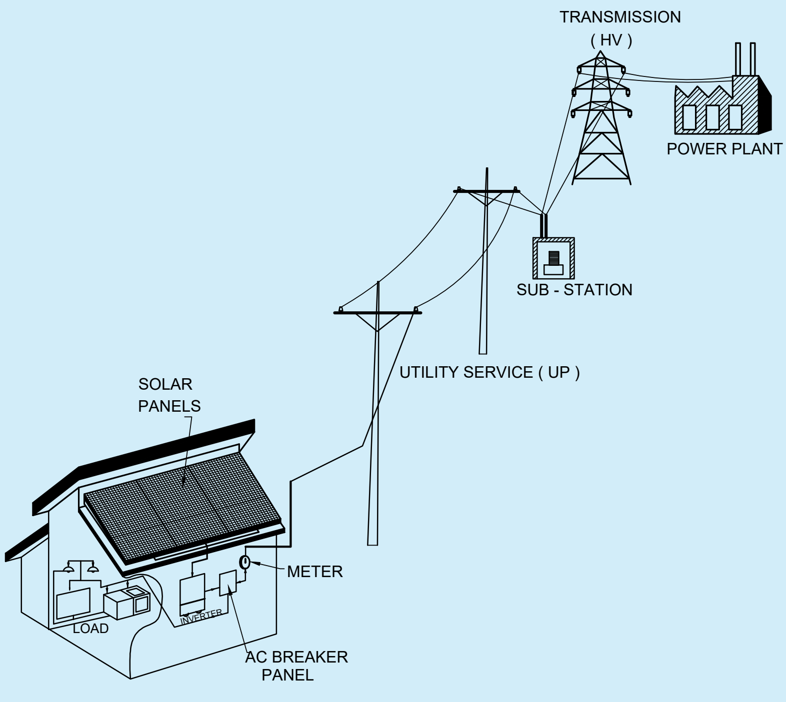


FIG. 13 GRID TIED SYSTEM

Off grid system is not connected to the grid and the loads are fully powered by solar. Since sunlight varies depending on the time of the day and the cloud levels and also since it is not available in the night, batteries are an inherent part of an off-grid system. These systems are used in places without electrical grid or by building owners desirous of being independent of grid connection (*see* Fig. 14).

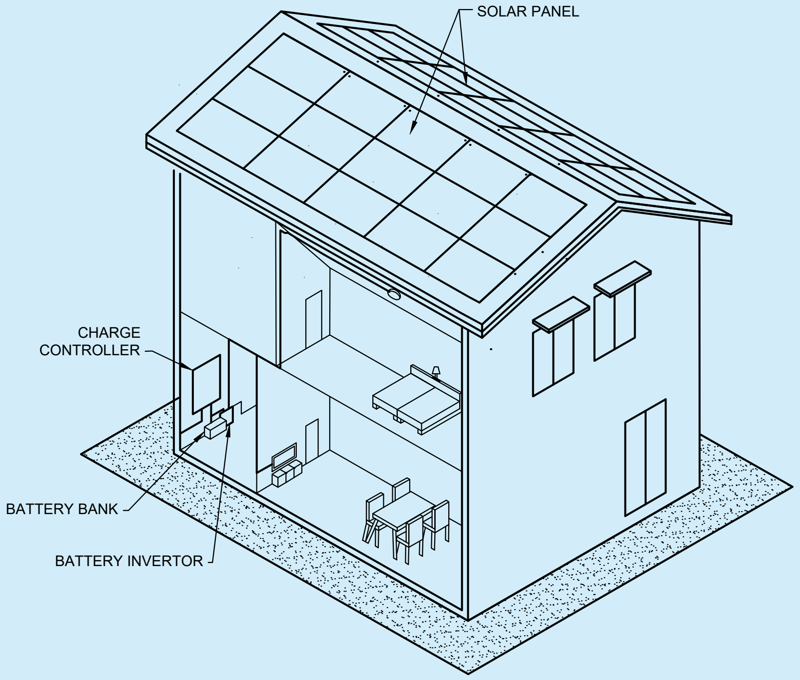


FIG. 14 AN OFF-GRID SYSTEM

10.8.3.3 *Protection system*

Since RTS is a power generating station on the topmost part of the building which is not in the view of occupants, it shall be protected against surge voltage, short circuit, lightning strike, earth fault, electrical fire, touch potential, shadow effect, etc by incorporating suitable protections into the system to avoid any mishap due to electrical fire. Special care should be taken to provide separate earth systems for d.c. power and a.c. power including interconnections and the earths will be interconnected at the earth level. The provisions for earthing and lightning protection shall be provided as per relevant Indian Standards and good practice [D-2(1)] including provision of necessary surge protection devices, where considered necessary.

10.8.3.4 *Solar PV system design*, *installation and maintenance*

Basic designing of the RTS depends upon the annual data relating to weather and solar insolation, shadow free area, generation potential, space required for other services, etc. For installation, all safety measures besides deployment of qualified personnel are required. Proper access to roof and distance between the row of solar panels is required for maintenance. System has to be maintained for guaranteed generation by regular cleaning of solar panels and inspection of the system for any abnormality from time to time.

10.8.3.5 *Regulations and standards relating to rooftop solar systems*

The system components shall comply with the relevant Indian Standards as amended up to date and work shall comply with the provisions of *Central Electricity Authority* (*Measures Related to Safety and Electric Supply*) *Regulations*, 2023 and good practice [D-2(1)]. Additionally, where mandated the use of solar d.c. cable as per accepted standard [D-2(67)] to prevent electrical fire on account of insulation failure due to exposure to UV radiations and weathering effect shall also be complied.

10.8.4 *Building Integrated Photovoltaic System* (*BIPV*)

BIPV system is an approach where conventional building components such as roofs and facades are replaced with multi-functional parts of a building envelope that integrate PV to generate electricity while still fulfilling their structural role.

BIPV refers to solar electric (photovoltaic) components that are integrated into the building envelope replacing the conventional building materials. Instead of being added on like traditional solar panels, BIPV models are designed to be part of building structure such as roofs, facades or skylights. This integration allows BIPV to serve both as a power generator and building material improving aesthetics and reducing reliance on conventional energy. Thus, there shall be regular meeting and consultation among the architects, engineers and builders to ensure seamless integration of BIPV into building design.

Building-integrated photovoltaics (BIPV) is an innovative technology that integrates seamlessly into a building envelope such as roofs, facades, windows and skylights. BIPV materials function both as conventional building elements as well as electricity producers (*see* Fig. 15).

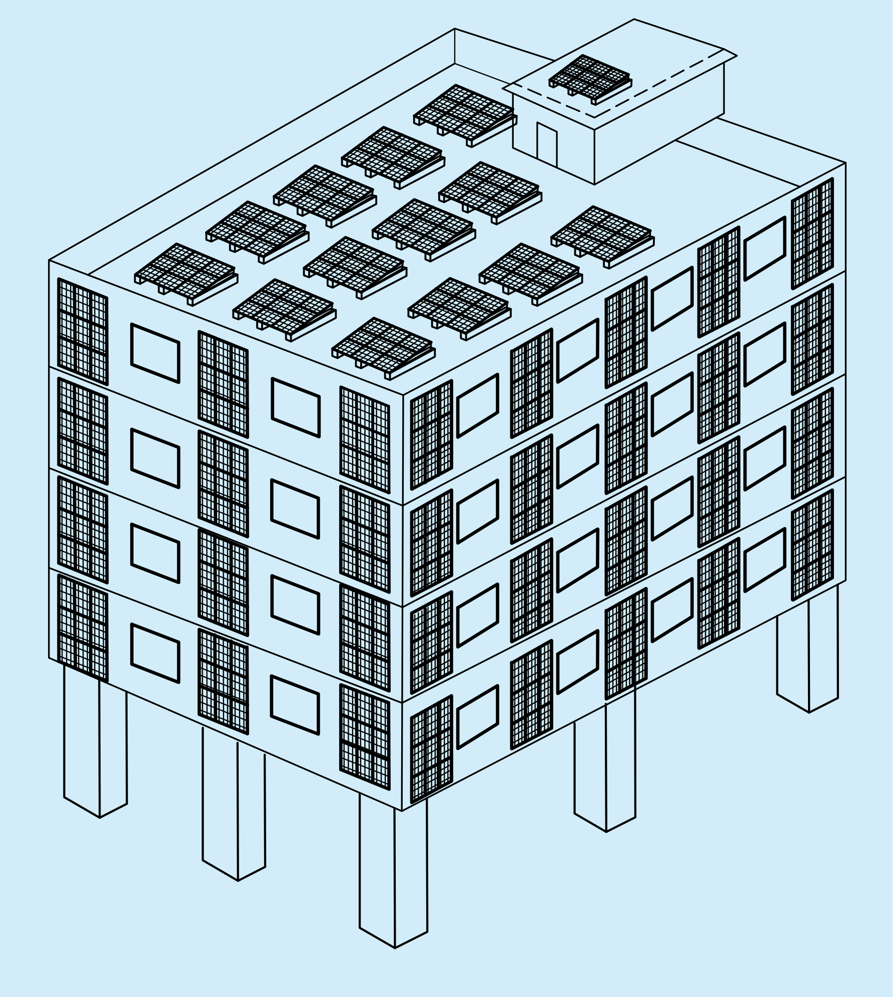


FIG. 15 BASIC STRUCTURE OF BIPV SYSTEM

10.8.4.1 *Design and planning of BIPV system*

a) *Building design integration* — Integration of BIPV technology into the building design is done, considering the factors like building orientation, shading, and structural integrity.

b) *Solar panel selection* — Solar panels are so selected to enable them to meet the structural and civil engineering requirements and are suitable for BIPV applications such as for facades, roof tiles, balustrades, window glazing/ atrium skylights.

c) *Energy generation* — The potential for energy generation is worked out based on system design optimization to achieve the maximum energy output for the components used.

10.8.4.2 *Installation and testing*

10.8.4.2.1 *Installation*

The installation of BIPV systems shall follow the manufacturer guidelines of the respective components and shall also comply with the relevant provisions of the structural requirements to ensure proper connectivity among the components. BIPV system design and installation should be optimized through BIM.

10.8.4.2.2 *Testing*

See accepted standards [D-2(67)].

a) *Performance testing* — After installation, the BIPV system shall be checked to ensure that the system generates power as envisaged in the planning stage and the energy produced shall be monitored for assessing the overall energy output and efficiency. Suitable solar power meter with data logger and performance monitoring software shall be used to monitor the performance of the BIPV system installed.

b) *Safety inspection* — Necessary earth test measurements, insulation resistance test, continuity test, measurement of voltage and current shall be measured using appropriate earth tester, insulation resistance tester and multi-meter to confirm that all connections have been made correctly and the electrical components function as planned in the planning stage.

c) *Electrical testing* — Confirm that electrical components function correctly and safely by performing insulation resistance test and continuity test and by measuring voltage and current, using a multimeter, insulation resistance tester and earth tester.

d) *Structural testing* — Necessary structural testing shall be carried out to ensure the BIPV system can withstand environmental stresses that may occur during the lifetime of the building and also withstand necessary sun, rain, wind, etc.

e) *Integration testing* — All systems of the building so installed may be integrated using integration software load testers and communication testing tools so that the complete building acts as a single envelope.

f) *Weatherproofing and sealing* — Necessary leakage deduction and sealant inspection shall be carried out as per weather proofing guidelines so that BIPV system is weatherproof and does not cause any leakage or damage to the building or to the structure as a whole.

10.8.4.3 *Safety and maintenance*

a) *Electrical safety* — Electrical safety shall be ensured by following relevant good practices and accepted standards.

b) *Fire safety* — Fire safety shall be ensured by using fire-resistant materials and following the good practices.

c) *Maintenance* — The BIPV system shall be inspected at least once in a month to ensure optimal performance and timely maintenance including replacement of damaged components if any without any delay.

10.9 Solar Water Heating System

10.9.1 *General*

This system is provided on the buildings to harness the energy of sun for heating water for kitchen and bathroom. Small systems from 100 litres to 300 litres are suitable for residential use while beyond this capacity are used for canteens, guest houses, hotels and hospitals. Located on top of building, the water is heated to 60 °C to 80 °C. However, in cloudy weather, back up electrical heater is sometimes required. Since it is provided on top of the building, it is necessary to provide equipotential bonding with the down conductor of LPS system as per relevant regulation and standards.

10.9.2 *Working*

When the sun's rays fall on the collector panel, which is a component of solar water heating system, a black absorbing surface (absorber) inside the collector absorbs solar radiation and transfers the heat energy to water flowing through it. Heated water is collected in a tank which is insulated to prevent heat loss. Circulation of water from the tank through the collectors and back to the tank continues automatically due to thermo-siphon system (*see* Fig. 16).

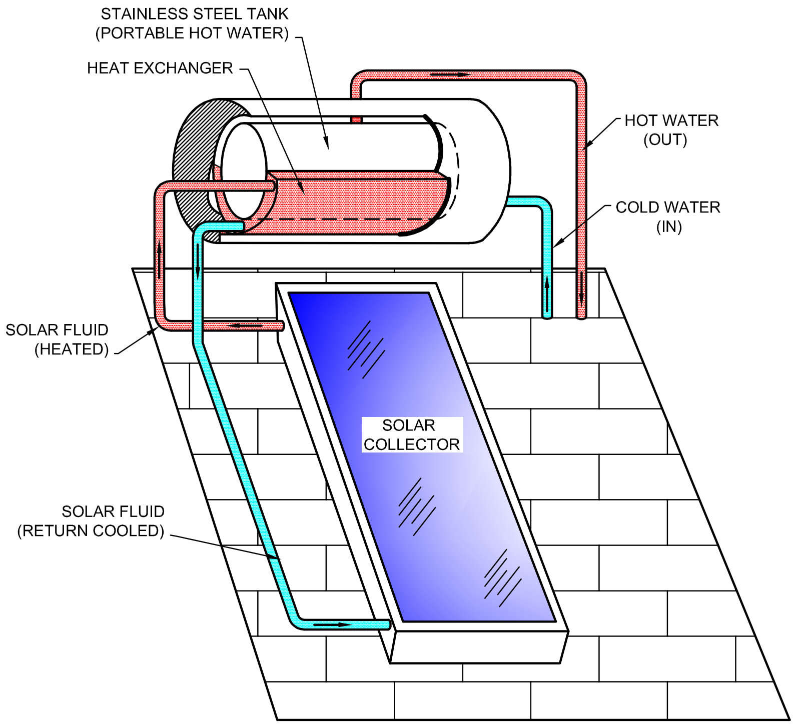
Solar water heating system are broadly categorized as active systems having pumps and controls, and passive systems relying on natural convection. Further, based on the collector system, solar water heaters can be of two types, namely flat plate collector type and evacuated tube collector type. In the first type, solar radiation is absorbed by flat plate collectors which consist of an insulated outer metallic box covered on the top with glass sheet. A black absorbing surface (absorber) inside the flat plate collectors absorbs solar radiation and transfers the energy to water flowing through it.

In the second type, the collector is made of double layer borosilicate glass tubes evacuated for providing insulation. The outer wall of the inner tube is coated with selective absorbing material. This helps absorption of solar radiation and transfers the heat to the water which flows through the inner tube.

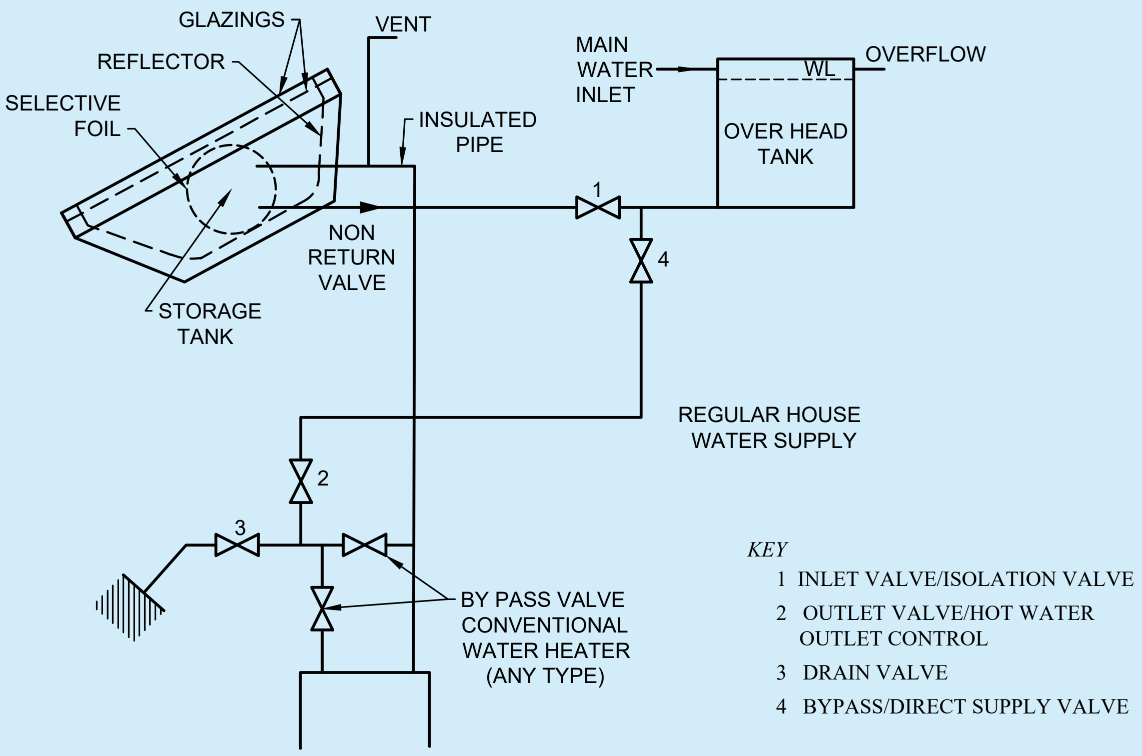
10.9.3 *Compliance with Regulations and Standards*

As mandated by the government solar collectors shall adhere to the accepted standard [D-2(68)]. Further, municipal corporation/local bodies may require solar water heating systems to be installed in new buildings in their territories including in renovated buildings.

Accepted standard [D-2(69)] caters to the requirements and components of flat plate collector type solar water heating system while accepted standard [D-2(70)] caters to glass evacuated tubes solar water heating system. Accepted standard [D-2(71)] caters to storage water tank for all glass evacuated tubes solar water heating systems.



16A SOLAR COLLECTOR BASED WATER HEATER



16B SCHEMATIC OF CONNECTION BETWEEN SOLAR BASED WATER HEATER OVERHEAD TANK BOTH CONNECTED TO THE CONVENTIONAL WATER HEATER

FIG. 16 WORKING OF SOLAR WATER HEATING SYSTEM

10.10 Aviation Obstacle Lights

High-rise buildings and structures such as chimneys and towers are potential hazards to aircraft. The provision of aviation obstacle lights (AOL) on tall buildings/structures is intended to reduce hazards to aircraft by indicating their presence. AOLs, low, medium or high intensity obstacle lights, or a combination of such lights, shall be provided on buildings of different heights as per the requirements of Annex 14 to the 'Convention on International Civil Aviation, Volume I Aerodrome Design and Operations', International Civil Aviation Organization (ICAO). A general arrangement of AOLs in case of group of buildings is given in Fig. 17.

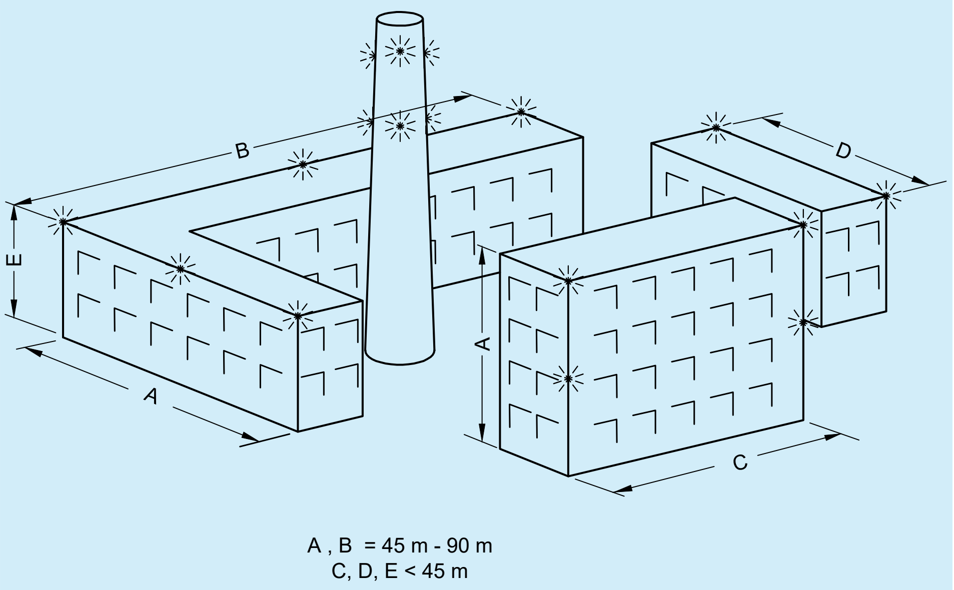


FIG. 17 TYPICAL ARRANGEMENT OF AVIATION OBSTACLE LIGHTS IN CASE OF GROUP OF BUILDINGS

10.11 Power/Electric Fence

10.11.1 *General*

Power/electric fence is provided in highly secured buildings to deter the intruders from climbing or sending a very low flying object, after proper risk assessment. It is provided on the boundary wall of the building and plays an important part in perimeter security by integrating with video surveillance software. This can also be used to protect the farms from the wild animals.

10.11.2 *Working*

This is a non-lethal fence consisting of wires carrying pulses of electric current and consists of an energizer producing regular current pulses. The supply lines conduct the pulses to the fence and takes back to the device from the earth stake. Wires should be good conductors and tied to the L-shaped posts securely over an insulator which separates the fence wire from the post and ensures that pulse current is not diverted to the ground. Height of the fence above the wall should be 1.5 m and spacing between wires shall be 100 mm. The distance between posts shall not be greater than 3 m and fence should be able to withstand the winds of 120 kmph when installed on boundary wall. A living object touching both wire and earth during pulse will get a shock and thus deter from entering the perimeter. The impact of shock depends upon the voltage, energy of the pulse, area of contact between the recipients, fence and ground, and the route of the current through the body. The intensity of shock can range from barely noticeable to uncomfortable or painful. It can also be solar powered. (*See* Fig. 18)

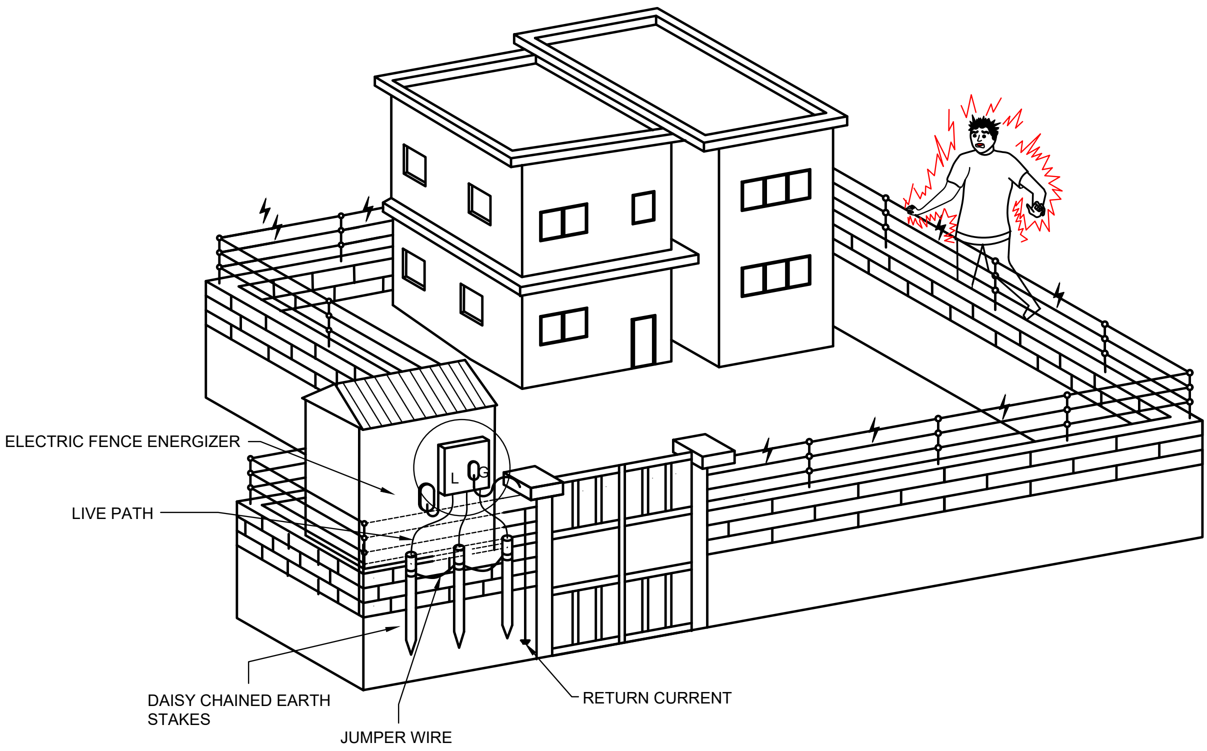


FIG. 18 POWER/ELECTRIC FENCE

10.11.3 *Compliance with Regulations and Standards*

It shall comply with specific safety and local regulatory standards. *See* accepted standard [D-2(72)] for safety and performance requirements. Prominent warning signs to alert people to the presence of the electric fence should also be displayed at regular intervals.

11 LIGHTNING PROTECTION OF BUILDINGS

11.1 General

Lightning flashes to, or nearby, structures (or lines connected to the structures) are hazardous to people, to the structures themselves, and to their contents and installations. Flashes of lightning discharges are of very high energy and can have devastating effects depending on the path of its discharge. Lightning strike is not predictable and is a random phenomenon both from the aspect of its occurrence and intensity or the energy discharge. There are no devices or methods capable of modifying the natural weather phenomena to the extent that they can prevent lightning discharges. Lightning protection measures therefore become essential. Lightning protection system (LPS) is the method of diversion of the lightning charges to the periphery of the building and to the ground so that the building, its occupants and sensitive equipment housed in the building, including connections from public systems to the building, are protected from the effects of lightning. The works carried out under this shall confirm to the relevant Indian Standards and the provisions of good practice [D-2(1)]. Brief details of the system are given hereunder.

NOTE — Reference is made to IEC 62305 (all parts) : 2024 instead of IS/IEC 62305 : 2010/11 since IEC 62305 : 2010/11 has been revised and is likely to be adopted by Bureau of Indian Standards shortly.

11.2 Steps for Deciding on Lightning Protection System

Lightning protection systems (LPS) are installed to avoid accidents, severe injuries and loss of life/deaths of humans and animals due to direct or indirect effects of lightning.

The following steps are to be followed to ensure protection from lightning in a building:

a) In order to utilize the building components such as reinforced cement concrete (RCC) columns and RCC foundation for embedded down conductor and earth termination system, the lightning protection system shall be considered and designed before starting the construction of the building;

b) Assess the need and level for lightning protection system based on risk and frequency of damage assessment as per IEC 62305-2 : 2024 'Protection against lightning — Part 2: Risk management' and the level of protection shall be identified along with coordinated SPDs to be connected to the power system and the telecommunication system;

c) Decide on the type of external LPS including usage of building components, corrosion problems, sizing of conductors and routing of conductors;

d) Calculate the separation distances, if isolated or electrically insulated LPS is used and find out the possibility of maintaining separation distance. Note that for high rise buildings, maintaining separation distance may not be practically possible. In such cases, alternate methods recommended in IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard' should be used. Equipotential bonding with the attached LPS shall be done at multiple levels and for exposed and extraneous metallic parts of the building;

e) Down conductors shall be placed to surround the building structure as uniformly as possible. Down conductors shall be installed vertically to provide the shortest and most direct path to the earth, and bends and joints should be avoided;

f) Decide on the type of earth termination system and the provisions for equipotential bonding at ground level. Type A earth termination (vertical or horizontal earth electrodes) is not efficient for large buildings. Lightning protection system without equipotential bonding (at ground level) to electrical installation is a safety hazard and shall be avoided;

g) In buildings where sensitive and large electronic systems are present such as datacentres, hospitals, telecommunication centres, etc, appropriate shielding measures shall be provided to mitigate the effects of electromagnetic disturbances; and

h) Measures to reduce touch and step potential outside the building, shall be adopted to ensure safety of the people.

While designing and execution of LPS, the following points shall be observed and implemented to prevent any unforeseen accidents due to lightning:

a) Where architectural or structural constraints exist, multiple down conductors shall not be routed collectively through vertical shafts or internal facades. In all such cases, appropriate separation distance between the conductors, continuous metallic bonding with accessibility for inspection and testing when required shall be ensured.

b) Equipotential bonding with the MET of the electrical installation and termination of SPDs shall be done.

c) Routing down conductors near metal installations such as facades, pipes, etc, without equipotential bonding should not be done.

11.3 Classification of Lightning Protection System

Class of LPS denotes the classification of an LPS according to the lightning protection level for which it is designed. Four classes of LPS I, II, III and IV are defined, based on the corresponding LPL I, II, III and IV. Each class of LPS is characterized by the rolling sphere radius, mesh size, protection angle, typical distance between down conductors, separation distance against dangerous sparking and minimum length of earth electrodes (IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'). Acceptable method to determine the position of air termination system includes the protection angle method, rolling sphere method and the mesh method. The rolling sphere method is suitable for all classes. The protection angle method is suitable for simple shaped buildings but is subject to limits indicated in Table 3. The mesh method is suitable for plane surface protection.

11.4 Planning and Designing of Lightning Protection System

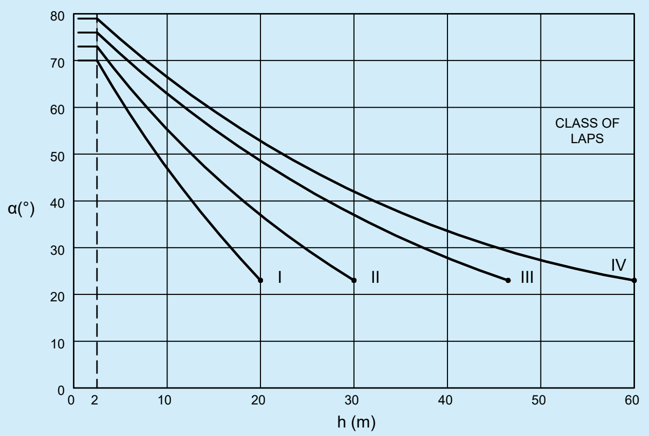
The lightning protection system (LPS) should be designed and installed by LPS designers or engineers experienced in LPS design. Planning, implementation and testing of an LPS encompasses a number of technical fields and demands for coordination by all parties involved with the construction and maintenance of the structure to ensure an efficient LPS.

It is essential to carry out the risk and frequency of damage assessment for a building based on IEC 62305-2 : 2024 'Protection against lightning — Part 2: Risk management' to decide the type of protection needed for lightning protection level and the level of coordinated SPDs required to be connected to the power line and telecommunication line. The lightning protection is divided into four levels (LPL I to IV) which helps in designing and implementing protection measures for an economical implementation. LPL I provides the maximum protection, whereas, LPL IV provides the least protection. Table 3 gives the values for different type of LPS.

Table 3 Maximum Values of Rolling Sphere Radius, Mesh Size and Protection Angle Corresponding to the Class of LPS

(*Clauses* 11.3, 11.4 *and* 11.6.1.7)

Sl No.	Class of LPS	Protection Method		
		Rolling Sphere Radius, r m	Mesh Size, W_m m	Protection Angle, α
(1)	(2)	(3)	(4)	(5)
i)	I	20	5 × 5	*See* Fig. 19 below
ii)	II	30	10 × 10	
iii)	III	45	15 × 15	
iv)	IV	60	20 × 20	



NOTES

1 NOT APPLICABLE BEYOND THE VALUES MARKED WITH • ONLY ROLLING SPHERE AND MESH METHODS APPLY IN THESE CASES.

2 h IS THE HEIGHT OF AIR-TERMINATION ABOVE THE REFERENCE PLANE OF THE AREA TO BE PROTECTED.

3 THE ANGLE WILL NOT CHANGE FOR VALUES OF h BELOW 2 m.

FIG. 19 PROTECTION ANGLE CORRESPONDING TO THE CLASS OF LPS

Complete system for protection of structures against lightning, including their internal systems and contents, be installed to avoid damages to structures, accidents, severe injuries to humans due to direct or indirect lightning. Both protection measures should complement each other.

The management of the LPS should be efficient if the steps given in **11.2** are followed. Quality assurance measures are of great importance, in particular, for structures including extensive electrical and electronic installations. *See* accepted standards [D-2(73)] and [D-2(74)].

During the planning, various parties involved shall deliberate and agree on the following points which influence lightning protection system:

a) Decide whether or not the structure needs protection and if so, the special requirements have to be decided by making necessary calculations {*see* accepted standards [D-2(75)]};

b) Special attention shall be provided by architects and civil engineers on routing of down conductors, separation distance, usage of natural components and foundation earthing (*see also* Note);

c) Electrical separation and equipotential bonding shall be carefully analysed. Equipotential bonding of electrical services by SPD is also mandatory to complete lightning protection;

d) Ensure a close liaison between the architect/engineer, the builder, the lightning protection system engineer and the appropriate authorities throughout the design stages (*see* Note); and

e) Agree the procedures and define the period for testing and future maintenance {*see* accepted standard [D-2(76)]}.

NOTE — Modern buildings with electronic equipment need protection from radiated surges of lightning. To achieve this, structural steel of the building is also used as a part of lightning protection system {*see* accepted standard [D-2(73)]}. In such cases, lightning protection measures shall be included in the structural steel, particularly in column foundation.

11.5 Use of Building Components in Lightning Protection System

Conductive components installed not specifically for lightning protection system which can be used in addition to LPS or in some cases could provide the functions of one or more parts of LPS like natural air termination, natural down conductor, natural earth electrode are known as natural components of LPS {*see* accepted standard [D-2(76)]}.

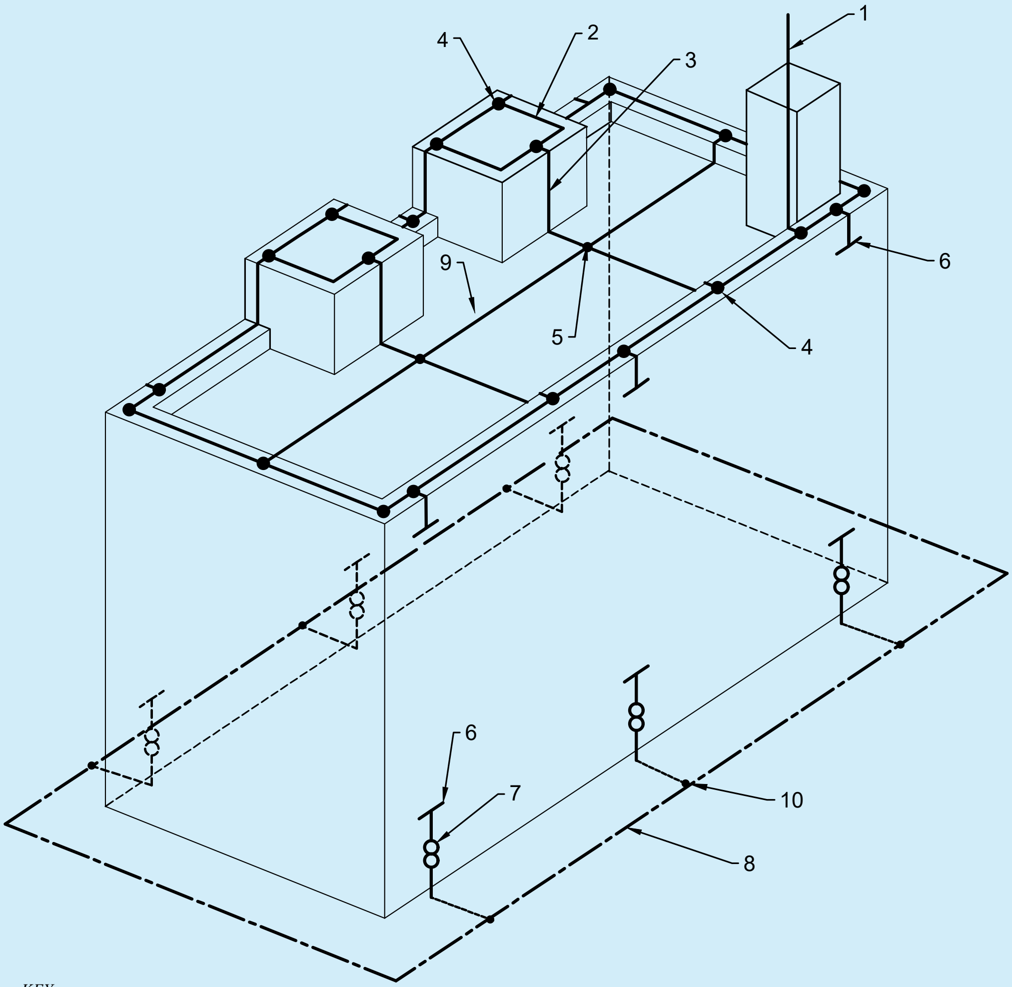
Steel work within RCC structure is considered to be electrically continuous provided that major interconnections of vertical and horizontal bars are welded or otherwise securely connected. For new structures the connections between the RCC elements in the columns of the building shall be easily achieved by the designer in cooperation with the builder and the civil engineer by providing one additional smoothly finished copper/copper coated steel/stainless steel conductor/rod of diameter as indicated in Table 8. From foundation to the terrace continuously welded with suitable clamping/welding to all intervening horizontal bars (in case if single welded rod is not possible then necessary lapping between two rods shall be made as per IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'). Thus, all external columns can act as down conductors for LPS and there will be no need for provision of separate external down conductors especially in architectural facia and glazed facia of the buildings. The overall electrical resistance of the reinforcement provided should not be greater than 0.2 Ω using test equipment suitable for this purpose. Suitable test joints shall be brought out from the structure at/near ground level for test purposes.

Interconnected reinforcement steel in concrete foundation buried directly in ground can be used as earth electrode. Special care shall be taken at the termination to prevent mechanical splitting of the concrete. Normally, a concrete raft foundation or pile foundation with pile caps of a building, provides reinforcement rods of not less than 25 mm having interconnected reinforcement steel duly clamped/welded, will provide a proper earth electrode and in closed loop formation. This will give a very good earth resistance for the down conductors and the overall value shall be measured to be less than earth resistance achieved normally.

The additional reinforcement provided for LPS shall be connected through welding to the foundation reinforcement and also with the roof conductors connecting the air termination through welding and suitably epoxy painted to prevent rusting. Thus, the LPS for the building shall become homogeneous after provision of air termination at the roof level based on rolling sphere/mesh system by interconnecting the top of the reinforcement from the columns with the terrace conductors (*see* Fig. 20).

Use of building components namely the cement concrete foundation and the columns, reduces not only the work of providing down conductors and earthing sets for the LPS and the cost thereof but also saves the difficulties of placing down conductors of LPS in respect of architectural facia of modern buildings. Also, test joints are not required for these cases {*see* **5.3.6** of accepted standard [D-2(73)]}.

NOTE — When incorporating a structure's natural components, such as rebar within the concrete foundation, specialized mechanical couplers or clamps {*see* accepted standard [D-2(77)]} may be specified during design to create secure equipotential bonding as per IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'.



KEY

1 AIR TERMINATION ROD
2 HORIZONTAL AIR TERMINATION CONDUCTOR
3 DOWN CONDUCTOR
4 T - TYPE JOINT
5 CROSS TYPE JOINT
6 CONNECTION TO STEEL REINFORCING RODS {*see* **E-4.3.3** AND **E-4.3.6** OF GOOD PRACTICE [D-2(73)]}
7 TEST JOINT
8 EARTH ELECTRODE TYPE B EARTHING ARRANGEMENT, RING EARTH ELECTRODE
9 FLAT ROOF WITH ROOF FIXTURES
10 T - TYPE JOINT - CORROSION RESISTANT

NOTE — THE STEEL REINFORCEMENT OF THE STRUCTURE SHOULD COMPLY WITH AVAILABLE INDIAN STANDARDS. ALL DIMENSIONS OF THE LPS SHOULD COMPLY WITH THE SELECTED PROTECTION LEVEL.

FIG. 20 CONSTRUCTION OF EXTERNAL LPS ON A STRUCTURE OF STEEL-REINFORCED CONCRETE USING THE REINFORCEMENT OF THE OUTER WALLS AS NATURAL COMPONENTS

11.6 Components of Lightning Protection System (LPS)

The main and most effective measure for protection of structures against physical damage is considered to be the lightning protection system (LPS). It usually consists of both external and internal lightning protection systems.

An external LPS which consists of air-termination system, down-conductor system and earthing system {*see* accepted standard [D-2(73)]} is intended to:

a) intercept a lightning flash to the structure (with an air-termination system);

b) conduct the lightning current safely towards earth (using a down-conductor system); and

c) disperse the lightning current into the earth (using an earth-termination system).

An internal LPS comprises equipotential bonding or a separation distance (and hence electrical insulation) between the external LPS components and other electrically conducting elements internal to the structure.

Both external and internal protection systems should complement each other.

Access to the ground and the proper use of foundation steelwork for the purpose of forming an effective earth-termination may well be impossible once construction work on a site has commenced. Therefore, soil resistivity and the nature of the earth electrode should be considered at the earliest possible preferably during the design stage of a project. This information is fundamental to the design of an earth-termination system and may influence the foundation design work for the structure.

Regular consultation between LPS designers and installers, architects/civil engineer and builders is essential in order to achieve the best result at minimum cost. If lightning protection is to be added to an existing structure, every effort should be made to ensure that it conforms to the principles of this NBCS. The design of the type and location of an LPS should take into account the features of the existing structure.

11.6.1 *Air-Termination System*

Air-termination system is a part of an external LPS using metallic elements such as rods, mesh conductors or catenary wires intended to intercept lightning flashes. The probability of penetration by a lightning current on a structure is considerably decreased by the presence of a properly designed air-termination system. Air-termination systems can be composed of any combination of the following elements:

a) vertical rods (offers certain angle of protection);

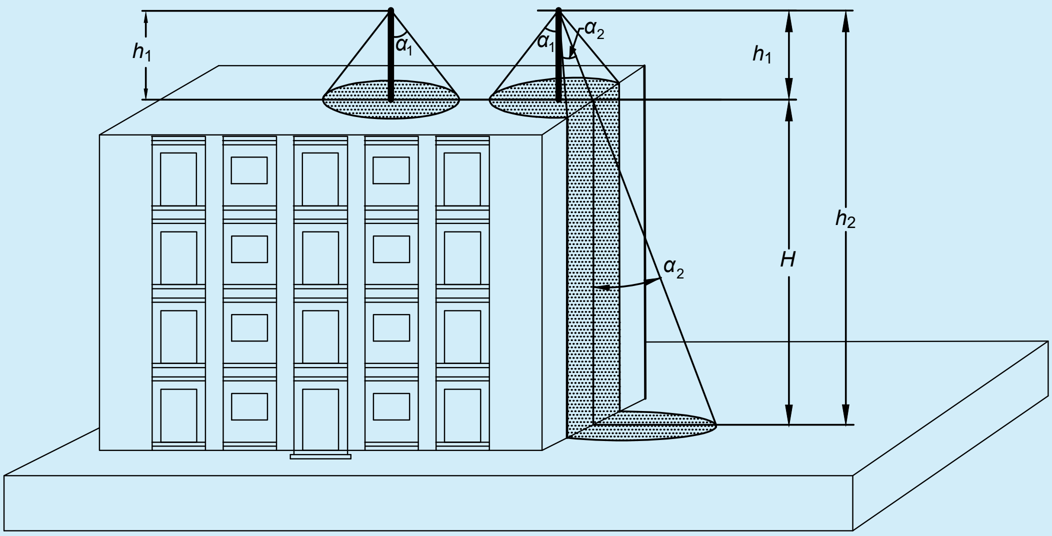
b) catenary wires; and

c) meshed/grid conductors.

Franklin rod (conventional) type of air-termination system shall be positioned in accordance with **11.6.1.1**. The individual air-terminations rods should be connected together at roof level to ensure current division. Radioactive air-terminals shall not be allowed. Any other kind of air-terminal like dissipation system/ESE air-terminal/CSE air-terminal or any other air terminal not approved by IEC 62305-3 : 2024 ‘Protection against lightning — Part 3: Physical damage to structures and life hazard’ shall not be acceptable.

11.6.1.1 *Positioning*

Air-termination components installed on a structure shall be located at corners, exposed points and edges (especially on the upper level of any facades) in accordance with one or more of the following methods {*see also* Fig. 21 to Fig. 25 in conjunction with good practice [D-2(11)]}:



KEY

H HEIGHT OF BUILDING OVER THE GROUND REFERENCE PLANE

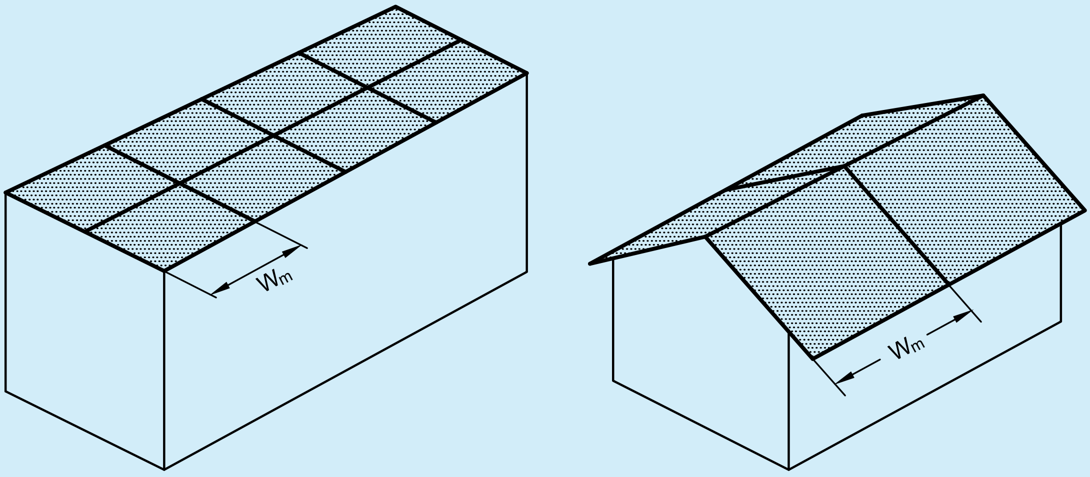
h_1 PHYSICAL HEIGHT OF AN AIR TERMINAL ROD

h_2 HEIGHT OF THE AIR TERMINATION ROD OVER THE GROUND = $h_1 + H$

α_1 THE PROTECTION ANGLE CORRESPONDING TO THE AIR TERMINATION HEIGHT, $h = h_1$, BEING THE HEIGHT ABOVE THE ROOF SURFACE TO BE MEASURED (REFERENCE PLANE)

α_2 THE PROTECTION ANGLE CORRESPONDING TO THE HEIGHT, $h_2 = H + h_1$

FIG. 21 PROTECTION ANGLE METHOD OF AIR TERMINATION DESIGN FOR DIFFERENT HEIGHTS



W_m – MESH SIZE

FIG. 22 DESIGN OF AIR-TERMINATION SYSTEM ACCORDING TO MESH METHOD

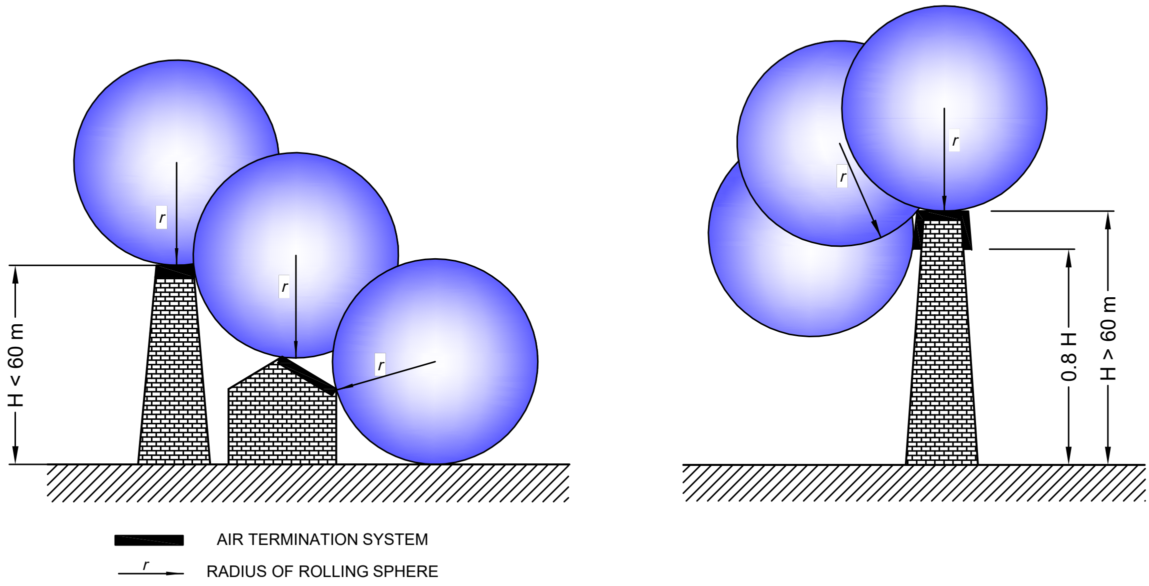
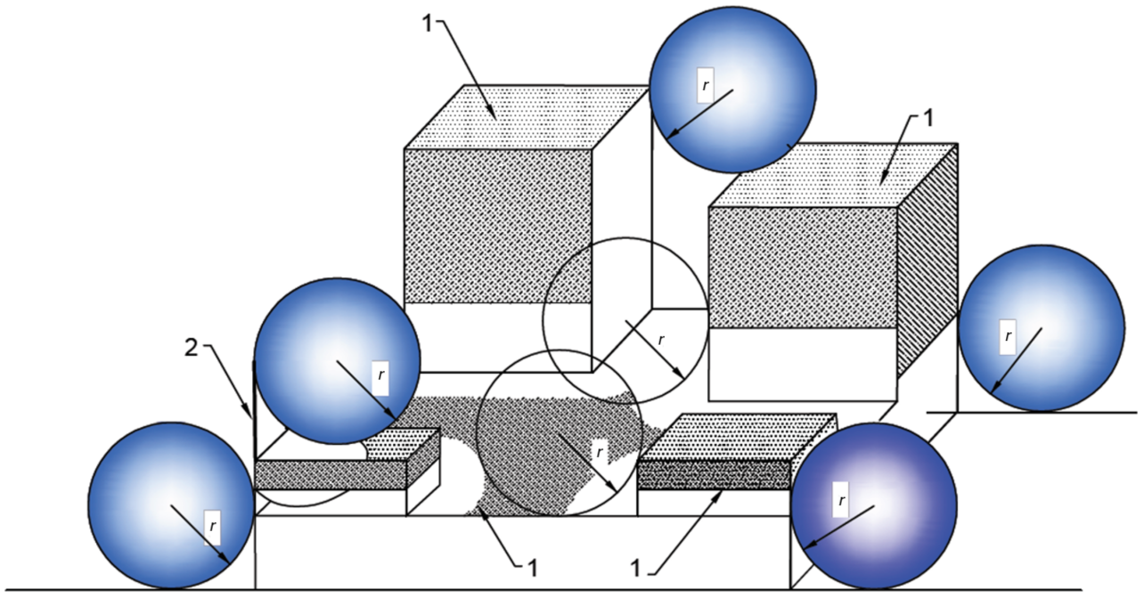


FIG. 23 DESIGN OF AIR-TERMINATION SYSTEM ACCORDING TO THE ROLLING SPHERE METHOD



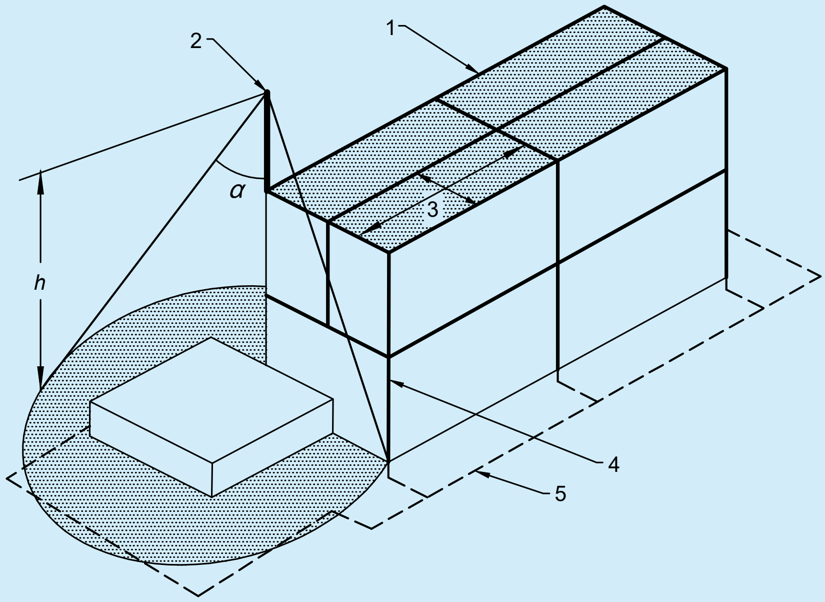
Key

1 Shaded areas are exposed to lightning interception and need protection

2 Mast on the structure

r Radius of rolling sphere

FIG. 24 DESIGN OF AIR-TERMINATION CONDUCTOR NETWORK FOR A STRUCTURE WITH COMPLICATED SHAPE



KEY

1 AIR - TERMINATION CONDUCTOR (ALSO CALLED AS MESH/FARADAY CAGE)
2 AIR - TERMINATION ROD
3 MESH SIZE
4 DOWN CONDUCTOR
5 EARTHING SYSTEM WITH RING CONDUCTOR
h HEIGHT OF THE AIR TERMINAL ABOVE GROUND LEVEL
α PROTECTION ANGLE

FIG. 25 DESIGN OF AN LPS AIR-TERMINATION ACCORDING TO THE PROTECTION ANGLE METHOD, MESH METHOD AND GENERAL ARRANGEMENT OF AIR-TERMINATION ELEMENTS

11.6.1.2 Roof mounted electrical/electronic equipment (for example, chillers, antennas, cameras and bill boards) need vertical air-termination to avoid direct strike. Power and data connection to these equipment should have proper SPDs (*see* **11.7.4**) to avoid failures. Overhead cables such as cable TV lines from one building to the other should be avoided.

11.6.1.3 Unearthed metallic roofs should be avoided. Metallic roofs shall be connected either to steel reinforcement or to other earthed steel parts of the building satisfying the requirements of number of down-conductors (*see* **11.6.2**).

11.6.1.4 *Structures of height less than* 60 m *and more than* 60 m

Lateral impacts of flashes to the side of tall structures are considered as negligible for structures up to 60 m in height measured from ground level. However, elements significantly protruding the facade can be endangered (for example, balconies, cameras, antennas).

On structures taller than 60 m, flashes to the side can occur, especially to protruding parts and to corners and edges of facades.

All external metal parts that have a conductive connection with internal metal parts separated by a distance $d > s$ from a down conductor (or air terminations) should not be connected to the down conductor so as to keep any unnecessary partial lightning currents out of the structure.

If this is not possible, the metal part should be directly bonded to the LPS. The possibility of the occurrence of partial lightning current into the structure should be considered.

NOTE — In general, the risk due to these flashes is low because only a few per cent of all flashes to tall structures will be to the side and their parameters are significantly lower than those of flashes to the top of structures. However, electrical and electronic equipment on walls outside structures can be destroyed even by lightning flashes with low current peak values.

An air-termination system shall be installed to protect significant protrusions (such as balconies, viewing platforms) on vertical surfaces on the upper part of tall structures (that is, typically the topmost 20 percent of the height of the structure), as well as the equipment installed on them (*see* Annex D of IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'). For parts of the upper 20 percent that do not exceed 60 m above ground level, protection of the vertical surface is optional.

NOTE — Elements protruding through the facade of structures up to 60 m in height are rarely endangered.

11.6.1.5 *Roof top solar PV and water heaters on Buildings*

Vertical air-terminals are required for protecting roof mounted installations such as solar PV, water heaters, chillers as well as water tanks. Vertical air-terminals need to be connected to the air-termination mesh/down-conductors. Metal support structure of these installations shall be bonded to the air-termination mesh/down-conductors. Class I/Class II surge protection devices (SPDs) (*see* **11.7.4**) should be installed in the electrical lines to protect the installations inside the building (typically d.c. SPD for solar PV output at inverter or junction box level and a.c. SPD for inverter output and mains input). The roof top PV system or 230 V a.c. water heaters or any other equipment should not pose any safety risk related to lightning protection and protection of overall building may be reviewed by the expert after installation of roof top PV system. Necessary measures may then be required to complete the lightning protection arrangement.

Vertical air-terminals for PV modules based on LPL III/IV connected directly to the frame shall protect against direct lightning impact in solar PV power plant. A design according to rolling sphere method should be done for zone of protection (for example, 1 m rod at 0.5 m height from panel at four corners provides protection to approximately 12 m × 9 m area). Maximum height of the air-termination rod above the panel should be calculated considering the falling of shadow or umbra region of shadow not falling on any PV module. To reduce step potential, structures should be interconnected with underground earth mats/isolating spark gaps, wherever necessary (*see* Fig. 26 and Fig. 27).

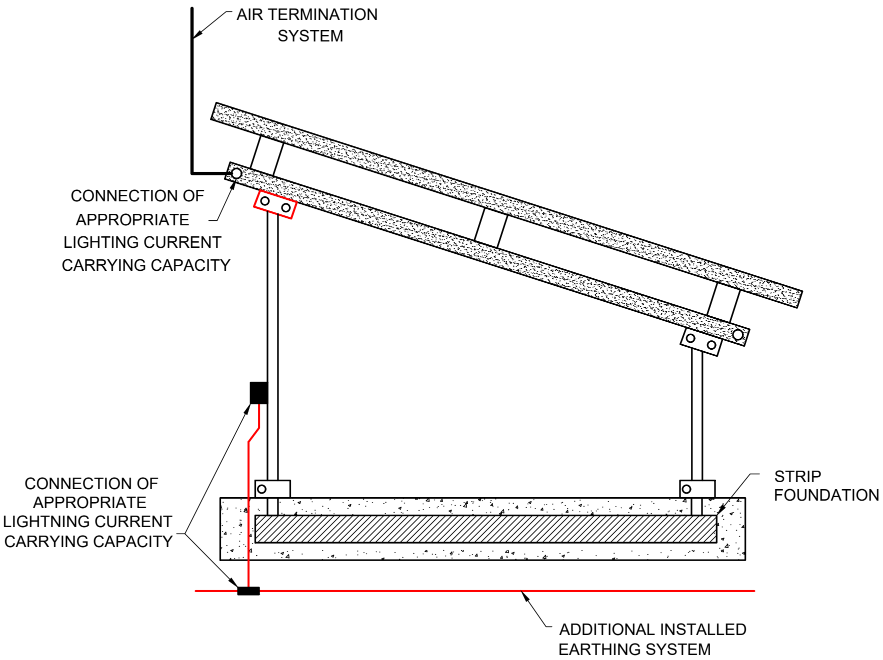


FIG. 26 TYPICAL INSTALLATION OF LIGHTNING PROTECTION SYSTEM IN SOLAR PV INSTALLATION

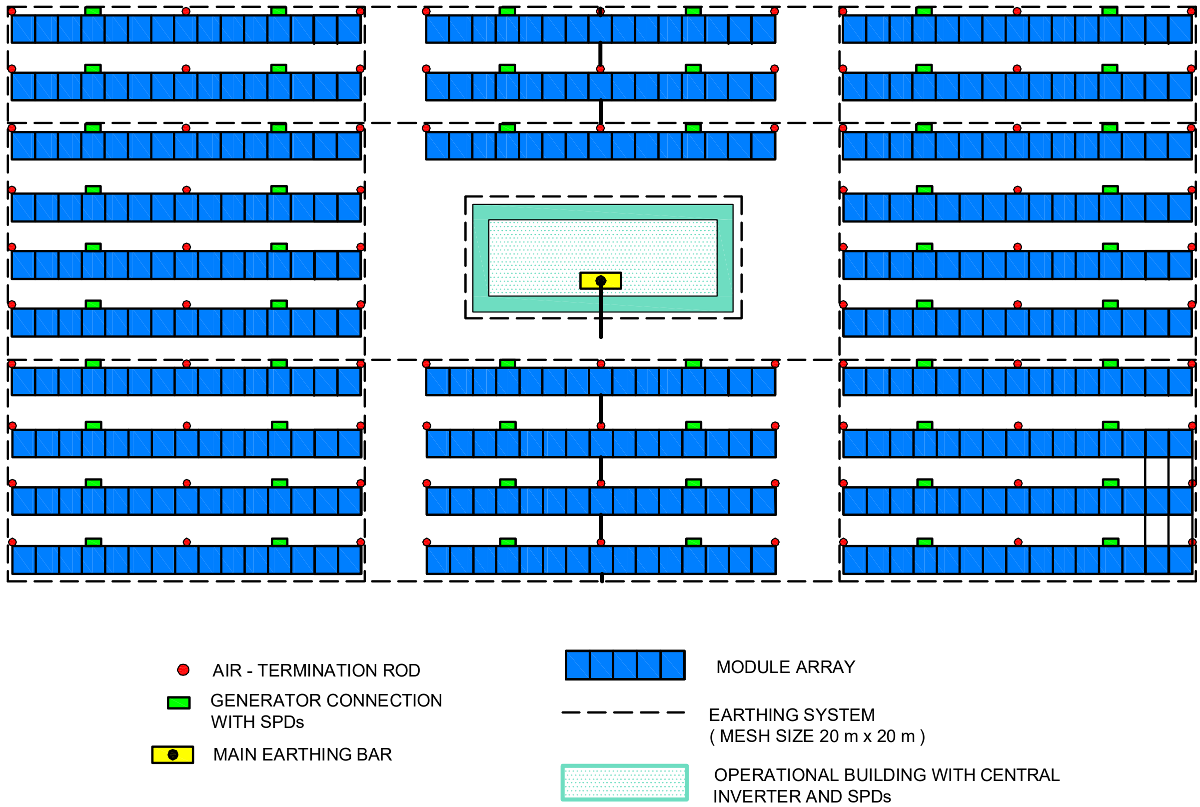


FIG. 27 TYPICAL ARRANGEMENT OF EARTHING AND SURGE PROTECTION DEVICES IN SOLAR PV INSTALLATION

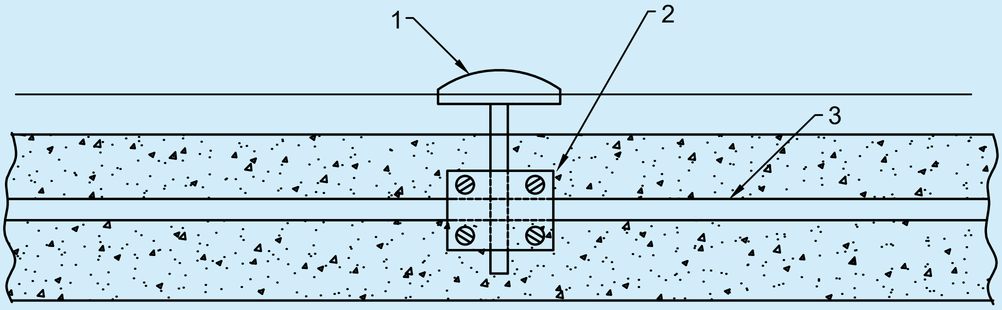
11.6.1.6 *Buildings with roof top telecom towers*

The metallic tower itself will act as air-termination. Antennas mounted above these towers need air-terminals connected to the main structure. The main structure shall be connected to the air-termination conductors for the balance of the building, if available. Two separate down-conductors with a size of minimum 150 mm^2 should be used in addition to regular down-conductors to make the bonding between tower and ring earthing. In order to avoid uncontrolled flash overs and also to protect equipment, which may be mounted on the tower itself, special cable with increased dielectric strength as down-conductor should be used in order to avoid separation distance. The area of cross section will be significantly lower than 50 mm^2 but not less than 28 mm^2, which is sufficient enough to discharge the lightning current where mechanical strength is not an essential requirement.

Every power, coaxial, data and other metallic lines connected between the telecom installation and the other parts of the building shall be protected with suitable SPD. It should be ensured that SPDs during their operation do not impede fire safety.

11.6.1.7 *Lightning protection for multi-storeyed car park roofs/helipads*

Air-termination studs (*see* Fig. 28) may be used for lightning protection for multi-storeyed car park roofs/helipads. Air-termination studs used can be connected to the reinforcement steel of a concrete roof. In the case of roofs where a connection to the reinforcement cannot be made, the roof conductors can be laid in the seams of the carriageway slabs and air-termination studs can be located at the mesh joints. The mesh width shall not exceed the value corresponding to the protection class given in Table 3. The persons and vehicles on this parking area are not protected against direct lightning.



KEY
1 AIR - TERMINATION STUD
2 STEEL CONDUCTOR CONNECTING TO SEVERAL BARS OF THE REINFORCEMENT STEEL
3 STEEL REINFORCEMENT TO CONCRETE

FIG. 28 TYPICAL AIR-TERMINATION STUDS USED ON CAR PARK ROOFS

11.6.2 *Down-conductor System*

Down-conductor system is a part of an external LPS intended to conduct lightning current from the air-termination system to the earth-termination system. The down conductor system shall confirm to **5.3** of accepted standard [D-2(73)]. In order to reduce the probability of damage due to lightning current flowing in the air-termination system, the down-conductors shall be arranged in such a way that from the point of strike to earth:

a) several parallel current paths exist;

b) the length of the current paths is kept to a minimum; and

c) equipotential bonding to conducting parts of the structure is performed.

11.6.2.1 Typical values of the distance between down-conductors are given in Table 4. These values can be used for horizontal ring conductors installed for a tall building more than 60 m height. The minimum number of down-conductors shall be 2 (diagonally opposite to each other) for building with an area less than 100 m^2.

Table 4 Minimum Distance between Down-Conductors

(*Clause* 11.6.2.1)

Sl No. (1)	**Class of LPS** (2)	**Distance** m (3)
i)	I	10
ii)	II	10
iii)	III	15
iv)	IV	20

11.6.2.2 Down-conductors shall be installed so that, as far as practicable, they form a direct continuation of the air-termination conductors. It shall be installed straight and vertical such that they provide the shortest and most direct path to earth. The formation of sharp bends and loops shall be avoided. Every down-conductor should be connected to a Type B ring/foundation earthing. Connection of down-conductor to a Type A earthing is allowed only in case of space constraints or existing buildings, where installation is difficult.

11.6.2.3 While routing the down-conductors, separation distance need to be calculated based on accepted standard [D-2(73)] and maintained from live parts/services.

11.6.2.4 Lateral connection of down-conductors at ground level and every 10 m to 20 m of height as a ring conductor, as per below table, is considered to be good practice. The installation of as many down-conductors as possible, at equal spacing around the perimeter interconnected by ring conductors, reduces the probability of dangerous sparking and facilitates the protection of internal installations {*see* accepted standard [D-2(73)]}. This condition is fulfilled in metal framework structures and in reinforced concrete structures in which the interconnected steel is electrically continuous.

11.6.2.5 Routing of down-conductors (insulated or uninsulated) through shafts with electrical and electronic system is not allowed. But routing of down conductors can be done through other service shafts by maintaining separation distance or with equipotential bonding of extraneous metal components.

11.6.2.6 Separation distance is the distance required between air-terminals/lightning down-conductor and any conductive/metallic/electrical/electronic part of a building to avoid uncontrolled flashover.

11.6.2.6.1 While calculating the separation distance, the parameter l (length, in m along the air-terminal and the down-conductor from the point where the separation distance is to be considered, to the nearest equipotential bonding point or earth termination system is not bonded to equipotential bonding point or the earth termination) is a major influencing factor. If the down conductors or the earth terminal system is not bonded to equipotential bonding system of the electrical installation in the building, the lightning protection system will be ineffective.

Cable with increased dielectric strength and tested for lightning current discharge may be used to avoid specific separation distance to live parts of the building {*see* accepted standard [D-2(73)]}.

11.6.2.7 The down-conductor shall be supported using suitable clamps or connectors or exothermic welding. The clamps or connectors or exothermic welding shall be tested for the lightning current as per selected LPL. Reference may be made to table given below and accepted standard [D-2(73)] for supporting details:

Sl No.	*Arrangement*	*Fixing Centres for Tape, Stranded and Soft Drawn Round Conductors* mm	*Fixing Centres for Round Solid Conductors* mm
(1)	(2)	(3)	(4)
i)	Horizontal conductors on horizontal surfaces	1 000	1 000
ii)	Horizontal conductors on vertical surfaces	500	1 000
iii)	Vertical conductors from the ground to 20 m	1 000	1 000
iv)	Vertical conductors from 20 m and thereafter	500	1 000

NOTES

1 This table does not apply to built-in type fixings, which may require special considerations.

2 Assessment of environmental conditions (that is, expected wind load) should be undertaken and fixing centres different from those recommended may be found to be necessary.

11.6.2.8 The wind speed shall be taken into account while mounting the air-termination and down-conductor system {*see* accepted standard [D-2(73)]}.

11.6.3 *Earth-termination System*

Earth-termination system is a part of an external LPS which is intended to conduct and disperse lightning current into the earth. When dealing with the dispersion of the lightning current (high frequency behaviour) into the ground, whilst minimizing any potentially dangerous overvoltages, the shape and dimensions of the earth-termination system are the important criteria. In general, a low earthing resistance (if possible lower than 10 Ω when measured at low frequency) is recommended. From the viewpoint of lightning protection, a single integrated structure earth-termination system is preferable and is suitable for all purposes (that is, lightning protection, power systems and telecommunication systems).

Type A earth-termination comprising of vertical/horizontal conductor or Type B earth-termination comprising of ring earthing/foundation earthing shall be used satisfying the requirements of this NBCS as well as accepted standard [D-2(73)]. In structures where only electrical systems are provided, a Type A earthing arrangement may be used, but a Type B earthing arrangement is preferable. In structures with electronic systems, a Type B earthing arrangement is recommended.

11.6.3.1 *Type A earthing*

Length of the earth electrode depends on the soil resistivity and class of LPS {*see* good practice [D-2(11)] for details and Table 5 for vertical earth electrode}.

Table 5 Minimum Length of Vertical Earth Electrode

(*Clauses* 11.6.3.1 *and* 11.6.3.2)

Sl No.	Class of LPS	Typical Length of Each Vertical Earth Electrode Based on Soil Resistivity m			
		Up to 500 Ω-m	1 000 Ω-m	2 000 Ω-m	3 000 Ω-m
(1)	(2)	(3)	(4)	(5)	(6)
i)	I	2.5	10	25	40
ii)	II	2.5	5	15	25
iii)	III	2.5	2.5	2.5	2.5
iv)	IV	2.5	2.5	2.5	2.5

11.6.3.2 *Type B earthing*

This type of arrangement comprises either a ring conductor external to the structure to be protected, in contact with the soil for at least 80 percent of its total length, or a foundation earth electrode. Such earth electrodes may also be meshed. For the ring earth electrode (or foundation earth electrode), the mean radius of the area enclosed by the ring earth electrode (or foundation earth electrode) shall be not less than the value of Type A earthing as given in Table 5 {*see* accepted standard [D-2(73)]}.

11.6.3.3 In industrial and commercial structures, the ring earth electrode around the structure or the ring earth electrode in the concrete at the perimeter of the foundation, should be integrated with a meshed network under and around the structure, having a mesh width of typically 5 m. This greatly improves the performance of the earth-termination system. If the basement's reinforced concrete floor forms a well-defined interconnected mesh and is connected to the earth-termination system, typically at every 5 m, the same will also be suitable {*see* accepted standard [D-2(73)]}.

11.6.3.4 Where large numbers of people frequently assemble in an area adjacent to the structure to be protected, further potential control for such areas should be provided. More ring earth electrodes should be installed at distances of approximately 3 m from the first and subsequent ring conductors. Ring electrodes further from the structure should be installed more deeply below the surface, that is, those at 4 m from the structure at a depth of 1 m, those at 7 m from the structure at a depth of 1.5 m and those at 10 m from the structure at a depth of 2 m. These ring earth electrodes should be connected to the first ring conductor by means of radial conductors.

11.6.3.5 For large buildings integrating structural steel as down-conductor and earth-termination, earth resistivity measurements are not required. Proper drawings should be made based on the actual installation and submitted to authorities if necessary {*see also* accepted standard [D-2(73)]}. However, provisions shall be provided for continuity resistance measurement.

11.6.3.6 Touch and step voltage safety measures outside structure shall be installed and maintained as per accepted standard [D-2(73)].

11.6.4 *Use of Natural Components*

11.6.4.1 Natural components installed not specifically for lightning protection system can be used in addition to LPS or in some cases could provide the functions of one or more parts of LPS, like natural air termination, natural down conductor, natural earth electrode. The use of building components such as columns and foundation have been explained in **11.5** separately, in order to give special attention for use of such systems in modern buildings.

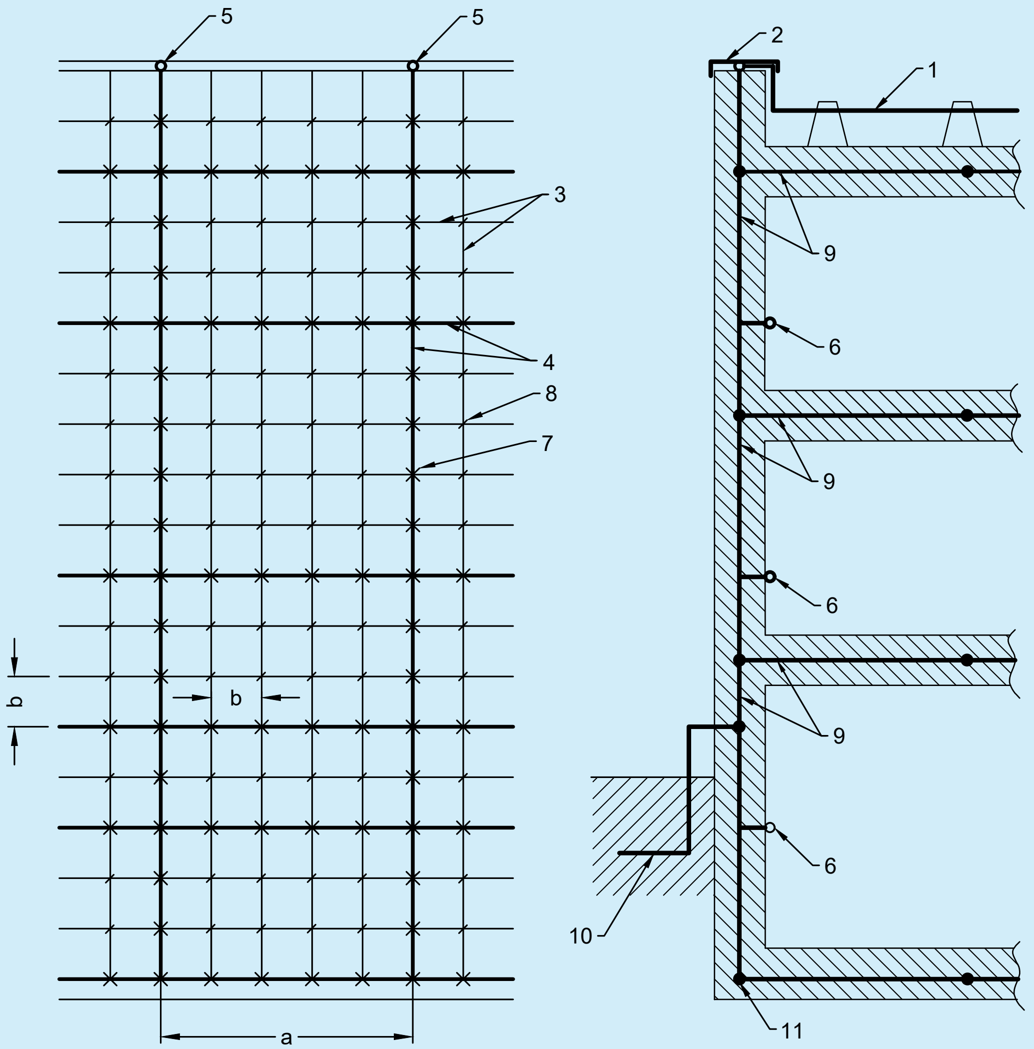
11.6.4.2 *Bonding network*

A low impedance bonding network is needed to avoid dangerous potential differences between all equipment inside the building. Moreover, such a bonding network also reduces the magnetic field, thereby reduces the radiated surges inside the building and provides more protection for electrical electronic equipment {*see* accepted standard [D-2(74)]}.

This can be realized by a meshed bonding network integrating conductive parts of the structure, or parts of the internal systems, and by bonding metal parts or conductive services at the boundary of each lightning protection zone (LPZ) directly or by using suitable SPDs.

The bonding network can be arranged as a three-dimensional meshed structure with a typical mesh width of 5 m (*see* Fig. 29 and Fig. 30). This requires multiple interconnections of metal components in and on the structure (such as concrete reinforcement, elevator rails, cranes, metal roofs, metal facades, metal frames of windows and doors, metal floor frames, service pipes and cable trays). Bonding bars (for example, ring bonding bars, several bonding bars at different levels of the structure) and magnetic shields of the LPZ shall be integrated in the same way.

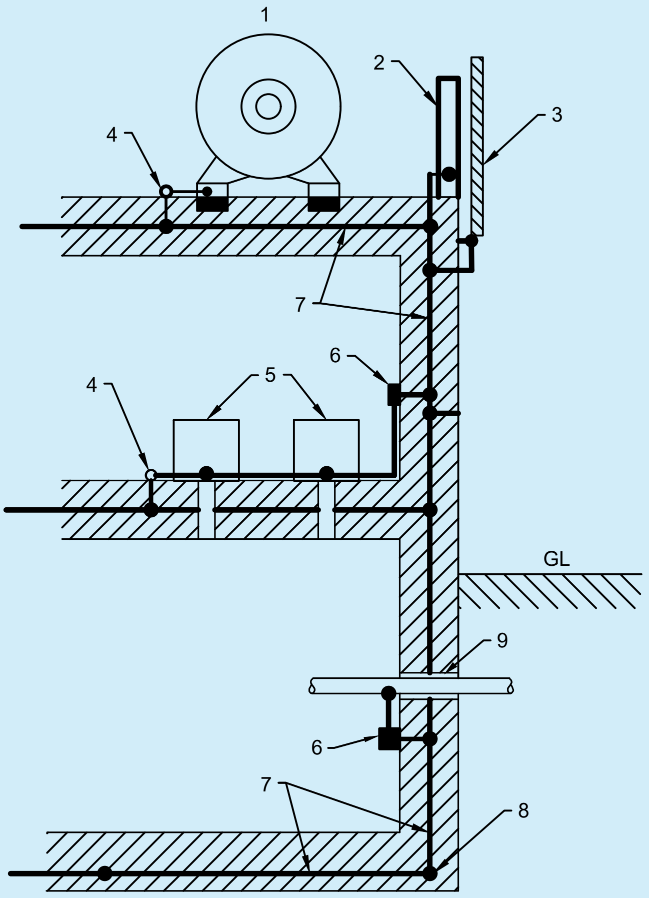
Conductive parts (for example, cabinets, enclosures, racks) and the protective earth conductor (PE) of the internal systems shall be connected to the bonding network {*see* accepted standard [D-2(74)]}.



KEY

1 AIR-TERMINATION CONDUCTOR
2 METAL COVERING OF THE ROOF PARAPET
3 STEEL REINFORCING RODS
4 MESH CONDUCTORS SUPERIMPOSED ON THE REINFORCEMENT
5 JOINT OF THE MESH CONDUCTOR
6 JOINT OF THE INTERNAL BONDING BAR
7 CONNECTION MADE BY WELDING OR CLAMPING
8 ARBITARY CONNECTION
9 STEEL REINFORCEMENT IN CONCRETE (WITH SUPERIMPOSED MESH CONDUCTORS)
10 RING EARTHING ELECTRODE (IF ANY)
11 FOUNDATION EARTHING ELECTRODE
a TYPICAL DISTANCE OF 5 m FOR SUPERIMPOSED MESH CONDUCTOR
b TYPICAL DISTANCE OF 1 m FOR CONNECTING THIS MESH WITH THE REINFORCEMENT

FIG. 29 UTILIZATION OF REINFORCING RODS OF A STRUCTURE FOR EQUPOTENTIAL BONDING



KEY

1 ELECTRIC POWER EQUIPMENT
2 STEEL GIRDER
3 METAL COVERING OF THE FACADE
4 BONDING JOINT
5 ELECTRICAL OR ELECTRONIC EQUIPMENT
6 BONDING BAR
7 STEEL REINFORCEMENT IN CONCRETE (WITH SUPERIMPOSED MESH CONDUCTORS)
8 FOUNDATION EARTHING ELECTRODE
9 COMMON INLET FOR DIFFERENT SERVICES

FIG. 30 EQUIPOTENTIAL BONDING IN A STRUCTURE WITH STEEL REINFORCEMENT

11.6.5 *Materials and Dimensions*

Copper and aluminium are recommended for exposed areas on installations required to have a long life. Galvanized steel may be preferred for temporary installations such as exhibition centres. Although it is a common practice to use material in the form of strip for horizontal air-terminations, down-conductors and bonds, it is more convenient to use round material, particularly as it facilitates the making of bends in any plane. *See* Table 6,

Table 7 and Table 8 for details for different materials and conditions of use, refer accepted standard [D-2(74)] for minimum dimension of materials used in LPS.

Table 6 LPS Materials and Conditions of Use[1)]

(*Clause* 11.6.5)

Sl No.	**Material**	**Use**			**Corrosion**		
		In Open Air	In Earth	In Concrete	Resistance	Increased by	May be Destroyed by Galvanic Coupling with
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Copper	Solid	Solid	Solid	Good in many environments	Sulphur compounds Organic materials	–
		Stranded	Stranded	Stranded			
		–	As coating	As coating			
ii)	Hot galvanized steel[2), 3) and 4)]	Solid	Solid	Solid	Acceptable in air, in concrete and in benign soil	High chlorides content	Copper
		Stranded[5)]		Stranded[5)]			
iii)	Steel with electro deposited copper	Solid	Solid	Solid	Good in many environments	Sulphur compounds	–
iv)	Stainless steel	Solid	Solid	Solid	Good in many environments	High chlorides content	–
		Stranded	Stranded	Stranded			
v)	Aluminium	Solid	Unsuitable	Unsuitable	Good in atmospheres containing low concentrations of sulphur and chloride	Alkaline solutions	Copper
		Stranded					

[1)] This table gives general guidance only. In special circumstances more careful corrosion immunity considerations are required.

[2)] Galvanized steel may be corroded in clay soil or moist soil.

[3)] Galvanized steel in concrete should not extend into the soil due to possible corrosion of the steel just outside the concrete.

[4)] Galvanized steel in contact with reinforcement steel in concrete may, under certain circumstances, cause damage to the concrete.

[5)] Stranded conductors are more vulnerable to corrosion than solid conductors. Stranded conductors are also vulnerable where they enter or exit earth/concrete positions. This is the reason why stranded galvanized steel is not recommended in earth.

Table 7 Material, Configuration and Minimum Cross-Sectional Area of Air-Termination Conductors and Rods, Earth Lead-in Rods and Down-Conductors[1)]

(*Clause* 11.6.5)

Sl No.	Material	Configuration	Minimum Cross-Sectional Area mm^2
(1)	(2)	(3)	(4)
i)	Copper, tin plated copper	Solid tape	50
		Solid round[2)]	50
		Stranded[2)]	50
		Solid round[3)]	176
ii)	Aluminium	Solid tape	70
		Solid round	50
		Stranded	50
iii)	Aluminium alloy	Solid tape	50
		Solid round	50
		Stranded	50
		Solid round[3)]	176
iv)	Copper coated aluminium alloy	Solid round	50
v)	Hot dipped galvanized steel	Solid tape	50
		Solid round	50
		Stranded	50
		Solid round[3)]	176
vi)	Copper coated steel	Solid round	50
		Solid tape	50
vii)	Stainless steel	Solid tape[4)]	50
		Solid round[4)]	50
		Stranded	50
		Solid round[3)]	176

1) Mechanical and electrical characteristics as well as corrosion resistance properties shall meet the requirements as per accepted standard [D-2(78)].

2) 50 mm^2 (8 mm diameter) may be reduced to 25 mm^2 in certain application where mechanical strength is not an essential requirement. Consideration should in this case, be given to reducing the space between fasteners.

3) Applicable for air-termination rods and earth lead-in rods. For air-termination rods where mechanical stress such as wind loading is not critical, a 9.5 mm diameter, 1 m long rod may be used.

4) If the thermal and mechanical considerations are important, then these values should be increased to 75 mm^2.

Table 8 Material Configuration and Minimum Dimensions of Earth Electrodes[1] and [2]

(*Clauses* 11.5 *and* 11.6.5)

Sl No.	Material	Configuration	Minimum Dimensions		
			Earth Rod Diameter mm	Earth Conductor mm^2	Earth Plate mm
(1)	(2)	(3)	(4)	(5)	(6)
i)	Copper, tin plated copper	Stranded	–	50	–
		Solid round	15	50	
		Solid tape	–	50	–
		Pipe	20	–	–
		Solid plate	–	–	500 × 500
		Lattice plate[3]	–	–	600 × 600
ii)	Hot dipped galvanized steel	Solid round	14	78	–
		Pipe	25	–	–
		Solid tape	–	90	–
		Solid plate	–	–	500 × 500
		Lattice plate	–	–	600 × 600
		Profile	[4]	–	–
iii)	Bare steel[5]	Stranded	–	70	–
		Sold round	–	78	–
		Solid tape	–	75	–
iv)	Copper coated steel	Solid round	14[6]	50	–
		Solid tape	–	90	–
v)	Stainless steel[7]	Solid round	15	78	–
		Solid tape	–	100	–

1) Mechanical and electrical characteristics as well as corrosion resistance properties shall meet the requirements as per accepted standard [D-2(78)]

2) In case of a Type B arrangement foundation earthing system, the earth electrode shall be correctly connected at least every 5 m with the reinforcement steel.

3) Lattice plate constructed with a minimum total length of conductor of 4.8 m.

4) Different profiles are permitted with a cross-section of 290 mm^2 and a minimum thickness of 3 mm, for example, cross profile.

5) Shall be embedded in concrete for a minimum depth of 50 mm.

6) 250 µm minimum radial copper coating, with 99.9 percent copper content.

7) Chromium ≥ 16 percent, nickel ≥ 5 percent, molybdenum ≥ 2 percent, carbon ≤ 0.08 percent.

11.6.6 Protection measures against injury to living beings due to touch voltage and step voltage shall be provided in accordance with accepted standard [D-2(74)].

11.6.7 Inspection of the LPS shall be done as per IEC 62305-3 : 2024 'Protection against lightning — Part 3: Physical damage to structures and life hazard'.

11.7 Protection of Electrical/Electronic Systems within Structures

11.7.1 The internal LPS shall avoid the occurrence of dangerous sparking within the structure to be protected due to lightning current flowing in the external LPS or in other conductive parts of the structure. Dangerous sparking between different parts can be avoided by means of equipotential bonding or electrical insulation between the parts.

Permanent failure of electrical and electronic systems can be caused by the lightning electromagnetic impulse (LEMP) *via*:

a) conducted and induced surges transmitted to equipment *via* connecting wiring; and

b) the effects of radiated electromagnetic fields directly into equipment itself.

11.7.2 *Equipment Protection Principles*

11.7.2.1 For protection against the effects of radiated electromagnetic fields impinging directly onto the equipment, SPM consisting of spatial shields and/or shielded lines, combined with shielded equipment enclosures should be used.

11.7.2.2 For protection against the effects of conducted and induced surges being transmitted to the equipment *via* connection wiring, SPM consisting of a coordinated SPD system should be used. SPD to be used according to their installation position are as follows:

a) At the line entrance into the structure (at the boundary of LPZ 1, for example at the main distribution panel):

   1) SPD tested with I_{imp} (typical waveform 10/350, for example, SPD tested according to class I); and

   2) SPD tested with I_n (typical waveform 8/20, for example, SPD tested according to class II).

b) Close to the apparatus to be protected (at the boundary of LPZ 2 and higher, for example, at secondary distribution board, or at a socket outlet):

   1) SPD tested with I_n (typical waveform 8/20 for example, SPD tested according to class II); and

   2) SPD tested with a combination wave (typical waveform 8/20 for example, SPD tested according to class II).

11.7.3 *Equipotentialization of Services to LPS*

Equipotentialization is achieved by interconnecting the LPS with structural metal parts, metal installations having no electrical power connection, internal systems, external conductive parts and lines connected to the structure. *See* Fig. 31.

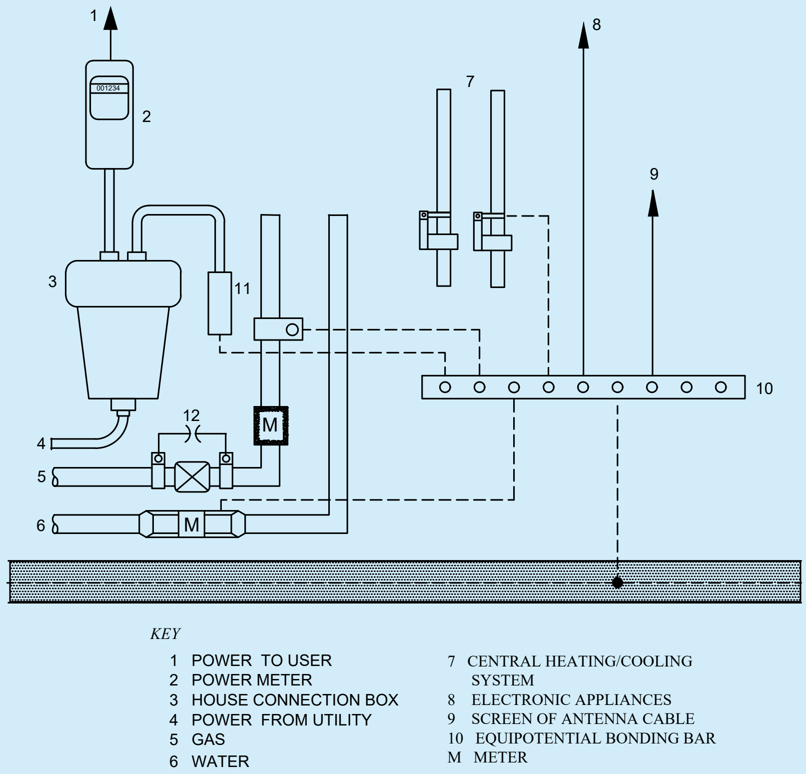


FIG. 31 EXAMPLE OF AN EQUIPOTENTIAL BONDING ARRANGEMENT

Interconnection can be done with bonding conductors, where the electrical continuity is not provided by natural bonding or by using surge protective devices (SPDs), where direct connections with bonding conductors is not feasible (for example, installation of SPDs for power, data, telecom lines, etc). It contains at least one non-linear component to ensure perfect equipotential bonding. All SPDs at the service entrance to an installation should be able to divert 10/350 µS impulse current depending upon selected level of protection. A three phase four wire system should be designed for 50 percent of the I_{imp} of selected LPS and single phase two wire system should be designed for 25 percent of the I_{imp} of selected LPS. The lightning current distribution for three phase four wire system is given in Fig. 32 {*see also* accepted standard [D-2(75)]}.

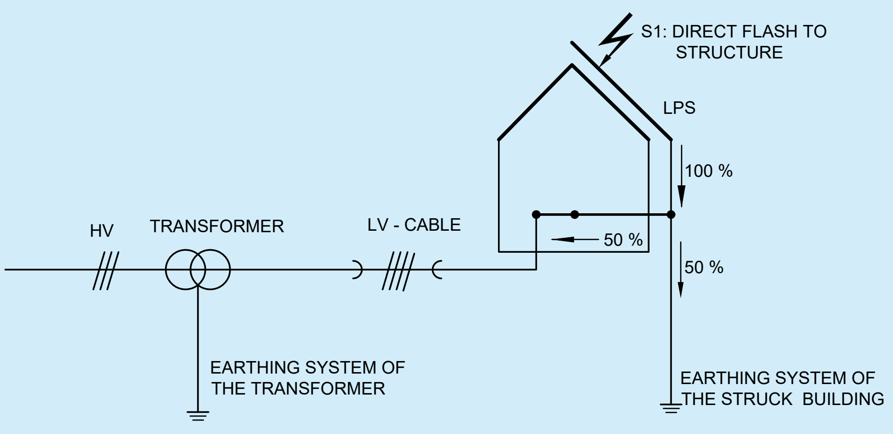


FIG. 32 BASIC EXAMPLE OF BALANCED CURRENT DISTRIBUTION

11.7.4 *Protection Measures with Surge Protection Devices* (*SPDs*)

11.7.4.1 Lightning surges frequently cause failure of electrical and electronic systems due to insulation breakdown or when overvoltages exceed the equipment's common mode insulation level. Power line protection is fundamental, however equal importance should be given to data, communication and instrumentation lines of the equipment that need protection. Equipment is protected if its rated impulse withstand voltage U_W at its terminals is greater than the surge overvoltage between the live conductors and earth. If not, an SPD shall be installed.

Implementing coordinated SPDs will provide protection against radiated surges for equipment {*see* accepted standard [D-2(74)]}. Shielding and routing of power and data lines, bonding of services and various lightning protection zones (LPZ) {*see* accepted standard [D-2(74)]} and earthing also plays a major role in protecting electrical and electronic equipment.

The SPDs are used to protect under specified conditions, electrical systems and equipment against various overvoltages and impulse currents such as lightning and switching surges. SPD shall be selected according to their environmental conditions and the acceptable failure rates of the equipment and the SPDs.

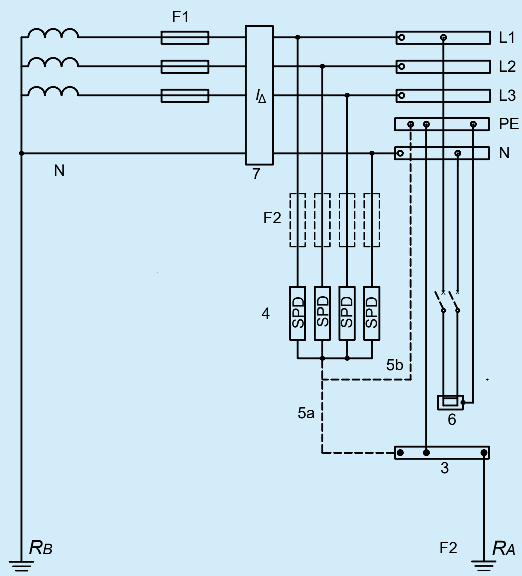
11.7.4.2 *Failure of SPDs*

Possibility of failure of any line to neutral or neutral to earth connected SPDs should not be ruled out, hence measures should be taken within the SPD for safe failure or withstand in worst conditions. SPD can fail in open or short modes. SPD should not create a fire hazard during failure. Safe failure mode is expected from the SPD.

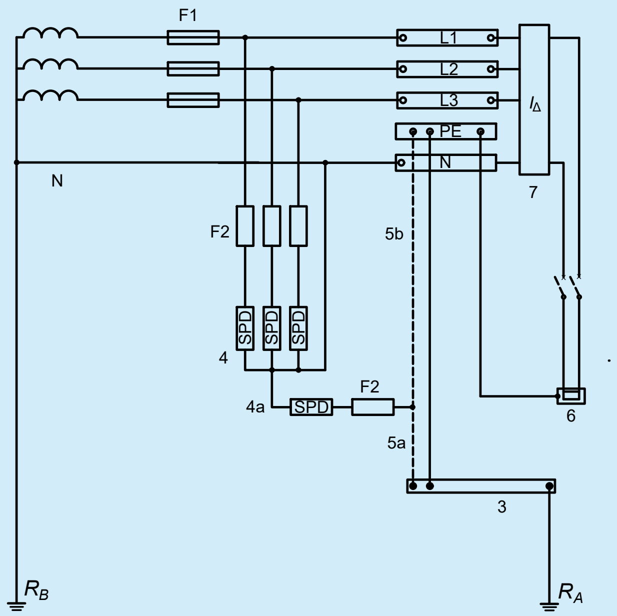
11.7.4.3 *Status indicators*

Each SPD should have inbuilt health indicator so as to show if protection is available. SPD should be installed in a way that visual inspection is easily possible. Failed SPD shall be replaced.

11.7.4.4 SPDs for power line need to be installed according to the type of service such as TN, TT, IT, etc. In general, the SPDs connected as per connection diagram given in Fig. 33 (informative) is suitable for TT connections. Reference may be made to relevant Indian Standard for such installations.



33A SPDs ON THE LOAD SIDE OF RCD



KEY

3 MAIN EARTHING TERMINAL OR BAR

4 SURGE PROTECTION DEVICES FOR SUB DB (CATEGORY II)

5 EARTHING CONNECTION OF SPDs

6 EQUIPMENT TO BE PROTECTED

7 RESIDUAL CURRENT PROTECTIVE DEVICE (RCD)

F1 PROTECTIVE DEVICE AT THE ORIGIN OF THE INSTALLATION

F2 PROTECTIVE DEVICE REQUIRED BY THE MANUFACTURER OF THE SPDs

R_A EARTHING ELECTRODE (EARTHING RESISTANCE) OF THE INSTALLATION

R_B EARTHING ELECTRODE (EARTHING RESISTANCE) OF THE SUPPLY SYSTEM

F2 IS REQUIRED IF F1 IS MORE THAN SPECIFIED VALUE OF BACK UP FUSE BY MANUFACTURER (OPTIONAL FOR N-PE CONNECTED SPD)

33B SPDS ON THE SUPPLY SIDE OF RCD

FIG. 33 TYPICAL INSTALLATION OF SPDS IN TT SYSTEM

11.7.4.5 *Selection of SPDs*

SPD(s) need to be selected based on the place of installation as well the impulse voltage withstanding capacity of the equipment. I_{imp} and I_n, are test parameters used to categorize Class I and Class II SPD(s). They are related to the maximum values of discharge currents, which are expected to occur at the LPL probability level at the location of installation (LPZ) of the SPD in the system. I_n is associated with Class II tests and I_{imp} is associated with Class I tests.

11.7.4.5.1 If no risk analysis according to accepted standard [D-2(76)] has been carried out or if the current value of SPD cannot be established, Class I SPDs should be installed with an I_{imp} not less than 12.5 kA (10/350µS) for connection according to Fig. 33A or between L to N or PE. For connection according to Fig. 33B, I_{imp} shall not be less than 50 kA (10/350µS) between N and PE for three-phase systems and 25 kA (10/350 µS) between N and PE for single-phase systems. Otherwise, the currents shall be considered as per level of protection.

11.7.4.5.2 *Follow current extinguishing capability of the SPD*

Follow current is the peak current (rms value) supplied by the electrical power system and flowing through the SPD after a discharge current impulse. Follow current extinguishing capability and short circuit withstand capacity of SPD shall be as per accepted standard [D-2(74)].

11.7.4.5.3 Where Class II tested SPD {*see* accepted standard [D-2(76)]} are required at or near the origin of installation, the value of I_n shall be not less than 5 kA for connections according to Fig. 33A or between L to N or PE. For Class II tested SPDs connected between neutral and PE for connection in Fig. 33B, I_n shall not be less than 20 kA for three-phase systems and 10 kA for single-phase systems.

11.7.4.5.4 *Voltage protection level of SPD*

The voltage protection level of SPD is the maximum voltage to be expected at the SPD terminals due to an impulse stress with defined voltage steepness and an impulse stress with a discharge current with given amplitude and wave shape. The protection level of SPDs shall be selected in accordance with impulse withstand voltage category as per Table 9.

Table 9 Rated Impulse Voltage for the Equipment Energized Directly from the Low Voltages Mains

(*Clause* 11.7.4.5.4)

Sl No.	Nominal Voltage of the Supply System[1] Based on on Accepted Standard [D-2(79)][2]		Voltage Line to Neutral Derived from Nominal Voltages a.c. or d.c. up to and Including	Rated Impulse Voltage[3]			
	Three Phase	Single Phase		Overvoltage Category [4]			
				I	II	III	IV
	V	V	V	V	V	V	V
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)			50	330	500	800	1 500
ii)			100	500	800	1 500	2 500
iii)		120 to 240	150	800	1 500	2 500	4 000
iv)	230/400 277/480		300	1 500	2 500	4 000	6 000
v)	400/690		600	2 500	4 000	6 000	8 000
vi)	1 000		1 000	4 000	6 000	8 000	12 000

[1] *See* accepted standard [D-2(80)].

[2] The '/' mark indicates a four-wire three-phase distribution system. The lower value is the voltage line-to-neutral, while the higher value is the voltage line-to-line. Where only one value is indicated, it refers to three-wire, while the higher value is the voltage line-to-line. Where only one value is indicated, it refers to three-wire, three-phase systems and specifies the value line-to-line.

[3] Equipment with these rated impulse voltage can be used in installations in accordance with IEC 60364-4-44 'Low-voltage electrical installations — Part 4-44: Protection for safety — Protection against voltage disturbances and electromagnetic disturbances'.

[4] *See* **4.3.3.2.2** of accepted standard [D-2(80)].

11.7.4.5.5 The preferred values of voltage protection level of SPDs, U_p is given below (*see* accepted standard [D-2(81)]):

0.08 kV, 0.09 kV, 0.10 kV, 0.12 kV, 0.15 kV, 0.22 kV, 0.33 kV, 0.4 kV, 0.5 kV, 0.6 kV, 0.7 kV, 0.8 kV, 0.9 kV, 1.0 kV, 1.2 kV, 1.5 kV, 1.8 kV, 2.0 kV, 2.5 kV, 3.0 kV, 4.0 kV, 5.0 kV, 6.0 kV, 8.0 kV and 10 kV

Hence, for protecting a 240 V/415 V connected equipment, selecting a $U_p \leq 1.5$ kV will be a safer choice. The U_p may be tested and certified by a third party as per the laid down test procedures. *See* Fig. 34 on connection diagram for SPD.

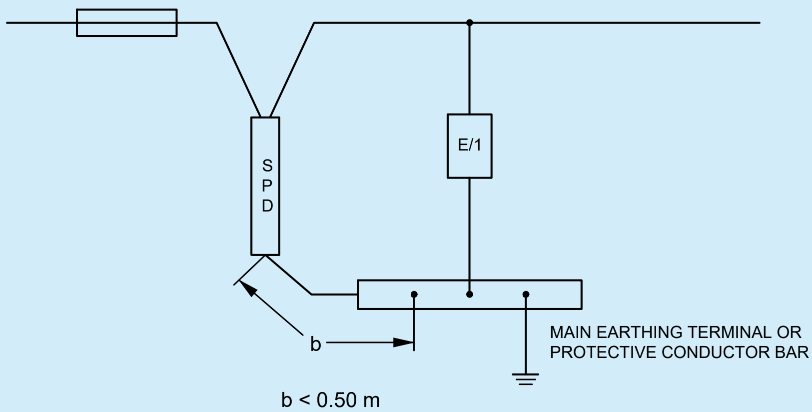


FIG. 34 EXAMPLE OF INSTALLATION OF SPDs AT OR NEAR ORIGIN OF INSTALLATION

11.7.4.6 For small residential buildings, power line SPD at the mains incoming panel will enhance the life of electronic equipment, such as TV, music system, refrigerators, LED lights, etc.

11.7.4.7 For large buildings, power line SPD is required at incoming panel as well as for sub distribution panels based on the LPZ principle {*see* accepted standard [D-2(74)]}.

11.7.4.8 For industrial and commercial buildings, critical and sensitive loads such as drives, PLC's, automation panels, etc, require protection with SPD in addition to SPD at incoming power panels and sub distribution boards. SPDs shall be selected to meet the requirements of relevant LPZs.

11.7.4.9 Lifts, escalators, moving walks and fire panels shall be protected with SPD in control panels. All electrical and control panels related to safety and security of building shall be protected with appropriate SPDs.

11.7.4.10 SPDs should be installed for outdoor equipment such as CCTV cameras, LED street lights, weighbridges, firefighting systems, roof top solar PV installations, etc. This will ensure availability of the vital services provided by these equipment as and when required.

11.7.4.11 Failure of equipment and chance of fire in electrical installations are more for buildings near tall structures (for example, telecom tower). These buildings should be protected with SPD at power incoming and ring earthing to avoid fire and equipment failure.

11.7.4.12 All SPDs should have status indication to show their healthy state for discharging the lightning current. The possibility of failure of L-N as well as N-PE connected SPD cannot be ignored.

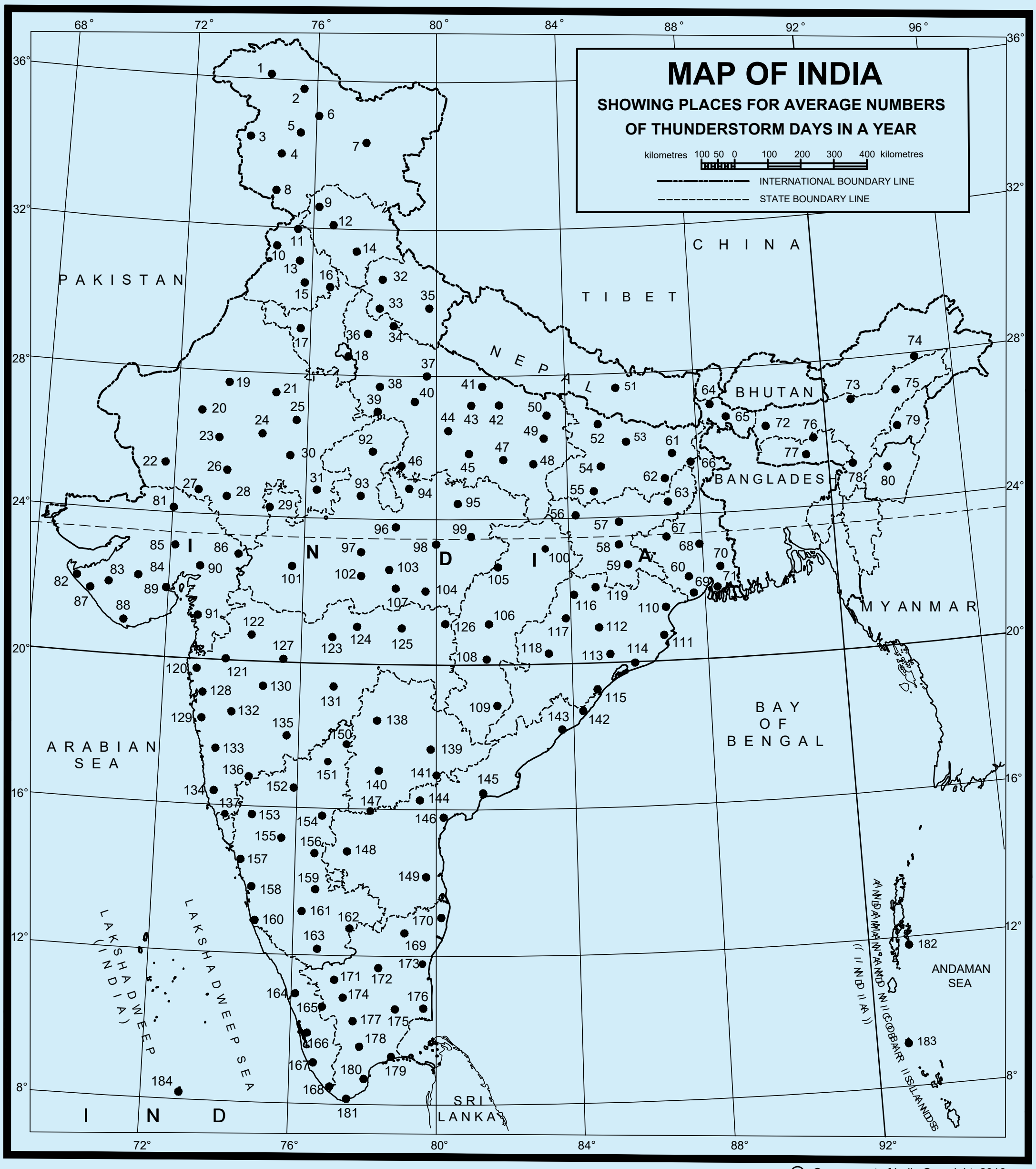
11.7.4.13 The SPD shall be installed at the entrance to ensure perfect bonding to the ground at the time of lightning {*see* good practice [D-2(11)] and accepted standard [D-2(74)]}. The let through energy of the protection system shall be less than the energy that equipment can withstand. As per accepted standard [D-2(74)], the energy coordination method is the best method to ensure the protection of the equipment. The let through energy details of the SPD shall be provided by manufacturer.

11.7.4.14 Maximum continuous operating voltage of the SPD (U_c) should not be less than 1.1 × $U_{nominal}$ = 1.1 × 230 = 253 V; temporary overvoltage (TOV) withstand values shall be as per accepted standard [D-2(81)].

11.7.4.15 Low-voltage SPDs for connection to low-voltage power systems shall be conforming to the accepted standard [D-2(81)].

11.8 Average Number of Thunderstorm Days

For the purpose of risk assessment, annual thunderstorm days in various places are provided in the table below read with Fig. 35.



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BASED UPON SURVEY OF INDIA POLITICAL MAP PRINTED IN 2002.

THE TERRITORIAL WATERS OF INDIA EXTEND INTO THE SEA TO A DISTANCE OF TWELVE NAUTICAL MILES MEASURED FROM THE APPROPRIATE BASELINE.

THE INTERSTATE BOUNDARIES BETWEEN ARUNACHAL PRADESH, ASSAM AND MEGHALAYA SHOWN ON THIS MAP ARE AS INTERPRETED FROM THE NORTH-EASTERN AREAS (REORGANIZATION) ACT, 1971, BUT HAVE YET TO BE VERIFIED.

THE EXTERNAL BOUNDARIES AND COASTLINES OF INDIA AGREE WITH THE RECORD/MASTER COPY CERTIFIED BY SURVEY OF INDIA.

THE RESPONSIBILITY FOR THE CORRECTNESS OF INTERNAL DETAILS REST WITH THE PUBLISHER.

FIG. 35 MAP OF INDIA SHOWING PLACES FOR AVERAGE NUMBER OF THUNDERSTORM DAYS IN A YEAR

Sl No.	Name of Place	Annual Thunder-storm Days
1	Gilgit	7
2	Skardu	5
3	Gulmarg	53
4	Srinagar	54
5	Dras	3
6	Kargil	2
7	Leh	3
8	Jammu	26
9	Dharmsala	13
10	Amritsar	49
11	Pathankot	4
12	Mandi	46
13	Ludhiana	12
14	Simla	40
15	Patiala	26
16	Ambala	9
17	Hissar	27
18	Delhi	30
19	Bikaner	10
20	Phalodi	14
21	Sikar	17
22	Barmer	12
23	Jodhpur	23
24	Ajmer	26
25	Jaipur	39
26	Kankroli	36
27	Mount Abu	5
28	Udaipur	38
29	Neemuch	28
30	Kota	27
31	Jhalawar	40
32	Mussorie	61
33	Roorkee	74
34	Najibabad	36
35	Mukteswar	53
36	Meerut	–
37	Bareilly	34
38	Aligarh	30
39	Agra	24
40	Mainpuri	23
41	Bahraich	31
42	Gonda	22
43	Lucknow	18
44	Kanpur	26
45	Fatehpur	24
46	Jhansi	20
47	Allahabad	51
48	Varanasi	51
49	Azamgarh	1
50	Gorakhpur	11
51	Katmandu	74
52	Motihari	38
53	Darbhanga	10
54	Patna	33
55	Gaya	38
56	Daltonganj	73
57	Hazaribagh	73
58	Ranchi	34
59	Chaibasa	70
60	Jamshedpur	66
61	Purnea	52
62	Sabour	76
63	Dumka	63
64	Darjeeling	28
65	Jalpaiguri	68
66	Malda	59
67	Asansol	71
68	Burdwan	39
69	Kharagpur	76
70	Kolkata	70
71	Sagar Island	41
72	Dhubri	8
73	Tezpur	27
74	Dibrugarh	98
75	Sibsagar	103
76	Shillong	75
77	Cherrapunji	49
78	Silchar	33
79	Kohima	34
80	Imphal	49
81	Deesa	7
82	Dwarka	5
83	Jamnagar	8
84	Rajkot	12
85	Ahmedabad	11
86	Dohad	17
87	Porbandar	3
88	Veraval	3
89	Bhavnagar	11
90	Vadodara	8
91	Surat	4
92	Gwalior	53
93	Guna	33
94	Nowgong	59
95	Satna	41
96	Sagar	36
97	Bhopal	44
98	Jabalpur	50
99	Umaria	37
100	Ambikapur	29
101	Indore	34
102	Hoshangabad	37
103	Panchmarhi	30
104	Seoni	51
105	Pendadah	56
106	Raipur	34
107	Chhindwara	27
108	Kanker	37
109	Jagdalpur	35
110	Balasore	81
111	Chandbali	75
112	Angul	81
113	Bhubaneswar	46
114	Puri	33
115	Gopalpur	34
116	Jharsuguda	85
117	Sambalpur	67
118	Titlagarh	24
119	Rajgangpur	1
120	Dahanu	1
121	Nasik	17
122	Malegaon	13
123	Akola	20
124	Amraoti	32
125	Nagpur	45
126	Gondia	10
127	Aurangabad	34
128	Mumbai	16
129	Alibag	12
130	Ahmadnagar	10
131	Parbhani	32
132	Pune	22
133	Mahabaleshwar	14
134	Ratnagiri	6
135	Sholapur	23
136	Miraj	25
137	Vengurla	39
138	Nizambad	36
139	Hnamkonda	43
140	Hyderabad	28
141	Khammam	26
142	Kalingapatam	20
143	Vishakapatnam	46
144	Rentichintala	42
145	Masulipatam	20
146	Ongole	25
147	Kurnool	29
148	Anantpur	22
149	Nellore	18
150	Bidar	15
151	Gulbarga	34
152	Bijapur	9
153	Belgaum	31
154	Raichur	17

155	Gadag	21
156	Bellary	22
157	Karwar	27
158	Honavar	5
159	Chikalthana	24
160	Mangaluru	36
161	Hassan	26
162	Bengaluru	46
163	Mysuru	44
164	Kozhikode	39
165	Palghat	35
166	Kochi	69
167	Alleppey	51
168	Thiruvanantha-puram	68
169	Vellore	25
170	Chennai	47
171	Udhagamandalam	24
172	Salem	69
173	Cuddalore	37
175	Trichchirapalli	41
176	Nagappattinam	15
177	Kodaikanal	82
178	Madurai	39
179	Pamban	5
180	Tuticorin	14
181	Cape Comorin	68
182	Port Blair	62
183	Car Nicobar 1	18
184	Minicoy 1	20

12 ELECTRICAL INSTALLATIONS FOR CONSTRUCTION AND DEMOLITION SITES

12.1 General

12.1.1 Electrical hazards are a major cause of serious injury and even death in construction sites. Accidents also cause loss of productivity and destroy the morale of workers. The need to use electricity and electrical/electronic equipment has been constantly increasing. Without these gadgets productivity and quality of work will suffer. Therefore, the use of electricity and the use of gadgets has to increase. Such increase requires a proper electrical distribution system in the work site.

12.1.2 To ensure continuous supply of power during the construction activity and maintain productivity, site security, etc the city power supply may be required to be supplemented by on-site standby power generation. Some gadgets require continuity of power supply without interruption, thereby requiring UPS systems. In a typical large construction site there may be a large temporary distribution network combined with more than one source of electricity, which can make the system quite complex from the safety point of view.

12.1.3 Problem may also arise in case of lack of required training to workers in the safe use of the tools and equipment that they are required to handle in a system with multiple sources of power supply. In case of use of imported equipment which may be manufactured to their own standards, problems may arise, such as, during connection and inter-connection of equipment and tools and mismatch of plugs and sockets. Appropriate action is to be taken by employing qualified/experienced electrical engineer.

12.1.4 Practical guidance to employers, designers, manufacturers, importers, suppliers (including hirers), electrical contractors and electricians on eliminating or reducing the risk of electrocution and electric shock to any person is necessary.

12.1.5 Even though awareness exist about good practices, the same may be compromised at times in the name of speed or economy or due to ignorance and neglect. The materials, equipment, tools, cables, switchgear used in the temporary installation face far more severe environmental working conditions because of use of discarded switchgear, cables, etc at the construction sites which compounds the risk to workmen from shock and fire. Hence, the laid down Indian Standards for permanent installations, need to be followed during construction and demolition meticulously. The complete installation shall be duly earthed as per *Central Electricity Authority* (*Measures related to safety and Electric Supply*) *Regulations*, 2023, good practice [D-2(11)] and [D-2(1)].

12.2 Installation and Removal of Construction Wiring

All construction wiring work shall be installed by an appropriately registered electrical contractor and by electrical workmen holding license of the appropriate level of competency depending on the voltage, local generation, etc, as provided in *Central Electricity Authority* (*Measures Related to Safety and Electric Supply*) *Regulations*, 2023 and *the Indian Electricity Act*, 2003. The installation should be inspected prior to commissioning and at regular intervals by the Engineer-in-Charge (if one is qualified to inspect electrical installations) or by representative, who should be competent to inspect an installation.

12.3 Provision of Indicating and Recording Instruments and Meters

Measurement is a prerequisite to analyse the performance. Construction site switchboards should have adequate instrumentation (such as, ammeters, PF meters, voltmeters, energy meters, etc) on different branches of the distribution system so that any abnormal condition or overload is noticed. Considering the nature of varying activity in a construction site, frequent visual, and instrument based inspections are necessary as well as keeping record of the same.

12.4 Residual-Current Circuit Breaker (RCCB)/Residual-Current Device (RCD)

The following shall be ensured in the construction and demolition sites:

a) Every electric supply to which electrical plant can be connected should incorporate an RCCB/RCD of appropriate capacity so as to protect persons who may come into contact with the electrical plant against electric shock;

b) The RCCB/RCD(s) should have a rated tripping current not exceeding 30 mA and should have the capacity to carry the load current required by the appliances permitted to be connected in that branch circuit or feeder;

c) Where construction work supply can only be obtained from a permanent wiring socket-outlet, RCCB/RCD should be connected at the socket-outlet;

d) Sub-mains supplying site sheds should incorporate an RCCB/RCD having a rated tripping current not exceeding 100 mA;

e) Every non-portable RCCB/RCD device on the worksite shall be trip tested by the built-in push button test every month, and performance tested for operation before being put into service and thereafter at least once every 12 months. It shall also be subjected to an imbalance of current not less than the rated residual current and shall trip in a time not exceeding 6 s;

f) Every portable RCCB/RCD device on the worksite shall be trip tested by the built-in push button test. The test shall be done prior to use and each day while in use; and it shall be performance tested for operation before being put into service and thereafter at least once every 3 months. It shall also be subjected to an imbalance of current not less than the rated residual current and shall trip in a time not exceeding 6 s;

g) Results of RCCB/RCD tests shall be recorded and kept on site or made available for audit and kept for a minimum period of 5 years or till the completion and handing over of the project, whichever is less [excluding the daily push button test for portable RCCB/RCD(s)];

h) Portable RCCB/RCD(s) when tested shall be fitted with a durable, non-reusable, non-metallic tag. The tag shall include the following information:

   1) The name of the person or company who performed the tests; and

   2) The test or retest date.

   The recommended colour coding for tags on tested RCCB/RCD, which is prescribed below should be indicated by its colour representing the period when the test was performed:

January to March	:	Red
April to June	:	Green
July to September	:	Blue
October to December	:	Yellow

j) Portable generators should be fitted with RCDs having a rated tripping current not exceeding 30 mA; and

k) Personal hoists used on construction sites shall be supplied from a separate final sub-circuit originating from the main switchboard; and be suitably identified by marking this supply on the RCD/MCB feeding it.

12.5 Temporary Supply Switchboards

All temporary supply switchboards used on building, construction and demolition sites shall be of robust construction and either securely attached to a pole, post wall or other structure which may be of stable freestanding design and:

a) where installed in outdoor locations, should be constructed and maintained to IP23 (or higher) rating so that safe operation is not affected by the weather;

b) switchboards should incorporate the support and elevation of cables and flexible extension cords;

c) switchboard enclosures shall be provided with an insulated or covered tie-bar or similar arrangement for the anchorage of the cables or flexible cords in order to prevent strain and mechanical damage at the termination of the cables or cords;

d) switchboards should be provided with a door and locking facility. The doors should be designed and attached in a manner that will not damage any flexible cord connected to the board and should protect the switches from mechanical damage;

e) the door should be provided with signs (in English, Hindi and at least one local language) stating, 'KEEP CLOSED – LEADS THROUGH BOTTOM';

f) switchboards should have an insulated slot (with edges suitably shaped or covered with plastic or rubber trims to avoid damaging the cable insulation/sheath) at the bottom for the passage of leads;

g) switchboards should be attached to a permanent wall or suitable portable or temporary structure in an elevated position suitable for easy access and having least interference with the activity in the area;

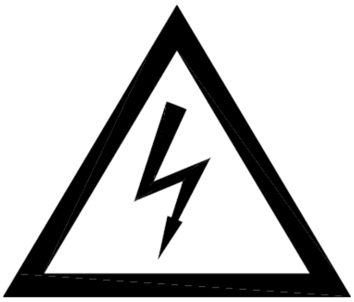
h) if the floor in that area is likely to be wet, additional precautions are necessary and an insulated platform should preferably be provided for access to operation of switches. Personal protective equipment such as helmet, electrician gloves, etc, should be available, placed near the switchboard, as an alternative;

j) a clearance of at least 1.2 m should be maintained in front of all switchboards;

k) the contractor or nominated persons should ensure that all power circuits are isolated or made inaccessible so as to eliminate the risk of fire, electric shock or other injury to persons after completion of the daily work;

m) switchboards shall be legibly and indelibly marked with a set of numbers or letters or both which uniquely identify the switchboard from others on a site;

switchboards shall be marked to indicate the presence of live parts in accordance with symbols such as the following drawing:



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p) chart indicating first aid measures for electric shock management should be placed near the switchboard;

q) 'Lock Out and Tag Out' procedures as adopted should be displayed;

r) the switchboard should be lighted properly to ensure its identification and safe working; and

s) all switchboards should have their identification names (or/and numbers) so that any communication/ instructions about them are unambiguous.

12.6 Cables Used in Worksite Installations

12.6.1 Worksite poses a number of hazards to the cables. The conditions are severe compared to those in permanent installations. Cables face dust, moisture, abrasives and even impact unlike those in permanent installations. Extra care should be taken to frequently inspect the cables and discard cables which show signs of damage as a damaged insulation combined with moisture can be dangerous.

12.6.2 Cables are likely to be run over by vehicles. Suitable protection is required by a steel pipe or hume pipe or steel plate, whichever is appropriate. Dragging of cables damages their sheath/insulation and may also cut a few strands of the conductor due to stretching, effectively reducing the current carrying capacity. Hence suitable care should be taken while laying the cables.

12.6.3 Overhead installation of cables is a common practice. Such installations should use a GI wire to carry the weight of the cable without causing the stretching of the cable. The carrier GI wire should be earthed. Unarmoured cables shall not be installed on metallic roofs or similar structures unless suitably protected against mechanical damage.

Overhead wiring should be positioned to avoid crossing roadways or accessways where cranes, high loads, or heavy machinery may travel. Where it is not possible to avoid accessways an effective means shall be provided to minimize the risk of the vehicular contact with the aerial wiring system. This condition may be satisfied by the placement of flagged catenary wires or cables of suitable material across the accessway, 6 m on either side of the overhead wiring and 0.6 m below the lowest point of the overhead electrical cables or lower.

All aerial conductors installed on construction and demolition sites shall be insulated. Cables supported by means of a catenary shall be stranded or shall be flexible cables affording double insulation or the equivalent of double insulation. Construction wiring (including switchboards) shall be visually inspected at intervals not exceeding 6 weeks.

12.7 Electrical Plant in Service Testing

Electrical plant shall be inspected and tested in accordance with the following:

a) Movable electrical plant that is hand held or portable during operation or moved between operations and is subject to damage or harsh environment shall be examined and tested every 3 months;

b) All other electrical plant used for construction purposes shall be inspected and tested at intervals not exceeding 6 months;

c) When any equipment inspected or tested in accordance with (a) and (b) is found to be unsatisfactory, it shall be withdrawn from service immediately and have a label attached to it, 'Warning against Further Use'. Electrical plant found to be unsatisfactory shall not be returned to service until it has been repaired and retested;

d) The inspection and testing specified should be carried out by an authorized/qualified person; and

e) The results of the inspection of electrical plant should be recorded and kept on site or made available for inspection by authorities. Information recorded shall include:

    1) the name of the person or company who performed the tests;

    2) the test or retest date; and

    3) identification of faulty equipment and action taken to repair or remove it from use.

12.8 Lighting

Following in respect of lighting shall be ensured in construction and demolition sites:

a) *Access lighting* — Adequate artificial lighting should be installed to illuminate the work area if there is insufficient natural lighting. Lamps in luminaires shall be protected against mechanical damage. Luminaires installed as part of the permanent electrical installation in site accommodation, may not require further mechanical protection;

   Sufficient battery powered lighting shall be installed in stairways and passageways to allow safe access and exit from the area if there is insufficient natural lighting. If there is a loss of supply to the normal lighting in the area, it should be ensured that battery powered lighting has sufficient capacity to operate for one hour to allow persons to exit the building safely.

   Temporary wiring supplying lighting circuits should be connected to the designated lighting circuits of the switchboard.

b) *Task lighting* — Portable luminaires shall be provided with the appropriate ingress protection (IP) rating. Task lighting in many construction site may have to move continuously in steps as the work progresses. The lighting system should also be shifted in suitable steps instead of simply extending the cables without any consideration for safety. Hand held additional lights, if any, should be taken from a socket protected by a RCCB/RCD of 30 mA setting just like hand held tools;

c) *Lift and service shaft lighting* — Lift and service shaft lighting may have either construction wiring or permanent wiring. Fluorescent lighting should be used. The lights should be located on the floor above or below the work area. It shall be ensured that the emergency lighting has sufficient battery capacity to operate for a minimum of 1 h if there is a loss of supply to the normal lighting in the area;

d) *Lighting in means of egress* — Sufficient battery powered lighting shall be installed in stairways and passageways to allow safe access and exit from the area if there is insufficient natural lighting. If there is a loss of supply to the normal lighting in the area it shall be ensured that battery powered lighting has sufficient capacity to operate for 1 hr to allow persons to exit the building safely; and

e) *Illumination of signs and warning boards* — Sufficient lighting should be available to illuminate the sign boards and warning boards. These lights may also be a part of the lighting set for means of egress.

12.9 Transportable Construction Buildings (Site Sheds)

Electrical installations to transportable construction buildings shall comply with the following:

a) If supply is by means of a flexible cord, the same shall not be taken from one transportable building to another transportable building.

b) The flexible cord supplying a transportable building should not be more than 15 m in length.

c) Each amenity in the building shall be connected/supplied by a flexible cord to a final sub-circuit protected by an RCCB/RCD device with a rated tripping current not exceeding 30 mA; these flexible cords should be protected from mechanical damage and power outlets in site sheds should be used to supply power to the equipment and lighting within the shed only.

d) Socket-outlets installed on the outside of transportable building shall be used only to supply power to the following:

   1) Electrical equipment and lighting immediately adjacent to those transportable buildings.

   2) Other transportable buildings when the socket-outlet is part of an interconnecting system and the flexible cord supplying those transportable buildings has a maximum length of 15 m.

12.10 Lock-Out and Tag-Out Practices

Whenever an electrician is to work on a branch feeder from the switchboard it becomes necessary that the switch feeding the feeder on which work is done is to be switched-off and should be kept switched-off till the same electrician decides to switch on after completion of work either for testing or for putting the section back into service. To ensure that during the period an electrician is working on a feeder or the equipment it feeds, the practice is to put a 'tag' on the concerned switch. The tag will carry details about which authorized worker has put the tag and when and who is authorized to remove it. This is a very critical operating practice for maintenance of the electrical system and following it is of particular importance in the case of temporary installations, where such requirements arise more often than in permanent installations.

Lock-out procedure shall be similar to the above but the difference is that the relevant switch is kept locked in off position and the key is kept by the person or the team leader who will be working on the feeder. The system should also maintain a register where the activities of this nature and the details of action (repair, transfer of the end equipment to a new location, addition of a new appliance on the feeder, etc) are recorded.

12.11 Standard Operating and Maintenance Practices in Sites with More Than One Source of Electricity

12.11.1 In medium and large construction sites it is common to use local generation of electricity by a diesel generating set. The reasons may be lack of dependable local/city distribution system for continuous power supply, high cost of temporary connection charges, restrictions on use of certain equipment like welding transformers on temporary connection, etc. Apart from DG sets for power requirement of site construction equipment, there may be office equipment with computers and site laboratory equipment which require un-interruptible power supply. Shock hazards depend on the voltage of the system and the consequent current flow through the body. As such shock risk associated with all these sources is the same.

12.11.2 Wherever an alternative power supply is provided, proper protocols shall be adopted for the supply from different systems and the associated change-over switch or contactors. Risk increases due to existence of more than one source of power supply. System schematics, operating practice and essential interlock between the different sources of power should be displayed and followed systematically. Earthing for each system should be provided. Changeover switches should be of 4 pole in order to ensure that earth leakage protections operate properly with each one of the power sources. The system should also maintain a register where the activities related to each source of power is recorded and the actions to be taken are provided.

12.12 Earthing or Grounding

12.12.1 Earthing or grounding is an essential pre-requisite for any electrical system from the aspect of safety of personnel, equipment, appliances and for avoidance of fire due to short-circuit or of feeding energy to a short circuit originating from any other cause consequent to damage to insulation by the fire.

12.12.2 The standard earthing practice is applicable to temporary installations also. The earth sets installed at the initial development of the site can be planned to be retained as earthing sets for use even after the completion of the building for the permanent installation.

12.12.3 The minimum requirement of earthing for any temporary electrical installation is given hereunder for easy adherence to the basic minimum. It is recommended to follow the earthing requirements given in **8** and accepted standard [D-2(11)].

12.12.3.1 The neutral of the system of each source or generator shall be having two distinct connection to two distinct earthing sets. All metal parts associated with electrical equipment are required to be connected to earth. The minimum requirement is the provision of two pipe earthing sets each with a pipe of 4.2 m length and separated by at least 2.5 m between them in accordance with good practice [D-2(11)] and connected to the source neutral by a conductor of cross-section more than half the size used for the phase conductor.

12.12.3.2 Earth continuity shall be maintained all over the site wherever electricity is made available and the earth continuity conductor shall have a cross-section at least more than that used for the phase conductor.

12.12.3.3 Wherever the requirements of earthing and use of RCD or RCCB cannot be satisfactorily met with at any site and electrical hand tools are required to be used, low voltage (< 50 V) appliances or self-contained battery operated tools shall be used as a safe alternative. This applies to work under damp or water logged areas also.

13 PROTECTION OF HUMAN BEINGS FROM ELECTRICAL HAZARDS

13.1 General

Danger to persons due to contact with live parts is caused by the flow of the current through the human body. The effects on the human body are dependent on the amount of the current flow and the duration of the flow. The current flow through the human body depends on the voltage and the resistance in the path. The contact resistance forms an important or significant contributor to the total resistance for the current flow. The contact resistance reduces by a factor of 100 to 1 000 due to the body part being wet and also the touch pressure at the point of contact. The works under this clause shall cover the relevant Indian Standards and good practice [D-2(1)].

13.1.1 *Physiological Effects of Electric Shock*

The following are the physiological effects of electric shock:

a) *Tetanization* — The muscles affected by the current flow involuntarily contract and letting go of gripped conductive parts is difficult;

   NOTE — Very high currents do not usually induce muscular tetanization because, when the body touches such currents, the muscular contraction is so sustained that the involuntary muscle movements generally throw the subject away from the conductive part.

b) *Breathing arrest* — If the current flows through the muscles controlling the lungs, the involuntary contraction of these muscles alters the normal respiratory process and the subject may die due to suffocation or suffer the consequences of traumas caused by asphyxia;

c) *Ventricular fibrillation* — The most dangerous effect is due to the superposition of the external currents with the physiological ones, which, by generating uncontrolled contractions, induce alterations of the cardiac cycle. This anomaly may become an irreversible phenomenon since it persists even when the stimulus has ceased; and

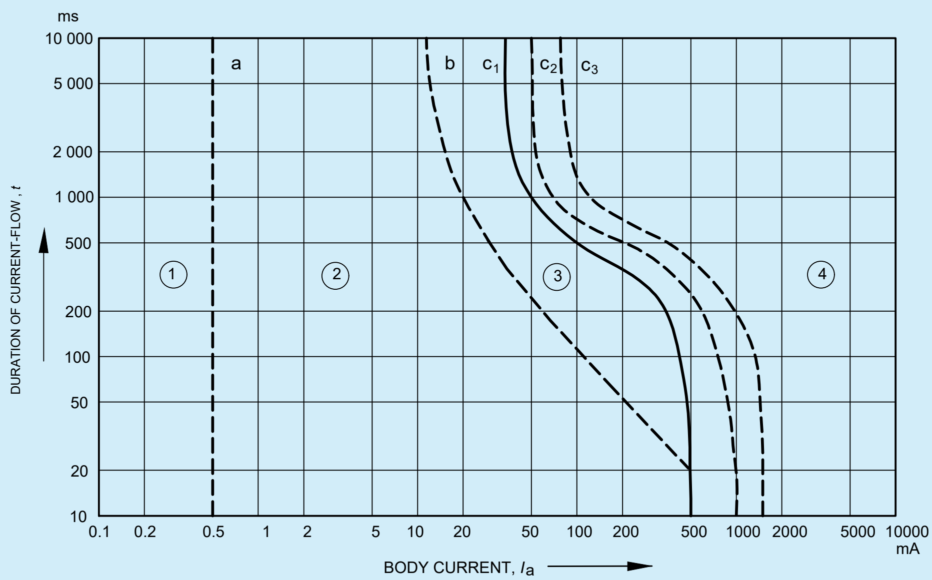
d) *Burns* — They are due to the heating deriving, by Joule effect, from the current passing through the human body.

13.1.2 On a time-current diagram (*see* Fig. 36), four zones to which the physiological effects of alternating current (15 Hz to 100 Hz) passing through the human body have been related as given below:

13.1.3 In general, 'special locations' involve one or more of the following environmental conditions or different risks, which in the absence of special arrangements, will give rise to an increased risk of electric shock:

a) wetness or condensation, that is, reduced skin resistance;
b) absence of clothing, that is, greater opportunity to make direct or indirect contact through increased area of bare skin and bare feet;
c) proximity of earthed metalwork, that is, increased risk of indirect contact; and
d) arduous or onerous site conditions, that is, conditions of use that may impair the effectiveness of the protective measures.

13.1.3.1 The additional requirements for these 'special locations' have been devised by assessing the relevant risks under each of the above categories and making adjustments to the protective measures accordingly. The intention is to remove or minimize the additional risks to users presented by the electrical installations within these 'special locations'. Designers, installers and operators of such installations should consider the particular requirements of the installation and design and install or operate the installation to reduce or protect against risks from these requirements. It should be however recognized, that all installations require adequate regular periodic inspection and testing, and any necessary maintenance or repair works be properly carried out.



NOTES

1 AS REGARDS VENTRICULAR FIBRILLATION, THIS FIGURE RELATES TO THE EFFECTS OF CURRENT WHICH FLOWS IN THE PATH 'LEFT HAND TO FEET'. FOR OTHER CURRENT PATHS, SEE 5 AND TABLE III OF ACCEPTED STANDARD [D-2(82)].

2 THE POINT 500 MA/100 MS CORRESPONDS TO FIBRILLATION PROBABILITY IN THE ORDER OF 0.14 PERCENT.

Zones	Physiological effects
Zone 1	Usually no reaction effects.
Zone 2	Usually no harmful physiological effects.
Zone 3	Usually no organic damage to be expected. Likelihood of muscular contractions and difficulty in breathing, reversible disturbances of formation and conduction of impulses in the heart, including atrial fibrillation and transient cardiac arrest without ventricular fibrillation increasing with current magnitude and time.
Zone 4	In addition to the effects of Zone 3, probability of ventricular fibrillation increasing up to about 5 percent (curve c_2), up to about 50 percent (curve c_3) and above 50 percent beyond curve c_3. Increasing with magnitude and time, pathophysiological effects such as cardiac arrest, breathing arrest and heavy burns may occur.

FIG. 36 TIME/CURRENT ZONES OF THE EFFECTS OF a.c. CURRENTS (15 Hz TO 100 Hz)

13.2 Protection Against Electric Shocks

13.2.1 Protective measures against electric shock are based on following two common dangers:

a) Contact with an active conductor, which is live with respect to earth in normal circumstances; and is referred to as a 'direct contact' hazard;

b) Contact with a conductive part of an apparatus which is normally dead, but which has become live due to insulation failure in the apparatus; and is referred to as an 'indirect contact' hazard; and

The third type of shock hazard exists in the proximity of HV or LV (or mixed) earth electrodes which are passing earth-fault currents.

c) This hazard is due to potential gradients on the surface of the ground and is referred to as a 'step-voltage' hazard; shock current enters one foot and leaves by the other foot, and is particularly dangerous for four-legged animals. A variation of this danger, known as a 'touch voltage' hazard can occur, for instance, when an earthed metallic part is situated in an area in which potential gradients exist. Touching the part would cause current to pass through the hand and both feet. Animals with a relatively long front-to-hind legs span are particularly sensitive to step-voltage hazards.

NOTE — Cattle have been killed by the potential gradients caused by a low voltage (240/415 V) neutral earth electrode of insufficiently low resistance.

13.2.2 Potential-gradient problems as mentioned in 13.2.1 are not normally encountered in electrical installations of buildings, provided that equipotential conductors properly bond all exposed metal parts of equipment and all extraneous metal (that is, those which are not part of an electrical apparatus or the installation, for example structural steelwork, etc) to the protective-earthing conductor.

Potential-gradient problems do exist at the switchboards of electrical installations. Even where components, such as the metallic cubicle and operating handle are earthed, there is a possibility that for very short duration when a fault current is flowing through the metal part, the potential of the earthed metal part may be increased due to the current flow. The practice of using an insulating mat at places where the switches are to be operated, is for protection against such transient over-voltages in earthed metal parts (if the metal part is not earthed the transient overvoltage will be beyond safe limits).

With the increased use of pumped water in bathrooms for spa-baths, power showers, etc, there is a need, on occasions, to provide for motive power within the area of the bath. In order to protect against direct and indirect contact, supplies for such equipment shall be by extra-low-voltage-system with the nominal voltage not exceeding 12 V r.m.s.

It is a requirement to provide supplementary equipotential bonding by bonding all simultaneously accessible exposed and extraneous conductive parts. This should include all in-coming metallic services (including metal wastes) and any structural metalwork, including a metallic floor grid, if provided.

Protection measures as given in 13.2.3 to 13.2.5 shall also be provided for protection against electric shock.

13.2.3 *Direct-Contact Protection or Basic Protection*

The main form of protection against direct contact hazards is to contain all live parts in housings of insulating material or in metallic earthed housings, by placing out of reach (behind insulated barriers or at the top of poles) or by means of obstacles. Where insulated live parts are housed in a metal envelope, for example, transformers, electric motors and many domestic appliances, the metal envelope is connected to the installation protective earthing system. Protection classification and the details of the different levels of protection are designated by IP XX and IK X {*see* accepted standard [D-2(36)] and [D-2(10)]. The metallic enclosure has to demonstrate an electrical continuity, then establishing a good segregation between inside and outside of the enclosure. Proper grounding of the enclosure further participates the electrical protection of the operators under normal operating conditions. For LV appliances this is achieved through the third pin of a 3 pin plug and socket. Total or even partial failure of insulation to the metal, can raise the voltage of the envelope to a dangerous level (depending on the ratio of the resistance of the leakage path through the insulation, to the resistance from the metal envelope to earth). The minimum standard for any equipment has to be IP2X to limit the possibility of a finger touching a live part of an electrical system.

13.2.4 *Indirect-Contact Protection or Fault Protection*

A person touching the metal envelope of an apparatus with a faulty insulation is said to be making an indirect contact. An indirect contact is characterized by the fact that a current path to earth exists (through the protective earthing (PE) conductor) in parallel with the shock current through the person concerned. The hazard in such cases is dependent on the earth resistance and the resistance of the earth continuity conductor, which would effectively be in parallel to the path of flow of current through the human body.

Extensive tests have shown that, providing the potential of the metal envelope not greater than 50 V with respect to earth, or to any conductive material within reaching distance, no danger exists.

If the insulation failure in an apparatus is between a HV/LV conductor and the metal envelope, it is not generally possible to limit the rise of voltage of the envelope to 50 V or less, simply by reducing the earthing resistance to a low value. The solution in this case is to create an equipotential 'earthing systems'. Insulation faults affecting the LV substation's equipment (internal) or resulting from atmospheric overvoltages (external) may generate earth currents capable of causing physical injury or damage to equipment.

a) Preventive measures essentially consist of all substation frames and connecting them to the earth bar;

b) Minimizing earth resistance; and

c) Fast tripping and isolation of the fault by the operation of the fuse or circuit breaker.

13.2.5 In order to provide protection against electric shock due to leakage current for human being, a 30 mA RCCB/RCD shall be installed at distribution board incomer of buildings, such as residential, schools and hospitals. For all other buildings, a 100 mA RCCB will suffice for protection against leakage current.

13.2.6 *Additional Protection Against Electric Shocks*

In a.c. systems, additional protection by means of a residual current protective device (RCD) with sensitivity not exceeding 30 mA shall be provided for:

a) socket-outlets with a rated current not exceeding 20 A that are for use by ordinary person and are intended for general use; and

b) mobile equipment with a current rating not exceeding 32 A for use outdoors.

The residual current devices shall be independent of the line voltage, except for installation operated, tested and inspected by skilled persons.

13.3 Hazards Due to Multiple Electrical Sources

The need for reliability of power supply and continued supply for different equipment leads to provision of redundant equipment chains as well as the provision of more than one source of power.

For critical installations it is normal to have more than one service connection, preferably from two different paths, so that at least one of the paths is in service at any time.

At the next level, to cover the failure of the area distribution network, local standby generating systems such as diesel generating sets are provided. For extremely critical loads which cannot tolerate even short break in service UPS systems which store energy in a battery bank and release it during the power failure become necessary.

As such, multiple sources have become a necessary common feature in most large buildings. Even in small buildings activities depending on computers, etc leads us to provide a standby system with an UPS or an inverter.

Where there is more than one source the shock hazards increase due to various reasons and a proper drill and display of instructions becomes necessary so that the users and in particular the operating and maintenance personnel are warned of the different sources and the associated protocol.

13.4 Care and Design of Electrical Installations for Human Safety

Location of switches and location of electrical equipment has to be done with a view to avoid electrical hazards to the users from the angle of human safety.

Most common hazard is of direct contact and that too when associated with wet surfaces. Wet surface reduces the contact resistance and thereby increases the current flow.

The second most common hazard comes up from old or/and damaged or under-sized flexible cords used for connection of portable equipment or devices.

The recommendations for typical areas requiring special attention are given in **13.4.1** to **13.4.5**.

13.4.1 *Bath Room Installations*

13.4.1.1 *Geyser*

The geyser or a storage water heater in not a portable device as it requires connection to the electricity line as well as to the water supply lines. The electrical connection will be from a 16 A socket-outlet located at least 500 mm away from the plumbing connections. The switch for the geyser should not be accessible from the bath area or the shower area, to avoid a person standing on the wet surface touching the switch. The switch should be located away from the wet floor area of a bath room and preferably outside near the entrance to the bath room. This switch should have a built in indicator to display the status.

13.4.1.2 *Shaver socket*

Shaver sockets are located near the mirror. The socket will be fed from an isolating transformer (which in addition to providing electrical isolation between the output to the shaver and the building electrical system, may also give a voltage choice selection) to avoid shock hazard.

Shaver sockets which do not have a built in isolating transformer should be protected at the back by a RCCB/RCD of 30 mA sensitivity.

13.4.1.3 *Protection against electric shocks in bathrooms*

All circuits in bathrooms shall be protected by a RCCB/RCD with a sensitivity of 30 mA. The RCCB/RCD shall be in compliance with its relevant Indian Standards.

13.4.1.4 *Protection against electric shocks in sauna heater*

All circuits in sauna shall be protected by a RCCB/RCD with a sensitivity of 30 mA. The RCCB/RCD shall be in compliance with the relevant Indian Standards.

13.4.2 *Bed Room Installations*

For the convenience of operating the lights, fan, nurses call bell system, etc from the bed, it is quite a common practice to provide a pendent switch wired with a flexible cord. The flexible cord used should be with an outer sheath and the sheath should be securely clamped at the pendent switch. Use of twisted flexible cables without double insulation is hazardous and should be avoided.

A bed-head control set for all outlets to be controlled, mounted either on the wall accessible from the bed or mounted on the head board of the bed would be a preferable and safer alternative. This alternative requires the termination of the permanently installed wiring at a box near the bed and a flexible multicore sheathed cable for connection from the wall termination box to the control switch set mounted on the bed-head.

13.4.3 *Kitchen Installations*

Today a large number of gadgets are in use in the kitchen and it would be desirable to provide a large number of combinations of 6 A/16 A sockets. As a general guideline any part of the kitchen counter should be within 1 m from a socket-outlet as most kitchen appliances come with a 1.5 m or 2 m flexible cord.

The sockets should be located on the basis of the equipment layout and the general-purpose outlets would be placed at 150 mm above the kitchen counter level.

For safety, all the socket-outlets should be from circuits with a RCCB/RCD protection of sensitivity 30 mA. If an instant geyser is provided the same has to be wired from circuit with a 100 mA RCCB/RCD as 30 mA may cause nuisance tripping for a geyser.

13.4.4 *Protection Against Electric Shocks in Swimming Pools and the Other Basins*

For swimming pools and other basins such as basins of swimming pools, paddling pools and their surrounding zones; areas in natural waters such as lakes in gravel pits, coastal and similar areas, specially intended to be occupied by persons for swimming, paddling and similar purposes and their surrounding zones, basins of fountains and their surrounding zones; all electrical systems provided shall be protected by residual current devices with a sensitivity not exceeding 30 mA. The RCD shall be in compliance with its product standard and the relevant Indian Standard. All circuits for lighting, etc in contact with water shall be at 24 V or less.

13.4.5 For protection against electric shock including those pertaining to medical and all other applications not mentioned above, reference shall be made to good practice [D-2(1)].

13.5 Earthing Requirements

Earthing is an essential need for protection against shock. Almost all devices require a proper earth connection through the earthing pin (and socket) and in turn through the protective earth conductor and shall comply with the requirements as per good practices [D-2(1)] and [D-2(11)].

13.6 Heating Appliances and Hot Appliances

Electrical appliance incorporating a heating element pose additional hazards from the possibility of overheating and inadvertent ignition of combustible material in the vicinity. While the installation by itself cannot be made failsafe, it is necessary to provide conspicuous indicators with the switches for the sockets used for such appliances, so that visible indication warns the user that the appliance is ON (many appliances which have their own built-in indication, may be in OFF position in their standby mode or in the cut-off mode. Under such conditions the indication associated with the switch would be a gentle reminder).

Almost all electrical appliances produce heat and require ventilation. The surface of the equipment is likely to be hot. Wherever the surface temperature is likely to be more than 45 °C, there should be a protection against the possibility of touching the surface and getting scalded or burnt.

13.7 Switches Getting Supply from Multiple Sources

Use of critical electrical equipment requiring continued power supply even in the absence of the commercial supply are on the increase both at home and in any general office, apart from special functional buildings where multiple or redundant sources is a necessity. Use of UPS systems has become common.

Where there are such points or switches fed from multiple sources, it is desirable to mark these switches or adopt a colour code so that they are identified readily.

Many UPS and inverters have the feature of tripping on overload followed by automatic restoration after a brief time delay. Such characteristics should be made known to the users, so that accidents do not happen by the uninformed user taking some action assuming that the power supply is not there. Shock hazards are dependent on the system voltage and the hazard level is the same whether it is from the power system or from a mini UPS of a few watts.

13.8 Protection Against Environmental Over-Voltages

Any building today has a number of metallic connections coming into the building or installed over (or near) the building such as lines for electricity, telephone, dish antenna, radio antenna, pipe lines for water supply, sewage, cable of cable TV, solar hot water system, solar photovoltaic array, cell phone tower, wind generator, etc. External disturbances such as lightning, faults in external systems, etc, can induce over-voltage in the installations within the building.

Buildings are to be provided protection from lightning strikes by a system of ‘Faraday cage’ on the basis of hazard assessment. But at the same time use of dish antenna for TV, cable TV (the cable for which is run from building to building on the top, apparently forming a line to catch a lightning stroke) are on the rise and are without any regulation. These installations pose hazards to the installations inside the building. Care should be taken while allowing such installations in buildings. In case the occupant is not in a position to avoid such incidence, suitable SPD should be provided in the circuit. Good practice [D-2(1)] may also be referred in this regard.

For sensitive and critical equipment, it is necessary to make a separate detailed study and build in appropriate protective device such as shield, suppressor or diverter, as may be appropriate for the equipment as well as the type of hazard assessed.

13.9 Flammable Atmosphere and Risk of Ignition by Electricity Leading to Fire or Explosion

13.9.1 Flammable atmospheres may result in electrical fire and explosion hazards. They are dealt with separately because of the complexities and problems peculiar to the subject. They occur when flammable gas, vapour, mist, aerosol or dust is present in a concentration in air between the upper (UEL) and lower (LEL) explosive limits. Such mixtures ignite if there is a source of ignition is present, such as incentive arcs or sparks from electrical equipment. The risk areas are divided into following zones for flammable gases/vapours:

a) *Zone* 0 Where the flammable atmosphere is always present;

b) *Zone* 1 Where the flammable material is present during normal operations and a flammable atmosphere is likely to occur; and

c) *Zone* 2 Where the flammable atmosphere is unlikely during normal operation and if it occurs, it will exist only for a short time.

13.9.2 A spark such as from a switch in an explosive atmosphere can trigger a fire or an explosion. In such areas special 'flame-proof' category switches, light fittings, cables, etc will have to be used. The installation practices are also precise. Zonal boundaries have to be determined by the occupier's specialist staff and marked on the site drawing in plan and elevation. It is a difficult task and requires a knowledge of the characteristics of the material and how it disperses from the source of emission.

To illustrate by an example, if LPG leaks from a domestic gas cylinder it creates a hazardous situation. LPG is heavier than air and will settle down at the bottom on the floor with gradually changing ratio of air/LPG mixture as you go up above the floor level and it would be an explosive mixture at a height of a few centimetres above the floor level, which can easily be ignited by a spark or a flame. LPG stored in a basement (particularly if it is without exhaust picked up from floor level) can be serious situation.

Armed with the zonal boundary plans, the next step is to determine the position of the electrical equipment. Wherever possible, the risk should be avoided or minimized by locating the electrical equipment outside the risk areas or in Zone 2 areas rather than in Zones 0 and 1.

Zone 0 is usually inside reaction vessels or storage tanks. There is not much electrical equipment available for use in Zone 0 and as the risk is high it should be avoided by locating the equipment outside, example, a container can have glazed port-holes so that the interior can be viewed through one and lit from a luminaire shining through another. Zone 1 areas often occur only in the immediate vicinity of where the flammable material is processed. They are invariably surrounded by Zone 2 areas where the electrical apparatus should preferably be situated. It is sometimes possible to light outside site zones by luminaires located at high level in safe areas above the vertical zone boundaries. For flammable interiors, lighting can often be provided by positioning the luminaires outside in a safe location and shining the light through glazed panels in the roof or walls. Again, motors can be outside and drive an internal machine through a long shaft extension passing through a gas-tight seal in the wall. In some plants, it is convenient to locate the motor starters together on a starter panel in a motor control room. This room may either be in a safe area or if within the flammable zone, fed with clean air, from outside the zone, maintained at a positive pressure to ensure that gas cannot enter. Conduits for wiring should be solid drawn or seam welded with screwed joints. Where flammable gases have to be excluded from equipment, barrier seals may be necessary. Flexible metallic conduit should be lined to prevent abrasion of the cable insulation. Aluminium and aluminium alloy conduits and accessories should not be used where frictional contact with oxygen-rich items such as rusty steel might occur and cause sparking.

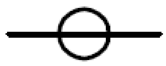
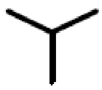
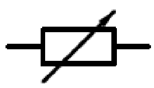
13.10 Protection against Fire Due to Leakage Current in the Building

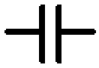
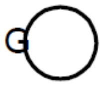
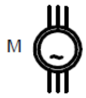
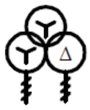
Large number of fires in the building occur due to faulty electrical installation. Mostly, the cause is assigned to over load or short-circuit fault. But there can be one more reason for causing the fire in the building. As the quality of insulation of the cables deteriorate due to ageing over a period of time, certain amount of leakage current flows in the system. This leakage current or tracking is difficult to detect and if it is allowed for a long period of time, say months or years, sparks occur which on a favourable condition, may lead to devastating fire in the building. Ignition of fire due to leakage current or tracking may be prevented by providing residual current devices of 300 mA as part of the main distribution board.

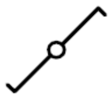
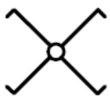
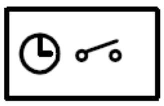
ANNEX A

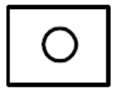
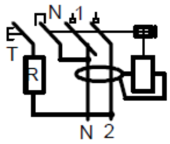
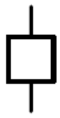
(*Clause* 2.2)

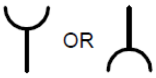
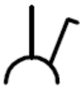
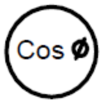
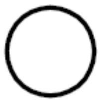
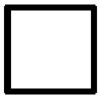
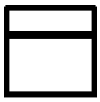
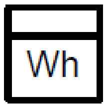
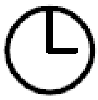
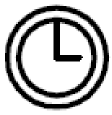
DRAWING SYMBOLS FOR ELECTRICAL INSTALLATIONS IN BUILDINGS

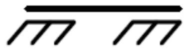
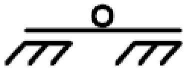
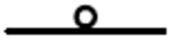
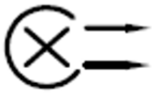
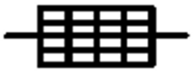
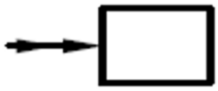
Sl No.	*Details*	*Symbol*
1.	Direct current	—
2.	Alternating current, general symbol	∼
3.	Alternating current, single-phase, 50Hz	∼ 50 Hz
4.	Alternating current, three phase, 415 V	3 ∼ 50 Hz 415 V
5.	Alternating current, three phase with neutral, 50 Hz	3N ∼ 50 Hz
6.	Neutral	N
7.	Positive polarity	+
8.	Negative polarity	−
9.	Direct current, 2 conductors 110 V	2 — 110 V
10.	Direct current 3 conductors including neutral 220 V	2N — 220 V
11.	Underground cable	
12.	Overhead Line	
13.	Winding, delta	△
14.	Winding, star	
15.	Terminals	○
16.	Resistance/Resis tor	
17.	Variable resistor	

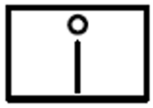
18.	Impedance	
19.	Inductance/Inductor	
20.	Winding	
21.	Capacitance, Capacitor	
22.	Earth	
23.	Fault	
24.	Flexible conductor	
25.	Generator	
26.	a.c. generator	
27.	d.c. generator	
28.	Motor	
29.	Synchronous motor	
30.	Mechanically coupled Machines	
31.	Induction motor, Three-phase, squirrel cage	
32.	Induction motor with Wound rotor	
33.	Transformers with two separate windings	
34.	Auto-transformer	
35.	3-Phase transformer with three separate windings-star-star-delta	
36.	Starter	
37.	Direct-on-line-starter for reversing motor	
38.	Star-delta starter	
39.	Auto-transformer starter	
40.	Rheostatic starter	
41.	Switch	

42.	Contactor	
43.	One - way switch, single pole	
44.	One - way switch, two pole	
45.	One - way switch, three pole	
46.	Single pole pull switch	
47.	Multiposition switch for different degrees of lighting	
48.	Two-way switch	
49.	Intermediate switch	
50.	Period limiting switch	
51.	Time switch	
52.	Pendant switch	

53.	Push button	
54.	Luminous push button	
55.	Restricted access push button	
56.	Relay	
57.	Circuit-Breaker	
58.	Isolator	
59.	Residual current circuit breaker	
60.	Surge protective device	
61.	Fuse	
62.	Signal lamp	
63.	Link	

64.	Distribution boards, cubicle box, main fuse board with switches	
65.	Socket outlet, 6A	
66.	Socket outlet, 16 A	
67.	Combined Switch and Socket Outlet, 6A	
68.	Combined switch and socket outlet, 16 A	
69.	Interlocking switch and socket outlet, 6A	
70.	Interlocking switch and socket outlet, 16A	
71.	Plug	
72.	Voltmeter	
73.	Ammeter	
74.	Wattmeter	
75.	Varmeter	
76.	Power factor meter	
77.	Ohmmeter	
78.	Indicating instrument (General symbol)	
79.	Recording instrument (General symbol)	
80.	Integrating meter	
81.	Watthour meter	
82.	Clock	
83.	Master clock	
84.	Current transformer	
85.	Voltage transformer	

86.	Wiring on the surface	
87.	Wiring under the surface	
88.	Conduit on surface	
89.	Concealed conduit	
90.	Wiring in conduit	
91.	Lamp	
92.	Lamp mounted on a ceiling	
93.	Lamp, mounted on a wall	
94.	Emergency lamp	
95.	Panic lamp	
96.	Water—tight lighting fitting	 WT
97.	Batten lamp holder	 BH
98.	Spot light	
99.	Flood light	
100.	Heater	
101.	Storage type water heater	
102.	Bell	
103.	Buzzer	
104.	Siren	
105.	Ceiling fan	
106.	Exhaust fan	
107.	Fan regulator	
108.	Aerial	
109.	Radio receiving set	

110.	Television receiving set	
111.	Manually operated fire alarm	
112.	Automatic fire detector switch	

ANNEX B

[*Clauses* 3.1.2.2 (Note) 3.3 (Note), 3.5.3(c), 4.2.1(q)(10), 4.9, 4.9.1.2, 4.9.1.4, 4.9.1.5, 4.9.3.1, 5.3.6.7, 5.4.1.1, 5.4.1.3, 6.4.6, 8.2.1, 8.3, 9.1.1, 9.3.2.6(d), 10.7.3 *and* 10.8.3]

EXTRACTS FROM *CENTRAL ELECTRICITY AUTHORITY* (*MEASURES RELATING TO SAFETY AND ELECTRIC SUPPLY*) *REGULATIONS*, 2023

B-1 The following are the extracts of some of the rules:

Regulation 3, Designated person to operate and carry out the work on electrical lines and apparatus

(1) The supplier or consumer, or owner of electrical installation, owner or agent or manager of a mine, or agent of any company operating in an oil-field or owner of a drilled well in an oil-field or a contractor who has entered into a contract with a supplier or a consumer, or owner of electrical installation, owner or agent or manager of a mine, or agent of any company operating in an oil-field or owner of a drilled well in an oil-field to carry out duties incidental to the generation, transformation, transmission, conversion, distribution or use of electricity shall designate person for the purpose to operate and carry out the work on electrical lines and apparatus.

(2) The supplier or consumer, or owner or agent or manager of a mine, or agent of any company operating in an oil-field or the owner of a drilled well in an oil-field or a contractor referred to in sub-regulation (1) shall maintain a record, in paper or electronic form, wherein the names of the designated person and the purpose for which they are designated, shall be entered.

(3) No person shall be designated under sub-regulation (1) unless-

(i) he possesses a certificate of competency or electrical work permit, issued by the Appropriate Government; and

(ii) his name is entered in the register referred to in sub-regulation (2).

Regulation 5, Electrical Safety Officer

(1) All suppliers of electricity including generating companies, transmission companies and distribution companies shall designate an Electrical Safety Officer for ensuring observance of safety measures specified under these regulations in their organisation for construction, operation and maintenance of electrical system of all generating stations, transmission lines, substations, distribution systems and supply lines.

(2) The Electrical Safety Officer shall possess a degree in Electrical Engineering with at least five years experience in operation and maintenance of electrical installations or a Diploma in Electrical Engineering with at least ten years experience in operation and maintenance of electrical installations:

Provided that the Electrical Safety Officer designated for mines shall possess educational qualification as mentioned in sub-regulation (2) with at least five years of experience in operation and maintenance of electrical installations relevant to mines.

(3) For every electrical installation including factory registered under the *Factories Act*, 1948 (63 of 1952) with more than 250 kW connected load and mines and oil-field as defined in the *Mines Act*, 1952 (35 of 1952), with more than 2 000 kW connected load, the owner of the installation or the management of the factory or mines, as the case may be, shall designate Electrical Safety Officer under sub-regulation (1) and having qualification and experience specified in sub-regulation (2), for ensuring the compliance of the safety provisions laid under the Act and the regulations made thereunder:

Provided that the Electrical Safety Officer shall carryout recommended periodic tests as per the relevant standards, and inspect such installations at intervals not exceeding one year, and keep a record thereof in Form I or Form II or Form III or Form IV, as the case may be, of Schedule II of these regulations; test reports and a register of recommendations in regard with safety duly acknowledged by owner; compliances made thereafter; and such records shall be made available to the Electrical Inspector, as and when required.

Regulation 6, Chartered Electrical Safety Engineer

(1) The Appropriate Government shall authorise Chartered Electrical Safety Engineer from amongst persons having the qualification and experience as per the guidelines issued by the Authority to assist the owner or supplier or consumer of electrical installations for the purpose of selfcertification under Regulation 32 and Regulation 45 of these regulations.

(2) The Appropriate Government shall upload the name of the Chartered Electrical Safety Engineer, as soon as any person is authorised as Chartered Electrical Safety Engineer, on the web portal of the Government or the Department dealing with matters of inspection of electrical installations for the information of the owner or supplier or consumer.

Regulation 8, Safety measures for operation and maintenance of transmission and distribution systems

(1) The Engineers or Supervisors engaged or appointed to operate or undertake maintenance of transmission and distribution systems shall hold degree or diploma in appropriate trade of Engineering from a recognised institute or university.

(2) The Engineers and Supervisors engaged or appointed to operate or undertake maintenance of transmission and distribution systems shall have successfully undergone the type of training specified in guidelines as per sub- regulation (4), within two years from the date of engagement or appointment.

(3) The Technicians to assist Engineers or Supervisors shall possess a certificate in appropriate trade, preferably with a two years course from an Industrial Training Institute recognised by the Central Government or State Government and should have successfully undergone the type of training as specified in guidelines as per sub- regulation (4), within two years from the date of engagement or appointment:

Provided that the existing employees, as on the date of notification of these regulations, who are extending technical assistance to Engineers or Supervisors and do not have requisite qualification as mentioned in this regulation, shall have to undergo the training either from Power Sector Skill Council or from training institute recognised by the Authority for carrying out trade specific course as per the guidelines issued by the Authority and get certificate as mentioned above within two years from the date of notification of these regulations.

(4) The Authority shall issue guidelines for the training for operation and maintenance of transmission, distribution systems within six months of the notification of these regulations:

Provided that the duration and content of the training course shall be as specified in the guidelines.

(5) Owner of every transmission or distribution system shall arrange for training of their personnel engaged or appointed to operate and undertake maintenance of transmission and distribution system, in his own institute or any other institute recognised by the Authority or State Government as per the guidelines and shall maintain records of the assessment of these personnel issued by the training institute in the format prescribed in guidelines and such records shall be made available to the Electrical Inspector, as and when required.

Regulation 14, General safety requirements pertaining to construction, installation, protection, operation and maintenance of electric supply lines and apparatus

(1) All electric supply lines and apparatus shall be of sufficient rating for power, insulation and estimated fault current and of sufficient mechanical strength, for the duty cycle which they may be required to perform under the environmental conditions of installation, and shall be constructed, installed, protected, worked and maintained in such a manner as to ensure safety of human beings, animals and property.

(2) Save as otherwise provided in these regulations, the relevant standards including National Electrical Code and National Building Code shall be followed to carry out the purpose of these regulations and where relevant Indian Standards are not available, International standards shall be followed and in the event of any inconsistency, the provisions of these regulations shall prevail.

(3) The material and apparatus used shall conform to the relevant standards.

(4) All electrical equipment shall be installed above the Highest Flood Level and where such equipment is not possible to be installed above Highest Flood Level, it shall be ensured that there is no seepage or leakage or logging of water.

Regulation 17, Identification of earthed and earthed neutral conductors and position of switches and switchgear therein

Where the conductors include an earthed conductor of a two-wire system or an earthed neutral conductor of a multi-wire system or a conductor which is to be connected thereto, the following conditions shall be complied with-

(i) an indication of a permanent nature shall be provided by the owner of the earthed or earthed neutral conductor, or the conductor which is to be connected thereto, to enable such conductor to be distinguished from any live conductor and such indication shall be provided as per relevant standards, namely: –

(a) where the earthed or earthed neutral conductor is the property of the supplier, at or near the point of commencement of supply;

(b) where a conductor forming part of a consumer's system is to be connected to the supplier's earthed or earthed neutral conductor, at the point where such connection is to be made; and

(c) in all other cases, at a point corresponding to the point of commencement of supply.

(ii) no cut-out, link, switch or circuit breaker other than a linked switch arranged to operate simultaneously on the earthed or earthed neutral conductor and live conductors shall be inserted or remain inserted in any earthed or earthed neutral conductor of a two wire-system or in any earthed or earthed neutral conductor of a multi-wire system or in any conductor connected thereto:

Provided that the above requirement shall not apply in case of a link for testing purposes, or a switch for controlling a generator or transformer.

Regulation 18, Earthed terminal on consumer's premises

(1) The supplier shall provide and maintain on the consumer's premises for the consumer's use, a suitable earthed terminal in an accessible position at or near the point of commencement of supply as per relevant standards:

Provided that in the case of installation of voltage exceeding 250 V the consumer shall, in addition to the aforementioned earthing arrangement, provide his own earthing system with an independent electrode and the same shall be interlinked with the earthed terminal mentioned in sub-regulation (1) through a suitable link.

(2) The consumer shall take all reasonable precautions to prevent mechanical damage to the earthed terminal and its lead belonging to the supplier.

Regulation 20, Danger notices

The owner of every installation of voltage exceeding 250 V shall affix permanently in a conspicuous position a danger notice in Hindi or English and the local language of the district, with a sign of skull and bones of a design as per relevant standards on:

(a) every motor, generator, transformer and other electrical plant and equipment together with apparatus used for controlling or regulating the same;

(b) all supports of overhead lines of voltage exceeding 650 V which can be easily climbed upon without the aid of ladder or special appliances; and

(c) luminous tube sign requiring supply, X-ray and similar high frequency installations of voltage exceeding 650 V but not exceeding 33 kV:

Provided that where it is not possible to affix such notices on any generator, motor, transformer or other apparatus, they shall be affixed as near as possible thereto, or the word 'danger' and the voltage of the apparatus concerned shall be permanently painted on it:
Provided further that where the generator, motor, transformer or other apparatus is within an enclosure one notice affixed to the said enclosure shall be sufficient for the purposes of this regulation.

Explanation — For the purposes of clause (b) rails, tubular poles, wooden supports, reinforced cement concrete poles and pre stressed cement concrete poles without steps, I-sections and channels, shall be deemed as supports which cannot be easily climbed upon.

Regulation 21, Handling of electric supply lines and apparatus

(1) Before any conductor or apparatus is handled, adequate precautions shall be taken, by earthing or other suitable means, to discharge electrically such conductor or apparatus, and any adjacent conductor or apparatus if there is danger therefrom, and to prevent any conductor or apparatus from being accidentally or inadvertently electrically charged when persons are working thereon shall be followed as per the relevant standards.

(2) Every person who is working on an electric supply line or apparatus or both shall be provided with, –

(a) personal protective equipment, tools and devices such as rubber gloves and safety footwear suitable for working voltage, safety belts for working at height, nonconductive ladder, earthing devices of appropriate class, helmet, line tester, hand lines, voltage detector and hand tools as per the relevant standards; and

(b) any other device for protecting him from mechanical and electrical injury due to arc flash and such personal protective equipment, tools and devices shall conform to the relevant standards and shall always be maintained in sound working condition.

(3) No person shall operate and undertake maintenance work on any part or whole of an electrical plant or electric supply line or apparatus and no person shall assist such person on such work, unless he is designated in that behalf and observes the safety precautions given in Part-I, Part-II, Part-III and Part-IV, as the case may be, of Schedule I.

(4) Every telecommunication line on supports carrying an overhead line of voltage exceeding 650 V but not exceeding 33 kV shall, for the purpose of working thereon, be deemed to be a line of voltage exceeding 650 V:

Provided that prior permission shall be taken from the concerned licensee before laying telecommunication lines on electric supports.

(5) For the safety of operating personnel, all non-current carrying metal parts of switchgear and control panels shall be properly earthed and insulating floors or mat conforming to the relevant standards, of appropriate voltage level shall be provided in front and rear of the panels where such personnel are required to carry out operation, maintenance or testing work.

(6) All panels shall be painted with the description of its identification at front and at the rear.

Regulation 29, Provisions applicable to protective equipment

(1) Fire buckets filled with clean dry sand and ready for immediate use for extinguishing fires, in addition to fire extinguishers suitable for dealing with fires, shall be conspicuously marked and kept in all generating stations, enclosed substations and enclosed switching-stations in convenient location.

(2) Appropriate type of fire extinguisher conforming to the relevant standards, shall be installed, maintained, periodically inspected and tested as per the relevant standards for extinguishing and controlling fire and record of such tests shall be maintained.

(3) Sufficient number of first-aid boxes or cupboards conspicuously marked and equipped with such contents as the State Government may specify or as per the relevant standards, shall be provided and maintained at appropriate locations in every generating station, enclosed substation, enclosed switching

station and in vehicles used for maintenance of lines so as to be readily available and accessible at all the times and all such boxes and cupboards shall, except in the case of unattended substations and switching stations, be kept under the charge of responsible persons who are trained in first-aid treatment and one of such persons shall be available during working hours.

(4) Two or more gas masks shall be provided conspicuously and installed and maintained at accessible places in every generating station with capacity of five megawatt and above and enclosed substation with transformation capacity of five megavolt-ampere and above for use in the event of fire or smoke:

Provided that where more than one generator with capacity of five megawatt and above is installed in a power station, each generator shall be provided with at least two separate gas masks in an accessible and conspicuous place.

(5) In every generating station, substation or switching station, an artificial respirator, fire extinguishers, first-aid boxes and gas masks shall be provided and kept in good working condition and locations of the same shall be displayed in the control room and operator cabin.

(6) Address and contact number of the nearest Doctor, Hospital with a facility for first-aid treatment for electric shock and burns, ambulance service and fire service shall be prominently displayed near the electric shock treatment chart in control room and operator cabin.

Regulation 30, Display of instructions for resuscitation of persons suffering from electric shock

(1) Instructions, in English or Hindi and the local language of the District and where Hindi is the local language, in English and Hindi for the resuscitation of persons suffering from electric shock, shall be affixed by the owner in a conspicuous place in every generating station, enclosed substation, enclosed switching station, mines and in every factory as defined in the Factory Act, 1948 (63 of 1952) in which electricity is used and in such other premises where electricity is used as the Electrical Inspector may, by notice in writing served on the owner, direct.

(2) The owner of every generating station, enclosed substation, enclosed switching station and every factory or other premises to which these regulations apply, shall ensure that all designated persons or persons engaged or appointed to operate and maintain electrical plants or transmission or distribution systems are acquainted with and are competent to apply the instructions referred to in sub-regulation (1).

Regulation 31, Precautions to be adopted by consumers, owners, occupiers, electrical contractors, electrical workmen and suppliers

(1) No electrical installation work, including additions, alterations, repairs and adjustments to existing installations, except such replacement of lamps, fans, fuses, switches, domestic appliances of voltage not exceeding 250 V and fittings as in no way alters its capacity or character, shall be carried out upon the premises of or on behalf of any consumer, supplier, owner or occupier for the purpose of supply to such consumer, supplier, owner or occupier except by an electrical contractor licenced in this behalf by the State Government and on its behalf under the direct supervision of a person holding a certificate of competency and by a person holding a permit issued or recognised by the State Government:

Provided that in the case of works executed for or on behalf of the Central Government and in the case of installations in mines, oil-fields and railways, the Central Government and in other cases the State Government, may, by notification in the Official Gazette, exempt on such conditions as it may impose, any such work described therein either generally or in the case of any specified class of consumers, suppliers, owners or occupiers.

Provided further that in the case of works executed for or on behalf of the Central Government and in the case of installations in mines, oil-fields and railways, an electrical contractor having licence issued by any State Government or Union Territory administration shall not require licence from other State Government in which the works are to be executed.

(2) No electrical installation work which has been carried out in contravention of sub-regulation (1) shall either be energised or connected to the works of any supplier.

Regulation 32, Periodical inspection and testing of installations

(1) The periodic inspection and testing of installation of voltage above the notified voltage belonging to the owner or supplier or consumer, as the case may be, shall be carried out by the Electrical Inspector:

Provided that the electrical installation below or equal to the notified voltage shall be self-certified by the owner or supplier or consumer, as the case may be.

(2) The periodicity of electrical inspection by the Electrical Inspector or the self-certification by the supplier, owner or consumer shall be as directed by the Appropriate Government.

Provided that the periodicity of electrical inspection and self-certification shall not exceed five years.

Provided further that in respect of the electrical installation belonging to mines, oil-fields and railways, such direction shall be issued by the Central Government.

(3) The periodic inspection and testing of installation of voltage equal to or below the notified voltage belonging to the owner or supplier or consumer, as the case may be, shall be carried out by the owner or supplier or consumer and shall be self-certified for ensuring observance of safety measures specified under these regulations and the owner or supplier or consumer, as the case may be, shall submit the report of self-certification to the Electrical Inspector in the Form I or Form II or Form III or Form IV, as the case may be, of Schedule II:

Provided that the electrical installation so self-certified shall be considered as duly inspected and tested only after the report of self-certification is duly received by the office of Electrical Inspector and if not acknowledged by the Electrical Inspector within three working days, it shall be deemed to be received:

Provided further that the owner or supplier or consumer has the option to get his installation inspected and tested by the Electrical Inspector of the Appropriate Government.

(4) Notwithstanding anything contained in sub-regulation (3), every electrical installation covered under section 54 of the Act including every electrical installation of mines, oil-fields and railways shall be periodically inspected and tested by the Electrical Inspector of the Appropriate Government.

(5) Where the supplier is directed by the Central Government or the State Government, as the case may be, to inspect and test the installation, such supplier shall report on the condition of the installation to the consumer concerned in the Form I, Form II, Form III and Form IV as provided in Schedule II and shall submit a copy of such report to the Electrical Inspector.

(6) The Electrical Inspector may, on receipt of such report, accept the report submitted by the supplier or record variations as the circumstances of each case may require and may recommend that the defects may be rectified as per report.

(7) In the event of the failure of the owner of any installation to rectify the defects in his installation pointed out by the Electrical Inspector in his report and within the time indicated therein, such installation shall be liable to be disconnected under the directions of the Electrical Inspector after serving the owner of such installation with a notice for a period not less than forty eight hours:

Provided that the installation shall not be disconnected in case an appeal is made under sub section (2) of section 162 of the Act and appellate authority has stayed the orders of disconnection.

(8) It shall be the responsibility of the owner of all installations to maintain and operate the installations in a condition free from danger and as recommended by the manufacturer or by the relevant standards.

Regulation 33, Testing of consumer's installation

(1) Upon receipt of an application for a new or additional supply of electricity and before commencement of supply or recommencement of supply after the supply has been disconnected for a period of six months, the supplier shall either test the installation himself or accept the test results submitted by the consumer when same has been duly signed by the licenced electrical contractor:

Provided that in case of voltage level equal to or below the notified voltage, Chartered Electrical Safety Engineer can also test the installation on request of owner.

(2) The testing and verifications shall be carried out as per relevant standards.

(3) The testing equipment shall be calibrated by a Government authorised or National Accreditation Board for Testing and Calibration Laboratories accredited laboratory at periodical interval as per the periodicity specified by them.

(4) The supplier shall maintain a record of test results obtained at each supply point to a consumer, as per the forms provided in Schedule III.

(5) If as a result of such inspection and test, the supplier is satisfied that the installation is likely to be dangerous, he shall serve on the applicant a notice in writing requiring him to make such modifications as are necessary to render the installation safe and may refuse to connect or reconnect the supply until the required modifications have been completed.

Regulation 34, Generating units required to be inspected by electrical inspector

The capacity above which generating units including generating units producing electricity from renewable sources of energy shall be required to be inspected by the Electrical Inspector before commissioning, shall be as per the notification issued by the Appropriate Government in this regard.

Regulation 35, Precautions against leakage before connection

(1) The supplier shall not connect its works with the apparatus in the premises of any applicant seeking supply unless the supplier is satisfied that at the time of making the connection cause a leakage from that installation or apparatus of a magnitude detrimental to safety which shall be checked by measuring the installation or apparatus insulation resistance as stipulated in the relevant standards.

(2) If the supplier declines to make a connection under the provisions of sub-regulation (1) the supplier shall convey to the applicant the reasons thereof, in writing for so declining.

Regulation 36, Leakage on consumer's premises

(1) If the Electrical Inspector or the supplier has reasons to believe that there is leakage in the system of a consumer which is likely to affect injuriously the use of electricity by the consumer or by other persons, or which is likely to cause danger, he may give notice to the consumer in writing to inspect and test the consumer's installation.

(2) If after such notice, the consumer fails to provide access to its installation for inspection and testing, or an insulation resistance of the consumer's installation is so low as to prevent safe use of electricity, the supplier may, and if directed so by the Electrical Inspector shall discontinue the supply of electricity to the installation but only after giving to the consumer forty eight hours notice in writing for disconnection of supply and shall not recommence the supply until he or the Electrical Inspector is satisfied that the cause of the leakage has been removed.

Regulation 37, Supply and use of electricity

(1) The electricity shall not be supplied, transformed, converted, inverted or used or continued to be supplied, transformed, converted, inverted or used unless the conditions provided in sub- regulations (2) to (8) are complied with.

(2) The following controls of requisite capacity to carry and break the current shall be installed as near as possible after the point of commencement of supply so as to be readily accessible and capable of completely isolating the supply to the installation, such equipment being in addition to any control switch installed for controlling individual circuits or apparatus, namely:

Supplied at voltage	*Control*
Below 11 kV.	Switch fuse unit or a circuit breaker by consumers.
11 kV and above.	A circuit breaker by consumers.

(i) In case of every transformer the following shall be provided, namely: –on primary side of transformer, a linked switch with fuse or gang operated air break switch with fuse or circuit breaker of adequate capacity:

Provided that the linked switch with fuse on the primary side of the transformer may be of such capacity as to carry the full load current and to break only the magnetising current of the transformer:

Provided further that for transformer having capacity of 1 000 kVA and above, a circuit breaker shall be provided:

Provided also that the linked switch with fuse or gang operated air break switch with fuse or circuit breaker on the primary side of the transformer shall not be required for the unit auxiliary transformer and generator transformer;

(ii) on the secondary side of all transformers a circuit breaker of adequate rating shall be installed:

Provided that for supplier's transformers of capacity below 1 000 kVA, a linked switch with fuse or circuit breaker of adequate rating shall be installed on secondary side.

(4) Except in the case of composite controlgear designed as a unit each distinct circuit is to be protected against excess energy by means of a suitable fuse link or a circuit breaker of adequate breaking capacity, suitably located and so constructed as to prevent danger from overheating, arcing or scattering of hot metal when it comes into operation and to permit for ready renewal of the fuse link without danger.

(5) The supply of electricity to each motor or a group of motors or other apparatus meant for operating one particular machine shall be controlled by a suitable linked switch or a circuit breaker or an emergency tripping device with manual reset of requisite capacity placed in such a position as to be adjacent to the motor or a group of motors or other apparatus readily accessible to and easily operated by the person incharge and so connected in the circuit that by its means all supply of electricity can be cut off from the motor or group of motors or apparatus from any regulating switch, resistance of other device associated therewith.

(6) All insulating materials shall be as per their application and their mechanical strength shall be sufficient for the purpose so as to maintain adequately their insulating property under all working conditions in respect of temperature, moisture, salinity and pollution.

(7) Adequate precautions shall be taken to ensure that no live parts are exposed as to cause danger.

(8) Every consumer shall use all reasonable means to ensure that where electricity is supplied by the supplier, no person other than the supplier shall interfere with service lines and apparatus placed by the supplier on the premises of the consumer.

Regulation 38, Provisions for supply and use of electricity in multi-storeyed building more than 15 metres in height

(1) The connected load and voltage of supply above which inspection is to be carried out by an Electrical Inspector for a multi-storeyed building of more than fifteen metre height shall be notified by the Appropriate Government.

(2) Before making an application for commencement of supply or recommencement of supply after an installation has been disconnected for a period of six months or more, the owner or occupier of a multi-storeyed building shall give not less than thirty days notice in writing to the Electrical Inspector specifying therein the particulars of installation and the supply of electricity shall not be commenced or recommenced within this period, without the approval in writing of the Electrical Inspector.

(3) The following safety measures shall be provided in the multi-storeyed buildings of more than fifteen metre height and other premises such as airports, hospitals, hotels, places of entertainment, places of worship, cultural centres, stadium, academic buildings, test labs, industrial installations, installation with explosive or flammable material, railway or metro stations and other public buildings, namely:

(i) the supplier or owner of the installation shall provide at the point of commencement of supply a suitable isolating device with cut-out or breaker to operate on all phases except neutral in the three-phase, four-wire circuit and fixed in a conspicuous position at not more than 1.70 metre above the ground so as to completely isolate the supply to the building in case of emergency;

(ii) the owner or occupier of a multi-storeyed building shall ensure that electrical installations and works inside the building are carried out and maintained in such a manner as to prevent danger due to shock and fire hazards, and the installation is carried out as per the relevant standards;

(iii) no other service pipes and cables shall be taken through the ducts provided for laying of power cables and all ducts provided for power cables and other services shall be provided with fire barrier at each floor crossing;

(iv) the Fire Retardant Low Smoke and Low Halogen power cables shall be used in building of more than fifteen metre height as per relevant standards;

(v) Provided that Halogen Free Flame Retardant power cables as per the relevant standards shall be used in airports, hospitals and hotels irrespective of height;

(vi) distribution of electricity to the floors shall be done using busbar trunking system;

(vii) lightning protection of the building shall be as per the relevant standards;

(viii) verification of electrical wiring of the building shall be carried out as per the relevant standards; and

(ix) electricity meter shall not be installed in the passage of staircase.

Regulation 39, Conditions applicable to installations of voltage exceeding 250 Volts

The following conditions shall be compiled with where electricity of voltage above 250 V is supplied, converted, transformed or used, namely:

(i) all conductors, other than those of overhead lines, shall be completely enclosed in mechanically strong metal casing or metallic covering which is electrically and mechanically continuous and adequately protected against mechanical damage unless the said conductors are accessible only to a designated person or are installed and protected so as to prevent danger:

Provided that non-metallic conduits conforming to the relevant standards may be used for installations of voltage not exceeding 650 V;

(ii) all metal works, enclosing, supporting or associated with the installation, other than that designed to serve as a conductor shall be connected with an earthing system as per relevant standards and the provisions of regulation 43;

(iii) every switch board shall comply with the following, namely:

(a) a clear space of not less than one metre in width shall be provided in front of the switchboard;

(b) if there are any attachments or bare connections at the back of the switchboard, the space, if any, behind the switchboard shall be either less than twenty centimetre or more than seventy five centimetre in width, measured from the farthest protruding part of any attachment or conductor; and

(c) if the space behind the switchboard exceeds seventy five centimetre in width, there shall be a passage way from either end of the switchboard, clear to a height of 1.8 metre;

(iv) in case of installations provided in premises where inflammable materials including gases and chemicals are produced, handled or stored, the electrical installations, equipment and apparatus shall comply with the requirements of flame proof, dust tight, totally enclosed or any other suitable type of electrical fittings depending upon the hazardous zones as per the relevant standards;

(v) where an application has been made to a supplier for supply of electricity to any installation, the supplier shall not commence the supply or where the supply has been discontinued for a period of six months or more, recommence the supply unless the consumer has complied with the relevant provisions in these regulations;

(vi) where a supplier proposes to supply or use electricity at or to recommence supply of voltage exceeding 250 V but not exceeding 650 V after it has been discontinued for a period of six months, he shall, before connecting or reconnecting the supply, give notice in writing of such intention to the Electrical Inspector; and

(vii) if at any time after connecting the supply, the supplier is satisfied that any provision of these regulations have not been complied with, the supplier shall give notice of the same in writing to the consumer and the Electrical Inspector, specifying the defects and to rectify such defects in a reasonable time:

Provided that if the consumer fails to rectify such defects the supplier may discontinue the supply after giving the consumer a reasonable opportunity of being heard and recording reasons in writing and the supply shall be discontinued only on written orders of an officer duly notified by the supplier in this behalf and shall be restored with all possible speed after such defects are rectified by the consumer to the satisfaction of the supplier.

Regulation 40, Appeal to electrical inspector in regard to defects

(1) If any applicant for a supply or a consumer is aggrieved by the action of the supplier in declining to commence, to continue or to recommence the supply of electricity to his premises on the grounds that the installation is defective or is likely to be dangerous, he may appeal to the Electrical Inspector to test the installation and the supplier shall not, if the Electrical Inspector intimates that the installation is free from the defect or danger complained of, refuse supply to the consumer on the grounds aforesaid, and shall, within twenty four hours after the receipt of such intimation from the Electrical Inspector, commence, continue or recommence the supply of electricity.

(2) Any test for which application has been made under sub-regulation (1), shall be carried out within seven days after the receipt of such application.

Regulation 41, Precautions against failure of supply and notice of failures

(1) The layout of the electric supply lines of the supplier for the supply of electricity throughout his area of supply shall under normal working conditions be sectionalised and so arranged, and provided with switchgear or circuit-breakers, so located, as to restrict within reasonable limits the extent of the portion of the system affected by any failure of supply.

(2) The supplier shall take all reasonable precautions to avoid any accidental interruptions of supply, and also to avoid danger to the public or to any employee or designated person when engaged on any operation during and in connection with the installation, extension, replacement, repair and maintenance of any works.

(3) The supplier shall send to the Electrical Inspector a notice of failure of supply of such kind as the Electrical Inspector may from time to time require to be notified to him, and such notice shall be sent by the earliest mode of communication after the failure occurs or after the failure becomes known to the supplier and shall be in the Form given in Schedule IV.

(4) For the purpose of testing or for any other purpose connected with the efficient working of the supplier's installations, the supply of electricity may be discontinued by the supplier for such period as may be necessary, subject to not less than twenty four hours notice being given by the supplier to all consumers likely to be affected by such discontinuance:

Provided that no such notice shall be given in cases of emergency.

Regulation 42, Test of insulation resistance

Where any electric supply line for use at voltages not exceeding 650 V has been disconnected from a system for the purpose of addition, alteration or repair, such electric supply line shall not be reconnected to the system until the supplier or the owner has carried out the test.

Regulation 43, Connection with earth

The following conditions shall apply to the connection with earth of systems at voltage exceeding 50 V but not exceeding 650 V, namely:

(i) neutral conductor of a three phase, four-wire system and the middle conductor of a two-phase, three-wire system shall be earthed as per the relevant standards;

(ii) neutral conductor shall also be earthed at one or more points along the distribution system or service line in addition to any connection with earth which shall be at the consumer's premises;

(iii) in the case of a system comprising electric supply lines having concentric cables, the external conductor or armour of such cables shall be earthed by two separate and distinct connections with earthing system;

(iv) in a direct current system, earthing and safety measures shall be as per the relevant standards;

(v) every building shall have protective equipotential bonding by interconnecting the exposed and extraneous conductive parts as per the relevant standards;

(vi) the alternating current systems which are connected with the earth as provided in this regulation shall be electrically interconnected:

Provided that each connection with the earth is bonded to the metal sheathing and metallic armouring, if any, of the electric supply lines;

(vii) the frame of every generator, stationary motor, portable motor, and the metallic parts, not intended as conductors, all transformers and any other apparatus used for regulating or controlling electricity, and all electricity consuming apparatus, of voltage exceeding 250 V but not exceeding 650 V shall be earthed by two separate and distinct connections with earth by the owner as specified in the relevant standards;

(viii) all metal casing or metallic coverings containing or protecting any electric supply line or apparatus shall be connected with the earth and shall be so joined and connected across all junction boxes and other openings as to provide good mechanical and electrical connection throughout the length:

Provided that the conditions mentioned in this regulation shall not apply, where the supply voltage does not exceed 250 V and the apparatus consists of wall tubes or brackets, electroliers, switches, ceiling fans or other fittings, other than portable hand lamps and portable and transportable apparatus, unless provided with the earth terminal and to class-II apparatus and appliances of the relevant standards:

Provided further that where the supply voltage is not exceeding 250 V and where the installations are either new or renovated, all plug sockets shall be of the three pin type, and the third pin shall be permanently and effectively earthed;

(ix) All earthing systems shall:

(a) consist of equipotential bonding conductors capable of carrying the prospective earth fault current without exceeding the allowable temperature limits as per relevant standards in order to maintain all non- current carrying metal works reasonably at earth potential and to avoid dangerous contact potentials being developed on such metal works;

(b) have earth fault loop impedance sufficiently low to permit adequate fault current for the operation of protective device within the time stipulated in the relevant standards; and

(c) be mechanically strong, withstand corrosion and retain electrical continuity during the life of the installation and all earthing systems shall be tested to ensure effective earth bonding as per the relevant standards, before the electric supply lines or apparatus are energised.

(x) all earthing systems belonging to the supplier shall in addition, be tested for resistance on dry day during the dry season at least once in a year;

(xi) earth fault loop impedance shall be tested to ensure the automatic operation of the protective device and a record of every earth test made and the result thereof shall be kept by the supplier for a period of not less than two years after the day of testing and shall be available to the Electrical Inspector when required;

(xii) earth fault loop impedance of each circuit shall be limited to a value determined by the type and current rating of the protective device used such that, on the occurrence of an earth fault, disconnection of the supply shall occur before the prospective touch voltage reaches a harmful value; and

(xiii) the neutral point of every generator and transformer shall be earthed by connecting it to the earthing system not by less than two separate and distinct connections.

Regulation 46, Use of electricity at voltage exceeding 650 V.

(1) The Electrical Inspector where the supply voltage exceeds the notified voltage shall not authorise the supplier to commence supply or recommence the supply, where the supply has been discontinued for a period of six months or more, or the supplier, where the supply voltage is equal to or below the notified voltage but exceeds 650 V, shall not commence supply or recommence the supply where supply has been discontinued for a period of six months or more, to any consumer unless:

(a) all conductors and apparatus situated on the premises of the consumer are so placed as to be inaccessible except to the designated person;

(b) the consumer has provided and agreed to maintain a separate building or a locked weather proof and fire proof enclosure of agreed design and location, to which the supplier at all times shall have access for the purpose of housing his apparatus and metering equipment, or where the provision for a separate building or enclosure is impracticable, the consumer has segregated the aforesaid apparatus of the supplier from any other part of his own apparatus:

Provided that the segregation shall be made by the fire walls, if the Electrical Inspector considers it to be necessary:

Provided further that in the case of an outdoor installation the consumer shall suitably segregate the aforesaid apparatus belonging to the supplier from his own;

(c) all pole type substations are constructed and maintained in accordance with regulation 52.

(2) Where electricity at voltage exceeding 650 V is supplied, converted, transformed or used, the owner shall:

(i) maintain safety clearances for electrical apparatus as per relevant standards specification so that sufficient space is available for easy operation and maintenance without any hazard to the operating and maintenance personnel working near the equipment and for ensuring adequate ventilation:

Provided that in case of mines, the safety clearances for electrical apparatus to be as per relevant mining regulations;

(ii) not allow any encroachment below such installation:

Provided that where the Electrical Inspector comes across any such encroachment, he shall direct the owner to remove such encroachments;

(iii) maintain minimum safety working clearances specified in Schedule V for the bare conductors or live parts of any apparatus in outdoor substations excluding overhead lines of installations of voltage exceeding 650 V;

(iv) ensure that the live parts of all apparatus within the reach from any position in which a person may require to be, are suitably protected to prevent danger;

(v) ensure that where the transformer is used, suitable provision shall be made, either by connecting with earth, a point of the circuit at the lower voltage or otherwise, to guard against danger by reason of the said circuit becoming accidentally charged above its normal voltage by leakage from or contact with the circuit at the higher voltage;

(vi) not install a substation or a switching station with apparatus having more than 2 000 litres of oil in the basement where proper oil draining arrangement cannot be provided;

(vii) undertake the following measures, where a substation or a switching station with oil-filled apparatus, such as transformer, static condenser, switchgear or oil circuit breaker having more than 2 000 litres of oil is installed, whether indoor or outdoors:

(a) the separation wall or fire barrier walls of thickness and dimensions as specified in the relevant standards shall be provided between the apparatuses and between the apparatus and adjacent building if building wall adjacent to the apparatuses is not rated for four hours fire withstand rating;

(b) provisions shall be made for suitable oil soakpit and where use of more than 9000 litre of oil in any one oil tank, receptacle or chamber is involved, provision shall be made for the draining away or removal of any oil which may leak or escape from the tank, receptacle or chamber containing the same, and special precautions shall be taken to prevent the spread of any fire resulting from the ignition of the oil from any cause and adequate provision shall be made for extinguishing any fire which may occur;

(c) spare oil shall not be stored in the vicinity of any oil filled equipment in any such substation or switching station; and

(d) all the transformers and switchgears shall be maintained in accordance with the maintenance schedules prepared in accordance with the relevant standards;

(viii) without prejudice to the above measures, undertake adequate fire detection and protection arrangement for quenching the fire of the apparatus;

(ix) ensure that every transformer of 10 MVA or reactor of 10 MVAR and above rating shall be provided with automatic firefighting system as per relevant standards;

(x) undertake the following measures, where it is necessary to locate the substation, or switching station in the basement, namely:

(a) the transformer room be in the first basement at the periphery;

(b) the direct access to the transformer room be provided from outside and the surrounding walls of four hours fire withstand rating be provided as per relevant standards;

(c) the entrances to the transformer room be provided with fire resistant doors of two hour fire rating and the door shall always be kept closed and a notice of this effect be affixed on outer side of the door;

(d) a curb of a suitable height be provided at the entrance in order to prevent the flow of oil from a ruptured transformer into other parts of the basement;

(e) the cables to primary side and secondary side have sealing at all floors and wall opening of atleast two hours fire withstand rating; and

(f) Fire Retardant Low Smoke Low Halogen cable as per relevant standards be used;

(xi) ensure that oil filled transformers installed indoors in other than residential or commercial buildings are placed on the ground floor or not below the first basement;

(xii) ensure that only dry type transformer shall be used inside the residential and commercial buildings;

(xiii) ensure that cable trenches inside the substations and switching stations containing cables are filled with sand, pebbles or similar non-inflammable materials or completely covered with non-inflammable slabs; and

(xiv) ensure that unless the conditions are such that all the conductors and apparatus may be made dead at the same time for the purpose of cleaning or for other work, the said conductors and apparatus shall be so arranged that these may be made dead in sections, and that work on any such section may be carried on by the person designated or appointed or engaged or permitted under these regulations without danger.

(3) The minimum clearances specified in Schedule VI shall be maintained for bare conductors or live parts of any high voltage direct current apparatus in outdoor substations, excluding high voltage direct current overhead lines.

(4) There shall not be tapping of another transmission line from the main line for 66 kV and above class of lines:

Provided that during natural calamities, tapping may be allowed to ensure emergency power supply to affected areas till normalcy is restored.

Regulation 47, Inter-locks and protection for use of electricity at voltage exceeding 650 volts

(1) The owner shall ensure the following, namely-

(i) isolators and the controlling circuit breakers shall be inter-locked so that the isolators cannot be operated unless the corresponding breaker is in open position;

(ii) isolators and the corresponding earthing switches shall be inter-locked so that no earthing switch can be closed unless and until the corresponding isolator is in open position;

(iii) where two or more supplies are not intended to be operated in parallel, the respective circuit breakers or linked switches controlling the supplies shall be inter-locked to prevent possibility of any inadvertent paralleling or backfeed;

(iv) when two or more transformers are operated in parallel, the system shall be so arranged as to trip the secondary breaker of the transformer in case the primary breaker of that transformer trips;

(v) all gates or doors which provide access to live parts of an installation shall be inter-locked in such a way that these cannot be opened unless the live parts are made dead and proper discharging and earthing of these parts shall be ensured before any person comes in close proximity of such parts; and

(vi) where two or more generators operate in parallel and neutral switching is adopted, inter-lock shall be provided to ensure that the generator breaker cannot be closed unless one of the neutrals is connected to the earthing system.

(2) The following protection shall be provided in all systems and circuits to automatically disconnect the supply under abnormal conditions, namely:

(i) overcurrent protection to disconnect the supply automatically if the rated current of the equipment, cable or supply line is exceeded for a time which the equipment, cable or supply line is not designed to withstand;

(ii) earth fault or earth leakage protection to disconnect the supply automatically, if the earth fault current exceeds the limit of current for keeping the contact potential within the reasonable values;

(iii) buchholz relay, pressure relief device and winding and oil temperature protection with alarm and trip contacts shall be provided on all transformers of ratings 1 000 kVA and above;

(iv) transformers of capacity 10 MVA and above shall be protected against incipient faults by differential protection;

(v) all generators with rating of 100 kVA and above shall be protected against earth fault or leakage;

(vi) all generators of rating 1 000 kVA and above shall be protected against faults within the generator winding using restricted earth fault protection or differential protection or by both;

(vii) high speed busbar differential protection along with local breaker back up protection shall be commissioned and shall always be available at all 132 kV and above voltage substations and switching stations and generating stations connected with the grid:

Provided that in respect of existing 132 kV substations and switching stations having more than one incoming feeders, the high speed busbar differential protection along with local breaker back up protection, shall be commissioned and shall always be available; and

(viii) in addition to above, all electrical protection system for generating stations, substations and transmission lines shall be as per the regulations notified by the Authority under clause (e) of sub-section (2) of section 177 of the Act.

Regulation 48, Testing, operation and maintenance

(1) Before the approval is accorded by the Electrical Inspector under regulation 45, the manufacturer's test certificates shall, if required, be produced for all the type, acceptance and routine tests as required under the relevant standards.

(2) No new apparatus, cable or supply line of voltage exceeding 650 V shall be commissioned unless such apparatus, cable or supply line are subjected to site tests as per relevant standards.

(3) No apparatus, cable or supply line of voltage exceeding 650 V which has been kept disconnected for a period of six months or more from the system for alterations or repair, shall be connected to the system until such apparatus, cable or supply line are subjected to the site tests as per relevant standards.

(4) Notwithstanding the provisions of this regulation, the Electrical Inspector may require certain tests to be carried out before or after charging the installations.

(5) All apparatus, cables and supply lines shall be maintained in healthy conditions and tests shall be carried out periodically as per the relevant standards.

(6) Records of all tests, trippings, maintenance works and repairs of all apparatus, cables and supply lines shall be duly kept in such a way that these records can be compared with the past records.

(7) It shall be the responsibility of the owner of all installations of voltage exceeding 650 V to maintain and operate the installations in a condition free from danger and as recommended by the manufacturer or by the relevant standards.

(8) Failures of any 220 kV and above voltage level transformer, reactor and transmission line towers shall be reported by the owner of electrical installation, within forty eight hours of the occurrence of the failure, to the Authority and the reasons for failure and measures to be taken to avoid recurrence of failure shall be sent to the Authority within one month of the occurrence in the forms provided in Schedule VII:

Provided that in case of mines and oil-fields, the failure of 10 MVA or above transformers shall be reported to Electrical Inspector of mines.

Regulation 50, Connection with earth for apparatus exceeding 650 volts

(1) The entire switchyard or substation equipment and buildings including all non-current carrying metal parts associated with an installation shall be effectively earthed to an earthing system or mat which shall –

(i) limit the touch and step potential to tolerable values as per relevant standards;

(ii) limit the earth potential rise to tolerable values as per relevant standards, so as to prevent danger due to transfer of potential through ground, earth wires, cable sheath, fences, pipe lines or other such equipment; and

(iii) maintain the resistance of the earth connection to such a value as to make operation of the protective device effective.

(2) In the case of star connected system with earthed neutrals or delta connected system with earthed artificial neutral point:

(i) the neutral point of every generator and transformer shall be earthed by connecting it to the earthing system not by less than two separate and distinct connections:

Provided that the neutral point of a generator may be connected to the earthing system through an impedance to limit the fault current:

Provided further that in the case of multi-machine systems, neutral switching may be resorted to, for limiting the injurious effect of harmonic current circulation in the system;

(ii) the generator or transformer neutral shall be earthed through a suitable impedance where an appreciable harmonic current flowing in the neutral connection causes interference with the communication circuits; and

(iii) in case of the delta connected system, the neutral point shall be obtained by the insertion of a earthing transformer and current limiting resistance or impedance wherever considered necessary at the commencement of such a system.

(3) In case of generating stations, substations and other installations of voltage exceeding 33 kV, the system neutral earthing and protective frame earthing may be, if system design so warrants, integrated into common earthing grid provided the resistance to earth of combined mat does not cause the step and touch potential to exceed the values as per relevant standards.

(4) Single phase systems of voltage exceeding 650 V shall be effectively earthed.

(5) In the case of a system comprising electric supply lines having concentric cables, the external conductor shall be connected with the earth.

(6) Where a supplier proposes to connect with earth an existing system for use at voltage exceeding 650 V which has not hitherto been so connected with earth, he shall give not less than fourteen days notice in writing together with particulars of the proposed connection with earth to the telegraph authority established under the *Indian Telegraph Act*, 1885 (13 of 1885).

(7) Where the earthing lead and earth connection are used only in connection with earthing guards laid under overhead lines of voltage exceeding 650 V but not exceeding 33 kV where they cross a telecommunication line or a railway line, and where such lines are equipped with earth leakage protective device, the earth resistance shall not exceed 25 Ω and the project authorities shall obtain no objection certificate from Railway Authorities and Power and Telecommunication Coordination Committee before energisation of the facilities.

Every earthing system belonging to either the supplier or the consumer shall be tested for its resistance to earth on a dry day during dry season not less than once in a year and records of such tests shall be maintained and produced, if so required, before the Electrical Inspector.

Regulation 60, Clearance in air of the lowest conductor of overhead lines

(1) The minimum clearance above ground and across road surface of National Highway or Expressway or State Highway or other road or highest traction conductor of railway corridor or navigational or non-navigational river of the lowest conductor of an alternating current overhead line, including service lines, of nominal voltage shall have the values specified in Schedule VIII A.

(2) The minimum clearances regarding high voltage direct current line shall be as per Schedule VIII B.

(3) In case of Electric lines of 33 kV and below passing through the protected areas (National Parks, Wildlife Sanctuaries, Conservation Reserves, Community Reserves), Eco-sensitive zones around the protected areas and Wildlife Corridors, only underground cable shall be used.

(4) No tower footing or structure of an overhead line of voltage 33 kV or above or high voltage direct current, shall be closer than twenty five metre from the edge of the right of way of a Petroleum or Natural Gas pipeline.

(5) Wherever overhead line of voltage 33 kV or above or high voltage direct current intending to cross the right of way of a Petroleum or Natural Gas pipeline, the angle of crossing of the overhead line with respect to the pipelines shall preferably be at right angles and, in any case, the crossing angle shall not be less than seventy five degrees.

Regulation 62, Clearance from buildings of lines of voltage and service lines not exceeding 650 Volts

(1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

(2) Where an overhead line of voltage not exceeding 650 V passes above or adjacent to or terminates on any building, the following minimum clearances from any accessible point, on the basis of maximum sag, shall be observed, namely–

(i) for any flat roof, open balcony, *verandah* roof and lean-to-roof:

(a) when the line passes above the building, a vertical clearance of 2.5 metre from the highest point; and

(b) when the line passes adjacent to the building, a horizontal clearance of 1.2 metre from the nearest point;

(ii) for pitched roof:

(a) when the line passes above the building, a vertical clearance of 2.5 metre immediately under the line; and

(b) when the line passes adjacent to the building, a horizontal clearance of 1.2 metre.

(3) Any conductor so situated as to have a clearance less than that specified in sub-regulation (2) shall be replaced with Aerial Bunched Cable and to be attached at suitable intervals to a bare earthed bearer wire having a breaking strength of not less than 350 kgf.

(4) The horizontal clearance shall be measured when the line is at a maximum deflection from the vertical due to wind pressure.

(5) The vertical and horizontal clearances shall be measured as per illustration provided in Schedule VIII C.

Explanation — For the purposes of this regulation, the expression 'building' shall be deemed to include any structure, whether permanent or temporary.

Regulation 63, Clearance from buildings of lines of voltage exceeding 650 V

(1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

(2) Where an overhead line of voltage exceeding 650 V passes above or adjacent to any building or part of a building it shall have on the basis of maximum sag a vertical clearance above the highest part of the building immediately under such line, of not less than:

(i)	for lines of voltage exceeding 650 Volts up to and including 33kV	-	3.7 m
(ii)	for lines of voltage exceeding 33 kV	-	3.7 m plus 0.30 m for every additional 33 kV or part thereof.

(3) The horizontal clearance between the nearest conductor and any part of such building shall, on the basis of maximum deflection due to wind pressure, be not less than:

(i)	for lines of voltage exceeding 650 V up to and including 11 kV	-	1.2 m
(ii)	for lines of voltage exceeding 11 kV and up to and including 33 kV	-	2.0 m
(iii)	for lines of voltage exceeding 33 kV	-	2.0 m plus 0.3 m for every additional 33 kV or part thereof.

(4) For high voltage direct current systems, the vertical and horizontal clearances, on the basis of maximum deflection due to wind pressure, from buildings shall be maintained as below:

Sl No.	*High Voltage Direct Current*	*Vertical Clearance* m	*Horizontal Clearance* m
1.	100 kV	4.6	2.9
2.	200 kV	5.8	4.1
3.	300 kV	7.0	5.3
4.	400 kV	7.9	6.2
5.	500 kV	9.1	7.4
6.	600 kV	10.3	8.6
7.	800 kV	12.4	10.7

(5) The vertical and horizontal clearances shall be as measured as illustrated in Schedule VIII C.

Regulation 66, Transporting and storing of material near electric lines

(1) No rods, pipes or similar materials shall be taken below, or in the vicinity of any bare overhead conductors or lines:

Provided that if these materials contravene the Regulations 62 and 63, such materials shall be transported under the direct supervision of a person designated or appointed or engaged or permitted under these regulations.

(2) No rods, pipes or other similar materials shall be brought within the flash over distance of bare live conductors or overhead lines.

(3) No material or earth work or agricultural produce shall be dumped or stored, no trees grown below or in the vicinity of bare overhead conductors or lines in contravention to the Regulations 62 and 63.

(4) No flammable material shall be stored under the electric line.

(5) No fire shall be allowed below overhead lines and above the demarcated underground cables.

Regulation 67, General clearances

(1) For the purpose of computing the vertical clearance of an overhead line, the maximum sag of any conductor shall be calculated on the basis of the maximum sag in still air and the maximum temperature as specified under Regulation 59 and computing any horizontal clearance of an overhead line the maximum deflection of any conductor shall be calculated on the basis of the wind pressure specified under Regulation 59.

(2) No blasting for any purpose shall be done within three hundred metre from the boundary of a substation or from the electric supply lines of voltage exceeding 650 V or tower structure thereof without the written permission of the owner of such substation or electric supply lines or tower structures; and in case of mining lease hold area, without the written permission of the Electrical Inspector of mines.

(3) No cutting of soil within ten metre from the tower structure of 110 kV and above voltage level shall be permitted without the written permission of the owner of tower structure.

(4) No person shall construct brick kiln or other polluting units near the installations or transmission lines of 220 kV and above within a distance of 500 metre.

Regulation 76, Safety and Protective Devices

(1) Every overhead line which is not being suspended from a dead bearer wire, not being covered with insulating material and not being a trolley-wire, is laid over any part of a street or other public place or in any factory or mine or on any consumer's premises shall be protected with earth guarding for rendering the line electrically harmless in case its conductor breaks.

(2) An Electrical Inspector may, by notice in writing, require the owner of any such overhead line, wherever it may be laid, to protect it in the manner specified in sub-regulation (1) of this regulation.

(3) To prevent bird dropping on the suspension insulator strings, suitable bird guards as per relevant standards, shall be provided on cross arms of suspension tower or suspension pole structures, over the suspension insulator strings.

Regulation 119, Additional safety requirements for renewable generating stations

The regulations under this chapter shall be applicable to renewable generating stations which shall be in addition to the regulations provided from chapter I to VII.

Regulation 121, Safety Requirements for Solar Installations

(1) The following general safety requirement for solar installations shall be ensured, namely –

(i) clear pathways of minimum seventy five centimetre in width with hand rails for roof access and emergency exit shall be provided for roof top system;

(ii) there shall be clear pathways, walkways between the rows or columns of solar panels which is necessary for cleaning and maintenance;

(iii) cables shall be laid in trenches for ground based photovoltaic installation;

(iv) ground mounted solar installations shall be protected by fencing or other means not less than 1.8 metre in height so as to prevent unauthorised entry;

(v) disconnection switches or circuit breakers provided in combiner boxes to disconnect the photovoltaic system from all other conductors of the system shall be located at a readily accessible location;

(vi) three phases on the alternating current side, and positive and negative conductor on the direct current side shall be marked and identified with different colours;

(vii) inverter unit for solar photovoltaics shall be installed in the periphery of the building and as near as the solar panel:

Provided that the direct current cable shall be ultraviolet protected or routed through ultraviolet protected pipe;

(viii) there shall be a manual disconnection switch to isolate the system from grid and shall be situated outside the alternating current combiner box; and

(ix) protection shall be deployed (for both input and output) on site for overload, surge current, surge voltage, short circuit, high temperature, over voltage, under voltage and over frequency, under frequency, reverse polarity and lightning.

(2) The following earthing requirements for solar installations shall be ensured, namely –

(i) solar earthing shall be as per the relevant standard;

(ii) the frame of inverter cabinet shall be connected with the earthing bus bar through the earthing terminals using flexible braided copper wire;

(iii) all metal casing and shielding of the plant, each array structure of the photovoltaic yard, equipment, inverters and control systems shall be earthed through proper earthing;

(iv) earthing system shall connect all non-current carrying metal receptacles, electrical boxes, appliance frames, chasis and photovoltaic module mounting structures in one long run and the earth strips shall be interconnected by proper welding and shall not be bolted;

(v) there shall be adequate number of interconnected earth pits provided in each location and minimum required gap shall be provided in between earth pits as per relevant standard.

(3) The following protection, testing and interlocking requirement for solar installations shall be ensured; namely:

(i) the solar photovoltaic power plant shall be provided with lightning and over voltage protection by deploying required number of lightning arresters as per the relevant standards;

(ii) every combiner box shall be provided with suitable surge protective device with arc extinguishing capability as per the relevant standards to avoid any risk of fire;

(iii) the input circuits of combiner box shall be provided with over current protection as per the relevant standards;

(iv) the output circuits of combiner box shall be provided with isolation protection;

(v) earth fault protection and insulation monitoring for photovoltaic arrays and inverters shall be provided;

(vi) all photovoltaic modules safety qualification shall comply with relevant standards.

(4) *Requirement to prevent fire for solar installations* — A fire detection system and automatic fire suppression system shall comply with the relevant standards.

(5) *Additional safety requirements for floating solar photovoltaic energy installations* — Photovoltaic modules and associated structure of floating solar power plant shall comply with the relevant standards for tests such as salt mist, ammonia corrosion, environmental stress cracking of High Density Poly Ethylene, stress cracking resistance of High Density Poly Ethylene and standard test method for tensile properties of plastics.

Regulation 123, Additional safety requirements for electric vehicle charging station

The regulations under this chapter shall be applicable to electric vehicle charging stations which shall be in addition to the regulations provided from chapter I to VII.

Regulation 124, General safety requirements for electric vehicle charging station

(1) Electric vehicle charging stations shall be provided with separate protection against the overload of input supply and output supply as per relevant standards.

(2) Socket-outlet of supply of electric vehicle charging points shall be installed at least 800 millimetre above the finished ground level.

(3) A cord extension set or second supply lead shall not be used in addition to the supply lead for the connection of the electric vehicle to the charging point and shall not be used as a cord extension set.

(4) No adaptor shall be used to connect a vehicle connector to a vehicle inlet.

(5) The distance between the charging point and the connection on the electric vehicle shall not be more than five metre during charging.

(6) The portable socket outlets shall not be permitted for electric vehicle charging.

(7) The lightning protection system shall be provided for the electric vehicle charging stations as per relevant standards.

(8) The electric vehicle charging station shall be equipped with a protective device against the uncontrolled reverse power flow from electric vehicle to the charging point.

(9) One second after disconnecting the electric vehicle from the supply mains, the voltage between accessible conductive parts or any accessible conductive part and earth shall be less than or equal to 42.4 Vpeak (30 Vrms), or 60 V DC, and the stored energy available shall be less than 20 J:

(10) Provided that, if the voltage is greater than 42.4 Vpeak (30 Vrms) or 60 V DC, or the energy is 20 J or more, a warning label shall be attached in a conspicuous position on the charging stations.

(11) A vehicle connector used for direct current charging shall be locked on the vehicle inlet if the voltage is higher than 60 V DC and in case of charging system malfunction, a means for safe disconnection shall be provided.

(12) The electric vehicle charging point shall disconnect supply of electricity to prevent overvoltage at the battery, if the output voltage exceeds maximum voltage limit permissible for the vehicle.

(13) The electric vehicle charging points shall not energise the charging cable when the vehicle connector is in unlock position.

(14) The electric vehicle connector shall not unlock if the voltage between the vehicle connector and the earth is more than 60 V.

(15) Safety clearance between the oil or gas dispenser and electric vehicle charging point shall be as per the order issued by the Authority.

(16) Only four core cable shall be used for charging points which require three phase power supply.

(17) Underground cables shall not cross the underground oil tank or oil pipeline.

(18) Underground cables through the charging area or vehicles passage shall be avoided and if provided shall be at a minimum depth of one metre from the finished ground surface.

Regulation 125, Earth Protection System for the Charging Station

(1) Each electric vehicle charging points shall be supplied individually by a dedicated sub-circuit protected by an overcurrent protective device complying with the relevant standards and the overcurrent protective device shall be part of a switchboard.

(2) Co-ordination of all protective devices in the charging stations shall be ensured.

(3) All electric vehicle charging stations shall be provided with an earth continuity monitoring system that disconnects the supply in the event of the earthing connection to the vehicle becomes ineffective.

(4) The charging lead shall be fitted with an earth-connected metal shielding and the cable insulation shall be wear resistant and maintain flexibility over the operating temperature range.

(5) A protective earth conductor shall be provided to establish an equipotential connection between the earth terminal of the supply and the conductive parts of the vehicle which shall be as per the relevant standards.

Regulation 126, Requirement to Prevent Fire for Electric Vehicle Charging Station

(1) The enclosure of electric vehicle supply equipment shall be made of fire retardant material with self-extinguishing property and free from halogen.

(2) The fire detection, alarm and control system shall be provided as per relevant standards.

Regulation 127, Testing of Charging Station

The owner of the charging station shall ensure that the tests as specified in the manufacturer's instructions for the residual current devices and the charging station have been carried out.

Regulation 128, Maintenance of records

(1) The owner of the charging station shall keep records of design, construction and labelling to be compatible with a supply of standard voltage at a nominal frequency of 50 Hz of the charging station.

(2) The owner of the charging station shall keep records of the relevant test certificates as indicated in these regulations and as per the relevant standards.

(3) The owner of the charging station shall keep records of the results of every inspection, testing and periodic assessment and details of any issues observed during the assessment and any actions required to be taken in relation to those issues.

(4) The owner of the charging station shall retain a copy of all records, as specified in sub-regulation (1), (2) and (3) of above, either in hard form or in electronic form, for at least seven years and shall provide a copy of the records to the officials during the inspection.

ANNEX C

[*Clauses* 4.2.1(ff), 4.2.2.2 *and* D-1 (Note)]

AREA REQUIRED FOR TRANSFORMER ROOM AND SUBSTATION FOR DIFFERENT CAPACITIES

C-1 The requirement for area for transformer room and substation for different capacities of transformers is given below for guidance:

Sl No.	*Capacity of Transformer(s)*	*Total Transformer Room Area, Min*	*Total Substation Area (Incoming, HV Panels, MV Panels, Transformer Roof but Without Generators), Min*	*Suggested Minimum Face Width*
	kVA	m^2	m^2	m
(1)	(2)	(3)	(4)	(5)
i)	1 × 160	14.0	90	9.0
ii)	2 × 160	28.0	118	13.5
iii)	1 × 250	15.0	91	9.0
iv)	2 × 250	30.0	121	13.5
v)	1 × 400	16.5	93	9.0
vi)	2 × 400	33.0	125	13.5
vii)	3 × 400	49.5	167	18.0
viii)	2 × 500	36.0	130	14.5
ix)	3 × 500	54.0	172	19.0
x)	2 × 630	36.0	132	14.5
xi)	3 × 630	54.0	176	19.0
xii)	2 × 800	39.0	135	14.5
xiii)	3 × 800	58.0	181	14.0
xiv)	2 × 1 000	39.0	149	14.5
xv)	3 × 1 000	58.0	197	19.0

NOTES

1 The areas given in respect of the different categories of rooms hold good, if they are provided with independent access doors in accordance with local regulations.

2 The minimum height of substation room/HV switch room/MV switch room shall be arrived at considering 1200 mm clearance requirement from top of the equipment to the below of the soffit of the beam. In case cable entry/exit is from above the equipment (transformer, HV switchgear, MV switchgear), height of substation room/HV switch room/MV switch room shall also take into account requirement of space for turning radius of cable above the equipment height

3 Additional space will be required in cases where the load requirement calls for redundancy for enhanced reliability through addition of switchgear, such as, bus couplers, etc.

4 For transformers of other capacity, it may lead to some minor changes in dimensioning.

ANNEX D

(*Clauses* 3.5.2(h) *and* 4.3.2)

ADDITIONAL AREA REQUIRED FOR GENERATOR IN ELECTRIC SUBSTATION

D-1 The requirement of additional area for generator in electric substation for different capacities of generators is given in the table below for guidance.

NOTES

1 The space requirements vary for specific installations due to factors such as derating due to site conditions (temperature, altitude, etc); type of cooling (radiator, heat exchanger, cooling tower, etc); type of ventilation; noise suppression system and special characteristics of loads, if any.

2 If the generator is located away from the substation, then additional switchgear will be required, area requirement for which can be estimated from the norms given in Annex C.

3 The area requirement suggested below covers the space requirement for day-tank and not the space for the bulk fuel storage.

Sl No.	*Capacity*	*Area*	*Clear Height below the Soffit of the Beam*
	kVA	m^2	m
(1)	(2)	(3)	(4)
i)	25	56	3.6
ii)	48	56	3.6
iii)	100	65	3.6
iv)	150	72	3.6
v)	248	100	4.2
vi)	350	100	4.2
vii)	480	100	4.2
viii)	600	110	4.6
ix)	800	120	4.6
x)	1 010	120	6.5
xi)	1 250	120	6.5
xii)	1 600	150	6.5
xiii)	2 000	150	6.5

NOTE — The area and height required for generating set room given in the above table are for general guidance only and may be finally fixed according to actual requirements.

ANNEX E

(*Clauses* 9.3.1 *and* 9.4)

CHECKLIST FOR INSPECTION, HANDING OVER AND COMMISSIONING OF VARIOUS EQUIPMENT OF SUBSTATION

E-1 Typical format for checklist for inspection, handing over and commissioning of HV cables is given below:

NOTE — Format given below covers a basic minimum check list; it should be augmented for specific and special cases. The checklist has to be repeated for each HV cable.

HV CABLE INSPECTION, HANDING OVER AND COMMISSIONING DETAILS

A. DETAILS OF WORK

1) Scope of works :
2) Handed over by :
3) Taken over by :
4) Date of commissioning :
5) Date of handing over :
6) Details of enclosures :

Sl No.	*Description*	*Applicable*	*Not Applicable*
(1)	(2)	(3)	(4)
i)	Quality check list		
ii)	Site test report		
iii)	As built GA drawing/layout		
iv)	Manufacturer's test certificates		
v)	Operation and maintenance manual		

Handed Over By

Taken Over By

Authorized Signatory

Authorized Signatory

B. QUALITY CHECK LIST

Item: Installation of HV Cables				
Make:				
Sl No.	*Tests Parameters*	*Bench Mark*	*Actual Observations*	*Remark*
(1)	(2)	(3)	(4)	(5)
i)	Cable size			
ii)	Voltage grade			
iii)	Type of material			
iv)	Check for routing			
v)	Meggering using 2.5 kV insulation tester			
	R-Y			
	Y-B			
	B-R			
	R-E			
	Y-E			
	B-E			
vi)	Minimum width of trench			
vii)	Minimum depth of trench			
viii)	Hume pipes used			
ix)	Hume joints with collar properly aligned and packed with 75 mm of cement concrete			
x)	Cable laying with suitable rollers			
xi)	Bending radius			
xii)	Cable tagging			
xiii)	Hi-Pot test (18 kV)			
xiv)	Sealing of cable ends, if not being terminated immediately			
xv)	Trench closing			

C. HI-POT TEST REPORT

1) Date of testing:

2) Equipment details:

Cable details

Location :
Panel No. :
Size :

3) Insulation Resistance Test (Value in MΩ, using 5 kV Megger):

Reference	*Measured Values* MΩ	
	Before	After
R-E		
Y-E		
B-E		
R-Y		

Y-B		
B-R		

4) Winding Resistance Test:

Applied Voltage kV	*Measured Current* mA
R = Y + B + E	
Y = B + R + E	
B = R + Y + E	

5) Remarks

E-2 Typical format for checklist for handing over and commissioning of HV panels is given below:

NOTE — Format given below covers a basic minimum check list; it should be augmented for specific and special cases. The checklist has to be repeated for each HV panel.

HV PANEL INSPECTION, HANDING OVER AND COMMISSIONING DETAILS

Project:

Owner:

Package:

Contractor:

A. HANDING OVER DETAILS

1) Scope of works :
2) Panel name and number :
3) Location :
4) Handed over by :
5) Taken over by :
6) Date of commissioning :
7) Date of handing over :
8) Details of enclosures :

Sl No.	Description	Applicable	Not Applicable
(1)	(2)	(3)	(4)
i)	Quality check list		
ii)	Site test report		
iii)	As built GA drawing/layout		
iv)	Manufacturer's test certificates		
v)	Operation and maintenance manual		

Handed Over By　　　　　　　　Taken Over By

Authorized Signatory　　　　　　　　Authorized Signatory

B. QUALITY CHECK LIST

Item: HV panel
Make:
Relevant Indian Standard:
As-built Drawing No:

Sl No.	*Test Parameters*	*Bench Mark*	*Actual Observations*
(1)	(2)	(3)	(4)
	Physical Checks for:		
	Circuit Breaker		
i)	Dimension enclosure		
ii)	Gauge		
iii)	Paint		
iv)	Degree of protection		
v)	Front doors		
vi)	Back cover		
vii)	Extension type enclosure		
viii)	Type of circuit breaker		
ix)	Number of circuit breakers		
x)	Type of surge arresting device included in the circuit breakers		
xi)	Earthing terminal		
xii)	Front plate with view glass		
xiii)	Space for mounting of current transformers		
xiv)	Space for mounting of potential transformers		
xv)	Cable entry		
xvi)	Interlocking with isolator		
xvii)	Earthing switch for cable in cable chamber		
xviii)	Earthing switch interlock with circuit breaker only it can be operated in 'OFF' condition		
xv)	Low voltage plug and socket		
xvi)	Vents for breaker/busbar/cable chambers		
xvii)	Insulation level		
xviii)	Toggle switch		
xix)	Ammeter selector switch		
xx)	Voltmeter selector switch		
xxi)	Trip/Neutral/Close		
xxii)	Local/remote selector switch		
xxiii)	LED lamps		
xxiv)	Mechanical operation		
xxv)	Remote operation		
xxvi)	Local operation		
xxvii)	Interlocking with isolator		
	Meters		
xxviii)	Voltmeter		

Sl No.	*Test Parameters*	*Bench Mark*	*Actual Observations*
(1)	(2)	(3)	(4)
xxix)	Ammeter		
xxx)	Tri vector meter		
xxxi)	Power factor meter		
xxxii)	Frequency meter		
	Specification Checks for		
xxxiii)	Rated current		
xxxiv)	Rated voltage		
xxxv)	Rated short circuit breaking capacity		
xxxvi)	Contact resistance		
xxxvii)	Control wiring		
xxxviii)	All control circuits Alarm and trip for OTI/WTI/Buchholz/ PRV		
xxxix)	SF6 pressure alarm and trip operation test		

C. COMMISSIONING REPORT

Customer:	
Project:	
Contractor:	
Panel Name:	
Location:	
Breaker Details:	

INSULATION TEST (MΩ)

R-Y	Y-B	B-R	R-E	Y-E	B-E	R-N	Y-N	B-N

RELAY SETTINGS

Overcurrent, Earth Fault and Undervoltage Relay	
CT ratio: Earth fault >Settings: I_o> Settings: Under voltage relay:	

GENERAL CHECKS

Breaker ‘ON’/‘OFF’	
Meters reading	
Indicating lamps	
Control supply	

E-3 Typical format for checklist for handing over and commissioning of transformers is given below:

NOTE — Format given below covers a basic minimum check list; it should be augmented for specific and special cases. The checklist has to be repeated for each transformer.

TRANSFORMER INSPECTION, HANDING OVER AND COMMISIONING DETAILS

Project:
Owner:
Package:
Contractor:

A. HANDING OVER DETAILS

1) Scope of works :
2) Transformer No. :
3) Location :
4) Handed over by :
5) Taken over by :
6) Date of commissioning :
7) Date of handing over :
8) Details of enclosures :

Sl No.	*Description*	*Applicable*	*Not Applicable*
(1)	(2)	(3)	(4)
i)	Quality check list		
ii)	Site test report		
iii)	As built GA drawing/layout		
iv)	Manufacturer's test certificates		
v)	Operation and maintenance manual		

Handed Over By

Taken Over By

Authorized Signatory

Authorized Signatory

B. QUALITY CHECK LIST

Item: Transformer
Make:
Relevant Indian Standard:
As-built Drawing No:

Sl No.	*Test Parameters*	*Bench Mark*	*Actual Observation*
(1)	(2)	(3)	(4)
a)	*Physical Checks for:*		
	i) Transformer Sl No.		
	ii) Dimension of enclosure		
	iii) Enclosure – Degree of protection		
	iv) Gauge		
	v) Paint		
	vi) Provision for lifts and jacking		
	vii) Wheels		
	viii) Locking facility for wheel		
	ix) Enclosure door (if applicable)		
	x) Enclosure door interlock wiring		
	xi) Removable case for access of taps		
	xii) Earthing terminal		
	xiii) Disconnecting type chamber for cable termination		
	xiv) LT side flanges (Bus ducts/Cables/ Overhead)		
	xv) Oil temperature indicator (relay unit)		
	xvi) Winding temperature indicator (relay unit)		
	xvii) Marshalling box winding and door switch		
	xviii) Thermistors for alarm and trip		
	xix) Separate neutral earthing chamber		
	xx) Rating and diagram plate		
	xxi) Tap changing options name plate		
	xxii) Oil level in conservator tank		
	xxiii) Oil level in breather cup		
	xxiv) Operation of PRV		
	xxv) Oil leakage (if applicable)		
	xxvi) Clearances around transformer		
	xxvii) Body earthing resistance		
	xxviii) Neutral earthing resistance		
b)	*Specification checks for*:		
	i) Rating of transformer		
	ii) Type		
	iii) Winding conductor		
	iv) Primary voltage		
	v) HV winding connections		

Sl No.	*Test Parameters*		*Bench Mark*	*Actual Observation*
(1)	(2)		(3)	(4)
	vi)	Secondary voltage		
	vii)	LV side connections		
	viii)	Vector symbol		
	ix)	System of supply		
	x)	Impedance percentage		
	xi)	Oil temperature rise		
	xii)	Winding temperature rise		
	xiii)	Tap changing		
	xiv)	Tapping range		
	xv)	Insulation type		
	xvi)	Type of cooling		
	xvii)	Vector group test		
	xviii)	Polarity test		
	xix)	Magnetizing test		
	xx)	Tan delta test (as per capacity)		
	xxi)	Breakdown voltage test: *Oil sample – I* (*Top*) *Oil sample – II* (*Bottom*)		

C. COMMISSIONING REPORT

Customer:	
Project:	
Contractor:	
Transformer Number:	
Location:	
Rating:	

INSULATION TEST (MΩ)

R-Y	Y-B	B-R	R-E	Y-E	B-E	R-N	Y-N	B-N

RELAY SETTINGS

Transformer Protection Relay (*WTI Scanner*): Winding temperature alarm set Winding temperature trip set (*OTI Scanner*): Oil temperature alarm set Oil temperature trip set	
Buchholz Relay	

GENERAL CHECKS

Breaker 'ON'/'OFF'	
Meters reading	
Indicating lamps	
Control supply	

E-4 Typical format for checklist for handing over and commissioning of MV/LV panel is given below:

NOTE — Format given below covers a basic minimum check list; it should be augmented for specific and special cases. The checklist has to be repeated for each MV/LV panel.

MV/LV PANEL INSPECTION, HANDING OVER AND COMMISIONING DETAILS

Project:
Owner:
Package:
Contractor:

A. HANDING OVER DETAILS

1) Scope of works :
2) Panel name :
3) Location :
4) Handed over by :
5) Taken over by :
6) Date of commissioning :
7) Date of handing over :
8) Details of enclosures :

Sl No.	*Description*	*Applicable*	*Not Applicable*
(1)	(2)	(3)	(4)
i)	Quality check list		
ii)	Site test report		
iii)	As built GA drawing		
iv)	Manufacturer's test certificates		
v)	Operation and maintenance manual		

Handed Over By

Taken Over By

Authorized Signatory

Authorized Signatory

B. QUALITY CHECK LIST

Item: MV/LV Panel
Make:
Relevant Indian Standard:
As-built Drawing No:

Sl No.	*Test Parameters*	*Bench Mark*	*Actual Observation*
(1)	(2)	(3)	(4)
a)	*Physical Checks for:*		
	i) Dimension enclosure		
	ii) Paint		
	iii) Degree of protection		
	iv) Front doors		
	v) Back cover		
	vi) Earthing terminal		
	vii) Front plate with view glass		
	viii) Cable entry		
	ix) Indication lamps		
	x) Check for any damages		
	xi) Check for rubber beading for the panel sections		
	xii) Coupling of panel		
	xiii) Coupling of busbars		
	xiv) Coupling of earth bus		
	xv) Number of shipping sections		
	xvi) Fish plate tightening		
	xvii) Any damages in busbar supports		
	xviii) Make of MCCB		
b)	*Specification Checks for:*		
	i) Rated current		
	ii) Rated voltage		
	iii) Rated short circuit capacity		
c)	*Circuit Breaker*:		
	i) Current rating		
	ii) Interlocks with other CB/relay		
	iii) Compartment		
	iv) End terminals segregation		
d)	*Busbar*:		
	i) Earth busbar		
	ii) Main busbar		
	iii) Link busbar		
	iv) Busbar sleeve		
e)	3 *Phase – Potential transformer*:		
	i) Make		
	ii) Ratio		

Sl No.	*Test Parameters*	*Bench Mark*	*Actual Observation*
(1)	(2)	(3)	(4)
	iii) Class		
	iv) Serial Nos.		
f)	*Current transformer*:		
	i) Make		
	ii) Ratio		
	iii) Class		
	iv) Serial No's		
g)	*Multi digital meter*:		
	i) Make		
	ii) Input voltage		
	iii) Input current		
	iv) Wiring		
	v) For potential transformer		
	vi) For current transformer		
	vii) Incomer		
	viii) Outgoing		

C. COMMISSIONING REPORT

Customer:	
Project:	
Contractor:	
Panel Name:	
Location:	
Breaker Details:	

INSULATION TEST (MΩ)

R-Y	Y-B	B-R	R-E	Y-E	B-E	R-N	Y-N	B-N

RELAY SETTINGS

GENERAL CHECKS

Breaker 'ON'/'OFF'	
Meters reading	
Indicating lamps	
Control supply	

ANNEX F

(*Clause* 9.5)

FORM OF COMPLETION CERTIFICATE

I/ We certify that the installation detailed below has been installed by me/us and tested and that to the best of my/our knowledge and belief, it complies with *Central Electricity Authority* (*Measures Relating to Safety and Electricity Supply*) *Regulations*, 2023 as amended up-to-date and also the relevant provisions of Indian Standards and National Electrical Code of India, 20xx.

Electrical Installation at ____

A. INSTALLATION DATA

Voltage and system of supply _______________________________________

a) *Particulars of Works*:

1) Internal electrical installation:

Sl No.	*Description*	*No.*	*Total Load*	*Type of System of Wiring*
i)	Light point			
ii)	Fan point			
iii)	Plug point			
	3-pin 6 A			
	3-pin 16 A			
	5-pin 6 A			
	6-pin 16 A			

2) Others:

Sl No.	*Description*	*HP/kW*	*Type of Starting*
i)	Motors		
	1)		
	2)		
	3)		
ii)	Other plants		
	1)		
	2)		
	3)		

3) If the work involves installations of overhead line and/or underground cable:

i)	1) Type and description of over headline	
	2) Total length and No. of spans	
	3) No. of street lights and its description	
ii)	1) Total length of underground cable and its size	
	2) No. of joints:	
	End joint:	
	Tee joint:	
	Straight through joint:	

b) *Earthing*:

i)	Description of earthing electrode	
ii)	No. of earth electrodes	
iii)	Size of main earth lead	
iv)	Earth electrode resistance	

The date on which the measurements for item (iv) was taken has to be recorded.

NOTE — The above data should be supported by an appropriate drawing and tables indicating the reference points at which the measurements were taken at the construction/installation stage.

B. PRE-COMMISSIONING TEST DATA

a) *Test Results*:

1) Appropriate drawings showing the points at which the measurements were taken should accompany this report.

2) Environmental notes pertaining to the day on which the tests are conducted:

i) Temperature:

ii) Date and time:

iii) Previous day/days history from the aspect of rain (in as much as it would affect the test results):

3) List of test instruments used:
(Please record the calibration certificate date, calibration authority and the validity date of each instrument listed.)

i) -------------------

ii) -------------------

4) Results:

i) Insulation resistance

a)	Insulation resistance of the whole system of conductors to earth	………… Megaohms
b)	Insulation resistance between the phase conductor and neutral:	
	Between phase R and neutral	………… Megaohms
	Between phase Y and neutral	………… Megaohms
	Between phase B and neutral	………… Megaohms
c)	Insulation resistance between the phase conductors in case of polyphase supply	
	Between phase R and phase Y	………… Megaohms
	Between phase Y and phase B	………… Megaohms
	Between phase B and phase R	………… Megaohms
d)	Insulation resistance between neutral conductor and earth in case of systems with neutral solidly earthed at the source	………… Megaohms

ii) Polarity test:

Polarity of non-linked single pole branch switches:

iii) Earth continuity test:

Maximum resistance between any point in the earth continuity conductor including metal conduits and main earthing lead ……………Ohms.

iv) Earth electrode resistance:

a) Resistance of each earth electrode:

1) …………….. Ohms
2) …………….. Ohms
3) …………….. Ohms
4) …………….. Ohms

b) Resistance of each earth grid:

1) …………….. Ohms
2) …………….. Ohms
3) …………….. Ohms
4) …………….. Ohms

v) Lightning protective system:

a)	Designed lightning protection level (LPL)	…..
b)	Total roof area (L X W)	….. m^2
c)	Air-termination mesh size (L X W)	….. m x m
d)	Number of vertical rods/terminal (Down-conductors)	…..
e)	Distance between down-conductors	….. m
f)	Impulse current rating of SPD, if any	…… kA
g)	Voltage protection level of incoming SPD	…….kV
h)	Resistance of down-conductor for reinforced concrete structure/PEB structures, if natural components are used as per NBCS (Part D/ Section 2) 'Building Services, Electrical and Allied Installations'	……. ohms
j)	Resistance of the whole of lightning protective system to earth before any bonding is effected with earth electrode and metal in/on the structure	……. ohms

Signature of Supervisor	Signature of Contractor
Name and Address	Name and Address
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Persons witnessing the test:

Name	Designation/Address	Signature	Remarks
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ANNEX G

(*Clause* 3.5)

DECLARATIONS

G-1 DECLARATION FOR DESIGN

I/We being the person(s) responsible for the design of the electrical installation (as indicated by my/our signatures below), particulars of which are described below, having exercised reasonable skill and care when carrying out the design, hereby DECLARE that the design work for which I/we have been responsible is to the best of my/our knowledge and belief conform to the *CEA* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 and in accordance with the National Electrical Code of India, 20xx, National Building Construction Standards, 2026 and relevant standards of the products used and except for the departures, if any, detailed as follows:

Departures

Project/Site details

(Sign)
Electrical Consultant
Name and Address

(Sign)
Electrical Engineer on site

G-2 DECLARATION FOR CONSTRUCTION

I/We being the person(s) responsible for the construction of the electrical installation (as indicated by my/our signatures below), particulars of which are described below, having exercised reasonable skill and care when carrying out the construction, hereby DECLARE that the construction work for which I/we have been responsible is to the best of my/our knowledge and belief conform to the *CEA* (*Measures relating to Safety and Electric Supply*) *Regulations,* 2023 and in accordance with the relevant Indian Standards, National Electrical Code of India, 20xx, and National Building Construction Standards, 2026 and except for the departures, if any, detailed as follows:

Departures

Project/Site details

(Sign)
Electrical Contractor
Name and Address

(Sign)
Electrical Engineer on site

G-3 DECLARATION FOR INSPECTION AND TESTING

I/We being the person(s) responsible for the inspection and testing of the electrical installation (as indicated by my/our signatures below), particulars of which are described below, having exercised reasonable skill and care when carrying out the inspection and testing, hereby DECLARE that the construction work for which I/we have been responsible is to the best of my/our knowledge and belief conform to the *CEA* (*Measures Relating to Safety and Electric Supply*) *Regulations*, 2023 and in accordance with the relevant Indian Standards, National Electrical Code of India, 20xx and National Building Construction Standards, 2026 and except for the departures, if any, detailed as follows:

Departures

Project/Site details

(Sign)
Electrical Safety Verifier
L. No./Registration No.
Name and Address

(Sign)
Electrical Engineer on site

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in the NBCS.

In the following list, the number appearing in the first column within parenthesis indicates the number of the reference in this Section.

	IS No.	*Title*
(1)	SP 30 : 2023	National Electrical Code of India, 2023 (*second revision*)
(2)	IS 8270 (Part 1) : 2023/ IEC 61082-1 : 2014	Preparation of documents used in electrotechnology: Part 1 Rules (*first revision*)
	IS 1885 (Part 16) : 2023/ IEC 60050-845 : 2020	Electrotechnical — Vocabulary: Part 16 Lighting
	IS 1885	Electrotechnical vocabulary:
	(Part 17) : 2024/ IEC 60050-441 : 1984	Switchgear and control gear (*second revision*)
	(Part 32) : 2019/ IEC 60050-461 : 2008	Electrical cables (*second revision*)
	(Part 78) : 1993/ IEC Pub 50 (601) : 1985	Generation, transmission and distribution of electricity — General
	IS 19127 : 2025/ IEC 60617 : 2024	Graphical symbols for diagrams in the field of electrotechnology
(3)	IS 17505 (Part 1) : 2021	Specification for thermosetting insulated fire survival cables for fixed installation having low emission of smoke and corrosive gases when affected by fire for working voltages upto and including 1 100 V a.c. and 1500 V d.c.: Part 1 Requirements for armoured cables
(4)	IS 694 : 2010	Polyvinyl chloride insulated unsheathed and sheathed cables/cords with rigid and flexible conductor for rated voltages up to and including 1 100 V (*fourth revision*)
(5)	IS 17048 : 2018	Halogen free flame retardant (HFFR) cables for working voltages up to and including 1 100 volts — Specification
(6)	IS/IEC 61537 : 2006	Cable management — Cable tray systems and cable ladder systems
(7)	IS/IEC 62271-202 : 2022	High-voltage switchgear and controlgear: Part 202 a.c. prefabricated substations for rated voltages above 1 kV and up to and including 52 kV (*first revision*)
(8)	IS/IEC 60947-1 : 2020	Low-voltage switchgear and controlgear: Part 1 General rules (*second revision*)
	IS/IEC 60947-2 : 20xx	Low-voltage switchgear and controlgear: Part 2 Circuit-breakers (*under publication*)

	IS/IEC 60947-3 : 2020	Low-voltage switchgear and controlgear: Part 3 Switches, disconnectors, switch-disconnectors and fuse-combination units (*second revision*)
	IS/IEC 60947-4-1 : 2023	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 1 Electromechanical contactors and motors-starters (*third revision*)
	IS/IEC 60947-4-2 : 2020	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 2 Semiconductor motor controllers, starters and soft-starters (*second revision*)
	IS/IEC 60947-4-3 : 2020	Low-voltage switchgear and controlgear: Part 4 Contactors and motor-starters, Section 3 Semiconductor controllers and semiconductor contactors for non-motor loads (*third revision*)
	IS/IEC 60947-5-1 : 2016	Low-voltage switchgear and controlgear: Part 5 Control circuit devices and switching elements, Section 1 Electromechanical control circuit devices (*second revision*)
	IS/IEC 60947-5-2 : 2019	Low-voltage switchgear and control gear: Part 5 Control circuit devices and switching elements, Section 2 Proximity switches (*first revision*)
(9)	IS/IEC 60309-1 : 2021	Plugs, fixed or portable socket-outlet and appliance inlets for industrial purposes: Part 1 General requirements (*second revision*)
	IS/IEC 60309-2 : 2021	Plugs, fixed or portable socket-outlets and appliance inlets for industrial purposes: Part 2 Dimensional compatibility requirements for pin and contact-tube accessories (*second revision*)
(10)	IS 17050 : 2023/ IEC 62262 : 2021 +AMD1 2021	Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code) (*first revision*)
(11)	IS 3043 : 2018	Code of practice for earthing (*second revision*)
(12)	IS 17036 : 2018	Distribution system supply voltage quality
(13)	IS 18732 : 2023	Guide for implementation of electrical installation standards in building
(14)	IS 7752 (Part 1) : 1975	Guide for improvement of power factor in consumers' installation: Part 1 Low and medium supply voltages
(15)	IS 5216	Recommendations on safety procedures and practices in electrical work:
	(Part 1) : 1982	General (*first revision*)
	(Part 2) : 1982	Life saving techniques (*first revision*)
(16)	IS 10028 (Part 2) : 1981	Code of practice for selection, installation and maintenance of transformers: Part 2 Installation

(17)	IS 10118 (Part 3) : 1982	Code of practice for selection, installation and maintenance of switchgear and controlgear: Part 3 Installation
(18)	IS 732 : 2019	Code of practice for electrical wiring installations (*fourth revision*)
(19)	IS 16996 : 2018/ IEC 60364-8-1 : 2014	Low-voltage electrical installations — Energy efficiency
(20)	IS 10118 (Part 2) : 1982	Code of practice for the selection, installation and maintenance of switchgear and controlgear: Part 2 Selection
(21)	IS 1646 : 2015	Fire safety of buildings (general): Electrical installations — Code of practice (*third revision*)
(22)	IS/IEC 61439 – All parts/ Sections	Low-voltage switchgear and controlgear assemblies — All parts/Sections
(23)	IS 1255 : 1983	Code of practice for installation and maintenance of power cables (up to and including 33 kV rating) (*second revision*)
(24)	IS 8084 : 2025	Interconnecting bus-bars for a.c. voltage above 1 kV up to and including 36 kV — Specification (*first revision*)
(25)	IS 1180 (Part 1) : 2014	Outdoor type oil immersed distribution transformers upto and including 2 500 kVA, 33kV — Specification: Part 1 Mineral oil immersed (*fourth revision*)
	IS 2026	Power transformers:
	(Part 1) : 2011	General (*second revision*)
	(Part 2) : 2010	Temperature-rise (*first revision*)
	(Part 3) : 2018/ IEC 60076-3 : 2000	Insulation levels, dielectric tests and external clearances in air (*fourth revision*)
	IS 2026 (Part 4) : 1977	Specification for power transformers: Part 4 Terminal marking, tappings and connections (*first revision*)
	IS 2026	Power transformers:
	(Part 5) : 2011	Ability to withstand short circuit (*first revision*)
	(Part 7) : 2009/ IEC 60076-7 : 2005	Loading guide for oil-immersed power transformers
	(Part 8) : 2009/ IEC 60076-8 : 1997	Applications guide
	(Part 10) : 2025/ IEC 60076-10 : 2016	Determination of sound levels (*first revision*)
	(Part 11) : 2021 IEC 60076-11 : 2018	Dry-type transformers
(26)	IS/IEC 61439-2 : 2020	Low-voltage switchgear and controlgear assemblies: Part 2 Power switchgear and controlgear assemblies (*first revision*)

	IS/IEC 60947-1 : 2020	Low-voltage switchgear and controlgear: Part 1 General rules (*second revision*)
	IS/IEC 60947-2 : 20xx	Low-voltage switchgear and controlgear: Part 2 Circuit-breakers (*under publication*)
	IS/IEC 60947-3 : 2020	Low-voltage switchgear and controlgear: Part 3 Switches, disconnectors, switch-disconnectors and fuse combination units (*second revision*)
	IS/IEC 60947-4-1 : 2018	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 1 Electromechanical contactors and motors-starters (*second revision*)
	IS/IEC 60947-4-2 : 2020	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 2 Semiconductor motor controllers starters and soft-starters (*second revision*)
	IS/IEC 60947-4-3 : 2020	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 3 Semiconductor controllers and Semiconductor contactors for non-motor loads (*third revision*)
	IS/IEC 60947-5-1 : 2024	Low-voltage switchgear and controlgear: Part 5 Control circuit devices and switching elements, Section 1 Electromechanical control circuit devices (*third revision*)
	IS/IEC 60947-5-2 : 2019	Low-voltage switchgear and control gear: Part 5 Control circuit devices and switching elements, Section 2 Proximity switches (*first revision*)
	IS/IEC 61439 (Part 0) : 2014	Low-voltage switchgear and controlgear assemblies: Part 0 Guidance to specifying assemblies
	IS/IEC 61439-1 : 2020	Low-voltage switchgear and controlgear assemblies: Part 1 General rules (*first revision*)
(27)	IS 11353 : 2023/ IEC 60445 : 2021	Basic and safety principles for man-machine interface, marking and identification — Identification of equipment terminals, conductor terminations and conductors (*first revision*)
(28)	IS/IEC 61439-6 : 2012	Low-Voltage switchgear and controlgear assemblies Part 6 Busbar trunking systems (Busways)
(29)	IS 13010 : 2002	ac watthour meters, class 0.5, 1 and 2 — Specification (*first revision*)
	IS 13779 : 2020	a.c. static watthour meters, class 1 and 2 — Specification (*second revision*)
	IS 14697 : 2021	a.c. static transformer operated watthour meters (class 0.2 s and 0.5 s) and var-hour meters (class 0.2 s, 0.5 s and 1 s) — Specification (*first revision*)
	IS 15884 : 2024	Alternating current direct connected static prepayment meters for active energy (class 1 and class 2) — Specification (*first revision*)
	IS 16444 : 2015	a.c. static direct connected watthour smart meter class 1 and 2 — Specification

(30)	IS 15707 : 2006	Testing, evaluation, installation and maintenance of ac electricity meters — Code of practice
(31)	IS/IEC 60079-1 : 2014	Explosive atmospheres: Part 1 Equipment protection by flameproof enclosures “d” (*first revision*)
(32)	IS 5578 : 1984	Guide for marking of insulated conductors (*first revision*)
	IS 11353 : 2023/ IEC 60445 : 2021	Basic and safety principles for man-machine interface, marking and identification — Identification of equipment terminals, conductor terminations and conductors (*first revision*)
(33)	IS/IEC 60309-1 : 2021	Plugs, fixed or portable socket-outlets and appliance inlets for industrial purposes: Part 1 General requirements (*second revision*)
	IS/IEC 60309-2 : 2021	Plugs, fixed or portable socket-outlets and appliance inlets for industrial purposes: Part 2 Dimensional compatibility requirements for pin and contact-tube accessories (*second revision*)
(34)	IS/IEC 60079 (Part 1) : 2014	Explosive Atmospheres: Part 1 Equipment Protection by Flameproof Enclosures "d"
	IS 10322 (Part 5/Sec 5) : 2026/IEC 60598-2-5 : 2015	Luminaires: Part 5 Particular requirements, Section 5 Flood lights (*second revision*)
(35)	IS 12640 (Part 1) : 2024/ IEC 61008-1 : 2013	Residual current operated circuit-breakers without integral overcurrent protection for household and similar uses (RCCBs): Part 1 General rules (*third revision*)
	IS 12640 (Part 2) : 2016/ IEC 61009-1 : 2012	Residual current operated circuit-breakers with integral overcurrent protection for household and similar uses (RCBOs): Part 2 General rules (*second revision*)
	IS 14614 : 2023/ IEC 61543 : 2022	Residual current-operated protective devices (RCDs) for household and similar use — Electromagnetic compatibility (*first revision*)
	IS/IEC 60898-1 : 2015	Electrical accessories – Circuit-breakers for overcurrent protection for household and similar installations: Part 1 Circuit-breakers for a.c. operation (*first revision*)
	IS/IEC 60947-1 : 2020	Low-voltage switchgear and controlgear: Part 1 General rules (*second revision*)
	IS/IEC 60947-2 : 20xx	Low-voltage switchgear and controlgear: Part 2 Circuit-breakers (*under publication*)
	IS/IEC 60947-3 : 2020	Low-voltage switchgear and controlgear: Part 3 Switches, disconnectors, switch-disconnectors and fuse combination units (*second revision*)
	IS/IEC 60947-4-1 : 2023	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 1 Electromechanical contactors and motors-starters (*third revision*)
	IS/IEC 60947-4-2 : 2020	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 2 Semiconductor motor controllers starters and soft starters (*second revision*)

	IS/IEC 60947-4-3 : 2020	Low-voltage switchgear and controlgear: Part 4 Contractors and motor-starters, Section 3 Semiconductor controllers and semiconductor contactors for non-motor loads (*third revision*)
	IS/IEC 60947-5-1 : 2024	Low-voltage switchgear and controlgear: Part 5 Control circuit devices and switching elements, Section 1 Electromechanical control circuit devices (*third revision*)
	IS/IEC 60947-5-2 : 2019	Low-voltage switchgear and control gear: Part 5 Control circuit devices and switching elements, Section 2 Proximity switches (*first revision*)
(36)	IS/lEC 60529 : 2001	Degrees of protection provided by enclosures (IP code)
	IS 17050 : 2023 IEC 62262 : 2021 +AMD1 2021	Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK Code) (*first revision*)
(37)	IS 3961	Recommended current ratings for cables:
	(Part 2) : 2017	PVC insulated and PVC sheathed heavy duty cables (*first revision*)
	(Part 3) : 1968	Rubber insulated cables
	(Part 5) : 1968	PVC insulated light duty cables
	(Part 6) : 2016	Crosslinked polyethylene insulated PVC sheathed cables
(38)	SP 72 : 2025	National Lighting Code of India 2025 (*first revision*)
(39)	IS 4648 : 1968	Guide for electrical layout in residential buildings
(40)	IS 900 : 2019	Code of practice for storage, installation and maintenance of induction motors (*third revision*)
(41)	IS 9537	Specification for conduits for electrical installations:
	(Part 1) : 1980	General requirements
	(Part 2) : 1981	Rigid steel conduits
	(Part 3) : 1983	Rigid plain conduits of insulating materials
(42)	IS 16205	Conduit systems for cable management:
	(Part 1) : 2017	General requirements
	(Part 24) : 2017	Particular requirements — conduit systems buried underground
(43)	IS 14297 : 2024/ ISO 9226 : 2012	Corrosion of metals and alloys — Corrosivity of atmospheres — Determination of corrosion rate of standard specimens for the evaluation of corrosivity (*first revision*)
(44)	IS/IEC 61537 : 2006	Cable management — Cable tray systems and cable ladder systems
(45)	IS 9537	Specification for conduits for electrical installations:

(Part 1) : 1980	General requirements
(Part 3) : 1983	Rigid plain conduits of insulating materials
(46) IS 9537	Specification for conduits for electrical installations:
(Part 1) : 1980	General requirements
(Part 2) : 1981	Rigid steel conduits
(47) IS 3837 : 1976	Specification for accessories for rigid steel conduits for electrical wiring (*first revision*)
IS 14768	Conduit fittings for electrical installations — Specification:
(Part 1) : 2000	General requirements
(Part 2) : 2003	Metal conduit fittings
IS 14772 : 2020	Boxes and enclosures for electrical accessories for household and similar fixed electrical installations — General requirements (*first revision*)
(48) IS 3419 : 1988	Specification for fittings for rigid non-metallic conduits (*second revision*)
IS 14772 : 2020	Boxes and enclosures for electrical accessories for household and similar fixed electrical installations — General requirements (*first revision*)
(49) IS 18925 (Part 7) : 2025/ IEC 62561-7 : 2024	Lightning protection system components (LPSC): Part 7 Requirements for earthing enhancing compounds
(50) IS 18925 (Part 5) : 2025/ IEC 62561-5 : 2023	Lightning protection system components (LPSC): Part 5 Requirements for earth electrode inspection housings and earth electrode seals
(51) IS 18925 (Part 1) : 2025/ IEC 62561-1 : 2023	Lightning protection system components (LPSC): Part 1 Requirements for connection components
(52) IS/IEC 62305-1 : 2010	Protection against lightning: Part 1 General principles
IS/IEC 62305-2 : 2010	Protection against lightning: Part 2 Risk management
IS/IEC 62305-3 : 2010	Protection against lightning: Part 3 Physical damage to structures and life hazard
IS/IEC 62305-4 : 2010	Protection against lightning: Part 4 Electrical and electronic systems within structures
(53) IS 16242	Uninterruptible power systems (UPS):
(Part 1) : 2025/ IEC 62040-1 : 2017 + AMD1 : 2021 + AMD2 : 2022 CSV	Safety requirements (*first revision*)

	(Part 2) : 2020/ IEC 62040-2 : 2016	Electromagnetic compatibility EMC requirements (*first revision*)
	(Part 3) : 2025/ IEC 62040-3 : 2021	Method of specifying the performance and test requirements (*first revision*)
	(Part 5/Sec 3) : 2025/ IEC 62040-5-3 : 2016	d.c. output UPS, Section 3 Performance and test requirements
(54)	IS 17017 — All parts/sections	Specification for electric vehicle conductive charging system — All parts/sections
(55)	IS 17896 (Part 1) : 2022/ IEC TS 62840-1 : 2016	Electric vehicle battery swap system: Part 1 General and guidance
	IS 17896 (Part 2) : 2022/ IEC 62840-2 : 2016	Electric vehicle battery swap system: Part 2 Safety requirements
(56)	IS 17017 (Part 23) : 2021	Electric vehicle conductive charging systems: Part 23 d.c. electric vehicle supply equipment
(57)	IS 17017 (Part 24) : 2021	Electric vehicle conductive charging system: Part 24 Digital communication between a DC electric vehicle supply equipment and an electric vehicle for control of DC charging
(58)	IS/ISO 15118 — All parts/sections	Road vehicles — Vehicle to grid communication interface — All parts/sections
(59)	IS 17017 (Part 2/Sec 3) : 2020	Electric vehicle conductive charging system: Part 2 Plugs, socket — outlets, vehicle connectors and vehicle inlets, Section 3 Dimensional compatibility and interchangeability requirements for d.c. and a.c./d.c. pin and contact-tube vehicle couplers
(60)	IS 14286 : 2023 IEC 61215 : 2021 — All parts/sections	Terrestrial photovoltaic (PV) modules — Design qualification and type approval — All parts/sections
	IS 16228 : 2024/ IEC 62108 : 2022	Concentrator photovoltaic (CPV) modules and assemblies — Design qualification and type approval (*second revision*)
	IS 17210 (Part 1) : 2019/ IEC TS 62804-1 : 2015	Photovoltaic (PV) modules — Test methods for the detection of potential-induced degradation: Part 1 Crystalline silicon
	IS 16664 : 2018/ IEC 62716 : 2013	Photovoltaic (PV) modules — Ammonia corrosion testing
	IS/IEC 61730 — All parts	Photovoltaic (PV) module safety qualification — All parts
	IS/IEC 61701 : 2011	Salt mist corrosion testing of photovoltaic (PV) modules
(61)	IS 14286 : 2023 IEC 61215 : 2021 — All parts/sections	Terrestrial photovoltaic PV modules design qualification and type approval — All parts/sections
	IS 16170 (Part 1) : 2014/ IEC 61853-1 : 2011	Photovoltaic (PV) module performance testing and energy rating: Part 1 Irradiance and temperature performance measurements and power rating

	IS/IEC 61730 — All parts/sections	Photovoltaic (PV) module safety qualification — All parts/sections
	IS 17210 (Part 1) : 2019/ IEC TS 62804-1 : 2015	Photovoltaic (PV) modules — Test methods for the detection of potential-induced degradation: Part 1 Crystalline silicon
(62)	IS 2062 : 2011	Hot rolled medium and high tensile structural steel — Specification (*seventh revision*)
	IS 4759 : 1996	Hot-dip zinc coatings on structural steel and other allied products — Specification (*third revision*)
(63)	IS 2062 : 2011	Hot rolled medium and high tensile structural steel — Specification (*seventh revision*)
	IS 1079 : 2017	Hot rolled carbon steel sheet, plate and strip — Specification (*seventh revision*)
	IS 1161 : 2014	Steel tubes for structural purposes — Specification (*fifth revision*)
	IS 1239 (Part 1) : 2004	Steel tubes, tubulars and other wrought steel fittings — Specification: Part 1 Steel tubes (*sixth revision*)
(64)	IS/IEC 61683 : 1999	Photovoltaic systems — Power conditioners — Procedure for measuring efficiency
	IS/IEC 60068-2-1 : 2007	Environmental testing: Part 2 Tests, Section 1 Test A: Cold
	IS 16169 : 2019/ IEC 62116 : 2014	Utility-interconnected photovoltaic inverters — Test procedure of islanding prevention measures (*first revision*)
	IS/IEC 60068-2-2 : 2007	Environmental testing: Part 2 Tests, Section 2 Test B dry heat
	IS/IEC 60068-2-14 : 2023	Environmental testing: Part 2 Tests, Section 14 Test N: Change of temperature (*first revision*)
	IS/IEC 60068-2-30 : 2005	Environmental testing: Part 2 Tests, Section 30 Test Db: Damp heat cyclic (12 h + 12 h cycle)
	IS 14700 — All parts	Electromagnetic compatibility (EMC) — All parts
	IS 16221 (Part 2) : 2015/ IEC 62109-2 : 2011	Safety of power converters for use in photovoltaic power systems: Part 2 Particular requirements for inverters
(65)	IS 16221 (Part 2) : 2015/ IEC 62109-2 : 2011	Safety of power converters for use in photovoltaic power systems: Part 2 particular requirements for inverters
(66)	IS 17293 : 2020	Electric cables for photovoltaic systems for rated voltage 1 500 V d.c.
(67)	IS 14286 (Part 1/Sec 1) : 2023/ IEC 61215-1-1 : 2021	Terrestrial photovoltaic (PV) modules — Design qualification and type approval: Part 1-1 Special requirements for testing of crystalline silicon photovoltaic (PV) modules (*third revision*)

	IS/IEC 61730-1 : 2023	Photovoltaic (PV) module — Safety qualification: Part 1 Requirements for construction (*second revision*)
(68)	IS 12933	Solar flat plate collector — Specification:
	(Part 1) : 2025	Requirements (*third revision*)
	(Part 2) : 2025	Components (*third revision*)
	IS 16544 : 2016	All glass evacuated tubes solar water heating system
	IS 16542 : 2016	Direct insertion type storage water tank for all glass evacuated tubes solar collector — Specification
(69)	IS 12933	Solar flat plate collector — Specification:
	(Part 1) : 2025	Requirements (*third revision*)
	(Part 2) : 2025	Components (*third revision*)
(70)	IS 16544 : 2016	All glass evacuated tubes solar water heating system
(71)	IS 16542 : 2016	Direct insertion type storage water tank for all glass evacuated tubes solar collector — Specification
(72)	IS 302 (Part 2/Sec 76) : 1999	Safety of household and similar electrical appliances: Part 2 Particular requirements, Section 76 Electric fence energizers
(73)	IS/IEC 62305-3 : 2010	Protection against lightning: Part 3 Physical damage to structures and life hazard
(74)	IS/IEC 62305-4 : 2010	Protection against lightning: Part 4 Electrical and electronic systems within structures
(75)	IS/IEC 62305-1 : 2010	Protection against lightning: Part 1 General principles
(76)	IS/IEC 62305-2 : 2010	Protection against lightning: Part 2 Risk management
(77)	IS 16172 : 2023	Reinforcement couplers for mechanical splices of steel bars in concrete — Specification (*first revision*)
(78)	IS 18925 — All parts/sections	Lightning protection system components (LPSC) — All parts/sections
(79)	IS 12360 : 2026	Voltage bands for electrical installations including preferred voltage and frequency (*first revision*)
(80)	IS 15382 (Part 1) : 2022/ IEC 60664-1 : 2020	Insulation coordination for equipment within low-voltage systems: Part 1 Principles, requirements and tests (*second revision*)
(81)	IS 16463 (Part 11) : 2016/ IEC 61643-11 : 2011	Low-voltage surge protective devices: Part 11 Surge protective devices connected to low-voltage power systems — Requirements and test methods
(82)	IS/IEC 60479-1 : 2018	Effects of current on human beings and livestock: Part 1 General aspects

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 3 AIR CONDITIONING, HEATING AND MECHANICAL VENTILATION

BUREAU OF INDIAN STANDARDS

CONTENTS

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Section 3) deals with the planning, selection, design considerations, and installation of air conditioning, heating and mechanical ventilation system for different types of building application and in towns and cities under all the climatic zones of India. It covers all aspects including the goals and objectives, basis of design, input parameters for design, guidelines for design of system, performance parameters, available system options, pre-planning considerations, range of equipment and system components, building management system, installation of the system, testing, commissioning and handing over and also operation and maintenance of the air conditioning, heating and mechanical ventilation system.

Indian construction industry is poised to add an-order-of-magnitude built foot-print in the coming years. With the advent of information technology and computers becoming part of our life-style, the requirement of air conditioning of built environment is being increasingly felt and met with. The challenge faced for this unprecedented development is the lack of material resources and natural resources like energy, water and clean air. Therefore, the selection of air conditioning and mechanical ventilation system, optimally suited for the specific type of building application and its climatic zone, becomes critical. It is necessary for the owner, designer, builder and developer to understand the provisions of this Section, and consult an air conditioning engineer at the planning stage.

Sustainable buildings movement in the country has gained tremendous momentum. The relevant 'Handbook on Sustainability in Buildings', deals with all aspects of sustainable buildings, from selection of site, building design, energy and water efficient systems, use of recycled material resources, construction, third party commissioning, operation and maintenance of sustainable buildings. By following the provisions of this Section in conjunction with those given in relevant 'Handbook on Sustainability in Buildings', India can successfully adopt sustainable buildings as the way of life, as has been the practice for centuries, prior to the onslaught of industrial revolution in India.

Computerized weather data is now available and has been included for around 145 locations across the country, covering all the five climatic zones. This data is based on the latest values obtained from India Metrological Department, Government of India.

This Section was first formulated in 1970 and was subsequently revised in 1983, 2005 and 2016 as a part of National Building Code of India. The major modifications made in 2005 version of this code included addition of several new terms and their definitions; incorporation of a new clause on design criteria; inclusion of 'indoor air quality' as one of the factors that needed to be controlled in the conditioned space; incorporation of recommendation for independent air handling unit rooms for each floor of large and multi storeyed buildings; inclusion of inside design conditions for various applications replacing earlier Table 2 and Table 3; revision of provisions on minimum outside fresh air in the light of the then accepted international norms thus covering a wider variety and a larger number of applications; addition of new details on temperature, humidity, vibration and noise; updation of provisions on application considerations, covering a wide variety of commercial applications, such as, offices, hotels, restaurants and computer rooms; inclusion of a new clause on statutory regulation/safety considerations; description of various system options available, under the clause on design considerations; updation of provisions on the characteristics and application of options available in piping, water distribution systems and piping layout; revision of provisions on air filters; revision of the clause on energy conservation and energy management to include concepts like energy targets, demand targets and consumption targets, the factors to be considered in system design that influence energy aspects, the need for analysis of operation of systems during various seasons of the year, and the need to incorporate energy recovery strategies. Apart from above, 'Automatic Controls' included in the 1983 version was replaced by building management system, which addressed not only the control function, but also had a telling impact on operation and maintenance as well, most importantly on the opportunities afforded to implement various energy conservation strategies; provisions on packaged air conditioners and room air conditioners were elaborated; provisions on heating were revised; provisions on symbols, units, colour code and identification of services, pipe work services, duct work services, valve labels and charts, and also on inspection, commissioning and testing were updated; and list of various parameters to be checked for performance of air handling unit, hydronic system balancing, and finally, the hand-over procedure were included.

The 2016 revision of this part introduced several important updates to align HVAC systems with modern needs and sustainable practices. It added new definitions to reflect emerging concepts and technologies and updated refrigerant guidelines to include eco-friendly options like zero-ozone-depleting and low-global-warming-potential refrigerants. Planning considerations were expanded to incorporate advanced systems such as VRF, district cooling, and hybrid central plants, supported by tools like energy modelling and solar shade analysis to optimize energy efficiency. The outdoor and indoor design conditions were updated using adaptive comfort standards and comprehensive weather data from across India. Key advancements in HVAC equipment included provisions for chilled beams, radiant floors, and geothermal systems were made. Specialized applications were addressed in greater detail, covering data centers, metro stations, and cold storage facilities critical for food security. Ventilation received a significant focus, emphasizing sustainability, demand control, and advanced fan designs. Installation practices were refined for ease of maintenance and better integration with building automation systems. The revision also standardized symbols and colour coding, introduced web-based monitoring for building management systems, and enhanced testing and commissioning protocols to ensure state-of-the-art performance.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

This revision of this Part D/ Section 3 of this NBCS focuses on improving air conditioning, heating, and ventilation systems for better efficiency and sustainability. New features include advanced refrigerants, geothermal systems, and tools for energy-saving designs. It also addresses specialized needs like data centres and metro stations. These changes help ensure better performance, lower energy use, and reduced environmental impact.

The significant changes incorporated in this revision include the following:

a) Additional details on systems like variable refrigerant flow (VRF) and inverter-based cooling for better efficiency, in **4.1.4**.

b) Added new eco-friendly refrigerants with no harm to the ozone layer and very low impact on global warming, in **5.3**.

c) Addition of modern low-energy heating and cooling solutions, including evaporative cooling, radiant cooling, and solar-assisted systems. Additionally, it provides improved guidelines for system selection, installation, and applications across residential, commercial, and industrial sectors, in **6**.

d) Inclusion of new methods for cooling and heating using the earth's natural temperature are included, in **6.4**.

e) Addition of efficient heating solutions like solar-powered systems and ground heat pumps for cold regions, in **6.5**.

f) Addition of guidelines for selecting air-conditioning and ventilation systems across various building typologies, including residential, commercial, and transport hubs, in **12**.

g) New guidelines for cold storage facilities to improve food preservation and reduce waste, in **13.2**.

h) Inclusion of enhanced guidelines for using automation systems to monitor and control building performance, in **15.2**.

j) Introduction of uniform standards for hydronic and refrigerant piping in HVAC systems to enhance performance covering material selection, sizing, installation, testing, and maintenance for efficient fluid transport, in **16**.

k) Inclusion of new procedures for testing and setting up HVAC systems to ensure they work efficiently, in **17**.

m) Introduction of comprehensive smoke mitigation guidelines for substructures, superstructures, and exit pathways, in **19**.

n) Terminology clause has been updated with new terms and definitions.

The information contained in this section is based largely on the following Indian Standards:

IS 1391	Room air conditioners specification
(Part 1) : 2023	Unitary air conditioners (*fourth revision*)
(Part 2) : 2023	Split air conditioners (*fourth revision*)
IS 2379 : 2024	Colour code for pipeline identification — Code of practice (*second revision*)
IS 3315 : 2024	Direct evaporative air cooler — Specification (*fourth revision*)
IS 8148 : 2018	Ducted and package air-conditioners – Specification (*second revision*)
IS 7896 : 2023	Air conditioning outdoor design conditions data for Indian cities (*second revision*)

Assistance has also been derived from the following publications in the preparation of this Section:

Handbooks of Indian Society of Heating, Refrigerating and Air Conditioning Engineers (ISHRAE)

India Model for Adaptive Comfort, CEPT University, Ahmedabad

ISO 16484-1: 2024 Building automation and control systems (BACS) — Part 1: Project specification and implementation

The provisions on natural ventilation are given in Part D 'Building Services, Section 1 Lighting and Ventilation' of this NBCS.

The provisions of this Section are without prejudice to the various Acts, Rules and Regulations including *The Factories Act*, 1948 and the rules and regulations framed thereunder.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties using this Section are encouraged to investigate the possibility of applying the most recent edition of the standards.

As the regulation of land development and buildings is a state subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **'good practice'** in relation to design, constructional procedures or other related information, and where reference is made to **"accepted standard"** in relation to material specification, testing, or other related information, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [D-3(1)] refers to the Indian Standard given at serial number (1) of the list of standards given at the end of this Section, that is, IS 3615 : 2020 'Glossary of terms used in refrigeration and air conditioning (*second revision*)'.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 3 AIR CONDITIONING, HEATING AND MECHANICAL VENTILATION

1 SCOPE

This NBCS (Part D/Section 3) covers the planning, design considerations, installation, testing, commissioning and handing over also operation and maintenance of air conditioning, heating and mechanical ventilation systems for buildings. It also covers refrigeration for cold storages.

The provisions of this Section aim to ensure an air conditioning, heating and mechanical ventilation system which shall provide comfort by managing air temperature, humidity, indoor air quality and distribution of conditioned air for the specific use and occupancy of built space while giving due consideration to minimizing energy consumption and other resources.

The provisions on natural ventilation are covered in Part D 'Building Services, Section 1 Lighting and Ventilation' of this NBCS.

The provisions in respect of air conditioning, heating and mechanical ventilation system in sustainable buildings are covered in relevant 'Handbook on Sustainability in Buildings'.

2 TERMINOLOGY

For the purpose of this Section the following definitions shall apply in addition to those given in the accepted standard [D-3(1)] and the following shall apply.

2.1 Air Conditioning — The process of treating air so as to control simultaneously its temperature, humidity, purity, distribution and movement and pressure to meet the requirements of the conditioned space.

2.2 Air System Balancing — Adjusting air flow rates through air distribution system devices, such as fans and diffusers, by manually adjusting the position of dampers, splitter vanes, extractors, etc, or by using automatic control devices, such as constant-air-volume or variable-air-volume (VAV) boxes.

NOTE — Air system should be balanced in order to minimize throttling losses. For fans, its speed should be adjusted to meet design flow conditions. By creating correct air flow at fans and outlet, system performance shall be increased.

2.3 Atmospheric Pressure — The force per unit area exerted against a surface by the weight of the air above that surface. It is the pressure indicated by a barometer. Standard atmospheric pressure or standard atmosphere or barometric pressure is the pressure of 760 mm of mercury column having a density of 13 600 kg/m^3 under standard gravity of 9.8 m/sec^2 (10.332 mm of water column/101.325 kPa).

NOTE — Generally atmospheric pressure is used as a datum for indicating the system pressures in air conditioning and accordingly, pressures are mentioned above the atmospheric pressure or below the atmospheric pressure considering the atmospheric pressure to be zero. A 'U' tube manometer will indicate zero pressure when pressure measured is equal to atmospheric pressure.

2.4 Building Management System (BMS) — An energy management system relating to the overall operation of the building in which it is installed. It often has additional capabilities, such as equipment monitoring, protection of equipment against power failure, and building security. It may also be a direct digital control (DDC) system where the mode of control uses digital outputs to control processes or elements directly.

NOTE — Mechanical and electrical equipment installed in the building, such as, air conditioning, ventilation, lighting, lifts, power, pumping stations, firefighting systems, security systems are controlled and managed through BMS.

2.5 Building Energy Simulation — Use of computer models for design and optimization of building's energy performance, to compare the cost-effectiveness of energy conservation measures in the design stage as well as assessing various performance optimization measures during the operational stage.

2.6 Building Integrated Renewable Energy — Integration of renewable energy application in parts of the building envelope such as the roof, skylights, or facades.

2.7 Buildings Related Illnesses (BRI) — The diagnosable illness attributed directly to the specific air-borne building contaminants, like legionnaire's disease, occupational asthma, etc.

NOTE — Indicators of BRI include complaints of cough, chest tightness, chills, fever and muscle aches. Occupants may need prolonged recovery times after leaving the building.

2.8 Coefficient of Performance, Compressor, Heat Pump — Ratio of the compressor heating effect (heat pump) to the rate of energy input to the shaft of the compressor, in consistent units, in a complete heat pump, under designated operating conditions.

2.9 Coefficient of Performance, Compressor, Refrigerating — Ratio of the compressor refrigerating effect to the rate of energy input to the shaft of the compressor, in consistent units, in a complete refrigerating plant, under designated operating conditions.

2.10 Coefficient of Performance (Heat Pump) — Ratio of the rate of heat delivered to the rate of energy input, in consistent units, for a complete operating heat pump plant or some specific portion of that plant, under designated operating conditions.

2.11 Coefficient of Performance (Refrigerating) — Ratio of the rate of heat removal to the rate of energy input, in consistent units, for a complete refrigerating plant or some specific portion of that plant, under designated operating conditions.

2.12 Cooling Load — Amount of cooling per unit time required by the conditioned space or product; or heat that a cooling system shall remove from a controlled system over time.

2.13 Cooling Tower — An enclosed device, often tower like, for evaporatively cooling water by contact with air.

2.14 Dedicated Outdoor Air System (DOAS) — A unit that is used to separately condition outdoor air brought into the building for ventilation or to replace air that is being exhausted.

2.15 Demand Based Ventilation — Intelligent airflow management that adjusts outside ventilation air based on the number of occupants and the ventilation demands that those occupants create.

2.16 Design Pressure Difference — The desired pressure difference between a given space and an adjacent space measured at the boundary of the given space under a specified set of conditions, such as, that required in various spaces of hospital, clean rooms, protected space in case of smoke control operation, etc.

2.17 Dew Point Temperature — The temperature at which condensation of moisture begins when the air is cooled at same pressure.

2.18 Dry-Bulb Temperature — The temperature of the air, read on a thermometer, taken in such a way as to avoid errors due to radiation.

2.19 Duct System — A continuous passageway for the transmission of air which, in addition to the ducts, may include duct fittings, dampers, plenums, grilles and diffusers.

2.20 Economizer, Air — It consist of duct, damper and control system that allow outside air to cool the building when outside air is cooler than inside.

2.21 Economizer, Water — In this system the supply air of a cooling system is cooled indirectly with water that is itself cooled by heat transfer or mass transfer to the environment without the use of mechanical cooling.

2.22 Effective Temperature — Combined effects of air temperature, humidity, air movement, mean radiant temperature, clothing and activity on the sensation of warmth or cold felt by the human body. Numerically equivalent to the temperature of still air producing similar thermal sensation as produced by combination of above six parameters of thermal comfort.

2.23 Energy Efficiency Ratio (EER) — Ratio of net cooling capacity in Btu/h to total rate of electric input in watts under designated operating conditions.

2.24 Energy Recovery Unit — A heat exchanger assembly for transferring energy between two isolated fluid sources. The recovery system may be of air-to-air design or a closed loop hydronic system design. The system will include all necessary equipment such as fans and pumps, associated ducts or piping and all controls (operating and safety), and other custom-designed features

2.25 Evaporative Cooling — The process of evaporating part of a liquid by supplying the necessary latent heat from the sensible heat of the main bulk of the liquid which is thus cooled.

2.26 Fire Damper — The damper, normally held open, installed in an air distribution system or in a wall or floor assembly and designed to close automatically in the event of a fire and maintain fire compartmentation.

2.27 Geothermal Heat Pump — A heating and/or cooling system utilizes the earth's crust, as heat source (in the winter) or a heat sink (in the summer) by using fluid to be pumped into the earth and circulated in order to exchange heat for the purpose of heating or cooling applications

2.28 Global Warming Potential (GWP) — The relative measure of how much a given mass of a refrigerant contributes to global warming over a given time period compared to the same mass of carbon dioxide over the same period.

NOTE — The GWP value of carbon dioxide is taken to be 1.0. The GWP value of a refrigerant is calculated over a time horizon. The time horizon shall greatly affect the numerical value of GWP. Usually the GWP values are reported over a 100 years' time horizon.

2.29 Heating Load — Heating rate required to replace heat loss from the space being controlled.

2.30 Heat Pump — A thermodynamic heating/refrigerating system to transfer heat. The condenser and evaporator may change roles to transfer heat in either direction. By receiving the flow of air or other fluid, a heat pump is used to cool or heat. Heat pumps may be the air source with heat transfer between the indoor air stream to outdoor air or water source with heat transfer between the indoor air stream and a hydronic source (ground loop, evaporative cooler, cooling tower, or domestic water).

2.31 Heat Recovery — Use of heat that would otherwise be wasted from a system or process, for example, heat-recovery chiller which uses hot waste gases as a heat source.

2.32 Hybrid Building — A building which contains both active and passive systems of heating or cooling. It requires small amount of non-renewable energy to maintain required amount of coefficient of performance (COP).

2.33 Hydronic Systems — The water systems that transfer heat to or from a conditioned space or process with hot or chilled water. The water flows through piping that connects a chiller or the water heater to suitable terminal heat transfer units located at the space or process.

2.34 Hydronic System Balancing — Adjusting water flow rates through hydronic distribution system devices, such as pumps and coils, by manually adjusting the position valves or by using automatic control devices, such as automatic flow control valves.

2.35 Indirect-Direct Cooling — The cooling which involves two stages:

a) The first stage, in which the air is made to pass through heat exchanger for sensible cooling (no direct contact of air and water), whereby the leaving air dry-bulb temperature (DBT) as well as the wet-bulb temperature (WBT) are reduced; and

b) The second stage, in which the air after the first stage is made to pass through the evaporative air-cooling (adiabatic cooling) application where water and air are in direct contact and there is simultaneous removal of sensible heat and the addition of moisture to the air giving lower DBT.

The resultant of this two-stage cooling is that the leaving air DBT is lower than ambient WBT.

NOTE — The first stage cooling is through indirect cooling. In this method, air-dry bulb and wet-bulb temperature are reduced without direct contact of air with water and through heat exchange only.

2.36 Indoor Air Quality (IAQ) — Air quality that refers to the nature of unconditioned or conditioned air that circulates throughout the space/area where one works or lives, that is, the air one breathes when indoors.

2.37 Infiltration/Exfiltration — The phenomenon of air leaking into (infiltration) or leaking out (exfiltration) out of an air-conditioned space.

2.38 Latent Heat — Change of enthalpy during a change of state, usually expressed, in kcal/kg. With pure substances, latent heat is absorbed or rejected at constant temperature at any pressure.

2.39 Latent Heat Load — Cooling load required to remove latent heat.

2.40 Mean Radiant Temperature — The uniform temperature of an imaginary enclosure in which the radiant heat transfer from the human body is equal to the radiant heat transfer in the actual non-uniform enclosure.

2.41 Mixed Mode Building — A hybrid approach to space conditioning that uses a combination of natural ventilation and mechanical systems. These buildings utilize mechanical cooling only when and where it is necessary to supplement the natural ventilation.

2.42 Naturally Conditioned Building — A building in which the ventilation system rely on opening and closing of window of the space to maintain the thermal comfort of the space rather than mechanical systems.

2.43 Operative Temperature — A uniform temperature of a radiantly black enclosure in which an occupant would exchange the same amount of heat by radiation plus convection as in the actual non-uniform environment. It is the combined effects of the mean radiant temperature and air temperature calculated as average of the two. It is also known as dry resultant temperature or resultant temperature.

2.44 Ozone Depletion Potential (ODP)—A relative capability of a refrigerant or a gas to degrade ozone in the atmosphere as compared to trichlorofluoromethane [R-11 or Clorofluorocarbon-11(CFC-11)]. The ODP of CFC-11 is taken to be 1.0.

2.45 Passive Cooling — A building design approach that focuses on heat gain control and heat dissipation in a building in order to improve the indoor thermal comfort with low or zero energy consumption by using natural ventilation, air cooling and shades.

2.46 Passive Heating — Passive heating uses the energy of natural source such as the sun, to keep the occupants of the building comfortable by design approach of building without the use of mechanical or electrical heating systems.

2.47 Plenum — An air compartment connected to one or more distributing ducts.

2.48 Positive Ventilation — The supply of outside air by means of a mechanical device, such as a fan.

2.49 Psychrometric Chart — A chart graphically representing the thermodynamic properties of moist air.

2.50 Recirculated Air — The return air that has been passed through the conditioning apparatus before being re-supplied to the space.

2.51 Refrigerant — The fluid used for heat transfer in a refrigerating system, which absorbs heat at a low temperature and a low pressure of the fluid and rejects heat at a higher temperature and a higher pressure of the fluid, usually involving changes of state of the fluid.

2.52 Relative Humidity — Ratio of the partial pressure of actual water vapour in the air as compared to the partial pressure of maximum amount of water that may be contained at its dry-bulb temperature.

NOTE — When the air is saturated, dry-bulb, wet-bulb and dew point temperatures are all equal, and the relative humidity is 100 percent.

2.53 Return Air — Air returned from conditioned or refrigerated space.

2.54 Sensible Heat — Heat which is associated with a change in temperature; in contrast to a heat interchange in which a change of state (latent heat) occurs.

2.55 Sensible Cooling — The process of removing sensible heat (lowering the dry-bulb temperature) from the air passing through it under specified conditions of operation.

2.56 Shade Factor — The ratio of instantaneous heat gain through the fenestration with shading device to that through the fenestration.

2.57 Sick Building Syndrome (SBS) — A term used to describe the presence of acute non-specific symptoms in the majority of people, caused by working in buildings with an adverse indoor environment.

NOTE — SBS could be a cluster of complex irritative symptoms like irritation of the eyes, blocked nose and throat, headaches, dizziness, lethargy, fatigue irritation, wheezing, sinusitis, congestion, skin rash, sensory discomfort from odours, nausea, etc. These symptoms are usually short-lived and experienced immediately after exposure; and may disappear when one leaves the building.

2.58 Smoke Barrier — A continuous membrane, either vertical or horizontal, such as a wall, floor, or ceiling assembly, that is designed and constructed to restrict the movement of smoke in conjunction with a smoke control system.

2.59 Smoke Damper — A damper similar to fire damper, however, having provision to close automatically on sensing presence of smoke in air distribution system or in conditioned space.

2.60 Smoke Management — A smoke control method that utilizes natural or mechanical systems to maintain a tenable environment for the means of egress from a large-volume space or to control and reduce the migration of smoke between the area on fire and communicating spaces.

2.61 Stack Effect — The vertical airflow within buildings driven by air buoyancy due to temperature-induced density differences between the building interior and exterior or between two interior spaces.

2.62 Static Pressure — The normal force per unit area that would be exerted by a moving fluid on a small body immersed in it if the body were carried along with the fluid. Practically, it is the normal force per unit area at a small hole in a wall of the duct through which the fluid flows (piezometer) or on the surface of a stationary tube at a point where the disturbances, created by inserting the tube. It is supposed that the thermodynamic properties of a moving fluid depend on static pressure in exactly the same manner as those of the same fluid at test depend upon its uniform hydrostatic pressure.

2.63 Supply Air — The air that has been passed through the conditioning apparatus and taken through the duct system and distributed in the conditioned space.

2.64 Terminal Devices — Devices fixed in the air-conditioned space for distribution of conditioned supply air and return of air such as, supply and return air grilles and diffusers.

2.65 Thermal Adaptation — The gradual diminution of the people's response to repeated environmental stimulation and subsumes all processes which building occupants undergo in order to improve the fit of the indoor climate.

2.66 Thermal Comfort – That condition of mind which expresses satisfaction with the thermal environment and is assessed by subjective evaluation.

2.67 Thermal Insulation Material — A material used over the conducting material to retard the flow of heat energy in the form of heat loss or gain to facilitate the temperature control as the process and prevent permeability of moist vapour and reduces condensation on cold surfaces.

2.68 Thermal Energy Storage — Storage of thermal energy, sensible, latent or combination thereof for use in central system for air conditioning or refrigeration. It uses a primary source of refrigeration for cooling and stored thermal energy for reuse at peak demand or for backup as planned.

2.69 Velocity Pressure — The pressure exerted by movement of air which makes the air to travel to longer distance in ducts. This pressure is always in the direction of flow.

2.70 Variable Refrigerant Flow (VRF) System — A heating and/or cooling system in which the flow of the refrigerant shall be varied according to the load.

2.71 Water Hardness — The hardness in water represented by the sum of calcium and magnesium salts in water, which may also include aluminium, iron, manganese, zinc, etc.

2.72 Water Treatment — The treatment of water circulating in a hydronic system, so that it shall be used without creating undue corrosion or scaling to the piping systems and other deleterious effects.

2.73 Wet-Bulb Temperature — The temperature at which liquid or solid water, by evaporating into air, shall bring the air to saturation adiabatically at the same temperature. Wet-bulb temperature is the temperature indicated by a wet-bulb thermometer constructed and used according to specifications.

2.74 Definitions/Terminology Reference to Refrigerants — Refer to the accepted standard [D-3(2)] for all the definitions. In addition, below are some definitions.

2.74.1 *Glide* — The difference in the dew and bubble point of a zeotropic refrigerant at constant pressure.

2.74.2 *Occupational Exposure Limit* (*OEL*) — The threshold amount of time-weighted average (TWA) concentration a normal 8 hour shift and a 40 hour work week to which nearly all workers shall be repeatedly exposed without any adverse effects.

3 BASIS OF DESIGN

3.1 Planning and Design Considerations

In any building, clear definition of design considerations holds the key for a good HVAC system. This includes, appropriate system selection, system sizing, layout and location planning, planning for operation, maintenance and controls as explained in following pages.

3.2 Fundamental Planning Requirements

The objective of installing air conditioning, heating and mechanical ventilation in buildings shall be to provide comfortable conditions without compromising on health and safety of occupants. Air conditioning, heating and mechanical ventilation installation shall aim at controlling the following factors in the building:

a) thermal comfort;

b) air quality, air quantity and air movement;

c) sound and vibration;

d) through meeting the minimum energy efficiency levels;

e) ensuring electrical; and

f) fire safety requirements.

3.2.1 All plans, design, drawings, specifications and data for air conditioning, heating and mechanical ventilation systems of all buildings and serving all occupancies within the scope of this NBCS, shall be supplied to the Authority, where called for.

3.2.2 The plans, design and drawings for air conditioning, heating and mechanical ventilation (HVAC) systems shall include all details and data necessary for detailed load calculation through computer simulation such as:

a) building name, type and location, address;

b) owners' name;

c) HVAC designers' name;

d) use of building;

e) building orientation: north direction on plans and design drawings;

f) general plans, dimensions and height of all rooms in SI unit;

g) intended use of internal spaces;

h) detail or description of wall construction, including insulation and finish;

j) detail or description of roof, ceiling and floor construction, including insulation and finish;

k) detail or description of windows and outside doors, including sizes, thermal properties, weather stripping, chajjas or projections;

m) internal load, such as people, equipment, computer/server load and lighting load;

n) layout showing the location, size and components of the HVAC equipment being installed;

p) information regarding air distribution system;

q) information on air and water systems and their flow rates and pressure drops;

r) information regarding location, size and accessibility of shafts;

s) information regarding type and location of dampers (both volume control and fire/smoke dampers) used in air conditioning system, such as, whether motorized or manually operated and rating;

t) location and grade of the required fire separations;

u) water treatment and softening arrangement;

v) information on presence of any chemical fumes or gases; and

w) HVAC automation design, description and piping and instrumentation (generally referred to as P&I drawing) drawing.

3.3 Fundamental Design Requirements

The following guidelines address critical factors such as load calculations, system adjustments for off-peak hours, and specialized needs for various applications, ensuring optimal performance and energy conservation across diverse scenarios.

3.3.1 Heat load calculation shall take into account the following factors:

a) recommended indoor temperature, relative humidity, air velocity, mean radiant temperature, clothing, activity and IAQ parameters;

b) outside and indoor design conditions as specified;

c) details of building construction and orientation of exposures of building components;

d) fenestration area, thermal properties and shading factors;

e) occupancy: number of people and their schedule of activities;

f) ventilation: requirement for fresh air;

g) infiltration, air leakage;

h) internal load: Equipment, computer/server and lighting;

j) effective volume; and

k) occupancy, lighting and equipment schedule.

3.3.2 The design of air conditioning, heating and mechanical ventilation system and its associated controls shall also consider the following:

a) application;

b) permissible control limits;

c) fire safety;

d) opportunities for heat recovery, condensate drain recovery;

e) energy efficiency;

f) filtration standard;

g) hours of use;

h) suitable diversity factor based on usage; and

j) outdoor and indoor air quality.

3.3.3 Due consideration shall be given to air conditioning load encountered during off-peak hours including night, weekend, holidays. The system shall be designed for the maximum operational efficiency to achieve lowest annual energy consumption. Consideration shall be given to the anticipated future changes (permanent or temporary) in building load. The system shall be designed for maximum operational efficiency is achieved throughout.

3.3.4 The air conditioning system for special applications like hospitals, operating theatres, computer rooms, data centres and telecommunication rooms, clean rooms, laboratories, libraries, museums, art galleries, sound recording studios, etc shall be designed as per specific requirement.

3.3.5 Computer based hourly load calculation and energy simulation tools should be used for HVAC equipment sizing and to identify effect of various energy conservation measures on energy consumption.

3.3.6 An uncertainty analysis should be carried out based on the input parameter accuracy level. The system design should be determined by the uncertainty analysis considering the parameters like variation in climatic conditions, material properties, usage pattern, internal loads over design or under design of the should be avoided.

3.3.7 The system sizing shall be based upon outdoor and indoor design conditions, considering a safety factor, together with capacity deration due to local conditions as well as aging of equipment. Performance maps of major equipment at different climatic conditions and loads should be considered for accessing their suitability, sufficiency and energy efficiency.

3.4 System Layout and Space Planning

Each type of HVAC system has peculiar space and layout requirement. Major requirements for planning an efficient and serviceable layout are explained below.

3.4.1 *Equipment Room for Central Air Conditioning Plant*

These guidelines address the provisions of equipment room for central air conditioning plant.

3.4.1.1 This room shall be located preferably within the building being air conditioned and closer to external wall for facilitating ventilation and equipment movement. The equipment may also be installed in a separate service block which should also be located as close as possible to the load/building being conditioned. The clear headroom below soffit of beam should be minimum 4.5 m for larger capacity chillers (500 TR and above) and minimum 3.6 m for smaller plants.

3.4.1.2 The floors of the equipment rooms should be compacted and finished. For floor loading, the air conditioning engineer should be consulted (*see* Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS).

3.4.1.3 Supporting of pipe within plant room spaces should be normally from the floor. However, outside plant room areas, structural provisions shall be made for supporting the water pipes from the floor/ceiling slabs. All floor and ceiling supports shall be isolated from the structure to prevent vibration transmission.

3.4.1.4 Equipment rooms, wherever necessary, shall have provision for mechanical ventilation. In hot and dry climate, evaporative air cooling shall be considered to maintain plant room temperature.

3.4.1.5 Plant machinery in the plant room shall be placed on levelled plain/reinforced cement concrete foundation block and provided with anti-vibratory supports or alternatively on inertia bases. Supports for appliances shall be designed and constructed to sustain vertical and horizontal loads within the stress limitations specified in the Part C 'Structural Design' of this NBCS. All foundations should be protected from damage by providing epoxy coated angle nosing. Seismic restraints requirement should also be considered (*see* **17.4.3**).

3.4.1.6 Appropriate sound insulation and noise control measures shall be taken in plant room space as per Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS. Acoustic treatment as may be required shall be provided in plant room space to prevent noise transmission to adjacent occupied areas.

3.4.1.7 In case air conditioning plant room is located in basement, equipment movement route shall be planned to facilitate future replacement and maintenance. Service ramps or hatch in ground floor slab should be provided in such cases. Fire egress and emergency battery backup lighting shall be provided for the plant room operator.

3.4.1.8 Floor drain channels or dedicated drain pipes in slope shall be provided within plant room space for effective disposal of waste water, if necessary, by automatic level-controlled sump pumps. Fresh water connection may also be provided in the air conditioning plant room.

3.4.1.9 Thermal Energy Storage should be used for limiting maximum demand, by controlling peak electricity load through reduction of chiller capacity, and by taking advantage of high system efficiency during low ambient conditions. Thermal storage will also help in reducing operating cost by using differential time-of-the day power tariff, where applicable. In case of central plant designed with thermal energy storage, its location shall be decided in consultation with the air conditioning engineer. For roof top installations, structural provision shall take into account load coming on the building/structure due to the same. For open area surface installation, horizontal or vertical system options shall be considered and approach ladders for manholes provided. Buried installation shall take into account loads due to movement of vehicles above the area. Provision for adequate expansion tank and its connection to thermal storage tanks shall be made.

3.4.2 *Equipment Room for Air Handling Units and Package Units*

These guidelines ensure optimal placement, safety, and functionality of equipment rooms for air handling and package units within the building's HVAC system**.**

3.4.2.1 This shall be located as centrally as possible to the conditioned area and contiguous to the corridors or other service areas for carrying air ducts in ceiling spaces.

3.4.2.2 In case of special and high-rise buildings, air handling units shall be provided in accordance with fire and life safety considerations, given in Part F 'Fire and Life Safety' of this NBCS. Air handling unit rooms should preferably be located vertically one above the other.

3.4.2.3 Provision shall be made for the entry of outdoor ventilation air into air handling unit room. For energy conservation, it is desirable to install mechanism for modulating the outdoor air quantity based on demand.

3.4.2.4 Exterior openings for outdoor air intake and also exhaust outlets shall have louvers with rain protection profile, volume control damper, pre-filter and bird screen.

3.4.2.5 In all cases, outdoor air intakes shall be located to avoid contamination from exhaust outlets and pollution sources in concentration higher than normal in the building locality. The separation distance between fresh air intake and any exhaust outlets shall be 8 m preferably.

3.4.2.6 Exhaust air from any dwelling unit shall not be circulated/ingress directly or indirectly to any other dwelling unit, to public corridor or into public stairway.

3.4.2.7 All air handling rooms should have floor drains. The trap in floor drain shall provide a water seal between the air-conditioned space and the drain line.

3.4.2.8 Supply/return air duct serving other areas shall not be taken through fire exits.

3.4.2.9 Waterproofing of air handling unit rooms shall be carried out to prevent damage to floor below.

3.4.2.10 The floors should be finished smooth. For floor loading, the air conditioning engineer should be consulted (*see* Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS).

3.4.2.11 Structural design should avoid beam obstruction to the passage of supply and return air ducts. Adequate ceiling space should be made available outside the air handling unit room to permit installation of supply and return air ducts and fire/smoke dampers at compartment wall crossings.

3.4.2.12 Appropriate sound insulation and noise control measures shall be taken in air handling unit rooms, if located in close proximity to occupied areas, as per Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS. The air handling unit rooms shall be acoustically treated in accordance with the noise control requirements if located in close proximity to occupied areas.

3.4.2.13 Access door to air handling unit room shall be single/double leaf type, air tight, opening outwards and should have a sill to prevent flooding of adjacent occupied areas. It is desired that access panels in air-conditioned spaces should be provided with tight sealing, gaskets and self-closing devices for air conditioning to be effective.

3.4.2.14 It should be possible to isolate the air handling unit room in case of fire. The door should be fire resistant (*see* Part F 'Fire and Life Safety' of this NBCS) and fire/smoke dampers shall be provided in supply/return air duct at air handling unit room wall crossings. Annular space between the duct and the wall should be fire sealed using appropriate fire resistance rated material.

3.4.2.15 Fire isolation shall be provided for vertical fresh air duct, connecting several floors.

3.4.2.16 It is desirable that individual air handling unit should be installed for each fire compartment, alternately, fire barrier should be provided at each fire separation for air handling units serving more than one compartment.

3.4.3 *Pipe Shafts*

These guidelines address the placement, safety, and accessibility requirements for pipe shafts to ensure effective distribution and maintenance of building services.

3.4.3.1 The shafts carrying chilled water pipes should be located adjacent to air handling unit room or within the room.

3.4.3.2 Shaft carrying condensing water pipes to cooling towers located on terrace should be vertically aligned.

3.4.3.3 All shafts shall be provided with fire barrier at floor crossings following the fire safety requirements.

3.4.3.4 Access to shaft shall be provided at every level, if there is any serviceable component in the shaft. In case of tall buildings, care shall be taken for expansion/contraction of pipes while planning the supports.

3.4.4 *Supply Air Ducts and Return Air Ducts*

These guidelines ensure efficient and safe design of supply and return air ducts, focusing on airflow distribution, energy conservation, and compliance with safety standards.

3.4.4.1 The duct supports shall be designed to handle the load and also to take into account seismic considerations. The support material should be galvanized/coated steel or aluminium and facilitate ease of installation at site using alternatives such as fully threaded rod/angle, chromium plated steel threaded rods or channel section/wire support systems using stud anchors provided in the ceiling slab from drilled holes or anchors suitable for threaded rods, wires without damaging the slab or structural member. Wire Rope when used shall be rated to take seismic loads. Steel sections shall be analysed for both tensile and compressive loads for strength and serviceability for the maximum length of element being used for bracing and other supports. Appropriate loads and its combinations, as per relevant clauses in Indian standards should be considered in design. Each straight run of the service shall be braced with a minimum of one transverse braces and one longitudinal-transverse brace with distance not greater than 12 m between them.

3.4.4.2 If false ceiling is provided, the supports for the duct and the false ceiling, shall be independent. Collars for grilles and diffusers shall be taken out only after false ceiling/boxing framework is done and frames for fixing grilles and diffusers have been installed. Flexible ducts may be used for making the final connections.

3.4.4.3 Where a duct penetrates the masonry wall, it shall either be suitably covered on the outside to isolate it from masonry, or an air gap shall be left around it to prevent vibration transmission. Further, where a duct passes through a fire resisting compartment/barrier, the annular space shall be sealed with fire sealant to prevent smoke and fire transmission as per relevant clauses of this NBCS.

3.4.5 *Cooling Tower*

These guidelines focus on the efficient design, placement, and maintenance of cooling towers to optimize cooling performance, minimize energy use, and ensure safe operation.

3.4.5.1 Cooling towers are used to dissipate heat from water cooled refrigeration, air conditioning and industrial process systems to the atmosphere. Cooling is achieved by evaporating a small proportion of recirculating water into outdoor air stream. Cooling towers shall be installed at a place where free flow of ambient air is available.

3.4.5.2 Range of a cooling tower is defined as temperature difference between the entering and leaving water. Approach of the cooling tower is the difference between leaving water temperature and the ambient air wet-bulb temperature.

3.4.5.3 *Selection of cooling tower*:

Following factors shall be considered for selection of cooling tower:

a) design wet-bulb temperature and approach of cooling tower. The designer shall endeavour for reducing the approach for maximizing the energy conservation potential;

b) height limitation and aesthetic requirement;

c) location of cooling tower considering possibility of easy drain back from the system;

d) placement with regard to adjacent walls, windows and other buildings, and effects on these from any water carried over by the air stream;

e) vibration and noise levels, particularly during silent hours;

f) material of construction for the tower;

g) direction and flow of prevailing wind;

h) quality of water used for make-up;

j) maintenance and service space availability; and

k) ambient air quality.

3.4.5.4 The recommended floor area requirement for various types of cooling tower is as given below:

a) Natural draft: 0.15 to 0.20 m^2/3.516 kW cooling tower; and

b) Mechanical draft: 0.07 to 0.10 m^2/3.516 kW cooling tower.

3.4.5.5 Structural provision for the cooling tower shall be taken into account while designing the building. Vibration isolation shall be an important consideration in structural design.

3.4.5.6 Special care should be taken in design where noise transmitted to the adjoining building shall be of serious concern. Special vibration control and sound attenuation devices may be required in that case. Appropriate sound insulation and noise control measures shall be taken in such cases in accordance with relevant requirements of this NBCS. System may be designed to have sound absorption ≥ 0.8 and sound insulation of ≥ 50 dB.

3.4.5.7 Certain amount of water is lost from circulating water in the cooling tower, as given below:

a) *Evaporation loss* — It is usually about 1 percent of the rate of water circulation.

b) *Drift loss* — The drift loss shall be below 0.1 percent of rate of water circulation.

c) *Blow-down/bleed-off* — The amount of blowdown shall be below 0.8 percent of the total water circulation. If simple blow-down is inadequate to control scale formation, chemicals may be added to inhibit corrosion and limit microbiological growth. Provision shall be made to make-up for the loss of circulating water.

3.4.5.8 Provision for make-up water tank to the cooling tower shall be made. Make-up water tank to the cooling tower shall be separate from the tank serving drinking water. Makeup water should be sourced from treated water from sewage treatment, waste water treatment plant, or from rain water harvesting. Make-up water having contaminants or hardness, which shall adversely affect the refrigeration plant life, shall be treated. Treated water where hardness as ppm of $CaCO_3$ is reduced to 50 ppm or below is recommended for air conditioning applications. Water with *p*H value less than 5 shall also need to be treated.

3.4.5.9 Cooling tower should be so located as to eliminate nuisance from drift to adjoining structures.

3.4.6 *Building Envelope*

The envelope of the building including wall, roof and fenestration may be planned as per relevant guidelines/best practices (‘Handbook on Sustainability in Buildings’).

3.4.7 *Fire Safety*

For fire safety in case of special and high-rise buildings, provisions of Part F ‘Fire and Life Safety’ of this NBCS should be applicable.

3.4.8 *Sound Insulation and Noise Control*

Sound insulation and noise control measures for HVAC system shall be in accordance with Part D ‘Building Services, Section 4 Acoustics, Sound Insulation and Noise Control’ of this NBCS.

3.4.8.1 *Designing for background noise control and vibration*

The background noise and vibration created by air distribution through the HVAC system, combined with noise from equipment and outdoor sources of noise and vibration may causes significant disturbance to building users. Adoption of measures for adhering to indoor noise limits and vibration control should be ensured in the design of equipment, their foundation, supports and air distribution system. Threshold values for indoor sound levels, are given in Table 1 and Table 2 below. The buildings/indoor spaces will be classified as Class A, B and C where Class A refers to aspirational, Class B refers to acceptable and Class C refers to minimum acceptable.

3.4.9 *Energy Conservation*

Designers shall aim for energy efficiency in buildings with the right blend of passive and active design strategies in accordance with relevant guidelines/best practices (‘Handbook on Sustainability in Buildings’), so as to minimise the energy use while ensuring comfort.

3.5 Outside design conditions

The outdoor design conditions shall be considered in accordance with accepted standard [D-3(3)]. Table 3 presents design conditions taken from this standard as ready reference. For cities not included in this table, it is recommended that extrapolation may be done, from the data of the nearby listed city, but keeping into consideration the specific topographical and climatic conditions of the concerned location. Values of ambient dry-bulb temperatures and wet-bulb temperatures, against the various annual percentiles, represent the value that is exceeded on average by the indicated percentage of the total number of hours. The 0.4 percent, 1.0 percent, 2.0 percent values are exceeded on average 35 h, 88 h and 175 h respectively in a year. The 99.0 percent and 99.6 percent values are defined in the same way but are usually reckoned as the values for which the corresponding weather elements are less than the design conditions of 88 h and 35 h respectively.

Mean coincidental values are the average of the indicated weather element occurring concurrently with the corresponding design value. After the calculation of design dry-bulb temperatures, the program locates the values of corresponding wet-bulb temperatures from the database for that particular station, the average of these values is computed, which are then called mean of coincidental wet-bulb temperature. In the same way, design wet-bulb temperatures and coincidental dry-bulb temperatures are evaluated.

The design values of 0.4 percent, 1.0 percent and 2.0 percent annual cumulative frequency of occurrence may be selected depending upon application of air conditioning system. For normal comfort conditions, values under 1.0 percent column should be used for cooling loads and 99 percent column for heating loads. For critical applications, values under 0.4 percent column should be used for cooling loads and 99.6 percent column for heating loads.

Table 1 Requirements of Acoustic Comfort

(*Clause* 3.4.8.1)

SI No.	**Types of Buildings**	**Noise Criterion**			**Reverberation Time**			**Speech Transmission Index**		
		A	B	C	A	B	C	A	B	C
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	**Residences, apartments, condominiums**	30	35	40	< 0.8 s	NA	NA	*0.6	*0.5	*NA
								Applicable for atriums and corridors		
ii)	**Hospitality**									
a)	Individual rooms or suites	30	35	40	< 0.8 s	< 0.6 s	< 1.0	0.6	0.5	NA
b)	Meeting/banquet rooms	30	35	40	< 0.5 s			0.7	0.6	0.5
c)	Corridors, lobbies	40	45	45	NA			NA	NA	NA
d)	Service/Support areas	40	45	50	NA			0.5	NA	NA
e)	All day dining and restaurants	45	50	50	<1.2s	<1.5s	<1.75s	0.6	0.5	NA
iii)	**Office buildings**									
a)	Executive and private offices	30	35	40	< 0.6 s	< 0.7 s	< 0.8 s	0.6	0.5	NA
b)	Conference rooms	30	35	35	< 0.6 s			0.7	0.6	0.5
c)	Teleconference rooms	25 (max)			< 0.6 s			0.7	0.6	0.5
d)	Open – plan offices	35	40	40	< 0.8 s			0.6	0.5	NA

Table 1 (*Concluded*)

SI No.	Types of Buildings	Noise Criterion			Reverberation Time			Speech Transmission Index		
		A	B	C	A	B	C	A	B	C
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
e)	Corridors and lobbies	40	45	45	NA			NA	NA	NA
f)	Collaborative space				< 0.8 s			0.5	NA	NA
g)	Cafeteria/Town hall	40	45	45	< 1.0 s	< 1.2 s	< 1.2 s	0.6	0.5	NA
iv)	**Education**									
a)	Class rooms up to 70 m^2	40 (*Max*)			< 0.6 s	< 0.8 s	< 1.0 s	0.6	0.5	NA
b)	Class rooms over 70 m^2	35 (*Max*)						0.6	0.5	NA
c)	Large lecture rooms, without speech amplification	35 (*Max*)						0.6	0.55	0.5
v)	**Libraries**	35	40	40	< 1.0 s	< 1.2 s	< 1.2 s	0.6	0.5	NA
vi)	**Health care***									
a)	Open Ward	35	35	45	< 1.0s	< 0.8 s	< 1.0 s	0.7	0.6	0.5
b)	Individual rooms or suites	30	35	40	< 0.8s			0.7	0.6	0.5
c)	Meeting rooms /Conference room	25	30	35	< 0.6s			0.5	NA	NA
d)	Teleconferencing rooms	25	NA	NA	< 0.6s			0.7	0.6	0.5
e)	Operating rooms	35	40	45	< 0.8s			0.6	0.5	NA
f)	Corridors, lobbies	35	45	45	< 1.0s			NA	NA	NA
g)	Testing/research lab, minimal speech	45	50	55	< 0.8s			0.5	NA	NA
h)	Research lab, extensive speech	40	45	50	< 0.8s			0.7	0.6	0.5
j)	Auditoria, large lecture rooms	25	30	NA	< 0.8s			0.6	0.5	NA

NOTES

1 In this table are given the performance requirements in terms of recommended dBA noise level (ideal), the not to exceed noise level (max), and the maximum acceptable reverberation time (T60) for many spaces including offices.

2 For the Class A and Class B, a building should score 90 percent and 80 percent respectively on the occupant satisfaction survey. There is no such requirement to achieve Class C.

3 Any other type of space not mentioned in the Table above shall meet requirements of the nearest representative category of that space.

4 For more details on Reverberation Time, refer graph with respect to volume and application in annexure.

5 * values measured in unoccupied condition.

Table 2 Threshold Noise Isolation Criteria Level Depending Upon Type of Space

(*Clause* 3.4.8.1)

Sl No.	Building	Type of space	D_w/NIC		
			Class A	Class B	Class C
(1)	(2)	(3)	(4)	(5)	(6)
i)	Office	Between two offices	45	40	40
		Where privacy is important			
		Cellular offices			
		Between enclosed office and circulation area	40	35	30
		Between two meeting or conference rooms	50	45	45
		Between meeting or conference room and circulation area	40	35	30
		Between two training rooms	50	45	45
		Between training room and circulation area	40	35	30
ii)	Residential	Partitions separating a water closet (WC) from a noise sensitive room	45	40	35
iii)	Hospitality	Partitions and floors between rooms and corridors			
		Walls and floor between two guestrooms/suites	55	50	45
		Between guestrooms/suites and circulation area	40	40	40
		Walls and floor between banquet halls and guestrooms/suites	55	50	50
		Between banquet hall and circulation area or pre functions	45	40	40
iv)	Entertainment	Walls of cinemas, auditoriums, studios, pubs	60	55	50
v)	Education	Between classrooms, labs, lecture halls	50	50	45
vi)	Hospital or Health care*	Between two patient rooms and circulation area	40	40	35
		Between patient room and circulation area (with entrance)	35	30	25
		Between Patient Room and Service Area	50	45	45
		Between Consultation Room and Public Space/Patient Rooms	40	40	40
		Between Consultation Room and Circulation Area (with entrance)	35	30	25

NOTE — Weighted level difference (D_w) a single integer number found by comparing the measured spectrum with the 'standard' curves for airborne and impact insulation. The D_w value is where the curve meets the 500 Hz curve and the unfavourable deviation is 32 dB. D_w will be identical to $D_{Nt,w}$ when T = 0.5 seconds. Noise Isolation Class NIC a single number rating of the degree of speech privacy achieved through the use of an acoustical ceiling and sound absorbing screens in an open office..

Table 3 Summary for Outdoor Design Conditions

(*Clause* 3.5)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
1)	Agartala	10.8	11.6	35.2	25	34.5	25	33.8	27	28.5	33	28.1	32	27.8	32
2)	Agra	4.6	6.1	42.1	21	40.8	22	39.8	22	28.7	31	28.3	30	28	31
3)	Ahmedabad	12.3	13.2	42.7	25	41.7	22	41	22	30.2	35	29.3	34	28.4	33
4)	Aizawl	10.5	11.3	30.5	20	29.5	21	28.9	21	24.8	27	24.5	26	24.1	27
5)	Ajmer	4.8	5.6	41.5	22	40.7	21	39.4	21	25.2	34	24.9	34	24.7	33
6)	Akola	12.4	13.9	43.1	23	42.2	21	41.4	22	26.9	33	26.5	32	26.2	31
7)	Aligarh	3.9	5.3	42.9	23	41.8	23	40.4	24	28.3	33	27.9	34	27.6	33
8)	Allahabad	9.8	10.6	43.1	23	41.7	23	40.3	23	28.7	32	28.3	33	28	32
9)	Amaravati	19.2	19.8	42	27	39.7	27	38.1	26	28.4	33	28.2	32	27.8	33
10)	Amritsar	3	4.5	42	23	41	23	39	23	29.2	32	28.6	32	28.2	31
11)	Anantapur	16.9	18.1	40	24	39.5	24	38.7	23	26.3	35	25.8	33	25.5	34
12)	Aurangabad	11.8	12.9	40.1	23	39.3	22	38.3	22	26.8	34	26	32	25.5	31
13)	Balasore	12.9	13.9	36.8	26	35.5	26	34.6	27	28.8	33	28.4	33	28.1	33
14)	Bareilly	6.4	7.2	39.6	23	38.7	26	37.7	25	29.2	34	28.8	33	28.4	33
15)	Barmer	10.6	11.5	43.2	23	42	22	40.4	24	28.3	36	27.9	35	27.5	34

Table 3 (*Continued*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
16)	Belgaum	13.5	14.6	36.7	19	35.5	18	34.2	19	24.1	29	23.7	29	23.4	28
17)	Bengaluru	15.6	16.2	34.1	21	33.4	20	32.7	19	23.5	29	23.1	29	22.8	28
18)	Bhagalpur	13.1	13.8	38.3	22	36.8	25	35	27	29.3	34	28.9	33	28.7	32
19)	Bhavnagar	12	12.8	42.4	25	41.3	27	39.8	25	28.4	30	28.1	32	27.8	32
20)	Bhilai	12.7	14	42.6	23	41.7	24	40.4	23	26.8	31	26.5	32	26.2	31
21)	Bhopal	9	10.2	42	21	40.7	25	39.5	21	26	30	25.7	30	25.3	29
22)	Bhubaneswar	14.1	15.1	38.9	27	37.8	27	36.7	26	30.1	35	29.5	33	29.1	33
23)	Bhuj	12.4	13.3	40.5	23	39.5	24	38.3	23	28.1	36	27.6	33	27.3	33
24)	Biharsharif	7.8	8.7	40.1	24	38.7	24	37.6	24	28.4	31	28	32	27.7	31
25)	Bikaner	7.6	9.1	44.8	22	43.3	23	41.9	23	27.6	34	27.2	32	26.8	33
26)	Bilaspur	11.8	13	43.2	22	42.3	23	41.3	23	26.9	31	26.6	31	26.4	31
27)	Chandigarh	2.8	4	43	22	40.8	20	39.2	18	27.2	33	26.8	32	26.5	32
28)	Chennai	20.5	21	39	26	38	26	37	26	28.5	34	28.1	34	27.8	35
29)	Chitradurga	16.2	16.9	37.4	21	36.5	21	35.6	20	23.8	27	23.5	26	23.2	26
30)	Coimbatore	18.6	19.5	36.5	23	35.8	24	34.9	24	25.4	33	25.1	31	24.9	29

Table 3 (*Continued*)

Sl No.	**City**	**Heating DB**		**Cooling DB/MCWB**						**Evaporation WB/MCDB**					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
31)	Cuddalore	20.1	20.8	37.9	26	37.1	26	36.1	27	29	33	28.7	33	28.3	34
32)	Dahod	8.6	9.8	41.1	20	40.4	23	39.3	21	27	30	26.5	32	25.7	33
33)	Daltonganj	8.8	9.9	43.3	24	42	25	41	24	28.1	32	27.7	31	27.5	32
34)	Davanagere	16.9	17.7	37.4	21	36.5	21	35.7	21	24.2	29	23.8	29	23.5	29
35)	Dehradun	5.8	6.6	37.7	22	36.2	23	34.5	22	27	30	26.5	30	26.2	30
36)	Dhanbad	8	9.3	41.5	23	40.5	22	39.7	22	27.4	32	27	32	26.7	31
37)	Dharmsala	3.7	4.7	32.1	20	31.1	21	30.2	18	23.7	28	23.4	26	22.9	25
38)	Dibrugarh	10.2	11	35	28	34	27	33.2	27	29.2	34	28.6	33	28.1	32
39)	Diu	10.5	11.4	35.9	24	34.6	23	33.6	22	25.1	30	24.8	29	24.6	29
40)	Erode	18.3	18.9	40.3	23	39.3	23	38.3	24	26.5	31	25.8	30	25.4	31
41)	Etawah	3.8	5.3	44.2	24	43.1	25	42	25	28.9	36	28.2	37	27.6	34
42)	Ganganagar	4.6	6.1	44.2	22	42.7	26	40.8	25	29.1	35	28.7	34	28.4	35
43)	Gangtok	1.4	2.3	23.1	19	22.7	18	22.3	19	19.8	22	19.6	22	19.4	22
44)	Gaya	7	9	43.3	20	41.8	23	40.2	21	28.5	33	28	33	27.7	32
45)	Gorakhpur	8.5	9.6	40.7	23	38.9	23	37.7	22	28.1	32	27.8	33	27.5	33

Table 3 (*Continued*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
46)	Gulbarga	19.4	20.4	41.5	24	40.6	25	39.8	25	27.4	36	27	34	26.6	35
47)	Guna	8.5	10	42.8	22	41.9	22	40.9	22	27	33	26.7	31	26.4	32
48)	Guntur	16.6	17.7	41.1	25	39.6	25	38.5	25	28.6	31	28.3	31	28.1	31
49)	Guwahati	11.2	12.1	35.4	27	34.5	29	33.9	27	29.2	34	28.9	33	28.6	33
50)	Gwalior	4.9	5.7	43.2	24	42.2	24	40.9	23	28.5	34	28.2	35	27.8	33
51)	Hissar	4.1	5.1	43.8	25	42.6	24	41.2	23	29	37	28.7	34	28.3	33
52)	Honavar	18.8	19.8	34.3	24	33.6	24	33.1	25	27.8	32	27.4	32	27.2	31
53)	Hubballi	12.5	13.2	41.8	24	40.7	23	39.4	24	26	26	25.5	29	25.2	30
54)	Hyderabad	15.3	16.3	41	22	39.6	22	38.4	24	25.4	34	25	31	24.7	31
55)	Imphal	4.9	5.9	32	21	31.3	23	30.7	23	25.7	31	25.2	30	24.8	29
56)	Indore	10.4	11.3	39.9	19	38.9	21	37.9	21	25.6	31	25.1	29	24.8	29
57)	Itanagar	8.8	9.7	35.2	29	34.3	29	33.4	29	29.8	34	29.5	34	29.1	32
58)	Jabalpur	8.4	9.8	41.6	20	40.7	20	39.5	20	26.3	30	25.9	31	25.6	30
59)	Jagdalpur	9.6	11.1	39.2	20	38.1	23	37	23	26.6	36	25.9	32	25.5	31
60)	Jaipur	8.4	9.3	42.4	20	41	21	40	21	27.6	33	27.2	31	26.9	32

Table 3 (*Continued*)

Sl No.	**City**	**Heating DB**		**Cooling DB/MCWB**						**Evaporation WB/MCDB**					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
61)	Jaisalmer	8.3	9.3	44.3	24	42.8	23	41.7	25	28.1	38	27.6	34	27.2	36
62)	Jalandhar	4.3	5.5	43.7	21	42.4	21	40.7	21	25.6	33	25	36	24.6	33
63)	Jammu	4.8	5.8	41.9	21	40.1	20	38.9	21	28.6	32	27.9	30	27.4	32
64)	Jamnagar	7.9	8.9	38.7	24	37.5	23	36.5	24	27.7	30	27.5	31	27.3	32
65)	Jamshedpur	9.8	10.7	41.4	23	39.8	23	38.2	24	27.6	33	27.3	33	27	32
66)	Jhansi	6.1	7	44.9	24	43.7	22	42.8	22	27.6	34	27.1	31	26.9	33
67)	Jharsuguda	11	12	42	24	40.8	24	39.5	23	27.8	31	27.4	32	27.1	31
68)	Jodhpur	9	10.4	42.1	22	40.7	23	39.8	23	27.9	33	27.4	34	27	32
69)	Jorhat	8.5	9.2	33.1	27	32.6	26	31.6	27	27.7	31	27.5	30	27.2	30
70)	Kakinada	18.1	19.4	42.2	26	39.4	25	37.3	28	29.2	35	28.8	33	28.4	34
71)	Kalingapatam	15.8	17.6	35.3	29	34.6	29	34	28	29.7	34	29.5	33	29.1	33
72)	Kannur	20.6	21.2	36.4	24	35.6	24	34.9	24	26.8	30	26.5	31	26.3	30
73)	Kanpur	6.4	7.2	43.4	23	42	23	40.9	23	28.5	34	28.2	34	28	33
74)	Karaikal	21.8	22.7	37	26	36.2	26	35.3	26	29.1	34	28.9	34	28.4	32
75)	Karimnagar	13.4	14.7	42.7	24	41.9	26	40.7	25	28.2	37	27.6	34	27.2	35

Table 3 (*Continued*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
76)	Karnal	4.7	5.6	40.8	22	39.7	24	38.4	22	27.8	33	27.3	31	26.9	32
77)	Karwar	20.6	21.6	35.9	27	35.4	26	34.9	26	29.6	34	29.2	34	28.8	33
78)	Kavaratti	23.3	24	33.8	28	33.4	27	32.9	27	28.6	33	28.2	33	27.9	32
79)	Kochi	22.8	23.2	32.9	26	32.6	26	32.3	26	27.6	32	27.3	31	27.1	31
80)	Kodaikanal	7	7.7	24.4	15	23.4	14	22.6	14	15.8	22	15.4	22	15.1	21
81)	Kohima	5.9	6.8	27.1	20	26.3	21	25.6	20	21.6	23	21.3	24	21.1	24
82)	Kolhapur	19.4	20.3	35.6	20	35.1	20	34.5	20	27.4	28	26.6	27	25.5	29
83)	Kolkata	12	13	37	28	36.3	28	35.7	28	29.6	33	29.2	34	28.8	34
84)	Kollam	21.8	22.3	34.2	25	33	23	32.4	23	26.8	30	26.6	30	26.4	29
85)	Kota	10	11.2	43.5	21	42.3	21	41	22	27.1	32	26.8	31	26.5	31
86)	Kozhikode	22.6	23.1	34.8	29	34.3	28	33.9	27	29.2	34	28.9	33	28.6	33
87)	Kurnool	17.1	18	41.1	22	40.1	24	39.2	24	26.1	32	25.8	34	25.5	33
88)	Leh	-20.7	-19	26.6	10	25.8	11	24.8	11	14	23	13.5	21	12.9	20
89)	Lucknow	6.7	8	42	23	41	23	40	24	29.4	34	29	36	28.6	34
90)	Ludhiana	2.6	3.6	40.7	22	39.3	24	38.1	22	28.1	30	27.6	32	27.2	32

Table 3 (*Continued*)

Sl No.	**City**	**Heating DB**		**Cooling DB/MCWB**						**Evaporation WB/MCDB**					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
91)	Machilipatnam	20.4	20.9	40.3	28	38.7	27	37.1	27	29.7	36	29.3	36	28.8	34
92)	Madurai	18.1	18.8	40.9	24	39.6	25	38.8	25	28.1	32	27.4	32	27	32
93)	Mangaluru	21.3	22	34.2	24	33.7	25	33.3	25	27	33	26.7	31	26.5	30
94)	Minicoy	23.9	24.4	33.3	28	32.9	28	32.5	28	28.7	32	28.3	32	28.1	32
95)	Moradabad	5.2	6.6	42.1	23	40.7	24	39.4	25	28.8	35	28.3	33	27.8	33
96)	Mumbai	19	20.2	35	23	34.7	24	33.9	23	27.9	31	27.7	31	27.3	31
97)	Muzaffarpur	8.5	9.3	38.9	25	37.9	26	36.8	27	29.5	35	29.2	34	28.9	32
98)	Mysuru	14.6	15.1	36.7	21	35.5	22	34.4	22	24	25	23.6	29	23.2	27
99)	Nagappattinam	22.7	23.2	36.9	26	36.2	26	35.4	27	29.1	32	28.9	32	28.6	32
100)	Nagpur	12	13	44	22	43	22	42	22	27.5	31	27.1	29	26.7	31
101)	Nashik	9.5	10.1	37.3	19	36.5	20	35.6	19	26.3	28	25	27	24.3	27
102)	Nellore	21	21.6	41	28	39.6	27	38.4	28	28.8	36	28.5	38	28.2	35
103)	New-Delhi	5	6.5	43.6	21	42.6	22	41	23	28.2	36	27.8	35	27.5	30
104)	Nizamabad	15.4	16.6	43.3	24	41.9	24	40.3	23	26.2	30	25.9	30	25.6	31
105)	Ongole	21.5	21.8	41.9	27	39.2	26	37.7	27	28.7	36	28.3	36	27.8	34

Table 3 (*Continued*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
106)	Panaji	19.8	20.5	34.1	26	33.8	27	33.3	26	28.2	33	27.9	32	27.6	32
107)	Pasighat	11	11.9	34.1	26	33.3	27	32.4	26	27.7	32	27.3	31	27	31
108)	Patiala	6.2	7	42.2	26	40.4	24	38.8	25	29.4	34	29	34	28.6	33
109)	Patna	7	8.2	41	24	39.6	26	37.7	23	28.6	33	28.3	33	28	32
110)	Pendra-road	9	10.2	39.7	21	38.7	21	37.6	23	26.2	31	25.7	30	25.3	29
111)	Port-Blair	22.1	22.9	33.1	27	32	27	31.3	26	27.5	32	27.2	31	26.9	30
112)	Pune	10.2	11.2	38.7	19	37.5	20	36.3	19	24.6	30	24	30	23.8	29
113)	Raipur	13.1	14.2	42.1	24	41.1	23	40	24	27.1	31	26.8	32	26.5	31
114)	Rajkot	12	13	41.3	21	39.9	23	38.9	24	27.4	29	27	30	26.7	31
115)	Ranchi	8.9	10	38.9	20	37.7	20	36.3	20	25.9	30	25.4	29	25.2	29
116)	Ratnagiri	17.9	18.6	34.1	22	33.7	24	33.1	23	27.1	31	26.9	30	26.7	30
117)	Raxaul	8.3	9.3	38.6	25	36.3	26	34.8	25	29	32	28.6	32	28.3	32
118)	Roorkee	4.4	5.3	40.4	24	39.1	23	37.9	23	27.9	32	27.4	31	27.2	32
119)	Sagar	8.1	9.5	41.9	24	41.1	23	40	21	26.5	37	25.9	33	25.5	30
120)	Saharanpur	4.2	4.9	43.2	22	42	23	40.7	24	28.5	33	28.2	33	27.9	34

Table 3 (*Continued*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
121)	Salem	17.5	18.2	39.9	24	39.1	22	38.1	23	25.6	30	25.2	31	25	30
122)	Sangli	7.8	8.5	40.1	20	38.9	19	37.8	18	24.5	27	24.2	27	23.9	28
123)	Satna	7.6	8.6	43.4	23	42.2	22	40.5	23	27.5	33	27	31	26.7	30
124)	Shillong	3.3	4.3	26.1	21	25.5	21	24.9	20	21.5	24	21.1	24	20.9	24
125)	Shivamogga	13.7	14.7	39	21	38	20	36.9	20	24	29	23.8	29	23.6	28
126)	Siliguri	8.6	9.6	36.3	30	35	28	33.9	28	30.8	32	30	31	29.1	32
127)	Silvassa	9	10.6	36.8	22	35.4	18	34.6	20	25.7	29	25.4	29	25	28
128)	Solapur	15.2	16.4	40.7	21	39.9	22	39.1	21	26.1	34	25.5	34	25.1	34
129)	Srinagar	-3.7	-2.4	31	20	29.9	20	28.9	20	22.4	30	21.9	28	21.3	26
130)	Surat	13.3	14.7	37.5	22	36.1	23	35.1	24	28.4	31	28.1	31	27.8	30
131)	Tezpur	11.2	12.1	34.5	29	33.8	26	33.1	27	29	32	28.6	32	28.2	32
132)	Thanjavur	18.8	19.8	41.5	26	40.5	25	39.3	25	27.4	35	27.1	32	26.9	31
133)	Thiruvananthapuram	22.5	23	34.3	26	33.6	25	33.1	26	27.8	32	27.4	31	27.2	32
134)	Thoothukudi	20.2	21	39.2	25	38.4	25	37.6	26	27.8	32	27.5	31	27.2	31

Table 3 (*Concluded*)

Sl No.	City	Heating DB		Cooling DB/MCWB						Evaporation WB/MCDB					
		99.6 percent	99 percent	0.40 percent		1 percent		2 percent		0.40 percent		1 percent		2 percent	
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)		(9)		(10)	
135)	Thrissur	19.3	20.3	37.8	26	36.1	27	34.8	25	28.8	31	28.6	32	28.2	31
136)	Tiruchrapalli	20	20.7	39	26	38.4	26	37.8	26	27.1	35	26.8	34	26.5	34
137)	Tirunelveli	20.3	21	41.8	24	40.3	25	39.1	24	26.5	30	26.2	30	26	31
138)	Tiruppur	16.4	17	31.9	20	30.8	20	29.9	18	21.9	25	21.6	24	21.4	24
139)	Tumakuru	14.7	15.5	37.8	19	36.4	20	35.2	18	22.3	30	21.9	30	21.5	28
140)	Udaipur	6.5	8.4	40.7	22	39.9	20	38.6	21	26.2	31	25.9	30	25.6	31
141)	Varanasi	8	9.9	43	24	42	24	41	24	28.7	35	28.4	33	27.9	33
142)	Vellore	15.9	16.7	42.1	23	40.3	23	38.8	22	25.6	29	25.3	31	25.1	30
143)	Veraval	15.5	16.5	35.2	22	34	26	33.2	28	28.8	33	28.5	32	28.3	32
144)	Vishakhapatnam	20	21.2	34	27	33.4	27	33	28	29.3	32	28.8	32	28.4	32
145)	Warangai	13.8	14.6	43.9	23	42.9	23	41.6	23	27.5	32	27.2	32	26.9	31

3.6 Inside Design and Operating Conditions

One of the primary objectives of designing indoor environment is to ensure thermal comfort and indoor air quality of all occupants. Air conditioning, heating and mechanical ventilation (HVAC) system is employed to achieve thermal comfort inside building, when means to achieve the same only through building design is limited.

According to the internationally accepted definition, thermal comfort is that state of mind which expresses satisfaction with the indoor environment. The above definition clearly highlights the importance of both the psychological (mental) and physiological (physical) factors in determining acceptable thermal comfort conditions. The following five environmental parameters are important for thermal comfort and indoor air quality while designing the HVAC system because these are directly controllable by an HVAC system**:**

a) air temperature (ºC);

b) radiant temperature (ºC);

c) air speed (m/s);

d) relative humidity (percent); and

e) indoor air quality.

In addition to the above five environmental parameters, there are two personal or behavioural parameters that also affect thermal comfort but are not controllable by the HVAC system and they are:

1) activity rate (W/m^2); and

2) clothing insulation ($m^2.K/W$).

There are number of secondary parameters also which are important to define thermal comfort conditions such as, radiative temperature asymmetry, temperature gradient, and draught rate.

Air conditioning systems for interior spaces intended for human occupancy shall be sized for not more than 26 °C for cooling and for not less than 18 °C for heating at occupied level.

HVAC systems should also be designed to control the humidity. Relative humidity in occupied spaces should not fall below 30 percent and should not exceed 70 percent. Adequate measures should be provided in system design itself for keeping it in the prescribed range including humidification and dehumidification as needed.

Air velocity control should be used to facilitate thermal adaptation as well as to save energy as described in **3.6.1**. However, air velocity should not exceed 1.5 m/s at the work plane to avoid discomfort due to draft and fluttering of papers etc.

While the system sizing needs to be done for design conditions mentioned above, the systems should be provided with features that facilitate thermal adaptation to be practiced by occupants. Buildings should provide users and facility managers to control air velocity in spaces and around individuals so that higher temperature set point shall be kept for saving energy. Buildings should provide spaces to control the set point individually as the temperature at which set point is to be kept varies as per the change in outdoor conditions over the time.

In conditioned buildings, the values for quality of thermal environment parameters for representative occupant of a space shall be as specified in Table 4 below.

Table 4 Conditions for Thermal Comfort

(*Clause* 3.6)

Sl No.	Air Velocity	Weather Condition	Level of Activity	Reference Table or Threshold Values
(1)	(2)	(3)	(4)	(5)
i)	Up to 0.2 m/s	Summer/winter	Met value ≤ 1.2	Table 5
ii)	Above 0.2 m/s	Summer/winter	Met value ≤ 1.2	Table 5 + Fig. 1 or Table 6
iii)	-	Summer	-	Relative humidity: 30 to 70 percent

Table 5 Acceptable Range of Operative Temperature with Air Velocity up to 0.2 m/s

(*Table* 4)

Sl No.	Level of Activity	Operative Temperature (°C)	
		Summer (Cooling Season) ~ 0.5 clo	Winter (Heating Season) ~ 1.0 clo
(1)	(2)	(3)	(4)
i)	Met > 1 and up to 1.2	23.0 ± 3.0	19.0 ± 4.0
ii)	Met ≤ 1.0	24.5 ± 2.5	22.0 ± 3.0

NOTE — At still air condition, operative temperature is the average of air temperature and mean radiant temperature. For details ISHRAE IEQ Standard 10001 can be referred.

High air velocity shall give opportunity of keeping higher air temperature without compromising thermal comfort. However, it is also suggested to keep under consideration noise and other effect of high indoor air velocity.

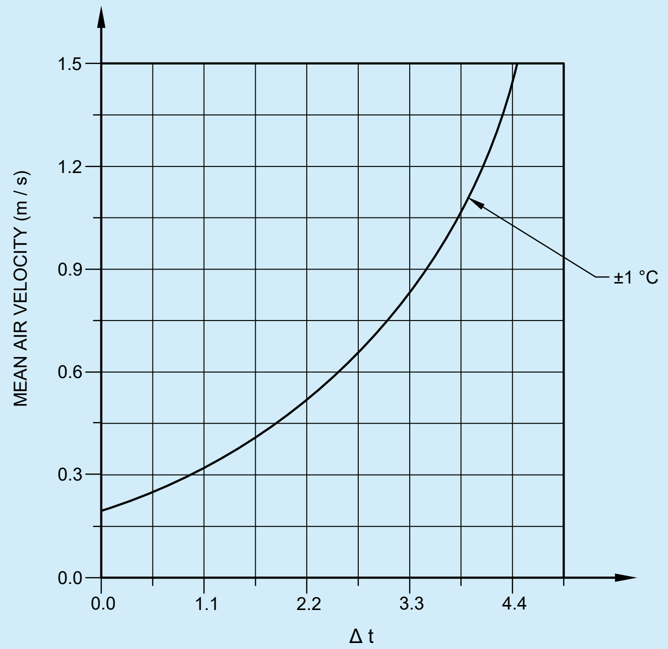


FIG. 1 REQUIRED AIR SPEED TO OFFSET INCREASED OPERATIVE TEMPERATURE (IN CELSIUS)

3.6.1 *Designing for Using Adaptive Thermal Comfort in Operation*

Design based on use of an adaptive thermal comfort model during operation of building plays a major role in reducing energy use whilst maintaining the comfort, productivity and well-being of occupants. This approach recognizes that people's thermal comfort needs depend on their past and present context and that these needs vary with the outdoor environmental conditions of their location. For more details, refer Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS.

3.6.1.1 *Path A*

The general requirements concerning predominantly dry bulb temperature (DBT) and mean radiant temperature (MRT) based adaptive thermal comfort model which differentiates the thermal response of occupants in air conditioned and mixed-mode buildings. For more details, refer Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS.

3.6.1.2 *Path B*

Since in many buildings, especially where microclimatic interventions or where occupants from wide range of ambient/indoor conditions (outside of the period of occupancy in the building in consideration) are likely to be present, it may be difficult to determine their running mean monthly outdoor temperature and they are likely to display different order of thermal adaptation. This alternate path for estimation of thermal comfort is through calculation of standard effective temperature (SET) using the look up Table 6 and Table 7, given below provided option for determining set of indoor conditions that are likely to provide thermal comfort.

The recommended range of SET for Indian conditions over all types of buildings is 23.5 ºC to 25.5 ºC, with 80 percent acceptability. The term SET includes combined effect of all five parameters that govern thermal comfort, including various adaptive actions such as air velocity, adjusting clothing, adjusting temperature set point, adjusting activity level, and humidity. Same method shall be used in unconditioned buildings as well for assessment of thermal comfort.

The desired value of SET shall be achieved at different values of air temperatures depending upon the values of other parameters. This Table 6 and Table 7 shall also be used in designing radiant cooling systems, personalised air conditioning system etc.

Table 6 Calculated Standard Effective Temperature for Adaptive Thermal Comfort

(*Clause 3.6.1.2 and Table 4*)

Standard Effective Temperature for Activity Level 1 Met (Office Work Type Activity) and Clothing 1.0 Clo (Winter Clothing)

AT (°C) ↓	MRT (°C) →	14				16				18				20				22				24				26				28				30				32			
	AV (m/s) → RH (percent) ↓	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4
14	20	16.1	13.9	12.8	12.1	17	14.6	13.5	12.7	18	15.4	14.1	13.4	19	16.2	14.8	14	19.9	16.9	15.5	14.6	20.9	17.7	16.2	15.3	21.9	18.5	16.9	15.9	23	19.3	17.7	16.6	24	20.2	18.4	17.3	24.9	21	19.1	17.9
	40	16.2	14	13	12.3	17.2	14.8	13.6	12.9	18.1	15.5	14.3	13.5	19.1	16.3	15	14.2	20.1	17.1	15.7	14.8	21.1	17.9	16.4	15.4	22.1	18.7	17.1	16.1	23.1	19.5	17.8	16.7	24.1	20.3	18.5	17.4	25	21.1	19.3	18.1
	60	16.4	14.2	13.1	12.4	17.3	14.9	13.8	13.1	18.3	15.7	14.5	13.7	19.2	16.5	15.2	14.3	20.2	17.2	15.8	15	21.2	18	16.5	15.6	22.2	18.8	17.3	16.2	23.2	19.6	18	16.9	24.3	20.5	18.7	17.6	25.2	21.3	19.4	18.2
	80	16.5	14.3	13.3	12.6	17.4	15.1	13.9	13.2	18.4	15.8	14.6	13.9	19.4	16.6	15.3	14.5	20.3	17.4	16	15.1	21.3	18.2	16.7	15.8	22.3	19	17.4	16.4	23.4	19.8	18.1	17.1	24.4	20.6	18.9	17.7	25.4	21.4	19.6	18.4
16	20	17	15.1	14.2	13.7	17.9	15.9	14.9	14.3	18.9	16.6	15.6	14.9	19.8	17.4	16.2	15.5	20.8	18.2	16.9	16.1	21.8	18.9	17.6	16.8	22.8	19.7	18.3	17.4	23.8	20.5	19	18.1	24.7	21.4	19.8	18.7	25.6	22.2	20.5	19.4
	40	17.2	15.3	14.4	13.9	18.1	16	15.1	14.5	19	16.8	15.7	15.1	20	17.6	16.4	15.7	21	18.3	17.1	16.3	22	19.1	17.8	17	23	19.9	18.5	17.6	24	20.7	19.2	18.3	24.9	21.5	19.9	18.9	25.8	22.4	20.7	19.6
	60	17.3	15.5	14.6	14	18.2	16.2	15.3	14.7	19.2	17	15.9	15.3	20.1	17.7	16.6	15.9	21.1	18.5	17.3	16.5	22.1	19.3	18	17.2	23.1	20.1	18.7	17.8	24.1	20.9	19.4	18.4	25.1	21.7	20.1	19.1	26.1	22.5	20.8	19.8
	80	17.5	15.6	14.8	14.2	18.4	16.4	15.4	14.8	19.3	17.1	16.1	15.5	20.3	17.9	16.8	16.1	21.3	18.7	17.5	16.7	22.3	19.5	18.2	17.3	23.3	20.3	18.9	18	24.3	21.1	19.6	18.6	25.4	21.9	20.3	19.3	26.4	22.7	21	19.9
18	20	17.9	16.4	15.7	15.2	18.8	17.1	16.3	15.8	19.8	17.8	17	16.4	20.7	18.6	17.6	17	21.7	19.4	18.3	17.7	22.7	20.2	19	18.3	23.7	20.9	19.7	18.9	24.6	21.8	20.4	19.6	25.4	22.6	21.1	20.2	26.3	23.4	21.8	20.9
	40	18.1	16.5	15.9	15.4	19	17.3	16.5	16	19.9	18	17.2	16.6	20.9	18.8	17.8	17.2	21.9	19.6	18.5	17.9	22.9	20.4	19.2	18.5	23.9	21.1	19.9	19.1	24.8	21.9	20.6	19.8	25.7	22.8	21.3	20.4	26.6	23.6	22	21.1
	60	18.2	16.7	16.1	15.6	19.2	17.5	16.7	16.2	20.1	18.2	17.4	16.8	21.1	19	18.1	17.4	22	19.8	18.7	18.1	23	20.5	19.4	18.7	24.1	21.3	20.1	19.3	25	22.1	20.8	20	26	23	21.5	20.6	26.9	23.8	22.2	21.3
	80	18.4	16.9	16.3	15.8	19.3	17.7	16.9	16.4	20.3	18.4	17.6	17	21.2	19.2	18.3	17.7	22.2	20	18.9	18.3	23.2	20.7	19.6	18.9	24.3	21.5	20.3	19.5	25.4	22.3	21	20.2	26.4	23.1	21.7	20.8	27.4	24	22.4	21.5
20	20	18.8	17.6	17.1	16.7	19.7	18.3	17.7	17.3	20.6	19.1	18.4	17.9	21.6	19.8	19	18.5	22.6	20.6	19.7	19.1	23.6	21.4	20.4	19.8	24.5	22.2	21.1	20.4	25.3	23	21.8	21	26.1	23.8	22.5	21.7	27	24.4	23.2	22.3
	40	19	17.8	17.3	17	19.9	18.6	17.9	17.6	20.8	19.3	18.6	18.2	21.8	20.1	19.3	18.8	22.8	20.8	19.9	19.4	23.8	21.6	20.6	20	24.7	22.4	21.3	20.6	25.6	23.2	22	21.3	26.5	23.9	22.7	21.9	27.3	24.6	23.4	22.5
	60	19.2	18	17.5	17.2	20.1	18.8	18.2	17.8	21	19.5	18.8	18.4	22	20.3	19.5	19	23	21	20.2	19.6	24	21.8	20.8	20.2	25	22.6	21.5	20.9	26	23.4	22.2	21.5	26.9	24.1	22.9	22.1	27.8	24.9	23.6	22.8
	80	19.4	18.3	17.8	17.5	20.3	19	18.4	18	21.2	19.7	19.1	18.6	22.2	20.5	19.7	19.2	23.2	21.3	20.4	19.8	24.3	22	21.1	20.5	25.4	22.8	21.8	21.1	26.4	23.6	22.4	21.7	27.4	24.4	23.1	22.3	28.4	25.2	23.8	23
22	20	19.7	18.8	18.5	18.3	20.6	19.6	19.1	18.8	21.5	20.3	19.8	19.4	22.5	21	20.4	20	23.5	21.8	21.1	20.6	24.3	22.6	21.8	21.2	25.2	23.4	22.4	21.9	26	24	23.1	22.5	26.8	24.7	23.8	23.1	27.6	25.3	24.3	23.7
	40	19.9	19.1	18.7	18.5	20.8	19.8	19.4	19.1	21.8	20.5	20	19.7	22.7	21.3	20.7	20.3	23.7	22.1	21.3	20.9	24.6	22.8	22	21.5	25.5	23.6	22.7	22.1	26.4	24.3	23.4	22.7	27.3	24.9	24	23.4	28.1	25.6	24.6	23.9
	60	20.1	19.3	19	18.8	21	20.1	19.6	19.4	22	20.8	20.3	20	22.9	21.5	20.9	20.5	24	22.3	21.6	21.1	25	23.1	22.3	21.8	26	23.8	22.9	22.4	26.9	24.6	23.6	23	27.9	25.3	24.3	23.6	28.8	26	24.9	24.2
	80	20.3	19.6	19.3	19.1	21.3	20.3	19.9	19.6	22.2	21	20.5	20.2	23.2	21.8	21.2	20.8	24.4	22.6	21.8	21.4	25.5	23.3	22.5	22	26.6	24.2	23.2	22.6	27.6	25	23.9	23.2	28.6	25.8	24.6	23.9	29.6	26.6	25.3	24.5
24	20	20.6	20.1	19.9	19.8	21.5	20.8	20.5	20.3	22.4	21.5	21.1	20.9	23.4	22.3	21.8	21.5	24.3	23	22.4	22.1	25.1	23.7	23.1	22.7	25.9	24.4	23.8	23.3	26.7	25	24.3	23.9	27.5	25.6	24.8	24.3	28.3	26.2	25.3	24.8

Table 6 (*Continued*)

Standard Effective Temperature for Activity Level 1 Met (Office Work Type Activity) and Clothing 1.0 Clo (Winter Clothing)

AT (°C) ↓	MRT (°C) →	14				16				18				20				22				24				26				28				30				32			
	AV (m/s) → RH (percent) ↓	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4
	40	20.8	20.3	20.2	20.1	21.7	21.1	20.8	20.6	22.7	21.8	21.4	21.2	23.6	22.5	22.1	21.8	24.6	23.3	22.7	22.4	25.5	24	23.4	23	26.4	24.7	24	23.6	27.3	25.3	24.6	24.1	28.1	26	25.1	24.6	28.9	26.6	25.7	25.1
	60	21.1	20.6	20.5	20.4	22	21.4	21.1	20.9	22.9	22.1	21.7	21.5	24	22.8	22.4	22.1	25	23.6	23	22.7	26	24.3	23.7	23.3	27	25.1	24.3	23.8	28	25.8	24.9	24.4	28.9	26.5	25.6	25	29.8	27.2	26.2	25.6
	80	21.3	20.9	20.7	20.6	22.3	21.6	21.4	21.2	23.3	22.4	22	21.8	24.5	23.1	22.7	22.4	25.7	24	23.3	23	26.8	24.9	24.1	23.6	27.9	25.7	24.8	24.3	28.9	26.5	25.5	24.9	29.9	27.3	26.2	25.6	30.8	28.1	26.9	26.2
26	20	21.5	21.3	21.3	21.3	22.4	22	21.9	21.8	23.3	22.7	22.5	22.4	24.2	23.5	23.2	23	25	24.1	23.7	23.5	25.8	24.7	24.2	24	26.6	25.3	24.8	24.5	27.4	25.9	25.3	24.9	28.2	26.5	25.8	25.4	29	27.1	26.3	25.9
	40	21.7	21.6	21.6	21.6	22.7	22.3	22.2	22.1	23.6	23.1	22.8	22.7	24.6	23.7	23.5	23.3	25.5	24.4	24	23.8	26.4	25.1	24.6	24.3	27.3	25.7	25.2	24.8	28.1	26.4	25.7	25.3	29	27	26.3	25.8	29.8	27.7	26.8	26.3
	60	22	21.9	21.9	21.9	23	22.6	22.5	22.5	24.1	23.4	23.2	23	25.1	24.2	23.8	23.6	26.2	24.9	24.5	24.2	27.2	25.7	25.1	24.8	28.1	26.4	25.7	25.3	29.1	27.1	26.4	25.9	29.9	27.8	27	26.5	30.8	28.5	27.6	27
	80	22.3	22.2	22.2	22.2	23.5	23	22.9	22.8	24.8	23.9	23.6	23.5	26	24.8	24.4	24.1	27.2	25.7	25.1	24.8	28.3	26.5	25.9	25.5	29.3	27.4	26.6	26.1	30.3	28.2	27.3	26.8	31.3	28.9	28	27.4	32.2	29.7	28.7	28
28	20	22.4	22.5	22.6	22.7	23.3	23.2	23.3	23.3	24.1	23.8	23.8	23.7	25	24.4	24.3	24.2	25.8	25	24.8	24.6	26.6	25.6	25.3	25.1	27.4	26.2	25.8	25.5	28.1	26.8	26.3	26	28.9	27.3	26.8	26.4	29.7	27.9	27.3	26.9
	40	22.7	22.9	23	23.1	23.7	23.6	23.6	23.6	24.6	24.2	24.1	24.1	25.5	24.9	24.7	24.6	26.4	25.6	25.3	25.1	27.3	26.2	25.8	25.6	28.2	26.8	26.4	26.1	29	27.5	26.9	26.6	29.8	28.1	27.5	27.1	30.6	28.7	28	27.6
	60	23.1	23.3	23.4	23.5	24.2	24.1	24.1	24.1	25.3	24.8	24.7	24.7	26.4	25.6	25.4	25.2	27.4	26.4	26	25.8	28.4	27.1	26.6	26.4	29.3	27.8	27.2	26.9	30.2	28.5	27.9	27.5	31.1	29.2	28.5	28	31.9	29.9	29.1	28.6
	80	23.8	24	24.1	24.2	25.2	25	24.9	24.9	26.5	25.9	25.7	25.6	27.8	26.8	26.5	26.3	28.9	27.6	27.2	27	30	28.5	27.9	27.6	31	29.3	28.6	28.2	32	30	29.3	28.9	32.9	30.8	30	29.5	33.7	31.5	30.6	30.1
30	20	23.3	23.6	23.8	24	24.1	24.2	24.3	24.4	24.9	24.8	24.8	24.8	25.8	25.4	25.3	25.3	26.6	26	25.8	25.7	27.3	26.5	26.3	26.2	28.1	27.1	26.8	26.6	28.9	27.7	27.3	27	29.6	28.2	27.7	27.5	30.3	28.8	28.2	27.9
	40	23.7	24.1	24.3	24.5	24.7	24.8	24.9	25	25.6	25.4	25.4	25.4	26.5	26.1	26	25.9	27.4	26.7	26.5	26.4	28.3	27.4	27.1	26.9	29.2	28	27.6	27.4	30	28.6	28.1	27.9	30.8	29.2	28.7	28.4	31.5	29.8	29.2	28.8
	60	24.5	24.9	25.1	25.3	25.6	25.7	25.8	25.9	26.7	26.4	26.4	26.4	27.8	27.2	27	27	28.8	27.9	27.7	27.5	29.8	28.6	28.3	28.1	30.7	29.3	28.9	28.6	31.5	30	29.5	29.2	32.4	30.7	30	29.7	33.2	31.3	30.6	30.2
	80	25.9	26.4	26.6	26.8	27.3	27.3	27.4	27.5	28.7	28.2	28.2	28.1	29.9	29.1	28.9	28.8	31	29.9	29.6	29.4	32	30.7	30.3	30	33	31.5	30.9	30.6	33.9	32.2	31.6	31.2	34.7	32.9	32.2	31.8	35.5	33.5	32.8	32.3
32	20	24.1	24.6	24.9	25.1	24.9	25.2	25.4	25.5	25.7	25.8	25.9	25.9	26.5	26.3	26.3	26.4	27.3	26.9	26.8	26.8	28.1	27.5	27.3	27.2	28.8	28	27.8	27.6	29.6	28.6	28.2	28.1	30.3	29.1	28.7	28.5	31	29.6	29.2	28.9
	40	24.8	25.4	25.7	25.9	25.8	26	26.2	26.4	26.7	26.7	26.8	26.9	27.6	27.3	27.3	27.3	28.5	28	27.8	27.8	29.3	28.6	28.4	28.3	30.2	29.2	28.9	28.7	31	29.8	29.4	29.2	31.7	30.4	29.9	29.7	32.5	30.9	30.4	30.1
	60	26.1	26.7	27	27.3	27.2	27.5	27.7	27.8	28.3	28.2	28.3	28.4	29.4	28.9	28.9	28.9	30.3	29.7	29.5	29.4	31.3	30.3	30.1	30	32.2	31	30.7	30.5	33	31.6	31.2	31	33.8	32.3	31.8	31.5	34.5	32.9	32.3	32
	80	28.7	29.3	29.7	29.9	30.1	30.2	30.4	30.5	31.4	31.1	31.1	31.2	32.5	31.9	31.8	31.8	33.5	32.6	32.4	32.3	34.5	33.4	33	32.9	35.3	34	33.6	33.4	35.6	34.7	34.2	33.9	36.1	35.3	34.7	34.4	36.7	35.3	34.7	34.9

Table 6 (*Continued*)

Standard Effective Temperature for Activity Level 1 Met (Office Work Type Activity) and Clothing 0.5 Clo (Summer Clothing)

AT (°C) ↓	MRT (°C) →	14				16				18				20				22				24				26				28				30				32			
	AV (m/s) → RH (percent) ↓	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4
14	20	11	7.2	5.3	4	12.1	8.2	6.2	4.9	13.3	9.1	7.1	5.7	14.4	10.1	8	6.5	15.6	11.1	8.9	7.4	16.8	12.2	9.8	8.3	18	13.2	10.7	9.1	19.2	14.2	11.7	10	20.4	15.2	12.6	10.9	21.6	16.3	13.6	11.8
	40	11.1	7.4	5.5	4.3	12.3	8.4	6.4	5.1	13.4	9.4	7.3	5.9	14.6	10.3	8.2	6.8	15.8	11.3	9.1	7.6	17	12.4	10	8.5	18.1	13.4	11	9.4	19.3	14.4	11.9	10.2	20.5	15.4	12.8	11.1	21.7	16.5	13.8	12
	60	11.3	7.6	5.8	4.5	12.4	8.6	6.6	5.4	13.6	9.6	7.5	6.2	14.8	10.6	8.4	7	15.9	11.6	9.3	7.9	17.1	12.6	10.3	8.7	18.3	13.6	11.2	9.6	19.5	14.6	12.1	10.5	20.7	15.6	13.1	11.3	21.9	16.7	14	12.2
	80	11.5	7.8	6	4.8	12.6	8.8	6.9	5.6	13.8	9.8	7.8	6.4	14.9	10.8	8.7	7.3	16.1	11.8	9.6	8.1	17.3	12.8	10.5	9	18.4	13.8	11.4	9.8	19.6	14.8	12.3	10.7	20.8	15.8	13.3	11.6	22	16.9	14.2	12.5
16	20	12.1	8.8	7.2	6.1	13.2	9.8	8.1	6.9	14.3	10.7	8.9	7.8	15.5	11.7	9.8	8.6	16.7	12.7	10.7	9.4	17.8	13.7	11.6	10.3	19	14.7	12.6	11.1	20.2	15.7	13.5	12	21.4	16.7	14.4	12.9	22.6	17.7	15.3	13.7
	40	12.2	9	7.5	6.4	13.4	10	8.3	7.2	14.5	11	9.2	8	15.7	11.9	10.1	8.9	16.8	12.9	11	9.7	18	13.9	11.9	10.5	19.2	14.9	12.8	11.4	20.4	15.9	13.7	12.2	21.5	17	14.7	13.1	22.7	18	15.6	14
	60	12.4	9.3	7.7	6.7	13.6	10.2	8.6	7.5	14.7	11.2	9.5	8.3	15.9	12.2	10.3	9.1	17	13.2	11.2	10	18.2	14.2	12.1	10.8	19.3	15.2	13.1	11.7	20.5	16.2	14	12.5	21.7	17.2	14.9	13.4	22.9	18.2	15.8	14.2
	80	12.6	9.5	8	7	13.7	10.5	8.8	7.8	14.9	11.4	9.7	8.6	16	12.4	10.6	9.4	17.2	13.4	11.5	10.2	18.3	14.4	12.4	11.1	19.5	15.4	13.3	11.9	20.7	16.4	14.2	12.8	21.9	17.4	15.1	13.6	23	18.4	16.1	14.5
18	20	13.1	10.4	9.1	8.2	14.3	11.3	9.9	9	15.4	12.3	10.8	9.8	16.5	13.3	11.7	10.6	17.7	14.3	12.6	11.4	18.8	15.2	13.4	12.3	20	16.2	14.3	13.1	21.2	17.2	15.2	13.9	22.3	18.2	16.2	14.8	23.5	19.2	17.1	15.6
	40	13.3	10.7	9.4	8.5	14.5	11.6	10.2	9.3	15.6	12.6	11.1	10.1	16.7	13.5	12	10.9	17.9	14.5	12.8	11.7	19	15.5	13.7	12.6	20.2	16.5	14.6	13.4	21.4	17.5	15.5	14.2	22.5	18.4	16.4	15.1	23.7	19.4	17.3	15.9
	60	13.6	10.9	9.7	8.8	14.7	11.9	10.5	9.6	15.8	12.8	11.4	10.4	16.9	13.8	12.2	11.2	18.1	14.8	13.1	12	19.2	15.7	14	12.9	20.4	16.7	14.9	13.7	21.6	17.7	15.8	14.5	22.7	18.7	16.7	15.4	23.9	19.7	17.6	16.2
	80	13.8	11.2	9.9	9.1	14.9	12.2	10.8	9.9	16	13.1	11.7	10.7	17.1	14.1	12.5	11.5	18.3	15	13.4	12.3	19.4	16	14.3	13.2	20.6	17	15.2	14	21.7	17.9	16.1	14.8	22.9	18.9	17	15.6	24	19.9	17.8	16.5
20	20	14.2	12	10.9	10.2	15.3	12.9	11.8	11	16.4	13.9	12.6	11.8	17.6	14.8	13.5	12.6	18.7	15.8	14.3	13.4	19.9	16.7	15.2	14.2	21	17.7	16.1	15	22.2	18.7	17	15.9	23.3	19.6	17.9	16.7	24.3	20.6	18.7	17.5
	40	14.4	12.3	11.3	10.6	15.6	13.2	12.1	11.4	16.7	14.2	12.9	12.2	17.8	15.1	13.8	12.9	18.9	16.1	14.7	13.7	20.1	17	15.5	14.6	21.2	18	16.4	15.4	22.4	18.9	17.3	16.2	23.5	19.9	18.1	17	24.5	20.8	19	17.8
	60	14.7	12.6	11.6	10.9	15.8	13.5	12.4	11.7	16.9	14.5	13.3	12.5	18	15.4	14.1	13.3	19.2	16.3	15	14.1	20.3	17.3	15.8	14.9	21.4	18.2	16.7	15.7	22.6	19.2	17.6	16.5	23.7	20.1	18.4	17.3	24.8	21.1	19.3	18.1
	80	14.9	12.9	11.9	11.3	16	13.8	12.7	12	17.1	14.8	13.6	12.8	18.3	15.7	14.4	13.6	19.4	16.6	15.3	14.4	20.5	17.6	16.1	15.2	21.6	18.5	17	16	22.8	19.5	17.9	16.8	23.9	20.4	18.7	17.6	25.1	21.3	19.6	18.4
22	20	15.3	13.6	12.8	12.3	16.4	14.5	13.6	13	17.5	15.4	14.4	13.8	18.6	16.3	15.3	14.6	19.7	17.3	16.1	15.4	20.9	18.2	17	16.1	22	19.2	17.8	16.9	23.1	20.1	18.7	17.7	24.2	21	19.5	18.5	25.1	22	20.3	19.3
	40	15.5	13.9	13.1	12.6	16.6	14.8	14	13.4	17.7	15.7	14.8	14.2	18.9	16.7	15.6	14.9	20	17.6	16.5	15.7	21.1	18.5	17.3	16.5	22.2	19.5	18.1	17.3	23.4	20.4	19	18.1	24.4	21.3	19.8	18.8	25.3	22.2	20.6	19.6
	60	15.8	14.2	13.5	13	16.9	15.1	14.3	13.8	18	16.1	15.1	14.5	19.1	17	16	15.3	20.2	17.9	16.8	16.1	21.3	18.8	17.6	16.9	22.5	19.7	18.5	17.6	23.6	20.7	19.3	18.4	24.7	21.6	20.1	19.2	25.7	22.5	20.9	19.9
	80	16.1	14.6	13.9	13.4	17.2	15.5	14.7	14.2	18.3	16.4	15.5	14.9	19.4	17.3	16.3	15.7	20.5	18.2	17.1	16.4	21.6	19.1	18	17.2	22.7	20	18.8	18	23.9	20.9	19.6	18.7	25	21.9	20.4	19.5	26.2	22.8	21.2	20.2
24	20	16.3	15.1	14.6	14.2	17.4	16	15.4	15	18.5	16.9	16.2	15.7	19.6	17.8	17	16.5	20.7	18.7	17.8	17.2	21.9	19.7	18.6	18	23	20.6	19.5	18.8	24	21.5	20.3	19.5	24.9	22.4	21.1	20.3	25.8	23.3	21.9	21

Table 6 (*Continued*)

Standard Effective Temperature for Activity Level 1 Met (Office Work Type Activity) and Clothing 0.5 Clo (Summer Clothing)

AT (°C) ↓	MRT (°C) →	14				16				18				20				22				24				26				28				30				32			
	AV (m/s) → RH (percent) ↓	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4	0.2	0.6	1	1.4
	40	16.6	15.5	15	14.7	17.7	16.4	15.8	15.4	18.8	17.3	16.6	16.1	19.9	18.2	17.4	16.9	21	19.1	18.2	17.6	22.1	20	19	18.4	23.2	20.9	19.8	19.1	24.2	21.8	20.6	19.9	25.2	22.7	21.4	20.6	26.2	23.6	22.2	21.3
	60	16.9	15.9	15.4	15.1	18	16.7	16.2	15.8	19.1	17.6	17	16.5	20.2	18.5	17.8	17.3	21.3	19.4	18.6	18	22.4	20.3	19.4	18.7	23.5	21.2	20.2	19.5	24.6	22.1	20.9	20.2	25.7	23	21.7	20.9	26.7	23.8	22.5	21.7
	80	17.2	16.2	15.8	15.5	18.3	17.1	16.6	16.2	19.4	18	17.3	16.9	20.5	18.9	18.1	17.7	21.6	19.7	18.9	18.4	22.7	20.6	19.7	19.1	23.8	21.5	20.5	19.8	25.1	22.4	21.3	20.6	26.3	23.3	22	21.3	27.4	24.1	22.8	22
26	20	17.4	16.6	16.4	16.2	18.5	17.5	17.1	16.9	19.6	18.4	17.9	17.6	20.6	19.3	18.7	18.3	21.7	20.2	19.5	19.1	22.8	21.1	20.3	19.8	23.9	21.9	21	20.5	24.8	22.8	21.8	21.2	25.6	23.7	22.6	21.9	26.5	24.3	23.4	22.6
	40	17.7	17.1	16.8	16.6	18.8	17.9	17.6	17.3	19.9	18.8	18.3	18	21	19.7	19.1	18.8	22	20.5	19.9	19.5	23.1	21.4	20.6	20.2	24.2	22.3	21.4	20.9	25.1	23.1	22.2	21.6	26.1	23.9	22.9	22.3	27.1	24.6	23.7	23
	60	18	17.4	17.2	17.1	19.1	18.3	18	17.8	20.2	19.2	18.7	18.5	21.3	20	19.5	19.2	22.3	20.9	20.3	19.9	23.4	21.8	21	20.6	24.6	22.6	21.8	21.3	25.7	23.4	22.5	21.9	26.7	24.2	23.3	22.6	27.8	25	23.9	23.3
	80	18.4	17.8	17.6	17.5	19.4	18.7	18.4	18.2	20.5	19.5	19.1	18.9	21.6	20.4	19.9	19.6	22.6	21.2	20.6	20.2	23.9	22.1	21.4	20.9	25.2	22.9	22.1	21.6	26.5	23.8	22.9	22.3	27.7	24.8	23.6	23	28.8	25.7	24.4	23.6
28	20	18.4	18.1	18.1	18	19.5	19	18.8	18.7	20.6	19.8	19.6	19.4	21.7	20.7	20.3	20.1	22.7	21.6	21.1	20.8	23.8	22.4	21.8	21.5	24.6	23.2	22.6	22.1	25.5	24	23.3	22.8	26.4	24.6	23.9	23.5	27.2	25.2	24.5	24
	40	18.8	18.6	18.5	18.5	19.9	19.4	19.3	19.2	20.9	20.3	20	19.9	22	21.1	20.7	20.5	23.1	21.9	21.5	21.2	24.1	22.8	22.2	21.9	25.1	23.6	22.9	22.5	26.1	24.3	23.6	23.2	27	25	24.2	23.8	27.9	25.7	24.8	24.3
	60	19.2	19	19	19	20.2	19.8	19.7	19.6	21.3	20.7	20.4	20.3	22.3	21.5	21.2	21	23.4	22.3	21.9	21.6	24.6	23.1	22.6	22.3	25.8	24	23.3	22.9	26.9	24.8	24	23.5	27.9	25.6	24.7	24.1	28.9	26.4	25.4	24.8
	80	19.5	19.4	19.4	19.4	20.6	20.2	20.1	20.1	21.6	21.1	20.8	20.7	22.7	21.9	21.6	21.4	24.1	22.7	22.3	22	25.5	23.6	23	22.6	26.8	24.6	23.8	23.3	28.1	25.6	24.6	24.1	29.3	26.5	25.4	24.8	30.4	27.4	26.2	25.5
30	20	19.5	19.6	19.7	19.8	20.5	20.4	20.4	20.5	21.6	21.3	21.2	21.1	22.6	22.1	21.9	21.8	23.7	22.9	22.6	22.4	24.5	23.6	23.3	23.1	25.4	24.3	23.9	23.6	26.3	24.9	24.4	24.1	27.1	25.5	24.9	24.6	28	26.1	25.5	25.1
	40	19.9	20.1	20.2	20.3	20.9	20.9	20.9	21	22	21.7	21.6	21.6	23	22.5	22.3	22.2	24	23.3	23	22.8	25.1	24	23.6	23.4	26.1	24.7	24.2	24	27	25.4	24.8	24.5	28	26.1	25.4	25	28.9	26.8	26	25.6
	60	20.3	20.5	20.7	20.8	21.3	21.3	21.4	21.4	22.3	22.1	22.1	22	23.5	22.9	22.7	22.6	24.7	23.7	23.4	23.2	25.9	24.6	24.1	23.9	27.1	25.4	24.8	24.5	28.2	26.2	25.5	25.1	29.2	27	26.2	25.8	30.2	27.8	26.9	26.4
	80	20.7	21	21.1	21.2	21.7	21.8	21.8	21.9	22.7	22.5	22.5	22.5	24.4	23.6	23.3	23.1	26	24.7	24.2	24	27.4	25.7	25.1	24.8	28.8	26.7	26	25.5	30	27.7	26.8	26.3	31.2	28.6	27.6	27	32.3	29.4	28.4	27.7
32	20	20.5	21	21.3	21.5	21.5	21.8	22	22.1	22.6	22.6	22.7	22.7	23.6	23.4	23.3	23.3	24.5	24	23.8	23.8	25.3	24.6	24.4	24.2	26.2	25.2	24.9	24.7	27.1	25.8	25.4	25.2	27.9	26.4	25.9	25.6	28.7	27	26.4	26.1
	40	21	21.5	21.8	22	22	22.3	22.5	22.6	23	23.1	23.1	23.2	24.1	23.8	23.7	23.7	25.1	24.5	24.3	24.2	26.1	25.2	24.9	24.8	27.1	25.9	25.5	25.3	28	26.6	26.1	25.8	29	27.3	26.7	26.3	29.9	27.9	27.3	26.9
	60	21.4	22	22.3	22.5	22.4	22.7	22.9	23.1	23.6	23.6	23.7	23.7	25	24.5	24.4	24.4	26.2	25.4	25.1	25	27.4	26.2	25.9	25.7	28.6	27	26.5	26.3	29.6	27.8	27.2	26.9	30.7	28.6	27.9	27.5	31.6	29.4	28.6	28.1
	80	21.8	22.4	22.9	23.3	23.1	23.7	24	24.2	25.1	25	25	25.1	26.9	26.2	26	25.9	28.5	27.3	26.9	26.7	29.9	28.3	27.8	27.5	31.2	29.3	28.6	28.2	32.4	30.2	29.4	29	33.5	31.1	30.2	29.7	34.5	31.9	30.9	30.3

NOTE — AT: Dry bulb temperature of indoor air, AV: Air velocity around occupant, MET: Metabolic rate, Clo: Thermal resistance due to clothing, MRT: Mean radiant temperature

Table 7 Minimum Ventilation Rates in Breathing Zone (*See* Notes 1 to 5)

(*Clause* 3.6.1.2)

(This table is not valid in isolation; it shall be used in conjunction with the accompanying notes)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
i)	Correctional facilities:						
	a) Cell	2.5	0.6	–	25	4.9	2
	b) Dayroom	2.5	0.3	–	30	3.5	1
	c) Guard Stations	2.5	0.3	–	15	4.5	1
	d) Booking/waiting	3.8	0.3	–	50	4.4	2
ii)	Educational facilities:						
	a) Daycare (through age 4)	5	0.9	–	25	8.6	2
	b) Daycare sickroom	5	0.9	–	25	8.6	3
	c) Classrooms (ages 5-8)	5	0.6	–	25	7.4	1
	d) Classrooms (age 9 plus)	5	0.6	–	35	6.7	1
	e) Lecture classroom	3.8	0.3	–	65	4.3	1
	f) Lecture hall (fixed seats)	3.8	0.3	–	150	4.0	1
	g) Art classroom	5	0.9	–	20	9.5	2
	h) Science laboratories	5	0.9	–	25	8.6	2
	j) University/college laboratories	5	0.9	–	25	8.6	2

Table 7 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1)] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
	k) Wood/metal shop	5	0.9	–	20	9.5	2
	m) Computer lab	5	0.6	–	25	7.4	1
	n) Media centre	5	0.6	*See* Note 6	25	7.4	1
	p) Music/theatre/dance	5	0.3	–	35	5.9	1
	q) Multi-use assembly	3.8	0.3	–	100	4.1	1
iii)	Food and beverage service:						
	a) Restaurant dining rooms	3.8	0.9	–	70	5.1	2
	b) Cafeteria/fast-food dining	3.8	0.9	–	100	4.7	2
	c) Bars, cocktail lounges	3.8	0.9	–	100	4.7	2
iv)	General:						
	a) Break rooms	2.5	0.3	–	25	5.1	1
	b) Coffee stations	2.5	0.3	–	20	5.5	1
	c) Conference/meeting	2.5	0.3	–	50	3.1	1
	d) Corridors	-	0.3	–	-	-	1
	e) Storage rooms	-	0.6	*See* Note 7	-	-	1

Table 7 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1)] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
v)	Hotels, motels, resorts, dormitories:						
	a) Bedroom/living room	2.5	0.3	–	10	5.5	1
	b) Barracks sleeping areas	2.5	0.3	–	20	4.0	1
	c) Laundry rooms, central	2.5	0.6	–	10	8.5	2
	d) Laundry rooms within dwelling units	2.5	0.6	–	10	8.5	1
	e) Lobbies/pre-function	3.8	0.3	–	30	4.8	1
	f) Multipurpose assembly	2.5	0.3	–	120	2.8	1
vi)	Office buildings:						
	a) Office space	2.5	0.3	–	5	8.5	1
	b) Reception Areas	2.5	0.3	–	30	3.5	1
	c) Telephone/data entry	2.5	0.3	–	60	3.0	1
	d) Main entry lobbies	2.5	0.3	–	10	5.5	1
vii)	Miscellaneous spaces:						
	a) Bank vaults/safe deposit	2.5	0.3	–	5	8.5	2
	b) Computer (not printing)	2.5	0.3	–	4	10.0	1

Table 7 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
	c) Electrical equipment rooms	–	0.3	*See* Note 7	–	–	1
	d) Elevator machine rooms	–	0.6	*See* Note 7	–	–	1
	e) Pharmacy (preparation area)	2.5	0.9	–	10	11.5	2
	f) Photo studios	2.5	0.6	–	10	8.5	1
	g) Shipping/receiving	–	0.6	*See* Note 7	–	–	1
	h) Telephone closets	–	0.0	–	–	–	1
	j) Transportation waiting	3.8	0.3	–	100	4.1	1
	k) Warehouses	–	0.3	*See* Note 7	–	–	2
viii)	Public assembly spaces:						
	a) Auditorium seating area	2.5	0.3	–	150	2.7	1
	b) Places of religious worship	2.5	0.3	–	120	2.8	1
	c) Courtrooms	2.5	0.3	–	70	2.9	1
	d) Legislative chambers	2.5	0.3	–	50	3.1	1
	e) Libraries	2.5	0.6	–	10	8.5	1

Table 7 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
	f) Lobbies	2.5	0.3	–	150	2.7	1
	g) Museums (children’s)	3.8	0.6	–	40	5.3	1
	h) Museums/galleries	3.8	0.3	–	40	4.6	1
ix)	Residential:						
	a) Dwelling unit	2.5	0.3	*See* Notes 8 and 9	*See* Note 8	–	1
	b) Common corridors	–	0.3	–	–	–	1
x)	Retail:						
	a) Sales (except as below)	3.8	0.6	–	15	7.8	2
	b) Mall common areas	3.8	0.3	–	40	4.6	1
	c) Barbershop	3.8	0.3	–	25	5.0	2
	d) Beauty and nail salons	10	0.6	–	25	12.4	2
	e) Pet shops (animal areas)	3.8	0.9	–	10	12.8	2
	f) Supermarket	3.8	0.3	–	8	7.6	1
	g) Coin-operated laundries	3.8	0.3	–	20	5.3	2
xi)	Sports and entertainment:						
	a) Sports arena (play area)	–	1.5	*See* Note 10	–	–	1

Table 7 (*Continued*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, *R*p**	**Area Outdoor Air Rate, *R*a**	**Notes**	**Default Values**		**Air[1)] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)
	b) Gym, stadium (play area)	–	1.5	–	30	–	2
	c) Spectator areas	3.8	0.3	–	150	4.0	1
	d) Swimming (pool and deck)	–	2.4	*See* Note 11	–	–	2
	e) Disco/dance floors	10	0.3	–	100	10.3	1
	f) Health club/aerobics room	10	0.3	–	40	10.8	2
	g) Health club/weight rooms	10	0.3	–	10	13.0	2
	h) Bowling alley (seating)	5	0.6	–	40	6.5	1
	j) Gambling casinos	3.8	0.9	–	120	4.6	1
	k) Game arcades	3.8	0.9	–	20	8.3	1
	m) Stages, studios	5	0.3	*See* Note 12	70	5.4	1

[1)] Air Class Characteristic

1 Air with low contaminant concentration, low sensory-irritation intensity, and inoffensive odour.

2 Air with moderate contaminant concentration, mild sensory-irritation intensity, or mildly offensive odours. Class 2 air also includes air that is not necessarily harmful or objectionable but that is inappropriate for transfer or recirculation to spaces used for different purposes.

3 Air with significant contaminant concentration, significant sensory-irritation intensity, or offensive odour.

4 Air with highly objectionable fumes or gases or with potentially dangerous particles, bio aerosols, or gases, at concentrations high enough to be considered harmful.

Table 7 (*Concluded*)

Sl No.	**Occupancy Category**	**People Outdoor Air Rate, R_p**	**Area Outdoor Air Rate, R_a**	**Notes**	**Default Values**		**Air[1] Class**
					Occupant Density (*see* Note 4)	Combined Outdoor Air Rate (*see* Note 4)	
		l/s.person	l/s.m^2		Persons per 100 m^2	l/s.person	
(1)	(2)	(4)	(6)	(7)	(8)	(10)	(11)

NOTES

1 The rates in this table are based on all other applicable requirements being met.

2 This table applies to no-smoking areas only. Rates for smoking-permitted spaces shall be determined using other methods.

3 Volumetric airflow rates are based on an air density of 1.2 kg of dry air/m^3, which corresponds to dry air at a barometric pressure of 1 atm (101.3 kPa) and an air temperature of 21°C. Rates may be adjusted for actual density but such adjustment is not required for compliance with this standard.

4 Actual occupant density should be considered, the default occupant density shall be used only when actual occupant density is not known. Default combined outdoor air (per person) rate is based on the default occupant density.

5 If the occupancy category for a proposed space or zone is not listed, the requirements for the listed occupancy category that is most similar in terms of occupant density, activities and building construction shall be used.

6 For high school and college libraries, use values shown for public assembly spaces- libraries.

7 The prescribed value may not be sufficient when stored materials include those having potentially harmful emissions.

8 Default occupancy for dwelling units shall be two persons for studio and one-bedroom units, with one additional person for each additional bedroom.

9 Air from one residential dwelling shall not be recirculated or transferred to any other space outside of that dwelling.

10 When combustion equipment is intended to be used on the playing surface, additional dilution ventilation and/or source control shall be provided.

11 The prescribed value does not allow for humidity control. Additional ventilation or dehumidification may be required to remove moisture.

12 The prescribed value does not include special exhaust for stage effects, for example, dry ice vapours, smoke.

3.6.2 *Designing for Air Quality*

The HVAC system shall be designed with adequate filtration for achieving following levels of indoor pollutants:

The threshold values for IAQ pollutants will be as mentioned in as per the Table 8 provided below. The buildings/indoor spaces will be classified as Class A, B and C where Class A refers to aspirational, Class B refers to acceptable and Class C refers to minimum acceptable.

Table 8 Threshold Values for IAQ Pollutants

(*Clause* 3.6.2)

Sl No.	Pollutants	Units	Classification		
			A	B	C
(1)	(2)	(3)	(4)	(5)	(6)
i)	CO_2	ppm	ambient + 350	ambient + 500	ambient + 700
ii)	$PM_{2.5}$	ug/m^3	< 15	< 25	< 25
iii)	PM_{10}	ug/m^3	< 50	< 100	< 100
iv)	O_3	ug/m^3	< 50	< 100	-
v)	TVOC (equivalent to isobutylene)	ug/m^3	< 200	< 400	< 500
vi)	CH_2O	ug/m^3	< 30	< 100	-
vii)	SO_2	ug/m^3	< 40	< 80	-
viii)	NO_2	ug/m^3	< 40	< 80	-
ix)	CO	ppm	< 2	< 9	< 9

NOTE — PM1.0 (sub-micron particles less than 1.0 µm in size) poses various health risks. PM1 particles may penetrate deep into the lungs and bloodstream, and prolonged exposure is associated with serious illnesses. Many bacteria and viruses are also within the PM1 size range, facilitating their spread indoors. The designers may consider filtration and air purification strategies that are effective in removing PM1 particles, thereby improving occupant well-being, productivity, and overall indoor air quality.

3.6.3 *Designing for Background Noise Control and Vibration*

The background noise and vibration created by air distribution through the HVAC system, combined with noise from equipment and outdoor sources of noise and vibration may causes significant disturbance to building users. Adoption of measures for adhering to indoor noise limits and vibration control should be ensured in the design of equipment, their foundation, supports and air distribution system.

4 HVAC SYSTEM TYPES

HVAC system requirements for the design, construction, installation, operation, replacement, repair, alteration, maintenance and end of life of mechanical cooling systems. It encourages the adoption of products designed with recyclability, reusability, and remanufacturing in mind, aiming to minimize waste, reduce environmental impacts, and promote resource efficiency throughout the entire lifecycle of the systems.

Following points to be considered for selection of cooling systems:

a) safeguard people from illness, injury or loss (including loss of amenity) due to the failure of air-conditioning installation;

b) it may be ensured that air-conditioning installation is suitable to meet the sizing requirement as per **3**;

c) conserve water and energy;

d) safeguard the environment;

e) safeguard public and private infrastructure; and

f) ensure that air-conditioning installation is designed and is capable of being maintained so that throughout its serviceable life it will continue to satisfy objectives (a) to (e).

Mechanical services, plant and equipment that are used for cooling, heating and/or ventilation of a building shall be adequate for the purpose meeting the basis of design. A building's HVAC system installation and maintenance shall support energy efficient outcomes and minimise any adverse impact on building occupants or occupants of adjoining places, the network utility operator's infrastructure, property and the environment.

Compliance with the requirement shall be verified either:

1) by satisfying the required criteria when tested in accordance with a specified test method endorsed by a recognised certification body; or

2) by calculation and certification by persons or organisations with recognised credentials in the testing of ventilation and air conditioning systems.

4.1 General

Systems for air conditioning may be broadly grouped as air-to-air, air-to-water, and liquid-to-liquid type. Suitability of system type shall not be generalized; hence it shall need to be building specific decision, considering initial cost, efficiency, maintenance, effect on building aesthetics, sound level, service life and other factors. Lower operating cost, central maintenance and control shall be of primary considerations.

It may be noted that central systems often require a higher initial cost than distributed systems but often result in annual energy savings. They also have a potential of reducing the total installed capacity by using load diversity. Over the life cycle, the centralized cooling systems usually prove to be more cost-effective when the total building load exceeds 350 kW (approximately 100 TR), depending on climate and patterns of occupancy use.

When low initial cost and simplicity are primary concerns, designers may select zone-by-zone distributed systems, incorporating either cooling or both heating and cooling facility. This approach shall normally be used for smaller buildings, however, larger buildings with multiple tenancy may also consider this as an option.

NOTES

1 Distributed systems, also known as unitary systems, individually consists of fan, cooling coil, compressor and outdoor condenser. Examples of distributed systems include unitary air conditioners, packaged rooftop air conditioners, variable refrigerant flows (VRFs), heat pumps, as well as refrigerant-based split-system fan-coil units (single or multi-unit).

2 Water-source heat pumps (WSHPs) are also considered as distributed systems in which the compressor is located close to the occupied space, but they are collectively served by a centralized water system with auxiliary heat rejection devices.

Comparative advantages, disadvantages and constraints of each option should be carefully evaluated before zeroing down to final HVAC system selection. Table 9 lists out various system characteristics of different HVAC systems; it may be used as a guiding tool for system selection.

Table 9 HVAC System Analysis and Selection Matrix

(*Clause* 4.1)

Sl No.	System Characteristics	Fixed Speed Unitary Systems (Window ACs/Split ACs/ Package ACs)	Variable Speed/Flow/ Frequency/capacity Unitary Systems (Split ACs/ Package ACs)	Variable Speed Multi or Modular Systems (VRF)	Central Systems
(1)	(2)	(3)	(4)	(5)	(6)
i)	Temperature	No uniform and effective control	Minimal uniform and effective control possible	Reasonably uniform and effective control possible	Uniform and effective control possible
ii)	Humidity	Effective control not possible	Effective control not possible	Effective control not possible	Effective control possible
iii)	Space pressure	Effective control not possible	Effective control not possible	Effective control not possible	Effective control possible
iv)	Capacity requirements	Capacity to suit zone peak, No diversity	Capacity to suit zone peak, No diversity	Capacity to suit zone peak, Limited diversity shall be considered	Allows the design engineer to consider HVAC diversity factors that reduce installed equipment capacity
v)	Redundancy	Does not have the benefit of back-up or standby equipment	Does not have the benefit of back-up or standby equipment	It has the benefit of partial back-up or standby equipment	Back-up or standby equipment shall easily be accommodated
vi)	Facility management	Allows minimal provision to maximize performance using good facility management techniques in operation	Allows minimal provision to maximize performance using good facility management techniques in operation	Limited possibilities to maximize performance using good facility management techniques in operation	Allows maximize performance using good facility management techniques in operation

Table 9 (*Continued*)

Sl No.	System Characteristics	Fixed Speed Unitary Systems (Window ACs/Split ACs/ Package ACs)	Variable Speed/Flow/ Frequency/capacity Unitary Systems (Split ACs/ Package ACs)	Variable Speed Multi or Modular Systems (VRF)	Central Systems
(1)	(2)	(3)	(4)	(5)	(6)
vii)	Spatial requirements	No plant/ equipment room required. Compromises building elevation	No plant/ equipment room required. Compromises building elevation	No plant/equipment room required. Outside units shall be located on roof or on adjacent ground. Very small depth but larger linear length (based on number of modular outdoor units) for shaft is required for refrigerant piping.	Equipment rooms/ spaces and accessible shaft required for chilled water piping
viii)	Electric supply	Distributed electric supply required	Distributed electric supply required	Zone wise distributed electric supply required	Minimal distribution cost by centralised supply near the substation
ix)	First cost	Minimum first cost	First cost marginally higher than fixed speed system	Moderate first cost but marginally higher than variable speed unitary product	Even with HVAC diversity, a central system may be of a bit higher or similar cost as a decentralized HVAC system
x)	Operating cost	Higher operating cost Strategic scheduling of multiple pieces of equipment shall save marginal operating cost. But equipment less efficient	Strategic scheduling of multiple pieces of equipment shall save reasonable operating cost. Higher peak energy requirement	Strategic scheduling of equipment shall save operating cost better than unitary product. Higher peak energy requirement	More energy-efficient primary equipment and multiple pieces of HVAC equipment allow staging of operation to match building loads while maximizing operational efficiency
xi)	Maintenance cost	Comparatively less maintenance cost	Maintenance cost marginally higher than fixed speed	Maintenance cost higher than both fixed and variable speed unitary system but less than central system	Comparatively higher since centralised equipment room requires operator with no access to occupant workspace, but with

Table 9 (*Concluded*)

Sl No.	System Characteristics	Fixed Speed Unitary Systems (Window ACs/Split ACs/ Package ACs)	Variable Speed/Flow/ Frequency/capacity Unitary Systems (Split ACs/ Package ACs)	Variable Speed Multi or Modular Systems (VRF)	Central Systems
(1)	(2)	(3)	(4)	(5)	(6)
					fewer pieces of HVAC equipment to service
xii)	Reliability	Reliable equipment but low service life	Reliable equipment but low service life	Reliable equipment but moderate service life	Reliable equipment with much longer service life
xiii)	Flexibility	Has to be placed at fixed locations	Has to be placed at fixed locations	Can be placed at distributed locations	Flexibility available in terms of alternative locations
xiv)	Level of control	Limited control level available	Limited control level available	Moderate control level available	Close control level available
xv)	Noise and vibration	Noise and vibration within/adjacent to occupied spaces for unitary window type air conditioners which is substantially reduced in case of split system	It is generally available in split system only for which noise and vibration is substantially reduced in occupied and adjacent spaces as compared to a window type air conditioner	Noise and vibration on roof terrace or ground away from occupied spaces, however the same shall be reduced with good installation practices	Noise and vibration away from occupied spaces
xvi)	Constructability	Multiple and similar-in-size equipment makes standardization a construction feature	Multiple and similar-in-size equipment makes standardization a construction feature	Multiple and similar equipment makes standardization a construction feature	Require more coordinated installation with added benefit of consolidated primary equipment in a central location.

4.1.1 *Window Air Conditioners*

Window Air Conditioners, as the name itself suggests, are usually fitted in a window or a recess created in a wall of a building. These are unitary direct expansion (DX) systems, having all the components packaged in a single box. The front side is designed to deliver the conditioned air to the connected space, and rear side shall be exposed to ambient for easy discharge of heat without any direct adverse effect on anyone else.

4.1.1.1 *Suitability*

Window air conditioners are suitable for residences, office cabin, general office area, and similar applications, where rigid comfort conditions are not required to be maintained.

4.1.1.2 *Limitations*

a) *Sound level* — Noise level of window air conditioner inside the conditioned room presents a serious problem due to close proximity of compressor to the occupied space;

   The noise level of window air conditioner shall be in accordance with the levels specified in Table 5 of accepted standard [D-3(4)];

b) *Fresh air for ventilation* — Most window air conditioners have a very small opening for entry of outdoor air for ventilation. As a result, the conditioned space, especially if occupied by more than 2-3 persons, may have higher levels of CO_2, as compared to centralised system; and

c) *Location* — Window air conditioner shall be mounted preferably at the window sill level on an external wall, where hot air from air-cooled condenser shall be discharged to the outdoor space without causing inconvenience to others. There should not be any obstruction for the inlet air to, nor for discharge air from, the condenser.

While deciding location of the window air conditioner, care shall be taken to ensure that the condensate drain water is piped to the ground level and does not drip in open space, causing nuisance.

Window air conditioner shall not be used for special applications like sterile rooms for hospitals and clean room applications where high filtration efficiency is desired, nor in applications where high amount of fresh air is desired for ventilation. It shall also not be used for areas that require close control of both the indoor temperature and relative humidity.

For detailed information regarding constructional and performance requirements and methods for establishing ratings of unitary (window) air conditioners, reference shall be made to the accepted standard [D-3(4)] and [D-3(5)].

4.1.1.3 For detailed information regarding constructional and performance requirements and methods for establishing rating of window type room air conditioners, reference shall be made to the accepted Indian standards.

4.1.2 *Split Air Conditioner*

Split air conditioners comprise of an indoor unit and an outdoor unit. Depending on its type, the Indoor unit may be mounted on either on floor, on wall, or at the roof. The outdoor unit consisting of compressor, heat exchanger, fan and motor; is installed in a separate independent cabinet. The indoor unit is an air handling system, designed primarily to provide conditioned air to an enclosed space, room or zone (conditioned space). It includes a prime source of refrigeration for cooling and dehumidification/heating and means for the circulation and filtering of air.

Various types of split air conditioners may be categorized based on type of compressor for outdoor unit and air-distribution for indoor unit, as below:

a) *Outdoor unit with variable speed compressor* — Also called an inverter AC or variable speed AC, works on part load depending on the demand for the conditioned space. This often uses a variable-frequency drive to control the frequency and thereby the speed of the compressor motor.

b) Outdoor unit with fixed speed compressor.

c) *Free-blow indoor unit* — It could be high wall mounted, ceiling suspended cassette (exposed type), or floor-mounted.

d) *Indoor unit* (*ceiling suspended*) — It is mounted in the ceiling void and provided with a duct collar and grille. This is commonly used in the vestibule of a hotel room.

e) *Ducted indoor unit* — It is generally mounted in ceiling void and requires duct work for air distribution.

4.1.2.1 *Suitability*

Split air conditioners are suitable for wide range of applications including residence, small office, club, restaurant, showroom, departmental store and others except for locations having high occupancy and requirement of high ventilation rates.

4.1.2.2 *Operating parameters*

In general, nominal capacity of the split air conditioner is rated as per accepted standard [D-3(6)] for an ambient temperature of 35 °C. Split ACs gets de-rated at higher ambient temperature (above 35 °C) in summer months, in most Indian cities, and locations where for significant duration ambient temperature is likely to be above 35 °C, information related to capacity deration and reduction in energy efficiency shall be the basis of system sizing.

At locations having high voltage fluctuation and inferior power quality, a voltage stabilizer shall be required to get stabilized rated voltage as per specifications defined by manufacturer. For 3 phase unit, it is recommended to use phase reversal protection device in power supply to the outdoor unit.

4.1.2.3 *Location*

Split air conditioner indoor units are to be mounted within the air-conditioned space or above the false ceiling from where the air distribution ducts are to be taken to the conditioned space for distribution of the conditioned supply air. When the indoor unit is mounted in the false ceiling, inspection panel shall be kept in the false ceiling to attend to the indoor unit, and periodic cleaning requirement of air filter. Outdoor unit shall be mounted at the nearest open area where unobstructed flow of outside air is available for the air-cooled condenser.

4.1.2.4 *Installation*

Wall mounted unit and similar exposed indoor unit are to be provided with installation plate for ease in installation. Care shall be taken to ensure clearance space as specified by the manufacturer is provided. The intake of return air shall not have any restrictions.

Ceiling suspended indoor unit shall be provided with vibration absorbers and isolators to minimize vibration transmission to the ceiling.

Outdoor unit shall be mounted on metal frame with anticorrosive coating or made of corrosion resistant material in an open area so that the condenser outlet air is thrown out with no obstruction and possible return of the air to condenser inlet side. Precaution shall also be taken that hot air from any one outdoor unit does not mix with the outdoor air intake of any other air-cooled condenser. This is generally referred as short cycling of the air and shall be avoided. The spacing around the condenser unit shall be ensured as per the manufacturer's installation guideline and recommendation. In case of multiple outdoor units in the same premise, the installations of all the condensing units shall be such that the outlet air from condensers shall not mix with the air inlet of any of the condensing units.

Service valves shall be provided on the outdoor unit for pressure testing of the system for leakages, for evacuation of moisture and air from the system, and to carry out predetermined gas quantity charge into the system. In case of maintenance of any part, it shall be provided with service valves to isolate and store the entire refrigerant charge within the outdoor unit by pumping down, thus saving the entire charge of refrigerant.

4.1.2.5 *Limitations*

Split air conditioners are not recommended for:

a) where distance between indoor and outdoor – Horizontal, vertical and total exceeds the limits as specified by manufacturers. Generally, the maximum of 30 m or higher as per recommendation of manufacturer from the outdoor unit for units up to 10 500 W (3 TR approximately), the horizontal distance between the indoor unit and outdoor unit should not exceed 10 m;

b) area requiring close and higher accuracy control of both the indoor temperature and relative humidity which is also referred as precision air conditioning;

c) special applications like sterile rooms for hospitals and clean room applications where high filtration efficiency is desired; and

d) spaces with large occupancy or high fresh air requirements.

4.1.2.6 For detailed information regarding constructional and performance requirements and methods for establishing rating of split type room air conditioners, reference shall be made to the accepted Indian standards.

4.1.3 *Ducted Split Air Conditioner*

Ducted split air conditioner (packages DX) units are ducted and packaged systems including rooftop and indoor packaged units in commercial air conditioning segment. Also known as unitary and light commercial systems, these typically should be used for small-to-medium commercial buildings to avoid the complexities associated with chilled water air conditioning systems. Packaged air conditioner shall be a self-contained unit suitable for floor mounting, or ceiling suspended ducted indoor units, designed to provide conditioned air through ducting in to the moderate sized conditioned spaces. It shall include a prime source of refrigeration for cooling and dehumidification, distinct facility for drawing of fresh air and mixing with return air, means for cleaning of mixed return and fresh air, and provision for external ducting for uniform air distribution in the conditioned spaces. If required, it shall also include winter heating and humidification package for winter operation. The machine shall be equipped with compressor, evaporator, expansion device and remote air-cooled condenser or water-cooled condenser with interconnected copper refrigerant piping.

Water cooled condenser packaged unit gives higher cooling capacity with lower power consumption as compared to an air-cooled condenser packaged unit which gets considerably de-rated in capacity for high (beyond 35 °C) ambient temperature in summer months for most of the cities in India. Air cooled package unit consequently also consumes more power in peak summer months. On the other hand, water-cooled condenser unit requires cooling tower with necessary piping and pump sets for circulating condenser-cooling water but is far more energy efficient at locations where ambient wet bulb temperatures are low in dry-summer months.

Water cooled systems shall be used where water for cooling is available. Treated wastewater, with proper TDS control shall be used in water cooled systems where available.

Packaged units with vertical air discharge should be used; however, if horizontal air discharge unit is procured then heat island effect would have to be avoided by providing clear unblocked space ahead of the condenser discharge.

4.1.3.1 *Suitability*

Packaged units are suitable for wide range of applications including office, club, restaurant, showroom, departmental store, banquet halls and others.

4.1.3.2 *Location*

Ducted split unit shall be mounted within the air-conditioned space with discharge air plenum or in a separate room from where the air distribution duct is taken to the conditioned space. While deciding location for the packaged unit, provision shall be kept for ease of servicing of the unit.

4.1.3.3 *Installation*

Ducted split unit shall be normally mounted on resilient pads which prevent vibration of the unit from being transmitted to the floor.

4.1.3.4 *Limitations*

Packaged air conditioners are not recommended for:

a) large multi-storey buildings, where multiplicity of compressors may entail subsequent maintenance problems;

b) where the length of air distribution ducting may exceed 30 m. However, duct length limitation depends upon permissible external static pressure specified by the manufacturer;

c) where the vertical distance of air-cooled condenser from the packaged unit exceeds about 20 m. The sum of horizontal and vertical distance should generally be kept within 25 m;

d) special applications like sterile rooms for hospitals and clean room applications where high filtration efficiency is desired. For these applications, special packaged units are available with high micron efficiency filters; and

e) operation theatres, where 100 percent fresh air is needed and fire hazard exists depending on the type of anaesthesia being used.

4.1.4 *Variable Refrigerant Flow System*

Variable refrigerant flow (VRF) systems are direct expansion (DX) multi-split system with compressors having ability to vary the refrigerant mass flow rate and, capable of delivering capacity according to variable load requirement. By delivering the variable refrigerant flow, VRF unit delivers required cooling/heating capacity allowing for substantial energy savings at partial-load conditions. VRF system has an outdoor unit connected with multiple number of indoor units via refrigerant piping, an attribute that distinguishes VRF system from other DX system. Other advantages of VRF system are its scalability, variable capacity, distributed temperature control, and the simultaneous cooling/heating mode application in different connected zones.

VRF system achieves temperature control on a zone-by-zone basis primarily by using refrigerant side control. As compared to the basic DX system, VRF Indoor unit has an additional electronic expansion device to control refrigerant flow passing through evaporator coil, to control indoor temperature, precisely based on the instantaneous load requirement. Similar to other air-cooled systems, air cooled VRF systems get derated at temperatures higher than rating temperature 35 °C. Therefore, sizing calculation shall be carried out based upon the information provided by the manufacturer about cooling/heating and energy performance at elevated temperatures.

VRF system is available with a dedicated air handling unit for supply of treated fresh air (TFA). In order to cater to fresh air needs, VRF system outdoor unit should be connected with TFA-AHU. A control box equipped with electronic expansion device and communication PCB is needed for TFA-AHU so that it shall communicate seamlessly with the VRF system outdoor unit.

VRF technology comes in two pipe or three pipe system. In a 2-pipe heat pump system, all of the zones shall either be in cooling mode or in heating mode. Application requiring simultaneous heating and cooling should have 3 pipe systems for enabling heat recovery.

4.1.4.1 *Configuration*

VRF system is available with air cooled or water-cooled condenser. Outdoor unit in VRF system could have multiple compressors, with one or more compressor provided with variable speed control. Generally, for air cooled VRF system, outdoor unit is available both with top discharge and with side discharge.

4.1.4.2 *Suitability*

VRF systems are suitable for a wide range of applications including residence, apartment, villa, small and big office, club, restaurant, showroom, departmental store, healthcare facility, hospitality, cultural facility, educational facility and industrial facility.

In VRF system, depending on application, same or different type of indoor units shall be connected to a single outdoor unit or to multiple outdoor units. VRF Indoor units and air handling units for fresh air or for special treatment of re-circulated air, shall be connected to the same refrigerant line, to facilitate more flexible system design, in mid and large size applications. However, usage of bigger AHU in VRF systems may have evident effect on the part load energy efficiency advantage of the system.

4.1.4.3 *Controls*

VRF systems shall have factory-packaged integral controls in each component; these communicate through their system specific protocol, to ensure that all system components operate collectively. VRF indoor and outdoor units include refrigerant and air side sensing and control devices which allow the system to optimize its output (compressor speed, discharge temperature, fan speed) based on inputs from controllers. Depending on the application and design, a VRF systems shall able to operate with at least one of the three different levels of controls: individual/group controller, centralized controller with remote monitoring and interface with building management system (BMS).

a) *Individual/group controller*

VRF indoor units shall be controlled individually with wired or wireless individual remote controller. Control options include ON/OFF temperature change, mode change, fan speed change and integration with the BMS system of the building. Applications such as a large common area in which multiple indoor units are running simultaneously, at the same condition, shall be controlled by a single remote controller by combining these indoor units as a group. All setting conditions remain same for all indoor units connected in the group. Each of the grouped indoor unit may operate according to the sensed return air temperature.

b) *Centralized controller with remote monitoring and control*

Centralized controller allows user to operate and monitor multiple indoor units centrally. Besides having all functions of individual remote controller, centralized controller has the additional facility of scheduling energy saving by operation restrictions, and web monitoring. In some applications, single outdoor unit is used by different tenants connected to different indoor units. In such cases, centralized controller offers facility of tenant billing through software application, depending on individual usage through proportional power distribution.

c) *Interface with building management system*

VRF system shall be integrated with building management system if available, with the use of different interface gateways that communicate with different protocols. With this interfacing, it is easy to operate and monitor air conditioning with user's building management system.

4.1.5 *Liquid Chilling System*

Liquid chilling system may be based on chilled water, brine, or other secondary coolant for air conditioning or refrigeration. The most frequent application of water chilling is for air conditioning. However, brine cooling systems for low temperature refrigeration requirement and chilling fluids for thermal storage processes, are also common.

The basic components of a vapour-compression, liquid-chilling system shall include a compressor, liquid cooler (evaporator), condenser – which could be air cooled or water cooled), compressor drive, liquid-refrigerant expansion or flow control device and control centre; it also normally includes a receiver, economizer, expansion turbine and/or sub cooler. In addition, auxiliary components may be used, such as a lubricant cooler, lubricant separator, lubricant-return device, purge unit, lubricant pump, refrigerant transfer unit, refrigerant vents, and/or additional control valves. Liquid chilling units are available with air cooled or water-cooled condensers. Wherever water is available for cooling, water cooled chillers shall be opted for.

4.1.5.1 *Vapour compression water chiller*

The unit shall be installed on to a solid foundation on appropriately designed resilient mountings as per the manufacturer recommendation. Pipe connections shall have flexible couplings; these should be considered in conjunction with the design of the pump mountings and the pipe supports.

Capacity control is required in all the chillers that shall be capable to maintain an approximately constant temperature of the chilled water leaving the evaporator. This may be adequate for one or two chiller packages, but a more elaborate central control and plant optimization system is desirable for a large number of chillers. The design of the refrigeration control system should be integrated, or be compatible, with the control system for the heat transfer medium circulated within the cooler.

Multiple chiller configuration, same or different size and types shall be used to achieve the highest energy efficiency and adequate system capacity availability at all conditions. System COP shall be the basis for design and selection of equipment.

Power consumption during operation should be reduced by taking advantage of a fall in the ambient dry bulb and wet bulb temperatures, which permit a corresponding increase in chilled water temperature and decrease in the condenser water temperature, and consequent considerable reduction in the compressor power consumption. It is important to optimize operating cost by optimum equipment selection and careful design of the automatic control system.

The water chilling packages are classified based on the type of compressor, as given below:

a) *Centrifugal compressor* — These compressors use impellers to impart pressure energy to the refrigerant. These may be modulated down to approximately 40 percent of full load capacity, with careful control of the condensing pressure. Part load operation of centrifugal compressor should be restricted to 40 percent or above in order to avoid surging of compressor that may damage it if continues to occur.

   In order to save energy, centrifugal compressors with variable speed drives (VFD’s) should be preferred over fixed speed chillers. VFD’s are used on chillers not only to enhance the part load performance but also to act as an alternate to auto-tap change starter for minimizing the starting current for the compressor.

   Modern variable-speed oil-free centrifugal chillers have the shaft with impeller levitated during rotation, using digitally controlled magnetic bearings. This unique feature reduces the friction and heat caused by conventional bearings, adding to the overall high efficiency. It also eliminates the need for lubricating oil and the ancillary components required to support the lubricating oil system, which further improves efficiency.

b) *Screw compressor* — Screw compressors are available in open and semi-hermetic form and are generally coupled directly to two-pole motors. The capacity of the compressor may be modulated down to about 20 percent of full load capacity.

   Similar to centrifugal compressors, screw compressors with VFD’s may be used as an effective way to enhance the part load performance and also minimize the starting current for the compressor.

   These are generally specified and used for capacities up to 1 400 kW (approximately 400 TR).

   In systems using a direct expansion evaporator (DX chiller), oil is trapped in the evaporator and an oil recovery system is necessary. With some systems, oil cooler is required in the oil circulation system, to remove the heat gathered by oil during compression cycle.

c) *Scroll Compressor* — Scroll compressors are generally used in residential and small/light commercial air-conditioning. Scroll chillers may be used in smaller sizes up to 425 kW. They offer relatively higher efficiency over other types of compressors, however, scroll compressors are not suitable for higher capacity range.

d) *Reciprocating compressor* — Due to their relatively lower efficiency, reciprocating compressors are nearly phased out from the comfort air-conditioning application. However, for cold storage and other special purpose applications, these still have importance due to their capacity to offer high pressure ratio.

4.2 Vapour Absorption/Adsorption System

The absorption cycle uses a solution which, typically has two working fluids: an absorber and an absorbent. These fluids typically in liquid state, have significant change in solubility of the absorbent at different temperatures. Function of a vapor compressor is replaced by a liquid pump. The absorbent/refrigerant mixture is pumped to a higher pressure as a solution, where the refrigerant is boiled off by the application of heat and is subsequently condensed in the condenser.

Absorption machines are extensively used in liquid-chilling applications. These are most suitable for applications where waste heat is readily available. Solar energy assisted air-conditioning/cooling systems also use absorption system.

4.2.1 *Indirect Firing*

The lithium bromide/water absorption system shall be powered by medium or high temperature hot water and low or medium pressure steam. Water is the refrigerant and lithium bromide the absorbent. The four compartments enclosing the heat exchanger tube bundles for the condenser, evaporator, generator and absorber shall be in a single or multiple pressure vessel arrangement. The whole assembly has to be maintained under a high vacuum for the correct functioning of the unit. Water and absorbent solutions are circulated within the unit by electrically driven pumps.

Capacity control down to 10 percent of full load capacity is achieved by modulating the flow of the heating medium in relation to the cooling demand. There is some loss in performance at part load, which shall be compensated by refinements in the system design and control.

4.2.2 *Direct Firing*

Direct fired lithium bromide/water absorption systems have become common, by incorporating precise control of generator temperature necessary to avoid crystallization.

Ammonia/water systems are generally direct fired, but are rarely used for water chilling duties, except for small sized units, which are installed outside the building.

There are two reasons for this; firstly, capital costs are higher and secondly the danger to personnel in the event of leakage of the refrigerant.

Direct firing has the advantage that the losses in an indirect heating system are avoided, but in an air conditioning installation where a boiler system is installed to provide heating, the advantage is not of much concern.

Adsorption machines differ from absorption machines in the manner the two working media interact. Adsorption is a surface phenomenon, unlike absorption. More details about such systems are provided under **6**.

5 REFRIGERANTS

The refrigerant is a working fluid used in the refrigeration cycle of a refrigeration and air conditioning system. Due to its deteriorating impacts on climate (especially ozone depletion and global warming) overview of the requirements for the safe use of refrigerants, except for ammonia, in mechanical or vapor-compression refrigeration and air conditioning systems that are partly or completely installed inside buildings shall be provided. It is intended for the guidance of the engineers, contractors, installers, and service technicians on the safe use of these systems and summarizes the guidelines applicable for the safe design, construction, installation, and operation of systems thereby providing insight of environmental impact of refrigerant emission and its management to minimize the emission of environmentally damaging refrigerants.

The refrigerant is a working fluid that is used in the refrigerating cycle of a refrigeration and air conditioning system {*see* accepted standards [D-3(2)], [D-3(5)], [D-3(7)], [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)]}. Refrigerant molecules shall be broadly classified into three categories: halocarbons, organic compounds, and inorganic compounds. The halocarbons shall be further subdivided into chlorofluorocarbon (CFC), hydrochlorofluorocarbon (HCFC), hydrofluorocarbon (HFC), hydrofluoroolefin (HFO) and hydrochlorofluoroolefin (HCFO). The organic compounds consist mostly of hydrocarbons (HC). Inorganic compounds include carbon dioxide and ammonia. Both zeotropic and azeotropic mixtures of two or more pure refrigerants are also used as refrigerants.

5.1 Numbering of Refrigerants

Refrigerants are numbered with an R-, followed by the identification number. The identifying numbers for hydrocarbons (for example, assigned identification of propane is R-290) and halocarbons (for example, assigned identification of difluoromethane is R-32) are explicitly related to their chemical formula. Isomers are identified by appending lowercase letters at the end of the identifying number (for example 1,1,2,2-Tetrafluoroethane and 1,1,1,2-Tetrafluoroethane are isomers, which have assigned identification of R-134 and R-134a, respectively). Alkenes are identified by adding "1" at the beginning of the number determined by the above method. The structural isomers are identified by appending lower case letters at the end. The suffixes "(Z)" and "(E)" are used to differentiate cis- and trans- isomers. For example, R-1234yf, R1234ze(E) and R1234ze(Z) are all isomers of tetrafluoropropene. An inorganic compound is identified by the digit 7 at the hundreds place of the identifying number followed by its molecular weight (for example, assigned identification of Ammonia is R-717). Refrigerant blends having the same pure components are assigned the same number, but different compositions are identified with an uppercase letter after the number. Series 400 (for example, R-410A is a blend of 50 percent by weight of R-32 and 50 percent by weight of R-125) and 500 refrigerant (for example, R-507A is a blend of 50 percent by weight of R-143a and 50 percent by weight of R-125) blends are zeotropic and azeotropic, respectively.

5.2 Safety Group Classification

The Table 10 provides the toxicity and flammability-based classifications for refrigerant safety.

Table 10 Refrigerant Classification Based on Toxicity and Flammability

(*Clause* 5.2)

Sl No.	Flammability	Lower Toxicity Safety Group	Higher Toxicity Safety Group
(1)	(2)	(3)	(4)
i)	Higher Flammability	A3	B3
ii)	Flammable	A2L	B2L
iii)	Lower Flammability	A2L	B2L
iv)	No Flame Propagation	A1	B1

The accepted standard [D-3(2)] has also established the maximum refrigerant concentration limit (RCL) in the air to avoid escape-impairing effects such as asphyxiation, acute toxicity, and flammability in normally occupied enclosed spaces. The RCL is established based on the most conservative limit between asphyxiation, acute toxicity, and flammability. This safety group classification is referred to in the installation and maintenance standards for HVAC equipment {*see* accepted standard [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)]}.

Product safety standards for AC equipment, such as accepted standard [D-3(5)] and [D-3(7)] specify safety requirements for electrically operated refrigeration and air conditioning appliances that have an incorporated compressor. Accepted standard [D-3(5)] deals with the safety of electrical heat pumps, air-conditioners, and dehumidifiers, while accepted standard [D-3(7)] deals with the safety of commercial refrigerating appliances.

Refrigerants should preferably be non-flammable or have lower flammability. The flammable refrigerant system shall be built with necessary protection system. Similarly, refrigerants with lower toxicity are preferable. Suitable risk mitigation processes and instruments should be implemented to mitigate risk to the occupants and the property per the accepted standards. Accepted standards [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)], set forth requirements for safeguarding people and property where refrigeration facilities are located and prescribes safety requirements.

5.3 Environmental Impact of Refrigerants

Refrigerants shall be released from the system during installation, minimizing refrigerant leaks, proper refrigerant recovery, and recycling during installation, operation, and disposal. Commercially used refrigerants shall be either ozone-depleting substances or greenhouse gases or shall be both. Releasing such refrigerants into the atmosphere shall be damaging to the environment. CFCs and HCFCs are ozone-depleting substances.

NOTE — India ratified the Montreal Protocol Agreement on Substances that Deplete the Ozone Layer in 1992 and CFCs in RAC units manufactured beyond 1 January 2003 have been phased out. The HCFCs are being phased out under the HCFC Phase-Out Management Plan (HPMP), and their use in RAC units manufactured beyond 1 January 2025 will be prohibited. The Kigali Amendment to the Montreal Protocol calls for the phase-down of HFCs due to their high GWP. India will complete its phase-down of HFCs in 4 steps from 2032 onwards with a cumulative reduction of 85 percent in 2047. The HFC phase-down schedule for India is scheduled to begin in 2032 and it will be completed in 2047. The baseline will be the average production and consumption of HFCs for the years 2024, 2025 and 2026. The quota will be frozen at the baseline level in 2028. The phase down will be achieved in four steps, with the cumulative reductions of production and consumption by 10 percent in 2032, 20 percent in 2037, 30 percent in 2042 and 85 percent reduction in 2047.

5.4 Classification of Commonly Used Refrigerants

The refrigerants shall be classified in accordance with accepted standard [D-3(2)] as listed in Table 11. This list is not exhaustive. The users must refer to tables from accepted standard [D-3(2)] for a complete list of refrigerants and accepted standards [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)] for RCL values.

Table 11 Classification of Commonly Used Refrigerants

(*Clause* 5.4)

Sl No.	Refrigerant	Type	Refrigerant Safety Group Classification	Amount of Refrigerant in Occupied Space	
				RCL (g/m^3)	LFL (g/m^3)
(1)	(2)	(3)	(4)	(5)	(6)
i)	R11	CFC	A1	6.1	–
ii)	R12	CFC	A1	90	–
iii)	R22	HCFC	A1	210	–
iv)	R32	HFC	A2L	77	306
v)	R123	HCFC	B1	57	–
vi)	R134a	HFC	A1	210	–
vii)	R290	HC	A3	9.5	38
viii)	R404A	HFC	A1	500	–
ix)	R407C	HFC	A1	290	–
x)	R410A	HFC	A1	420	–
xi)	R448A	HFO	A1	390	–
xii)	R449A	HFO	A1	370	–
xiii)	R454A	HFO	A2L	52	293.9
xiv)	R454B	HFO	A2L	49	352.6
xv)	R454C	HFO	A2L	71	289.5
xvi)	R455A	HFO	A2L	79	432.1
xvii)	R513A	HFO	A1	320	–
xviii)	R514A	HFO	B1	14	–
xix)	R600a	HC	A3	9.5	38

Table 11 (*Continued*)

Sl No.	Refrigerant	Type	Refrigerant Safety Group Classification	Amount of Refrigerant in Occupied Space	
				RCL (g/m^3)	LFL (g/m^3)
(1)	(2)	(3)	(4)	(5)	(6)
xx)	R744	CO_2	A1	72	–
xxi)	R1234yf	HFO	A2L	75	289
xxii)	R1234ze(E)	HFO	A2L	76	303
xxiii)	R1233zd(E)	HFO	A1	85	–
xxiv)	R1336mzz(Z)	HFO	A1	84	–

NOTE — The RCL/LFL data is taken from accepted standard [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)].

5.5 Occupancy Based Classification

Refrigerating systems shall have different locations based on how the space connected to them is occupied, which shall depend on how people in that space react to possible refrigerant exposure.

5.5.1 *Institutional Occupancy*

Portion of a building which occupants shall not readily leave without assistance. For example, prisons, hospitals etc.

5.5.2 *Public Places/Public Assembly Occupancy*

Where large number of people gather, and the occupants shall not quickly leave the space due to large gathering. For example, restaurants, movie theatres, auditoriums etc.

5.5.3 *Residential Occupancy*

Where people/occupants live either permanently or temporarily. For example, home/apartment/hotel/hostel etc.

5.5.4 *Commercial Occupancy*

Where people/occupants gather to do commercial transactions like small shops/offices etc.

5.5.5 *Large Mercantile Occupancy*

Where occupants/people gather in large quantities like 70+ people gather to a big grocery store like Big Bazaar.

5.5.6 *Industrial Occupancy*

Occupancy where general public is not allowed. Only controlled people are allowed which shall be used for manufacturing or as a warehouse

5.6 Classification as per Leakage Probability

These guidelines classify systems based on their likelihood of leakage, ensuring appropriate safety measures and maintenance practices are in place for each classification.

5.6.1 *Low Probability Systems* (*Indirect Systems*)

Any system where the design or placement of parts makes it unlikely that refrigerant will leak from a broken connection, seal, or component into the occupied space. Examples of low-probability systems include water-cooled chillers in machinery rooms.

5.6.2 *High Probability Systems* (*Direct Systems*)

Those systems where the components are designed or located in a way that makes it likely for refrigerant to leak into an occupied space from a broken connection, seal, or component. Examples of high-probability systems include direct expansion (DX) split systems, packaged DX rooftop units, and variable refrigerant flow (VRF) systems.

5.7 Classification as per System Locations

These guidelines categorize mechanical equipment by location to ensure safety and accessibility, from units within occupied spaces to those in dedicated machinery rooms or ventilated enclosures.

5.7.1 *Class* I – *Mechanical Equipment Located within the Occupied Space*

If the refrigerating system or any of its refrigerant-containing parts is located indoors in occupied space or non-occupied space that is not sealed from the occupied space or shall leak into the occupied space, then the system is considered to be of Class I unless the system complies with the requirements for other classes. Integral display units are typical examples of this category.

5.7.2 *Class* II – *Compressors and Pressure Vessels Outside the Occupied Space*

If all compressors and pressure vessels are outside the occupied space and refrigerant leaks from compressors or pressure vessels shall not flow into the occupied space then the requirements for a class II location shall apply to any refrigerant containing part located in the occupied space or that shall leak into the occupied space. Coil-type heat exchangers and pipework, including valves, may be located in an occupied space.

Refrigerant shall leak into the occupied space through heat transfer fluid if, for example:

a) the heat transfer fluid is ventilation air;

b) the heat transfer fluid is an open spray system;

c) leakage of the refrigerant into the occupied space is caused by leakage of the refrigerant into the heat transfer fluid; or

d) there is a heat transfer fluid relief valve or an automatic purge point vents into the space.

5.7.3 *Class* III – *Machinery Room or Open Air*

The requirements for a Class III location shall apply to any refrigerant containing part located in the open air or in a machinery room that is as per the requirements of accepted standard [D-3(10)]. Water or air-cooled chillers are typical examples of this type.

5.7.4 *Class* IV – *Ventilated Enclosures*

If, with the intent of preventing migration of leaked refrigerant to the surrounding area, refrigerant-containing parts are located in a ventilated enclosure that is as per the requirements of accepted standard **5.2.17** of [D-3(9)] and **4.6** of [D-3(10)] then the requirements for a Class IV location shall apply.

Heat pumps within a ventilated enclosure are typical examples.

5.8 Classification as per Access to Occupied Spaces, Machinery Rooms and Open Air

Access classification shall be determined according to Table 12.

The access category is defined according to Table 12 by consideration of the group of occupants who are least familiar with the safety precautions. For example, in a supermarket, department store or transport terminus an unregulated number of customers who are not familiar with the location or its safety precautions shall gather. In a general office the number of people in the building is more easily regulated and occasional visitors will be accompanied by a regular occupant who shall advise on the safety precautions in the event of an emergency. In a manufacturing facility all of the occupants will be familiar with the safety requirements and access to the workplace will be restricted.

A complex building shall contain several access categories, for example the public areas and the plant rooms in a hospital shall be classified as access Category A and access Category C respectively.

Table 12 Categories of Access

(*Clause* 5.8)

Sl No.	Categories	General Characteristics	Examples
(1)	(2)	(3)	(4)
i)	A	General access (least restrictive access to the space) Parts of buildings where a) Sleeping facilities are provided, b) People are restricted in their movement, c) An uncontrolled number of people are present, or d) To which any person has access without being personally acquainted with the necessary safety precautions.	Hospitals, courts or prisons, theatres, supermarkets, schools, lecture halls, public transport termini, hotels, dwellings, restaurants, car parks, footpaths, gardens
ii)	B	Supervised access Parts of buildings where only a limited number of people shall be assembled, some being necessarily acquainted with the general safety precautions of the establishment.	Business or professional offices, laboratories, places for general manufacturing, where people work, delivery zones and some non-public areas in supermarkets
iii)	C	Authorized access (most restrictive access to the space) Parts of buildings where only authorized persons have access, who are acquainted with general and special safety precautions of the establishment and where manufacturing, processing, or storage of material or products take place.	Manufacturing facilities, for example, for chemicals, food, cold stores, some non-public areas in supermarkets, fenced outdoor compounds or roofs with restricted access

Open air locations including roofs, car parks and outdoor equipment compounds are considered to be parts of the building. Machinery rooms shall not be considered as occupied space except as defined in **5.1** of accepted standard [D-3(10)].

Refrigerant quantity limits shall be as per accepted standard [D-3(8)] expect for:

a) *Institutional Occupancy* — Consider a safety factor of 0.5 (limiting the charge to 50 percent of the calculated charge) while calculating the charge limits;

b) *Residential Occupancy* — Listed self-contained systems limited to 3 kg charge limit;

c) *Commercial Occupancy* — Listed self-contained systems limited to 10 kg charge limit; and

d) Industrial occupancy and large mercantile occupancy spaces to follow requirements of charge limits in accepted standards [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)].

5.9 Mixing of Refrigerants

Refrigerants with different refrigerant designations as per accepted standard [D-3(2)] shall not be mixed.

5.10 Purity

Refrigerants used in refrigeration systems shall be new, recovered or reclaimed refrigerants as per below.

5.10.1 *New Refrigerants*

Refrigerants shall be of purity level specified by the equipment or appliance manufacturer as per accepted standard [D-3(12)].

5.10.2 *Recovered Refrigerants*

Shall not be used in the refrigeration system as it may not meet the quality requirements of a system.

5.10.3 *Recycled Refrigerants*

Refrigerants that are recovered from refrigeration and air-conditioning systems shall not be used in other than the system from which they were recovered and in other systems of the same owner.

5.10.4 *Reclaimed Refrigerants*

Used refrigerants shall not be reused in a different owner's equipment or appliances unless tested and found to meet the purity requirements. Contaminated refrigerants shall not be reused unless reclaimed and found to meet the purity requirements. For additional information on purity requirements of refrigerants refer to accepted standard [D-3(12)].

5.10.5 High probability direct systems for human comfort applications shall only use A1 and A2L refrigerants. If the equipment is listed, it shall follow the requirements of accepted standard [D-3(5)] and [D-3(7)]. If the system is designed and installed at site, the requirements of accepted standards [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)] will follow.

5.11 Machinery room/plant room systems shall follow accepted standards [D-3(8)], [D-3(9)], [D-3(10)] and [D-3(11)] requirements.

5.12 Lifecycle Management of Refrigerants

Lifecycle management of refrigerants shall encompass all phases from design, manufacturing, installation, operation, servicing, to decommissioning and end-of-life. In the design stage, appropriate selection of corrosion-resistant materials, robust packaging for transport and handling, protective coatings on brazed joints, and dimensioning of interconnecting tubes to prevent fractures shall be ensured. Manufacturing processes such as brazing shall be carried out under controlled conditions to minimize leakage risks. During installation and commissioning, only trained and certified technicians shall be permitted, using calibrated tools (for example, torque wrenches) and considering environmental conditions at site. In line with the *Kigali Amendment*, the phasedown of HCFCs and high-GWP HFCs shall be undertaken in accordance with national schedules, with preference given to adoption of lower-GWP alternatives such as olefins (HFOs).

At the operational and end-of-life stages, systems shall incorporate leak detection, regular monitoring, and timely repair. Recovery, recycling, reclamation, or safe destruction of refrigerants shall be carried out in accordance with the *E-Waste* (*Management*) *Rules*, the *Hazardous Waste Management Rules*, and relevant CPCB guidelines. Venting of refrigerants during maintenance, servicing, or decommissioning shall be strictly prohibited. All systems using flammable refrigerants shall be evacuated with vacuum pumps and purged with nitrogen as per manufacturer's specifications. Producers and manufacturers shall bear extended producer responsibility (EPR) for the collection and environmentally sound management of appliances containing refrigerants at end-of-life. Certified recovery and destruction facilities shall be used, with proper record-keeping and reporting to CPCB/state pollution control boards. Awareness among consumers, technicians, and service providers shall be promoted to ensure safe handling and environmentally responsible lifecycle management of refrigerants.

5.12.1 *Refrigerant Recovery*

When decommissioning an appliance that contains refrigerant, it should be properly recovered using certified equipment before servicing or disposing of the appliance.

5.12.2 *Certified Technicians*

Only individuals who are certified under Ozone Cell Regulations are allowed to handle and remove refrigerant from appliances.

5.12.3 *Responsible Disposal*

The final person or entity responsible for the disposal of the appliance (for example, scrap metal recyclers, landfills) should ensure that the refrigerant has been recovered beforehand.

5.12.4 *Documentation*

If the final disposer accepts an appliance that has had its refrigerant charge removed, they should maintain a signed statement from the person who removed the refrigerant. This statement should include details like the name and address of the person who performed the recovery and the date of recovery.

5.12.5 *Recovered Refrigerant Usage*

Recovered refrigerants should generally be reused only in the system from which they were recovered or in other systems owned by the same entity in line with **5.10.2** and **5.10.3**.

5.12.6 *Reclaiming Contaminated Refrigerant*

Contaminated refrigerants cannot be reused unless they are reclaimed to meet the purity requirements as per the accepted standard [D-3(12)] and in line with **5.10.4**.

5.12.7 *Filtering and Drying*

Recovered refrigerants should be filtered and dried before reuse.

Venting of refrigerants to the atmosphere should be avoided. During commissioning, any purging, if necessary, should be done outside or in a well-ventilated area to avoid formation of flammable or toxic atmosphere.

5.13 Total Equivalent Warming Impact (TEWI)/Life Cycle Climate Performance (LCCP)

There are 2 metrics used to calculate the greenhouse impact of a product/equipment over its lifetime. This is given in terms of its equivalent CO_2 emissions; in kg or Tonnes depending upon its absolute value.

The second metrics an extension of the first one in that it includes even the warming impact due to the manufacturing of the product as well as the indirect emission due to the atmospheric decomposition of the refrigerant including the energy consumed during the manufacture of the raw materials. For example, consider a Window unit. The coils are made of copper tubes and aluminium fins. Thus, if the copper tube manufacturing process and/or aluminium fin stock manufacturing process is highly energy inefficient then it will reflect badly in the LCCP of the Window unit. LCCP is more comprehensive as it has a "cradle to grave" approach and it has its merit in evaluating the overall impact.

The following equation gives the expression for calculating TEWI for a particular product (it shall not be generalized; it is application/product specific):

$$\text{TEWI} = \text{GWP} \times L \times n + \text{GWP} \times m \times (1-\alpha) + n \times E \times \beta$$

where

GWP	=	Refrigerant global warming potential (kg of CO_2/kg of refrigerant);
L	=	Annual leakage rate (kg/year);
n	=	System operating life time (years);
m	=	Refrigerant charge (kg);
α	=	Recycling factor (percent);
E	=	Annual energy consumption (kWh/year); and
β	=	CO_2 emissions on energy generation [kg CO_2/kWh].

The above equation is considering both the direct (due to refrigerant leakage/non recycling) and the indirect effect (due to energy consumption). TEWI is completely product/system/location dependent and shall not be generalized. However, it does give an idea as to how the indirect impact is much more than the direct impact. With the new low-GWP refrigerants being introduced, the value(s) of TEWI would be different than what they were with the old high-GWP refrigerants. Also, an important parameter is the β factor. This is based on the location/country where the system is operating. It is directly dependent on the fuel source/type of power generation plants being used to supply the electrical power. Naturally if the major source of power generation is fossil fuel. Coal-based thermal power generation then this factor will be high, and which turn will result in higher TEWI.

The following equation gives the expression for calculating LCCP for a particular product:

$$LCCP = Direct\ Emissions + Indirect\ Emissions$$

$$Direct\ Emissions = C \times (L \times ALR + EOL) \times (GWP + Adp.GWP)$$

$$Indirect\ Emissions = L \times AEC \times EM + \sum(M \times mm) + \sum(mr \times RM) \times +C(1 + L \times ALR) \times RFM + C \times (1 - EOL) \times RFD$$

where

C	=	Refrigerant charge (kg);
L	=	Average lifetime of equipment (year);
ALR	=	Annual leakage rate (percent of refrigerant charge);
EOL	=	End of life refrigerant leakage (percent of refrigerant charge);
GWP	=	Global warming potential (kg CO_{2e}/kg);
Adp. GWP	=	GWP of atmospheric degradation product of the refrigerant (kg CO_{2e}/kg);
AEC	=	Annual energy consumption (kWh);
EM	=	CO_2 Produced /kWh (kg CO_{2e}/kg);
M	=	Mass of unit (kg);
MM	=	CO_2 Produced/material (kg CO_{2e}/kg)
mr	=	Mass of recycled material (kg);
RM	=	CO_{2e} Produced/recycled material (kg CO_{2e}/kg);
RFM	=	Refrigerant manufacturing emissions (kg CO_{2e}/kg); and
RFD	=	Refrigerant disposal emissions (kg CO_{2e}/kg).

The above equation is as per the Guideline for LCCP recommended by International Institute for Refrigeration.

When it comes to selecting refrigerants and equipment, both choices are crucial due to their direct impact on a user's scope 1 and scope 2 emissions. Scope 1 emissions encompass direct greenhouse gas (GHG) emissions, which, in the case of refrigerants, primarily result from gas leakage at the premises. To mitigate these emissions, it is vital to adopt low global warming potential (GWP) refrigerants that minimize the environmental impact.

However, as we push for the rapid adoption of these low-GWP refrigerants, it is equally important to ensure that the efficiency and capacity of the equipment are not compromised. This balance is required because any reduction in equipment performance shall increase scope 2 emissions, which are indirect GHG emissions from the energy consumed by the equipment.

Leveraging digitally enabled and AI-driven solutions shall significantly enhance our approach to managing refrigerants. These advanced technologies provide predictive insights that help prevent gas leakages, reducing the need for technician visits and minimizing downtime for repairs. By integrating generative AI, we shall proactively address potential issues, further reducing scope 1 emissions by preventing leaks before they occur. Moreover, reducing technician visits not only lessens service providers' tailpipe emissions lowering the end user's scope 3 emissions, which are indirect emissions from third-party activities.

When selecting equipment and refrigerants, it’s crucial to consider the Total Equivalent Warming Impact (TEWI), which factors in both scope 1 and scope 2 emissions. This comprehensive approach ensures that the chosen solutions are sustainable over their entire lifecycle. Additionally, implementing digitization, advanced leak detection systems, and remote monitoring of refrigerants shall effectively map and manage the operational emissions associated with the equipment’s deployment at a user's premise. These technologies ensure that emissions are minimized, aligning with broader sustainability goals and regulatory requirements.

In summary, achieving a balanced and strategic selection of refrigerants and equipment, underpinned by AI and digital solutions, for reducing scope 1, scope 2, and scope 3 emissions. This integrated approach supports the fast-tracking of low-GWP refrigerants while maintaining operational efficiency and capacity, ultimately contributing to a more sustainable and compliant future in building operations.

6 LOW ENERGY COOLING

In addition to the popular air-conditioning options and traditional passive cooling options, various types of modern low energy heating/cooling solutions have also emerged which may be used to provide thermal comfort to occupants using alternative strategies. These include direct, indirect and hybrid evaporative cooling, radiant cooling/heating, structural cooling, solar assisted cooling/heating, and few others.

6.1 Evaporative Air Coolers

Evaporative coolers use the process of evaporation of water to lower air temperature. This method of comfort cooling not only conserves energy but also offers numerous benefits for environment. The principle for evaporative air cooling is straightforward: when water evaporates, it absorbs latent heat of evaporation from the surrounding air, thereby cooling the air. This is also termed as adiabatic cooling. This process is most effective in arid and semi-arid climates where the humidity is low, making evaporation more efficient.

An evaporative cooler typically consists of a fan, absorbent pads, a water reservoir and some kind of water distribution system including pump to distribute water on the pads, and often some set of louvers/deflectors to guide air to desired direction.

The process is:

a) the fan draws warm, dry air from the outside into the cooler;

b) the air passes through the water-saturated pads, where the water evaporates, and thereby absorbing heat from the air; and

c) the cooler but more humid air is then blown into the living space, lowering the temperature.

6.1.1 *Advantages of Evaporative Cooling*

These points highlight the key benefits and positive aspects of the specified system or approach, emphasizing efficiency, performance, and practicality.

a) *Energy efficiency* — Evaporative coolers consume significantly less energy compared to conventional air conditioners. They use only the power needed to run the fan and water pump, resulting in lower electricity bills and reduced carbon footprint;

b) *Environmental benefits* — Traditional air conditioners use refrigerants like CFCs and HFCs, which are harmful to the environment. Evaporative coolers, on the other hand, do not use these chemicals, making them an eco-friendly alternative;

c) *Cost-effective* — The initial purchase cost and maintenance expenses of evaporative coolers are significantly lower than those of air conditioning systems for a given cooling area. Evaporative air coolers have fewer mechanical parts, which means there are fewer components that shall break down;

d) *Improved air quality* — Evaporative coolers add moisture to the air, which shall be beneficial in dry climates where low humidity shall cause respiratory problems and dry skin. They also filter out dust and pollen as the air passes through the pads, improving indoor air quality;

   At times, several processes need humidity (for example, paper mill, garment industry, etc) and hence evaporative air cooling serves dual purpose of cooling and providing humidity; and

e) *Easy installation and maintenance* — These coolers are relatively simple to install, requiring only a water source and an electrical outlet. Maintenance is straightforward, typically involving the periodic cleaning or replacement of pads and ensuring the water reservoir is filled.

6.1.2 *Applications of Evaporative Cooling*

These points describe the various uses and suitable scenarios for implementing evaporative cooling, focusing on its effectiveness in specific environments and conditions. Following are some of different uses:

a) *Residential use* — They are ideal for cooling homes in dry climates. Portable models shall be moved from room to room, providing flexible cooling solutions. Even table top models are available for personal cooling purposes. Window or wall mounted units are available too;

b) *Commercial and industrial use* — Large evaporative coolers shall be used in showrooms, restaurants, institutes, warehouses, factories, and workshops where traditional air conditioning may be impractical or too expensive especially in semi-open spaces;

c) *Outdoor cooling* — Evaporative coolers are popular for outdoor events, patios, and agricultural applications. They shall create comfortable environments in open spaces where conventional cooling systems are ineffective; and

d) *Greenhouses* — These coolers are beneficial in horticulture for maintaining optimal temperature and humidity levels, promoting healthy plant growth. Other related uses are poultry farms, pig farms, etc.

6.1.3 *Sizing of Evaporative Cooling*

Air coolers are used either for spot cooling or space cooling:

a) *For spot cooling* — An important parameter to consider is the air throw the distance to which the air cooler will throw air with adequate velocity and air quantity;

b) *For space cooling* — Important parameter is air changeover. Generally, 20 to 40 air changeovers per hour (ACH) are required depending upon application and heat load; and

c) *The residential air coolers* — It shall be as per accepted standard [D-3(13)]. The large commercial air evaporative air-cooling system shall be as per the special publication.

   NOTES

   1 Further, ISHRAE simulation tool called EvapCal shall be used for sizing guidelines based upon city, heat load, required thermal comfort to correctly size the system.

   2 Special publication may be ISHRAE Standard 2005 : 2024.

6.1.4 *Limitations of Evaporative Cooling*

These points outline the challenges and constraints associated with evaporative cooling, highlighting factors that may affect its efficiency and suitability in certain conditions.

a) *Climate dependence* — The efficiency of evaporative coolers decreases significantly in humid climates. When the air is already saturated with moisture, it shall not absorb much more, making evaporation less effective and reducing the cooling capacity of the device;

b) *Limited to cooling only* — Evaporative coolers are designed to only cool the space and shall not condition it like air conditioner. Hence shall not be used for heating or dehumidification;

c) *Increased humidity* — While added humidity shall be beneficial in dry climates, it shall be problematic in areas that are already humid. High humidity levels shall create a sticky, uncomfortable indoor environment and potentially promote mould growth if not designed properly; and

d) *Water usage* — Evaporative coolers require a continuous supply of potable quality of water to operate, which shall be a concern in regions facing water scarcity or unavailability. However, their overall water consumption is relatively low compared to other water-intensive appliances.

6.1.5 *New Trends/Technologies in Evaporative Cooling*

Smart Controls, efficient cooling media and two stage, multi stage hybrid evaporative cooling system enhances the efficiency and comfort level. These technologies should be considered based on the climate conditions and application. Two stage are:

a) *Evaporative cooling systems* — Advanced version of direct evaporative cooling system, many times referred as dual stage evaporative cooling combines use of direct and indirect evaporative cooling system to provide supply air temperature which are much lower than wet bulb temperature and with lower specific humidity making the space temperature very comfortable in dry conditions; and

b) *Hybrid systems* — Some systems combine single or multiple stage evaporative cooling with traditional air conditioning, optimizing cooling performance while minimizing energy use. These hybrid systems switch between modes based on ambient conditions, providing efficient cooling in various climates. Indirect (air cooling by passing over a cool coil or heat exchanger) cooling is also often combined with a direct evaporative cooling process such devices are often termed as indirect direct evaporative coolers (IDEC). Cool coil may in turn use chilled water, evaporatively cooled water, pre-cooled air, or any such other cooling media.

6.2 Radiant Cooling and Heating

Radiant cooling system is a temperature-controlled system in which the temperature of the building surface facing indoor (ceiling, floor, wall etc) is reduced by the means of cool/chilled water which flows through a network of pipes embedded in the surfaces.

Any system shall only be called a radiant system if more than 50 percent of heat transfer takes place by radiative heat exchange by the temperature-controlled surface. Reduced surface temperature causes radiative heat exchange between human body and cold/hot surfaces and provide sensible cooling/heating inside the building. A separate air system shall be used for the ventilation requirements and handling the latent load.

The radiant cooling system design and installation shall ensure no condensation in the conditioned space.

6.2.1 *Advantages of Radiant Cooling System*

These points highlight the key benefits of radiant cooling, focusing on energy efficiency, comfort, and improved indoor air quality:

a) *Energy efficiency* — In radiant cooling system, cooling is provided through water which has a higher heat capacity than air which loses more energy in transportation. In radiant cooling system less fan energy is required than conventional all-air system due to very significant reduction in amount of air handled;

b) *Indoor air quality* — As a radiant cooling system is a hybrid system that also includes a separate air system to provide 100 percent outdoor air, the indoor air quality is enhanced;

c) *Thermal comfort* — Uniform temperature shall be maintained with radiant cooling system ensuring better thermal comfort;

d) *Integration with low grade energy/passive cooling sources* — In radiant cooling system a higher temperature 15 ºC to 20 ºC water temperature shall be used. The low-grade energy sources like cooling tower, geo-thermal and others shall be used; and

e) *Noise and reduced duct work* — This cooling system reduces noise, as they do not rely on fan and compressor for cooling, and this system use water pipes instead of the ducts, thereby having reduced duct work.

6.2.2 *Limitations of Radiant Cooling*

Radiant cooling is useful in the building where sensible load is higher and have humidity control. There are several areas like restaurant, lobby, and kitchen, etc where humidity control is very difficult, at such places, its application shall attract a problem of condensation. Similarly, if the latent load in a zone is significant then the use of radiant cooling system is limited. The radiant cooling system also has a limitation of cooling capacity in order to counter the condensation problem. Hence, the surfaces have to be maintained at a higher temperature as compared to the dew point temperature of indoor air.

6.2.3 *Types of Radiant Cooling Systems*

Following three types of Radiant systems are used:

a) *Radiant cooling panels system* (*RCP*) — The most extensively used system of radiant systems is the PANEL Systems. The insulation over the aluminium panel plays an important role in avoiding the heat transfer to the building structure. Poor connection shall lead to less heat transfer resulting in an inefficient system;

b) *Thermally activated building system* (*TABS*) — In thermally active building system (TABS) the conditioning plant operation is varied based on thermal loads. TABS has a high thermal inertia, which enables it to hold a large amount of heat within it during the day time and release this heat during night time to the hydronic system; and

c) *Embedded surface cooling systems* (ESC) — In embedded surface cooling system, an insulation cover on the pipes mounted on the side adjacent to the building structure minimizing thermal energy exchange of heat carrier with the building structure. Some floor systems are made by extruded plastic plates which forms capillary channels. In this case the heat exchange surfaces is very wide and at the same time the thickness of the system is very less which allows very low temperature differences.

6.2.4 *Requirement of Fresh Air Delivery*

Radiant cooling systems require a dedicated an outdoor air system responsible for supply of fresh/ventilation air into the radiant cooled space operating in parallel to the radiant cooling.

In case of high latent load and in case of oversized DOAS, energy savings from a radiant cooling system is reduced.

6.2.5 *Control Requirements*

Following controls are required to efficiently operate a radiant cooling system:

a) radiant surface temperature minimum two degree centigrade higher than indoor dew point temperature to avoid condensation on radiant surfaces. In case of occurrence;

b) water supply temperature or flow rate or both need to be controlled; and

c) modulate ventilation air supply as per the CO_2 monitoring.

6.3 Structural Cooling

Due to the high solar insolation during the daytime on built structures, heat gets embedded in the structure or the thermal mass of the buildings and is transmitted indoors. This heat raises the temperature of the indoor surfaces resulting in higher mean radiant temperatures experienced by occupants making them uncomfortable.

This heat has to be flushed out before it shall enter the living space.

This approach significantly reduces the cooling loads on the air conditioning system, in case the indoors are air conditioned.

Structural cooling generally employs the use of cooled water between 15 °C to nearly 30 °C, which is circulated in embedded piping network laid within the structure.

For air-conditioned areas, the minimum water temperature is 15 °C. Depending on the requirements this water temperature shall be higher up to 24 °C. This generally requires mechanical refrigeration similar to the chilled water supply in radiant cooling system.

Water between 20 °C and 30 °C can be generated employing natural cooling processes, which make use of the ambient conditions of air.

Naturally cooled water in ponds, rivers, or from geothermal sources should be considered. Properly designed and selected cooling towers make use of the lower wet bulb temperatures to achieve this cooling of water to slightly above prevailing wet bulb temperatures.

NOTES

1 In a conducive weather condition, cooled water shall be obtained by utilising the night sky cooling and storing the cooled water in an insulated and properly sized tank during the night. The same water shall be used to cool the structure during the harsh times of the day.

2 The water is circulated by employing a small pump, in the closed loop system. Principally, these systems are quite similar to thermally activated buildings covered under radiant cooling section.

6.3.1 *Advantages of Structural Cooling*

Structural cooling impedes solar heat from roof and floors by absorbing it before it causes thermal discomfort to occupants. Following are the advantages of structural cooling:

a) floor temperatures are maintained below body skin temperature, allowing the body to radiate the heat to the cooler surfaces;

b) reduces the cooling load on the building and lower size of ancillary cooling systems are required, leading to smaller pipes and ducting;

c) because of the large thermal mass of the structure, it shall act as a storage of cooling effect; and

d) as the Structure is cooled, it is maintained in a small band of temperature variations, thus eliminating the heat stress on the structure. The thermal expansion and contraction of the structure are very low, as compared to a building which is not structurally cooled.

6.3.2 *Applications*

Structural cooling is very versatile and shall be used to cool places like public utility areas, malls, hospitals, residential areas, office buildings, commercial spaces, educational institutes and many more applications to reduce air conditioner loads and provide comfort.

6.4 Geothermal for Heating and Cooling

Geothermal heating and cooling systems, also known as ground source heat pumps (GSHPs), uses the stable temperatures of the earth's subsurface to provide efficient climate control for buildings. By using the ground as a heat source in the winter and a heat sink in the summer, these systems shall achieve high efficiencies compared to conventional HVAC systems and needs to balance the earth as source.

NOTE — Geothermal systems shall be used in residential, commercial, and industrial buildings. They are particularly beneficial for new constructions but shall also be retrofitted into existing structures. Large-scale applications include district heating systems where multiple buildings are connected to a central geothermal system.

6.4.1 *Advantage of GSHP Systems*

GSHP systems offer significant energy efficiency, environmental benefits, and long-lasting performance. Following are the advantages of GSHP systems:

a) *Energy efficiency* — GSHPs achieves high efficiencies on the coldest winter nights, meaning they produce 3 units to 6 units of heat for every unit of electricity consumed;

b) *Cost savings* — GSHP systems provide substantial cost savings by delivering multiple units of heat for each unit of electricity, reducing energy bills over time;

c) *Environmental impact* — Reduce greenhouse gas emissions and reliance on fossil fuels; and

d) *Longevity* — Ground loops life of 50 years and above, and indoor components life of 20 years and above.

6.4.2 *Challenges with GSHP*

GSHP systems face certain challenges related to installation, maintenance, and initial investment:

a) *High initial cost* — Installation are expensive due to drilling and trenching requirements depending on terrain;

b) *Site-specific design* — Soil conditions, land availability, and local climate affect feasibility and design; and

c) *Maintenance* — While minimal, the system requires periodic maintenance, especially the indoor components. Geothermal heating and cooling systems represent a sustainable and efficient technology for managing building climate control. Despite the high initial investment, their long-term benefits in terms of energy savings, environmental impact, and system longevity make them an attractive option for a wide range of applications. As technology advances and installation costs decrease, geothermal systems are likely to become an increasingly common choice for sustainable building design.

6.5 Solar Cooling

There are different ways to utilise solar energy for producing cooling effect:

a) Solar thermal cooling systems

b) Solar photovoltaic cooling systems

Solar photovoltaic cooling systems is a vapour compression-based systems powered by electricity generated by solar photo voltaic system. Solar thermal based cooling systems use heat-based cooling systems that do not need electricity driven mechanical compressor.

6.5.1 *Solar Thermal Cooling Systems*

There are two different types of solar thermal cooling systems absorption type and desiccant type. All of these are driven by the heat from solar energy as explained below.

6.5.1.1 *Absorption cooling*

Vapour absorption machines (VAMs) are used for cooling using waste steam or hot water as a byproduct of process in various industries. VAMs use a salt-based solution (typically lithium bromide solution) to alternative evaporate water out and then condense it back into the solution. Types of VAMs depending on the temperature of the heat source:

a) *Single effect* — Uses hot water source between 80 °C to 90 °C to drive the cooling cycle.

b) *Double effect* — Uses 120 °C to 140 °C steam to drive the cooling cycle.

c) *Triple effect* — Uses 180 °C to 210 °C steam to drive the cooling cycle.

For most buildings, single effect VAM and double using solar hot water is the viable configuration. Double effect VAMs offer higher COP of 1.2 to 1.4, it is recommended to use single effect VAMs it is recommended to design the VAM around this temperature even though the COP of such system is only 0.7 to 0.8. Higher operating hours in a year make such a system more viable at lower COP.

Double and triple effect VAMs shall be used in buildings where waste heat or low-cost heat >120 °C is available in immediate vicinity. This would typically be in industrial settings.

6.5.1.2 *Adsorption cooling*

Adsorption cooling systems use solid desiccant material like silica gel to enable a heat driven cooling cycle. Water is used as a refrigerant in such systems. The cycle works by alternating 'adsorption' and 'desorption' of water from a solid desiccant. Heat between 60 °C to 80 °C is required for 'desorption' and heat from solar collectors shall be used.

The COP of adsorption chiller is between 0.35 to 0.55 depending on the chilled water temperature required, hot water source temperature and condenser water temperature. While lower COP increases the solar field size, adsorption chiller's ability to operate even at 60 °C source temperature allows.

6.5.1.3 *Desiccant cooling*

This typically refers to liquid desiccant cooling where solar heat is used to regenerate the desiccant. Typical regeneration temperature is 60 °C that be consistently achieved using solar collectors.

6.5.2 *Advantages of Solar Thermal Cooling Systems*

Solar thermal cooling systems offer sustainable and energy-efficient solutions by harnessing solar energy for cooling purposes, reducing reliance on conventional energy sources. Following are the advantages of solar thermal cooling systems:

a) *Energy efficiency* — Overall cooling system electricity consumption shall be 0.4 kW/Ton to 0.6 kW/Ton compared to 1 kW/Ton to 1.4 kW/Ton for conventional air conditioning systems; and

b) *Lower carbon footprint* — Lower electricity consumption leads to lower carbon footprint of the system. This shall be useful for achieving decarbonization goals of the building or organization.

6.6 Personal Environmental Control Systems

Personal environmental control systems (PECS) includes low-energy, low-powered personalised HVAC systems as well as controls and sensors. The PECS largely shall be categorised into the following five categories and further detailed in Table 13:

a) Heating only PECS;

b) Cooling only PECS;

c) Ventilation only PECS;

d) Heating and ventilation PECS;

e) Cooling and ventilation PECS; and

f) Heating, cooling and ventilation PECS.

NOTE — Heating PECS and cooling PECS rely only on conduction and radiative heat transfer without convection (air movement). While all PECS with ventilation, rely on air as a heat transfer medium, hence air movement in ventilation PECS is inherent.

Table 13 Personal Environmental Control Systems

(*Clause* 6.6)

a) Heating only PECS Devices

Warmed Seat	**Device Description**	**Warmed Seat Heated Using Electrical Elements or Embedded Water Tubes**	The warmed seat is recommended to be used in combination with a foot heater.
	T_{room}	> 5 °C	
	$T_{PECS\text{-}surface}$	< 44 °C	
	V_{PECS}	N/A	
	Target body parts	Back, buttocks, thighs	
	PECS position	As the occupant's seat	
	Restriction of movement	Occupant shall remain in contact with the seat fabric	
	Manual control	Temperature	
	Background ventilation	Required	
Radiant Foot Heater		**Radiant Foot Heater heated using electrical elements**	The radiant foot heater should be operated in the range of T_{room}+10 °C to T_{room}+20 °C for optimum comfort.
	T_{room}	> 10 °C	
	$T_{PECS\text{-}surface}$	< 30 °C	
	V_{PECS}	N/A	
	Target body parts	Feet	
	PECS position	~20 cm from the feet	
	Restriction of movement	Feet shall remain adjacent to the heated surface	
	Manual control	Temperature	
	Background ventilation	Required	
Radiant Heating Panel		**Radiant Heating Panels placed as desktop partitions heated using electrical elements or embedded water tubes**	The exposed surface area of the radiant heating panel to the occupant should be optimized by placing more panels to the left and right sides of the occupant.
	T_{room}	> 19 °C	
	$T_{PECS\text{-}surface}$	< 23 °C	
	V_{PECS}	N/A	
	Target body parts	Face, chest, arms, legs	
	PECS position	~50 cm from the occupant, as the desk partition	

Table 13 (*Continued*)

	Restriction of movement	Occupant should remain in the field of view of the Radiant Panel	
	Manual control	Temperature	
	Background ventilation	Required	
Palm Warmer		**Desktop-based Palm Warmers heated using electrical elements**	The device shall be placed such that it acts as a resting spot for the hands while typing.
	T_{room}	> 18 °C	
	$T_{PECS\text{-}surface}$	< 35 °C	
	V_{PECS}	N/A	
	Target body parts	Palms	
	PECS position	On the desktop, between the computer keyboard and the occupant.	
	Restriction of movement	Palms shall be in direct contact with the device surface	
	Manual control	None	
	Background ventilation	Required	
b) Cooling only PECS Devices			
Cooled Seat	**Device Description**	**Seat cooling using embedded water pipes cooled using compressive cooling**	It is recommended to use a movable desktop fan in combination with the cooled seat.
	T_{room}	< 45 °C	
	$T_{PECS\text{-}air}$	> 18 °C	
	V_{PECS}	N/A	
	Target body parts	Back, buttocks, thighs	
	PECS position	As the occupant's seat	
	Restriction of movement	Occupant shall remain in contact with the seat fabric	
	Manual control	Temperature	
	Background ventilation	Required	
Cooled Garment		**Jacket-like garment using phase change materials or compressive cool air supply**	Typical cooled garments are battery-operated, with an average charge lasting for ~ 6 hours; they weigh around 3 kg to 5 kg; cooled garments with phase change materials typically perform better than those with embedded fans.
	T_{room}	< 34 °C	
	$T_{PECS\text{-}air}$	> 21 °C	
	V_{PECS}	N/A	
	Target body parts	Chest, back, shoulders	
	PECS position	As the occupant's vest	
	Restriction of movement	None	
	Manual control	Temperature	
	Background ventilation	Required	

Table 13 (*Continued*)

Radiant Cooling Panel		**Radiant Cooling Panels placed as desktop partitions cooled using embedded water pipes using compressive cooling**	The exposed surface area of the radiant cooling panel to the occupant should be optimized by placing more panels to the left and right sides of the occupant.
	T_{room}	< 32 °C	
	$T_{PECS\text{-}air}$	> 17 °C	
	V_{PECS}	N/A	
	Target body parts	Face, chest, arms, legs	
	PECS position	~ 50 cm from the occupant, as the desk partition	
	Restriction of movement	Occupant should remain in the field of view of the radiant panel	
	Manual control	Temperature	
	Background ventilation	Required	
c) Ventilation only PECS Devices			
Desktop-based Devices	**Device Description**	**Desk-mounted air-terminal device supplying (fresh or recirculated) air at room temperature**	The airflow shall not strike the desktop and the airflow pattern should preferably be kept as close to the natural wind pattern in terms of its turbulence and frequency of variability.
	T_{room}	< 30 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 1.5 m/s	
	Target body parts	Face, neck, chest, arms	
	PECS position	~50 cm from the occupant, on the desktop	
	Restriction of movement	Occupant shall move while manually changing the airflow direction of the fan/device	
	Manual control	Air velocity, direction	
	Background ventilation	Optional	
Movable Panel		**Desktop based movable air-terminal device supplying (fresh or recirculated) air at room temperature**	The air stream should enter the occupant's breathing zone parallel to the cheek, avoiding direct contact with the eyes.
	T_{room}	< 28 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 0.7 m/s	
	Target body parts	Face, neck, chest, arms	
	PECS position	~50 cm from the occupant, on the desktop	
	Restriction of movement	Occupant is free to move within the flow region while adjusting the direction of the panel	
	Manual control	Air velocity, direction	
	Background ventilation	Optional	
Headrest-embedded Device		**Seat headrest-based air-terminal device supplying recirculated air at room temperature**	Air terminal devices with a large opening area require low air

Table 13 (*Continued*)

	T_{room}	< 26 °C	velocity and vice versa.
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 0.6 m/s	
	Target body parts	Face	
	PECS position	As the seat headrest, next to the cheeks	
	Restriction of movement	Occupant shall keep the head on the headrest.	
	Manual control	Air velocity	
	Background ventilation	Required	
Pedestal Fan		**Pedestal fan supplying recirculated air at room temperature**	The fan direction should be kept such that it does not cause discomfort by displacement of light-weight objects on the desktop.
	T_{room}	< 30 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 2.5 m/s	
	Target body parts	Whole body	
	PECS position	~ 1.5 m from the occupant	
	Restriction of movement	Occupant is free to move within the flow region while adjusting the direction of the fan	
	Manual control	Air velocity, direction	
	Background ventilation	Optional	
Ceiling Fan		**Ceiling-mounted fan supplying recirculated air at room temperature**	To optimize the effect of the fan, it should be placed on the ceiling right above the desktop, with the rotating blades at an angle of 30° to 45° from the ceiling.
	T_{room}	< 30 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 1 m/s	
	Target body parts	Head, shoulders	
	PECS position	~ 1.5 m above the occupant, on the ceiling	
	Restriction of movement	Occupant shall remain in the flow direction	
	Manual control	Air velocity	
	Background ventilation	Optional	
Ventilated Seat		**Ventilated seat embedded with fans supplying recirculated air at room temperature**	The occupant's body should not block the complete airflow and it should reach the breathing zone.
	T_{room}	< 32 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	< 2 m/s	
	Target body parts	Buttocks, back, thighs	
	PECS position	As the occupant's seat	
	Restriction of movement	Occupant shall remain in contact with the seat fabric	
	Manual control	Air velocity	

Table 13 (*Continued*)

	Background ventilation	Required	
Ventilated Garment		**Jacket-like garment embedded with low-wattage fans supplying recirculated air at room temperature**	A typical battery-operated ventilated garment operates up to 7 hours on 60 percent output power.
	T_{room}	< 34 °C	
	$T_{PECS\text{-}air}$	Same as T_{room}	
	V_{PECS}	Equivalent to a flow rate of < 22 L/s	
	Target body parts	Chest, Back	
	PECS position	As the occupant's vest	
	Restriction of movement	No restriction in movement	
	Manual control	Air velocity	
	Background ventilation	Required	
d) Heating and Ventilation PECS Devices			
Desk-mounted Device	**Desk-mounted air-terminal device supplying air warmed through compressive heating**		The device should be used in combination with a Foot-heating Panel; the airflow should not directly strike the eyes of the occupant as it might lead to dry-eye discomfort.
	T_{room}	> 19 °C	
	$T_{PECS\text{-}air}$	< 25 °C	
	V_{PECS}	< 1 m/s	
	Target body parts	Face, arms, chest, front thighs	
	PECS position	~ 50 cm from the occupant, on the desktop	
	Restriction of movement	Occupant shall remain in the flow direction	
	Manual control	Temperature, air velocity, direction	
	Background ventilation	Optional	
Ventilated and Warmed Seat	**Warmed seat with embedded fans – the seat is warmed using electrical elements or embedded water pipes**		The device should be used in combination with a foot-heating panel
	T_{room}	> 20 °C	
	$T_{PECS\text{-}air}$	< 45 °C	
	V_{PECS}	< 0.7 m/s	
	Target body parts	Neck, back, buttocks, back thighs	
	PECS position	As the occupant's seat	
	Restriction of movement	Occupant shall remain in contact with the seat	
	Manual control	Temperature, air velocity	
	Background ventilation	Required	
Nozzle	**Nozzle-based air terminal device supplying air warmed through compressive heating**		The air stream should enter the occupant's breathing zone parallel to the cheek, avoiding direct contact with the eyes.
	T_{room}	> 19 °C	
	$T_{PECS\text{-}air}$	< 50 °C	
	V_{PECS}	< 1 m/s	

Table 13 (*Continued*)

	Target body parts	Head, neck	
	PECS position	~ 1 m from the occupant, directed towards the head	
	Restriction of movement	Occupant shall remain in the flow direction	
	Manual control	Temperature, air velocity	
	Background ventilation	Optional	
Movable Panel	**Desk-mounted movable air terminal device supplying air warmed through compressive heating**		The air stream should enter the occupant's breathing zone parallel to the cheek, avoiding direct contact with the eyes.
	T_{room}	> 20 °C	
	$T_{PECS\text{-}air}$	< 26 °C	
	V_{PECS}	< 1 m/s	
	Target body parts	Head, neck, chest, arms	
	PECS position	~ 50 cm from the occupant, directed towards the head	
	Restriction of movement	Occupant is free to move within the flow region while adjusting the direction of the panel	
	Manual control	Temperature, air velocity, direction	
	Background ventilation	Optional	
e) Cooling and Ventilation PECS Devices			
Desk-mounted Device	**Desk-mounted air-terminal device supplying air cooled through compressive cooling**		The air stream should enter the occupant's breathing zone parallel to the cheek, avoiding direct contact with the eyes.
	T_{room}	< 30 °C	
	$T_{PECS\text{-}air}$	> 15 °C	
	V_{PECS}	< 1.5 m/s	
	Target body parts	Face, neck, chest	
	PECS position	~50 cm from the occupant, on the desktop	
	Restriction of movement	Occupant shall remain in the flow direction	
	Manual control	Temperature, air velocity	
	Background ventilation	Optional	
Desk-edge based Device	**Desk-edge mounted horizontal air terminal slits supplying air cooled through compressive cooling**		The device should preferably provide a vertical airflow directly under the occupant's chin and should have adjustable vanes for manual adjustment of airflow direction.
	T_{room}	< 26 °C	
	$T_{PECS\text{-}air}$	> 20 °C	
	V_{PECS}	< 1 m/s	
	Target body parts	Face, neck	
	PECS position	~ 10 cm from the occupant, on the desk edge	

Table 13 (*Continued*)

	Restriction of movement	Occupant shall remain directly above the device outlet	
	Manual control	Temperature, air velocity, direction	
	Background ventilation	Optional	
Movable Panel	**Desk-mounted movable air terminal device supplying air cooled through compressive cooling**		The air stream should enter the occupant's breathing zone parallel to the cheek, avoiding direct contact with the eyes.
	T_{room}	< 30 °C	
	$T_{PECS\text{-}air}$	> 20 °C	
	V_{PECS}	< 1 m/s	
	Target body parts	Face, neck, chest, shoulders, arms	
	PECS position	~50 cm from the occupant, on the desktop	
	Restriction of movement	Occupant is free to move within the flow region while adjusting the direction of the panel	
	Manual control	Temperature, air velocity, direction	
	Background ventilation	Optional	
Armrest-embedded Device	**Seat armrest-mounted air terminal device supplying air cooled through compressive cooling**		The airflow should strike the occupant at a 45° angle from below.
	T_{room}	< 25 °C	
	$T_{PECS\text{-}air}$	> 20 °C	
	V_{PECS}	< 0.7 m/s	
	Target body parts	Face, neck, chest, arms	
	PECS position	As the armrest of the occupant's seat	
	Restriction of movement	Occupant shall remain seated and the armrests shall not be blocked	
	Manual control	Temperature, Air Velocity	
	Background ventilation	Required	
Floor-mounted Device	**Floor-mounted air terminal device supplying air cooled through compressive cooling, also known as underfloor air distribution**		The device should be used in combination with a desk-mounted or movable panel device for better control
	T_{room}	< 27 °C	
	$T_{PECS\text{-}air}$	> 16 °C	
	V_{PECS}	< 1 m/s	
	Target body parts	Feet, legs, thighs, buttocks	
	PECS position	~50 cm from the occupant's seat, on the floor	
	Restriction of movement	Occupant shall move radially around the device	
	Manual control	Temperature, air velocity	
	Background ventilation	Optional	
	Ceiling-mounted air terminal device supplying air cooled through compressive cooling		The airflow should have a low turbulence

Table 13 (*Concluded*)

Ceiling-mounted Device	T_{room}	< 29 °C	intensity and long range.
	$T_{PECS\text{-}air}$	> 21 °C	
	V_{PECS}	< 0.8 m/s	
	Target body parts	Head, neck, shoulders	
	PECS position	~1.5 m above the occupant's head, on the ceiling	
	Restriction of movement	Occupant shall move around the vertical flow	
	Manual control	Temperature, air velocity	
	Background ventilation	Optional	
Radiant Cooling Panel with Fan	**Radiant panels placed as desktop partitions with embedded water pipes cooled using compressive cooling - used in combination with low-power desktop fans**		The exposed surface area of the radiant panel to the occupant should be maximized; the airflow from the fan should not strike the desktop.
	T_{room}	< 28 °C	
	$T_{PECS\text{-}air}$	> 17 °C	
	V_{PECS}	< 1.5 m/s	
	Target body parts	Face, neck, chest, arms	
	PECS position	Radiant panels as the desk partition ~ 50 cm from the occupant; Fan on the desktop, ~ 40 cm from the occupant	
	Restriction of movement	Occupant shall remain exposed to the radiant surface and shall move fan as per comfort	
	Manual control	Temperature, air velocity, direction	
	Background ventilation	Optional	

7 DISTRICT COOLING SYSTEMS

District cooling systems (DCS) provides and distributes cooling energy from a central plant to a surrounding cluster of residences, commercial spaces, industries, or institutions occupied by tenants or clients.

In a district cooling system, a large central plant produces chilled water for supply to multiple buildings through an insulated closed-loop underground piping network. Traditional AC systems consume a majority of a building's peak electricity demand, usually at peak cost. With district cooling, peak demand on the grid, as well as operating energy consumption, decreases. It avoids the capital costs of installing multiple HVAC systems that is chillers and cooling towers and associated mechanical and electrical equipment at each building to a central location, freeing up valuable rooftop and building space. By aggregating the cooling needs of multiple buildings, district cooling generates economies of scale. Further, DCS also contributes to sustainability by ensuring reuse of STP water and reducing refrigerant requirement and leakages.

In India, DCS is ideal for mixed-use developments, large infrastructure projects such as aero cities, industrial clusters, residential clusters, educational campuses, IT parks, smart city infrastructure or retrofit of a DCS system for an existing cluster of individually cooled buildings or a redevelopment of existing buildings.

DCS provide several economic benefits to end-users, including reduced capital and operating costs, less frequent maintenance, space savings, and lower electricity usage than in more traditional AC systems (*see* Table 14).

Table 14 Benefits of DCS Systems

(*Clause* 7)

Environmental	Societal	Economic
(1)	(2)	(3)
		For DCS Service Providers/Investors:
a) Energy saving b) Peak energy demand reduction c) Carbon and GHG emission saving d) Enabling transition to low or zero GWP refrigerants e) Water conservation f) Improved micro-climatic conditions by reduced urban Heat Island Effect g) Improved air quality	a) Enhanced aesthetics b) Local business development c) Skill development and improved employability d) Security of supply for cooling e) Access to affordable cooling for achieving thermal comfort f) Improved quality of life g) Reduced noise from cooling equipment	a) New business venture with competitive advantage b) Long-term stable and profitable revenues c) Employment generation Avoided investments in summer electricity peak production, transmission, and distribution Higher energy and resource utilization **For end-users of DCS:** a) Avoided capital expenditure (CAPEX) and space requirement for cooling equipment like chillers and cooling towers and related costs b) Reduced O&M cost

7.1 Key Components of District Cooling System

District cooling systems (DCS) shall utilize recycled treated water from sewage effluent as cooling tower makeup water. Municipalities should provide this recycled water for in-house treatment, addressing water shortages and promoting water circularity initiatives in cities. Following are the requirements of DCS:

a) DCS benefit from technologies like thermal energy storage (TES) and free cooling. TES reduces peak electricity costs by shifting cooling production to off-peak hours, while free cooling leverages ambient conditions to lower overall electricity demand.

b) In some cases, district cooling systems may directly use seawater or fresh surface water for heat rejection, subject to local laws.

c) Large central plants should use waste heat or geothermal energy for absorption refrigeration in district cooling systems. This approach is more feasible at the district scale than for individual buildings.

7.1.1 *District Cooling Plant*

The district cooling plant (DCP) generates chilled water for cooling services. It includes chillers, cooling towers, pumps, and auxiliary systems. Thermal energy storage (TES) shall enhance chiller capacity. DCPs play a crucial role in efficient and sustainable cooling solutions for urban areas. All new plants with capacity equal to or greater than 10 000 TR shall include a TES system. All new plants with capacity equal to or greater than 10 000 TR shall carry out due feasibility analysis to include a TES system. Further, for all cluster development of multiple buildings with cumulative cooling demands above 5 000 TR, setting up of District Cooling System, if feasible, should be considered.

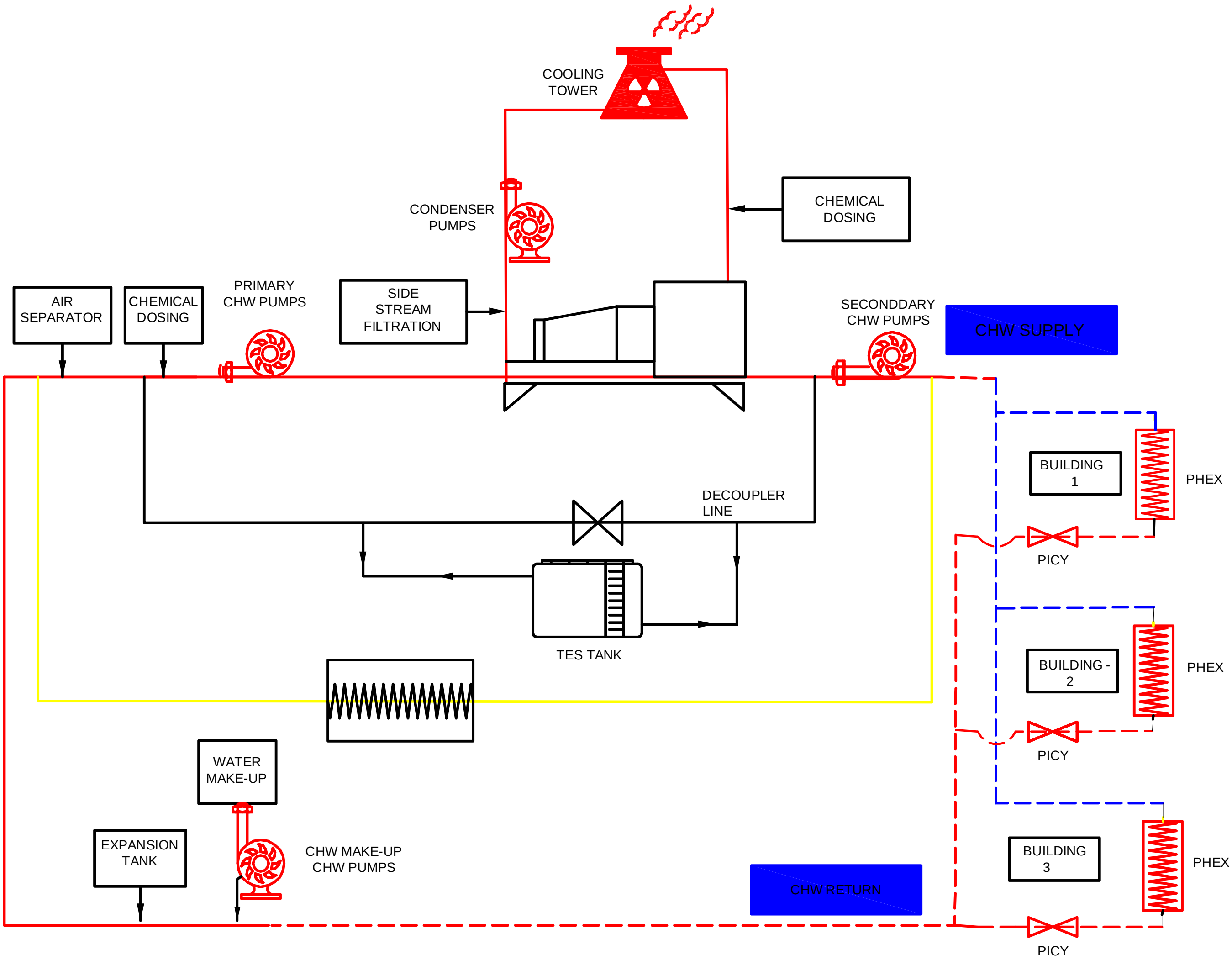


FIG. 2 LAYOUT OF A DISTRICT COOLING SYSTEM

7.1.2 *Distribution Network*

Based upon ease of operation, geographical constraints, safety, and applications, there are two kinds of distribution networks underground pre-insulated and above-ground site insulated chilled water distribution systems. Carbon steel or mild steel pipes should be used for chilled water and condenser water distribution piping, with proper protective coating wrap for preventing thermal losses. In the Indian context, an indicative DC system design (including the distribution system) is presented in Fig. 2. However, the actual design specifications may vary depending on a multitude of factors. Distribution pipelines should be designed with a life expectancy of minimum 40 years. Following shall be the default design conditions for the distribution network, unless specified by the designer/user:

a) Ambient temperature (for design): based on where the DC system is located;

b) Chilled water supply temperature at DCP: 5 ºC ± 1 ºC;

c) Temperature rise in the chilled water supply distribution network: 0.5 ºC to1 ºC per unit length (0.5 ºC up to 5 000 m);

d) Approach across Energy Transfer Station: </= 1.1 ºC;

e) Chilled water supply temperature at consumer end: 6 ºC ± 1 ºC;

f) Chilled water return temperature at consumer end: 15 ºC ± 1 ºC;

g) Temperature drop in the chilled water return distribution network: 0.5 ºC to 1 ºC;

h) Chilled water return temperature at DCP: 14 ºC ± 1 ºC; and

j) Design pressure of the chilled water distribution network: 5 bar to 15 bar (g) (depends on the overall DCS design and building elevation).

7.1.3 *Energy Transfer Station*

An energy transfer station (ETS) serves as a bridge between a district cooling (DC) service and consumers' air conditioning systems. Energy transfer shall be direct or indirect transfer. In district cooling plants where there is uniform time of operation of all spaces and the cooling service provider and end user have no need for decoupling then direct transfer that is supply of chilled water to all air handling units shall be applied. In all other cases indirect heat transfer via an ETS shall be applied.

The ETS transfers thermal energy generated at the central plant to buildings. Within the ETS, heat exchangers allow buildings to utilize this energy for space conditioning. Key components include pumps, isolation valves, and control elements. Proper sizing ensures meeting cooling demand without redundancy. Consideration for equipment access, maintenance, and transport pathways shall be done. Controls, such as pressure independent control valves, help in maintaining efficiency.

7.1.4 *Instrumentation, Monitoring and Control System* (*SCADA*)

A DC instrumentation and control system shall be designed, installed, and commissioned to monitor and control various system components for enhancing operational efficiency of the DC system. The main components of an instrumentation and control system are a programme logic controller (PLC), human machine interface (HMI), and distribution control system. The control system shall be designed to maintain the entire DC plant operations and functionality, and integration with ETSs to maintain transparency in operations. The control system shall monitor comprehensive chiller plant parameters, cooling tower parameters, TES parameters, chemical treatment equipment parameters, building management system (BMS) of the DC plant building, firefighting status of the plant building, etc and may be integrated on SCADA.

7.1.5 *Measurement and Billing System*

The measurement and monitoring system shall facilitate real time monitoring of kW/TR of the plant. Infrastructure for continuous measurement and logging of cooling production and consumption (TR-h) and complete breakdown of all associated power demand (kW, voltage, ampere, power factor), and energy consumption (kWh) is crucial for district cooling (DC) systems. Separate BTU meters for each cooling consumer, installed at the Energy Transfer Station (ETS), generate monthly cooling bills. Individualizing BTU metering by consumer or apartment enhances responsibility for thermal energy consumption. Properly sized BTU meters accurately measure the entire cooling load range. All BTU meters shall be calibrated annually.

7.2 Design Approach for DCS

7.2.1 *Site Selection Criteria*

Site selection involves careful evaluation of various aspects with different degrees of importance. A methodology should be prepared to ensure that the critical aspects are not overlooked. The aim is to define a set of criteria that helps in shortlisting the most optimum site, which has the following characteristics:

a) resource-efficient;

b) minimum environmental impact;

c) cost-effective;

d) adequate in terms of logistics;

e) ensures maximum efficiency;

f) offers possibility of easy future expansion;

g) proximity to utilities (power and STP infrastructure) and drainage; and

h) proximity to load centre of large multi tower developments.

The first stage of site selection is determining the study area and indicators. The second stage involves the diagnosis process, wherein indicators are ranked and weighted according to benchmarks, guidelines, and preferences. Furthermore, this stage involves analysis of data collected for various indicators. The last stage is decision making, where potential sites are ranked, followed by the development of a feasibility plan. The process shall take place via the traditional method of site suitability analysis or using computer-based applications and the major factors influencing DCS implementation are given in Table 15.

7.2.2 *Sizing Philosophy for DCS*

The sizing of both mechanical and electrical systems shall ensure optimal performance, energy efficiency, and reliability of the district cooling system.

7.2.2.1 *Mechanical system sizing* (*chiller sizing, network and delta T*)

In sizing a chiller for a district cooling system, following critical factors shall be considered.

a) First, the required cooling capacity based on the peak cooling load and combined load diversity. This load shall vary significantly depending on the specific application (for example, residential, commercial, or industrial) and local climate conditions.

b) The desired temperature difference between the inlet and outlet water is to be decided. This difference affects the chiller's efficiency and overall performance.

c) The ambient conditions, such as the local climate and ambient temperature. Hotter climates may necessitate larger chillers to handle higher cooling demands effectively.

d) Assess installation constraints, such as available space, and any limitations related to the chiller's physical placement within the district cooling network proper chiller sizing is crucial for achieving efficient and reliable district cooling operations.

The successful implementation of district cooling systems depends greatly on the ability of the system to obtain high-temperature differentials that is delta temperatures (T) between the supply and return water. It is crucial to ensure that the district cooling system shall operate with reasonable size distribution piping and pumps to minimize the pumping energy requirements.

The pipe sizing is governed by four key factors:

1) the system's delta T;

2) maximum allowable flow velocity;

3) distribution network pressure at the design load conditions; and

4) minimum differential pressure requirements to service the most remote customers.

NOTES

1 Generally, it is most cost-effective to design for a high delta T in the district cooling system because this allows for smaller pipe sizes in the distribution system. The system operating efficiency also increases with an increased delta T due to the reduced pumping requirements caused by the reduced flow rates and the reduced heat gains/losses in the distribution system. The system's delta T is typically limited to 8 ºC to 11 ºC.

2 The maximum allowable flow velocities are governed by pressure drop constraints and critical system disturbances caused by transient phenomena (that is water hammer effects). Generally, velocities higher than 2.5 m/s to 3.0 m/s should be avoided unless the system is specifically designed and protected to allow for higher flow velocities.

Table 15 Major Factors Influencing DC System Implementation

(*Clause* 7.2.1)

Building Design and Use (1)	**Building Energy Demand** (2)	**Local Context** (3)	**Resource Requirement** (4)	**Policy and Governance** (5)
a) Availability of space for distribution centre or possibility of extension b) Floor Area Ratio or Floor Space Index c) Carpet area d) Total floor area e) Diverse use of buildings f) Ownership g) Availability of water and power to meet current and future demand h) Electricity and water cost	a) Peak demand b) Demand density c) Delivery density d) Time distribution of energy demand over days, weeks, months and years. e) Potential future cooling demand	a) Topography b) Slope and contours c) Existing and proposed utility lines d) Water e) Drainage f) Telecom, PNG line, electricity line, etc. g) Access to natural cold-water body h) Access to renewable energy and/or waste energy source j) Road network	a) Financial assistance b) Capital Investment c) Rate of return d) Payback period e) Labour and other resources f) Availability of skilled labourers g) Availability of machinery h) Time j) Project timeline	a) Existing policies, programmes, and missions b) Proactiveness of local government c) Administrative capacity d) Ease of getting permissions and clearances

7.2.2.2 *Electrical system sizing (transformers, diesel generators and substation)*

For efficient and safe functioning of DCS, the electrical supply system plays a critical role. The sizing of electrical supply system components such as transformers, diesel generator and substation shall be designed considering the parameters listed below:

a) *Transformers* — Transformers play a critical role in stepping up or down voltage levels within the distribution network. Sizing shall be done considering:

   1) system voltage level requirements;

   2) capacity for transformer selection;

   3) diversity considerations;

   4) efficiency/losses; and

   5) redundancy/outage considerations.

b) *Diesel generators* — DG is a power backup during emergency when main power supply not available. Sizing shall be done considering:

   1) backup voltage level requirements;

   2) quantum of backup power required when the main power is not available;

   3) allowable loading during continuous operation;

   4) redundancy/outage considerations; and

   5) allowable noise considerations as per CPCB guidelines.

c) *Substation design* — Substation design involves planning, engineering, and constructing electrical substations for efficient power transmission and distribution. It needs to be done based on district cooling plant (DCP) capacity requirement. Accurate sizing ensures efficient and reliable electrical operations in district energy systems. Here are some key points about substation design:

   1) voltage and power capacity requirements;

   2) component selection;

   3) CapEx and OpEx; and

   4) footprint requirements.

7.3 General Requirements

These requirements ensure that DCS installations meet operational efficiency, safety, and performance standards.

7.3.1 *Specific Power Consumption for DCS*

The specific power consumption shall be measure as ratio of kW electric power to kW thermal output the system efficiency of a DC chiller plant shall not be less than 0.85 input kilowatts per TR (ikW/TR). However, depending on system size and detailed economic analysis, there are opportunities to improve the efficiency levels of DCS.

7.3.2 *Testing and Commissioning Procedures for Hydrotesting and Pipe Conditioning*

Necessary hydro testing shall be carried out for the distribution network. The distribution pipes should be subjected to pressure testing as per the design specifications and guidelines approved by the owner and their design consultant representatives. Where required for certain equipment like chillers, factory testing will be required to ensure design conditions are being complied with. Once the individual equipment is all put together as a system, proper testing of the complete system should be carried out.

The procedures for hydrotesting and flushing of pipes during the testing and commissioning phase for district cooling systems:

7.3.2.1 *Hydrostatic testing*

The strength and integrity of piping system shall be tested by hydrostatic testing at water pressure equivalent to 1.5 times the design operating pressure for stipulated time.

a) *Hydrostatic pressure testing procedure* – The hydrostatic pressure test shall be conducted as mentioned below:

   1) *Step* 1 – Filling the System

      i) Fill the entire piping network with water.

      ii) Ensure that all air pockets are removed.

   2) *Step* 2 – Pressurization

      i) Gradually increase the water pressure to at least 1.5 times the design pressure.

      ii) Maintain this pressure for a specified duration (usually a few hours).

   3) *Step* 3 – Inspection

      i) Inspect the entire system for leaks.

      ii) Pay attention to joints, welds, and connections.

   4) *Step* 4 – Record Keeping

      i) Document the test results, including pressure readings and any observed leaks.

   5) *Step* 5 – Depressurization

      i) Gradually reduce the pressure and drain the system.

b) *Safety considerations*

   1) Ensure safety protocols are followed during pressurization and depressurization.

   2) Personnel should be aware of potential hazards.

c) *Pipe conditioning*

   Pipe conditioning removes debris, dirt, and contaminants from the pipes and to form a protective anti corrosive layer on the inner surface of the pipes. The procedure for pipe conditioning is as follows:

   1) *Step* 1 – Preparation

      i) Ensure the system is clean and free from construction debris.

   2) *Step* 2 – Flushing Process

      i) Flush the pipes with water to remove loose particles.

      ii) Use stipulated high-velocity flow to dislodge any remaining debris.

      iii) Passing criteria to next step is to get clean water after flushing.

   3) *Step* 3 – Chemical cleaning

      i) Flush the pipes with water mixed with cleaning chemicals.

ii) Use stipulated high-velocity flow to dislodge any remaining debris.

iii) Passing criteria to next step is to get clean water and to meet certain water parameters at the end of chemical cleaning.

4) *Step* 4 – Passivation

i) Circulate the water mixed with passivation chemicals for stipulated period

ii) Passing criteria is to get clean water and to meet certain water parameters at the end of passivation

iii) Minimum quantity of passivation chemicals shall be ensured at the end of passivation and throughout the operation period.

5) *Step* 5 – Verification

i) Verify that the system is clean and ready for operation.

6) *Step* 6 – Record Keeping

i) Document the pipe conditioning process.

7.3.2.2 *Performance testing of DCS*

Performance tests fall under two categories, operational performance and availability performance. Operational performance is to ensure that during operation plant is meeting the designed electrical and water performance. Availability test is to ensure that availability of the plant equipment is meeting the availability and reliability conditions set forth over a stipulated time without any downtime.

Following indices shall be used to establish performance of a DCS:

Overall plant efficiency	kWh/TR-h
Water consumption	litre/h
Cooling tower water discharge	litre/h

Table 16 shall be used for benchmarking of performance of DCS.

Table 16 Recommended Performance of DCS

(*Clause* 7.3.2.2)

Sl No.	DCS Plant Type	TES	Condenser Cooling	Water Source	Max kW/ TR-h
(1)	(2)	(3)	(4)	(5)	(6)
	Energy Consumption				
i)	New	Chilled Water	Cooling Tower	Recycled/City supplied	0.87
ii)	New	None	Cooling Tower	Recycled/City supplied	0.91
iii)	New	Ice	Cooling Tower	Recycled/City supplied	1.17
iv)	New	None	Cooling Tower	Sea Water	0.96
v)	Existing	Chilled Water	Cooling Tower	Recycled/City supplied	1.00
	Condenser Water Consumption				litres/TR-h
i)	New/Existing	Ch Water/None	Cooling Tower	City Supply	8.00
ii)	New/Existing	Ch Water/None	Cooling Tower	Recycled	11.00
	Cooling Tower Water Discharge				
i)	New/Existing	Ch Water/None	Cooling Tower	City Supply	1.5 to 2.0
ii)	New/Existing	Ch Water/None	Cooling Tower	Recycled	4.5 to 5.0

7.4 Operation and Maintenance of DCS – Best O&M Practises

Practices as given below shall be followed for operating and maintaining a district cooling system:

a) Regular inspections and maintenance:

   1) Conduct routine inspections of equipment, pipelines, and components;

   2) Address minor issues promptly to prevent major failures; and

   3) Implement PPM strategies

b) Water treatment and quality:

   1) Maintain water quality in chilled water and hot water systems:

      i) Regularly clean and treat cooling towers to prevent scaling, corrosion, and biological growth.

c) Energy efficiency measures:

   1) Optimize the operations of chillers, pumps, cooling towers etc;

   2) Monitoring and control strategies; and

   3) Implement variable flow systems and energy-efficient equipment.

d) Leak detection and repair:

   1) Monitor for leaks in distribution networks and fix them promptly; and

   2) Leaks waste energy and reduce system efficiency.

e) Thermal storage optimization:

   1) Use thermal energy storage (TES) effectively to balance supply and demand; and

   2) Charge TES during off-peak hours and discharge during peak demand.

f) Metering and billing accuracy:

   1) Ensure accurate metering for billing purposes; and

   2) Individualize metering for end-users where possible.

7.5 Economics of DCS [Business Model for DCS – CaaS (Cooling as a Service)]

Cooling as a service (CaaS) is an innovative business model where end-users subscribe to cooling services without owning the infrastructure. The district cooling provider finance, build and maintains the cooling plant, promoting energy efficiency and sustainability. Users benefit from reduced upfront costs and simplified operations, while the provider ensures reliable service delivery. CaaS shifts the focus from ownership to access, fostering efficient and sustainable cooling solutions

7.5.1 *Tariff Structure - Fixed and Variable Charges + Connection Charges*

The tariff structure for district cooling systems is monthly and typically includes the following components. The recovery mechanism shall be tailored based on the various models in the real estate market say lease, sale, hybrid model etc and hence various contract strictures shall be tailored between real estate developer, DCS developer and end users:

a) *Connection charge* — It’s a one-time fee paid by users when connecting to the district cooling system.

b) *Contract capacity charges* — These charges are effectively availability fees, covering capital expenditures (CapEx) and fixed operation and maintenance (O&M) costs. They ensure that the district cooling provider maintains the capacity to deliver cooling services.

c) *Consumption charge* — The consumption charge covers variable O&M costs, including chemicals, electricity, and water (potable or treated sewage effluent) consumed by the end users.

7.5.2 *Service Level Agreement* (*SLA*) *and Key Performance Indicators* (*KPI*)

These ensure clear expectations and measurable standards for the performance and maintenance of district cooling systems:

a) *Service level agreement* (*SLA*) — A Service level agreement (SLA) outlines the expected level of service between a service provider (district cooling plant) and its customers. Key components of an SLA for a district cooling plant might include:

   1) cooling capacity commitment;

   2) reliability and availability; and

   3) penalty terms basis performance matrix.

b) Key Performance Indicators (KPIs):

   1) Coefficient of performance (COP);

   2) Chilled water supply and return temperature (CHWST/CHWRT);

3) Chilled water return temperature (CHWRT); and

4) Water usage efficiency (Litre/TRh).

8 SUPPLY AIR DUCTS AND RETURN AIR DUCTS

Following provisions to be considered for supply air ducts and return air ducts construction:

a) duct Systems are a continuous passageway for the transportation of air which, in addition to the ducts, may include duct fittings, dampers, plenums, grilles and diffusers and such other accessories as may be required. Leak tightness, adequate supporting and proper installation are important requirements;

b) due to aerodynamic considerations, it is recommended that the aspect ratio of rectangular ducts (height to width ratio) be kept within 1 : 4;

c) designer shall specify the material of ducts, appropriate standard to be followed, maximum permitted velocity of air in the duct and the required pressure class as indicated in Table 17 and Table 18;

d) the duct supports shall be designed to handle the load and also to take into account seismic considerations;

e) if false ceiling is provided, the supports for the duct and the false ceiling, shall be independent. Collars for grilles and diffusers shall be taken out only after false ceiling/boxing framework is done and frames for fixing grilles and diffusers have been installed; and

f) where a duct penetrates the masonry wall, it should either be suitably covered on the outside to isolate it from masonry, to prevent vibration transmission. Refer Part F 'Fire and Life Safety' of this NBCS for safety requirements.

For more details on duct construction and installation, refer to the latest edition of accepted standard [D-3(14)].

Table 17 Classification by Internal Pressure and Pressure Range

(*Clause* 8)

Sl No.	**Classification by Pressure**	**Normal Service Pressure** Pa		**Pressure Limits** Pa	
		Positive Pressure	Negative Pressure		
(1)	(2)	(3)	(4)	(5)	(6)
i)	Low Pressure Duct	$P \leq +500$	$-500 \leq P$	+ 1 000	– 750
ii)	Medium Pressure Duct	$+500 < P \leq +1\,000$	$-500 > P \geq -1\,000$	+ 1 500	– 1 500
iii)	High Pressure Duct	$+1\,000 < P \leq +2\,500$	$-1\,000 > P \geq -2\,000$	+ 3 000	– 2 500

Table 18 Recommended *Max* Duct Velocities

(*Clause* 8)

Sl No.	Application	Controlling Factor Noise Generation Main Ducts	Controlling Factor - Duct Friction			
			Main Ducts		Branch Ducts	
			Supply	Return	Supply	Return
		m/s	m/s	m/s	m/s	m/s
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Residences	3	5	4	3	3
ii)	Apartments	5	8	7	6	5
iii)	Hotel Rooms	5	8	7	6	5
iv)	Hospital Rooms	5	8	7	6	5
v)	Private Offices	6	10	8	8	6
vi)	Directors Rooms	6	10	8	8	6
vii)	Libraries	6	10	8	8	6
viii)	Theatres	4	7	6	5	4
ix)	Auditoriums	4	7	6	5	4
x)	General Offices	8	10	8	8	6
xi)	Restaurants	8	10	8	8	6
xii)	Stores	8	10	8	8	6
xiii)	Banks	8	10	8	8	6
xiv)	Cafeterias	9	10	8	8	6
xv)	Industrial **	13	15	9	11	8

NOTE — ** For specific industrial applications, refer to customer specific needs and choose a velocity to optimize energy consumption requirements.

8.1 Space Air Distribution

Space air distribution shall be classified by:

a) their primary objective; and

b) the methods by which the objectives are attempted to be accomplished.

The objectives of space air distribution shall be classified as one of the following:

1) providing occupant thermal comfort;

2) to support processes within the space; or

3) a combination of (i) and (ii).

The occupied zone is a space in any location where occupants normally reside and may differ from project to project. It may be application-specific and should be carefully defined by the designer. The occupied zone is generally considered to be the room volume between the floor level and 1.8 m above the floor level. For additional information special publication may be referred.

NOTE — Special publication maybe ANSI/ASHRAE Standard 55 and Standard 62.1 for further information.

Space Air Distribution methods shall be classified as one of the following:

i) fully mixed systems (for example, overhead air distribution) which have zero or very little thermal stratification of air in the occupied zone or process space.

ii) fully stratified systems (for example, thermal displacement ventilation) have practically no air mixing in the occupied zone or Process Space.

iii) partially mixed systems (for example, most underfloor air distribution designs) provide limited air mixing in the occupied zone or process space.

iv) unidirectional air flow system (for example a laminar flow system) is designed to ensure no mixing and maintain a consistent air flow pattern, ensuring that air always moves from a clean to less clean areas.

v) personalized system (for example task/ambient air distribution systems) is an air distribution strategy that combines two different air supply methods to optimize IAQ, thermal comfort and energy efficiency.

8.1.1 *Fully Mixed Systems*

In mixed air systems, comfort is maintained by mixing of high velocity supply air with the room air. The air mixing, heat transfer and velocity reduction should occur outside the occupied zone. The three fundamental characteristics of the system are:

a) the supply air outlets are located far away from the occupied zone;

b) the supply air temperature is considerably lower than the room temperature (the normal ΔT between supply air and room air would be of the order of 11 °C to 12 °C). When supplying heated air from overhead outlets, ensure that the ΔT between supply air temperature and room air temperature is not more than 8 °C to 9 °C to avoid stratification; and

c) the supply air (SA) outlet velocity should be much higher than the desired occupied zone air velocity. The supply air outlet velocity should be of the order of 3.5 m/s to 5.0 m/s. The resultant occupied zone velocity should be between 0.2 m/s to 0.25 m/s for cooling, and not more than 0.15 m/s for heating. When SA outlets are used at the perimeter with vertical projections for heating and/or cooling, they should be located near the perimeter surface. Outlet selection shall be done on the basis of the isothermal terminal velocity of 0.75 m/s extends at least half way down the surface or 1.5 m above the floor (whichever is lower) thus ensuring that the warm air mixes with the cool downdraft on the perimeter surface, to reduce draft in the occupied zone.

8.1.2 *Fully Stratified Systems*

Systems that discharge cool air at low sidewall or floor locations with very little entrainment of room air into the primary air stream create vertical thermal stratification throughout the space. The three fundamental characteristics of the system are:

a) the supply air outlets are located directly in the occupied zone;

b) the supply air temperature is much closer to the room temperature (the normal ΔT between supply air and room air would be of the order of 3 °C to 6 °C); and

c) the supply air (SA) outlet velocity should be of the order of 0.25 m/s to 0.35 m/s.

In spaces where the internal heat gain density exceeds 120 W/m^2, care should be taken to ensure comfort by providing supplementary localized cooling terminals to address the higher heat gain densities. Care should be taken to limit downward drafts driven by cold walls or windows. The vertically stratified temperature field needs to be stable for proper functioning.

Ensure that the design provides for appropriate measures for maintaining the RH within the specified levels.

Stationary occupants should not be subjected to discharge velocities exceeding about 0.2 m/s. SA outlets shall not be located closer than 1 m from any occupant. SA outlets should not be located behind furniture and other obstructions that impede airflow.

Stratified systems shall not be used in conjunction with mixed air systems because mixing destroys natural stratification that drives displacement ventilation.

8.1.3 *Partially Mixed Systems*

Supply air (SA) (usually from the floor) is discharged vertically at relatively high velocities and entrains room air akin to SA outlets in fully mixed systems. Underfloor air distribution systems (UFAD) shall be classified as partially mixed systems. The cavity below the access floor tiles is generally pressurized and used as SA Plenum.

As with fully stratified systems, the partially stratified systems in humid climates should ensure that the design provides for appropriate measures for maintaining the RH within the specified levels.

SA outlets should be of the swirl type with a high-induction core which promotes quick reduction in velocities and temperature gradients.

SA outlets should be located far enough from stationary occupants to ensure that they are not exposed to drafts and noise causing discomfort.

SA temperature in the access floor cavity should be kept at or above 16 °C to minimize the risk of condensation and consequent fungal growth.

Access floor plenums should be well sealed to prevent air leakages. Exterior walls should be well insulated and have proper vapour retarders. Night/holiday temperature set-backs should be avoided (or at least reduced) to minimize plenum condensation and thermal mass effect problems.

Return static pressure drop should be relatively equal throughout the space being served by the plenum. This reduces the chances of unequal pressurization within the Plenum.

Partially stratified systems shall not be used in conjunction with mixed air systems because mixing destroys natural stratification that drives displacement ventilation.

8.1.4 *Unidirectional Air Flow Systems*

A unidirectional air flow system is a type of ventilation system that provides a controlled, one-way flow of air through a facility or room. This system is designed to minimize contamination, reduce air turbulence, and improve indoor air quality.

Key features of this system are:

a) air flows in a single direction, from the cleanest area to the most contaminated area;

b) air flows in a smooth, parallel stream, reducing turbulence and mixing (laminar flow);

c) air is filtered to remove particles and contaminants before entering the space;

d) the space is maintained at a positive pressure to prevent outside air from entering; and

e) contaminated air is exhausted directly outside, without recirculation.

The key benefits of the unidirectional air flow system is improved air quality, reduced risk of cross-contamination, enhanced particle count control thus leading to easier regulatory compliance in required applications like cleanrooms, laboratories, healthcare facilities (operating room, isolation rooms), food processing facilities and the like.

8.1.5 *Personalized Systems*

Task/Ambient air distribution systems is an air distribution strategy that combines two different air supply methods to optimize IAQ, thermal comfort and energy efficiency.

Following are the two methods:

a) *Task air* — Supplies conditioned air directly to the breathing zone (directly via special outlets inbuilt in the seating systems/workstations). The SA is very precisely tempered to the correct temperature and humidity; and

b) *Ambient air* — Supplies to the overall space above and out of the breathing zone so as to maintain a comfortable thermal environment and air quality.

Such personalized systems, by separating the air components into task and ambient, offer several benefits including improved air quality, enhanced thermal comfort and increased energy efficiency.

8.1.6 *Supply Air Outlet* (*Airflow Mixing*) *Types and Ventilation Effectiveness*

The Table 19 and Table 20 below provides an understanding of the various types of systems, their ventilation efficiencies.

Table 19 Various Types of Systems and Their Ventilation Efficiencies

(*Clause* 8.1.6)

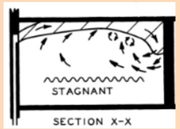
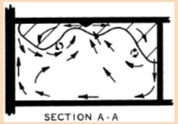
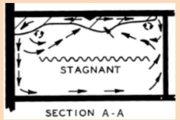
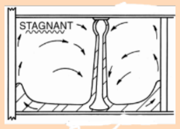
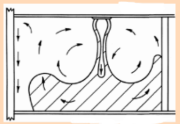
Sl No.	**Methods**	**Suitable Air Outlet**	**Zone Air Distribution/Change Effectiveness (E_z)**		
			Cooling	Heating	Features
(1)	(2)	(3)	(4)	(5)	(6)
i)	Fully Mixed System, Entrainment from Sidewall	High Sidewall Grilles, Sidewall Diffuser, Nozzle and similar outlet providing ceiling attached flow	 E_z=1 (Ideal)	 E_z =0.8 (Ideal)	Suitable for summer cooling and minor winter heating (temperature difference +10K warm air). Velocity in the occupied zone should be less than 0.35 m/s
ii)	Fully Mixed System, Horizontal Discharge from Ceiling	Ceiling Diffuser, Linear Ceiling Diffuser or similar outlets with Horizontal diffusion pattern	 E_z =1 (Ideal)	 E_z =0.8 (Ideal)	Suitable for summer cooling and minor winter heating (temperature difference +10K warm air). Velocity in the occupied zone should be less than 0.35 m/s
iii)	Fully Mixed System, Vertical Downward Air Projection from Ceiling	Adjustable Grille, Jet Nozzle, Ceiling Spot Diffuser, Linear Ceiling Diffuser or similar outlets with vertical downward air projection	 E_z =1 (Ideal)	 E_z =1 (Ideal)	Suitable for either cooling or heating with increased terminal velocity in order to project warm air up-to the occupied zone.
iv)	Partially Mixed System, Non Spreading Vertical Jet in or near floor	Floor Diffuser, Sidewall Grilles, Sidewall Diffuser or similar outlets with fixed vertical spread pattern.			Suitable for cooling and heating both at terminal velocity ~ 0.75 m/s (ideal). Outlet shall be placed at 150 mm distance from wall.

Table 19 (*Concluded*)

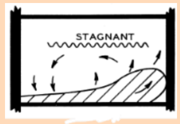
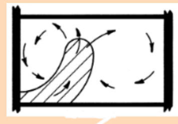
Sl No.	**Methods**	**Suitable Air Outlet**	**Zone Air Distribution/Change Effectiveness (E_z)**		
			Cooling	Heating	Features
(1)	(2)	(3)	(4)	(5)	(6)
			E_z =1 (Ideal)	E_z =1 (Ideal)	
v)	Partially Mixed System, Spreading Vertical Jet in or near floor	Floor Diffuser, Sidewall Grilles, Sidewall Diffuser with adjustable spread pattern or similar outlets			Suitable for severe heating and minor cooling requirements (temperature difference -8K cool air). Velocity at 1.4m above floor i.e. occupied region should be less than 0.25 m/s
			E_z =1 (Ideal)	E_z =1 (Ideal)	
vi)	Fully Stratified System, Low Horizontal Discharge near floor	Baseboard or Low sidewall Grille, Displacement Diffuser or similar outlets			These outlets direct high velocity total air directly into the occupied zone and therefore not recommended for particularly summer cooling. Terminal Velocity <1.5 m/s and mostly suitable for process installations where controlled air velocities are desired.
			E_z =1.2 (Ideal)	E_z =1 (Ideal)	
vii)	Unidirectional Airflow System, Plug type from ceiling exhausted through wall	Laminar Flow Diffuser with perforated/helical face plate			Generally, a preferred choice for use in a cleanroom for manufacturing, healthcare and research. Other performance parameters such as colony forming unit and room class affects its choice and selection

Table 20 Recommended Supply Air Outlet Selection

(*Clause* 8.1.6)

Outlet Types	**Fully Mixed**			**Fully Stratified**		**Partially Mixed**		
	Outlet Location							
	Ceiling	Wall	Floor	Wall	Floor	Ceiling	Wall	Floor
Grilles								
Adjustable Blade	☑	☑	NR	NR	NR	NR	NR	NR
Fixed Blade	NR	☑	NR	☑	NR	NR	NR	NR
Linear Bar Grille	NR	☑	☑	☑	☑	NR	NR	☑
Jet Nozzle	NR	☑	NR	NR	NR	NR	NR	NR
Diffusers								
Round	☑	NR	NR	NR	NR	NR	NR	NR
Square	☑	NR	NR	NR	NR	NR	NR	NR
Perforated Face	☑	NR	NR	NR	NR	NR	NR	NR
Louvered Face	☑	NR	NR	NR	NR	NR	NR	NR
Plaque Face	☑	NR	NR	NR	NR	NR	NR	NR
Laminar Flow	☑	NR	NR	NR	NR	NR	NR	NR
Slot Diffuser	☑	☑	NR	NR	NR	NR	NR	NR
Swirl Diffuser	☑	NR	NR	NR	☑	NR	☑	☑
Displacement	NR	NR	NR	☑	☑	NR	NR	NR

NOTES

1 Obstructions, vane adjustments, and other ceiling elements influencing airflow patterns shall affect comfort and efficiency.

2 Personalised system will increase E_z when used in combination with the above listed methods up to 1.5

3 During partial load, VAV systems shall affect E_z by reducing the air velocity at the outlet. Minimum airflow should be considered so that velocity in the occupied region does not go below 0.1 m/s while designing such systems and other performance criteria such as outdoor air intake and critical space CO_2 concentration should be monitored and controlled (demand control ventilation). Additionally, a dual duct system, fan powered and reheat terminal may be used in combination with VAV systems where precise and wide climate control range is required without affecting the E_z.

8.1.7 *Return Air Inlets*

Return air intake affects room air motion only immediately around the inlet. the selection of return air inlets should be based on performance data and not merely based on aesthetic and architectural considerations. The Table 21 below provides the recommended face velocities across Inlets based on their location.

Table 21 Recommended Face Velocities Across Inlets Based on their Location

(*Clause* 8.1.7)

Sl No.	Inlet Location	Velocity Across Inlet m/s
(1)	(2)	(3)
i)	Above occupied zone	> 4
ii)	In occupied zone (not near seats)	3 to 4
iii)	In occupied zone (near seats)	2 to 3
iv)	Door/wall louvers	1 to 1.5
v)	Through door under-cuts	2 to 1.5

8.2 Airflow Measurements in Space Air Distribution Systems

Air flow measurement requirements and methods of air distribution systems aim to ensure adequate indoor air quality, energy efficiency, and occupant comfort in the occupied space. These requirements vary depending on the specific standard in use, but there are common elements that are often addressed. Understanding and adhering to these requirements and methods ensures that air distribution systems in buildings provide adequate ventilation, maintain indoor air quality, and operate efficiently. It is recommended that the airflow measurement should be conducted to meet the following requirements.

8.2.1 *System Balancing*

Air distribution systems shall be balanced to ensure each space receives the designed air flow. This typically involves testing, adjusting, and balancing (TAB) procedures to verify system performance. Airflow measurements using one of the following methods should be conducted in each space to achieve the designed values within a band of ± 5 percent. Proper documentation of all measurements, adjustments and maintenance activities shall be maintained. The following describes various airflow measurement methods to comply with the above requirements of system balancing:

a) *Anemometers* — Anemometers are used to measure the velocity of air flowing through ducts or at registers/diffusers.

b) *Barometers/flow hoods* — Barometers or Flow hoods are used to measure the airflow directly at the supply air outlets.

c) *Duct traverse method* — Duct traverse method involves measuring air velocity at multiple points across the duct cross-section using a pitot tube or hot-wire anemometer and calculating the average velocity to determine airflow rate.

d) *Pressure measurement* — Differential pressure measurements across known resistances (for example, orifice plates, flow hoods) shall be used to determine airflow rates.

e) *Tracer gas method* — Trace gas method introduces a tracer gas into the airflow and measures its concentration downstream to determine the airflow rate based on dilution principles.

For airflow measurements within ducts, it is recommended to use an air flow measuring station. Such air flow measuring station should ideally be located with a clear straight duct length of at least 4 times the equivalent duct diameter (of the duct in which they are installed), at the inlet of the air flow measuring station.

8.3 Flow Control – Volume Control Dampers

A damper is a valve or plate that stops or regulates the flow of air inside a duct, and other air-handling equipment.

Manual dampers are operated by a handle on the outside of a duct. Automatic dampers are used to regulate airflow constantly and are operated by electric or pneumatic motors, in turn controlled by a thermostat or the building automation system.

Dampers are customized in terms of size to suit the openings, where they are to be installed. A right choice of volume control dampers/s shall result in energy conservation.

8.4 Variable Air Volume (VAV)

Variable air volume terminal units are devices that regulate the amount of air supplied to a specific zone or space in a building. They are an important component of fully mixed systems, which help to optimize energy efficiency and indoor air quality.

By optimizing air flow and conditioning, VAV terminal units play a crucial role in creating a comfortable, efficient, and sustainable indoor environment. When using VAV terminal units, observe the following guidelines to ensure safe and efficient operation:

a) Wherever possible, a straight duct inlet connection with a minimum length of three duct diameters and the same internal diameter as the inlet should be provided.

b) Never insert a duct inside the Inlet or control calibration will be adversely affected.

c) Ensure accurate sensor calibration to maintain precise control and avoid oscillations.

d) Balance airflow to prevent uneven distribution, reduced efficiency, and increased energy consumption.

e) Inlet air velocity should not exceed 12 m/s at max flow and should not be below 2 m/s. Ensure unit sizing is adequate to meet flow range and also meets noise criterion for the space.

f) When using terminals equipped with re-heat elements, to avoid compromising zone ventilation effectiveness, the discharge temperature should not be more than 8 °C warmer than the zone it serves. If electrical re-heat is being used, ensure all safety controls (like air stats/flow switch and other safety cut outs) as required by local codes are provided.

g) Ensure proper access is provided for the unit (like adequately sized access doors).

8.5 Thermal Beams

Chilled beams are a type of convection HVAC system used to heat or cool rooms they are air distribution devices with integral coils designed to heat or cool large buildings. Chilled beams shall be classified into two main categories, passive and active.

Passive chilled beams work using the phenomenon of natural convection, and they do not require any primary air supply.

Active chilled beams consist of a primary air supply connection that provides conditioned/dehumidified air and enhances air induction through the coil.

8.5.1 *Application Considerations*

Chilled beams are intended to operate without condensation. Consequently, active chilled beam supply water temperatures should be maintained at or above the room dew point temperature to prevent condensation on the coil. Passive chilled beam should be kept slightly above the room dew point temperature. In both cases, the chilled water piping shall be adequately insulated to prevent condensation on the pipe itself. Where adequate control of space humidity levels shall not be ensured, higher supply water temperatures and/or suitable condensation control should be considered.

Heating with passive beams is not effective. For heating, the hot water temperature serving the beam's coil shall be selected to limit the discharge air temperature to less than 8 °C above the room design set point.

8.6 Air Curtains

An air curtain unit acts as a controlled barrier for environmental and thermal separation and wind resistance when a building's doors or windows are opened. Air curtains create an invisible barrier at openings and doors which separates different temperature zones without limiting access for people and vehicles. Air curtains are used where it is not feasible to have an engineered air lock.

8.6.1 *Application Considerations*

Air curtain unit selection depends on the opening's width and height. To maximize effectiveness, the air curtain unit shall at least cover or slightly overlap the entire opening and have a minimum velocity projection of 2 m/s at the target surface. The air curtain unit discharge shall have a free and clear path to the entire opening for optimum performance.

9 AIR FILTRATION AND ALTERNATE AIR CLEANING METHODS

With people spending a significant portion of their time indoors, ensuring good indoor air quality (IAQ) is paramount for occupant health, productivity, and overall well-being. The provisions related to indoor spaces that are regularly occupied and are either naturally or mechanically ventilated are covered. Following are the aspects to be considered for air filtration and alternate air cleaning methods.

9.1 Threshold Levels for Pollutants

Threshold values are made available for different pollutants depending on the severity of the health hazard that a pollutant shall create for the occupants. For details of threshold values for the different pollutants encountered in indoor air refer to **3.6.2**.

9.2 IAQ Monitoring

IAQ is a critical component in ensuring the health, comfort and productivity of occupants in commercial buildings. Monitoring IAQ involves the assessment of various air quality parameters. Some critical pollutants and air quality parameters are monitored continuously, while some are monitored/tested periodically. Continuous monitoring and periodic testing data are required for interpreting the indoor environment and initiating necessary mitigation measures.

9.2.1 *Continuous Real Time Monitoring*

Pollutants for continuous monitoring include CO_2, TVOC, PM2.5, and ozone. These pollutants should be monitored continuously in all regularly occupiable areas of the building. Regularly occupiable areas may be defined as spaces intended for human occupancy within commercial buildings, excluding areas like the pantry, toilets and store rooms. When determining compliance, missing measurements and data loss will be interpreted as a threshold/range exceedance for that period.

9.2.1.1 *Sampling frequency*

For continuous monitoring of temperature, relative humidity, CO_2, TVOC, PM2.5 and O3, the sampling frequency should be at least once every 5 minutes. Monitoring of these parameters should be done on a 24 h basis to establish a trend. This provides insights into indoor air quality, leading to possible actions for improvement. PM1.0 (particulate matter $\leq$ 1.0 µm) is increasingly recognized for its potential health impacts and may also be considered for monitoring as an emerging best practice.

9.2.1.2 *Deployment of sensors/monitoring devices*

Deployment refers to the position and the number of sensors/monitoring devices installed in the indoor space. Ideally, the sensors/monitoring devices should be installed in the breathing zone. The breathing zone is defined as that region of the indoor space between the horizontal planes at 1.1 m and 1.7 m height above the finished floor level. Sensors/monitoring devices may be installed under the ceiling if the ceiling height is restricted to 2.13 m. In such cases, it is necessary to ensure that the air distribution in the room is uniform so that the air parameters at the ceiling will be similar to those in the breathing zone. Sensors/monitoring devices should be installed at least 0.9 m from exits, window openings and air conditioning vents. Every enclosed place, such as cabins or meeting rooms occupied for an average of 50 percent of the daily working hours, shall have dedicated monitoring devices. General office and common areas shall have at least one device for every 450 m^2 floor area.

9.2.1.3 *Calibration of sensors*

All sensors and pumps used in the monitoring devices shall be calibrated to ensure that the accuracy levels are maintained as specified by the manufacturer and to identify any measurement errors. Calibration should cover the parametric range the sensors are expected to experience in real service. Sensors measuring PM2.5 and TVOC values should be recalibrated or replaced annually. CO_2 sensors utilising the automatic baseline correction method shall be recalibrated or replaced once every three years. Sensors measuring thermal comfort parameters shall be recalibrated or replaced once every three years. Field calibrations using a reference sensor are acceptable, provided the master sensor is similar in range, accuracy, least count and resolution to the sensor being calibrated. Moreover, the calibration procedure should be such that the master sensor will perform within the manufacturer's specified range and accuracy. The master sensor should have been calibrated at a laboratory. The calibration process shall capture a sufficient range and contaminant concentration using known span gases or exposure to ambient pollution to perform adjustments accurately. The calibration activity should be properly documented, and traceability certificates shall be provided wherever applicable. Appropriate calibrations of PM1.0 sensors may be done if provided.

9.2.1.4 *Display of IAQ data*

All IAQ parameters, under the purview of continuous monitoring, should be prominently displayed in a format that provides easy viewing and understanding to all occupants of the building. Having at least one common electronic data display on each floor/section showing the real-time snapshot of all the devices on that particular floor/section is recommended. Alternatively, there could be one common display at the main entrance that provides a real-time representation of the IAQ data for the entire building.

9.2.1.5 *Recommended sensor specifications for continuous monitoring*

The selection of the sensors is a crucial requirement for obtaining reliable and accurate measurements. Improper sensor selection will lead to erroneous data on IAQ. The Table 22 provides details about the type of sensor to be used and their recommended range, accuracy, and resolution to capture meaningful IAQ data.

Table 22 Required Range and Accuracy of Sensors Used for Continuous Monitoring

(*Clause* 9.2.1.5)

Sl No.	Parameters	Unit	Range	Accuracy	Resolution
(1)	(2)	(3)	(4)	(5)	(6)
i)	PM10	µg/m^3	0 µg/m^3 to 1 000 µg/m^3	± 10 percent of reading	1 µg/m^3
ii)	PM2.5	µg/m^3	0 µg/m^3 to 1 000 µg/m^3	± 10 percent of reading	1 µg/m^3
iii)	CO_2	ppm	400 ppm to 5 000 ppm	± 50 ppm or ± 5 percent of reading (whichever is greater)	1 ppm
iv)	TVOCs	ppb	0 ppm to 5 000 ppb	± 20 ppb or ± 15 percent of reading (whichever is greater)	1 ppb
v)	Temperature	°C	0 °C to 100 °C	± 1 °C	0.1 °C
vi)	Relative humidity	percent	5 percent to 95 percent	± 5 percent of reading	1 percent
vii)	Ozone	ppb	20 ppb to 500 ppb	± 10 ppb at 0-100 ppb	5 ppb

9.2.2 *Periodic Monitoring/Testing*

Periodic monitoring refers to measuring pollutants at a reasonably low frequency, typically several months apart. Pollutants for periodic monitoring include CO, SOx, NOx and HCHO. Sampling locations for periodic monitoring shall be representative of occupied spaces and should be located in the breathing zone as defined in 9.2.1.2. Testing should be carried out under regular building conditions. Any test conducted without occupants or when the occupancy is considerably lower than on regular days shall be avoided.

9.2.2.1 *Sampling Frequency during periodic monitoring/testing*

Based on the pollutant's characteristics, the constraints in their monitoring technology, and their occurrence indoors, some pollutants are assessed periodically. The periodicity of such tests may be defined as "at least once every six months" (may be considered by NABL Labs). Building operators may resort to a more frequent periodicity if the test results warrant the same.

9.2.2.2 *Procedure for periodic monitoring/testing*

a) HCHO — The test shall be in accordance with best practices[1]. The sampling duration should be a minimum of one continuous hour or the equivalent duration of the sampling procedure special publication[2] may be referred.

b) CO — The sampling shall be carried out for at least one continuous hour (10 minutes of acclimation followed by 50 minutes of measurement time), with measurements recorded at least once every minute. Recommended procedures are the special publications[3] may be referred. (CO by gas bag sampling).

c) NOx — NOx is measured gravimetrically using the modified Jacob and Hochheiser method and the chemiluminescence test. The monitoring of NOx shall be carried out in accordance with best practices[4].

d) SOx — SOx shall be monitored gravimetrically using the improved West and Gaeke method and Ultraviolet Fluorescence in accordance with best practices[5].

[1] best practices maybe ISO 16000-3 for HCHO.

[2] Special publication maybe ISO16000-3 for sampling procedure in case of HCHO.

[3] Special publication for recommended procedures for CO maybe NIOSH Manual of Analytical Methods (NMAM), Fifth Edition,1997, OSHA Methods ID 209: 1993 (CO by direct-reading monitor) and OSHA Methods ID 210: 1991.

[4] best practices maybe ISO 16000-15 for NOx.

[5] best practices maybe ISO 16000-1 for SOx.

9.2.2.3 *Recommended sensor specifications for periodic monitoring/testing*

During periodic monitoring/testing, appropriate measuring instrument shall be used to obtain reliable and accurate measurements. The following Table 23 provides details about the type of instrument to be used and their recommended range, accuracy, and resolution to obtain factual test results.

Table 23 Sensor Specifications for Periodic Monitoring/Testing

(*Clause* 9.2.2.3)

Sl No.	Parameters	Unit	Range	Accuracy	Resolution
(1)	(2)	(3)	(4)	(5)	(6)
i)	CO	ppm	0.1 to 25	± 3 percent or ± 1 ppm (whichever is greater)	0.1 ppm
ii)	NOx	ppb	0 to 500	± 20 ppb or ± 15 percent of reading (whichever is greater)	1 ppb

Table 23 (*Concluded*)

Sl No.	Parameters	Unit	Range	Accuracy	Resolution
(1)	(2)	(3)	(4)	(5)	(6)
iii)	SOx	ppb	0 to 500	± 5 percent of reading	1 ppb
iv)	Formaldehyde	ppb	20 to 1 000	± 20 ppb	1 ppb

9.3 Air Filters for Removing Pollutants in Indoor Spaces

Air filters shall be used in HVAC systems to remove airborne pollutants from indoor air and prevent outdoor pollutants from entering the building through outdoor air intakes.

The filters shall be as per accepted standards [D-3(15)], [D-3(16)], [D-3(17)] and [D-3(18)] for classification and performance requirements may be referred.

9.3.1 *Classification of Air Filters*

Air filters are classified mainly based on their ability to filter out pollutants. Due to the differences in the methods for determining this ability, air filters shall be classified in multiple ways. Air filters are classified based on their gravimetric efficiency, particulate matter efficiency or fractional particle size efficiency.

9.3.1.1 *Classification of air filters*

The accepted standard [D-3(15)] shall be adopted to classify air filters based on particulate matter efficiency. This method of specifying the performance of air filters applies to fine filters. It is a preferred method of classification while specifying air filters for general human health and wellness applications (comfort applications), as particulate matter concentrations inside buildings directly correlate with the adverse health effects on the occupants. The test method mentioned in the standard determines the initial efficiency of the air filter in both the "as-manufactured" condition and the discharged condition to calculate the minimum and average particulate matter efficiency. The filter class is reported as ISO ePMx (y percent), where x is the particle size referred to (1 µm, 2.5 µm or 10 µm) and y is the particulate matter efficiency. Classification of air filters based on particulate matter efficiency as per accepted standard [D-3(15)] is given Table 24.

NOTE — The symbol ePMx describes the efficiency of an air cleaning device to particles with an optical diameter between 0.3 µm and x µm.

Table 24 Classification of Air Filters based on Particulate Matter Efficiency

(*Clause* 9.3.1.1)

Sl No.	Group Designation	Requirement $ePM_{1,\ min}$	Requirement $ePM_{2.5,\ min}$	Requirement $ePM_{10,\ min}$	Class Reporting Value
(1)	(2)	(3)	(4)	(5)	(6)
i)	ISO Coarse	NA	NA	< 50 percent	Initial gravimetric arrestance
ii)	ISO *e*PM10	NA	NA	≥ 50 percent	ePM_{10}
iii)	ISO *e*PM2.5	NA	≥ 50 percent	NA	$ePM_{2.5}$
iv)	ISO *e*PM1	≥ 50 percent	NA	NA	ePM_{1}

9.3.2 *Resistance to Airflow in Air Filters*

The resistance to airflow is measured as the differential pressure across the air filter in a test setup. The resistance vs airflow rate characteristics are established at 25 percent, 50 percent, 75 percent, 100 percent and 125 percent of the rated flow of the air filter. Procedures mentioned in accepted standard [D-3(17)] shall be followed to determine the same. The resistance as a function of airflow rate provides useful information on the selection and performance of an air filter.

9.3.3 *Air Filters Using Electrostatically Charged Fibrous Media*

Some air filters use fibrous media that carries an electrostatic charge. The charge helps in improving the filtration efficiency of smaller particles. This will help achieve higher filtering capabilities at a lower initial resistance for a given airflow rate, thereby reducing operating costs. However, should the charge be compromised for any reason, the filtering capability of these air filters will be drastically reduced. Hence, when electrostatically charged fibrous media air filters are used, it is recommended that suitable monitoring systems be used for measuring the indoor particle concentrations so that any reduction in the filtering capability of the air filter shall be noticed and suitable actions shall be taken.

9.3.4 *High-Efficiency Filters*

High-efficiency filters shall be used in critical industrial, scientific and research applications that demand stringent control of airborne solid and liquid phase pollutants. They are classified based on their filtration efficiency at most penetrating particle size (MPPS). The MPPS is defined as the particle size of the challenge aerosol that will offer the lowest efficiency when challenged onto an air filter. The accepted standard [D-3(19)] shall be used to classify high-efficiency filters. This standard classifies high-efficiency filters based on their overall efficiency and local efficiency (leak scan test). (*see* Table 25)

Table 25 Classification of High-Efficiency Air Filters

(*Clause* 9.3.4)

Sl No.	Filter class	Overall Efficiency at MPPS percent	Local Efficiency at MPPS percent
(1)	(2)	(3)	(4)
i)	ISO 15 E	≥ 95	NA
ii)	ISO 20 E	≥ 99	NA
iii)	ISO 25 E	≥ 99.5	NA
iv)	ISO 30 E	≥ 99.90	NA
v)	ISO 35 H	≥ 99.95	≥ 99.75
vi)	ISO 40 H	≥ 99.99	≥ 99.95
vii)	ISO 45 H	≥ 99.995	≥ 99.975
viii)	ISO 50 U	≥ 99.999	≥ 99.995
ix)	ISO 55 U	≥ 99.999 5	≥ 99.997 5
x)	ISO 60 U	≥ 99.999 9	≥ 99.999 5
xi)	ISO 65 U	≥ 99.999 95	≥ 99.999 75
xii)	ISO 70 U	≥ 99.999 99	≥ 99.999 9
xiii)	ISO 75 U	≥ 99.999 995	≥ 99.999 9

9.3.5 *Air Filters for Removing Airborne Molecular Contamination*

A gaseous state pollutant in the air is called an airborne molecular contamination. Many gas phase pollutants are detected in the indoor air of buildings, and they shall be broadly classified as acid gases, base gases, aldehydes, oxidising gases, VOCs and other miscellaneous gases. Adsorption and chemisorption filters are used in HVAC systems and outdoor air intakes to remove these pollutants (for example H_2S, NH_3, HCHO, O_3, VOCs, NOx, SOx, Cl_2, CO etc.) from indoor air. These filters use adsorbents, known as molecular sieves, to remove the pollutants from the air stream through physical adsorption. Activated carbon, activated alumina and zeolites are the commonly used adsorbents. Additionally, the adsorbents are impregnated with suitable chemicals that react with the adsorbed pollutant to prevent their desorption. This process is called chemisorption. Molecular sieves are available as granules, pellets, and powders coated on fibrous media; hence, they shall be constructed as packed bed filters and extended surface pleated filters. For additional information special publications may be referred to assess the performance of these air filters.

NOTE — Special publications maybe ISO 10121-2 : 2013 and ISO 10121-3 : 2022.

The fundamental criteria for air filter selection is based on its:

a) filtration efficiency at the particle size range of interest;

b) gravimetric, particulate matter, or fractional particle size efficiencies;

c) resistance the air filter offers at its rated airflow; and

d) test dust capacity.

9.3.6 *Installation of Air Filters in HVAC Systems*

Most air filter installations in HVAC systems comprise of multiple air filters in a single system. The installation of air filters is vital since improper installation could lead to filter bypass, compromising the overall filtration system's efficiency even though the individual air filters may be good. Proper mounting arrangements and usage of gaskets to avoid filter bypass for ensuring the proper installation of an air filter.

9.3.7 *Field Testing of Air Filters*

Since the test results from standard laboratory tests shall vary from performance results in real service, conducting a filtration performance test in situ under real service conditions will help understand the actual performance. For *in-situ* testing purpose special publications may be referred, which provides the method for *in-situ* field testing of air filters and systems, shall be adopted.

NOTE — Special publication maybe ISO 29462 : 2022.

9.4 Alternate Air Cleaning Technologies

To ensure the indoor concentration of these pollutants within the safe levels as mentioned in **3.6.2**. The following text provides functional principle and instructions for use of various types of air cleaning technologies.

9.4.1 *Electrostatic Precipitators*

Electrostatic precipitators (ESP) use the principle of electrostatic attraction to trap particles. They are commonly used in industrial applications with high particle concentrations where mechanical (fibrous media extended surface) air filters may not be practical. ESPs have been used in comfort HVAC/AHU applications to reduce indoor PM2.5 concentration. ESPs also efficiently remove oil mist, such as in kitchen smoke and certain other industrial applications.

Ionization of the air stream may lead to ozone generation. When using ESPs, it shall be ensured that the ozone concentration in occupied spaces does not exceed the values mentioned in **3.6.3**. To ensure this, it is recommended to use suitable "ozone removal/reduction" technologies as a part of the system. ESPs shall be designed with safety interlocks to avoid electrical shocks and to stop the power module in case of less airflow.

The pressure loss of the electrostatic precipitator is very low and stays nearly constant. However, the filtration efficiency decreases as the collection plates accumulate with particles. Hence, the ionization and collection sections need to be cleaned regularly. ESPs are preferably located upstream of AHU cooling coils.

9.4.2 *Ultraviolet Germicidal Irradiation*

Ultraviolet germicidal irradiation (UVGI) systems in buildings shall be installed inside air handling units (AHUs) or air ducts. Additionally, they shall be installed as upper-zone UVGI systems (UZ-UVGI). The upper zone units have uniquely designed baffles that irradiate the upper zone of the room towards the ceiling. A gentle air movement in the room makes the upper zone unit more effective. For UZ-UVGI systems, the UV equipment/lamp shall be out of the line of sight for the occupants from all parts of the room. To meet the safe exposure limit of 60 J/m^2 for an 8-hour exposure period, the UV radiation intensity at the occupied zone shall be less than 0.002 W/m^2.

The use of UVGI for the heat transfer coils of air conditioning equipment is found to be very effective in eliminating biofilm build-up on coils and condensate drain pans, thereby improving heat transfer efficiency and hygiene inside air conditioning equipment.

Direct and visual contact of humans with UVGI systems shall be avoided. Service personnel shall wear the required safety goggles and aprons while attending to any service issue in AHU fitted with UVGI systems. The HVAC equipment fitted with UVGI shall have visible stickers on the equipment mentioning 'UVGI installed inside'.

NOTE — The degradation of materials used in HVAC equipment like AHUs due to UV exposure is another important aspect to consider when installing UVGI systems. Material compatibility with UVGI shall be assessed during design or subsequent retrofits.

9.4.3 *Photocatalytic Oxidation*

Photocatalytic oxidation (PCO) is a light-mediated redox (oxidation and reduction) reaction of gases and biological matter. This is achieved by impinging ultraviolet rays on a catalyst material and giving out hydroxyl radicals and super oxides, commonly referred to as reactive oxidative species (ROS).

9.4.4 *Photo-Hydro Ionization*

Photo-hydro ionization (PHI) is a light-mediated active air cleaning method similar to PCO. Removes bioaerosols (bacteria, viruses, fungi, etc), VOCs, formaldehyde and certain odour-causing chemicals from the air.

9.4.5 *Air Ionizers*

The use of air ionizers helps easier removal of particles by agglomeration.

9.4.6 *Plasma Air Ionization*

Ions are formed as air passes over the ionization tubes. Pollutants, including bioaerosols coming in contact with the ions, are oxidized and destroyed. Additionally, the generated ions encourage the agglomeration of particles, allowing them to be more easily removed through gravitational settlement and by the air filtration systems.

10 OTHER COMPONENTS OF CENTRAL HVAC SYSTEM

These components are required for the efficient functioning and regulation of heating, ventilation, and air conditioning in a centralized system.

10.1 Cooling Tower

Cooling towers are used to dissipate heat from water cooled refrigeration, air conditioning and industrial process systems to the atmosphere. Cooling is achieved by evaporating a small proportion of the recirculating water to the atmosphere. Cooling towers shall be installed at a place where free flow of atmospheric air is available.

Range of a cooling tower is defined as temperature difference between the entering and leaving water. Approach of the cooling tower is the difference between leaving water temperature and the ambient air wet-bulb temperature.

10.1.1 *Selection of Cooling Tower*

Following factors shall be considered for selection of cooling tower:

a) recirculating water flow rate;

b) design wet-bulb temperature;

c) approach, the designer shall endeavour for reducing the approach for maximizing energy conservation in the chiller;

d) range;

e) water quality and;

f) outside air quality for cooling tower;

g) height limitation and aesthetic requirement;

h) location of cooling tower considering possibility of easy drain back from the system;

j) placement with regard to adjacent walls, windows and other buildings, and effects on these from any water carried over by the air stream, also known as drift;

k) sound levels, particularly during silent hours;

m) material of construction of the tower; particularly considering the nature/quality of ambient air and recirculating water;

n) direction and flow of prevailing wind; to minimise recirculation and interference; and

p) quality of water used for make-up; maintenance and service space availability.

10.1.2 *Selection Criteria*

These factors ensure the optimal performance, efficiency, and suitability of cooling towers for specific HVAC system requirements.

10.1.2.1 *Location of the cooling tower*

Location of cooling tower shall be determined by one or more of the following:

a) architectural compatibility;

b) rigging limitations;

c) structural support requirements;

d) cost of bringing auxiliary services to the cooling tower; and

e) noise transmission, plume, and drift considerations.

These are best handled by proper site selection during the planning stage. Cooling tower shall be installed raised above the mounting surface by 0.45 m to 1.25 m to permit installation of pot strainer in condenser water line from cooling tower sump; also, to provide easy drainage and maintenance of mounting surface as follows:

1) Cooling tower accessories shall be project-specific and shall include such items as walking platform; stairs and ladder safety cage; bird screen; tower loading and supporting structure; and variable speed drive fan motor. However, variable speed drives should be used only in very specific instances as allowing the cooling tower fan to run at full speed during periods of low load is generally more beneficial due to the lower water temperature achieved by the cooling tower that results in higher condenser efficiency and, hence, lower power consumption in the air-conditioning plant.

2) Cooling tower installation shall include installation of conductivity controller, flow meter on the makeup water line, overflow alarm and low-level alarm.

3) The recommended floor area requirement for various types of cooling tower is as given below:

Mechanical draft cooling tower : 0.014 to 0.28 m^2/kW (0.05 to 0.10 m^2/TR)

NOTE — The recommended floor area above does not include the area required around the tower for unobstructed air flow.

4) Structural provision for the cooling tower shall be taken into account while designing the building. Vibration isolation shall be an important consideration in structural design.

5) Special care should be taken in design where noise transmitted to the adjoining building shall be of serious concern. Special vibration control and sound attenuation devices may be required in that case. Appropriate sound insulation and noise control measures shall be taken in such cases in accordance with Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS.

10.1.2.2 *Water quality*

Water is commonly used as a heat transfer medium to remove heat from the refrigerant vapour in the condenser. A cooling tower is used to dissipate the heat from the air-conditioning system to atmosphere. A cooling tower shall cool condenser water to within 2 °C to 3 °C of the ambient wet-bulb temperature, with modern cooling towers even achieving approach as low as 1 °C. Therefore, a water-cooled system operates at a condenser temperature that could be more than 20 °C in a corresponding air-cooled system, resulting in considerably improved efficiency and significantly reduced energy consumption.

Where make-up water is expensive, or not available at all or in limited quantities only, the use of an adiabatic cooler should be considered. An adiabatic cooler allows the use of water-cooled air-conditioning plant with water consumption 70 percent to 90 percent lower than that of a conventional cooling tower.

Certain amount of water is lost from circulating water in the cooling tower, as given below:

a) *Evaporation loss* — It is approximately 1 percent of the rate of water circulation per 6 °C of range.

b) *Drift loss* — The drift loss shall be below 0.1 percent of the rate of water circulation.

c) *Blow-down/bleed-off* — The amount of blow-down shall be below 0.8 percent of the total water circulation. If simple blow-down is inadequate to control scale formation, chemicals may be added to inhibit corrosion and limit microbiological growth.

Provision shall be made to make-up for the loss of circulating water. Provision for make-up water tank to the cooling tower shall be made. Make-up water tank to the cooling tower shall be separate from the tank serving drinking water. Makeup water should be sourced from treated water from sewage treatment, wastewater treatment plant, or from rain water harvesting.

Make-up water having contaminants or hardness, which shall adversely affect the refrigeration plant life, shall be treated. Treated water where hardness as ppm of $CaCO_3$ is reduced to 50 ppm or below is recommended for air conditioning applications. Water with pH value less than 5 shall also need to be treated. For treatment of water for cooling towers good practice [D-3(20)] shall be referred.

Cooling tower should be so located as to eliminate nuisance from drift to adjoining structures. Cooling tower approach and minimum efficiency shall be as per Table 26.

Table 26 Cooling Tower Approach and Minimum Efficiency

(*Clause* 10.1.2.2)

Sl No.	Building	Chiller kWr	HWT °C	CWT °C	WBT °C	Efficiency kW/LPS
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	All others	Any	36.4	30.8	28.3	0.35
ii)	24 × 7 facilities	Any	35.6	30.0	28.3	0.35

10.1.2.3 *Installation of cooling tower*

The cooling tower shall be mounted on a set of four or six numbers of reinforced cement concrete pillars (structural foundation) as per manufacturer's recommendations. Height of these pillars shall be not less than 1 m, actual height is decided at site as per space required to install a pot strainer below the water level in cooling tower sump, drain pipe and valve for complete drainage of sump, and to permit maintenance of slab upon which structural foundation is installed.

Cooling tower should be located at a well-ventilated place, preferably on the terrace of the building, in consultation with the structural consultant. Dynamic structural loading on the terrace shall be considered. Cooling tower shall be installed in such a way that its dynamic load is transferred directly to the building structural columns, for which necessary mild steel I-section may have to be provided. Anticorrosive coating such as epoxy coating shall be applied required for these mild steel I-sections. Suitable thickness of vibration isolation pads shall be placed between the tower and the I-sections to avoid transfer of vibration to building structure. Sufficient space shall be left all around the cooling tower support structure for efficient operation of the cooling tower.

When cooling tower is installed at ground level, contiguous to the utility services block, great care has to be exercised to prevent users and visitors from coming in close proximity to the cooling tower. This is necessary to avoid their exposure to legionellae bacteria which shall hibernate in the cooling tower sump, if operator does not follow strict instructions for regular bleed-off and chemical treatment of condenser water.

Also, the bleed-off tap should be given in the return line to cooling tower, instead of only providing in the sump to ensure the desired water quality.

10.1.2.4 *Safety aspects* (*while designing tower installation*)

a) Access to and from maintenance access doors, top of tower access platform with handrail system should be provided;

b) Ladders (either portable or permanent) to gain access to the discharge level or maintenance access doors should be provided;

c) Surrounding area should be clear;

d) Provision for lockout of mechanical equipment should be provided;

e) Safety cages around ladders should be provided;

f) Cooling tower shall be with appropriate electrical earthing;

g) Motor terminals shall be provided with appropriate enclosure for appropriate ingress protection; and

h) Bird cages shall be provided from all sides to prevent birds entering into the cooling tower.

10.1.2.5 *Testing of cooling tower*

Upon completion of installation, the cooling tower shall be tested for performance. The entering and leaving water temperatures, the entering air wet-bulb temperature, and the flow rate of water through the condenser water circuit shall be measured. The fan power consumption shall be checked from measurements at the fan motor. The tower capability shall then be calculated using the measured values and the manufacturer’s performance curves given with the technical submittal of the cooling tower and shall not be less than 95 percent.

10.1.2.6 *Critical parameters for cooling tower*

For cooling tower system following are the critical parameters:

a) Water flow rate;

b) Water temperature:

 1) *Inlet temperature* — Measures the temperature of the water entering the tower; and

2) *Outlet temperature* — Measures the temperature of the water leaving the tower.

c) Cooling range measures the difference between inlet and outlet water temperatures;

d) Wet-bulb approach measures the difference between the outlet water temperature and the ambient wet-bulb temperature;

e) Fan power consumption monitors the energy used by the fan;

f) Water level;

g) *p*H level;

h) Conductivity/Total dissolved solids(TDS); and

j) Drift rate.

10.1.2.7 *Water quality management in HVAC system*

The objective of cooling water treatment is to:

a) increase production;

b) reduce downtime;

c) lower operating and chemical costs; and

d) reduce your operation's impact on the planet

Treatment of cooling water will be different depending upon the kind of system in use. Here are the basic types:

1) Once-through;

2) Open recirculating;

3) Closed recirculating; and

4) Specialized.

NOTE — A specialized cooling system might utilize compression or absorption-type refrigeration or air conditioning systems or it might include the use of industrial air washers in which air is filtered, sprayed and then forced through a series of mist eliminators.

Water quality is crucial in HVAC systems to prevent scaling, corrosion, and microbiological growth, which shall lead to reduced efficiency, equipment damage, and health risks.

Water treatment should be considered to prevent scaling, corrosion, and biological fouling of the condenser and circulating system.

Large system shall be provided with fixed continuous-feeding chemical treatment system in which chemicals, including acids for *p*H control, are diluted, blended and pumped into the condenser water system.

Corrosion-resistant materials should be required for the treatment of system components that come in contact with these chemicals. In piping system design, provision for feeding the chemicals, blow down, drain, and on-line testing should be included.

Treatment system design should incorporate mainstream filtration without separate pumping system.

Additionally, side stream filtration incorporation in the system design should be considered especially when the primary make-up water is from an un-clarified water source (bore well-tanker water, STP effluent etc) that is high in suspended solids and/or iron.

10.1.2.8 *Recommended methods to control scale formation*

a) *Elimination at source* — Limit the concentration of scale-forming minerals by controlling cycles of concentration, or by removing the minerals before they enter the system;

b) *Monitoring cycle of concentration* — The ratio of makeup water rate to the sum of blow down and drift rate. The cycle of concentration shall be monitored by calculating the ratio of chloride ion, which is highly soluble in the system water, to that in the makeup water;

c) Mechanical changes in the system, like increasing water flow and providing exchanger with larger surface area, to reduce the chances of scale formation;

d) Feed acid to keep the common scale forming minerals, like, calcium carbonate, in dissolved state. Organic acid based specialty chemical formulations (*p*H reducers) are best suited for *p*H reduction in cooling water systems in terms of reducing the risk of acid induced corrosion; and

e) Treat with chemicals designed to prevent scale. (for example, scale inhibitors, dispersants and crystal modifiers).

For treatment of water for cooling towers, good practice [D-3(20)] shall be referred.

10.2 Closed-Loop Water System Treatment

Cooling and heating systems harness the power of water and refrigerant to transfer heat. Open circuits use cooling towers, while closed loops enclose water in pipes, cooling it through tubes and refrigerant, ensuring efficient temperature control in three precise steps.

Closed-loop systems have a reduced chance of contamination, but they are not maintenance free. Neglecting maintenance of a closed system promotes the growth of sludge and deposits. The most common problem with these systems is black magnetic iron oxide mud deposits. Over time, magnetic iron oxide particles bind together and collect on narrowed parts of the system. The most common place for this sludge is narrowed pipes, heat-transfer surfaces, cooling or heating coils and AHU and fan coils

The *p*H level of the water drops leads to bacterial growth or a system leak. Bacteria shall lead to pinhole leaks in the pipes, which shall affect both the water levels and chemistry. In addition to leaks, bacteria or mineral deposits shall reduce the ability of the system to transfer heat by coating the surfaces of the parts. Regular care of your components through water testing and chemical maintenance prevents the corrosion of the interior components of the system. In the closed system, effect of corrosion shall not be diagnosed.

While open systems need monitoring of water levels and watching for particulate matter from the environment in the water, closed circuit systems need control of the water’s chemistry.

10.2.1 *Chemical Treatment of Closed Loop Systems*

To prevent corrosion in carbon steel components, the water shall maintain a high, or alkaline, *p*H. To keep the correct chemical balance, water treatment professionals have several options at their disposal:

a) *Polymers* — Prevent corrosion and scale by dispersing harmful components in water, without environmental impact;

b) *Corrosion inhibitors* — Form a protective film on system surfaces, reducing corrosion, buffering *p*H, and reacting with oxygen; and

c) *Biocides* — Eliminate bacteria, preventing organic growth, biofilms, and Legionella bacteria, which shall contribute to corrosion.

10.3 Chilled and Condenser Water Pumps

Radial flow centrifugal pumps are recommended for use in HVAC systems for circulation or transfer of water or water/glycol solutions. Direct coupled pump assembly and inline pump should be used for smaller flow applications. Higher flow rate requiring motor size above 7.5 kW, pumps shall be coupled with high efficiency (IE3 and above) motors. Mono-block pumps shall be used up to 7.5 kW.

The split case horizontal pump shall be used in larger applications. Pumps shall be provided with the tapping on bearing housing for mounting of Vibration and temperature sensors, which shall be used for condition monitoring. For these pumps minimum IE3 (high efficiency class) motors are recommended.

Cavitation shall be avoided by selecting suitable location of pump with respect to the cooling tower, so as to always meet the net positive suction head (NPSH) requirement of condenser water pump.

10.3.1 *Selection Criteria and Types of Pumping System*

Pump shall be selected for optimized efficiency over the load profile, to operate at or near the best efficiency point (BEP) not more than 5 percent from the maximum efficiency curve to achieve energy efficiency. Pump shall be selected for optimized efficiency over the load profile, to operate at or near the best efficiency point (BEP) not more than 5 percent from the maximum efficiency curve to achieve energy efficiency. Wherever there is variable speed application, the pumps shall be with motor mounted VFDs or integrated drives with inbuilt intelligent IOT controls. IoT based controls shall have pump predictive analysis function with alert mechanism through SMS/email for various operational parameters.

Pump motor shall be energy efficient, having minimum IE3 (high efficiency class) motors, totally enclosed, fan-cooled, Class-F insulation and suitable for operation on AFD. Motor shall be specially designed for quiet operation. The motor rating shall be such as to ensure non overloading of the motor throughout its capacity range. Motor shall be suitable for 3-phase 415 percent + 10 percent volts, variable frequency power supply.

For water pumps, available net positive suction head (NPSH) shall exceed required NPSH to avoid pump cavitation.

The possible types of chilled water recirculation pumps are as given below:

a) Constant speed pumping system;

b) Variable speed pumping system (VSPS):

   1) Parallel VSPS configuration;

   2) Zoned VSPS configuration; and

   3) Primary secondary tertiary pumping system (P-S-T).

c) Primary only variable speed pumping system (PVF).

In order to improve energy efficiency of larger systems, where pumping energy amounts to 5 percent to 9 percent of total energy consumption in a building, glass flake coatings for large size pumps may be considered, based on life cycle cost assessment as it reduces friction improves and sustain efficiency.

Chilled water circulation system having pump motor larger than 3.7 kW rating are generally designed for variable fluid flow to offer large savings in operating cost.

10.3.2 *Installation of Pump*

The vertical in-line pumps do not require concrete pump foundation as they are of pipe mounted design. They become integral part of the piping system when installed as per manufacturer's instructions. Installation of efficient suction guide and triple duty valves shall also eliminate many piping accessories, resulting smaller installed footprint, as well as lowest embodied and life cycle carbon.

The horizontal base mounted pump base frame shall be mounted on concrete block which in turn shall be mounted on machinery isolation cork or any other equivalent isolation material. More than one pump set shall not be installed on a single base or on a single cement concrete block. Foundation bolts are required, shall be embedded correctly. Before the bolts are grouted and the coupling bolted, the base frame level shall be checked before proceeding with work.

The pump motor shall then be mounted on base frame, alignment checked and shall then be connected to the pump with flexible coupling and with guard, both for the condenser and chilled water pumps.

The insulation for chilled water pump shall be carried out in a manner so that it allows maintenance of the pump without causing damage to the insulation. After installation of the complete system and before testing, the pump shall be lubricated in strict accordance with the manufacturer's instructions.

10.3.3 *Testing of Pump*

Upon completion of installation, capacity of the pump shall be checked by measuring water flow, using the balancing valve in fully open position, motor current and pressure differential between the inlet and outlet. The readings shall be compared with actual performance identified in the technical submittal of the pump.

10.3.4 *Critical Parameters for Pumping System*

a) Pumps for condenser/chilled water:

1) Water flow rate as established from pump curves based on suction/discharge pressure;

2) *Head pressure* — Measures the pressure generated by the pump;

3) *Current and power consumption* — Monitors the current drawn and energy used by the pump;

4) *Vibration* — Tracks the mechanical vibration of the pump;

5) *Temperature* — Monitors the pump's and motor's operating temperature;

6) *Discharge pressure* — Measures the pressure at the pump outlet;

7) *Suction pressure* — Measures the pressure at the pump inlet; and

8) *Efficiency* — Calculates the pump's efficiency based on flow rate and power consumption.

10.4 Air Handling Unit (AHU)

All-air system is commonly used for comfort, healthcare, pharma, process applications. The provisions do not apply to AHUs used for data-center application.

10.4.1 *Selection Criteria*

The designer shall select the air handling unit on the basis of supply air temperature and volume; outdoor air requirement; desired space pressure; heating and cooling coil capacity; humidification and dehumidification capacity; return, relief, and exhaust air volume requirement; filtration levels, and the corresponding pressure capability of the fan(s). Controls may be added as required for project specific requirements.

Air handling unit shall be as per accepted standard [D-3(21)] of AHU. It should be verified or certified for the compliance of the standard through an third part certifying agencies for compliance as per the accepted standard [D-3(21)]. Selection criteria should include the sound levels of the AHU like casing radiated, supply and return to help acoustic engineers.

10.4.2 *Fan*

Fan selection should be based on efficiency and sound power level throughout the anticipated range of operation, as well as on the ability of the fan to provide the required flow at the anticipated static pressure.

Fans used in AHUs, with motor power exceeding 0.37 kW shall be of minimum mechanical efficiency and minimum fan motors efficiency requirements as specified in below table or as per ESCBC whichever is stringent:

System Type	*Fan Type*	*Mechanical Efficiency*	*Motor Efficiency* {*As per* [D-3(22)]}
(1)	(2)	(3)	(4)
Air- Handling unit	Supply, return and Exhaust	65 percent	IE3 and above

In addition, all fans used in AHUs having shaft power 2.5 kW or higher shall meet or exceed the FEI requirements of > 1.1.

a) Fan speed modulation control should be considered to save energy using duct static pressure signal; and

b) The power consumed by the AHU to be declared in the selection output and that should be part of the critical parameter for validation and measurement.

10.4.3 *Cooling Coil*

Cooling coil shall be designed as per the sensible and latent load. Number of rows of coils shall be determined as per the dehumidification requirements. Provisions shall be made to modulate the chilled water flow rate in the cooling coil for achieving high energy efficiency.

10.4.4 *Reheat Coil*

Reheat coil is heating coil placed downstream of a cooling coil. Reheat system is strongly discouraged, unless recovered thermal energy is used. Reheat coil is used to provide precise humidity control, which is generally not required for comfort conditions in most applications. Either reheat or desiccant is usually required to dehumidify outdoor air.

Reheating is necessary for specific applications like, laboratory, health care, or similar applications where temperature and relative humidity shall be controlled accurately. Hot-water coil, which provides a very controllable source of reheat energy, and thereby accurate relative humidity control is the preferred option for reheat coil. Condenser water recirculation through reheat coils may be considered for energy saving.

10.4.5 *Humidifier*

Humidifier shall be installed as part of the air handling unit only where close humidity control of selected spaces is required. For comfort installations requiring humidity control, moisture shall be added to the air by mechanical atomizer or point-of-use electric or ultrasonic humidifier placed in the conditioned area.

10.4.6 *Air to Air Energy Recovery Devices*

All hospitality and healthcare occupancies with energy recovery systems shall have air-to-air heat recovery equipment with minimum 60 percent recovery effectiveness at design conditions.

To prevent the wheel media from transferring exhaust air to the supply air duct, the heat wheel shall be fitted with an adjustable purging sector, with adjustable angle provision. The center of the wheel media shall consist of a hub with a shaft and bearings with efficient seal provided in the clearance between the rotor and the casing to minimize leakage between the supply air and exhaust air. Recommended higher face velocity limit is 3.5 m/s.

The outdoor air correction factor (OACF) and exhaust air transfer ratio (EATR) have direct impact on the energy consumption of the fans. The OACF should be in the range of 0.9 to 1. The EATR > 5 percent should not be acceptable.

10.4.7 *Fan Coil Unit* (*FCU*)

FCU shall be in accordance with accepted standard [D-3(21)].

10.4.8 *Air Filters*

Air filters used in AHUs shall be in accordance with accepted standards [D-3(15)], [D-3(16)], [D-3(17)] and [D-3(18)]. It is recommended to use ISO ePM1 50 percent in combination with ISO coarse and ISO ePM1 80 percent as second stage, for comfort applications (apart from critical applications like healthcare and cleanroom etc).

AHU filters shall be tested for filter bypass leakage in accordance with the accepted standard [D-3(21)].

10.4.9 *Vibration Isolation*

Where floor mounted fans are used, the AHU shall have an internal vibration isolation system by mounting fan, motor and drive assembly on suitable deflection type anti-vibration mountings as well the fan discharge shall be connected to the AHU casing through canvas connection to prevent vibration transfer.

10.4.10 *Dampers*

The exhaust air damper/s shall not allow inflow of outdoor air under any circumstances. Dampers should be modulated via an automated control system.

10.5 Installation of Air Handling Unit (AHU)

Following provisions to be considered of installation of air handling units:

a) Floor mounted air handling unit is generally installed on a set of precast PCC blocks to raise it off the mounting surface to permit easy drainage of the AHU drain pan and cleaning of the mounting surface. A set of 4 or 6 numbers of 0.20 m × 0.20 m × 0.20 m blocks are generally used for AHUs up to capacity of 10 000 m^3/h airflow, and 6 or more 0.30 m × 0.30 m × 0.30 m blocks for higher capacity AHU;

b) Floor trap shall be provided in the AHU room for disposal of condensate which accumulates in the drain pan during operation of AHU. AHU room floor should slope in the direction of floor trap to avoid the accumulation of water in AHU room. The condensate drain piping shall be connected with U-trap of the floor drain to avoid odour carry over with the return air;

c) AHU location shall be marked on the AHU floor as per approved shop drawings/manufacturer’s details. Co-ordination with contractors for civil works and other services shall be checked prior to installation;

d) Easy accesses and sufficient clearance shall be ensured for servicing and maintenance, that is, for cleaning of filters, maintenance of strainer/valve packages, tightening of fan belts, and repair as well as possible replacement of fan motor;

e) Duct flexible connection made of fire-proof material, shall be fixed on air outlet of the AHU, and if possible, also in perpendicular direction in main ducts within AHU room, to avoid vibration transmission along the ducts beyond the AHU room;

f) The valve package and piping connections shall be completed as per approved shop drawings;

g) Installation of ceiling suspended air handling unit shall be done with rod and fasteners as per the procedure stated above, barring the PCC blocks required for floor mounted AHU, and that insulated condensate drain pipe shall be laid in slope within ceiling space terminating into the U-trap of the nearest floor drain; and

h) Since the AHUs have the tendency to vibrate, both floor mounted and ceiling suspended AHUs shall be isolated from the structure by suitable rubber or spring-based isolators.

10.5.1 *Installation of Air Washer Unit (AWU)* – Air washer unit (AWU) components are similar to those of AHU, except for water spray section with in-line recirculation pump, which replaces the cooling coil. Installation of AWU shall follow the same steps as described above for AHU. Ceiling suspended AWU should be avoided.

10.6 Thermal Insulation

10.6.1 Air conditioning and water distribution systems, carrying chilled or heated fluids/air shall be thermally insulated to prevent undue heat gain or loss and to prevent internal and external condensation. Vapour seal shall be provided on chilled water pipes to avoid possibility of condensation.

10.6.2 The thermal insulation material shall be selected based on following physical characteristics:

a) *Fire properties* — Insulating materials shall be fire resistant and, in case of fire, shall not produce noxious smoke and toxic fumes;

b) Materials and their finishes should inherently prohibit rotting, mould and fungal growth, attack by vermin, and should be non-hygroscopic;

c) Material should not give rise to objectionable odour at the temperature at which they are to be used;

d) It should not cause a known hazard to health during application, while in use, or on removal, either from particulate matter or from toxic fumes;

e) It should have a low thermal conductivity throughout the entire working temperature range;

f) It should have good mechanical strength and rigidity, otherwise it should be cladded for protection; and

g) The insulation material selected should be non-hygroscopic and naturally resist the ingress of water vapour without any vapour barriers.

10.6.3 *Material*

Selection of material shall be as per design requirement, such as, closed cell flexible elastomeric foams, expanded/extruded polystyrene (EPS/XPS) and phenolic foam. Insulation should be flexible in natural to ease of handling, should be able to take contours of the surface on which it is installed, and thickness of insulation should be calculated for each project. For additional information special publication may be referred. The fire rating of insulation material for Class 0 for fire propagation test and Class 1 for surface spread of flame test may be referred from special publication for additional information.

NOTE — Special publication maybe DIN EN 12667 and EN ISO 8497 for testing thermal conductivity, EN 13501 and BS 476 Part 6 : 1989 for fire rating Class 0 for fire propagation test and BS 476 Part 7, 1987 for fire rating Class 1 for surface spread of flame test.

Insulation should possess the self-extinguishing property and it should not drip or spread the flames. Insulation should have zero ozone depletion potential, zero global warming potential, dust and fibre free. Insulation should have good resistance to ozone, oil and chemical. Insulation material should be compatible to various covering options like metal, polymeric, glass cloth, silver polymeric laminate etc.

The guidelines for insulating with fibre glass are given below; for other insulation materials, manufacturers' recommendations for installation should be followed.

10.6.4 *Thermal Insulation of Duct*

Insulation material for ducts shall be anti-microbial to ensure the indoor air quality. For microbiological growth on insulation surface and bacterial resistance special publications may be referred.

NOTE — Special publications maybe ASTM G-21 for Microbiological growth and ASTM 2180 or EN ISO 846 Method A and Method C for bacterial resistance.

The insulation materials shall be phenolic foam {as per accepted standard [D-3(23)]} closed cell flexible elastomeric foam/ fibre glass based. The insulation materials should be non-hydroscopic. PUF/Polyurethane shall not be used.

Elastomeric foam tubes, due to its vulnerability to fire, should be used only with full protection fire against with full cladding to smaller size pipes, and not on ducts.

The thermal insulation material should be wrapped around the duct with facing on outer side. Joints of insulation should be properly sealed with either same type of material or tape of 50 mm width on all longitudinal/transverse joints. Finally, straps should be fixed at suitable interval to ensure that the insulation is properly fixed with the ducts. Thermal insulation of the ducts, surface of duct, on which the external thermal insulation is to be provided, shall be thoroughly cleaned with wire brush and rendered free from all dust and grease. Then, one coat of cold-setting adhesive, should be applied over the duct surface and one coat of adhesive to be applied on insulation.

The facing of the insulation should be determined based on the site conditions. Facings can be plain, alufoil, polymer based, Glass cloth, PVC based or non-metallic composite cladding in case of flexible elastomeric foams depending on application area. In case of fiber glass insulation, metal claddings after wrapping polyethylene as vapor barrier is to be used to prevent fiber migration and also regulate the water vapor ingress. Fiber Glass with only alufoil should not be used due to vulnerability of foil which is very thin and can be damaged during installation causing water vapor ingress and increased thermal conductivity over time.

The thermal insulation material should be installed around the duct with facing on outer side. Joints of insulation should be properly sealed with either same type of material or tape of 50 mm width on all longitudinal/transverse joints. Finally, PVC straps should be fixed at suitable interval to ensure that the insulation is properly fixed with the ducts in case of fiber glass insulation is being used.

10.6.5 *Acoustic Lining of Duct*

Acoustic lining of duct should be carried out as follows:

a) The inside surface of duct on which the acoustic lining is to be provided should be thoroughly cleaned with wire brush and rendered free from all dust and grease;

b) The materials to be used for duct lining should be 25 mm thick resin bonded fiberglass having density of 48 kg/m^3. Open cell flexible elastomeric foam having density of minimum 140 kg/m^3 specially formulated only for sound attenuation not for inside duct surfaces, due to its poor fire and smoke emission risks;

c) The insulation should be fixed inside the duct using suitable adhesive and covered with fibre glass tissue paper. Open cell flexible elastomeric Foam should be installed with eco-friendly low VOC adhesive elastomeric foam should pass the tests (for test procedure special publications may be referred) for 50 kW/m^2 exposure;

NOTE — Special publications may be ISO 5659-2.

d) The plain insulation materials used should have an air erosion test certified so as no particles/fibres are dislodged with high velocity air and mix with conditioned air compromising indoor air quality. Materials used should have antimicrobial properties;

e) The insulation should then be covered with 0.5 mm thick perforated aluminium sheet with at least 20 percent to 40 percent perforation;

f) The insulation and aluminium sheet should be secured with cadmium coated bolts, nuts and cup washers/steel screws;

g) Finally, the ends should be sealed completely, so that no lining material is exposed; and

h) The lining of initial length of the duct may have to be done as per the requirement at site.

10.6.6 *Acoustic Lining in Equipment Room*

Acoustic treatment in equipment room to prevent noise transmission to adjacent occupied areas should be provided on the walls and ceiling of equipment room with acoustic lining of thermal insulation material. Acoustic lining shall be provided with performance levels adequate to achieve effective sound absorption and sound insulation, ensuring compliance with the acoustic comfort and noise control requirements of the space.

Acoustic lining of equipment may be carried out as follows:

a) A 610 mm × 610 mm frame work of 25 mm × 50 mm × 25 mm or 50 mm × 50 mm × 50 mm ‘U’ shape channel, for 25 mm for open cell flexible elastomeric foam or and 50 mm thick fiber glass acoustic lining respectively, made of 0.6 mm thick galvanized steel sheet may be fixed on to walls leaving 610 mm gap above floor (to prevent damage from flooding) by means of rawl plug in walls. Similar frame work may also be fixed on ceiling by means of dash fasteners;

b) Resin bonded glass wool/ Open Cell Elastomeric foam as specified, cut to size may be fitted in the frame work and covered with fibre glass tissue paper;

c) Surfaces may be finished by covering with 0.5 mm thick perforated aluminium sheet having perforation 20 to 40 percent with brass screws;

d) All horizontal and vertical joints may be covered with at least 25 mm wide, 1 mm aluminium strips held in position by steel or brass screws; and

e) For, Open Cell Elastomeric Foams, metal frame work is not required. 1m x 1m sheets of insulation to be directly applied to wall with suitable adhesive and finally two galvanized stainless steel metal strips of 0.6mm to be fixed in diagonally crossing each other with fasteners.

10.6.7 *Insulation of CHW/HW Refrigerant Pipes*

Pipe insulation Standard material should be EPS/XPS/PUF/closed cell flexible elastomeric foam/fibre glass as per requirement, specified with suitable density and thickness. Adhesive used for setting the insulation should be non-flammable, vapour proof, cold-setting, eco-friendly compound.

The insulation should then be finished as per the specific requirement of the site, as given below:

a) *Inside the building* — Insulation over the pipe work exposed in the building should be finished with specified thickness of aluminium sheet cladding, over a vapour barrier, with 50 mm overlap and tied down with lacing wire. For flexible elastomeric foams, suitable covering for mechanical protection can be used;

b) *Outside the building* — Insulation over the pipe work exposed to weather should be finished with vapour barrier, and 12 mm thick cement-sand plaster in two layers of 6mm thick each, followed by curing of minimum 48 h. For flexible elastomeric foams, suitable covering for mechanical protection can be used for UV resistance as per specialist literature;

   NOTE — Specialist literature maybe DIN EN 4892-2 Method A artificial weathering test.

c) *Buried pipe insulation* — For pipes outside the building and laid underground, the insulation should be covered with suitable gauge polythene faced hessian, (the polythene facing outward), with 50 mm overlap. All joints should be sealed with bitumen. A layer of maximum 0.50 mm × 20 mm GI wire mesh netting should be provided over it butting all joint, and it should be laced down with GI wire. A 20 mm thick cement-sand plaster (1 : 4) should be provided in 2 layers of 10 mm thickness each and should be waterproofed by applying hot bitumen and fixing tar-felt over the plaster. It should be finally finished with a coat of hot bitumen;

d) *Pump insulation* — Chilled water pump should be insulated to the same thickness as the pipe to which they are connected, and application should be same as above. Care should be taken to apply insulation in a manner as to allow the dismantling of pumps without damaging the insulation;

e) *Insulation of valves and fittings in chilled water line* — All valves, fittings, strainers, etc, should be insulated to the same thickness and in the same manner as for the respective piping, taking care to allow operation of valves without damaging the insulation; and

f) The thermal insulation of VRF system should be flexible elastomeric rubber insulations with UV resistance.

10.7 Thermal Energy Storage for Demand Management

Thermal storage may be used for limiting maximum demand, by controlling peak electricity load through reduction of chiller capacity, and by taking advantage of high system efficiency during low ambient conditions. Thermal storage will also help in reducing operating cost by using differential time-of-the day power tariff, where applicable. TES system may be designed in accordance with specialist literature.

NOTE — Specialist literature maybe API-650/AWWA D-100.

TES systems should be carefully designed and installed, considering factors like:

a) *Location* — Central plant TES location shall be determined in consultation with the air conditioning engineer, ensuring optimal performance and efficiency;

b) *Structural Integrity* — Adequate structural provisions to accommodate TES system loads, open-area surface installations shall consider vertical system options, complete with approach ladders for manholes etc;.

c) *Buried Installations* — Designs shall account for vehicular loads and movements above the installation area. The tanks may be rectangular or Circular with RCC construction; and

d) *Insulation* — To meet the thermal performance specifications, the tank to be provided with adequate insulation and cladding.

Thermal storage tank should be technically evaluated for tank material selection, insulation, and sizing, as well as the selection of suitable thermal energy storage media like water, ice, or phase-change materials. System design and controls should also be optimized, including charging and discharging strategies, temperature control, and integration with building management systems. TES systems should be designed with scalability and flexibility, allowing for modular expansion and integration with various HVAC systems.

10.8 Thermal Energy Storage for Critical Facilities

During power outage, chilled water to be made available for the duration the DG set starts and chillers ramp up. TES tanks are to be installed in line with the chillers, so that chilled water flows through the TES tanks then to the load, this to ensure that the tanks are always in charged condition.

TES tanks to be pressurized cylindrical vertical, all welded CS vessels with upper and lower ellipsoidal dish heads. For designing, manufacturing, certification and testing of tanks special publications maybe referred.

NOTE — Special publication may be ASME Section VIII Div.1.

Pressure vessels TES tanks to be fitted with radial disk diffusers to maintain stratification of water to ensure that the chilled water is delivered at the designed temperature and duration.

Designing of TES system and radial disk diffusers may be referred from special publications.

NOTE — Special publication may be ASHRAE Handbook, HVAC Systems and Equipment, Chapter 51.

Pressure vessels to be adequately insulated to meet the thermal performance specifications. Insulation to be protected with cladding.

System controls should also be optimized for charging and discharging strategies, temperature monitoring, and integration with building management systems.

11 HEATING

The efficiency of a heating equipment shall never be more than 100 percent, while a refrigeration system for heating usage shall have the coefficient of performance (COP) than its cooling mode.

Heating Systems are mainly used for hot water generation and space heating for comfort applications and also for Industrial use. The various applications where heating shall be effectively used are:

a) sanitary hot water generation for residential, commercial and hospitality industry;

b) space heating for residential, commercial and industrial sector in winters/as required depending on location and climatic conditions; and

c) hot water generation for industrial requirements.

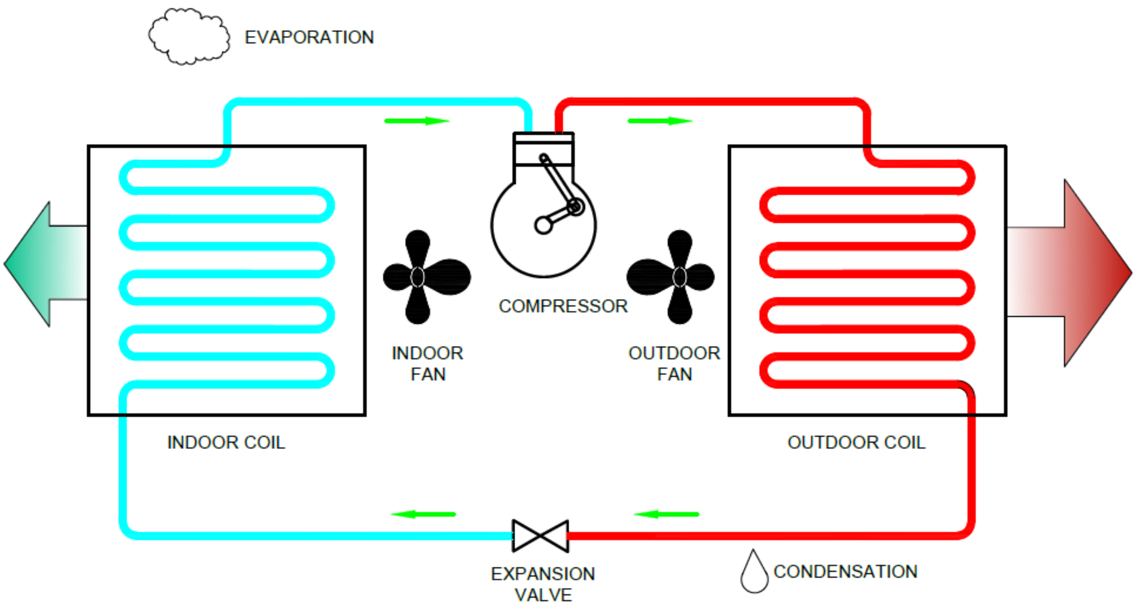
11.1 Various Heating Equipment/Systems

Various heating equipment/systems are described under 11.1.1 to 11.1.7. These heating equipment/systems shall be used for the building space and water heating applications based on their effectiveness, efficiency, and life cycle cost and installation process.

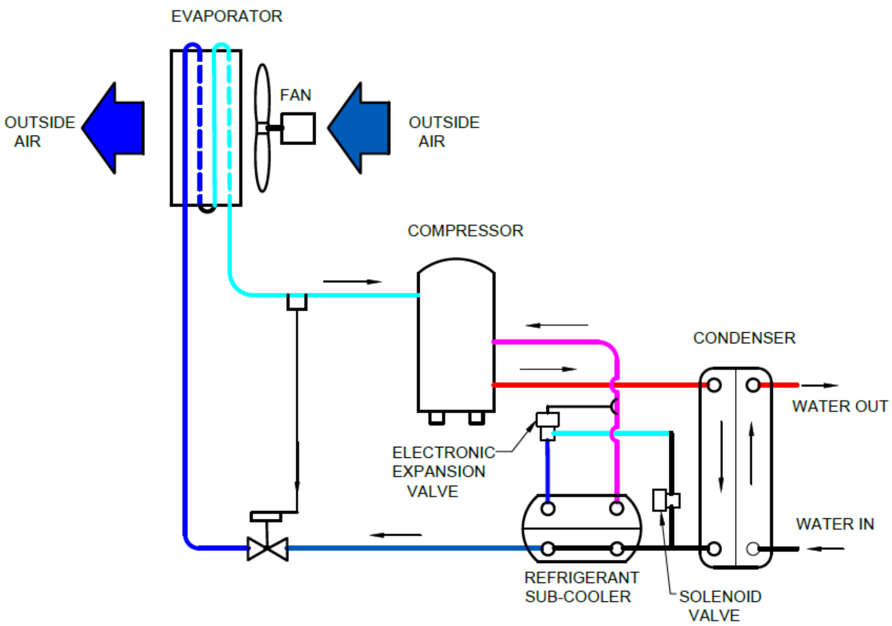
11.1.1 *Heat Pump – Air to Water and Air to Air and Water to Water Type*

There are two types of air source heat pumps. It extracts heat from the air and then transfers heat to either the inside or outside depending on the season is called air to air heat pump. The air-to-air heat pump shall be as per accepted standard [D-3(6)], [D-3(24)] or [D-3(25)] based on its capacity and type (*see* Fig. 3). The air to water heat pump shall be as per best practices given special publications. Water to water heat pump shall be as per standard [D-3(24)] (*see* Fig. 4).

NOTE — Special publication may be ISO 21978.



3A AIR -TO - AIR HEAT PUMP



3B AIR -TO - WATER HEAT PUMP

FIG. 3 AIR TO AIR AND AIR TO WATER HEAT PUMP

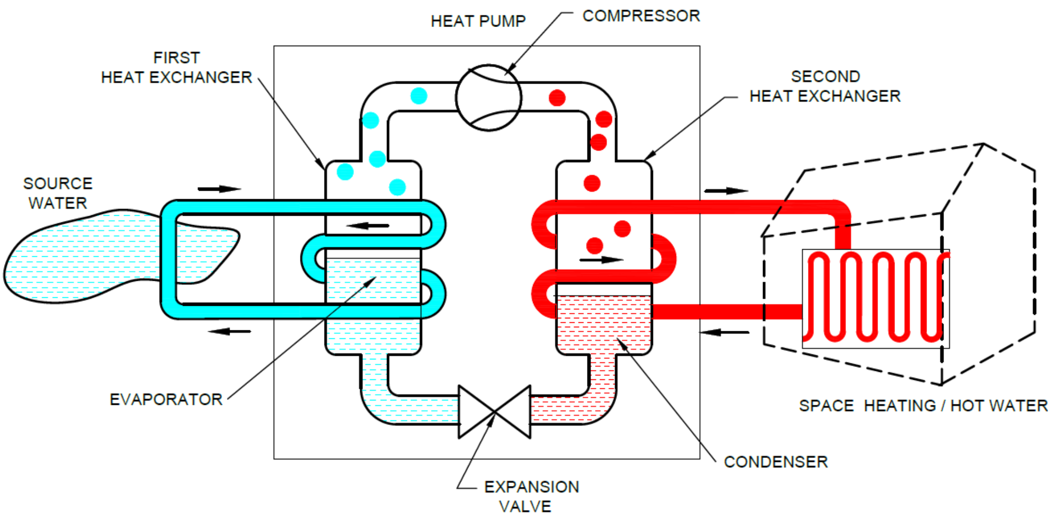


FIG. 4 WATER TO WATER HEAT PUMP

11.1.2 *Heat Pump – Ground Source*

Ground source heat pump uses the earth, or ground water, or both as the source of heat in the winter, and as the sink for heat removed from the conditioned areas in the summer. Heat is removed from the earth by using groundwater or an antifreeze solution; the liquid's temperature is raised by the heat pump; and the heat is transferred to indoor air. For year-round air conditioning, the process is reversed during summer months, heat is taken from indoor air and transferred to the earth, by the ground water. Fig. 5 illustrates the schematic of ground source heat pump. This shall be effectively used in locations having low temperature also.

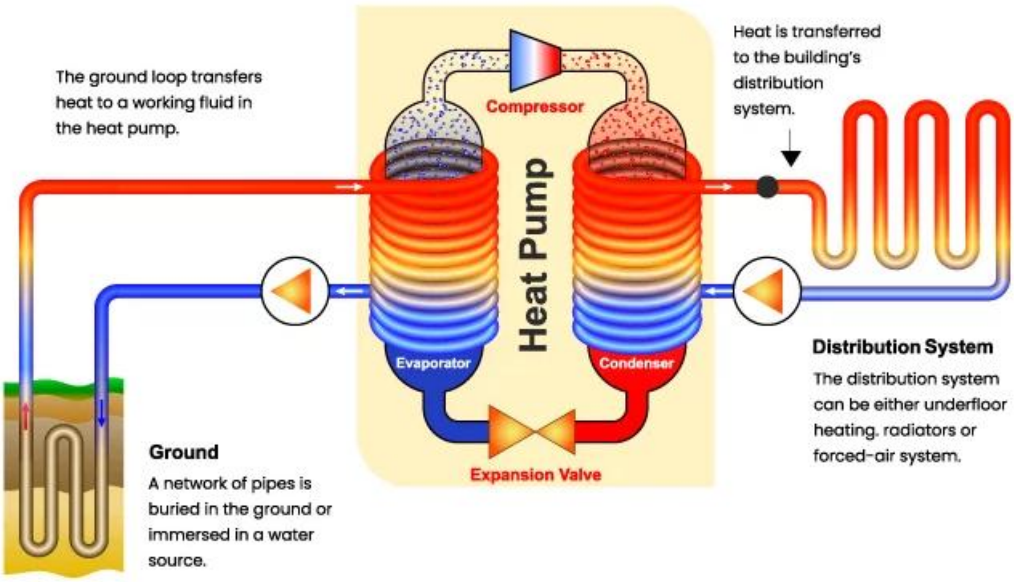


FIG. 5 GROUND SOURCE HEAT PUMP

11.1.3 *Reverse Cycle Systems*

Air conditioner window and split type, VRF system and DX package units, these systems are available with reverse cycle operation mode which shall supply cold air for cooling in summer and hot air for heating in winter. The performance of window air conditioners shall be as per accepted standard [D-3(4)], the split air conditioners shall be as per accepted standard [D-3(6)], the ducted split air conditioners shall be as per accepted standard [D-3(25)]. The variable refrigerant flow systems shall be as per accepted standard [D-3(26)] and chillers shall be as per accepted standard [D-3(24)]. The following Fig. 6 shows the reversible heat pump both in heating and cooling mode of operation.

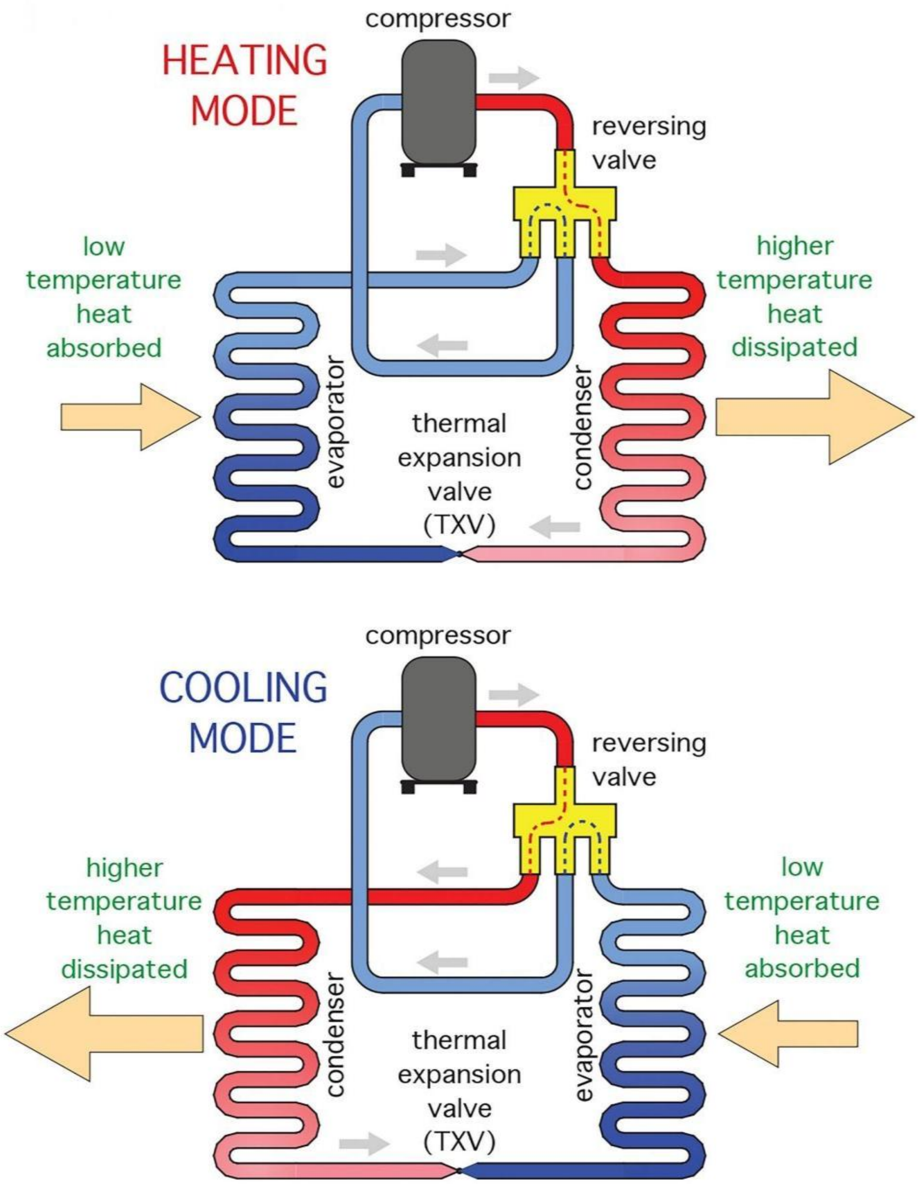


FIG. 6 REVERSIBLE HEAT PUMP

11.1.4 *Solar Water Heating (SWH) System*

SWH system converts energy content of solar radiation into heat energy for water heating using a solar thermal collector. In a close-coupled SWH system, the hot water storage tank is horizontally mounted immediately above the solar collectors on the roof. No pumping is required as the hot water naturally rises into the tank through thermal siphon flow. In a pump circulated system, the storage tank is ground or floor mounted and is below the level of the collectors; a circulating pump moves water or heat transfer fluid between the tank and the collectors.

SWH systems are designed to deliver hot water for most times of the year. However, in winter/monsoon, sometimes there may not be sufficient solar heat gain to deliver sufficient hot water. Therefore, alternative back up system is recommended to be used to meet hot water requirements.

Residential solar thermal installations fall into two groups; passive systems (sometimes called compact) and active systems (sometimes called pumped). Both typically include an auxiliary energy source (electric heating element or gas connection for the hot water storage tank, or fuel oil fired central heating system or heat pump) which is activated when the water in the tank falls below a minimum temperature setting such as 55 °C. However, it is recommended to use heat pump in active as well as passive systems to reduce carbon footprint and protect environment hence, hot water is always available. The combination of solar water heating and using the back-up heat from heat pump to heat water shall enable a hot water system to work all year round in cooler climates.

The amount of heat delivered by a solar water heating system depends primarily on the amount of heat delivered by the sun at a particular place, that is, the insolation. In tropical places, the insolation shall be relatively high, for example, 7 kWh/m^2 per day, whereas the insolation shall be much lower in temperate areas where the days are shorter in winter, for example, 3.2 kWh/m^2 per day. Even at the same latitude the average insolation shall vary a great deal from location to location due to differences in local weather patterns and the amount of overcast. The design, installation and performance evaluation of solar heating systems shall be in accordance with the good practice.

11.1.5 *Condenser Heat Recovery System*

Condenser heat recovery system captures energy that would otherwise be wasted in the atmosphere and converts this energy into useful heat. It is possible to capture this heat from the condenser and use it to generate hot water. By capturing this heat, overall system efficiency is considerably improved. Such a system should be used, wherever there is heating requirement concurrent with cooling demand. In a well-designed system this improves the performance of the cooling or refrigeration

11.1.6 *Solar Air Heating* (*SAH*) *System*

SAH system converts energy content of solar radiation into heat energy for air heating using a solar thermal collector. SAH systems shall be once through or of recirculating type. In the once through system fresh outdoor air is heated to a higher temperature of around 60 °C to 65 °C and supplied indoors through insulated ducting. This air mixes with the conditioned space air to maintain the indoor at a comfortable temperature typically between 18 °C to 22 °C. In the recirculating type, the conditioned space air is recirculated through the roof or wall integrated solar air heaters. In this case the air delivery temperatures shall be moderate at around 30 °C to 45 °C. Electric fans or blowers are used to enable circulation of the air. Solar air heaters shall offer solar collector efficiency in the range of 50 percent to 80 percent. Reversible air-to-air heat pumps powered with grid electricity shall be used as backup.

11.1.7 *Electric Heating*

Electric heating system are highly energy consuming and should be applied to specific application where other efficient Heating system shall not be applied due to technical/ location constraints. With all electric heating, care should be taken to preclude the risk of fire under abnormal conditions of operation, by the use of a suitably positioned temperature sensitive trip of the manual reset type, to cut off the electric supply in case of temperature rise beyond the specified limit.

It is recommended to avoid electrical heating to maximum extent possible for the building space or water heating applications.

12 BUILDING TYPOLOGY RELATED CONSIDERATIONS

The general guidance, for various selected building typologies, for the factors that usually influence the selection of the type, design and layout of the air conditioning or ventilating system, to be used are given in this section. The primary objective of these is to define the special provisions required for the air conditioning system, to ensure comfort conditions or defined conditions as the case may be for occupants in the respective applications.

These are the following typical building typologies normally envisaged in construction for different applications based on the usage factors:

a) Residential buildings;

b) Office buildings;

c) Campus buildings;

d) Hospitality buildings;

e) Healthcare buildings;

f) Underground metro stations;

g) Airport buildings;

h) Shopping complex buildings;

j) Industrial buildings; and

k) Tall buildings.

The following criteria shall be common to most of the building typologies:

1) *Building Layout and Orientation* — North-south orientation is often preferred to minimize direct sun exposure. The building orientation to be optimized to reduce heat gain. The design floor plans shall facilitate natural ventilation, air circulation and daylighting avoiding deep floor plans that limit these measures in case of naturally ventilated buildings.

2) *General Considerations* — Buildings may include both the outer and inner zones. The outer zone may be considered as extending from approximately 4 m to 6 m inwards from the outer wall of the building and is generally subjected to wide load variation owing to daily and annual changes in outside temperature and solar radiation. Ideally, the system(s) selected to serve an outer zone should be able to provide summer cooling and winter heating. During intermediate seasons, the outer zone of one side of the building may require cooling and at the same time, the outer zone on another side of the building may require heating. The main factors affecting load are usually window area and choice of shading devices.

   The other important factors are the internal gain owing to people, light, and other equipment loads. The choice of system may be affected by requirements to counteract down-draughts and chilling effects due to radiation associated with single glazing during winter. Cooling load for the inner zone is generally entirely from the heat gain from people, lights, and equipment, which is generally uniform throughout the year.

   Other important considerations in building applications may include requirements for individual room control, partitioning flexibility serving multiple tenants or residents, and the requirement of operating selected areas outside of normal working hours.

3) *Building Envelope* — The building envelope shall be designed to minimize energy use. Materials with low thermal transmittance (U-value) for walls, roofs, and selection of glass following a detailed study of solar heat gain coefficient (SHGC), U-value and visible light transmittance (VLT) shall be selected. For additional information on high-efficiency insulation materials, special publication may be referred.

   NOTE — Special Publication may be Energy Conservation and Sustainability Building Code (ECSBC) 2024

4) *Ventilation* — Minimum outdoor airflow rate based on occupancy and area requirements shall be provided as detailed in **3**. Comprehensive mechanical ventilation system with usage of energy/heat recovery system and CO_2 monitoring and control system to adjust airflow following indoor air quality norms shall be utilised. Also refer to **5.2.5** of Part D 'Building Services, Section 1 Lighting and Natural Ventilation' of this NBCS.

5) *Indoor Air Quality* (*IAQ*) *and Thermal Comfort* — For threshold values related to indoor air quality and requirements for thermal comfort, article on 'Basis of Design' to be referred.

12.1 Residential Building Typology

This classification helps in understanding the different types of residential buildings based on their design, usage, and structure.

12.1.1 *Room-Specific Design*

a) *Bedrooms* — Quiet AC operation and individual temperature control for better thermal comfort shall be ensured. Outdoor unit of the air-conditioner shall not be installed towards the room/living spaces/balconies. Hot air of the outdoor unit shall be directed toward open spaces/ambient.

b) *Living rooms* — Design of AC in living room shall be appropriate considering peak occupancy. For better thermal comfort and energy savings, AC and ceiling fan both shall be installed in living room.

c) *Kitchens* — Ventilation hoods to remove excess heat and gaseous contaminants shall be installed.

12.1.2 *Insulation and Thermal Mass*

a) *Insulation* — High-quality insulation in walls, roofs, and floors to reduce cooling loads shall be entertained.

b) *Thermal mass* — Materials with high thermal mass (for example brick) to stabilize indoor temperatures shall bring in better energy efficiency.

12.1.3 *Window Design and Shading*

a) *Glazing* — Glazing in residential houses refers to the installation of glass in windows, doors, skylights, and other transparent or translucent building components. The choice of glazing impacts energy efficiency, aesthetics, security, and comfort. Glass preferably of lower SHGC based on climatic zone shall be selected. Ratio of VLT/SHGC shall be more.

b) *Shading devices* — Install external permanent/movable shading devices (for example, shade, awnings, louvers) to block direct sunlight. Consider the use of trees and vegetation for natural shading. For large dwelling/projects, optimising shading shall lead to reduced cost and more benefits. Facade shall be designed with specific shading devices/solutions.

c) *Window placement* — Strategically placement of windows to maximize natural light while minimizing heat gain shall be encouraged.

12.1.4 *Ventilation*

a) *Natural ventilation* — Cross-ventilation by aligning windows and openings to allow air to flow through the building shall be envisaged. For a building to be naturally ventilated, it shall have operable windows that occupants shall adjust to maintain their thermal comfort. These windows may include dynamic components like louvers to regulate the amount of air exchange. Open door/window and operation of exhaust fan to ventilate during night, to cool the spaces and remove the heat that traps during daytime shall be worked out.

b) *Mechanical ventilation* — Mechanical ventilation systems shall be designed to complement AC systems, providing fresh air without compromising energy efficiency.

12.1.5 *Sustainability Measures*

a) *Passive cooling* — Passive cooling strategies such as orientation, providing buffer spaces, thermal chimneys, evaporative cooling, and night purging to reduce reliance on AC shall be included in the design.

b) *Landscaping* — Strategic landscaping (for example, planting trees, creating windbreaks) to naturally cool the building and reduce AC load shall be used.

12.1.6 *Cultural and Climatic Adaptations*

a) *Regional design adaptations* — Building design shall adapt to local climatic conditions, such as high humidity areas requiring enhanced dehumidification.

b) *Cultural preferences* — Cultural preferences in building layout and AC usage, such as preferred room temperatures and ventilation styles shall be investigated and included during design.

12.1.7 *Adaptive Reuse and Retrofitting*

a) *Historic buildings* — When retrofitting historic buildings, the preservation of architectural features with the integration of modern AC systems shall be balanced.

b) *Older homes* — Upgrading insulation, windows, and ductwork to improve the efficiency of new AC systems shall be envisaged.

12.2 Office Building Typology

Important considerations in office building applications may include requirements for individual room control, partitioning flexibility serving multiple tenants, and the requirement of operating selected areas outside of normal office hours. Areas such as conference rooms, board rooms, shall teens, etc., will often require independent systems.

12.2.1 *Specific Guidelines for Large/Small Office Buildings*

a) *Centralized HVAC systems* — Centralized HVAC systems shall be specifically considered for their high efficiency and flexibility in handling variable loads. Centralized systems shall provide cooling and heating based on the specific needs of different zones, enhancing overall energy efficiency;

b) *Demand-controlled ventilation (DCV)* — Implementing demand controlled ventilation (DCV) should be considered to optimize ventilation rates based on real-time occupancy and indoor air quality. This shall reduce energy consumption while maintaining a comfortable environment;

c) *Indoor air quality monitoring (IAQ)* — IAQ sensors shall be considered to continuously monitor and maintain optimal indoor air quality levels. These sensors shall measure parameters such as CO_2 levels, humidity, and volatile organic compounds (VOCs), there by maintaining and monitoring to ensure a healthy indoor environment;

d) *Air purification technologies* — Advanced air purification technologies such as UV-C light/bipolar ionization/other passive air purification methods should be used to reduce airborne contaminants of their levels exceed the threshold values given in **3.6.2**. These technologies shall be particularly beneficial in reducing the spread of airborne diseases and improving overall air quality;

e) *Predictive maintenance* — Predictive maintenance using IoT and smart sensors should be considered. This approach shall identify potential issues before they lead to system failures, reducing downtime and maintenance costs;

f) *Enhanced humidity control* — Desiccant dehumidification systems should manage high humidity levels, particularly in areas with high latent gains. Proper humidity control for maintaining comfort and improving indoor air quality;

g) *Energy recovery ventilation* (*ERV*) — ERV systems to improve indoor air quality inclusion shall enhance energy efficiency by recovering energy from exhaust air. ERVs shall significantly reduce the load on heating and cooling systems by pre-conditioning incoming fresh air. For additional information, special publication may be referred for the Energy recovery system;

NOTE — Special publication maybe ISO 16494.

h) *Advanced zoning strategies* — Implementing advanced zoning strategies for large office spaces should be considered by dividing office spaces into detailed zones based on occupancy and usage patterns and use of automated systems to adjust HVAC settings dynamically in different zones shall lead to energy efficiency to systems;

j) *Building management systems (BMS) and IoT integration* — HVAC systems should be integrated with building management systems (BMS) for centralized control and monitoring and utilizing IoT devices to enable real-time data collection and automated adjustments, shall optimize energy efficiency and system performance;

k) *Noise reduction measures* — Advanced soundproofing materials and design techniques to minimize the operating noise levels of HVAC equipment and further ensuring low noise levels enhances occupant comfort, especially in open office environments. Noise control measures shall be as per Part D 'Building services, Section 4 Acoustics, Sound Insulation and Noise control' of this NBCS; and

m) *Green building practices* — Green building practices such as incorporating renewable energy sources (for example, solar panels) to power HVAC systems and using sustainable building materials should be considered. This aligns with global trends towards environmental responsibility and sustainability.

12.3 Campus Building Typology

The provisions related to HVAC requirements for a large campus (facility consisting of multiple single or multi-use buildings or structures within a premise), including residential and (or) non-residential educational institutions, IT/ITES parks are covered. The design and implementation of heating, ventilation, and air conditioning (HVAC) systems for campus projects shall meet several key requirements to ensure efficient, effective, and sustainable operation.

12.3.1 *General Requirements*

These guidelines help in categorizing and designing buildings within a campus based on their function, layout, and structural needs.

a) *Scalability and flexibility* — The system design shall allow for future expansion as well as varying nature of usage of spaces by designing modular components for easy upgrades to ensure flexibility to adapt;

b) *Energy efficiency* — Central HVAC systems with advanced controls, with predictive maintenance and energy system for optimum energy use shall be designed. Centralized air-conditioning system with a diligent feasibility/viability study of a particular system invariably leads to better energy efficiency solutions. In case of mixed used buildings (residential, office, laboratories, etc) in the campus, the projects should consider choosing district cooling system for better resource efficiency and sustainability. The system design feasibility for district cooling systems or alternate energy cooling systems shall be referred in detail from **6** and **7**. Variable speed drives for motors and fans shall be used.

Life cycle analysis shall further support to demonstrate business case of selecting a particular system. Minimizing the ecological footprint of HVAC systems and use environmentally friendly refrigerants, implementation of systems that conserve water and other resources should be considered;

c) *Heating systems* — Heat pumps/high efficiency central heating system as detailed in **11** for space heating as well as water heating Integrate heating systems with building management/optimization systems (BMS) for optimal control;

d) *Zoning and control* — Multiple zones for different buildings or areas use advanced control systems for precise temperature and humidity levels and incorporate programmable thermostats and building management systems (BMS). Advanced zoning strategies for large campuses by dividing campus buildings into detailed zones based on occupancy and usage patterns, automated systems to adjust HVAC settings dynamically in different zones shall be standard practices. Occupancy diversity into zoning for better control and energy savings as well as thermal control strategies based on space type and usage shall be considered. For more details on building automation and zoning control refer to **15**;

e) *Occupancy and lighting schedules* — HVAC and lighting schedules with occupancy to optimize energy use shall be aligned. Automated lighting systems with daylight harvesting and occupancy sensors and scheduling HVAC operations before occupancy starts for optimal usage is preferred; and

f) *Urban heat island* (*UHI*) — Urban heat island (UHI) causes discomfort and increases energy consumption of air-conditioners. Mitigation measures to reduce the impact of UHI and its associated impacts shall be considered. Proper placement of HVAC equipment to avoid UHI and manage distance between equipment (especially air-cooled outdoor equipment) to release the heat and avoid creating local heat spots need a due consideration.

12.3.2 *Operation and Maintenance (O&M)*

These ensure the effective management, upkeep, and long-term performance of campus buildings and their systems.

a) *Energy management* — Energy management strategies within the BMS to minimize energy consumption by scheduling HVAC operations to match occupancy patterns and utilizing energy-efficient modes during non-peak hours and similarly system designed to showcase performance levels (square foot/TR) shall lead to better energy management;

b) *Maintenance* — As the energy consumption in the large campuses is huge, the predictive maintenance and cloud monitoring for advanced energy efficiency may be considered as part of the system for effective monitoring and for taking proactive measures. Advanced monitoring systems to track performance in real-time and scheduled maintenance based on predictive analytics will lead to lower down times. Detailed logs using a computerized maintenance management system (CMMS) shall be maintained; and

c) *Retro/re-commissioning* — Analysing HVAC system performance after one year of operation to compare design with actual performance. Necessary retro-re/commissioning to improve performance based on load profiles of several components for example fan/pump/motor, etc (following over/under load conditions) shall be undertaken.

12.3.3 *Specific Requirements for Campus Project*

These requirements address the unique operational needs and considerations for designing and managing campus buildings.

a) *Climate adaptation* — Implement HVAC system modifications for different climatic zones (hot and dry, warm and humid, composite, temperate, cold);

   Use adaptive thermal comfort models to enhance occupant comfort and energy savings.

b) *Water conservation* — Emphasize water-efficient HVAC systems, especially in water-scarce regions. Encourage the use of condensate recovery systems to reuse water in cooling towers or irrigation. Promote the use of water-efficient fixtures and fittings in all campus buildings;

c) *Sustainability practices* — Campus buildings are encouraged to use sustainability practices (green building guidelines) and sustainability standards shall follow for enhanced sustainability. Promote the use of recycled and sustainable materials in HVAC installations;

d) *Resilience and reliability* — HVAC systems that shall handle frequent power outages and voltage fluctuations shall be envisaged. The inclusion of backup systems to ensure continuous operation during power failures. Robust disaster recovery plans to maintain HVAC operations during emergencies shall be included; and

e) *Testing and commissioning* — Thorough testing, balancing, and commissioning procedures shall be followed. Standardized protocols for verifying system performance before handover shall be compulsory. For detailed guidelines refer 17.

12.3.4 *Educational Campus Buildings*

Special attention for the design of the HVAC systems shall be paid to meet the needs of every age group in this type of buildings. The equipment should be easy to operate and maintain and the design should provide no drafts. These facilities have two distinct occupant zones: (a) the floor level, where younger children play, and (b) normal adult height, for the teachers. The administrative and teachers' offices shall be considered as a separate zone.

Special consideration shall be given to operating schedule for setback. Supply air outlets should be positioned to avoid drafts. Proper ventilation and exhaust shall be provided for controlling odours and to prevent the spread of diseases. University and college campus having large diversity in cooling/heating loads, should be provided with large central utility plant or smaller mechanical rooms serving a cluster of buildings. The central utility plant shall supply chilled/hot water depending on the zonal requirement. The designer should consider site constraints, including geographic location. In addition to accommodating the mechanical and electrical equipment, central utility plant may also house engineering, operation and maintenance personnel. A central control room shall be provided for energy monitoring.

12.4 Hospitality Buildings Typology

It is critical to focus on guest comfort, energy efficiency, and system reliability. Each guest room or each zone shall be provided with not less than one thermostat capable of being set from 20 °C to 30 °C and capable of operating the system's cooling and heating. The thermostat or control system, or both, to have an adjustable dead band, the range of which includes a setting of 12 °C between cooling and heating, where automatic changeover is provided. Wall mounted temperature control shall be mounted preferably on an inside wall and shall meet the requirements given in **15**.

12.4.1 *Guest Rooms*

Guest room systems are required to be available for operation on a continuous basis. The room may be unoccupied for most of the day and therefore provision for operating at reduced capacity or switching off. Low operating noise level, reliability and ease of maintenance shall be taken care. Treated fresh air (TFA) is introduced through an independent TFA system, it is generally balanced with the bathroom extract ventilation, to ensure air circulation into the bathroom. Fan coil unit is generally found to be the most suitable for this kind of application with speed control for fan and motorized/modulating valve for chilled water control for cooling.

12.4.2 *Restaurant, Cafeteria, Bar and Night Club*

These applications have several factors in common; highly variable loads, with high latent gains (low sensible heat factor) from occupants and meals, and high odour concentrations (body, food and tobacco smoke odours) requiring adequate control of fresh air extract volumes, direction of air movement for avoidance of draughts and make up air requirement for associated kitchens to ensure an uncontaminated supply. This type of application is generally best served by the all-air type of system preferably with some reheat or return air bypass control, to limit relative humidity. Either self-contained packaged unit or split system, or air handling unit served from a central chilled system, may be used. Sufficient control flexibility to handle adequately the complete range of anticipated loads.

12.4.3 *Specific Criteria to be Adhered for Hospitality Buildings*

a) *Occupant (guest) comfort* — Air-conditioning system to maintain consistent temperature and humidity control in guest rooms, lobbies, restaurants, and common areas. There shall be individual temperature and ventilation controls in guest rooms for personalized comfort. Ensure rapid response to temperature adjustments for good comfort conditions;

b) *Energy efficiency* — Energy-efficient HVAC equipment such as high-efficiency chillers/cooling system, boilers, and heat pumps shall be needed. Occupancy sensors and smart thermostats reduce energy use in unoccupied rooms. Integrating energy recovery ventilation systems to reclaim and reuse waste heat or cooling shall bring in additional energy efficiency. Heat recovery system shall be sized based on floor/occupancy loading;

c) *Zoning and control* — HVAC systems cater to multiple zones with different activity areas of the hotel, such as guest rooms, dining areas, and conference rooms. Advanced control systems for precise temperature and humidity regulation as well as building management system (BMS) for centralized monitoring and control of HVAC operations shall be designed and used as detailed in **15**;

   It shall be necessary to avoid mixing of air of various type of zoning (corridor, guest rooms, guest areas where smoking allowed, kitchen ventilation, etc), designate smoking and non-smoking areas by pressure differentials between zones;

d) *Noise control* — Low-noise HVAC equipment shall be used to ensure a quiet environment. Necessary sound attenuation measures shall be installed in ductwork and mechanical rooms to reduce noise. Proper acoustic insulation in guest rooms and other sensitive areas for ensuring low/nil cross talk between areas shall be initiated. Noise control measures shall be as per Part D 'Building services, Section 4 Acoustics, Sound Insulation and Noise control' of this NBCS;

e) *Sustainability* — Integrating passive and active heating/cooling system to minimize greenhouse gas emissions and developing the hotel/hospitality project to incorporate sustainable design practices shall be followed;

f) *Reliability and maintenance* — Durable and reliable HVAC equipment to minimize downtime and maintenance issues shall be needed as the systems are generally required to function round the clock throughout the year. Systems should have easy-access for maintenance and repairs and a preventive maintenance schedule to ensure long-term system reliability and performance shall be implemented;

g) *Aesthetic integration* — HVAC design shall coordinate with the hotel's interior design to minimize visual impact and use discreet and aesthetically pleasing diffusers, grilles, and other HVAC components without compromising on functional effectiveness and air distribution patterns. HVAC components shall be selected to seamlessly integrated into the overall architectural design;

h) *Environmental impact* — The ecological footprint of HVAC systems shall be minimised by using environmentally friendly refrigerants and materials. Water conservation measures, particularly in cooling towers and other water-intensive components shall be a priority. Use of treated water for cooling tower and designing systems to reduce overall energy consumption and waste shall be adhered. Calculate environmental impacts of refrigerants used in air-conditioning/fire suppression system (calculate ODP – ozone depleting potential if applicable and GWP – global warming potential). Project to utilise new age refrigerant/natural refrigerant to minimize or avoid negative impact of refrigerants. For more details, refer to 5;

j) *Guest room control systems* — User-friendly interfaces for guests to control room temperature, ventilation, and other comfort settings shall be provided. Integrating with hotel management systems to optimize energy use based on occupancy and guest preferences and by offering customizable settings enhance the guest experience and also accommodates individual needs;

k) *Operation and maintenance of HVAC system* — Comprehensive training for facility managers and maintenance staff on the operation and maintenance of HVAC systems shall be provided. Ongoing support and service agreements with HVAC vendors for timely repairs and updates and constant skill upgradation for staff on energy-saving practices and the proper use of HVAC control systems shall be followed. Make provision to clean ducts periodically to offer hygiene conditions to guest, staff/occupants;

m) *Scalability* — HVAC systems shall be designed to accommodate future expansions or renovations, use modular components for easy upgrades and modifications. Ensure the system shall adapt to varying occupancy rates and seasonal changes; and

n) *Specialty areas* — HVAC systems shall be designed for specialty areas such as spas, kitchens, and conference rooms for meeting specific requirements. Adequate ventilation and humidity control in spas, pools, and fitness centres and robust exhaust systems in kitchens to manage odors, heat, and smoke effectively shall be provided.

12.5 Healthcare Building Typology

HVAC Systems in healthcare facilities provide a broad range of services in support of people who are vulnerable to an elevated risk of health, fire and safety hazard.

12.5.1 *Basic Classification of Health Care Facilities*

Health care facilities vary widely in the nature and complexity of services they provide and the relative degree of illness or injury to the patients treated-like clinic, diagnostic centers, medical centers, specialty hospitals, super specialty hospitals, etc. Environmental control requirements and the role of HVAC system in life safety and infection control becomes very important with the medical services provided.

Ventilation in healthcare buildings is crucial for maintaining air quality, controlling infections, and ensuring the comfort and safety of patients, staff, and visitors. Healthcare facilities have unique ventilation requirements due to the need to manage airborne pathogens, chemical contaminants, and varying needs of different areas.

12.5.2 *Design Criteria*

While designing HVAC for a healthcare facility, Patient therapy shall be the prime consideration, the plan shall include the following parameters which shall be taken into consideration:

a) temperature and humidity requirements of various spaces;

b) ventilation requirements/air changes requirements;

c) filtration requirements for contamination control;

d) restriction on air movement between adjoining spaces;

e) permitted tolerance on environmental conditions;

f) redundancy/backup;

g) fire and life safety;

h) efficiency bench marking on the equipment;

j) guidelines for parameters;

k) standards/ good practices; and

m) ease of installation and Maintenance friendly.

12.5.3 *Key Considerations*

These considerations ensure that healthcare buildings are designed to meet the specific functional, safety, and regulatory requirements for patient care and medical operations.

a) *Comfort conditioning* — Across health care facilities, health care practices often expose patients and staff to conditions that dictate unique environmental requirements. As in any other building, comfort of occupants is fundamental to overall well-being and productivity also it has a significant role in facilitating healing and recovery;

b) *Therapeutic conditioning* — Certain Medical functions, treatments or healing processes demand controlled environmental temperature and relative humidity conditions that deviate from the requirements for personal comfort. Operating rooms and other key areas require a range of room temperature spanning several degrees regardless of the season to best facilitate a given procedure or patient condition;

c) *Infection control* — Medical facilities are places where relatively high levels of pathogens are generated and concentrated by an infected patient population or by procedures that handle infected human tissues and bodily fluids. Filtration levels and air changes per hour (ACH) play a vital role in controlling the risk;

d) *Pressure differentials* — Implement negative pressure in airborne infection isolation (AII) rooms to contain airborne pathogens and positive pressure in operating rooms (OR)/protective environment (PE) rooms to protect patients from outside contaminants;

e) *Life safety* — HVAC systems are required to support special building compartmentation, opening protection, smoke control and smoke detection systems. Containment is facilitated by smoke and fire dampers and some cases restricting cross overs between smoke zones; and

f) *Noise control* — Noise control is of high importance because of negative impact of high noise levels on the patients and the need to safeguard patient privacy. Noise control measures shall be as per Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS.

12.5.4 *Energy Efficiency/Sustainability Considerations*

These guidelines focus on optimizing energy use and incorporating sustainable practices in the design and operation of buildings to reduce environmental impact.

a) *Role of building envelope* — The building envelope in healthcare facilities is a critical component that affects energy efficiency, indoor environmental quality, and overall functionality. Given the unique needs of healthcare environments, the building envelope shall be designed with stringent requirements to ensure patient safety, comfort, and operational efficiency;

b) *Energy-Efficient Window* — Windows with low solar heat gain may be used in order to minimize heat loss and gain shall be used. Windows shall have proper seals to prevent air infiltration;

c) *Air Tightness* — Building envelope shall be well-sealed to prevent air leaks by providing proper sealed building envelope. This shall be done for controlling indoor air quality and maintaining the efficacy of HVAC systems;

d) *Testing and Commissioning* — Apart from the procedures listed in**,** additionally shall conduct blower door tests and infrared thermography tests during commissioning to identify and rectify any air leakage points;

e) *Natural Light and Views Daylighting* — The building envelope shall be designed to maximize natural light, reducing the need for artificial lighting and creating a more patient friendly environment. This includes the strategic placement of windows, skylights, and light wells; and

f) *Glare Control* — Shading devices, low-E glass, or other technologies shall be incorporated to control glare and excessive solar heat gain, enhancing comfort and reducing cooling loads.

Guidelines for the above parameters for various functional spaces in a health care facility are given in Table 27 and Table 28.

Table 27 Guidelines for Parameters to be Considered for HVAC System Design for Health Care Facilities

(*Clauses* 12.5.4 *and* 12.5.5)

Sl No.	**Area/Functional Space**	**Temperature** °C	**Relative Humidity** percent	**Minimum Total Air Changes per Hour**	**Minimum Air Changes of Outdoor Air per Hour**	**Air Pressure in Relation to Surrounding Area/ All Room Air Directly Exhausted to Outdoors**
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Operating theatres	20 to 24	20 to 60	20	4	Positive/NR
ii)	Cath labs	21 to 24	*Max* 60	15	3	Positive/NR
iii)	Delivery rooms	20 to 24	20 to 60	20	4	Positive/NR
iv)	Recovery room, ICU, treatment rooms	21 to 24	*Max* 60	6	2	Equal/NR
v)	Endoscopy, bronchoscopy	20 to 23	*Max* 60	12	2	Negative/NR
vi)	Patient rooms	21 to 24	*Max* 60	6	2	Equal/NR

Table 27 (*Concluded*)

Sl No.	Area/Functional Space	Temperature °C	Relative Humidity percent	Minimum Total Air Changes per Hour	Minimum Air Changes of Outdoor Air per Hour	Air Pressure in Relation to Surrounding Area/ All Room Air Directly Exhausted to Outdoors
(1)	(2)	(3)	(4)	(5)	(6)	(7)
vii)	Toilets	–	–	–	10	Negative/Yes
viii)	Protective environment rooms (Immuno suppressed patients)	21 to 24	*Max* 60	12	2	Positive/NR
ix)	Isolation room (for patients with infectious disease)	21 to 24	*Max* 60	12	2	Negative/Yes
x)	Corridors	–	–	2	2	Negative/NR
xi)	X-ray/Radiology	21 to 24	*Max* 60	15	3	Positive/NR
xii)	Laboratories (other than biochemistry and serology)	22 to 26	*Max* 60	6	2	Negative
xiii)	Biochemistry and serology labs and pharmacy	22 to 26	*Max* 60	6	2	Positive/NR
xiv)	Admission/waiting rooms	22 to 24	*Max* 60	6	2	NR/NR
xv)	Diagnostic/treatment OPD	21 to 24	30 to 60	12	2	NR/NR
xvi)	Sterilizer equipment room	NR	NR	10	–	Negative/Yes
xvii)	Sterilizer storage	22 to 26	*Max* 60	4	2	Positive/NR

Table 28 Guidelines for Minimum Filter Efficiency Requirement in Health Care Facilities

(*Clause* 12.5.4)

Sl No.	Category	Area/Designation	Filter Class (Efficiency)	
			Filter Bed 1	Filter Bed 2
(1)	(2)	(3)	(4)	(5)
i)	Category 1	a) Orthopedic OT b) Bone marrow transplant OT c) Organ transplant OT d) PE rooms including Burn Units	ISO Coarse 60 percent	ISO ePM2.5 80 percent and HEPA at outlets

Table 28 (*Concluded*)

Sl No.	**Category**	**Area/Designation**	**Filter Class (Efficiency)**	
			Filter Bed 1	Filter Bed 2
(1)	(2)	(3)	(4)	(5)
ii)	Category 2	a) General OT delivery rooms b) ICU, patient care rooms	ISO Coarse 60 percent	ISO ePM2.5 80 percent
iii)	Category 3	a) Treatment rooms b) Diagnostic areas	ISO Coarse 60 percent	Not required
iv)	Category 4	a) Laboratories b) Sterile storage	ISO Coarse 60 percent	Not required
v)	Category 5	a) Food preparation b) Laundry c) Admin, etc	ISO Coarse 60 percent	Not required

12.5.5 *Air Distribution System*

The fresh air changes per hour as well as the total air changes shall be minimum as per the guidelines given in Table 27 above.

For critical care areas, variable air volume systems shall be used for energy conservation by controlling supply, exhaust and return air and maintaining the pressure relationships as per standards and inside design conditions without any compromise. Critical areas in general are operating theatres, ICUs, isolation rooms, sterile areas, post-operative patient care; any place which houses medical surgical procedures and some of the laboratories.

For patient rooms and non-critical areas, variable air volume system shall be considered for energy conservation. However, when variable air volume systems are used, care should be taken to ensure that minimum ventilation requirements are not compromised and pressure balance is maintained even at minimum flow.

Active smoke control systems shall be used along with fire and smoke partitions to limit the spread of smoke in the event of fire. Smoke and fire management should be done in accordance with Part F 'Fire and Life Safety' of this NBCS and **19**. For health care facilities which provide ambulatory care to patients, safety from fire and smoke is a paramount design consideration and should be provided in accordance with Part F 'Fire and Life Safety' of this NBCS.

A separate exhaust system shall be provided for removal of anaesthesia gases from the operation theatre. Alternatively, proper isolation dampers may be provided in the air circulation system for removal of anaesthesia gases or fumigation effect if this system of fumigation is followed in health care.

Air handling devices shall be designed to prevent water intrusion and permit access for inspection and maintenance. Draw through air handling systems shall be used as moisture ingression is minimized in these systems. Lower face velocity across the coil in the range of 1.27 m/s to1.52 m/s shall be considered for better heat transfer.

When humidifiers are provided, they shall be located with-in the air handling unit or duct work in such a way that moisture accumulation on downstream components do not take place. Evaporative pan type humidifiers shall not be used.

Fibrous acoustic insulating material shall not be used as duct lining for critical spaces. Supply air ducts shall be externally insulated as required.

Multiple direct drive plenum fans may be used in air handling units serving critical areas. This provides redundancy and minimizes shut down of critical areas in case of single fan failure.

For critical areas, both supply air and return air shall be ducted. Ducts shall be sized for medium velocity and medium pressure drop 12 m/s and 4 mm for 10 m maximum. For applications like operation theatres and intensive care units, the preferred material for ductwork should be of corrosion resistive metal sheets.

Ultraviolet germicidal irradiation is recommended for air handling units serving health care facilities. Lamps may be installed either upstream or downstream of the evaporator coil but away from the filter media.

12.5.6 *Air Flow and Filtration*

Outside air intakes shall be located minimum 8 m away from exhaust stacks, cooling tower and/or any other polluting source. Bottom of an outdoor air intake shall not be located less than 2 m above ground level and 1 m above any roof terrace level.

Exhaust outlets shall be located at a minimum height of 3 m away from ground level and away from doors, occupied areas and operable windows. Locating exhaust outlets above the roof, projecting upwards, is preferred.

While installing filters, care shall be taken to see that there is no scope for leakage between frame and filters and between filter segments. Filters shall be as per tested to accepted standards [D-3(15)], [D-3(16)], [D-3(17)] and [D-3(18)].

All openings in ducting/diffusers collars shall be sealed to prevent infusion of dust and dirt. All slab openings shall be terminated in enclosed rooms and airflow systems shall be designed and balanced to create positive or negative air pressure with-in specified areas.

It is recommended that supply air outlets shall be located at or near the ceiling and return/exhaust is collected near the floor level. This is to ensure that clean conditioned air moves through breathing and working space to the floor area, for return/exhaust.

For operating room and procedure room, where patients are highly susceptible to infection, such as orthopaedic and cardiac operating theatres, laminar air flow system shall be considered. A unidirectional air flow pattern at a max velocity of 0.13 m/s to 0.18 m/s should be aimed for to prevent the thermal plume at surgical site getting disturbed. A vertical laminar flow system which will wash the patient on the operating table and flow downwards, to be collected near floor level, shall be chosen for such rooms. The area of laminar flow grid shall extend by a minimum of 0.450 m beyond the footprint of the operating table on all sides. If required, additional supply diffusers may be provided for achievement of required temperature and humidity.

Minimum two (2) exhaust grilles shall be provided at opposite corners of the room at approximately 0.200 m above the finished floor level.

12.5.7 *Pressure Differential*

Operating rooms where highly infectious patients are treated and isolation is required, an anteroom shall be provided between the operating room and external area. Anteroom shall be maintained at positive pressure with respect to both the operating room and the surrounding areas.

All air from operating rooms shall be recirculated with adequate fresh air changes and Total air changes per hours and filtration as per tables above or exhausted after energy recovery. Similarly, from all airborne infection isolation rooms, all air shall be exhausted directly to outdoors through high efficiency particulate air (HEPA) filters. Such rooms shall be maintained at a minimum negative pressure of 2.5 N/m^2 with respect to surrounding areas.

Protective environment rooms, such as, bone marrow transplants and organ transplants, shall be maintained at a positive pressure of 2.5 N/m^2 with respect to surrounding spaces.

12.5.8 *Conditioning Equipment*

Cooling equipment shall be central or local chilled water systems, direct expansion type condensing units or variable refrigerant flow systems. Indirect cooling systems using chilled water are preferred and, if direct cooling systems are used, required safety measures should be adopted. Heating equipment shall include heat pumps and heat exchangers.

Sizing and arrangement shall give adequate consideration to minimum loading and standby facility for critical areas. It shall be possible for the facility to operate even when one of the systems is under break down or maintenance.

Where specific areas are to be maintained at a low temperature coupled with low humidity, an additional direct expansion coil may be introduced downstream of the regular evaporator coil. In such cases, the control strategy shall ensure that the DX coil is energized only after the main evaporator coil is at full load.

When variable refrigerant flow systems are used, care shall be taken to see that minimum air change and pressure differential considerations are not compromised.

12.5.9 *Installation and Maintenance*

The air distribution system shall be provided with access panels to allow inspection and cleaning. Duct systems shall be cleaned of construction debris before commissioning.

Surfaces of air terminals shall be suitable for cleaning.

Access to equipment rooms shall be planned so as to avoid intrusion of maintenance personnel into critical care areas. Equipment room layout shall allow easy access to equipment for its operation and maintenance.

Operation and maintenance records shall include indoor temperature and pressure requirements as well as permitted tolerances for all spaces. It shall also include standard operating procedures for emergencies, such as power failure, equipment breakdown and fire situation.

Pressure differentials of operating rooms, protective environment rooms, and air borne infection isolation rooms with respect to surrounding areas, shall be verified and recorded semi-annually.

HEPA filters shall be replaced periodically based on pressure drop. Filters of fan coil units and air handling units (AHU) shall be cleaned according to a regular maintenance schedule. AHU/fan coil drain pans shall be of stainless-steel construction and shall be cleaned monthly.

Supply and return air ducts for critical areas shall be tested for air leakage and an air blow down process shall be undertaken before loading of filters. All areas served by the AHU shall be cleaned and mopped before the AHU is started. Filters shall be loaded in sequence. Pre-filters shall be loaded first followed by microvee or fine filters and finally HEPA filters. AHU shall be run with each set of filters for 24 h and the conditioned area shall be cleaned and mopped each time before the next set of filters is loaded.

After air balancing, and adjusting of air quantity, a validation process shall be carried out by recording temperature, humidity and pressure differentials. For critical areas, validation process shall be repeated and recorded at least once every year.

For operating rooms where validation is needed the air handling devices shall be run 24 h. However, the system deliverables shall be limited to maintaining minimum pressure differentials considerations with turndown procedures to control and limit contamination (unoccupied turn down is permitted subject to maintaining pressure differentials).

12.6 Underground Metro Station Buildings

A typical underground metro station generally consists of three underground levels i.e. concourse level, platform level and undercroft level. The first level at minus 1 being the concourse with a height ranging from 4.8 m to 5.5 m followed by Platform level at minus 2 of height varying from 4.5 m to 5 m. The third level that is undercroft level provided below the platform having minimum clear height of 1.8 meter is mainly used for the services like cables, fire pipes, escalator pits, seepage and sewage sumps etc. The ancillary building at metro stations is also provided outside the station building to house either cooling towers, air cooled chillers, water cooled chillers, etc.

The underground metro station is provided with environment control system (ECS). The ECS system shall serve following function:

a) station public area and ancillary room air conditioning;

b) mechanical ventilation of plant rooms;

c) smoke management system; and

d) staircase Pressurisation

12.6.1 *Station Public Area and Ancillary Room Air Conditioning*

12.6.1.1 *Station air conditioning design philosophy*

The underground stations of the metro corridor shall be designed to provide forced ventilation for removal of the large quantity of heat dissipated from train equipment, station electrical accessories and equipment, passenger body heat, obnoxious odours, and harmful gases such as CO_2. The station heat load shall be assessed by taking into account the provision of platform screen doors (PSDs) at the station. PSDs, which act as barriers between the station platform and the trackway, are provided for enhancing passenger safety, improving indoor air quality, and achieving energy savings. PSDs may be of two types, full Height PSD or half Height PSD, and their installation significantly influences the overall heat load profile of the station. The station heat load generally consists of a combination of unsteady train heat loads, passenger heat load, lighting and miscellaneous equipment loads, fresh air loads, transmission heat loads, and infiltration/exfiltration loads. The station heat load shall be considered as

a) *Unsteady heat load due to train movement* — The unsteady heat load at a station due to train movement is almost negligible when the full height PSD are provided between station trackway and station platform. In case of no PSD or half height PSD the unsteady heat load is the major source of heat for the station is from the train air conditioning systems and the train traction and braking systems. The air that gets pushed in to the tunnel due to the piston effect gets exhausted through separate gravity dampers provided at the top of tunnel. The simulation studies also evaluate the tunnel temperatures during the normal operation of the train movements. The underground stations with provision of full height PSD doesn't allow the cool air of the station to go into the tunnel and the tunnel temperatures tends to be higher. The simulation studies inform the amount of tunnel cooling required for normal and congested operation of the train in the tunnel sections.

b) *Steady heat load*:

1) *Passenger heat load* — The passenger occupancy at station is taken from peak hour peak direction traffic (PHPDT) data given in detail project report (DPR). Passenger heat gain shall be computed from the above data;

2) *Lighting and miscellaneous equipment load* — It is based on lighting in public area and equipment like lifts, escalators and AFC Gates contributing to the heat load;

3) *Fresh air loads* — Fresh air is supplied to the platform and concourse to meet the physiological requirements of the passengers. Fresh air quantity shall be calculated as per **3**;

4) *Transmission heat loads* — The transmission of heat through the station walls between public and non-public areas, through concourse floor slab and heat transmission through PSD glass doors to be considered; and

5) *Infiltration and exfiltration heat load* — Infiltration or ex-filtration of air at platform takes place through PSD doors from the trackway to platform and vice versa. The source of heat on trackway is heat rejected from train condenser, traction, braking, and airflow due to train piston effect. In order to maintain the required conditions in the station, the loss of cooling at station platform due to infiltration/exfiltration of air may be considered in station heat load estimation

12.6.1.2 *Design criteria*

Since the passengers stay in the stations only for a short period, a reasonable degree of comfort conditions, just short of discomfort conditions are considered appropriate. Station air conditioning system shall be designed to maintain specific design conditions.

Station air conditioning system design conditions are:

a) Platform, Concourse : 27 °C at 55 percent relative humidity

b) Outdoor : Summer and monsoon condition as per location

12.6.1.3 *System description*

The station environment control shall be by a central air conditioning system. Station shall have provision of fresh air intake and exhaust shaft. The station shall be supplied with conditioned air to concourse and concourse area via ductwork from air handling unit (AHU) located in plant room either at concourse or platform level. AHU shall be supplied with chilled water from water cooled/air-cooled chillers and its associated pumping system. The AHU shall be as per accepted standard [D-3(21)] and the liquid chillers shall be as per accepted standard [D-3(24)].

The trackway exhaust system shall be installed in the trainways of each station to capture both excessive tunnel airflows and the heat rejected by the vehicle propulsion/braking/air-conditioning systems as the train dwells in the station. An over track exhaust (OTE) duct is normally utilized to capture heat that is rejected from the vehicle propulsion/braking as well as the heat that is rejected from the above car area rooftop mounted air-conditioning units. An under platform exhaust (UPE) duct can also be utilized to capture heat from the train's undercarriage heat sources if the heat generated is significant. Openings in the OTE and UPE shall be located to be near the heat generating sources on the train.

For systems designed without PSD or half height PSD, the extracted air from the station trackway is used as a return air for the air-conditioning system during the close mode of operation. For stations with full height PSDs, the heat extracted through station trackway can be directly exhausted to the atmosphere or can also be used as return air as per the system requirements.

12.6.1.4 *Modes of operation*

There are namely two types of mode of operation:

a) *Open mode operation* — In open mode, 100 percent outside air is circulated in the stations. It is an economical mode of operation when the outside air enthalpy is relatively low and favourable for the inside condition (the best-case scenario is when the outside air enthalpy is same as the design AHU coil leaving air enthalpy). AHU shall draw the air directly from outside via fresh air shaft and deliver the air at platform level and concourse level. Trackway exhaust fans (TEF) shall extract the air from OTE and UPE and discharge it outside directly via exhaust shaft. The centralized chillers units shall remain shut down in this mode. This mode is known as 'Economiser Cycle' and is generally available during winter season;

b) *Closed mode operation* — In close mode, air is extracted from the public areas of the station and returned to the air handling units to be cooled and delivered back to the platform and concourse. Closed mode operation is proposed when the outside air temperature and humidity are high. In this operation, AHU shall recirculate the air of OTE and UPE with the addition of fresh air, brought in by fresh air fans (FAF). This mode is generally used during summer and monsoon season; and

c) During close mode operation in station with full height PSDs, the return air is taken from the station concourse and public areas and then again recirculated by the AHUs provided in the ECS plant rooms.

12.6.2 *Mechanical Ventilation of Plant Rooms*

The back of house area rooms are provided with mechanical ventilation supply, ventilation exhaust or both according to their use and particular requirements.

12.6.2.1 *Ancillary area air conditioning*

Air conditioning of back of the house areas and ancillary areas, that are occupied by staff and non-travelling persons, who are going to stay in these areas for considerable duration, shall be same as air conditioning system for commercial buildings.

12.6.3 *Smoke Management System*

The and trackway exhaust fans (TEF) are used for concourse and platform public areas either TEF can be used or separate smoke extract fans can be provided. In case of full height PSD as TEF is not available for concourse and platform smoke extraction, separate smoke extraction ducts and fans are compulsorily required for these areas. The fans with related equipment shall be suitably located at each end of the station. Where a smoke extract ducting passes through protected areas, it shall be rated to the same fire resistance as the walls of the compartments. Station smoke removal fans to provide a positive indication of the fan operation at the station control rooms (SCR) as well as operations control centres (OCC).

12.6.4 *Staircase Pressurization*

Pressurization of the firemen's staircase and emergency staircase should be done in accordance with Part F 'Fire and Life Safety' of this NBCS.

12.7 Airport Buildings

HVAC systems for airports shall address unique requirements due to the complexity, scale, and diverse usage of these facilities. The key considerations for airport typology as given below shall be followed for energy efficient and sustainable building.

12.7.1 *Passenger Comfort*

These recommendations focus on maintaining optimal air quality, temperature, and humidity levels to enhance the comfort and experience of passengers in airport terminals:

a) *Temperature control* — Consistent and comfortable temperatures to provide thermal comfort in passenger areas, considering high traffic and varying external conditions as detailed in **3** shall be ensured;

b) *Humidity control* — Optimal humidity levels to enhance comfort and protect electronic equipment shall be ensured. (*see* **3**); and

c) Indoor air quality shall be as per the requirements as specified in **3**.

12.7.2 *Energy Efficiency*

The equipment like variable air volume (VAV) Systems, energy recovery ventilation (ERV) systems, and BMS as specified in **15**, should be considered.

12.7.3 *Ventilation and Air Exchange*

The ventilation and air exchange rate shall refer to the **3** and **8**.

12.7.4 *Load Management*

These guidelines focus on optimizing energy consumption and ensuring efficient operation of HVAC systems in response to fluctuating passenger traffic and varying environmental conditions:

a) *Peak load handling* — Systems shall be designed to handle peak loads during busy periods without compromising comfort; and

b) Demand control ventilation should be implemented to adjust ventilation rates based on occupancy to reduce energy use during off-peak hours.

12.7.5 *Redundancy and Reliability*

These considerations ensure the continuous, uninterrupted operation of HVAC systems by incorporating backup systems and fail-safe mechanisms to handle critical airport operations:

a) *Backup systems* — Backup HVAC systems and power supplies shall be incorporated to ensure continuous operation during power outages or system failures as the systems are expected to operate continuously all 365 days of the year; and

b) *Maintenance access* — Easy access to HVAC components for regular maintenance and emergency repairs shall be envisaged.

12.7.6 *Special Considerations for Specific Areas*

For the following specific areas, special considerations should be taken:

a) *Control towers* — HVAC system shall be designed for precise temperature and humidity control along with high air quality. Design should be fail safe system with redundancy;

b) *Baggage handling areas* — Maintain moderate temperatures to protect mechanical systems and baggage handling equipment; and

c) *Retail and Dining Areas* — Adequate ventilation and temperature control to ensure comfort for patrons and staff shall be provided.

12.7.7 *Noise Control*

These recommendations focus on minimizing noise pollution from HVAC systems to ensure a quiet and comfortable environment for passengers and staff in airport terminals:

a) The acoustic design of HVAC system should be considered to lower the noise level;

b) *Sound attenuation* — Sound attenuators and low-noise equipment to control noise levels in critical areas should be used; and

c) Noise control measures shall be as per Part D 'Building Services, Section 4 Acoustics Sound Insulation and Noise control' of this NBCS.

12.7.8 *Advanced HVAC Technologies*

These technologies aim to enhance energy efficiency, air quality, and overall system performance, ensuring a more sustainable and comfortable environment for passengers and staff.

a) *Smart sensors* — Occupancy and air quality sensors to optimize HVAC operations dynamically should be used;

b) *Advanced controls* — Advanced control systems to automate and optimize HVAC performance based on real-time data should be implemented 15; and

c) *High volume low speed fans* (*HVLS fans*) — These fans should be used in conjunction with air conditioning systems to provide better thermal comfort in high ceiling areas there by enhancing passenger comfort and energy efficiency.

12.7.9 *Scalability*

The system should be designed with flexibility in mind to accommodate future expansion needs. This includes providing sufficient capacity, space, and infrastructure to allow for easy integration of additional components, equipment, or systems as the airport's operations grow. It should also consider scalability in terms of power, cooling, and airflow requirements, ensuring that the system can be upgraded or modified without significant disruption.

12.8 Shopping Complex Buildings

HVAC systems for shopping complex buildings like malls shall be designed to handle the unique demands of large, complex spaces with varying loads and occupancy levels.

For a small shop and store, unitary split type air conditioning system shall offer many advantages, including low initial cost, minimum space requirement and ease of installation. For large department store a very careful analysis of the location and requirement of individual department shall be required as these may vary widely, for example, for lighting department, food halls, restaurants, etc. Some system flexibility to accommodate future changes should be attempted in design.

Generally, internal loads from lighting and people are predominate considerations shall be given to initial and operating costs, system space requirements, ease of maintenance and type of operating personnel operating the system.

The economical options of all-air type of system, with variable volume distribution from local air handling unit, shall be considered. Facilities to take all outside air, for 'free cooling' under favourable conditions, should be provided.

a) *Load Calculations* — The following considerations should be taken into account for load calculation:

   1) *Diverse spaces* — Malls include various spaces such as retail shops, food courts, fine dining restaurants, cinemas, play zones and common areas, each with different HVAC needs;

   2) *Peak occupancy* — Systems shall accommodate peak load times, such as weekends and holidays; and

   3) *Heat loads* — Due consideration for heat loads from lighting, equipment, and occupants shall be accounted.

b) *Zoning* —Proper zoning is required for ensuring optimal comfort and energy efficiency in HVAC systems which shall be ensured by adhering to following:

   1) *Multiple zones* — Separate zones for different areas to ensure comfort and efficiency shall be designed. For example, food courts, theatres, and individual stores should have their own control zones served by common or individual air handlers with distinct functioning options; and

   2) *Independent controls* — Each zone should have independent temperature and humidity controls.

c) *Ventilation* — The following considerations should be taken into account for ventilation:

   1) *Fresh air requirements* — Adequate ventilation to ensure indoor air quality, with proper intake of fresh air as detailed in **3** shall be followed; and

   2) *Exhaust systems* — Effective exhaust systems for areas with specific requirements, such as food courts and restrooms, shall be needed for controlling odour, and pollutants.

d) *Air Distribution* — Effective air distribution is crucial for maintaining consistent comfort and air quality across all areas of the space:

   1) *Even distribution* — The system should be designed for uniform air distribution; and

   2) *Diffusers and grilles* — These shall be appropriately selected and placed for efficient air distribution.

e) *Energy Efficiency* — The system should be designed for energy-efficient operation, aiming to minimize energy consumption while maintaining optimal comfort levels. Proper system sizing, effective load management, and routine maintenance practices to ensure the HVAC system operates at peak efficiency, reducing overall energy costs and environmental impact;

f) *Humidity Control* — The following considerations should be considered for humidity control:

   1) *Dehumidification* — Proper dehumidification to maintain comfort and prevent mould growth shall be planned in humid climates or areas located in zones with high humid ambient conditions; and

   2) *Humidification* — In colder climates, additional humidification strategies in air handlers shall be added to maintain comfort.

g) *System Redundancy* — To ensure system redundancy following recommendations to be considered:

   1) The design should consider backup systems for critical components to reduce downtime; and

   2) The system design should be for easy maintenance access.

h) *Building Management Systems* (*BMS*) — The design of BMS should focus on centralizing control and monitoring of various building systems to enhance operational efficiency and energy management by adhering to following:

   1) *Integrated control* — BMS should be planned for centralized control and monitoring and integrating various operating HVAC systems and other systems such as lighting controls, lift management, water pumping systems, DG sets and MCC panels as well as fire alarm system to provide comprehensive and energy efficient operations; and

   2) Real-time monitoring should be incorporated for feedback on the performance and provide opportunity for necessary alterations to achieve energy efficiency goals.

i) *Noise Control* — Systems should be designed to minimize noise as well as sound transfer from one zone to another, especially in common areas and near sensitive zones like screens in multiplex, conference areas etc;

j) *Monitoring and Diagnostics* — Should implement predictive maintenance strategies using data analytics to reduce downtime and maintenance costs;

m) *Commissioning and Re-Commissioning* — Thorough initial commissioning as well as ongoing recommissioning shall be carried out as detailed in **17**;

n) *Climate Resilience* — The systems shall be designed to withstand and operate efficiently during extreme weather conditions as per the weather data provided in this NBCS;

p) *Water Management* — Water management systems should be designed to optimize water usage, enhance efficiency, and support sustainability within the building's HVAC operations and overall facility management which shall be ensured through:

   1) *Condensate management* — This shall be ensured by efficiently managing and reusing condensate water from air conditioning systems; and

   2) *Cooling towers* — Principally all types of cooling towers need make up water as well as constant replenishment and to achieve better system efficiency lower approach, designing of cooling towers may be referred from special publication.

   NOTE — Special publication may be Energy Conservation and Sustainability Building Code 2024 (ECSBC).

q) *Operational Training* — Operational training to ensure that maintenance and operational staff are well-equipped, to manage the HVAC systems efficiently and respond effectively to emergencies and following may be adhered:

   1) *Staff training* — Comprehensive training shall be provided for maintenance and operational staff on system use and troubleshooting; and

   2) *Emergency procedures* — Emergency procedures shall be established with clear procedures for HVAC-related emergencies, such as system failures or fire/smoke instances.

r) *Tenant Coordination* — Effective coordination with tenants to ensure that their HVAC needs are addressed while maintaining overall system efficiency. The following points outline key aspects of this coordination:

   1) *Tenant integration* — Coordinate with tenants shall be done to ensure their HVAC needs are met without compromising overall system efficiency; and

   2) *Tenant fit-out* — Proper guidelines shall be provided for tenant fit-out to maintain compatibility with the central HVAC system installed.

s) *Other requirements* — Other requirements such as safety, system type and components, should be as per Part F 'Fire and Life Safety' of this NBCS. HVAC should be designed flexible to accommodate future expansion:

   1) The design shall ensure thermal comfort as defined in **3**. (Adaptive controls – Adaptive control algorithms that respond to real-time occupancy and environmental data should be incorporated);

   2) Demand control ventilation (DCV) should be considered as reference to the thermal comfort and Indoor air quality requirements defined in this section; and

   3) *Backup power* — Critical HVAC systems shall have backup power to maintain operation during power outages.

12.9 Industrial Facility Buildings

12.9.1 *Zoning*

Zoning of different spaces shall be planned according to the functionality. Based on the hazard classification of different entities, compartmentation measures and safe distances shall be maintained. Emergency exits and assembly points shall be planned in line with the local regulations.

Planning shall consider adequate ventilation and natural light ingress into majority of the working spaces.

12.9.2 *Indoor Conditions*

Indoor conditions of different spaces shall be planned in line with the functional requirements of the spaces in general. Temperature, relative humidity and air quality shall be planned as per guidelines.

Ventilation and air conditioning shall be planned based on the functional requirements of the industrial buildings in line with basis of design brought out in **3**.

In industrial buildings, acceptable working conditions shall be ensured rather than comfort as providing comfortable conditions in huge volumetric spaces will not be efficient in terms of energy usage and cost economics. However, the ancillaries and office spaces shall be planned for comfort conditions in line with the guidelines.

The ventilation systems may contain chemical compounds or toxic substances in concentrations crossing prescribed safe limits. Necessary precautionary arrangements shall be made to remove, contain, exhaust and maintain the hazardous pollutants within acceptable limits in occupied spaces.

Emergency preparedness includes fire safety measures, with designated smoke extraction and pressurization systems integrated for safe evacuation and emergency response capabilities. Additionally, industrial zones shall be equipped with specialized HVAC systems, ensuring operational continuity within sensitive areas such as equipment rooms, control centers, and personnel amenities.

12.9.3 *Energy Efficiency and Conservation*

Energy-saving initiatives such as variable air volume (VAV) systems in public and operational zones, optimizing airflow and energy consumption based on real-time demand, variable frequency drives (VFDs) control AHUs, pumps, air compressors and chillers, optimizing operational efficiency during fluctuating production cycles, enthalpy economizer cycle in 100 percent Fresh air scenario and adaptive thermal comfort approach should be considered.

12.9.4 *Fire Safety and Electrical Integration*

It should be as defined in the Part F 'Fire and Life Safety' of this NBCS.

12.10 Tall Buildings Typology

12.10.1 *General*

Tall buildings are defined as structures with a height greater than 100 m. Buildings that exceed 300 m are classified as super tall buildings. Furthermore, buildings taller than 600 m are categorized as mega tall buildings.

Important considerations for tall/super tall/mega tall buildings are:

a) The effect of ambient air temperature over the height of buildings especially for super tall and mega tall buildings;

b) Facades for tall Buildings, leakage rates and pressure resistance;

c) Natural ventilation for tall buildings;

d) Energy consumption/conservation and new trends in HVAC design for tall/super tall/mega tall buildings;

e) Stack effect; and

f) Green/Sustainable tall buildings.

12.10.2 *Architectural Consideration*

Following key points outline key aspects of architectural consideration:

12.10.2.1 *Core design*

Design of core areas determine the projects floor to floor heights. This reflects in the building cost and architectural as well as HVAC design. An important consideration for architecture shall be the core design comprising of the three commonly used cores – Central Core/Double Core/Single sided (or offset) core. For hot and humid as well as temperate climates the core is suggested to be located on the east and west facades to insulate the interior zones from ambient conditions. Also, the cooling loads are found to be minimal when the double core configuration with glazing is used on north south and south facade and core on east and west facades. The core incorporates the following minimum entities:

Fire escape stairs/vertical transportation elements/rest rooms/electrical closets/communication closets/local fan rooms/shafts for HVAC risers/shafts for HVAC and plumbing systems/electrical distribution and building management and fire alarm system cabling.

12.10.2.2 *Facade systems*

The orientation influences the either increased or decreased infiltration rates and solar loads through facades. This is specifically important for mega tall and super tall buildings as infiltration from wind loads shall be significant for very tall buildings since there may not be adjacent buildings to mitigate high winds.

A high-performance facade needs to be planned to minimize solar cooling load and allow for efficient performance of air-conditioning systems while maintaining thermal comfort.

Facade is to be designed to minimize solar gain to 10 percent to 20 percent of the incident radiation. A double skin facade with fan coil/chilled ceiling/chilled beam cooling systems with return air directed to the floor of the perimeter and up between exterior and interior windows could be used.

Different types of multi skin facades such as one story faced module/corridor facade/multiple story facade and shaft box facade shall be designed

12.10.3 *Shading Strategies*

Shading strategies are important especially for cooling load dominated climates. It is better to minimize window to wall ratio as shall be seen the difference between 40 percent window to wall ratio and 65 percent window to wall ratio as seen in most modern buildings (*see* Fig. 7).

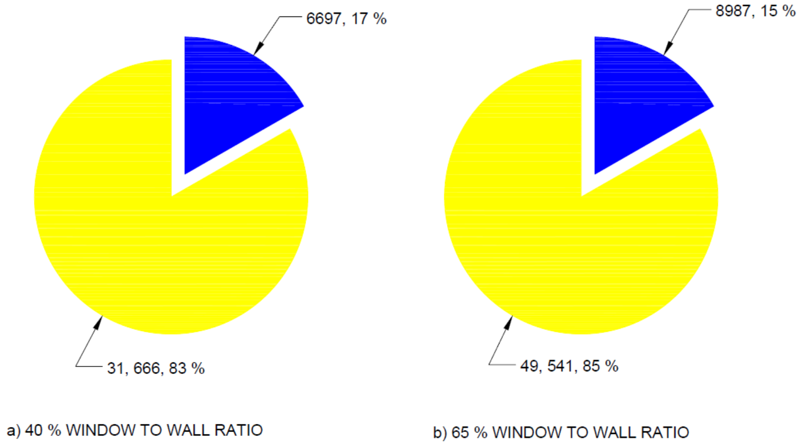


FIG. 7 IN 40 PERCENT THE HEAT TRANSFER THROUGH THE GLAZING IS 83 PERCENT OF THE LOAD TO THE SPACE AND IN 65 PERCENT IT IS 85 PERCENT TO THE SPACE.

12.10.4 *Facade Leakage Rates*

Envelope leakage depends on HVAC system operation. In pressurized buildings when HVAC systems operate the infiltration leakage could be very low. In accordance with the best practices of special publications recommended air leakage test of the completed building shall be with a leakage rate of not exceeding 2.0 $l/s.m^2$ at 75 Pa pressure differential.

As per special publication the recommended leakage rate value is 9.0 l/sm^2 at 75 Pa pressure differential.

The following are the two options as recommendations either as per special publications:

a) 2.0 $l/s.m^2$ at 75Pa; and

b) 9.0 $l/s.m^2$ at 75 Pa.

NOTE — Special publications maybe International Energy Conservation Code (IECC) and ASHRAE 90.1.

The leakage rate for tall building however shall vary with height. If the maximum leakage rate is achieved at a particular height, then at any height above this there will be facade leakage as infiltration load (*see* Fig. 8).

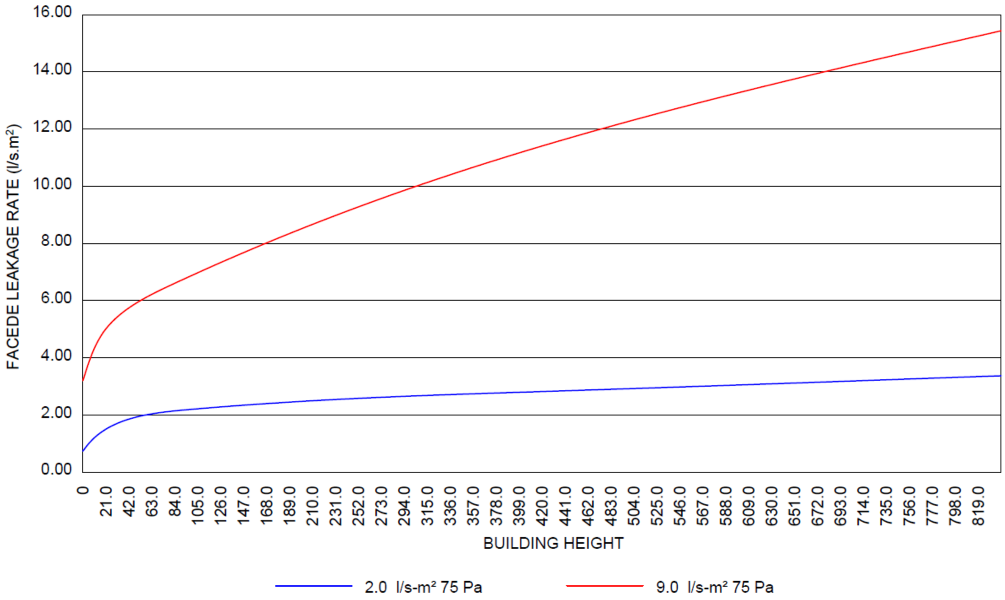


FIG. 8 CALCULATED LEAKAGE RATES FROM WIND SPEED IN CFM(L/S) OF FACADE AREA FOR TALL/MEGA TALL BUILDING

12.10.5 *Climate Data*

Generation of climate data at upper elevations is a shall. It could be achieved through:

a) Remote sensing technologies or ballon releases; and

b) Model using weather forecasting techniques.

Following are some of the calculation methods:

1) To obtain external temperature at different building heights use:

$$T = T_g - 0.006\ 49\ h$$

where

T = temperature at height h (m);

T_g = Temperature at ground level F (°C); and

h = height from ground level (m).

2) External Pressure vertically with the building height is

$$P = 101\ 325(1 - 2.255\ 77 \times 10^{-5} \times h)^{5.255\ 9}$$

where, P = pressure (kPa)

3) Density of air vertically with the height of the building shall be

$$\rho = \frac{P_{abs}}{0.286\ 9\ (T\ + 273.1)}$$

where

T = absolute temperature (K);

P_{abs} = absolute pressure (Pa); and

ρ = air density (kg/m^3).

4) Wind speed and pressure are generally obtained from meteorological stations located away from urban environments. This wind speed shall be corrected for terrain conditions and for building height relative to wind measurement at that height. Approximate correction factor is:

$$v_z = v_m k Z^a$$

where

v_z = wind speed at the building height (m/s);

v_m = wind speed measures in open country at a height of 10 m (m/s);

Z = building height (m); and

k and a = constants dependent on terrain as per table below:

Wind speed terrain constants:

Sl No.	*Terrain Category*	*k*	*a*
(1)	(2)	(3)	(4)
i)	Open, flat country	0.68	0.17
ii)	Country with scattered wind breaks	0.52	0.20
iii)	Urban	0.35	0.25
iv)	City	0.21	0.33

12.10.6 *Stack Effect*

By convention, the stack pressure differences are positive when building is pressurized relative to outdoors which causes flow out of building.

When indoors is warmer than outdoors the base building is depressurized, and top is pressurized relative to outdoors. When the indoors is cooler than outdoors then the reverse becomes true. At some elevation in building the pressure indoors is equal to the outdoors and this height is known a neutral pressure level (NPL). NPL is influenced by leakage distribution and not necessarily at mid height of building.

The location of NPL (h) is expressed by:

$$h = H/(1 + (A_1/A_2)^2 T_i/T_o)$$

where

A_1 and A_2 = lower and higher leakage areas;

T_o and T_i = outdoor and indoor absolute temperatures; and

H = Building height.

12.10.7 *Heating and Cooling Loads*

Cooling and heating loads shall be calculated by dividing the building into sections. For example, splitting the building as 100 m sections and calculate heating and cooling loads in view of variable outside temperature, density and pressure over the building height as compared to constant parameters for other building typologies.

12.10.8 *Indoor Air Quality and Thermal Comfort*

The procedures and standards mentioned in 3 shall be followed. If this ventilation procedure does not yield adequate results to contain a specific contaminant or if the contaminant levels shall be attained with lower ventilation rates, then as an alternative indoor air quality procedure (IAQP) may be used.

12.10.9 *HVAC Systems*

All HVAC system options available for any building typology shall be applicable for use in Tall Buildings. These include:

a) All air variable air volume (VAV) systems;
b) Low temperature air VAV systems;
c) Underfloor air systems;
d) Air-water systems;
e) Radiant ceilings;
f) VRF drive fan coil units; and
g) VRF fan coils with 100 percent outdoor air ventilation.

12.10.10 *Exhaust Fan Pressures*

There will be considerable wind pressures at the air intake and exhaust locations shall be calculated as follows:

P_s intake = C_p intake $[\frac{\rho_a U_H^2}{2g_c} / 2.152]$

P_s exhaust = C_p exhaust $[\frac{\rho_a U_H^2}{2g_c} / 2.152]$

(P_s intake – P_s exhaust) + $\Delta P_{fan} = F_{Sys}\frac{\rho Q^2}{A_L^2 g_C}$

$\Delta_{P_{wind}}$ = (C_p intake - C_p exhaust) $[\frac{\rho_a U_H^2}{2g_c} / 2.152]$

where

ρ_a = outdoor air density at height h;

U_H = approach wind speed at upwind height h (m/s);

F_{Sys} = system flow resistance;

A_L = flow leakage area and; and

Q = system volume flow rate.

12.10.11 *Water Distribution Systems*

A major consideration in the design of a piping system in a tall building is the hydrostatic pressure created by the height of the building. The design of piping shall take static pressure on the piping system due to the height of into consideration. The piping network is divided into zones vertically separated by pressure breaks and served by heat exchangers and pumps with main chiller located at the utility complex in basement or an adjacent service building. The building pressure break shall be at equal interval at different heights separated using heat exchangers and tertiary pumps. It is needed that consideration to be given for different vertical zones in chilled water temperatures at the inlet to heat exchangers in each zone. In a typical scenario of 4 zones to serve chilled water at 6 °C at top of building following temperatures at each Heat exchanger will have to be maintained:

I Zone	1.5 °C
II Zone	3.0 °C
III Zone	4.5 °C
IV Zone	6.0 °C

Suggested piping materials shall be steel pipes schedule 40 with standard wall thickness for pipes up to 250 mm in diameter and for pipes 300 mm or larger a wall thickness of 9.5 mm. This by and large would accommodate the working pressures in tall buildings. However, a cross verification with accepted standard [D-3(27)] is preferred. Based on the working pressure valves and fittings also need to be selected. For steam condensate piping or for condenser water piping where corrosion is a possible concern pipe with heavier wall thickness shall be considered.

12.10.12 *Expansion and Contraction in Piping*

Design of piping shall take other factors including expansion and contraction and static and dynamic loads of the piping should be taken into consideration.

13 SPECIAL PLACES IN BUILDINGS

13.1 Cleanroom

The basic considerations in a cleanroom design are:

a) Cleanroom class;

b) Temperature;

c) Relative humidity;

d) Sound level;

e) Vibration;

f) Light spectrum; and

g) Electrostatic charge/static electricity.

NOTE — Number d, e, f and g are important for only selective cleanroom applications such as semiconductor, electronics and others.

13.1.1 *General Requirement*

a) Areas surrounding the cleanroom shall be less pressured than the cleanroom to avoid ingress of non-clean air to classified areas;

b) Airlock rooms shall be provided before clean room to maintain cleanroom integrity and reduce entry of contamination in classified areas:

   1) Separate change rooms shall be provided for male and female workers. multiple change rooms should be designed in series as per criticalness of application with primary change rooms designed as controlled not classified (CNC) and secondary change rooms (opening in cleanroom areas) designed under classified category; and

   2) A suitable door interlocking shall be used to ensure that both doors of airlock do not open “simultaneously” during usual operation time (except during emergency situations).

c) Air shower shall be used to reduce particles entry during human movement;

d) Dedusting tunnels or pass box should be used to reduce particles entry during material movement;

e) Existing Windows, cleanroom wall/ceiling panel joints shall be closed permanently using suitable sealants to avoid cross contamination;

f) Good smooth and scratch resistant flooring with skirting shall be provided;

g) All Corners and edges (ceiling to wall, wall to wall and wall to floor) shall be provided with coving to avoid dust accumulation and for easy cleaning;

h) The room shall be considered a restricted area. Only authorized personnel shall be allowed to enter during operation;

j) No paper products/cotton buds shall be allowed inside the cleanroom;

k) All personnel entering the clean room facility shall use 'cleanroom compatible gowning and shoes';

m) All personnel/visitors shall wear cleanroom garments while entering cleanroom; and

n) Cleanroom garments shall be stored properly in a laminar air flow storage cabinet:

   1) The cleanroom cleaning protocol shall be designed as per the product nature and contaminations to be removed. The cleaning agents shall also be compatible with cleanroom architecture; and

   2) Movement of materials from and to the cleanroom shall be as optimum as possible.

13.1.2 Selection of airborne particulate cleanliness classes for cleanrooms and clean zones as per special publication is given in Table 29.

Table 29 Airborne Particulate Cleanliness Classes for Cleanrooms and Clean Zones

(*Clause* 13.1.2)

Sl No.	ISO Classification Number	Maximum Concentration Limits (Particles/m^3 of Air) for Particles Equal to and Larger than the Considered Sizes Shown below:					
		≥ 0.1 µm	≥ 0.2 µm	≥ 0.3 µm	≥ 0.5 µm	≥ 1 µm	≥ 5 µm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	ISO Class 1	10	2	–	–	–	–
ii)	ISO Class 2	100	24	10	4	–	–
iii)	ISO Class 3	1 000	237	102	35	8	–
iv)	ISO Class 4	10 000	2 370	1 020	352	83	–
v)	ISO Class 5	100 000	23 700	10 200	3 520	832	29
vi)	ISO Class 6	1 000 000	237 000	102 000	35 200	8 320	293
vii)	ISO Class 7	–	–	–	352 000	83 200	2 930
viii)	ISO Class 8	–	–	–	3 520 000	832 000	29 300
ix)	ISO Class 9	–	–	–	35 200 000	8 320 000	293 000

NOTE — Special publication may be ISO 14644-1.

13.1.3 The meanings of the cleanrooms classification is given in Table 30.

Table 30 Meanings of the Cleanrooms Classification

(*Clause* 13.1.3)

Sl No.	Class	Description
(1)	(2)	(3)
i)	Class 1	This Cleanrooms class is mainly used within the microelectronic industry when manufacturing integrated circuits which requires submicron resolution.
ii)	Class 10	This Cleanroom class is mainly used within the semiconductor industry using band widths below 2 micrometer.
iii)	Class 100	This cleanroom class is, according to many, the most useful critical cleanliness class. Cleanrooms class 100 are often, incorrectly, known as sterile rooms and are used when > bacterial free < and /or > particle free < environments are required. Cleanroom class 100 is used in aseptic manufacturing within the pharmaceutical industry, for example. This cleanroom class is frequently used during the manufacture of integrated circuits; and also during isolation and treatment of patients who are especially sensitive to bacterial infections, for example, after bone marrow transplantation.
iv)	Class 1 000	This cleanroom class is mainly used when producing high quality optics; When carrying out mounting work and testing of gyroscopes for aircrafts; and also, when mounting high quality miniature bearings.
v)	Class 10 000	Clean rooms of class 10 000 are used for mounting procedures in hydraulic or pneumatic equipment and on some occasions are also used within the food and beverage industry. Class 10 000 cleanrooms are also commonly used within the pharmaceutical industry.
vi)	Class 100 000	This classroom class is used by many industries, for example when working with optical products, when building large electronic systems based on smaller components, when building hydraulic and pneumatic systems and also within the food and beverage industry. The pharmaceutical industry also makes frequent use of this class of cleanrooms.

13.1.4 Air velocities and air changes in cleanrooms is given in Table 31.

Table 31 Air Velocities and Air Changes in Cleanrooms

(*Clause* 13.1.4)

Sl No.	Class of Cleanroom	Airflow Type	Average Velocity (m/h)	Air Changes per h
(1)	(2)	(3)	(4)	(5)
i)	ISO 8 (100 000)	N/M	18 to 146	5 to 48
ii)	ISO 7 (10 000)	N/M	183 to 274	60 to 90
iii)	ISO 6 (1 000)	N/M	457 to 732	150 to 240
iv)	ISO 5 (100)	U/N/M	732 to 1 463	240 to 480
v)	ISO 4 (10)	U	914 to 1 646	300 to 540
vi)	ISO 3 (1)	U	1 097 to 1 646	360 to 540
vii)	Better Than ISO 3	U	1 097 to 1 829	360 to 600

NOTES

1 Air Changes per hour = Average airflow velocity × room area × 60 min/h

2 Room Volume

3 N=non unidirectional; M= mixed flow room; U= Unidirectional flow.

13.1.5 The classification of filters given in Table 32 should be applicable.

Table 32 Classification of Filters

(*Clause* 13.1.5)

Sl No.	Filter Class	Overall Value Efficiency (percent)	Overall Value Penetration (percent)	Leak Test Efficiency (percent)	Leak Test Penetration (percent)
(1)	(2)	(3)	(4)	(5)	(6)
i)	H 10	85	15	–	–
ii)	H 11	95	5	–	–
iii)	H 12	99.5	0.5	–	–
iv)	H 13	99.95	0.05	99.75	0.25
v)	H 14	99.995	0.005	99.975	0.025
vi)	U 15	99.999 5	0.000 5	99.997 5	0.002 5
vii)	U 16	99.999 95	0.000 05	99.999 75	0.000 25
viii)	U 17	99.999 995	0.000 005	99.999 9	0.000 1

13.1.6 Schedule of tests to demonstrate continuing compliance is given in Table 33.

Table 33 Schedule of Tests to Demonstrate Continuing Compliance

(*Clause* 13.1.6)

Sl No.	Test Parameter	Class	Maximum Time Interval
(1)	(2)	(3)	(4)
i)	To demonstrate compliance by particle counting	≤ ISO 5	6 months
		> ISO 5	12 months
ii)	Schedule of additional	All Classes	12 months
iii)	Air flow velocity or volume		
iv)	Air pressure difference	All Classes	12 months
v)	Schedule of optional tests	All Classes	24* months
vi)	Installed filter leakage		
vii)	Airflow visualisation	All Classes	24* months
viii)	Recovery	All Classes	24* months
ix)	Containment leakage	All Classes	24* months

* = Suggested time interval

13.2 Refrigeration for Cold Stores

13.2.1 *Key Elements and Components of Cold Room*

a) *Product quality* — Design and operation of cold storages is governed by the level of quality control parameters as per the individual products.

b) *Temperature of cold room* — During the entire storage period, the temperature should remain as constant as possible:

   1) In case of some produces the rate of cooling for the freshly cooled products shall be considered, as per recommended cool down period. (Respiration loads, mainly from fruits and vegetables).

   2) Shall avoid putting into a room a large amount of goods at a temperature too different from that of goods already in the room. The cold-production equipment should be checked to make sure it shall handle this extra load. Within one room the temperature should be as uniform as possible, as this affects not just products that are sensitive to temperature variations, but also those with recommended temperatures near 0 °C. These optimum conditions shall be approachable without risk of freezing.

   3) Dead zones and regular drafts shall be avoided. Temperature sensors should be placed around the room on the basis of the accuracy, as indicated by the manufacturer, of the automatic devices. Cold store temperature shall be either recorded or read at regular intervals. Thermometers should be protected from impact and yet be easily accessible. Temperature should be read at more than one location. Moreover, the temperature of the products themselves should be taken.

c) *Measurement of temperature of food* — A thermometer/temperature measurement device that shall measure the internal temperature of food, like one with a probe that shall be inserted into the food is required. Food internal temperature shall be measured as the surface temperature may be warmer or cooler than the temperature of the rest of the food. The thermometer shall be accurate to ±0.5°C. The thermometer/temperature measurement device should be cleaned and sanitized before inserting it into food. The probe should be washed in warm water and detergent, sanitized according to the sanitizer instructions or the instructions that thermometer accompanies, and the probe should be allowed to air dry or thoroughly dried with a disposable towel.

For further details, refer [D-3(50)].

13.2.2 *Typical Design Inputs for a Standard Cold Room*

The criteria involved in designing a cold room are similar to any warehouse, that is, storage capacity, facilities for receiving and dispatching goods, interior operating space, location of cold room, selection and location of refrigeration units. While designing cold room, the difference between room temperature and product temperature should be kept at a minimum value.

13.2.3 *Heat Loads*

Refrigeration load includes:

a) Transmission load, which is heat transferred into the refrigerated space through its surfaces;

b) Product load, which is heat removed from and produced by products when these are brought into and kept in the refrigerated space;

c) Internal load, which is heat produced by internal sources (for example, lights, electric motors, defrost mechanism and people working in the space);

d) Infiltration air load, which is heat gain associated with air entering the refrigerated space; and

e) Equipment related load. For example, Air-cooling unit fan motors, fork lifts, etc.

The first four segments of load constitute the net heat load for which a refrigeration system is to be provided; the fifth segment consists of all heat gains created by the refrigerating equipment itself. Thus, net heat load plus equipment heat load is the total refrigeration load.

While other loads are quite similar to loads in regular HVAC system design, the product load includes the primary refrigeration loads from products brought into and kept in the refrigerated space, which includes:

a) Heat that shall be removed to bring products to storage temperature. In case of some produces the rate of cooling for the freshly cooled products shall be considered, as per recommended cool down period.

b) Heat generated by produces in storage such as respiration loads, mainly from fruits and vegetables.

In addition, another new aspect in heat load is the heat from defrosting where the refrigeration coil operates at a temperature below freezing and shall be defrosted periodically, regardless of the room temperature. In order to avoid system oversizing, and to take full advantage of load diversity, hour-by-hour load calculation or simulation method should be used.

13.2.4 *Considerations for Cold Room Selection*

The conditions within a closed refrigerated chamber shall be maintained to preserve the stored product. This refers particularly to seasonal, short shelf life, and long-term storage.

The four major concerns for product compatibility are temperature, moisture, ethylene, CO_2 and odour. Specific considerations include:

a) Uniform temperatures as much as possible in empty condition as well as with product loaded condition;

b) Length of air blow and impingement on stored product;

c) Effect of relative humidity;

d) Effect of air movement on employees working in cold store;

e) Controlled ventilation, if necessary;

f) Product entering temperature;

g) Expected duration of storage;

h) Required product shipping temperature when leaving cold store;

j) Traffic in and out of storage area;

k) Design of storage bins; and

m) Size of food product (L × W × H) or volume.

Rack systems are mostly related to retail store refrigeration and are not considered as part of building system.

13.2.5 *Installation and Maintenance of Cold Room*

All panels for thermal insulation of cold storage shall be as per the accepted standard [D-3(28)].

Refrigeration system has to be selected for a particular application based on a particular refrigerant keeping in mind environmental issues. Thus, the refrigerant shall be a natural refrigerant like ammonia, carbon dioxide, hydrocarbons, or synthetic refrigerants, with very low GWP. System shall have either a) unitary system or b) DX systems or c) integrated single or d) two-stage system depending on the application temperature. The entire design has to be done as per proper standards.

Refrigeration system shall be assembled as per proper specifications to ensure that no refrigerant leakage. It should have proper emergency measures in case of any accidental leaks. The building structure shall be designed with adequate safety factors and the thermal insulation has to be protected properly from any possible occurrence of fire. Emergency alarms should be provided in the cold store with switches in each chamber.

Safety is critical in the design, construction and operation of refrigeration systems for cold storage, especially with ammonia systems. Refrigeration system's safety standards shall meet requirements relating to safe design, construction, installation, and operation of refrigeration systems by establishing safeguards for life, limb, heath and property as per the applicable safety standards. This includes, but is not limited to, occupancy classification, restriction on refrigerant use, installation restrictions, design and construction of equipment and systems, and operation and testing. Equipment selection and their placement should be such so as to ensure safe and easy maintenance. A safety review of the engineered design and equipment layout is recommended with participation from the owner's site operations and maintenance (O&M) team. Personnel safety measures are required as part of the facility design. The designer should prepare a system safety plan with full hazard analysis.

13.2.6 *Cold Room Safety*

The cold temperatures inside a low temperature cold room shall cause increase in blood viscosity and risk to life unless proper precautions are taken. An audio alarm should be installed outside the cold room, powered through a UPS and the switch should be fixed prominently and within reach, inside the cold room. This will help any person trapped inside to alert those outside, for help.

The following safety provisions should be kept/taken care of, at the cold room site:

a) Firefighting equipment;

b) Safe handling of refrigerant leaks;

c) Safety devices, controls and alarm systems;

d) Emergency lighting in the cold chambers;

e) Lightning arrestor;

f) First aid kit;

g) Air filters/breathers;

h) Emergency assembly points;

j) Regular safety drills and training;

k) Water shower in ammonia plants; and

m) Avoiding ramps as they become slippery when wet or frozen.

13.2.6.1 *Precautions against getting trapped in freezing room*

In order to facilitate getting out of a person who may accidentally get locked in a cold room, emergency exits shall be provided. These shall have facility to be opened from inside, even when they are locked from the outside. In all cases, following shall be installed in the freezing room:

a) Doors which shall open manually both from the inside and the outside (for functioning during power shortages, etc). In case of rooms with large dimensions say more than 25 m, it is advisable to provide emergency doors with UPS/battery lit exit sign at both ends of the freezing room for easy exit;

b) A signage system outside the freezing room with a steady or blinking light, and a siren or buzzer, activated from the inside by pressing a UPS-lit button. This button should also be connected to the caretaker's apartment, where a monitoring board shows where the button has been pressed. These visible and audible alarms should always be connected to a self-contained electric circuit, operated with a permanently charged battery. The same circuit should also connect to an independent, battery-operated emergency lighting system or a phosphorescent signalling system, all the way to the nearest emergency exit from the area; and

c) Electric or pneumatic sliding or manual doors, capable of being operated both from outside and inside by push button or pull cords, easily accessible, in case of large freezing rooms, to the persons inside/outside.

13.2.6.2 *Precautions against refrigerant leaks*

Safety measures shall be taken against leak of liquid or gaseous refrigerant. Consideration should also be given to installing gas detectors and shut-off valves. The safety of the people should be considered by installing in traffic and public areas with which the refrigerant supply shall be cut long before the refrigerant concentration reaches the level which would affect the people or the stored products. The gaskets around the cold storage door should be periodically inspected and corrected for any damage.

In refrigeration applications with ammonia (if used) as per standard practice, there is a risk of a high concentration accumulating in an enclosed room. With 15 percent to 28 percent by volume, and an ignition source (of 582 °C), it is possible to create a fast-burning process which could be something like an explosion. So, a rapid dilution/containment system or an expulsion system that is emergency ventilation should be installed as per relevant standards.

Self-contained breathing apparatuses and protective clothing for entering a building with a concentration of ammonia greater than 50 ppm. Ammonia clouds shall be neutralized with dry ice (CO_2) or transformed by fog nozzle, but there is then a risk of water pollution. For safety requirements for cold stores as also for refrigerant gas related safety, relevant standards shall be followed.

13.2.7 *Cold Room in Various Segments and Requirements*

The recommended temperature ranges for various applications/product categories to be stored in modular cold rooms are given below (temperature range is to be finally decided based on the product intended to be stored):

Sl No.	*Application/Product Category*	*Recommended Temperature Range*
(1)	(2)	(3)
i)	Horticulture (fruits and vegetables)	+ 2 ºC to + 8 ºC
ii)	Floriculture	+ 2 ºC to + 6 ºC
iii)	Hotels, restaurants, fast food chains	+ 2 ºC to + 6 ºC
iv)	Pharmaceuticals	+ 8 ºC to – 25 ºC
v)	Dairy	+ 4 ºC to – 25 ºC
vi)	Ice cream	– 20 ºC to – 25 ºC
vii)	Ripening	+ 12 ºC to + 20 ºC
viii)	Controlled Atmosphere	+ 1 ºC to + 3 ºC

The above temperature ranges are indicative only, and may tend to vary, based on specific application requirements and other conditions. The relative humidity in most of the cases are above 85 percent to 90 percent range except pharmaceuticals, seeds, dry products and frozen products, or as required for a specific case by the user.

13.2.8 *Refrigeration Technologies for Cold Storage*

The prime considerations for selection should be life cycle cost, ease of maintenance and controls. Designers may select unitary or centralized systems that comprise unitary off the shelf products, DX units, centralized refrigeration system - air cooled or water-cooled.

13.2.9 *Ammonia Refrigeration System*

Ammonia is one of the most efficient applications, with the application range from high to low temperatures. With the ever-increasing focus on energy consumption, ammonia systems are a safe and sustainable choice for the future.

Ammonia is the most environmentally friendly refrigerant. It belongs to the group of so called "natural" refrigerants, and it has both global warming potential (GWP) and ozone depletion potential (ODP) equal to zero.

Ammonia is a toxic refrigerant, and it is also flammable at certain concentrations. That is why it has to be handled with care, and all ammonia systems have to be designed with safety in mind. At the same time, unlike most other refrigerants, it has a characteristic odour that shall be detected by humans even at very low concentrations. That gives a warning sign even in case of minor ammonia leakages. In case it is necessary to reduce ammonia charge, combination of ammonia and CO_2 (as cascade or as brine) could be a good and efficient option. For ammonia refrigeration system, good practice [D-3(29)] to be referred.

13.2.10 *Freon Refrigeration System*

The refrigeration system shall be unitary and centralised systems. Condenser cooling may be either water-cooled and air-cooled.

13.3 Supermarket Refrigeration

Supermarket refrigeration systems include various types of equipment and condensing units designed for both remote and direct applications. The further clauses cover the different systems and components used in supermarket refrigeration.

13.3.1 *Type of Supermarket Refrigeration Equipment*

The following types of refrigeration equipment are commonly used in supermarkets, offering various configurations such as remote and plug-in systems:

a) Open multideck chillers, with remote refrigeration system;

b) Glass door upright freezers and chillers with remote refrigeration system;

c) Display cold room (Wall replaced with glass doors);

d) Plug and play solutions;

e) Serve-over counters, plug-in type;

f) Self-service counters, plug-in type;

g) Stainless steel fish counters, plug-in type;

h) Plug-in island freezers;

j) Visi-coolers – plug-in type;

k) Visi-freezers – plug-in type;

m) Confectionery displays or cake showcase;

n) Four-sided visi-tower;

p) Four-sided visi-frozen tower/visi freezer; and

q) Glass top freezers with sliding glass (plug-in type).

13.3.2 *Remote Condensing Units for Remote Application*

Various types of condensing units are listed and explained below:

a) *Rack system for supermarkets* — The multi-compressor rack system comes with various types of reciprocating, scroll, or screw compressors. It should be compact, solidly built, and enhance performance with minimal pipe work. Also, the addition of an inverter in the rack enhances its efficiency and increases the life of the system;

b) *Individual condensing units* — Glass-door cold rooms, supermarket open or closed-door chillers and freezers shall be connected with a variety of condensing units. Their selection is based on cold room size, temperature and usage. The selection of these units is critical and should be verified ahead of installation.

Various types of Compressors are used in the condensing units as listed below:

1) *Air cooled systems* — These are generally for small to industrial applications, excessive pipe run or vertical lift distances and speciality blast chilling and freezing applications. They are highly efficient and perform tirelessly; and

2) *Water cooled systems* — These are generally for medium to industrial applications, these types of units are very effective in malls, where units are required to be installed in the basement or in a hidden area, were fresh air and exhaust facilities in not available. These are also suitable for chilling and freezing applications. They are highly efficient but need treated water and regular maintenance.

See accepted standard [D-3(7)] for more details.

13.4 Data Centre

13.4.1 *Location*

a) *Seismic considerations*:

1) *Seismic zone mapping* — Consult the latest seismic zone map of India {*see* accepted standard [D- 3(30)]} to accurately assess the risk level of the proposed location; and

2) *Structural design* — Engage qualified structural engineers experienced in seismic design for data centres. Adhere to the latest codes and standards, incorporating base isolation or energy dissipation systems if necessary.

b) *Proximity to utilities*:

1) *Power grid stability* — Evaluate the historical reliability and voltage stability of the local power grid. Prioritize locations with minimal outages and voltage fluctuations;

2) *Redundant power sources* — Having access to two independent power feeds from different substations for enhanced reliability;

3) *Water availability* — Assess water availability throughout the year, accounting for seasonal variations. Consider water recycling and rainwater harvesting to reduce dependency on municipal supply; and

4) *Distance to utilities* — Minimize the distance between the data centre and power/water sources to reduce transmission losses and infrastructure costs.

c) *Connectivity*:

1) *Carrier diversity* — Ensure access to at least two diverse fibre paths from different carriers to avoid single points of failure;

2) *Latency* — Consider the proximity to major internet exchange points (IXPs) and content delivery networks (CDNs) for low latency connections; and

3) *Future expansion* — Plan for future connectivity needs by choosing locations with ample capacity for additional fiber optic cables.

d) *Environmental factors*:

Flood risk — Refer to flood zone maps and avoid areas prone to flooding. If unavoidable, implement appropriate flood mitigation measures (elevated platforms, watertight barriers):

1) *Cyclone and wind risk* — Consider the historical cyclone tracks and wind speeds in the region. Design the building to withstand wind loads as per local codes;

2) *Air quality* — Analyse the ambient air quality data to understand the levels of particulate matter (PM), dust, and other pollutants. Employ high-efficiency air filtration systems to protect sensitive equipment if required;

3) *Electromagnetic interference (EMI)* — Evaluate potential sources of EMI (power lines, radio towers, etc.) and implement shielding if necessary; and

4) *Environmental clearances* — Understand and comply with all environmental impact assessment (EIA) and pollution control norms relevant to data center construction and operation.

e) *Temperature and humidity*:

1) *Climate zone analysis* — Refer to India's climatic zone map (as per Part D 'Building Services, Section 1 Lightning and Ventilation' of this NBCS) to identify regions with favourable temperature and humidity profiles for data centre operations;

2) *Microclimate considerations* — Within a climatic zone, local variations shall exist. Investigate microclimates around the proposed site, accounting for factors like proximity to water bodies, elevation, and prevailing wind patterns;

3) *Historical weather data* — Analyze at least 10 years of historical temperature and humidity data for the location. Pay close attention to peak summer temperatures and humidity levels, which will drive cooling requirements;

4) *Wet-bulb temperature* — Evaluate wet-bulb temperature data as a critical indicator of evaporative cooling potential. Higher wet-bulb temperatures limit the effectiveness of evaporative cooling technologies;

5) *Climate change projections* — Consult climate change models and projections for the region to anticipate future shifts in temperature and humidity patterns. Factor these projections into long-term planning and design; and

6) *Adaptive cooling strategies* — Explore the potential for free cooling or hybrid cooling systems that shall leverage favourable ambient conditions to reduce energy consumption.

f) *Accessibility and logistics*:

1) *Transportation infrastructure* — Evaluate the quality and capacity of road networks, proximity to airports/railway stations, and availability of reliable freight transport for equipment delivery and maintenance;

2) *Manpower access* — Consider the ease of commute for staff and the availability of skilled IT professionals in the region. Plan to be in vicinity of a public transport infrastructure or provide common shuttle service; and

3) *Expansion space* — Ensure sufficient space for future expansion of the facility or campus, including considerations for power and network infrastructure upgrades.

g) *Security and risk mitigation*:

1) *Physical security* — Evaluate the overall security environment of the location, including crime rates and potential threats. Implement robust physical security measures for the data centre premises;

2) *Cybersecurity* — Consider the proximity to potential cybersecurity risks and implement stringent network security protocols to protect data; and

3) *Disaster recovery* — Develop a comprehensive disaster recovery plan that considers natural disasters, power outages, and other disruptions.

13.4.2 *Redundancy for Cooling Infrastructure*

a) *Chillers*:

1) Implement N+1 redundancy for chillers, ensuring that the remaining chillers shall handle the full cooling load even if one unit fails. Consider 2N redundancy for critical facilities or those in hot climates;

2) Size chillers based on peak cooling load with a safety margin to accommodate future expansion or equipment upgrades; and

3) Implement preventive maintenance schedules and conduct regular inspections to ensure optimal performance and reliability.

b) *Precision air handling units* (*PAHUs*):

1) Deploy N+1 redundancy for PAHUs, ensuring sufficient airflow even if one unit is offline for maintenance or repair;

2) Consider zoning the data center and assigning dedicated PAHUs to each zone for better control and redundancy; and

3) Equip PAHUs with variable speed drives (VSDs) to optimize airflow based on real-time cooling demands, improving energy efficiency.

c) *Power distribution for cooling*:

1) Provide redundant power feeds (A+B) to critical cooling equipment, including chillers, PAHUs, and pumps;

2) Utilize dedicated PDUs for cooling infrastructure with branch circuit monitoring to track power consumption and identify potential issues; and

3) Implement emergency power off (EPO) functionality to safely shut down cooling systems in emergencies.

d) *Additional considerations for cooling redundancy*:

1) If the climate permits, explore the use of free cooling (air-side or water-side economizers) to reduce reliance on mechanical cooling and improve energy efficiency;

2) Design chilled water and condenser water piping systems with redundancy to prevent single points of failure and ensure continuous circulation;

3) Implement N+1 redundancy for cooling towers or consider alternative cooling technologies like dry coolers or adiabatic coolers for water-scarce regions; and

4) Deploy redundant chilled water and condenser water pumps with automatic switchover capabilities in case of a pump failure.

13.4.3 *Water Leak Detection* (*WLD*)

a) *Comprehensive coverage* — Install a water leak detection system with adequate coverage in all areas prone to leaks, such as under raised floors, near cooling systems, piping, and water entry points. Tailor the coverage to the specific layout and potential risks of each data centre;

b) *Sensor types* — Utilize a combination of sensor technologies to ensure comprehensive detection:

   1) *Spot Sensors* — Rope sensors or point sensors for localized detection in critical areas like under racks and near equipment; and

   2) *Zone sensors* — For broader coverage in areas like underfloor plenums or around piping systems.

c) *Sensitivity and response time*

   1) *Sensor selection* — Choose sensors with high sensitivity and rapid response times to enable early detection and minimize potential damage;

   2) *Calibration* — Regularly calibrate sensors and adjust sensitivity levels based on environmental conditions and potential risk factors;

   3) Alarms and Notifications;

   4) *On-site alerts* — Implement visual and audible alarms within the data centre to immediately notify onsite personnel of a leak; and

   5) *Remote Notifications* — Utilize remote notifications via email, SMS, or dedicated monitoring systems to alert responsible personnel even when they are off-site.

d) *BMS integration*

   1) *Centralized monitoring* — Integrate the WLD system with the BMS for centralized monitoring and control, enabling real-time visualization of leak locations and status; and

   2) *Automated responses* — Configure the BMS to trigger automated responses to leaks, such as shutting off water valves or activating backup systems.

13.4.3.1 A few additional considerations to be taken are:

a) *Regular testing and maintenance* — Conduct periodic testing and maintenance of the WLD system, including sensor checks, cleaning, and calibration;

b) *Documentation* — Maintain detailed documentation of the WLD system, including sensor locations, alarm thresholds, and maintenance records; and

c) *Emergency response procedures* — Develop and regularly review clear emergency response procedures for water leaks, ensuring staff are trained and prepared to act swiftly to mitigate damage

13.4.4 *Inside Acceptable Temperature and RH Condition*

a) *Temperature*:

   1) Maintain a temperature range between 18 °C to 27 °C for optimal performance and longevity of IT equipment;

   2) Consider higher temperature setpoints (closer to 27 °C) where possible to improve energy efficiency, ensuring equipment compatibility; and

   3) Minimize temperature fluctuations and avoid hot spots or cold spots through effective airflow management and cooling strategies.

b) *Relative humidity* (*RH*):

   1) Keep RH within the range of 8 percent to 60 percent to prevent condensation on equipment and electrostatic discharge (ESD) that shall cause damage;

      NOTE — The upper limit may be lower (50 percent or even 35 percent) depending on the presence of specific corrosive gases in the environment.

2) Monitor dew point temperature and ensure a safe margin between the dew point and the lowest surface temperature in the data centre; and

3) Employ humidification or dehumidification systems as needed to maintain RH within the desired range, especially in regions with fluctuating humidity levels.

c) *Monitoring and control*:

1) *Real-time visibility* — Implement a monitoring system that provides real-time visibility into all critical infrastructure components, including power, cooling, security, and environmental conditions.

2) *Data granularity* — Capture and store granular data at frequent intervals to enable in-depth analysis and trend identification.

3) *Multi-protocol support* — Ensure compatibility with various communication protocols (SNMP, Modbus, BACnet) for seamless integration of diverse equipment and systems.

4) *Centralized management*:

i) *Unified platform* — Deploy a centralized management platform (BMS, DCIM) to aggregate data from all monitored systems, enabling holistic oversight and control of the data center environment;

ii) *Remote access* — Provide secure remote access to the management platform, empowering authorized personnel to monitor and manage the data center from anywhere, anytime; and

iii) *User-friendly interface* — Choose a platform with an intuitive and user-friendly interface for easy navigation and efficient operation.

5) *Advanced analytics*:

i) *Predictive analytics* — Leverage AI-powered analytics to analyze historical and real-time data, enabling predictive maintenance, anomaly detection, and optimization of resource utilization;

ii) *Machine learning* — Utilize machine learning algorithms to learn from data patterns and proactively identify potential issues or inefficiencies; and

iii) *Data visualization* — Employ data visualization tools to present complex information in a clear and concise manner, facilitating decision-making and troubleshooting.

6) *Automated control*:

i) *Threshold-based actions* — Configure the system to trigger automated actions based on predefined thresholds or events, such as adjusting cooling setpoints, initiating failover procedures, or sending notifications; and

ii) *Workflow automation* — Streamline routine tasks through workflow automation, reducing manual intervention and human error.

7) *Alarms and notifications*:

i) *Prioritized alerts* — Implement a multi-tiered alarm system to prioritize critical events and ensure prompt attention to potential issues;

ii) *Multiple notification channels* — Utilize multiple communication channels (email, SMS, phone calls, mobile app) to ensure timely notifications reach relevant personnel; and

iii) *Escalation procedures* — Define clear escalation procedures for critical alarms to ensure swift and appropriate action.

8) *Additional Recommendations*:

   i) *Cybersecurity* — Implement robust cybersecurity measures to protect the monitoring and control systems from unauthorized access and potential threats;

   ii) *Data retention and analysis* — Retain historical monitoring data for long-term analysis and trend identification, aiding in capacity planning and optimization efforts; and

   iii) *Training and skill development* — Provide comprehensive training to data center staff on the use and management of monitoring and control systems.

13.4.5 *Air Distribution Pattern with Positive Pressure*

a) *Hot/Cold aisle containment*:

   1) *Mandatory implementation* — Implement hot/cold aisle containment systems for all data centers to ensure efficient cooling and prevent hot air recirculation;

   2) *Containment types* — Choose between physical barriers (for example, aisle doors, curtains, or rigid panels) or chimney rack systems based on space constraints, future flexibility requirements, and desired level of containment efficiency; and

   3) *Bypass airflow control* — Minimize bypass airflow (air leakage around containment systems) to the lowest achievable level, ideally below 5 percent. Employ blanking panels, brush strips, and proper sealing to achieve optimal containment.

b) *Positive pressure*:

   1) *Pressure differential* — Maintain a slight positive pressure differential of 2.5 Pa to 5 Pa between the data centre and surrounding areas. This effectively prevents dust, pollutants, and other contaminants from entering the data centre;

   2) *Airlock systems* — Implement airlock systems at all entrances to further enhance pressure control and minimize air infiltration during door openings; and

   3) *Pressure monitoring* — Continuously monitor pressure differentials using differential pressure sensors and integrate them with the BMS for automated control and alarms in case of deviations.

c) *Computational fluid dynamics (CFD) analysis*:

   1) *Design optimization* — Utilize CFD modelling during the design phase to comprehensively optimize airflow patterns, identify potential hot spots, and ensure adequate cooling for all equipment under various operating scenarios; and

   2) *Validation and troubleshooting* — Use CFD analysis to validate the effectiveness of the implemented containment and airflow management strategies. It shall also aid in troubleshooting cooling issues and optimizing airflow distribution as the data centre evolves and equipment densities change.

13.4.5.1 Following are some additional considerations to be taken for airflow management:

a) *Raised floor plenum* — Design the raised floor plenum with appropriate airflow tiles and perforated tiles strategically placed to distribute cold air efficiently to IT equipment intakes;

b) *Cable management* — Implement meticulous cable management within racks and under raised floors to avoid airflow obstructions and maximize cooling efficiency; and

c) *Equipment placement* — Consider the airflow requirements of different equipment types during rack layout and placement within the data centre to ensure optimal cooling for all components.

13.4.6 *Commissioning and Retrofitting Guidelines for Data Centres*

a) *General principles*:

1) *Adherence to standards* — Follow the commissioning process and relevant industry standards for commissioning and retrofitting processes;

2) *Documentation* — Maintain thorough documentation throughout the project, including design intent, test plans, procedures, and results;

3) *Independent commissioning agent* (*CxA*) — Engage a qualified and independent CxA to oversee the commissioning or retrofitting process. The CxA should be involved from the early design stages to ensure that the project meets its intended performance and efficiency goals; and

4) *Collaboration and communication* — Foster effective collaboration and communication among all stakeholders, including the owner, design team, contractors, and CxA.

b) *Commissioning phases*:

1) *Factory acceptance testing* (*FAT*):

i) *Purpose* — Verify that equipment and systems meet specified requirements before shipment to the site;

ii) *Scope* — Conduct functional and performance tests on major equipment (for example, chillers, UPS systems, generators, PAHUs) at the manufacturer's facility; and

iii) *Documentation* — Document test procedures, results, and any identified issues for resolution before shipment.

2) *Site acceptance testing (SAT)*:

i) *Purpose* — Verify that installed equipment and systems function correctly and meet performance specifications on-site;

ii) *Scope* — Conduct comprehensive testing of individual systems and components (electrical, mechanical, controls, fire protection) after installation but before integration; and

iii) *Documentation* — Document test procedures, results, and any deficiencies for rectification before proceeding to the next phase.

3) *Integrated system testing* (*IST*):

i) *Purpose* — Verify the proper interaction and performance of all systems and components as an integrated whole under various operating conditions;

ii) *Scope* — Conduct tests under normal, peak load, and simulated failure scenarios to assess system redundancy, failover capabilities, and overall performance; and

iii) *Documentation* – Document test procedures, results, and any identified issues for resolution before final handover.

13.4.6.1 *Specific guidelines for retrofitting*

a) *Thorough assessment* — Conduct a detailed assessment of the existing data centre infrastructure, including its current performance, energy efficiency, capacity limitations, and potential risks;

b) *Clear objectives* — Define clear objectives for the retrofitting project, focusing on areas such as energy efficiency improvements, capacity upgrades, reliability enhancement, or technology modernization;

c) *Phased approach* — Consider a phased approach to minimize disruption to ongoing operations, allowing for gradual implementation and testing of new systems while maintaining service continuity;

d) *Compatibility and integration* — Ensure that new equipment and systems are compatible with existing infrastructure and shall be seamlessly integrated into the current environment; and

e) *Thorough testing* — Conduct rigorous testing of all retrofitted systems and components, both individually and in an integrated manner, to validate performance and ensure compliance with design intent.

13.4.6.2 A few additional recommendations for retrofitting are:

a) *Energy efficiency focus* — Prioritize energy-efficient technologies and design strategies during commissioning and retrofitting to reduce operational costs and environmental impact;

b) *Monitoring and control* — Implement comprehensive monitoring and control systems to enable real-time visibility, proactive management, and optimization of data centre performance; and

c) *Documentation and training* — Provide detailed documentation and training to data centre staff on the operation and maintenance of new or retrofitted systems.

13.4.7 *Electrical Safety*

a) *Earthing and bonding* — Effective earthing and bonding are crucial for personnel safety and equipment protection:

   1) *Earth resistance* — Achieve low earth resistance as per good practice [D-3(31)] for effective fault current dissipation. The specific requirements depend on voltage levels and earthing methods. Use multiple earth electrodes and ground enhancement techniques if needed. Regularly test and inspect the earthing system;

   2) *Automatic disconnection of supply* — Fault loop impedance of every circuit shall be lower enough to ensure automatic disconnection of supply as per good practice [D-3(31)]. If fault loop impedance is higher than the requirement of OCPD, RCDs shall be used. For equipment with 1 phase VFD's Type F RCD shall be used and for equipment with 3 phase VFD's Type B RCDs shall be used;

   3) *Bonding* — Ensure proper bonding of all conductive elements to create an equipotential environment and prevent electrical hazards. Use low-impedance bonding conductors and regularly inspect connections;

   4) *Lightning protection* — Protect your data centre from lightning strikes with a robust system complying with accepted standard [D-3(32)]. The LPS should include air terminals, down conductors, and earth termination networks. Install surge protection devices (SPDs) to safeguard sensitive equipment and ensure effective equipotential bonding for shielding effectiveness; and

   5) *Water pipes* — Avoid using water pipes as earth electrodes. Use of multiple earth electrodes are in violation to the recommendation of good practice [D-3(31)] and shall be avoided.

b) *Circuit protection* — Prevent overcurrent's and short circuits with appropriate circuit protection devices:

   1) *Circuit breakers and fuses* — Install appropriate circuit protection devices (MCBs, MCCBs, fuses) with suitable ratings to protect against overloads and short circuits;

   2) *Selective coordination* — Ensure selective coordination of devices to minimize disruptions caused by faults; and

   3) *Arc flash protection* — Conduct an arc flash hazard analysis and implement appropriate protective measures including appropriate personal protective equipment (PPE) and arc-resistant switchgear (for example, arc-resistant switchgear, personal protective equipment) to safeguard personnel from arc flash incidents.

c) *Regular inspections* — Regular inspections are vital for identifying and addressing potential hazards.

   1) *Scheduled inspections* — Conduct comprehensive inspections at least annually. Depending on the criticality of the equipment;

2) *Thermography* — Utilize thermography to identify hotspots and potential issues; and

3) *Corrective actions* — Address any identified deficiencies or potential hazards promptly to maintain a safe electrical environment.

13.4.7.1 Some of the additional considerations to be taken for electrical safety are:

a) *Design and installation* — Engage qualified professionals for design and installation with accepted standards [D-3(33)] and [D-3(34)] and the National Building Construction Standards;

b) *Power quality* — Monitor and maintain power quality parameters (voltage, current, harmonics) within acceptable limits;

c) *Emergency power off (EPO)* — Implement EPO systems for critical areas; and

d) *Training and awareness* — Provide regular electrical safety training to staff, including arc flash safety and lockout/tagout procedures.

13.4.8 *Periodic Maintenance/Predictive Maintenance*

a) *Periodic maintenance*:

   1) *Comprehensive schedule* — Develop and adhere to a comprehensive preventive maintenance schedule for all critical equipment, including:

      i) *Electrical systems* — UPS, generators, switchgear, PDUs;

      ii) *Mechanical systems* — Chillers, cooling towers, pumps, CRAC/CRAH units;

      iii) *Fire suppression systems* — Sprinklers, gas suppression systems, fire alarm panels; and

      iv) *IT equipment* — Servers, storage arrays, network switches.

   2) *Frequency* — Base the maintenance frequency on manufacturer recommendations, equipment criticality, and operating conditions. Generally:

      i) *Quarterly or as per OEM recommendations* — CRAC/CRAH units, air filters, UPS batteries;

      ii) *Semi-annually or as per OEM recommendations* — Generators, electrical switchgear; and

      iii) *Annually or as per OEM recommendations* — Chillers, cooling towers, fire suppression systems.

   3) *Qualified technicians* — Engage qualified and certified technicians for maintenance activities to ensure proper procedures are followed and equipment warranties are maintained.

   4) *Documentation* — Maintain detailed records of all maintenance activities, including dates, tasks performed, and any observations or issues encountered.

b) *Predictive maintenance*:

   1) *Sensor deployment* — Install sensors on critical equipment to monitor parameters such as vibration, temperature, humidity, current, and power quality;

   2) *Data collection and analysis* — Collect and analyse sensor data using advanced analytics platforms and machine learning algorithms to identify trends, anomalies, and potential failures;

   3) *Early warning system* — Set up alerts and notifications to proactively identify equipment degradation or impending failures; and

   4) *Proactive maintenance* — Schedule maintenance activities based on predictive insights, minimizing unplanned downtime and optimizing equipment lifespan. Adhere to recommended maintenance intervals or as per OEM recommendations.

c) *Thermal imaging*:

1) *Regular inspections* — Conduct periodic thermal imaging inspections of electrical connections, switchgear, and IT equipment to identify hotspots and potential overheating issues;

2) *Qualified thermographers* — Engage certified thermographers to perform inspections and interpret results; and

3) *Corrective action* — Address any identified hotspots promptly to prevent equipment damage and potential fire hazards.

13.4.8.1 Some additional considerations to be taken for maintenance are:

a) *Spare parts inventory* — Maintain an adequate inventory of critical spare parts onsite to enable quick repairs and minimize downtime;

b) *Vendor support* — Establish strong relationships with equipment vendors and leverage their expertise for troubleshooting and support;

c) *Remote monitoring* — Utilize remote monitoring and management tools to gain real-time insights into equipment health and performance; and

d) *Training* — Provide ongoing training to data centre staff on maintenance procedures and predictive maintenance tools.

13.4.8.2 The specific considerations for the Indian context are:

a) *Dust and pollution* — In regions with high dust and pollution levels, increase the frequency of air filter replacements and AHU maintenance;

b) *High temperatures and humidity* — Pay close attention to the performance and maintenance of cooling systems during peak summer months; and

c) *Power fluctuations* — Implement measures to protect equipment from power fluctuations and surges, which shall impact equipment lifespan.

13.4.9 *Design for Future Expansion and Technology*

To consider for the future expansions scalability in below dimensions to be considered:

a) *Capacity planning*:

1) Conduct a detailed capacity planning exercise, factoring in:

i) Projected IT load growth based on business forecasts and technology roadmaps;

ii) Anticipated storage requirements, including data growth rates and retention policies;

iii) Network bandwidth demands considering increasing traffic and application needs; and

iv) Time horizon for expansion planning (typically 5 years to 10 years).

2) Utilize industry-standard tools and methodologies for capacity planning, such as data centre infrastructure management (DCIM) software and power/cooling modelling.

b) *White space and power provisioning*:

1) Design the data centre with adequate white space to accommodate future rack installations and expansions:

i) Consider factors like anticipated growth rate, rack densities, and desired aisle widths; and

ii) Leave room for future power and cooling distribution infrastructure.

2) Provision power capacity based on projected IT load growth and power densities:

   i) Factor in the power requirements of new technologies and potential increases in rack densities; and

   ii) Account for inefficiencies in power distribution and cooling systems.

c) *Modular power and cooling*:

1) Implement modular UPS systems, PDUs, and cooling units that shall be easily scaled up as IT load increases. This allows for incremental expansion and avoids upfront overinvestment; and

2) Consider using modular power skids and containerized cooling solutions for flexibility and rapid deployment.

d) *Network infrastructure*:

1) Design a flexible and scalable network architecture with ample capacity for future bandwidth requirements:

   i) Evaluate the need for higher-speed network technologies to accommodate future demands.

2) Deploy high-density fiber optic cabling and modular network switches to facilitate seamless upgrades and expansions.

14 VENTILATION SYSTEMS

Ventilation is the process of changing air in an enclosed space. A portion of the air in the space shall be continuously withdrawn and replaced by fresh air drawn from outside to maintain the required level of air purity, health, comfort and safety of building occupants.

Ventilation shall be done either by mechanical systems or through natural means. Mechanical ventilation usually consists of fans, filters {accepted standards [D-3(15)], [D-3(16)], [D-3(17)] and [D-3(18)]}, ducts, air diffusers and outlets for air distribution within the building. Natural ventilation and natural exhaust are covered in Part D 'Building Services, Section 1 Lighting and Ventilation' of this NBCS. The scope is therefore restricted to mechanical ventilation.

14.1 Types of Fans

Fans are broadly categorised as:

a) Axial fan; and

b) Centrifugal.

In axial fans, the direction of air flow is parallel to the fan's axis of rotation and in centrifugal fans, this direction is perpendicular. Each category encompasses further designs for example in centrifugal – forward curved, backward curved/inclined, aerofoil fans, plug/plenum fans and in axial – tube axial, vane axial and propellers. Fans have also been classified in terms of their application and special design for example roof ventilator, inline fan, jet fan, spark resistant fan, fume extract fan to name a few. These types may be available in both centrifugal and axial designs.

Recent advances in fan technology have led to the development of mixed-flow fans. These fans combine the high-pressure capability of centrifugal fans with the high flow rate of axial fans.

14.1.1 *Application*

Table 34 provides a general guideline on selecting the type of a fan for a given application, for example for kitchen exhaust, a backward curved centrifugal fan, for basement smoke ventilation, a high temperature rated fan and for laboratory ventilation, a fume extract fan.

Table 34 General Guideline for Fan Type and its Application

(*Clause* 14.1.1)

Sl No.	General Fan Selection Guide	Double Inlet Centrifugal			Single Inlet Centrifugal			Plenum	Tube Axial	Vane Axial	Propeller	Mixed Flow	Tubular Centrifugal	Roof Ventilator	Spark Resistant Construction	Fume Extract Fan	Jet Fan	Powerless Roof Ventilator	High Volume Low Speed (HVLS)
	Application	FC	BC/ BI	AF	FC	BC/ BI	AF												
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
i)	Air handling units - static less than 70 mm	X	X	X				X											
ii)	Air handling unit - static above 70 mm		X	X				X											
iii)	Fan filter units	X	X	X				X											
iv)	Cabinet fans	X	X	X				X											
v)	Basement ventilation-supply fans	X	X	X				X	X	X									
vi)	Basement ventilation-exhaust fans		X			X			X	X		X							
vii)	Ductless basement ventilation																X		
viii)	Kitchen exhaust					X	X	X	X										
ix)	Smoke extraction		X			X		X	X	X		X		X					
x)	Chemical fume extraction								X	X						X			
xi)	Wall mounted exhaust								X	X	X								
xii)	Roof exhaust													X				X	

Table 34 (*Concluded*)

Sl No.	General Fan Selection Guide	Double Inlet Centrifugal			Single Inlet Centrifugal			Plenum	Tube Axial	Vane Axial	Propeller	Mixed Flow	Tubular Centrifugal	Roof Ventilator	Spark Resistant Construction	Fume Extract Fan	Jet Fan	Powerless Roof Ventilator	High Volume Low Speed (HVLS)
	Application	FC	BC/ BI	AF	FC	BC/ BI	AF												
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
xiii)	Toilet exhaust (Ducted)											X	X						
xiv)	Toilet exhaust (Unducted)										X			X					
xv)	Ventilation of explosive environment														X				
xvi)	Ventilation of nuclear facilities		X	X				X	X										
xvii)	Ventilation of dusty environment		X			X		X	X										
xviii)	Tunnel ventilation								X	X							X		
xix)	Large volume recirculation																		X

14.1.2 Selection and Operation

A fan consumes most of the energy in a mechanical ventilation system. To meet a given requirement of air flow rate and system pressure drop, the primary fan selection criterion shall be the overall fan system power consumption and efficiency, that is by the fan, the drive train, the motor and any electric controller such as a variable frequency drive (VFD). This is shown schematically in Fig. 9.

The fan energy index (FEI) is a measure of the overall fan system efficiency and relates the electric power input to the fan motor (or to the motor drive) with the aerodynamic power imparted to the air by the fan (for additional information special publication may be referred). FEI is an index (ratio) that compares the overall energy performance of the selected fan to that of a reference fan. The reference fan is a conceptual fan capable of producing the required airflow and fan pressure at a specified shaft input power using a V-belt drive and a 4-pole IE3 efficiency classification electric motor. The reference fan does not include speed control. FEI of a selected fan at a desired duty point is defined as:

FEI = (Reference Fan Electrical Input Power)/(Selected Fan Electrical Input Power)

NOTE — Special publication maybe ISO12759-6 : 2024 'Efficiency Classification for Fans, Part 6 – Calculation of the Fan Energy Index'.

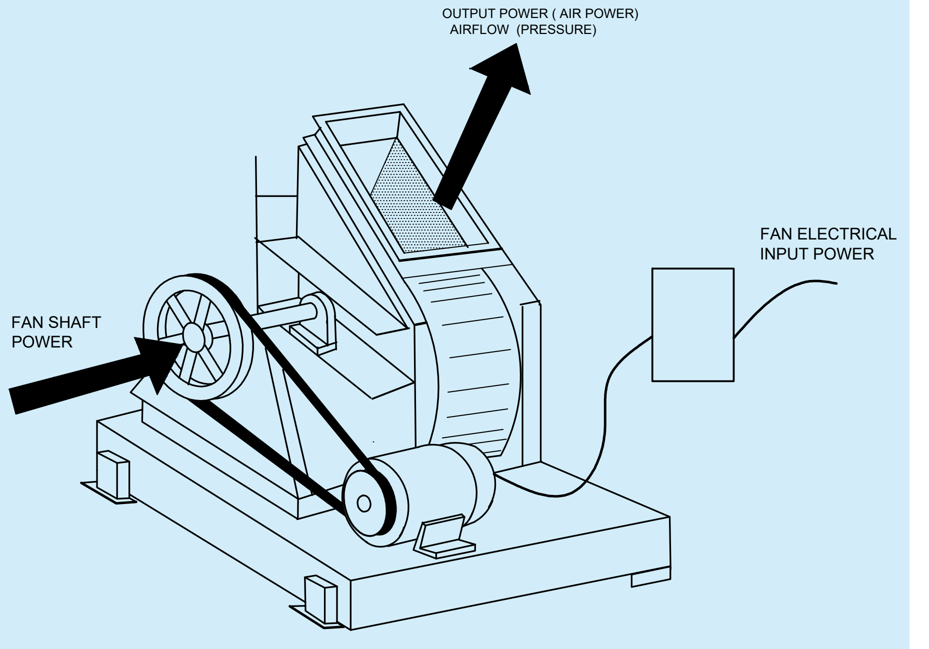


FIG. 9 FAN INPUT (ELECTRIC) POWER TO FAN OUTPUT (AIR) POWER

14.1.2.1 *Reference fan*

The Reference fan is a conceptual fan whose input power and efficiency values for a given duty point (flow rate Q and pressure P) are predefined and are used as a reference to calculate the FEI of any fan. The same reference fan is used to calculate FEI of various types of fans for a given duty point (Q, P).

Reference fan shaft power ($H_{i,ref}$) at a given duty point of flow rate Q and Pressure P is defined as:

For Ducted Outlet Fans $H_{i,ref} = (Q + 0.118) \times (P_t + 100)/660$ kW

For Free Outlet Fans $H_{i,ref} = (Q + 0.118) \times (P_s + 100)/600$ kW

where Q is in m^3/s and P_t is total pressure, P_s is static pressure, both in Pascals.

Reference transmission efficiency ($\eta_{t,ref}$) is defined as a function of reference fan shaft power $H_{i,ref}$ as:

$$\eta_{t,ref} = \frac{0.96}{[H_{i,ref}/(H_{i,ref} + 1.64)]^{0.05}}$$

Combining fan shaft power with transmission efficiency, the Reference motor output power shall be calculated as:

$$= H_{i,ref}/\eta_{t,ref}$$

Basline motor efficiency ($\eta_{m,ref}$) shall be calculated from the Reference motor output power from the graph in Fig. 10.

By reading the value of Reference motor efficiency corresponding to the reference motor output power, the reference motor input power (or Reference fan electric power input) shall be calculated as under:

$$= H_{i,ref}/(\eta_{t,ref} \times \eta_{m,ref})$$

The reference electric power input at a given duty point is the same irrespective of fan type and shall be used to calculate the FEI for any fan selected for the given duty point.

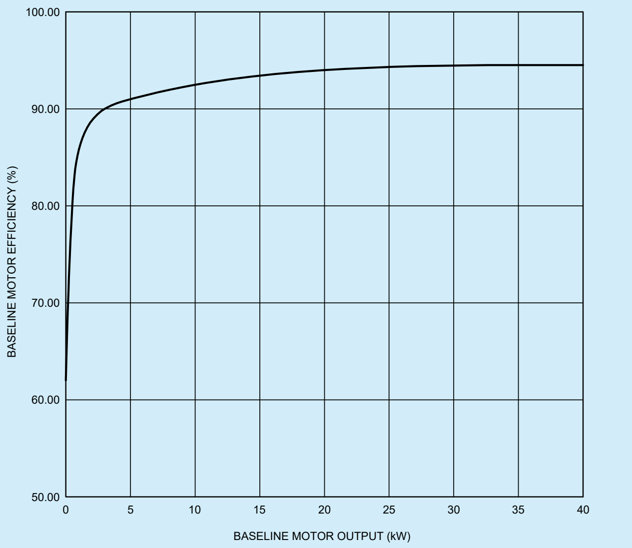


FIG. 10 MECHANICAL AND MOTOR EFFICIENCY REQUIREMENTS (MINIMUM) FOR FANS IN ECSBC BUILDINGS

14.1.2.2 *Fan selection criteria*

The following selection criterion shall be used to select fan for various type of mechanical ventilation applications for a new building:

a) For axial fans requiring a shaft power of 2.0 kW or more, its FEI shall be 1.00 or more;

b) For centrifugal fans and mixed flow fan requiring a shaft power of 2.0 kW or more, its FEI shall be 1.10 or more; and

c) The above prescribed minimum FEI values shall be applicable for all the duty points the fan is selected to operate at.

The choice of a fan type and its drive arrangement also depends on the nature of air to be handled and the type of application. If the fan has to handle dust laden or contaminated air, for example, industrial exhaust or kitchen exhaust, the fan drive arrangement (motor, pulleys, bearing and belts) shall be kept outside the air stream. The same applies to the temperature of air to be handled. For higher-than-normal air temperatures, for example in smoke ventilation, the fan shall be rated for high temperature application.

If fan noise is a concern, it shall be addressed by specifying acceptable limits of certified fan sound power ratings and not by arbitrary limits on fan speed or fan outlet velocity. Specifying fan noise in terms of fan sound pressure levels shall be discouraged.

14.2 Industrial Ventilation

In Industrial buildings, ventilation is needed not only to provide oxygen rich fresh air normally required for health and hygiene and to mitigate against thermal loads due to equipment, people and building heat gains, but also to remove and maintain the hazardous industrial pollutants within safe limits.

14.2.1 *Ventilation for Heat Control*

The required ventilation flow rate shall be calculated both for the sensible and latent heat load. In majority of cases the sensible heat load far exceeds the latent heat load, so the design rate usually shall be calculated on the basis of sensible heat alone. The sensible heat load includes solar heat gain, occupancy, lighting load, equipment load as well as other particular sources, if any.

The ventilation flow rate shall be calculated using the following equation:

$$Q_s = 3.462 \times \frac{H_s}{\Delta T}$$

where

Q_s = air volume flow rate, in m^3/h;

H_s = sensible heat load, in kcal/h; and

ΔT = allowable temperature rise, in °C.

Ventilation system shall be designed for air to flow through the hot environment in a manner that will effectively control the excess heat.

14.2.2 *General (Dilution) Ventilation Versus Local Exhaust Ventilation*

General exhaust ventilation (dilution ventilation) is appropriate when,

a) emission sources contain materials of relatively low hazard (*see* note);

b) emission sources are primarily vapours or gases, or small, respirable size aerosols (those not likely to settle);

c) emissions occur uniformly;

d) emissions are widely dispersed;

e) moderate climatic conditions prevail;

f) heat is to be removed from the space by flushing it with outside air;

g) concentrations of vapours are to be reduced in an enclosure; and

h) portable or mobile emission sources are to be controlled.

NOTE — The degree of hazard is related to toxicity, dose rate, and individual susceptibility.

Local exhaust ventilation is appropriate when:

1) emission sources contain materials of relatively high hazard;

2) emitted materials are primarily larger diameter particulates (likely to settle);

3) emissions vary over time;

4) emission sources consist of point sources;

5) employees work in the immediate vicinity of the emission source;

6) the plant is located in a severe climate; and

7) minimizing air turnover is necessary.

Local exhaust ventilation normally requires lower air flows than general (dilution) ventilation.

Dilution ventilation is used to reduce the concentration of vapours from a given liquid solvent in the air to a safe level known as the threshold limit value (TLV) of the solvent expressed in ppm (parts per million). For a given solvent, the volume of air required to dilute its vapour concentration to below TLV shall be calculated by the following equation:

$$\text{Air volume in m}^3\text{ per kg of evaporation} = \frac{(24 \times 10^6 \times k)}{M \times TLV}$$

where

k = constant varying from 3 to 10 depending on the solvent, uniformity of air distribution, dilution of vapours in air and location of exhaust hood; and

M = molecular weight of the solvent.

A local exhaust ventilation system normally consists of a hood, a duct system, air cleaner, a fan and an exhaust stack. Such a system captures the contaminants at the point of generation through a properly mounted exhaust hood. The exhaust flow rate is determined from the area of the hood opening and capture velocity sufficient to prevent outward escapement of the contaminant. Table 35 lists the recommended range of capture velocity for various types of industrial contaminants.

Table 35 Recommended Capture Velocities for Industrial Contaminants

(*Clause* 14.2.2)

Sl No.	Condition of Dispersion of Contaminant	Process Example	Recommended Capture Velocity m/s
(1)	(2)	(3)	(4)
i)	Released with practically zero velocity into still air	Evaporation from pickling tank, degreasing tank	0.25 to 0.5
ii)	Released at low velocity into moderately still air	Spray booth, intermittent container filling, welding, plating, low speed conveyor transfer	0.5 to 1.0
iii)	Active generation into zone of rapid air motion	Spray painting in shallow booth, barrel filling, conveyor loading, crushers	1.0 to 2.5
iv)	Released at high initial velocities into zone of very rapid air motion	Grinding, abrasive blasting, tumbling	2.5 to 10

The sizing of the ducts shall be determined considering the volume of air required and recommended duct velocity necessary to convey the contaminants with minimum possible pressure drop keeping in mind economics of installation and operation. Recommended duct velocity for exhaust ventilation system is given in Table 36.

Table 36 Recommended Duct Velocity for Exhaust Ventilation Systems

(*Clause* 14.2.2)

Sl No.	Nature of Contaminants	Examples	Recommended Duct Velocity m/s
(1)	(2)	(3)	(4)
i)	Vapours, gases, smoke	All vapours, gases and smoke	5 to 10
ii)	Fumes	Welding	10 to 12.5
iii)	Air laden with very fine dusts	Litho powder, wood flour, cotton lint	12.5 to 15
iv)	Dry dust and powders	Fine rubber dust, moulding powder dust, cotton dust, jute lint, soap dust, leather shaving	15 to 20
v)	Average industrial dusts	Grinding dust, general material handling, clay dust, brick cutting, lime stone dust, asbestos dust in textile industry, dry buffing lint, granite dust, silica flour, shoe dust	17.5 to 20
vi)	Heavy dusts	Metal turnings, saw dust, sand blast dust, C.I. boring dust, lead dust, foundry tumbling barrels and shakeout	20 to 22.5
vii)	Heavy and moist dusts	Lead dust with small clips, moist cement dust, sticky buffing lint, quick lime dust	22.5 and above

The selection, installation and operation of fan shall be in accordance with the nature of contaminants. For dust and other industrial contaminant laden air, the drive arrangement (motor, belts and pulleys) shall be kept outside the air stream.

14.3 Underground Car Park Ventilation

14.3.1 *Requirement*

Ventilation in enclosed car parking areas to dilute the level of toxic gases such as carbon monoxide (CO), oxides of nitrogen (NOx), presence of petrol/diesel fumes and smoke from engine exhaust. The ventilation rate will be such so as to ensure that the CO level is maintained within 40 mg/m^3 (35 ppm) for 1 h exposure, with a maximum of 29 mg/m^3 (25 ppm) for an 8 h exposure (as per best practices of special publication).

NOTE — Special publication may be ISO 10121-2.

14.3.2 *Ventilation Rate Requirement of Mechanically Ventilated Underground Car Parks*

For enclosed underground car parks without provision for natural ventilation, a minimum ventilation rate of 6 air changes per hour (ACH) shall be provided to keep contaminants within acceptable hygiene limits. In large basements, each compartment shall be independently ventilated at the minimum rate of 6 ACH.

14.3.3 *System Requirement*

The underground car park ventilation system shall be classified as supply-only, exhaust-only or a combination of the two. A system of ducts or impulse/induction fans (jet fans) may also be used for proper distribution of air in the car park.

For underground car parking, the fans and the ventilation system used for normal CO level ventilation are also used for smoke ventilation during a fire. The extraction fans, ancillaries and the system should therefore be rated for high temperature operation including air changes per hour requirement, as specified in Part F 'Fire and Life Safety' of this NBCS.

14.3.4 *Demand Control Ventilation Based on CO Level*

The ventilation air flow rate shall be varied according to CO level in order to conserve energy during off peak hours when vehicular movements is much lower than during peak hours. In multilevel basements as well as in large single level structures, independent fan system with individual control is required to take care of fire compartmentation requirements (*see* also Part F 'Fire and Life Safety' of this NBCS).

14.3.5 *Location of Sensors*

The sensors for car park ventilation shall be placed in the following manner:

a) Maximum distance of any corner in the car park to the nearest sensor should be less than 25 m;

b) Sensors should be grouped according to the zone covered by the exhaust fan. The coverage area of each sensor should typically be 400 m^2; and

c) Sensors should ideally be located between 0.9 m and 1.8 m above floor level. However, for practical reasons (in order to avoid vandalism), the sensors may be installed at just above 1.8 m height from floor.

14.4 Commercial Kitchen Ventilation

The basic purpose of a kitchen ventilation system (KVS) is to provide a comfortable environment in the kitchen and to ensure the safety of the people working in the kitchen and other building occupants, by effective removal of effluents which may include gaseous, liquid and solid contaminants produced by the cooking process and products of fuel and food combustion.

Heat and grease are the primary ingredient of kitchen effluents. 50 percent to 90 percent of the appliance energy input is released in the form of a rising convective thermal plume above the cooking surface; balance is released into the surrounding space through radiation. The exhaust hood shall be of sufficient size and placed at proper height to capture the whole plume. The hood exhaust flow rate shall be slightly higher than the plume flow rate. Extra exhaust capacity may be required to resist cross drafts.

14.4.1 *Hood Exhaust Flow Rate*

Kitchen hoods have been classified as two types, Type I and Type II. Type I hoods are used to collect and remove grease, smoke, steam and heat. Type II hoods only remove steam and heat. Thus, Type I hoods are fitted with some kind of grease collection device such as grease filters, baffles and a fire suppression system but a Type II hood typically does not have these devices. Appliances such as cooking ranges, fryers, broilers and griddles require Type I hoods whereas ovens, steamers and dishwashers shall work with Type II hoods.

Hood exhaust flow rates for different types of cooking equipment and exhaust hoods per linear meter of hood length shall be calculated as per Table 37. If more than one duty category appliance is placed under one hood, the hood exhaust flow shall be calculated on the basis of the heaviest duty appliance.

For Type II hoods, the prescribed exhaust flow rates are from 150 liters per second per linear meter of hood length to 460 liters per second per linear meter of hood length for oven hoods, and 460 liters per second to 770 liters per second for condensate hood.

Table 37 Hood Exhaust Flow Rates by Appliance Category

(*Clause* 14.4.1)

Sl No.	**Appliance Category Surface Temperature** °C	**Light** 200 °C	**Medium** 200 °C	**Heavy** 315 °C	**Extra Heavy** 370 °C
(1)	(2)	(3)	(4)	(5)	(6)
i)	Cooking equipment	a) Electric/gas ovens	a) Hot top/element	a) Open burner ranges	Appliances using solid fuels for example, wood, charcoal briquettes
		b) Electric/gas steamers	Ranges	b) Broilers	
		c) Cheese melters	b) Griddles	c) Wok ranges	
		d) Pizza ovens	c) Fryers		
		e) Food warmers	d) Pasta cookers		
			e) Conveyor ovens		
			f) Grill		
			g) Rotisseries		
ii)	Plume velocity (m/s)	0.25	0.43	0.75	0.93
iii)	Hood type	Hood exhaust flow rates per linear meter of hood length (litres per second)			
	a) Wall mounted canopy	309	463	618	850
	b) Single island	618	772	927	1080
	c) Double island (per side)	386	463	618	850
	d) Back shelf	463	463	618	–

14.4.2 *Good Hood Design and Installation Practices*

Wherever feasible, the following good hood design and installation practices shall be adhered to:

a) Increasing hood overhang increases capture volume which aids capture and prevents spillage. A minimum overhang of 150 mm on all open sides for all canopy hoods is prescribed. Increasing front overhang and use of inclined side panels (instead of side overhang) significantly reduces capture flow rates;

b) Deployment of side panels improves hood performance significantly. Side panels prevent the plume from spilling at the side, prevent cross drafts and increase velocity at the hood front;

c) Under a wall canopy hood, pushing the appliance towards the back wall significantly improves hood performance in two ways, increased front overhang and reduction in gap between the appliance and the back wall;

d) When using multiple duty category appliances in line under a single hood, the lowest capture rates are achieved when light duty appliance are at the end of the line. Therefore, hood performance is best when heavy duty appliances are placed in the middle of the line; and

e) Hood shall be mounted at as low a height as practical above the appliance surface.

14.4.3 *Oil/Grease Removal*

The removal of oil/grease from the exhaust airflow is a very important part of commercial kitchen operation. Collection of grease in the duct system, in the fan, in roof top equipment and its emission in the environment poses not only pollution problems but is also a serious fire hazard. Grease shall be effectively removed in the hood exhaust system through the use of proper filtration device. The design and selection of proper filtration device shall ensure that the exhaust air is clean and in accordance with the applicable pollution control norms.

14.4.4 *Exhaust Duct Design, Installation and Maintenance Practices*

Kitchen exhaust ductwork carries hot grease laden air. The following general guidelines shall be followed in their design, installation and maintenance:

a) Ducts should be round or rectangular;

b) Ducts shall be grease tight and should be free of traps that shall hold grease;

c) Minimum sheet gauge should be (16 gauge) mild steel or 18 gauge stainless steel;

d) All joints and seams shall be fully welded and made grease tight;

e) Ductwork shall lead directly to building exterior and should not be interconnected with any other type of building ductwork;

f) Horizontal duct runs should be minimized and pitch towards the hood or an approved reservoir for continuous drainage of liquid grease and condensate. The slope should be 2 percent for runs under 23 m. For horizontal runs greater than 23 m, 8 percent slope should be provided. A grease drain outlet shall be provided in form of a leg under a vertical riser;

g) Maximum velocities are limited by pressure drop and noise and should normally not exceed 12.5 m/s;

h) The minimum air velocity for exhaust ducts should be 2.5 m/s;

j) For new single speed fan system, a design duct velocity of 7.5 m/s to 9 m/s is appropriate;

k) Ducts should be as per the requirements given in Part F ‘Fire and Life Safety’ of this NBCS;

m) Access doors duly nut bolted with lead/fire rated gasket shall be provided for scavenging/grease removal during maintenance. This should be marked/provided in drawings as well as in actual duct work, so that nothing is built to block access to them; and

n) It is recommended that duct cleaning at regular intervals be carried out so that the grease film thickness inside ducts (measured with a wet film thickness gauge or equivalent device) does not exceed 180 microns. This will avoid accidental grease sparks and fire.

14.4.5 *Fan for Kitchen Exhaust*

Kitchen exhaust consists of hot, grease laden air with some solid particulate matter also. Fan shall be capable of handling this air and the motor and the drive train (shaft, bearings, belts, etc) shall be kept outside the air stream. The kitchen exhaust fan shall consist of a backward centrifugal impeller or backward SISW centrifugal impeller to capture grease particles.

14.4.6 *Terminations of Kitchen Exhaust System*

Kitchen exhaust systems shall be terminated so that:

a) Discharge direction shall be such as to minimize re-entry into fresh air intake. This not only requires a minimum separation between exhaust and fresh air intake but also knowledge of prevailing winds;

b) Grease shall be collected and drained into a closed container (a fire safety precaution);

c) Rainwater shall be kept out of the grease container;

d) Grease shall not be allowed to drain down the side of the building;

e) Discharge shall not be directed downward or towards pedestrian areas;

f) Roof top discharge shall de released a minimum 4 m above the roof surface; and

g) For discharge from building sides, it shall be ensured that the discharge air is clean, free from odours and in accordance with the applicable pollution control norms.

14.4.7 *Replacement* (*Make-up*) *Air Considerations*

The air exhausted through a kitchen hood shall be replaced 100 percent with clean outside air. Kitchen room pressure shall be maintained slightly lower than the adjoining building space (for example, dining room) to allow conditioned air to transfer into the kitchen and to contain heat and odours within the kitchen. For kitchens adjacent to a building exterior wall, the kitchen pressure shall be slightly higher than outside to prevent ingress of dust, heat and insects.

14.4.8 *Energy Management Considerations*

Significant energy savings are possible with demand control ventilation (DCV) because the power consumed by a fan is proportional to the third power of its speed. Exhaust and supply air flow rates shall be controlled to cater to peak and off-peak periods by installing variable frequency drives (VFD) on the fan motors.

14.5 Laboratory Ventilation

Most laboratories are the source of chemical and biological contaminations which may be very harmful for the occupants. Effective ventilation shall be provided to control the air quality and remove contaminants to protect the health and safety of laboratory personnel and surrounding environment.

14.5.1 *Ventilation of Chemical Laboratories*

a) The laboratory rooms shall have mechanically operated supply and exhaust air system. Exhaust air shall not be recirculated or short cycled through make up air;

b) Chemical laboratories shall have a minimum of 6 air changes per hour (as per best practices of special publication);

c) Fume hoods (*see* Fig. 11) and/or local extractors shall be deployed for effective exhaust from localized processes. Face velocity at fume hood sash shall be sufficient to maintain the maximum allowable escape of contaminants at 0.01 ppm. This shall be verified by tracer gas tests post commissioning;

d) Laboratories shall be maintained at negative pressure with respect to surrounding corridors or less hazardous areas;

e) Ventilation system shall be capable of shutting down in case of fire;

f) Exhaust duct air velocity shall be maintained at 5 m/s to 10 m/s to prevent condensation of fumes or accumulation of solid contaminants;

g) Eddies and cross drafts in the vicinity of fume hoods shall be minimized;

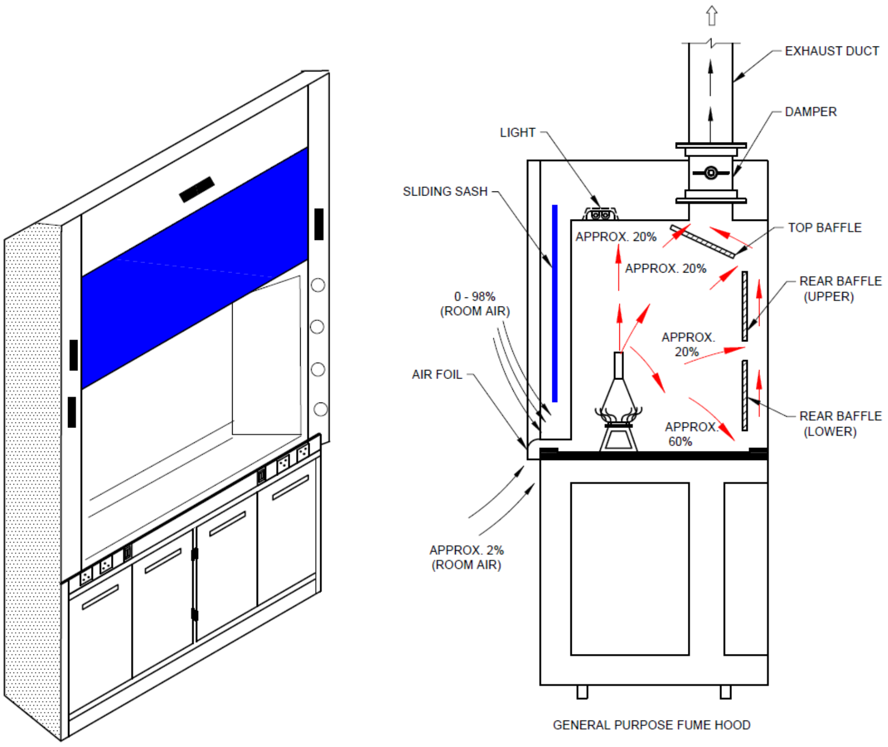


FIG. 11 GENERAL PURPOSE LABORATORY FUME HOOD

h) Certain minimum level of ventilation shall be provided even when the laboratories are not functional to evacuate the vapours generated from the chemicals in the laboratory;

j) Openable windows shall be prohibited to maintain containment and negative pressure in laboratory rooms;

k) Exhaust duct shall not be internally insulated for thermal or acoustic purpose. Sound attenuators or external insulation should provide adequate sound transmission class (STC) and weighted sound reduction index (R_w) ratings to ensure better isolation. Sound attenuators or external insulation may have STC and $R_w \geq 50$ dB for better isolation;

m) Fume hood using heated perchloric acid shall be designed to have the entire exhaust path in spark and corrosion resistant construction as well as with a washing system to avoid explosion hazard;

n) Adequate makeup air shall be provided to compensate the air exhausted by the hoods or other exhaust system; and

p) Air flow direction shall be from low hazard to high hazard areas.

NOTE — Special publication may be ISO 10121-2.

14.5.2 *Ventilation of Biological Laboratories*

Depending upon level of hazard and requirement of protection, biological labs are generally classified into four bio-safety levels (BSL), further categorized into BSL1, BSL2, BSL3 or BSL4 based on combination laboratories practices and techniques, safety equipment and laboratory facilities. General ventilation requirements of BSL3 labs are as prescribed below:

a) Minimum rate of 6 air changes per hour shall be maintained at all times;

b) Laboratories shall be maintained at negative pressure of 12.5 Pa or lower with respect to adjoining corridors, areas, other containment or pressure zones. Monitoring and control devices shall be installed to ensure that pressure differential is maintained at all time;

c) Ventilation system shall effectively remove heat, contaminants and odour from all equipment and laboratory spaces;

d) Visual readout devices and alarm should be provided at entry to containment space, in ante rooms and at entry to other individual rooms within the containment area;

e) Dedicated exhaust air system shall be provided. This system may not be connected to a common system or any other system serving the spaces outside the biocontainment space;

f) Exhaust system shall be provided with variable frequency drive with necessary airflow sensors and a standby fan for redundancy;

g) All duct work shall be with welded joints and in stainless steel construction;

h) It is highly recommended to provide the exhaust system for a BSL3 lab with high efficiency particulate air (HEPA) filters {accepted standard [D-3(19)], [D-3(35)], [D-3(36)], [D-3(37)] and [D-3(38)]}. This filtration shall be provided as close as possible to the biocontainment area. The filtration system shall include bag-in/bag-out type filter replacement system so as to replace the filters without shutting down the exhaust system. Necessary arrangement shall be provided for full face scanning of HEPA filter integrity;

j) Necessary equipment such as laminar flow units, bio safety cabinets of required class, glove box etc. shall be used to ensure the protection of biologically delicate material or to contain bio hazard material during the experiment; and

k) The airflow direction shall be from the clean area to the biocontainment spaces.

14.5.3 *Ventilation of Animal Research Labs*

Properly designed ventilation system provides an adequate amount of oxygen to the lab personnel and animals, removes thermal load generated by animals, removes biological contaminants including allergens and airborne pathogens, controls the humidity of animal housing systems, dilutes the gaseous contaminations, maintains pressure differential between adjoining spaces, maintains micro environment in primary enclosure (housing cage) and maintains animal health.

Animal research facility shall be classified as under with respect to housing system:

a) Conventional animal facility with open type pan; and

b) Modern animal facility with individually ventilated animal housing system.

14.5.3.1 *Animal lab with conventional system*:

a) Shall be provided with minimum of 10 to 15 outside air changes per hour for animal biosafety level (ABSL) 1 and ABSL 2 labs and higher for ABSL 3 facility. System should be once through. Recirculation of air is generally not permitted;

b) Both, supply and exhaust air system shall be provided with HEPA filter to protect the animal from outside contamination and also to protect the environment from biological contamination;

c) System shall be designed to maintain required hygroscopic conditions as per animal requirement; and

d) High velocity draft shall be avoided near animal housing pans to protect animals from wind chill effect.

14.5.3.2 *Animal lab with individually ventilated housing system*:

a) Shall be provided with fresh air supply depending upon the total cage volume and number of air changes required by the system;

b) Necessary adjustment factor shall be considered while designing the HVAC system as the relative humidity in close cage will be 8 percent to 10 percent higher than in the room;

c) Shall ensure no air movement from contaminated area to clean area; and

d) Ventilated caging system shall be selected and pressure gradients shall be adjusted in system itself according to different ABSL level requirements.

14.5.3.3 *In addition*:

a) The direction of air flow shall be from clean corridor to animal room to the service corridor. Necessary pressure gradient shall be maintained to ensure this all the time;

b) Both, supply and exhaust systems shall be designed for redundancy and shall be in operation all the time;

c) Animal experiment room shall be maintained under negative pressure with respect to clean area. Animal quarantine room and healthy animal rooms shall be maintained under positive pressure;

d) Use necessary isolation equipment to protect surrounding environment from infected animal or to protect immune compromised animals from surrounding environment.

14.6 Tunnel Ventilation

14.6.1 *General*

The tunnel ventilation system (TVS) for underground metro network is intended to provide:

a) An acceptable environment in the tunnel and station trackway for the operation of trains;

b) Pressure relief produced by the piston effect of train movement during normal operation;

c) Heat removal during congested/maintenance operation; and

d) An effective means of controlling smoke flows during emergency conditions for safe evacuation of passengers

14.6.2 *Operation Philosophies*

There are three design operating conditions for the tunnel ventilation system that is normal operation, congested operation and emergency operation.

During normal operation, that is when trains are running as per scheduled headway the main source of ventilation for the tunnel is airflow generated due to piston effect produced by moving train. Tunnel ventilation fans are not operated during train movement in tunnel. The trackway exhaust system (TES) at the station extract the heat from train air-conditioning or braking equipment when train is dwelling at the stations. The air extracted from the TES is either exhausted to the atmosphere or shall be used to recirculate to the station air-conditioning system as per the requirement. The trackway exhaust fans are generally provided with variable frequency drives to operate the fan at optimal speed to save energy.

In congested operation, the tunnel ventilation fans (TVFs) are activated for preventing the accumulation of warm tunnel air around idling train in the affected tunnel section. The tunnel ventilation design condition for congested train operation is maximum stratified tunnel air temperature (approximately 50 °C) as per the design of the rolling stock.

In emergency operation, the TVS is set to operate to control the movement of smoke and provide a smoke-free path for safe evacuation of the passengers and for firefighting purposes. The ventilation system is operated in a 'push-pull' supply and exhaust mode with tube axial flow fans/nozzles. This enables driving tunnel air flows such that the smoke is forced to move in one direction, enabling evacuation to take place in the opposite direction. A typical ventilation system in tunnel during fire on a train is shown in Fig. 12.

One of the most important factors in selection of the system is to generate the minimum critical velocities of air flow in the tunnel to prevent back-layering of smoke laden air in the tunnel.

14.6.3 *System Description*

The tunnel ventilation system (TVS) consists of tunnel ventilation fans (TVFs), trackway exhaust fans (TEFs), tunnel booster fans (TBFs), tunnel ventilation dampers (TVDs), tunnel ventilation nozzles and sound attenuators provided in the tunnel ventilation plant rooms at each end of the station and connected to both trackways and to outdoors through ventilation shafts.

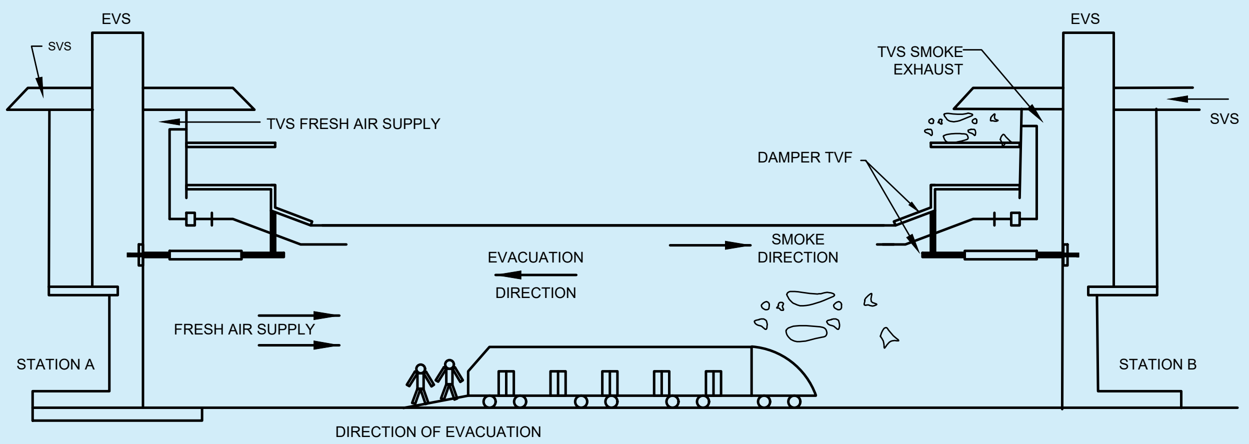


FIG. 12 VENTILATION IN TUNNEL DURING FIRE ON A TRAIN

Each tunnel ventilation installation has two fully reversible tunnel ventilation fans with fan isolation dampers. These dampers are closed when the fan is not in operation. In addition, there is a bypass duct with isolation dampers around the fan room, which acts as a pressure relief shaft when open during normal conditions and enables the flow of air to bypass the TVFs, allowing air exchange between tunnel and outdoor with flows generated by train movement.

The tunnel booster fans are installed at the crossover locations/at portals that is, where the train moves from underground to overground to provide thrust and to direct the flow in desired direction during congestion and emergency scenarios.

The trackway exhaust fans are located in separate plant rooms at each end of the station and connected to station trackway through under platform exhaust and over track exhaust ducts and to the outdoors through exhaust ventilation shafts, to enable heat removal from train AC system and braking system. The TEFs shall also be utilized for smoke extraction from the concourse/platform public areas with the help of dampers connecting the ducting system.

All the components of the tunnel ventilation system are rated minimum for 250 ºC for one hour. The fire rating of the system shall be increased based on the requirements.

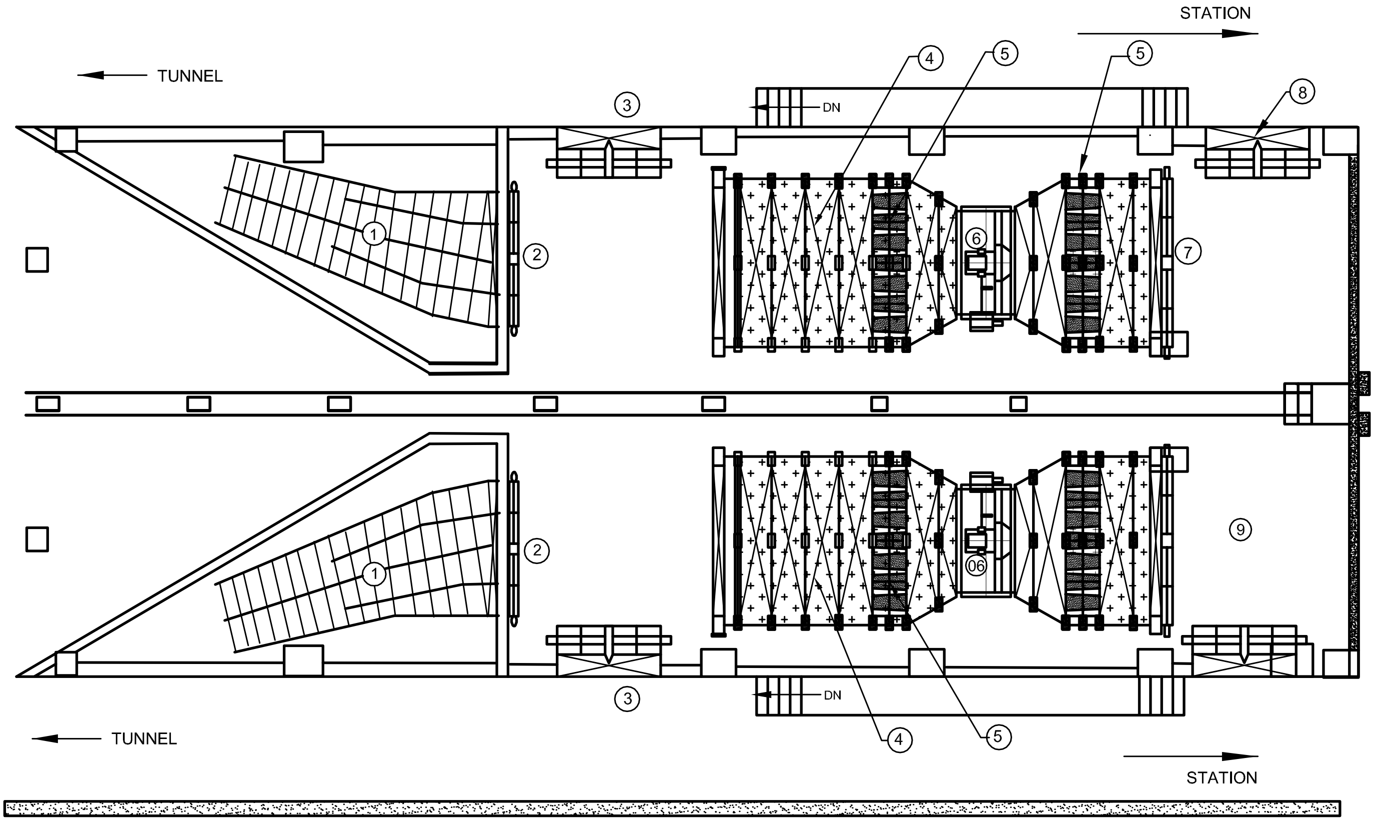
The specialized software tools such as subway environment simulation (SES), IDA tunnel software, computational fluid dynamics (CFD) analysis are used to evaluate the requirement and efficacy of the tunnel ventilation system.

A typical schematic of the TVS system installed at platform level is shown in Fig. 13.

14.6.4 *Control and Monitoring of TVS*

The TVS system shall be equipped with provisions for automatic, manual, local and remote controls so that the fans and damper motors shall be operated from a station control room (SCR) or from the operations control centre (OCC). At OCC, an integrated supervisory control and data acquisition system (SCADA), should be employed to control and monitor the TVS plant in each station, as the same are generally not attended locally.

The operation of the tunnel ventilation fans during congestion are integrated with the signalling system to identify the affected tunnel section or the ventilation zone. The fans shall be operated automatically or semi-automatic as per the requirements.



1. NOZZLE
2. NOZZLE DAMPER
3. TVD (SCHEMATIC)
4. DUCT
5. ATTENUATOR
6. TVF
7. FID
8. DRD
9. SD ABOVE

FIG. 13 TYPICAL SCHEMATIC OF THE TVS PLANT ROOM AT PLATFORM LEVEL

In case of unattended train operation, suitable means shall be provided to locate the fire on train for deciding the direction of operation of the ventilation system.

The SCADA for TVS shall be designed with a minimum safety integrity level that is SIL-2. The redundancy shall be provided for communication, processor and power supply.

14.7 General Ventilation Rates for Non-Air-Conditioned Areas

In non-air-conditioned areas, ventilation may be provided through natural or mechanical means to maintain acceptable indoor air quality and thermal comfort. The design may consider climatic conditions, building orientation, effective opening area, and occupancy pattern to ensure adequate air movement and dilution of indoor pollutants. For more details, refer to **5** of Part C 'Structural Design, Section 1 Loads and Load Effects' of this NBCS.

15 AUTOMATION AND CONTROLS

In commercial buildings, there are various equipment and systems that consume a lot of energy in applications while delivering occupant comfort and achieving their functionalities. There are multiple aspects for operating these systems to deliver design conditions due to which automation and controls become inevitable. Systems are supposed to function at the set parameters throughout the year whereas ambient conditions vary throughout the operating hours of the building. Systems and equipment normally get designed for peak loads and demands but they operate at part load conditions for a majority of their operating hours. This aspect of operations under various outside as well as inside conditions impacts the operating efficiencies of these systems. To maintain various functionalities within the design conditions, systems should have proper controls incorporated in design as well as in operations. Nowadays equipment come with a lot of built in intelligence and it becomes very necessary to integrate them to a higher level of BMS to ensure owners and end users shall monitor and control the various systems to achieve operating efficiencies, safety, comfort and health of the building and its occupants. In addition to these aspects, the speed with which the digital technology is moving forward, internet of things (IoT) is becoming an integral part of equipment. Buildings shall incorporate futuristic design aspects of BMS in building design. The various aspects of automation and control elements in a building for the various systems are addressed below.

15.1 Roles and Requirements of Automation and Controls in a Building for Various Systems

Given below are the required roles for automation and control in building for various system:

15.1.1 *Heating, Ventilation and Air Conditioning Systems (HVAC)*

Automation and Controls should be designed for:

a) *Monitoring* — Should monitor air quality and thermal comfort parameters as specified in section;

b) *Control* — Controls shall switch on/off of the various equipment based on demand, and open/close or modulate valve actuators and HVAC dampers based on time schedule, occupancy or control parameters like temperature or RH or pressure or flow based on the application;

c) *Energy data management* — Should Provide energy consumption data to monitor and help create energy saving strategies;

d) *Alarm management* — Should Provide better monitoring, control and early notification for predictive and preventive maintenance, optimize operational efforts and reduce downtime of the system; and

e) *Building parameter data management* — Should provide historical data for analysis of system performance.

15.1.2 *Fire and Life Safety*

There is a lot of data exchange required between the fire alarm system and HVAC system for monitoring and controlling the pressurization systems in a building from the BMS. Life safety systems are always stand-alone systems but monitoring of all these systems from the BMS should be deployed critical for proper operation and monitoring of the buildings:

a) Automation for fire and life safety in commercial buildings, residential communities, public places and FAS and BMS should be seamlessly integrated:

    1) *Monitoring* — Shall monitor status of various equipment - water tank levels, primary pumps, diesel pumps, jockey pumps, fire sensors, smoke sensors, pressure across the system, leakages, fire-panel alerts and diesel fuel tank level, fire extinguishers;

2) *Control* — It should have controls for automatic filling of water in fire tanks independently and proactive steps for managing alerts from the fire panel. should include predictive and preventive maintenance scheduling via automation for various components; and

3) The Wired or wireless technology for automation should be deployed based on the overall implementation of that project and budget.

b) Automation for life safety systems and monitoring of systems with interlinked logics and sharing of information with fire responders.

Application in buildings, communities, cities and towns, public places, industries, data centres, hospitality and hospitals, etc.

1) *Monitoring of fire alarm panels* – status monitoring – ON/OFF, alarm status/normal status this should be covered over IoT (Via Wi-Fi/4G/3G or wired). - Intend of this application – To have status monitoring locally and remote capability via IOT which shall be seen by fire responders;

2) *Monitoring of hydrant system* – Pressure in hydrant pipes should be continuously monitored and remotely displayed, Change of Pump switch from auto/manual or vice versa, change in ON/OFF status of pumps. Calls and SMS/Or mail alerts at all levels with escalation matrix defined; and

3) The Wired or wireless technology for automation should be deployed based on the overall implementation of that project and budget.

For real life saving impact, we need to look at early detection methods. Wired and wireless technology shall be employed. Wireless IoT sensors and applications should be programmed to provide 'Notifications' of critical events in small and medium sized buildings that do not have a complete BMS system.

15.1.3 *Electrical*

The following sub systems should be connected with the BMS for monitoring purposes. BMS should monitor:

a) critical parameters of diesel generator (DG);

b) fuel levels of day fuel tank;

c) electrical breakers for risk free operation of the system;

d) various parameters from energy meters to ensure that energy consumption levels are trended and logged for proper analysis of the usage of energy and taking corrective action at the appropriate time to save the energy consumption levels in buildings; and

e) vertical and horizontal transportation (VHTs) from a single platform.

The integrated building management system (IBMS) should create necessary dashboards for easy access to the operating personnel.

Medium size buildings should use independent wireless IoT based energy monitoring device to monitor and control energy usage of VRF and ducted split units.

15.1.6 *Controls Requirement for Significant Components of Building*

Based on the above details of requirements of various systems for control and monitoring functionalities, following Table 38 defines compliance requirements for significant components of BMS at equipment level and system level.

Table 38 Controls Compliance Requirement for Significant Components of Building

(*Clauses* 15.1.6, 15.2.1(1) *and* 15.2.3)

Sl No.	Discipline	Control and Monitoring Level	Equipment/System	Control/Monitoring Base Level	Control/Monitoring Energy Intensive Buildings	Control/Monitoring for Critical Buildings
(1)	(2)	(3)	(4)	(5)	(6)	(7)
	HVAC					
i)	HVAC	Equipment level	DX IDU/ODU	Stand alone	Stand alone	Stand alone
ii)	HVAC	Equipment level	DX VRF	Stand alone	Stand alone	Provide networked controllers
iii)	HVAC	Equipment level	CHW FCU	Stand alone	Individual time clock control using controller	Provide networked controllers
iv)	HVAC	Equipment level	CHW AHU	Stand alone	Individual time clock control using programmable controller	Provide networked controllers
v)	HVAC	Equipment level	CHW Pumping	Stand alone	Provide group controls for all the pumps	Provide networked controllers
vi)	HVAC	Equipment level	Cooling tower fan	Stand alone	Stand Alone - as per ECSBC	Provide networked controllers
vii)	HVAC	Equipment level	Extract fan	Stand alone	Stand alone	Provide networked controllers
viii)	HVAC	Equipment level	Pressure Control (Air side)	Stand alone	Stand Alone	Provide networked controllers
ix)	HVAC	Equipment level	CT Level Control	Stand alone	Stand alone	Provide networked controllers
x)	HVAC	Equipment level	Basement ventilation	Stand Alone - as per ECSBC	Stand Alone - as per ECSBC	Provide networked controllers with all monitoring points in the dashboard screens
xi)	HVAC	Equipment level	Energy recovery (Airside)	Stand Alone - as per ECSBC	Stand Alone - as per ECSBC	Provide networked controllers
xii)	HVAC	System level	CHW pumping	Stand alone	Stand alone	Provide networked controllers
xiii)	HVAC	System level	Variable air volume	Stand alone	Stand alone	Provide networked controllers

Table 38 (*Continued*)

Sl No.	Discipline	Control and Monitoring Level	Equipment/System	Control/Monitoring Base Level	Control/Monitoring Energy Intensive Buildings	Control/Monitoring for Critical Buildings
(1)	(2)	(3)	(4)	(5)	(6)	(7)
xiv)	HVAC	System level	Pressure control	Stand alone	Stand-alone	Provide networked controllers
xv)	HVAC	System level	Demand control ventilation	Stand alone as detailed in ECSBC	Stand alone as detailed in ECSBC	Provide networked controllers
xvi)	HVAC	System level	Economizer	Provide controls as per ECSBC	Provide controls as per ECSBC	Provide networked controllers
xvii)	HVAC	System level	Chillers and chiller plant control	Chiller plant control as per ECSBC details	Chiller plant control as per ECSBC details	Provide networked controllers with data for analysis
	LIGHTING					
xviii)	Lighting	Equipment level	Lux level control	as per details given in ECSBC	as per details given in ECSBC	as per details given in ECSBC
xix)	Lighting	System level	Lighting Management System (LMS)	-	-	Integrate LMS with BMS; share occupancy/unoccupancy mode data; based on based on which, VAVs to switch to occupied/unoccupied modes
	ELECTRICAL AND VERTICAL TRANSPORTATION					
xx)	Electrical and vertical transportation	Equipment level	Transformers, breakers, VHT	-	-	Monitor healthy status of the equipment
xxi)	Electrical and vertical transportation	Equipment level	Energy meters	Record energy value at all meters for monitoring purposes for all utilities	Digitally connect all utility energy meters; track energy consumption for analysis	Digitally connect all utility energy meters; track power and energy consumption data for analysis

Table 38 (*Concluded*)

Sl No.	Discipline	Control and Monitoring Level	Equipment/System	Control/Monitoring Base Level	Control/Monitoring Energy Intensive Buildings	Control/Monitoring for Critical Buildings
(1)	(2)	(3)	(4)	(5)	(6)	(7)
xxii)	Electrical and vertical transportation	System level	Building level	-	Comply as per ECSBC	Comply as per ECSBC
	WATER MANAGEMENT					
xxiii)	Water management	Unit/equipment level	PHE Equipment	Provide stand-alone control for equipment functioning	Provide stand-alone control for equipment functioning	Track parameters at the dashboards
xxiv)	Water management	Equipment level	STP System	Stand-alone control	Stand-alone control	Track parameters at the dashboards
xxv)	Water management	Equipment level	Water Meters	Recording of Water Consumption data;	Recording of Water Consumption data;	Recording and trending of water consumption data

NOTE — Details of for abbreviations mentioned in column under equipment/system are provided below - this matrix is part of Chapter 13 of ECSBC and shall be referred also from ECSBC.

DX IDU/ODU	DX Split Unit
DX VRF	DX Variable Refrigerant Flow Unit
CHW FCU	Chilled water Fan Coil Unit
CHW AHU	Chilled Water Air Handling Unit
CHW Pumping	Chilled water Pumping
CT Level Control	Cooling Tower Level Control
PHE equipment	Public Health and Engineering equipment
VHT	Vertical and Horizontal Transportation

15.2 Building Management System (BMS) and Internet of Things (IoT)

The following points outline the various components and functionalities associated with BMS and IoT in modern buildings

15.2.1 *BMS System Architecture*:

a) Control Systems in a building shall have a system architecture that ensures control and monitoring of various operating parameters at the following levels:

   1) Unit/equipment Level for equipment performance, monitoring, control and energy optimization as per the control logic developed to achieve the design objective. Necessary and appropriate field devices shall be used at this level to capture various parameters that need to be controlled and monitored (*see* Table 38);

   2) System and automation level shall ensure coordinated and integrated control logic between the various sub systems as per the design intents; and

   3) Management Level shall contain necessary workstations with software and dashboard screens provide macro level parameters of various high-level systems to monitor trend and analyse the various data.

b) System shall bring in all mechanical, electrical and public health engineering (MEP) equipment and systems under its architecture;

c) Control system shall be constructed to achieve energy optimization based on the level of intelligence and functionalities in-built in the equipment;

d) A matrix of equipment and parameters that shall be controlled and monitored shall be drawn (*see* Table 38 for various building systems). In addition to Table 38, critical parameters namely, temperature, pressure, voltage, current, energy, and others shall be recorded and monitored and shall be made available for the operations team for corrective action either through stand-alone controls or networked controls;

e) Based on point (d) above, detailed Input/output point summary (IOP Summary) shall be developed;

f) Control schematics and sequence of operation shall be developed for various control loops based on the respective design philosophy and intent of the building by the concerned design team; and

g) The controls system architecture shall be developed to effectively integrate and manage various building systems and technologies. The key requirements shall include:

   1) Scalability and flexibility – Allowing for future expansion and adaptation;

   2) Interoperability;

   3) Real-time data acquisition and analysis;

   4) Security and privacy;

   5) Redundancy and fault tolerance;

   6) Energy efficiency and sustainability; and

   7) Centralized management for control and monitoring.

15.2.2 *Protocols*

All unit level and system level controls shall have industry standard open protocols adopted for connecting various equipment/controllers/third party devices that have built-in intelligence in their controllers and factory installed field devices/equipment like chillers, diesel generating sets, energy meters, and variable speed drives. Other industry standard open and accepted protocols such as open platform communications (OPC), message queuing telemetry transport (MQTT), zigbee, Wi-Fi, modbus, meter-bus (M-Bus), global system for mobile communication/general packet radio services (GSM/GPRS), Z-wave, device language message specification (DLMS) shall be used for connecting devices.

15.2.3 *Controllers*

For various applications, controllers with built-in logic or programmable control logic shall be used to achieve the desired design intents.

Depending upon the building application and criticality, networked controls shall be used.

a) All networked controller systems shall comply with **15.2.2** for protocols used; and

b) Critical parameters namely, temperature, pressure, voltage, current, energy, and others shall be made available for effective monitoring and control of the system through networked centralized control system. (Table 38 shall be referred for various parameters to be considered based on the application.)

15.2.4 *Internet of Things* (*IoT*)

IoT devices and automation optimize building performance by providing data on core building operational systems and enabling automatic control of the building's main operating functions. IoT devices, when implemented, shall be able to connect via standard building and industry protocols as mentioned in **15.2.2**.

For IoT devices to be accessed and configured on the field and/or remotely shall have hardware interface and software or wireless interface for remote configuration.

Integrating IoT with BMS represents a significant advancement in building management, offering improved efficiency and enhanced user experiences. While implementing systems with IoT enabled controls, following aspects have to be taken care of in the design:

a) *Interoperability* — This is to ensure different IoT devices and systems work together seamlessly. For medium sized buildings, all applications for devices should be accessible from the same screen. Use of higher-level local servers running continuously are to be considered to achieve interoperability when stated in OPR;

b) *Data privacy* — Safeguarding the personal data collected by loT devices is important. Use of VLAN and Privacy Levels to be as per requirement detailed in OPR. For medium sized buildings WPA/WPA2 will otherwise be the minimum level;

c) *Scalability* — Ensure the system is scalable and adaptable to changing needs within the same platform; and

d) *Security* — Protect the system from cyber threats by ensuring that remote access to all IP and other applicable devices have USER configured usernames. Default usernames and passwords shall not be retained and device shall not be accessed by use of default username and password. Security levels to be as per requirement detailed in OPR. For medium sized buildings WPA/WPA2 otherwise accepted as the minimum level.

15.2.5 *Cyber Security*

Vulnerabilities in the IT based systems are diverse. Due to their physical location across all parts of a facility and connectivity with open protocols, systems are prone to technical and physical attacks at all architectural levels. System shall take care of all aspects of cyber security.

15.2.6 *Software*

The BMS workstation software that brings in all the data and the graphics including all the functionalities and sequence of operation and logic to the operator workstation shall have the following options depending upon the criticality and application of the building and also based on owner's project requirements (OPR).

a) Computer workstation software or server based software depending upon the number of data points for monitoring and control.

b) Optional server based on premise or remote depending upon the scale and application of the building.

c) IoT and cloud based system as per OPR.

15.2.7 *Integrators/Gateways*:

a) High level integrators/gateways shall be used to ensure conversion of various protocols mentioned under **15.2.7 (c)**.

b) Protocol conversion for 3rd party devices integration of various equipment in a building shall happen when various equipment is connected through soft link to bring in the various parameters residing at the equipment to the BMS platform without installing any additional field devices.

c) Conversion from one protocol to another protocol:

   1) BACnet MSTP to BACnet/IP;

   2) MODBus RTU– BACnet/IP;

   3) MODBus TCP/IP - BACnet/IP;

   4) M-Bus to BACnet-IP; and

   5) MODBus – GSM/GPRS.

   NOTE — The above are not exhaustive and other conversions may also be required as per the equipment and protocols installed.

15.2.8 *Dashboard*

High level parameters of various sub systems shall be captured on the dashboard screen and customized based on the OPR.

a) HVAC;

b) Electrical;

c) Energy management – consumption, generation;

d) Water management – levels, volume, consumption;

e) Sewage treatment plant;

f) Lighting management – power consumption, circuits status, luminaire fused status circuits status, luminaire fused status;

g) Fire – status of alarms/alerts;

h) IEQ/IAQ;

j) Waste management;

k) Occupancy status; and

m) UPS/battery management.

15.3 Instrumentation and Field Devices

Instrumentation in automation and controls shall be provided to measure various parameters and communicate the same to the next level in the system architecture:

a) *Monitoring* — Status monitoring of multiple aspects of buildings like,

   Utilities (water, lighting, energy, gas), water quality (WTP, STP, ETP), air quality, ventilation (HVAC, mechanical), VHT (lifts, elevators, escalators, travellators), fire and life safety, UPS (power consumption, battery management), DG set, parking, security (cameras, boom barriers, access control, intrusion detection), plumbing etc;

b) Control of critical/non-critical areas as defined in OPR;

c) *Instrumentation considerations* — These devices range from field devices, loggers, aggregators, controllers - sensors, PLCs, controllers (PID, IoT), mechanical (actuators, valves, dampers etc.), internal and external communication devices and any special purpose instruments. These shall be of analogue or digital type. Communication protocols like analogue (V, A), digital, serial, TCP/IP etc. based on the application shall be considered; and

d) *Technology* — A combination of wired and wireless field devices shall be used based on the overall application of the project and OPR.

15.4 Sequence of Operation and Control Logic

Sequence of operation and the control logic of any control loop shall be developed in the BMS based on the design logic of that particular discipline and application. This is to be developed by the concerned design team and the same will be used in developing the Input/output point summary as well as for determination of necessary field devices to be installed for capturing the parameters to be controlled and/or monitored.

15.5 Data Gathering for Analysis and Performance Improvements

This is a very important part of the BMS where all the data that are gathered through the field devices as well as those processed at the controllers, data stored through various trending and logging from the various equipment and system shall be stored in a PC workstation or a server depending upon the number of data points.

This data shall be made available for analysis by the facilities and operations team for making performance improvements in the system as well as comparing through a range of time spans to understand the behaviour of the building. These data will also be useful in making informed decisions like the life of the equipment, for drawing up replacement schedules, comparing performances for various buildings etc.

15.6 Energy Savings — Optimization for Decarbonization

To effectively use the data from the BMS for measuring the sustainable level of operations of a building, following dashboard screens, at the minimum shall be made in addition to those listed under **15.2.6**. These will not only help the facilities and operations team for trending and analysis but also for measuring the carbon footprint of the building and the decarbonisation levels of the building. These will help the owner in understanding the carbon emissions and help in migrating towards net zero goals.

Dashboard screens shall be developed for the following:

a) Energy;

b) Environment;

c) HVAC system;

d) Lighting control system;

e) Water management system; and

f) Electrical and VHT systems.

For example, energy consumption data to monitor and help create energy saving strategies, energy dashboards shall provide energy data to building and asset management team in graphical form for easy understanding and implementation.

15.7 Installation and Commissioning of Control Systems

15.7.1 *Field Devices*

Field Devices like temperature sensors, RH Sensors, pressure transmitters, flow meters and transducers, level sensors and switches, IAQ Sensors shall be installed as per manufacturers installation manual and recommendations. Necessary precautions shall be taken during installation of the field device for measuring the parameter that is detailed in the input/output point summary and the sequence of operation and control logic diagrams.

15.7.2 *Enclosure Panels*

The controllers, integrators and other components of BMS shall be housed in an enclosure panel that is appropriately rated (electrically) for the ambient conditions at which they are surrounded by so that the control components are neither damaged nor communicate incorrect data to the system. Since most of the controllers and field devices operate with 240V a.c./24V a.c./24V d.c. (as per manufacturers data sheets and installation manuals), necessary transformers have to be installed in the enclosure panel to supply the voltage as per the requirements. The transformers shall be rated to take care of the necessary VA ratings of the field devices and the relays being operated by the controller. Proper earthing etc, shall be provided as per the relevant sections of the accepted standard [D-3(39)] and Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS. These enclosure panels shall be suitably IP rated depending on their installation environment.

Relevant sections concerning electrical components in NBCS shall be followed for proper compliance.

15.7.3 *UPS Power Supply*

Necessary uninterrupted power supply shall be provided to the controllers and other control components of the BMS through proper electrical distribution system. Necessary earthing etc, shall be provided as per the relevant sections of the accepted standard [D-3(39)].

Relevant sections concerning electrical components in accepted standard [D-3(39)] shall be followed for proper compliance.

15.7.4 *Field Cabling and Containment*

Field cabling shall be provided between the various control components like the field devices, sensors, transducers, controllers, starter panels, 3rd party equipment for integration as per the relevant sections of this NBCS and as per the recommendations of the manufacturer of the control system. Control cables, integration cables and field cables shall be screened, twisted, shielded and the sizing shall be as per the recommendations of the controls' contractor and OEM. There shall be a detailed cable schedule drawn up based on the input/ output point summary before carrying out this work. Necessary earthing etc, shall be provided as per the relevant sections of the accepted standard [D-3(39)]. Containment of cabling shall be carried out as per the relevant sections of the accepted standard [D-3(39)] and as per the details of the projects approved documents. Necessary drawings have to be made for the entire system and approvals taken from concerned Engineer-in-charge

15.7.5 *BMS PC Workstation/Server*

The PC Workstations or servers, depending upon the number of data points and based on the application of the building and OPR, shall be installed in an air-conditioned room. These along with either the monitoring station or display units shall be accessible by the facilities and operations team. UPS power supply shall be provided for these components since these will have to handle data from the entire BMS online and have to store data as well for use.

Necessary network components like network switches and accessories shall be installed based on the control manufacturers recommendations and drawings.

15.7.6 *Programming and Commissioning of BMS*

The entire BMS system shall be properly installed, field cabling all checked for continuity prior to the commissioning of the system.

Based on the control logic and the applications, necessary programming shall be developed by the controls contractor and the same shall be downloaded to the various controllers or configured based on the type of control system (wireless or wired).

Various control parameters that are measured through the field devices are to be mapped onto the controllers and checked for consistency. Various set points need to be determined based on the applications and the programs have to be test run and observed for proper operations and necessary fine tuning needs to be done on various control parameters like – PID gains, range etc, for trouble free and accurate operation of the programs.

With all the above completed and as-built drawings, the BMS system needs to be handed over to the end user with as-built drawings and all IOMs for the entire system.

16 HYDRONIC PIPING SYSTEMS

This NBCS establishes uniform standards for the design, installation and maintenance of hydronic and refrigerant piping used in heating, ventilation and air conditioning (HVAC) systems. By adhering to these standards, practitioners shall enhance system performance, reduce operational risks and meet regulatory compliance.

The provisions for material selection, sizing criteria, installation practices, testing procedures and maintenance guidelines for HVAC and refrigerant piping are specified.

16.1 General Hydronic Piping Systems

In HVAC systems, piping systems play a crucial role in transporting fluids such as water and refrigerants throughout the building. Some general type of hydronic systems commonly found in HVAC system are:

a) *Chilled Water Piping System* — Network of pipes designed to distribute chilled water from a central plant to various points of use within a building or facility. Primarily used for air conditioning, refrigeration, and industrial cooling applications;

b) *Hot Water Piping System* — Network of pipes designed to distribute heated water from a central heating source to various points of use within a building or facility. Primarily used for space heating, domestic hot water supply, industrial processes, and other applications requiring heated water;

c) *Condenser Water Piping System* — Network of pipes designed to circulate water between a chiller's condenser and a cooling tower or other heat rejection equipment. It's a crucial part of a water-cooled HVAC system, commonly used in large commercial buildings for air conditioning and refrigeration; and

d) *Condensate Drain Piping System* — Network of pipes designed to collect and remove condensate water generated by cooling equipment, such as air conditioners, refrigerators, and dehumidifiers. Condensate forms when warm, moisture-laden air comes into contact with a cold surface, causing the water vapor in the air to condense into liquid water.

16.1.1 *Open and Closed HVAC Piping Systems*

a) *Open and closed system* — An open system is one in which the water enters a reservoir that is exposed to the atmosphere such as collecting tanks, air washers, and cooling towers. A closed system is one in which the water flow is never exposed to the outside environment except at expansion tank part; and

b) *Once – thru and recirculating* — Piping system shall be once through that is, when water is released from system, it only goes through the machinery once. Water circulates in a recirculating system by going through the refrigeration equipment, the heat exchanger, and back again instead of being discharged.

16.1.2 *Selection of Pipe and Fitting Materials*

Selection of right HVAC pipe material shall be done considering fluid type, pressure and temperature range, system layouts, space constraints and special requirements if any (flexibility/corrosion resistance):

a) *Material options for HVAC pipe*:

   1) *Steel pipe* — Carbon steel or stainless-steel pipes should be used in HVAC systems for their strength, durability, and resistance to corrosion. They are suitable for high-pressure applications such as chilled water, hot water, and steam distribution;

   2) *Copper pipe* — Copper pipes have excellent thermal conductivity, corrosion resistance, and ease of installation. They should be used for refrigerant lines, domestic water supply, and hydronic heating systems;

   3) *Polyvinyl chloride (PVC) pipe* — PVC pipes are lightweight, cost-effective, and resistant to corrosion. They may be used for low-pressure applications such as drainage, venting, and condensate lines in HVAC systems;

   4) *Cross-linked polyethylene (PEX) pipe* — PEX pipes are flexible, durable, and resistant to corrosion and scale buildup. They may be used for radiant heating systems, and hydronic piping systems in HVAC; and

   5) *High-density polyethylene (HDPE) pipe* — HDPE pipes are highly resistant to corrosion, abrasion, and chemical damage, making them suitable for outdoor or underground applications in HVAC systems, such as geothermal heat pump systems and chilled water distribution.

b) *Pipe fittings terminology*:

   1) *Elbows* — Elbows are used to change the direction of flow in a piping system. They are available in various angles (for example, 45°, 90°) to accommodate different piping layouts;

   2) *Tees* — Tees create branch connections in a piping system, allowing fluid to flow in multiple directions;

   3) *Reducers* — Reducers connect pipes of different sizes, allowing for a smooth transition between different sections of the piping system;

   4) *Couplings* — Couplings join two pipes together end-to-end, providing a leak-tight connection;

   5) *Flanges* — Flanges connect pipes to valves, fittings, or equipment. They provide a secure and leak-proof connection and allow for easy disassembly when required; and

   6) *Unions* — Unions are similar to couplings but allow for easy disassembly of pipe joints for maintenance or repairs.

   A significant portion of the pressure drop in the piping system is caused by elbows. Elbows with a longer radius shall be provided wherever feasible. Whenever possible, 45° elbows shall be provided over 90° elbows when arranging offsets.

c) *Piping connection methods*:

   1) *Soldering/brazing* — Soldering or brazing is commonly used to join copper pipes and fittings together in HVAC systems. It involves heating the joint and applying solder or brazing material to create a strong and leak-free connection;

   2) *Threaded connections* — Threaded connections are used with threaded pipes and fittings, allowing for easy assembly and disassembly without the need for soldering or welding. This is usually recommended for low pressure system up to 6.9 kPa. (*see* Fig. 14);

3) *Grooved connections* — Grooved connections use grooved pipes and fittings with rubber gaskets to create a tight seal. They are quick and easy to install and are commonly used with steel pipes in HVAC systems. (*see* Fig. 14);

4) *Flanged connections* — Flanged connections use flanges to connect pipes to valves, fittings, or equipment. They provide a secure and leak-proof connection and allow for easy disassembly when required. (*see* Fig. 14); and

5) *Welding* — Welding is used to create permanent connections between pipes and fittings in HVAC systems. It involves melting the edges of the pipe and fitting and fusing them together with heat. This type of fitting is recommended on high pressure system and larger diameter pipes.

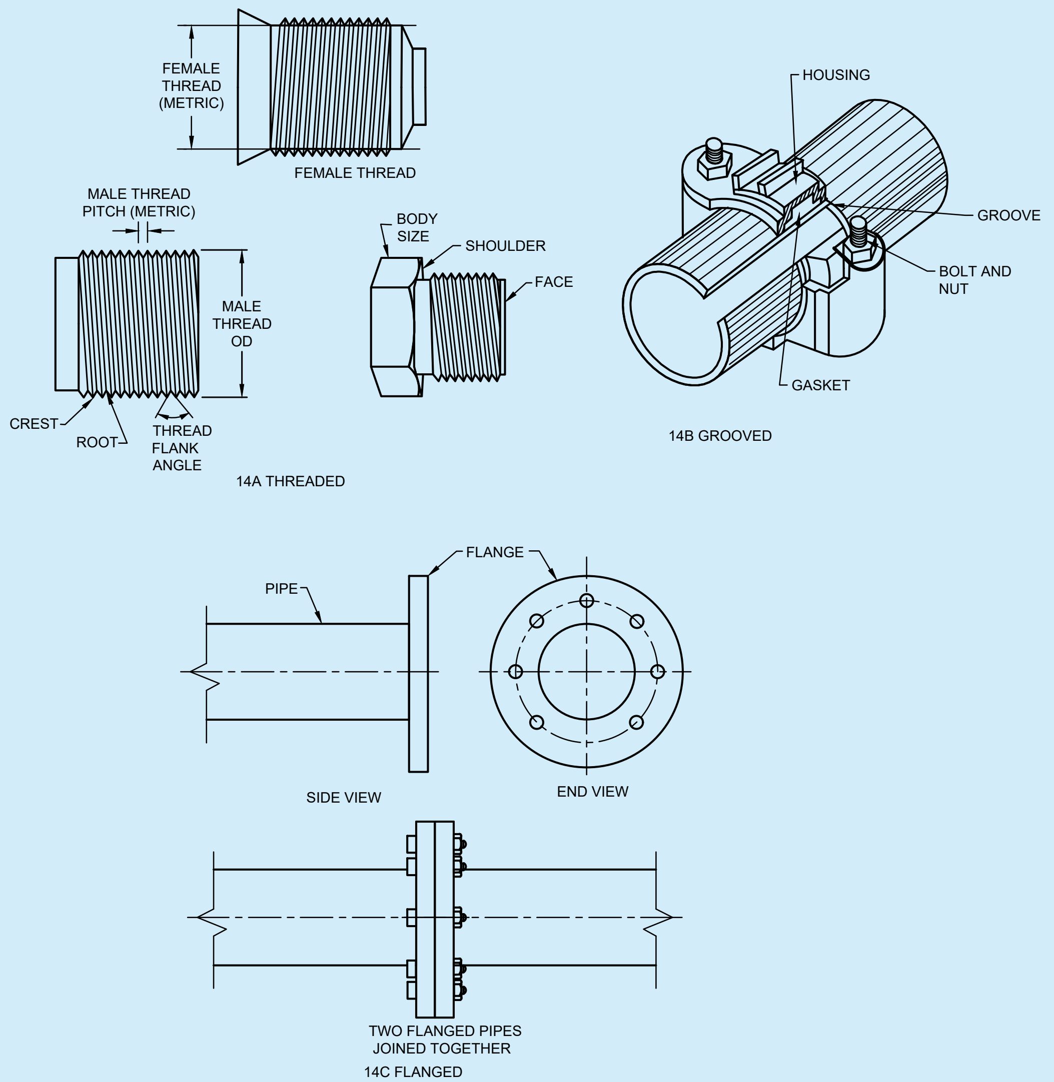


FIG. 14 PIPING CONNECTION METHODS

16.1.3 *Piping Supports*:

a) Pipe supports shall be selected to ensure strong anchoring, absorbing vibrations, providing flexibility and reducing noise transmission;

b) Rigid supports/spring hangers and supports/neoprene pads and inserts/ vibration isolators shall be selected based on load capacity to bear the combined weight of the insulation, pipe, fittings, valves, and fluid in the pipe accounting for static and dynamic loads;

c) Manufacturer's guidelines and specifications should be referred for recommended support type based on pipe material, size and system dynamics; and

d) Consultation with HVAC engineers, vibration specialists or manufacturers to tailor support selection is recommended for specific project requirements.

16.1.4 *Expansion Loops*:

a) Expansion loops should be installed in straight runs of pipe or at points where significant temperature changes occur, such as near equipment or in outdoor installations. They prevent damage to piping, equipment, and surrounding structures caused by thermal expansion; and

b) Expansion loops shall be fabricated from standard piping fittings such as elbows and tees, or they shall be custom-designed to meet specific project requirements.

16.1.5 *Valves – Types and Applications*:

a) *Gate valve* — Gate valves control flow by raising or lowering a gate (wedge-shaped disk) to allow or restrict fluid passage. When fully open, the gate is lifted entirely out of the flow path, minimizing flow restriction. Gate valves may be used for ON/OFF control in HVAC systems where full flow or shut-off is required. They may also be provided for isolating equipment/piping sections for maintenance;

b) *Globe valve* — Globe valves feature a globe-shaped body with a movable plug (the 'disc') that regulates flow by moving up and down. As the disc moves, it alters the size of the flow opening, allowing for precise flow control. Globe valves may be used for pressure balancing, throttling or regulating flow in HVAC systems, such as controlling the flow of water or steam in heating and cooling applications. They may also be used for isolation of equipment/piping sections;

c) *Ball valve* — Ball valves feature a spherical disc (the 'ball') with a hole through the center that controls flow. The ball rotates within the valve body to open or close the flow path. Ball valves may be used for on/off control, isolation, and emergency shut-off;

d) *Butterfly Valve* — Butterfly valves use a circular disc (the 'butterfly') mounted on a shaft to control flow. The disc rotates perpendicular to the flow direction to open or close the valve. Butterfly valves should be used for large-diameter pipes in HVAC systems for ON/OFF control;

e) *Check Valve* – Check valves allow fluid to flow in one direction only, preventing backflow or reverse flow. They typically consist of a movable disc or flap that opens to allow forward flow and closes to prevent reverse flow. Check valves should be used in HVAC systems to prevent damage to equipment, maintain system efficiency, and ensure proper flow direction in piping systems;

f) *Balancing Valve* — Balancing valves should be used in hydronic systems to achieve proper flow distribution and maintain system balance;

g) *Control Valve* — Control valves regulate fluid flow based on signals from a control system, such as a thermostat or building automation system. They modulate the flow by adjusting the position of a valve plug, disc, or ball in response to changing system conditions. Control valves should be used in HVAC systems for precise temperature control, flow modulation, and energy management in heating, cooling, and air distribution systems;

h) *Pressure Independent Control Valve (PICV)* — A PICV is an advanced type of control valve used in HVAC (Heating, Ventilation, and Air Conditioning) systems for precise flow control and temperature regulation. It combines the functions of a control valve, differential pressure regulator, and flow limiter into a single device. This is recommended to be used to replace balancing and control valve in the system. (*see* Fig. 15); and

j) *Triple Duty Valve* — A triple duty valve is a multifunctional valve and combines the functions of a check valve, balancing valve and an isolation valve into a single unit. Use of these valves simplify installation, save space and improve system efficiency. These are recommended to be used in hydronic chilled water and hot water systems (*see* Fig. 16).

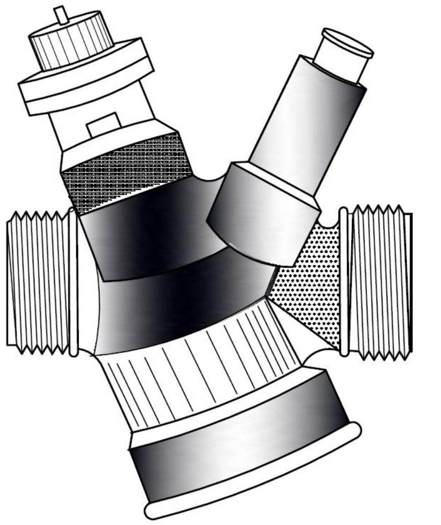


FIG. 15 PICV VALVE

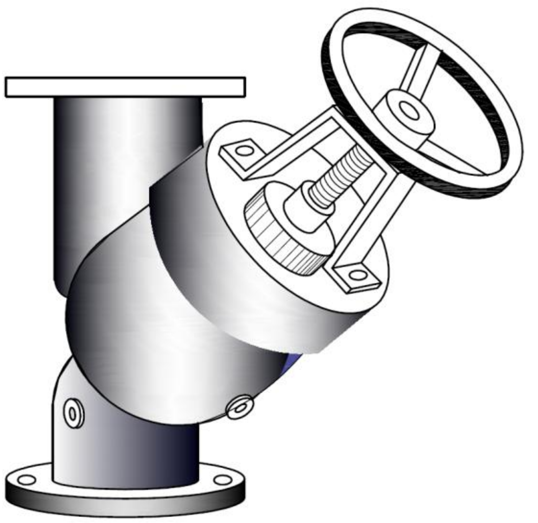


FIG. 16 TRIPLE DUTY VALVE

16.1.6 *Buried Piping Systems*

The following points cover the key aspects of buried piping systems:

a) *Design considerations* — Thermal expansion, corrosion protection and accessibility should be considered in buried piping design. Corrosion-resistant materials like HDPE, steel with protective coatings, ductile iron, or PVC should be selected. These materials shall withstand corrosive environments and prolong the service life of the piping system. Access points like valve boxes, manholes, or cleanouts should be planned along the pipeline route for maintenance;

b) *Material options* — These are material options recommended to be used:

   1) *High-Density Polyethylene* (*HDPE*) — HDPE pipes may be used for buried piping due to their lightweight nature, flexibility, and excellent resistance to corrosion, abrasion, and chemical damage;

2) *Steel* — Steel pipes with corrosion-resistant coatings like epoxy or polyethylene may enhance their longevity in buried applications; and

3) *Polyvinyl Chloride* (*PVC*) — PVC pipes are lightweight, cost-effective, and resistant to corrosion, making them ideal for low-pressure applications in buried piping systems.

c) *Installation considerations* — Proper installation is crucial to ensure the efficiency, safety, and longevity of systems and equipment. The following points highlight key considerations for installation:

1) *Trenching* — Excavation of trenches shall be carried out to the required depth and width to accommodate the piping system. Trench dimensions should be determined based on factors such as pipe size, soil conditions, and local building codes;

2) *Bedding and backfilling* — A layer of bedding material such as sand or gravel should be placed at the bottom of the trench to provide support and stability for the pipes. After installation, the pipes to be backfilled with soil, compacted in layers to prevent settling, and ensure proper support;

3) *Protection* — Buried pipes may be wrapped with protective materials such as geotextile fabric or insulation to shield them from damage due to abrasion, soil movement, or external forces. This protective layer helps preserve the integrity of the piping system over time; and

4) *Marking and identification* — Buried piping systems shall be marked and identified with durable markers or tape along the pipeline route to indicate the location, type of pipe, and purpose of the system. This facilitates future reference, maintenance, and repair activities.

16.1.7 *HVAC Piping in High Rise Buildings- Design Considerations*

Designing of HVAC Piping systems for high rise buildings should consider efficiency, reliability and safety aspects. Material selection shall be suitable for building's structural and environmental conditions. Factors such as corrosion resistance, thermal expansion and fire safety ratings should be considered. HVAC piping in high rise buildings should be designed to handle the required pressure ratings. Accounting for pressure drops due to friction losses, fittings and changes in elevations. Design should facilitate ease of maintenance and future expansion. Few recommendations are:

a) Ensure valve controls and access points are easily available for servicing and repair;

b) Implement measures to mitigate noise and vibration transmission through HVAC piping; and

c) Comply with fire safety regulations by using fire rated materials and design pipe routes to minimize fire spread and maximize compartmentation.

16.1.8 *HVAC Piping in Radiant Cooling Systems-Design Considerations*

Designing piping systems for radiant cooling involves several critical considerations:

a) *Pipe material selection and sizing* — Piping with high thermal conductivity such as cross-linked polyethylene, copper or specialized composite pipes should be used. Select pipe sizes based on flow rates required to meet cooling/ heating demands. Zoning strategies should be implemented to allow different areas to be heated or cooled independently based on occupancy and usage patterns;

b) *Floor construction and thermal mass* — The type of floor construction should be assessed (for example, concrete/wood) to determine its conductivity for efficiency of radiant system. Ensure proper insulation beneath the radiant pipes to minimize heat loss downward;

c) *Integration with building design* — Radiant piping layout to be coordinated with other services like electrical, plumbing and structural components; and

d) *Maintenance and accessibility* — Access to manifold stations and control valves should be planned for maintenance and repair.

16.1.9 *HVAC Piping Testing, Commissioning and Maintenance Guidelines*

Testing, commissioning, and regular maintenance of HVAC piping systems to ensure proper functioning and long-term reliability. The following points outline the guidelines for these processes:

a) *Testing procedures* — The periodic testing procedures recommended to ensure efficiency, longevity and reliability of HVAC piping system shall include pressure testing to check integrity of piping joints and connections; leak detection tests to identify and repair any leaks; flow tests to measure flow rates to check they meet design specifications and adjust as necessary.

b) *Balancing and commissioning process* — The piping system shall be balanced for even distribution of flow and pressure across all the branches. Verify system performance against design criteria and operational requirements. The commissioning procedure shall involve flushing, purging and filling the system with appropriate fluids.

c) *Regular inspection and maintenance* — A regular maintenance schedule should be established based on manufacturers recommendations and operational demands:

   1) This shall include but not limited to visual inspection for signs of corrosion, damage or leaks. Periodic cleaning of pipes should be done to remove any debris or build up to maintain optimal flow and efficiency; and

   2) Valves operation shall be tested to ensure proper functioning and prevent sticking or seizing. Inspect and adjust pipe supports and hangers for any sagging or stress in joints.

d) *Documentation* — Maintain comprehensive documentation of testing, balancing, commissioning activities including test results.

16.2 Refrigerant Piping

The following points outline key considerations for the design, installation, material, and maintenance of refrigerant piping

16.2.1 *Considerations in Refrigerant Piping*

The refrigerant when flowing through these pipes, imposes friction with the inner piping wall due to the velocities. The friction reduces the system efficiencies and as such shall be maintained as low as possible. The norm is to maintain the pressure drops in these pipes, not exceeding an equivalent of 1.10°C which for most of the commonly used refrigerant works out to 1.3 kPa to 1.9 kPa for suction line and 4.0 kPa to 6.0 kPa for hot gas and liquid lines. This shall however be verified with the refrigerant pressure-temperature charts.

The second consideration is oil migration during operation. When refrigerant flows out of the compressor in vapor form, some amount of compressor lubrication oil escapes along with the refrigerant. The piping system shall be designed to ensure that the escaped oil is made to return back to the compressor after travelling through the system to ensure that the oil levels in the compressor is maintained. Following practices should be followed to achieve this objective:

a) The refrigerant velocities in suction line as well as hot gas line be maintained high enough so that it impels oil to move along the direction of the flow. The velocities in horizontal pipes should be minimum 3.8 m/s and that in vertical lines be maintained at a minimum of 7.6 m/s;

b) The suction line and hot gas line in horizontal traverse should be pitched downwards in the direction of the flow to ensure that the oil does not flow back. The recommended gradient is 1 : 200;

c) When the suction line and hot gas line traverse in vertical direction, the U-shaped trap shall be provided at the bottom and inverse U-shaped trap shall be provided at the top. The bottom trap provides holding space for the oil in vertical pipe and not allowing it to travel backward. The top trap prevents the oil that has reached the top from traversing back;

d) In the event of vertical pipe for suction line or hot gas line travelling upwards exceeds 3.5 m, a U-shaped oil trap should be installed at every 3.5 m;

e) In the event the compressor has part load operation capabilities, the piping velocities in vertical risers may not be possible at the lowest capacity operations. In such events, a double suction riser should be installed in vertically upward pipes. One riser should be sized for the minimum capacity and the second riser should be sized for the balance capacity; and

f) Since liquid refrigerant is perfectly miscible with the oil, the refrigerant velocity is not needed for transportation of oil. However, sudden closing of refrigerant flow shall result in liquid hammer effect. To minimize this impact and consequent damage to the piping system, the refrigerant velocities in liquid line should not exceed 1.5 m/s.

16.2.1.1 The third consideration is migration of liquid refrigerant during the system off cycle. When the system is not operational, the piping layout shall ensure that the liquid does not migrate either to the evaporator or the compressor.

a) When the relative position of compressor is at lower elevation than the condenser, the inverted U-shaped riser should be provided at an elevation higher than the condenser;

b) When the evaporator is at the lower elevation than the condenser, once again a similar inverted U-shapes riser should be employed. It is also a good practice to employ controls such that the system is switched on only after a time delay after the evaporator pump/fan is switched on; and

c) Pump down control may be employed, particularly on the larger systems to shut off the flow of refrigerant by installation of solenoid valve so that when the system is off cycle, the most of the liquid refrigerant is stored in condenser or receiver.

16.2.2 *Piping Material*

Since the refrigerant in the system operate at high pressures, the piping material and thicknesses should be complaint with the subjected pressures. Following materials could be used for refrigerant piping.

a) Steel tubes and pipes;

b) Copper tubes and pipes;

c) Aluminium tubes; and

d) Copper alloy tubes.

NOTE — For copper tubes and pipes, When the pipes are laid in an environment where pipes may be subjected to damage, soft annealed copper tubes should be avoided.

Any other proprietary material may be used provided they are rated for the subjected pressures as well as the verified data of pressure drops is well documented.

Post installation of piping the entire piping system should be pressure tested at 1.5 times the operating high pressure and joints tested for leaks. Post the pressure testing the system should be vacuumized to a pressure of 500 microns with holding of vacuum for the period of 4 h to 6 h to ensure that all non-condensable vapours are removed from the system and any trace of remaining moisture is removed.

The suction line pipes shall be adequately insulated. The insulation thickness should be calculated minimize the heat loss as well as to avoid condensation over the surface. Hot gas line insulation may not be deployed unless the pipes are laid in an environment where the hot gas temperatures may cause burns to occupants. Liquid lines may not be insulated. However, if the equipment is with reversible cycle such as heat and cool mode, all pipes need to be insulated. The insulated pipes should be protected from possible damages as well as the ultra violet exposures.

The piping system should be adequately supported. The supporting system should not damage or compress the insulation system integrity. Alternatively, the piping may be rested on well supported continuous trays.

16.2.3 *Refrigerant Piping Accessories*

The piping system should incorporate accessories which will either aid in proper operation of the system or may assist in trouble shooting:

a) *Oil separators* — Oil separators may be mounted in the hot gas line at the outlet of the compressor when piping lengths are excessively high or when the refrigerant velocities are on the border line. The refrigerant piping velocities however, should not be compromised;

b) *Receivers* — Receivers should be installed where the condenser volume is not adequate to store the entire refrigerant charge of the system. Receivers should be properly designed so that no flash vapor shall enter the liquid line;

c) *Filter dryers* — Filter dryers should be installed in the liquid line to ensure that all solid impurities are arrested prior to the entry of refrigerant in the expansion device. These devices also aid in removing minor amount of water vapor from the piping system that may have been left behind;

d) *Sight glass* — Sight glass may be installed in the liquid line to visualize the adequate flow of refrigerant as well as for to view adequate system dryness; and

e) *Isolation valves* — Isolation valves either in form of hand shut-off valve or solenoid valve may be installed upstream and downstream locations of filter dryer for maintenance works.

17 INSTALLATION AND COMMISSIONING OF HVAC SYSTEMS/EQUIPMENT

The installation and commissioning of HVAC systems are critical to ensuring the system operates efficiently, safely, and in compliance with design specifications. The following points outline the objectives and planning considerations for the installation and commissioning process.

17.1 Objectives

The objectives for the installation and commissioning of HVAC systems and equipment shall be to fully meet the owner’s project requirements. The installation process must prioritize optimal energy efficiency, ensuring the systems operate reliably and maintain high-quality standards. Safety is a top priority, with all installations executed to the highest safety standards. Timeliness to meet project deadlines, while also ensuring the optimum life cycle cost is achieved. In addition, the systems should be designed and commissioned to minimize noise and vibration, providing a comfortable and efficient environment.

a) Shall fully meet owner’s project requirements;

b) Optimal energy efficiency;

c) Reliability/Quality;

d) Safety;

e) Speed/Timeliness;

f) Optimum life cycle cost; and

g) Minimum noise and vibration.

17.2 Planning the Installation

Meticulous planning results in a trouble-free installation at optimum time and cost with little chance of wastage and rework. Planning ensures safety during installation and commissioning.

Following factors are common across all types of equipment/systems:

17.2.1 Location of installation shall be selected with care. Location shall be such that there is adequate space to mount the equipment as well as to operate, service and maintain it. It shall be as close to the area to be served as possible. Good ventilation and natural lighting are important.

Where the installation is large, a separate utility block should be considered along with a service tunnel to house chilled and condenser water piping, electrical and control cabling and communication systems.

Equipment room floor shall be light coloured and finished smooth but non-slippery with epoxy paint coating. Where required, water proofing of plant/equipment room floor shall be done.

17.2.2 Noise generated by the equipment shall not be objectionable to surrounding spaces and any vibration generated shall not be transmitted to the building structure. Noise levels to be as per CPCB norms.

17.2.3 Equipment rejecting heat to surroundings shall be mounted in such a way that there is no obstruction to heat rejection and good ventilation and adequate top and side clearances are provided.

17.2.4 Location shall be such that proper access to material handling equipment such as crane, forklift etc is available not only during installation but also for its removal and replacement at a later date. When equipment has to be located in the basement of the building, service ramps or removable hatches shall be provided for this purpose.

17.2.5 Structure should be able to take the static and dynamic load of the equipment and transfer it to the ground.

17.2.6 Location shall be dust free and clean.

17.2.7 Equipment which needs drainage shall have the facility. For basement installations, provision shall be kept to pump out excess water from sumps provided. These sumps shall be coupled to alarms in case water level rises beyond set point. In no case shall the condensate water be left out in the open. It shall either be connected to the building drainage and if the installation is large, options for condensate recovery and re use shall be explored.

17.2.8 Locations of shafts, wall openings etc shall be planned in advance so that breakage/rework shall be avoided. Pipes may be supported from floor or ceiling slab as specified by designer. All floor and ceiling support for pipes, cable trays, ducts etc shall be suitably isolated from building structure to prevent transmission of vibration. The openings around service shafts shall be closed as specified. Openings in structural elements (beam, slab etc) shall not be made without the approval of structural engineer.

17.2.9 Equipment foundations shall be structurally designed and concrete foundation blocks shall be protected by epoxy coated angle nosing. Inertia blocks and seismic restraints shall be used where specified. Floor finish/slope shall be such that any water spillage does not stagnate around equipment foundations.

17.2.10 Fresh air intakes shall be planned taking into consideration required distance from exhaust outlets as well as other sources of contamination. All such intakes and exhausts which open outside the building shall be provided with wire mesh and bird screen on inside and louvres with rain protection or cowl piece on the outside. Where specified, intake air filters shall be used. When any duct is terminated on the roof, it shall have a swan’s neck to prevent rain ingress.

17.2.11 Access to equipment rooms shall be through double leaf doors or rolling shutters of adequate width and height and fan room doors shall be airtight. Doors shall open outside and a sill provided to prevent water ingress.

17.2.12 It shall be possible to isolate equipment rooms in case of fire or distress. For provisions for compartmentation in case of fire, *see* **19**. Escape routes shall be planned and clearly marked on site.

17.2.13 All equipment/materials delivered to site shall be compared with the documented and approved specifications and found correct. Test certificates shall be sought and verified for physical properties such as thickness, malleability, test pressure etc. Where independent testing is specified, third party test reports from labs [D-3(40)] accredited shall be produced. Agreed lot sizes shall be inspected at site before use. Where factory inspection is specified, same shall be carried out as provided in contract. Agreed quality assurance plan (QAP) shall be specified and test results recorded suitably. All instruments and gauges shall be calibrated, and valid calibration certificates shall be available.

17.2.14 All equipment's, materials and sub systems shall be cleaned prior to use. Installed sub systems such as duct and pipe work shall be cleaned as per an approved Standard operating procedure and shall be kept with all open ends suitably closed. All equipment shall be kept covered with tarpaulin /canvas cloth till they are ready to be commissioned. Shafts of electric motors and other rotating equipment, if stored for a long time, shall be rotated by hand once a fortnight so that grease applied on them do not solidify.

17.2.15 Testing of sub systems shall be carried out in sections as per an approved schematic drawing. Applicable standards and test pressures/duration of test shall be approved by the owner. Testing schedules shall be prior intimated to owner and tests shall be witnessed by owner's representative or by commissioning authority (CxA). Owner shall have the right to approve contractor's testing personnel regarding qualification and experience.

17.2.16 Number of test points for duct and pipe work shall be pre agreed between owner/CxA and the contractor. Once testing of individual sections are completed (with suitable isolation), the entire system shall be tested as a whole. Power and control cabling shall be tested as one single unit.

17.2.17 Receiving of equipment at site, its unloading, shifting to place of installation or interim storage shall be done with care and by using suitable tools and tackle. This requires advance planning and providing adequate lifting and shifting equipment at site. This is of special significance in case of heavy and bulky equipment such as chillers, fans etc to be installed in basement or building terrace. To the extent possible, heavy equipment shall be received at site only after its location of installation is fully ready and at the first instance the equipment should be located at the right place of installation. Multiple shifting of heavy equipment shall be avoided.

17.2.18 Source of power and water shall be provided as close to point of use as possible and shall be terminated with an isolator/valve. All electrical systems shall be properly grounded and electrical protections such as surge protectors/circuit breakers installed as required. When remote start/stop or control is specified, adequate provisions shall be made.

17.3 Specific Requirements for Equipment/Systems

17.3.1 *Room* (*Window*) *Air Conditioner*

There shall be an opening of adequate size in the outer wall of the room (can be a window) and a wooden frame of suitable size shall be fitted therein. The air conditioner unit shall be placed inside the frame with its outer casing firmly fitted inside the wooden frame with metal screws keeping a slope of 6 mm to 10 mm for drainage of condensate. Drain line of adequate size shall be fitted to the back of the unit. Unit shall be properly levelled and any gap between the frame and wall/window shall be sealed off. Adequately rated electric power socket with on/off switch shall be provided close to the unit. Before fixing the front grill on the unit, it shall be ensured that supply and return air openings are properly isolated and short cycling of air does not happen (*see* Fig. 17, Fig. 18 and Fig. 19).

In addition, the installation shall be as per the manufacturer specified installation requirements and instructions. In absence of manufacturer's installation instruction and requirement.

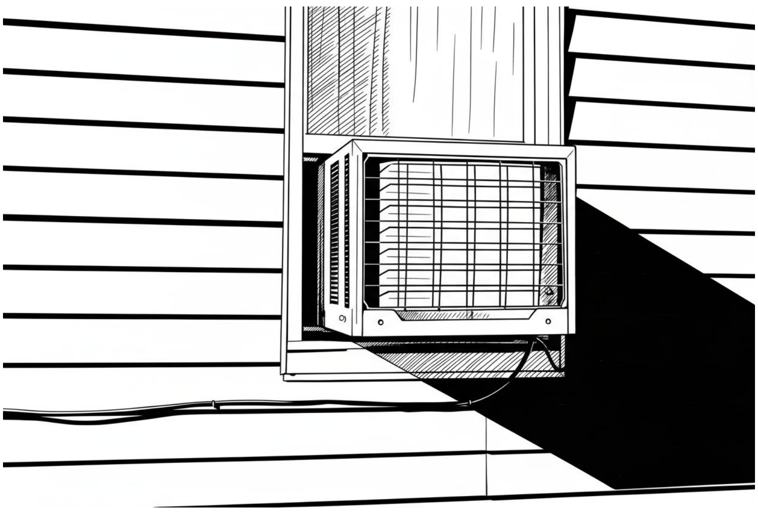


FIG. 17 ROOM AC INSTALLATION – WALL MOUNTED

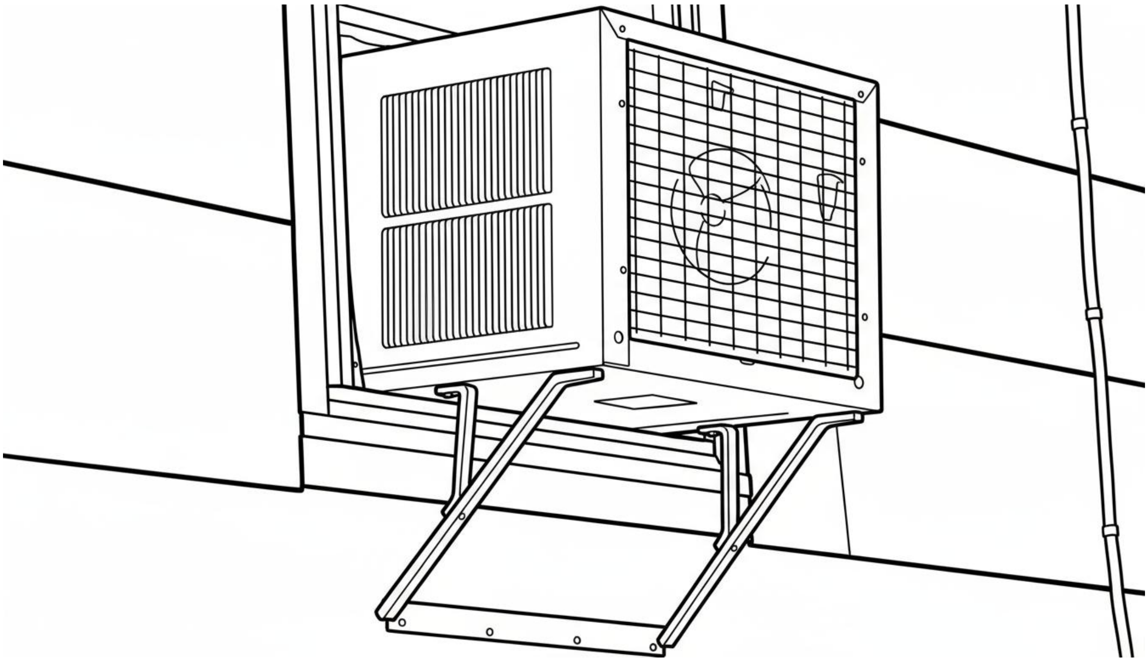


FIG. 18 ROOM AC INSTALLATION – SUPPORTED ON BRACKETS

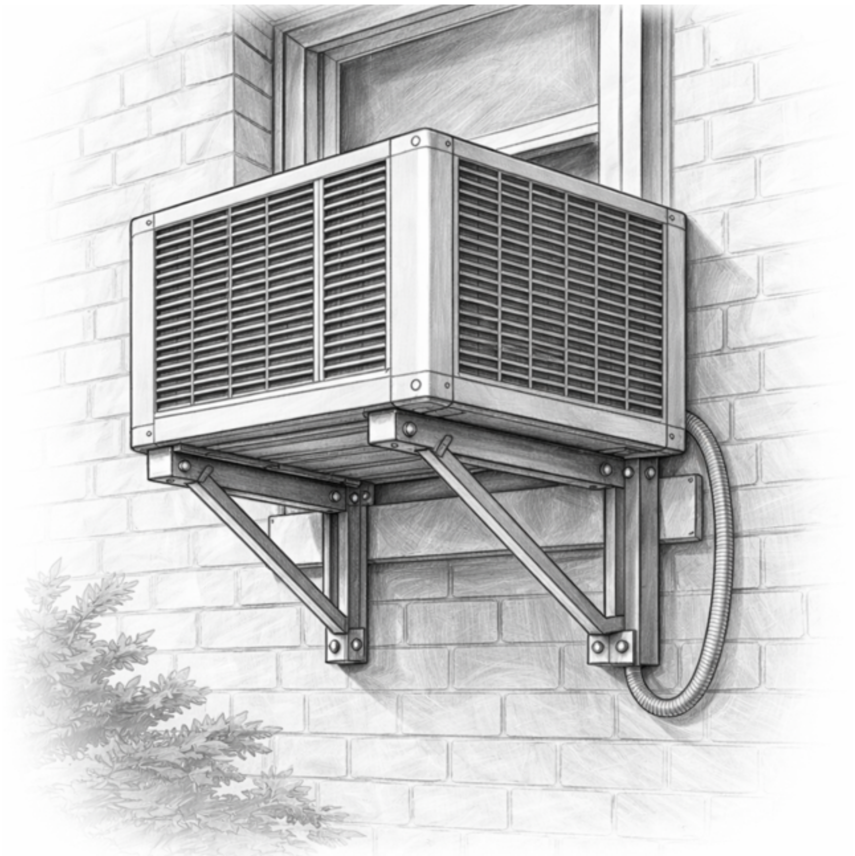


FIG. 19 WINDOW AIR CONDIONING UNIT INSTALLATION

17.3.2 *Split Air Conditioner*

The indoor unit (IDU) shall be located in the conditioned space at a suitable location away from any obstacles to airflow. IDU shall be accessible for regular filter cleaning and maintenance. Drilled holes in the wall to locate the IDU are best located with a template and a spirit level. A hole of adequate diameter shall be drilled in the wall close to the unit (adjacent to the refrigerant pipe connection side of the unit) through which supply and return refrigerant piping (both insulated), drain pipe to remove condensate water and power cabling to the outdoor unit (ODU) shall be carried. A PVC sleeve may be used to carry the lines and the annular space between the wall and the sleeve shall be filled with caulking material or sealant (*see* Fig. 20 and Fig. 21).

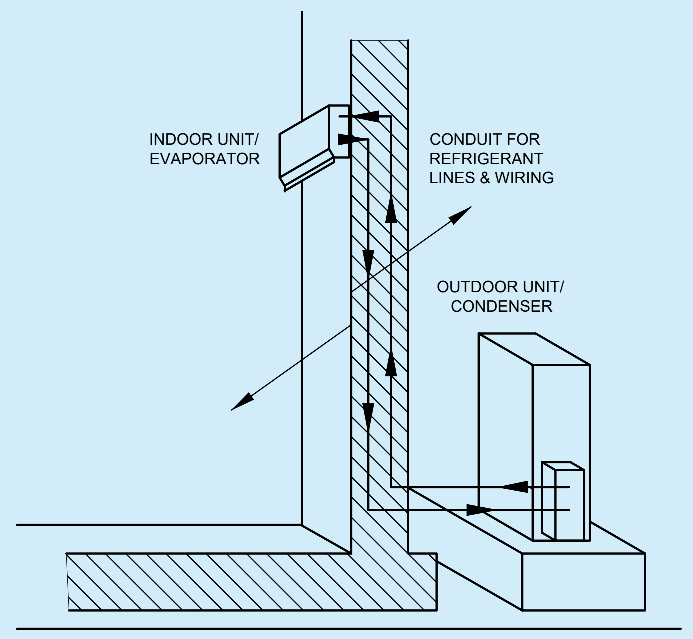


FIG. 20 TYPICAL SPLIT AC UNIT PIPING

FIG. 21 TEMPLATE FOR LOCATING SPLIT INDOOR UNIT (IDU)

Refrigerant piping between the IDU and ODU shall be of copper with minimum number of bends and the minimum radius of each bend shall be 10 mm. Maximum piping length between IDU and ODU, difference in height between IDU and ODU and any additional refrigerant charge/lubricating oil charge that may be required shall all be specified by the equipment supplier. In cases where the IDU is located at the same level or above the ODU level, the suction line shall rise to the top of the IDU. This prevents the refrigerant liquid from migrating to the compressor during off cycle. Horizontal suction lines shall be either straight or slightly sloped towards the ODU. A refrigerant trap shall be installed at the bottom of the vertical riser (*see* Fig. 22 and Fig. 23).

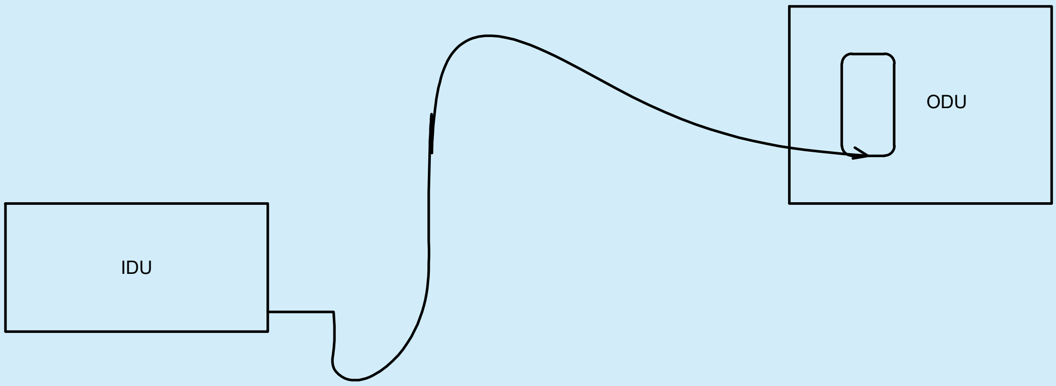


FIG. 22 SUCTION PIPE WITH U TRAP

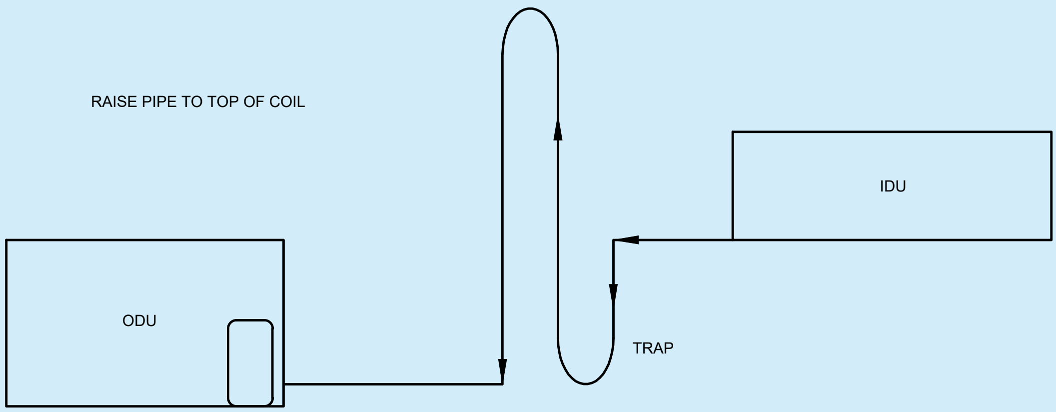


FIG. 23 SUCTION PIPE WHEN ODU IS BELOW IDU LEVEL

Where the ODU is located above the level of the IDU, a trap shall be installed at the bottom of the suction riser immediately after the IDU.

Pipes shall not be in direct contact with the building structure to prevent vibration transmission. Pipes shall be clamped to walls at suitable intervals and separation may be kept between pipe and clamp to prevent bi-metallic erosion.

ODU shall be placed on a flat firm surface which shall take the load of the unit. Vibration isolators or serrated rubber pads shall be placed below the unit. The ODU shall be located where air for cooling shall flow freely through the condenser coil. As far as possible, ODU shall not be exposed directly to sunlight. ODU shall be easily accessible for servicing. Where multiple ODUs are located, they should not face one another, and an adequate arrangement shall be made to prevent short cycling.

Once the piping between the ODU and IDU is completed, it shall be pressure tested at 21×10^4 Kg/m^2 on the high side and 10×10^4 Kg/m^2 (or 1.5 times the maximum operating pressure) on low side for minimum 6 h. Thereafter, the system shall be evacuated and a vacuum of 500 microns of Hg shall be maintained for 6 h before refrigerant is charged. Refrigerant charge/oil charge shall be as recommended by the manufacturer.

Condensate drain from the indoor unit tray shall be planned, routed and connected to the designated drain point (*see* Fig. 24).

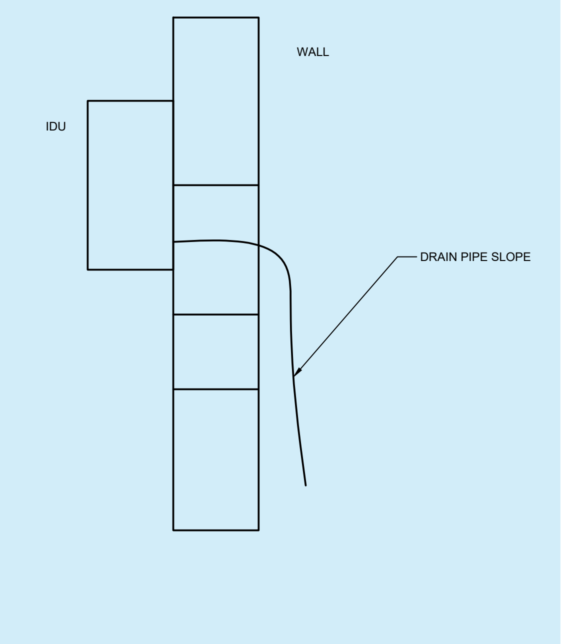


FIG. 24 TYPICAL DRAIN CONNECTION FOR IDU

Unit shall be started only by the supplier's authorised service personnel. All installation and start up checks shall be as per the manufacturer's/supplier's documented SOP.
In addition, the installation shall be as per the manufacturer specified installation requirements and instructions. In absence of manufacturer's installation instruction and requirement.

17.3.3 *Variable Refrigerant Flow* (*VRF*) *Systems or Multi Split Units*

All requirements detailed in **17.3.2** shall apply (*see* Fig. 25).

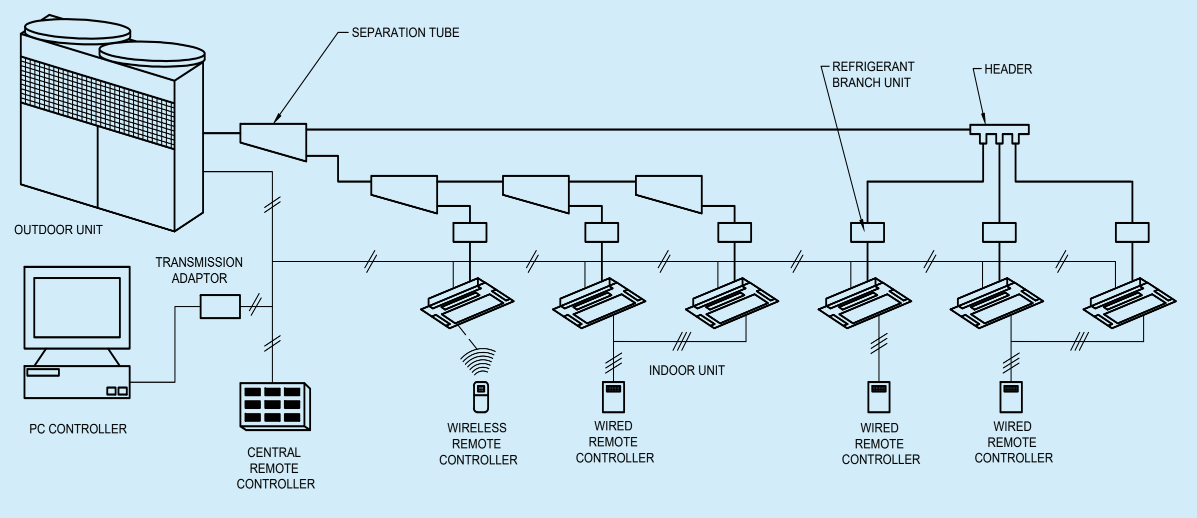


FIG. 25 TYPICAL VRF CONFIGURATION.

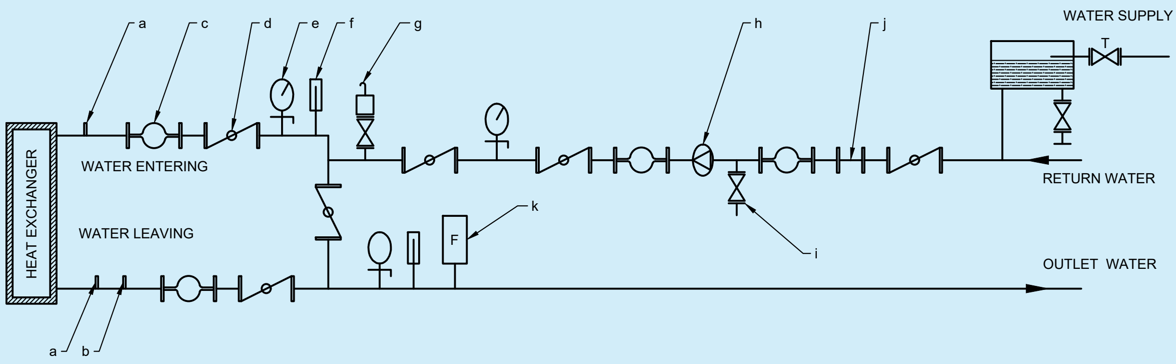
Since the same (set of) outdoor units serves multiple indoor units with varying lengths of piping, charging the correct amount of refrigerant and lubricating oil is important and manufacturer’s instruction shall be followed. Communication wiring between all indoor units and the corresponding ODU shall be connected as per the manufacturer’s instructions and communication cable shall be away at a distance minimum 300 mm to avoid interference.

Allowable length of refrigerant piping and height between various IDUs and between IDU and ODU shall follow manufacturer’s specification. Ball valves shall be installed for zoned refrigerant isolation. Evacuation shall be possible at each IDU and ODU. Start up by manufacturer/installer’s authorised personnel shall include configuring and setting up of all control devices.

17.3.4 *Chiller* (*Air Cooled and Water Cooled*)

Chillers are heavy and built with either rotating or reciprocating equipment. adequate foundation system shall be provided to transfer the load to ground thru beams/columns and this shall be done in consultation with structural specialist. Dynamic loading shall be considered while designing the structure and shall be as per the accepted standard [D-3(41)]. Vibration isolation mounts or serrated rubber pads (as recommended by manufacturer) shall be placed between the chiller frame and the foundation. Where open type compressors are used, the alignment of the compressor with its motor shall be checked with a dial gauge (*see* Fig. 26).

Piping connections to the chiller shall be made only after the chiller is finally located in place and levelled. Piping ends shall have offsets for flexibility and piping shall be supported independent of the chiller. Coolers and condensers shall be connected through victaulic/equivalent couplings and suitable flexible connections as recommended by the manufacturer. Provision for fixing temperature sensor and pressure gauges shall be made immediately after the condenser and cooler at a distance of minimum 150 mm from the outlet.



a	TEMP. SENSOR	b	THERMO SWITCH	c	SHOCKPROOF JOINT
d	BUTTERFLY VALVE	e	PRESSURE GAUGE	f	THERMOMETER
g	AUTOMATIC AIR RELIEF	h	WATER PUMP	i	DRAIN VALVE
j	Y - SHAPE FILTER	k	WATER FLOW SWITCH		

FIG. 26 TYPICAL CHILLER PIPE CONNECTIONS.

Temperature at the location where chiller is installed shall be within the limits specified by the manufacturer. Where air-cooled chillers are installed open to atmosphere, a suitable cover over the roof panels is recommended. All electrical enclosures in case of open installations shall be IP 55 or higher as per accepted standard [D-3(42)]. Space for air circulation, people movement and maintenance shall be provided on all four sides.

Where the chiller is mounted above grade, a structural steel platform, capable of supporting the dynamic load of the chiller, shall be provided. A maintenance platform providing access to all parts requiring maintenance/ attention shall also be provided. Service space shall include space for tube cleaning or for attaching the tube cleaning device.

Electrical isolation shall be provided not more than 2 m away from the chiller. Floor drain and water supply point near to the chiller shall be planned. Where accessories to chiller such as expansion tank, air separator unit etc are to be located in the plant room, adequate space for installing and servicing them shall be planned.

Where chillers are located inside the building, floor to floor height of the plant room shall be adequate to house the piping, valves, strainers etc. Height ranging from 3.6 m to 4.5 m is recommended depending on chiller type. A guide rail arrangement is recommended over the chiller for easy removal of components in case of failure. The chiller concrete foundation base should be 150 mm to 200 mm above floor level.

17.3.4.1 Quality of water used for condenser circulation and chilled water make up greatly influences chiller performance as well as chiller life. Make up supply of soft water may be provided at a suitable height above the chiller allowing flow by gravity. Manufacturer's recommendation in this regard shall be strictly followed even for the water used for pressure testing and commissioning. The Table 39 should be referred for the water quality requirements.

Table 39 Water Quality Requirements

(*Clause* 17.3.4.1)

Sl No.	Properties	Limit	Circulating Chilled Water	Make up Water (Chiller)	Condenser Water
(1)	(2)	(3)	(4)	(5)	(6)
i)	Total hardness as CaCo3	*Max*	80	50	200
ii)	Total alkalinity as CaCo3	*Max*	100	80	100
iii)	Chloride ion, ppm	*Max*	50	50	200
iv)	Iron as Fe, ppm	*Max*	0.3	0.3	1.0
v)	Silica as SiO2, ppm	*Max*	30	30	50
vi)	Ammonia, ppm	*Max*	0.2	0.2	1.0
vii)	*p*H	*Min* to *Max*	7.2 to 8.5	6 to 8	7.2 to 8.5

17.3.5 *Packaged Air Conditioner (Air Cooled and Water Cooled)*

Installation of water-cooled packaged air conditioners (PAC) shall be similar to installation of water-cooled chiller whereas installation of air-cooled chiller shall generally follow the installation guidelines for split air conditioners as detailed in **17.3.2**. Following additional points are specific to PAC installation

a) While lifting and hoisting the unit with a sling, spreader bars shall be used to prevent damage to sides and top;

b) In respect of air-cooled PAC where piping length is more than 7 m, additional refrigerant/oil may require to be charged;

c) Unit connections shall be through reinforced flexible connector;

d) Condensate drain with a water trap shall be installed next to the access panel of the unit; and

e) Pre-filters shall be anti-bacteria and shall be washable. Unit shall not be operated without filter in place.

17.3.6 *Pumps*

17.3.6.1 Pump base frame shall be mounted on an adequately sized concrete block and there shall be cork or serrated rubber pads or anti-vibration mounts between the concrete block and plant room floor and shall be as per the accepted standard [D-3(41)]. Load shall be transmitted to beam or column as per advice of structural engineer.

17.3.6.2 More than one pump shall not be installed on a single base frame or concrete block. Factory supplied pumping systems mounted on common base frame is an exemption to this.

17.3.6.3 Foundation bolts where required shall be embedded correctly. Before grouting the bolts, the level of the base frame shall be checked.

17.3.6.4 Pump motor shall be mounted on the base frame, alignment checked and then only coupled with the pump shaft. Pump shall not run without its coupling guard. Pump alignment shall be checked once again after completing the suction and discharge piping in order to ensure that alignment is not disturbed after connecting the piping.

17.3.6.5 Before being test run, the pump shall be lubricated as per manufacturer's instructions.

17.3.6.6 Suction and discharge lines shall be supported independent of the pump and shall be connected to the pump through flexible connectors. Isolation valves shall be provided on both sides (suction and discharge). Provision for pressure gauge connections at the inlet and outlet of the pump shall be made (*see* Fig. 27).

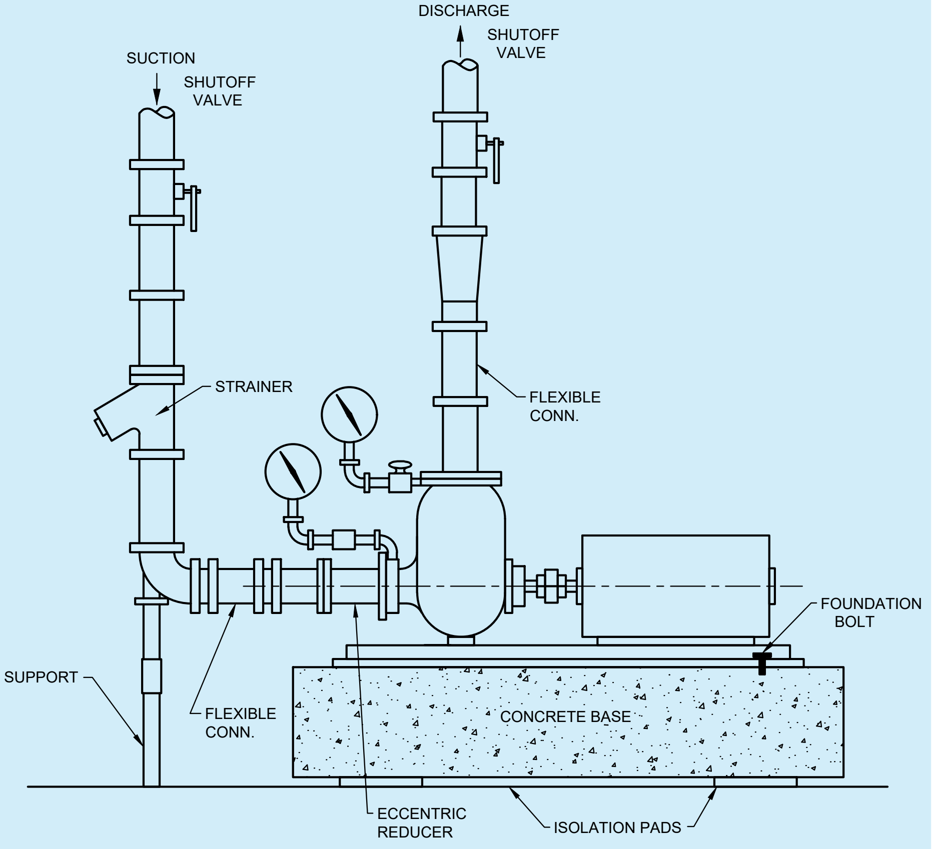


FIG. 27 TYPICAL PUMP INSTALLATION DETAILS.

17.3.6.7 In-line pumps may be installed directly in the piping with pipe hangars adequately sized to handle the load of the pump and piping. For larger size in -line pumps, a flange support or floor support with saddles shall be used. All supports to floor and ceiling shall be vibration isolated.

17.3.6.8 Correct direction of rotation of the pump (which shall be indicated on the housing) shall be checked before start up.

17.3.6.9 A factory installed drain connection with a site fitted installation valve shall be used to drain the pump reservoir.

17.3.6.10 Where low temperatures are encountered, freeze up precautions shall be taken.

17.3.7 *Cooling Tower*

17.3.7.1 Cooling tower shall be located in such a way that noise as well as water drift are not objectionable. Tower location should not be close to leaf shedding trees.

17.3.7.2 Location shall be open and well-ventilated and manufacturer's instructions regarding side clearances for air circulation shall be meticulously followed.

17.3.7.3 If located on building terrace, the operating load shall be transferred to beams/columns as per advice of structural engineer. Cooling tower shall rest on adequately sized MS I-sections. These I-sections shall be hot dip galvanized to accepted standard [D-3(43)]. All welded joints shall be primer with corrosion resistant coated such as epoxy. Vibration isolation shall be provided between tower and the MS I-Sections.

17.3.7.4 Installation shall be such that the space below the cooling tower shall accommodate a strainer below the water level in the sump as well as a drain pipe with a valve. It shall be possible to completely drain the cooling tower sump when required (*see* Fig. 28).

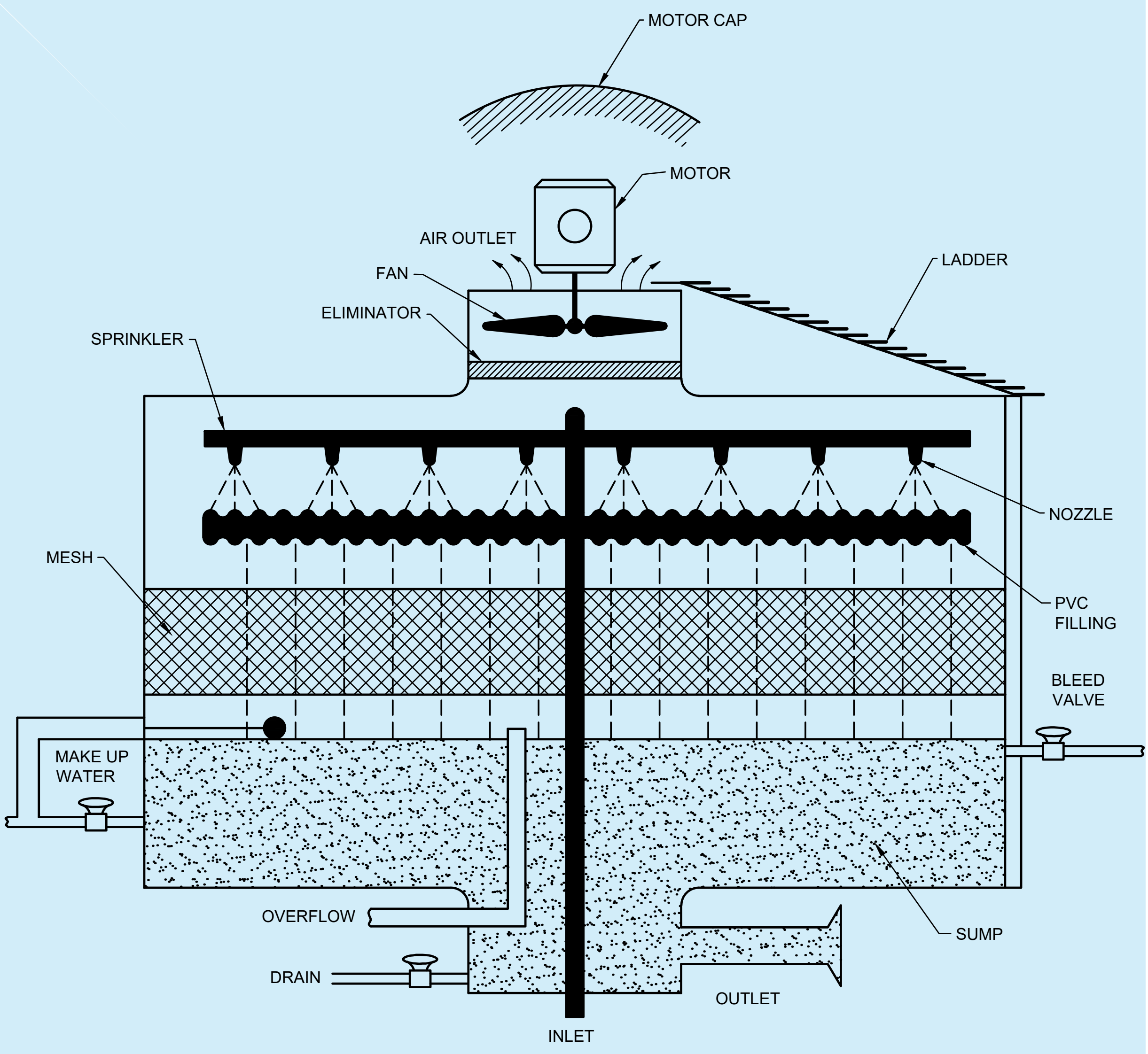


FIG. 28 TYPICAL COOLING TOWER INSTALLATION

17.3.7.5 A service platform may be constructed beside the tower so that all parts requiring cleaning or maintenance are easily accessible. Slope of the floor below the tower shall be such that water does not stagnate below the tower.

17.3.7.6 Make up and quick fill connections shall be done and uninterrupted water supply shall be made available. For water quality, water treatment and bleed off, refer 7.4.

17.3.7.7 In case of multiple cooling tower installation at the same location, equaliser connections shall be provided. In case the pumps are at the same location as the cooling tower, the cooling tower sump shall be minimum 300 mm above the pump suction to ensure positive pressure while starting the pump. The cooling tower basin shall be leak tested by filling up with water and holding it for minimum two hours.

17.3.8 *Air Handling Units* (*AHU*)/ *Heat Recovery Units* (*HRU*)/ *Fan Coil Units* (*FCU*)

Air handling units shall be either floor mounted or ceiling mounted. AHUs and FCUs shall be located close to or with in the area to be air conditioned so that duct runs are minimum. Location shall take into consideration the feasibility of running ducts, taking back return air to the unit (either through return air ducts, above the false ceiling or directly into the unit as well as provision of chilled water connections, provision of humidifiers, fresh air inlet as well as drain connections.

In a multi-storied building, it is advisable to locate the AHUs on above the other on each floor so that piping shall rise through the same shaft.

From fire safety considerations, individual AHUs shall not serve more than one floor. Similarly, it is advisable to have exclusive AHU for each fire compartment. In case, these are not possible, fire and smoke dampers shall be fitted on both the supply air and return air paths where floor crossing or fire compartment crossing takes place. Provision for pressure gauge and thermometer connections shall be made at inlet and outlet piping connections.

AHU/FCU shall be mounted in such a way that both the mechanical noise as well as the air noise does not disturb the occupants. This is specially emphasised for ceiling-suspended units. Duct mounted silencers and acoustic lining of supply air duct may be provided as specified. AHU/FCU shall not be mounted above electrical or IT rooms where water leakage shall lead to serious problems.

Adequate maintenance and service space shall be provided around AHU/FCU and trap doors for convenient access shall be available. Lamps inside the unit, belt guards, sight glass and door interlocks shall be ensured (*see* Fig. 29).

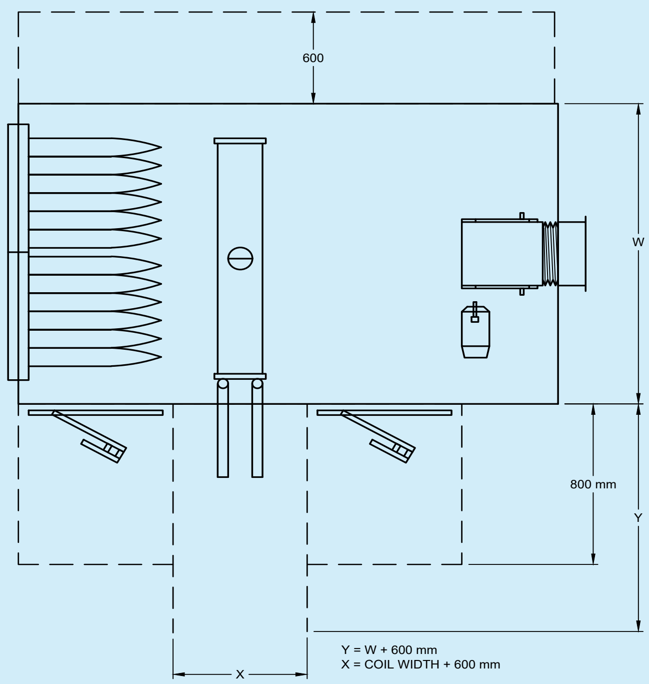


FIG. 29 TYPICAL SERVICE CLEARANCES AROUND AHU

Floor mounted AHU/heat recovery unit shall be mounted on top of pre-cast concrete block to permit easy drainage of condensate and to facilitate cleaning below the unit. Such supports shall be adequate in number and also located suitably so that transfer of dynamic load to structure safely takes place. Serrated rubber pads or other vibration isolation material shall be placed between the AHU/HRU frame and the concrete support. Insulated condensate drain connection shall be with a U- trap and the floor slope shall be towards the floor drain. Slope of the drain pan connection shall also be towards the drain pipe connection. Chilled water piping to AHU shall have a manual air vent at the top most point in the coil circuit and a drain plug at the bottom of the coil (*see* Fig. 30).

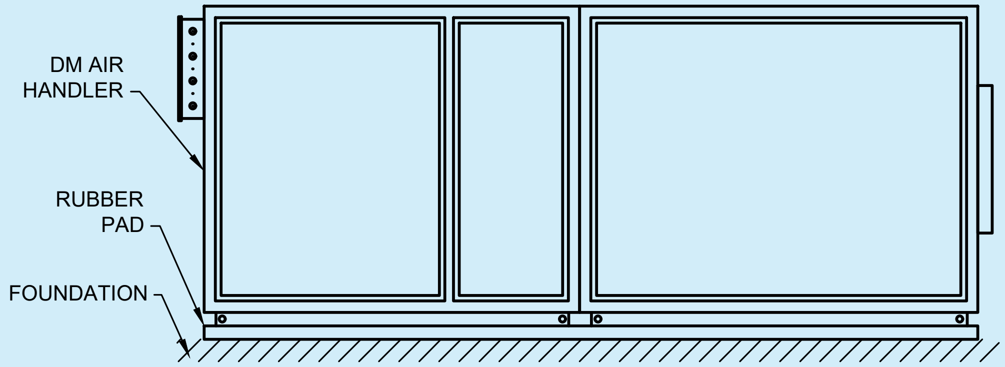


FIG. 30 AHU LOCATED ON FOUNDATION

Ducts shall be connected to the AHU/HRU with fire-proof connections to prevent transmission of vibration. Ceiling suspended AHUs/FCUs shall be hung from the ceiling with adequately sized rod and fasteners and shall be isolated from the structure by rubber or spring isolators. Illumination near the unit, drain point, water supply point and floor drain shall be provided.

17.3.9 *Air Washer Units* (*AWU*) *and Evaporative Cooling Units* (*ECU*)

Installation of AWU and ECU are similar to that of floor mounted AHU/HRU. They will have additional water connections to the spray- section with in-line recirculating pumps. Air washer units shall be preferably located on the summer wind-ward side. They shall be painted with reflective coating or thermally insulated. Large size air washers may be constructed with concrete basin. These shall have leak proof access doors as specified and basin shall be tested for water proofing integrity.

17.3.10 *Fans – Centrifugal, Axial Flow and Cabinet Type*

Fan performance is influenced by inlet and outlet conditions. 'Fan System Effect' which is the impact on fan performance due to inlet and outlet conditions as well as ways to mitigate them are described in **14**.

While rigging and hoisting heavy fans, manufacturer's instructions shall be carefully followed since the load is not uniformly distributed around the periphery.

Dynamic load of the fan, its distribution at different supporting points, vibration and noise shall be taken into account while designing the fan foundation.

Outlet connection to a duct (if provided) shall be by means of flexible fire-retardant connections.

17.3.10.1 *Centrifugal fans*

Where foundation bolts are provided, they shall be securely placed along with base plates while foundation is cast. Alternatively, fan may be mounted on suitable channel frame.

Vibration isolation base for fan and motor shall be an integral unit and shall be mounted on concrete foundation with anti-vibration mounts.

Safety screen shall be provided on fan suction side. Belt guard/coupling shall be provided and fan shall not be started without these protective devices.

17.3.10.2 *Axial flow fans*

Removable safety screens shall be provided on either side of the fan. Fans shall be properly grouted on the walls with vibration isolation material. Where the fan is suspended from the ceiling, vibration isolators shall be used.

17.3.10.3 *Cabinet fans*

A door interlock switch which shuts off the fan and switches on a bulk-head lamp mounted within the cabinet shall be provided. A water drain point shall also be provided within the cabinet. Transparent inspection windows of minimum 150 mm dia shall be provided.

17.3.10.4 For all type of fans, the sound generated is a major consideration which shall be taken care during the design/selection stage itself. Additionally, as specified by the fan manufacturer or system designer, acoustic silencers shall be mounted at inlet and outlet of the fan. Bird screen shall be provided at fresh air intake points and at exhaust outlet.

In all cases, adequate space for service and maintenance including removal and replacement of motor shall be provided.

Direction of rotation shall be verified on startup and field balancing shall be done where required. An electrical isolator or a push button switch shall be provided adjacent to the fan.

17.3.11 *Air Distribution and Ducting*

17.3.11.1 *Duct supporting/hanging systems*

Duct hanging system shall have three parts: Upper attachment to the building, the Hanger and Lower attachment to duct.

Upper attachments shall be concrete inserts which are cast into the structure, threaded concrete fasteners which are inserted into the cast slab or structural steel fasteners. Upper attachment method shall be selected with care and a safety factor of 5 is recommended. In case of steel structure, careful selection of primary and secondary supporting to be done on case-to-case basis.

Hangars shall be strips of galvanized steel or threaded round steel rods. Threaded portion of the rods shall be painted after installation.

Lower attachment is the connection between the hangar and duct section. Fasteners that penetrate the duct shall be sheet metal screws or rivets. All nuts, bolts and washers shall be zinc plated steel. Rivets shall be galvanized or made of magnesium-aluminium alloy.

Where ducts shall not be supported from the ceiling, wall brackets or other suitable support arrangements may be used.

Neoprene or other vibration isolation material of minimum 6 mm thickness shall be inserted between the ducts and the angle iron supports/brackets. Vertical ductwork shall be supported at each floor by suitable structural steel members.

Plenums where provided shall have access doors of 450 mm × 450 mm. Support details for different duct section is given in Fig. 31.

Supporting details for rectangular ducting for static pressure up to 750 Pa are given in Table 40 and for static pressure above 750 Pa are given in Table 41. Supporting details for low pressure (Up to 750 Pa) rectangular ducting system are given below:

Table 40 Supporting Details for Low Pressure (Up to 750 Pa) Rectangular Ducting System

(*Clause* 17.3.11.1)

Sl No.	Larger Side of Duct	Supporting Angle	Vertical Rod Diameter	Maximum Spacing between Supports
	mm	mm	mm	mm
(1)	(2)	(3)	(4)	(5)
i)	Up to 900	40 × 40 × 5	8	2 400
ii)	901 to 1 500	40 × 40 × 5	8	2 400
iii)	1 501 to 2 400	40 × 40 × 5	10	2 400
iv)	2 401 and above	65 × 65 × 5	12	2 400

Table 41 Supporting Details for High Pressure (Above 750 Pa) Rectangular Ducting System

(*Clause* 17.3.11.1)

Sl No.	Larger Side of Duct	Supporting Angle	Vertical Rod Diameter	Maximum Spacing between Supports
	mm	mm	mm	mm
(1)	(2)	(3)	(4)	(5)
i)	Up to 1 250	50 × 50 × 5	12	2 400
ii)	1 251 to 2 100	65 × 65 × 5	12	2 400
iii)	2 101 and above	75 × 75 × 5	15	2 400

Alternatively, slotted galvanized brackets attached to the top two bolts of the four-bolt duct connector shall be used.

For round, oval and flexible ducts, galvanized steel straps, wire or rods shall be used and support details shall be as provided by manufacturer/installer and approved by Engineer-in-Charge.

Sheet metal ducts shall be fabricated from galvanized iron sheets with minimum 180 GSM zinc coating and of lock forming quality. Fabrication of ducts shall be carried out at a factory from coil stock or at site from cut sheets. Where the overall duct area for the project exceeds 600 m^2 or where the external static pressure exceeds 250 Pa, lock forming machines shall be used for fabrication.

17.3.11.2 *Pre-insulated ducts*

Pre-Insulated duct made of polyisocyanurate sandwich panels comprising of a rigid foam board faced on both sides by stucco embossed aluminium foil shall be joined with tiger connectors or with male-female jointing system up to 500 mm. Higher sizes shall be joined by cover corners which has a holding pin which goes inside the flange. Ducts shall have suitable fixtures made of polymer plastic or extruded aluminium for jointing as well as for fixing of accessories such as grills, diffusers etc. Duct work shall be installed using supports as specified by the manufacturer.

17.3.11.3 *Fabric ducting*

Tools and accessories for installation shall be provided by the Fabric duct supplier. Fabric duct shall be connected to fan/AHU with GI duct connectors. While laying the duct, care shall be taken to prevent twisting of the fabric. A ratchet strap may be used to secure the fabric to duct connector.

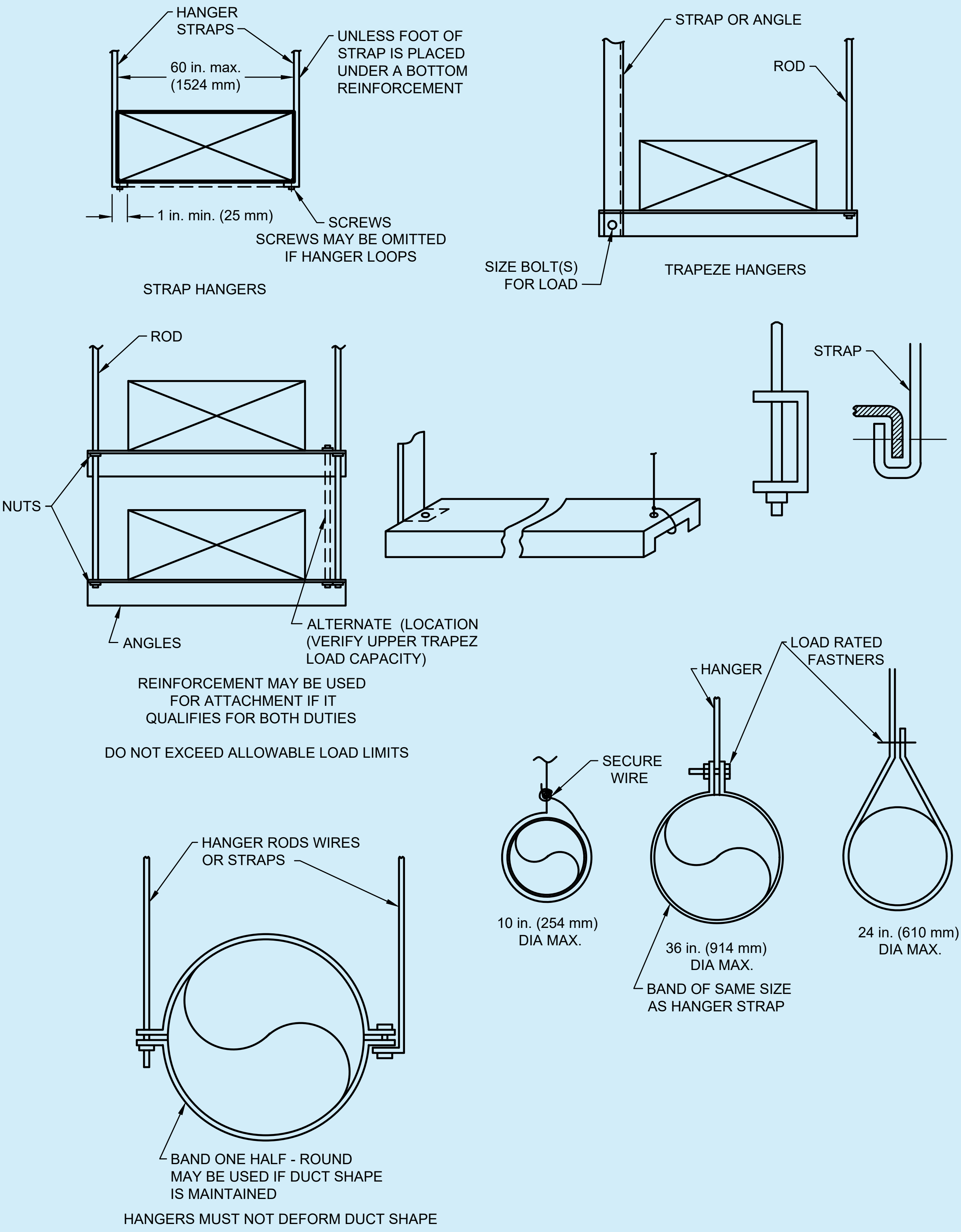


FIG. 31 SUPPORT DETAILS FOR DIFFERENT DUCT SECTIONS

17.3.11.5 Where duct penetrates brickwork or masonry openings, treated wooden framework shall be provided within the opening and the crossing duct shall be provided with heavy flanged collars on either side of the wooden frame so that the duct crossing is made leakproof. Alternatively, the duct section passing through the structure may be suitably wrapped so that vibration is not transmitted to the structure. Where duct passes through a fire compartment barrier, a fire barrier shall be created by installing a fire cum smoke damper.

17.3.11.6 When fans feed into a duct system, uniform straight flow conditions shall be ensured. Fan manufacturer's recommendation in regard to straight length of connecting duct both at inlet and outlet of the fan to obtain optimum fan performance needs to be followed. Similarly, pressure loss in a duct elbow or a volume control damper is minimum when air approaching the elbow/damper has a uniform velocity profile and positioning the elbow/damper after an adequate straight piece of duct is recommended. All elbows shall have adequate number of turning/guide vanes to ensure uniform air flow over the entire cross-section.

17.3.11.7 When duct is connected to an air handler or a fan, flexible connections made of rubberised canvas and 100 mm long shall be used between duct and AHU/fan and shall be securely bonded and bolted to the unit casing as well as the duct to prevent air leakage.

17.3.11.8 Turning vanes shall be provided at supply air outlet collars and at branch take-offs. They shall be fabricated from minimum 0.8 mm galvanized steel sheet and equally spaced on side runner and riveted or bolted to the duct (*see* Fig. 32).

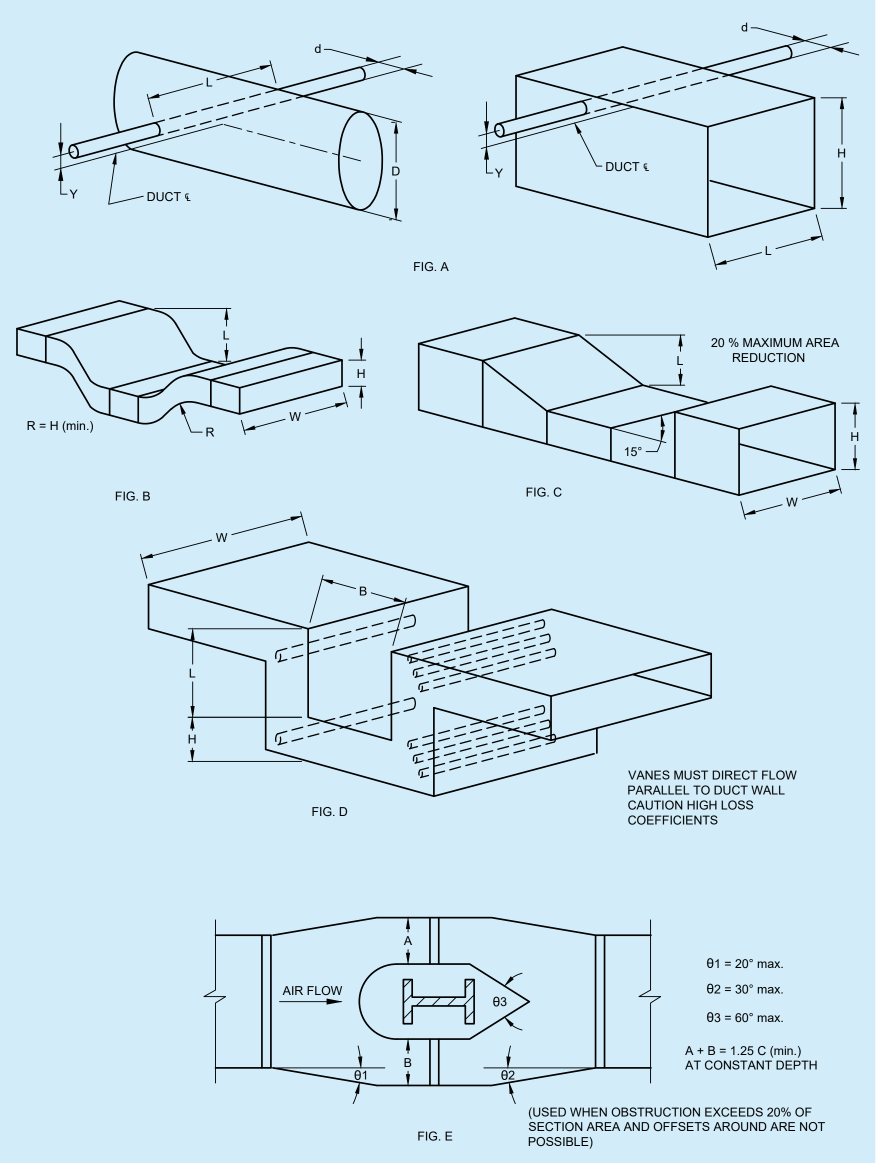


FIG. 32 TYPICAL TURNING VANES

17.3.11.9 Take offs allow air from main duct to be diverted to branch ducts and take offs are fitted into circular or rectangular openings cut into the wall of the main duct and small metal tabs cut at the start of the take-off are bent to connect the take off to main duct. Alternatively, a snap-in attachment with an adhesive foam gasket shall be used, outlet of the take-off connects to branch duct.

17.3.11.10 Dampers shall be manually controlled or automatic. Dampers shall be rated for operation against 1.5 times the rated pressure class of ducting. Suitable links, levers and quadrants shall be provided for proper operation, control and setting of the dampers. Linkages shall be blade to blade and concealed in a jamb out of the airstream. Every damper shall have an indicating device showing the damper position and a locking device. Where Dampers are operated via an external gear train, these are to be constructed out of engineering polymers with self-lubricating properties, and, located out of the air stream. Leak resistant access doors next to the dampers shall be provided to facilitate attending to the dampers and duct cleaning.

17.3.11.11 Splitter dampers shall be installed in branches where a split takes place. They shall be made of aerofoil blades of minimum 1.6 mm thickness and hinged at downstream edge. Operation rod shall be accessible from outside with an airtight hub and locking screw. Splitter damper shall be enclosed in sheet steel ducting with flange at both ends and shall be aligned with the main ducting. Damper section shall be made of one size higher thickness than the upstream duct.

17.3.11.12 Louvre dampers shall be provided in all branches they shall be multiblade type of aerofoil construction or in triple vee grove construction, rotating in lubricated ball/roller bearings/delrin bushes. Blades shall be connected by a linkage for simultaneous operation by a rod extending outside the duct and terminating in a locking quadrant with damper position indicator. Damper shall be enclosed in sheet steel box made of one size thicker than upstream duct and shall have flanges at both ends for connecting to main duct.

17.3.11.13 Fire dampers and combination fire and smoke dampers shall be installed wherever a fire compartment barrier exists. They shall be installed in ducts, return air passages and at all floor crossings. Access door shall be provided for each fire damper. Fire dampers may be motorized (activated by an electrical impulse from the fire alarm panel or a thermal trip element) or operate on a fusible link and shall shut off airflow completely. Wiring from AHU panel to the damper (in case of electrically operated damper) shall be carried out by damper installer. Dampers shall be suitable for vertical or horizontal mounting. Damper shall be installed maintaining fire integrity of the partition and all dampers at wall/floor crossings shall have a factory fitted sleeve. In such cases, the damper shall be a free float and the annular space shall be free to allow for expansion through a smoke seal. *See also* Part F 'Fire and Life Safety' of this NBCS.

17.3.11.14 Any type of electrical heaters shall not be mounted inside ducts. The electrical heaters shall be allowed to be installed in locations identified and approved by competent authority.

17.3.11.15 Variable air volume (VAV) boxes shall be installed at specified locations. They may be supported from floor, wall or ceiling as specified and may be mounted in vertical or horizontal direction. Boxes shall be factory set for minimum airflow of 10 percent and maximum airflow of 110 percent of design flow. Access to the box shall be from below and shall be completely sealed but easily removable. Electrical supply point with an isolator shall be provided by others and step-down transformer wherever needed shall be part of the unit.

17.3.11.16 Grilles, diffusers and registers act as terminal units for supplying air from the duct system to the room or for collecting back air from the room either to exhaust or for return to AHU/fan. Terminal units may be fitted with air filters. Square or rectangular wall outlets shall have a flanged frame with outside edges turned and fitted with a flexible gasket between the concealed face of the flanges and the finished wall face. Core of supply air register shall have adjustable front loves parallel to the longer side capable of deflection and adjustable back louvres parallel to shorter side to achieve horizontal spread of air up to 45°. Outer framework shall be made of 1.6 mm minimum thickness of aluminium sheet and louvres shall be of aero foil design of extruded aluminium section with minimum thickness of 0.8 mm. The vane ratio (ratio of gap between the vanes and vane depth) should not be less than 1 and should not exceed 1.2.

17.3.11.17 Square and rectangular ceiling outlets /intakes shall have a flange flush with the ceiling on which it is fitted. Outlets shall have an outer ring with duct collar and removable core assembly. Inner as well as outer rings shall be minimum 1.1 mm thick and diffuser assembly shall be 0.8 mm thick extruded aluminium section. Circular ceiling outlets shall be either flush or with an outer cone. Flush cones shall have the lower edge not more than 5 mm below the finished ceiling. All outlets/inlets shall be designed to reduce dirt deposit on the ceiling adjacent to the outlet.

17.3.11.18 Linear diffusers shall have a flanged frame with the outer edge turned 3.5 mm and shall have 1 to 8 slots as specified in design. These shall be constructed out of 1.6 mm thick extruded aluminium sheet. All supply air outlets shall have volume control device made of extruded aluminium.

17.3.11.19 Locations of air outlets and intakes shall be shown in the drawing and necessary openings and the wooden framework for fixing the grilles/diffusers shall be provided by the contractor.

17.3.11.20 While installing fresh air intakes, no fixing device shall be visible from the face of the frame. Where louvres are to be fixed to brick, masonry or concrete, fixing shall be with expanding plugs or raw plugs. Where louvres are to be fixed to wood or metal, nonferrous screws or bolts shall be used. A flanged frame made of rolled steel sections shall be provided on the front face to cover the gap between the louvres and the adjoining wall face. Frame shall be structurally rigid and corners shall be welded. Where the louvre length exceeds 1 250 mm additional intermediate and equally spaced supports shall be provided to prevent sagging/vibration. Louvres shall be made of 1.6 mm thick 100 mm wide extruded aluminium sections and fitted in an extruded aluminium frame. A bird screen made of 12 mm mesh in 1.6 mm steel wire held in angle or channel frame shall be fixed to the rear face of the louvre frame. Louvres shall be fixed 45 degrees to vertical and shall provide 60 percent minimum net opening. Louver design and spacing shall ensure that there is no water ingress and the lowest louver shall be suitably overlapped to prevent water being drawn in.

17.3.11.21 Test probes which shall be installed at every fan discharge and suction duct shall be of heavy-duty zinc alloy casting. Expansion plugs shall be neoprene and capable to withstand 80 °C and 300 kPa pressure.

17.3.11.22 Kitchen exhaust ducting shall be of 16 G MS CRCA sheets and shall be of welded construction. Access doors shall be provided at 3 m intervals and the duct shall have a mild slope towards the kitchen hood. Laundry and dish washer extract ducts shall be of air and watertight construction and shall be of Aluminium sheets in accordance with accepted standard [D-3(44)]. Gasket used to be fire rated. Access doors to be provided at regular intervals and at critical points. The ducting shall be wrapped with high temperature grade EPDM, nitrile rubber or with fire wrap.

17.3.12 *Filters and Air Cleaning Devices*

The installation and commissioning of filters and air cleaning devices require strict adherence to these guidelines to ensure optimal performance and safety.

17.3.12.1 Filter elements shall be kept stored in original packing till they are ready to be installed with all upstream areas such as AHU rooms and upstream ducting is cleaned and made dust free.

17.3.12.2 Filters shall be installed with directional arrows pointing towards air flow direction. Filters shall be tightly fitted in the frames provided using gaskets with no gap between the filter and the frame.

17.3.12.3 Differential pressure gauge/switch is recommended to be installed across filters. After initial running of 48 h, if required, replace the filters (In case of pre filters only).

17.3.12.4 HEPA filters shall be handled with special care since the media is fragile and shall be installed with the gasket side of the filter facing the frame.

17.3.12.5 ESP filters shall be firmly fitted in a specially fabricated frame and wiring of the ESP filters shall be as per manufacturer’s recommendation. A clear sticker ‘Caution – High Voltage Filters’ shall be prominently displayed in front of ESP filters.

17.3.12.6 While installing ultra violet germicidal irradiation (UVGI) lamps, the layout provided by the manufacturer shall be followed. In AHUs, UVGI lamps may be installed downstream of cooling coil. Mounting frame for UVGI lamps may be at a distance of 300 mm from the cooling coil for optimum performance. After installation, the lamp surface shall be cleaned with a wick wetted with spirit.

17.3.12.7 Ballast of the lamps may be accommodated with in the UVGI electrical panel or may be tied to the UVGI mounting brackets as per manufacturer’s recommendation and IP rating of ballasts.

17.3.12.8 Follow manufacturer’s instruction for any shielding of drain pan, cable sheaths and conduits.

17.3.12.9 Safety guidelines provided by manufacturer shall be closely followed. If for any reason, AHU door needs to be opened while the lamps are 'ON' person/s in AHU room shall wear PPE kit and protective goggles. Bare eye or skin shall not be exposed.

17.3.12.10 Upper zone irradiation units may be mounted on walls or hung from ceiling well above occupied zone (minimum 2.2 m from floor level). Baffles/ louvres shall be directed in such a way that the rays are not directed towards occupied zone. In respect of upper zone irradiation units, the occupied zone UVGI shall be measured periodically to ensure that it does not exceed 1 000 UV/m^2 to 2 000 UW/m^2.

17.3.13 *Installation of Piping Works*

The following are key points for the proper installation of pipeworks, ensuring alignment, sealing, support, and compliance with safety:

17.3.13.1 It shall be verified that pipes of correct specifications such as pipe class, length, thickness, straightness and surface finish are received at the project site. Pipe routing as per 'Good for Construction' drawing shall be verified at site and pipes may be cut to required lengths.

17.3.13.2 Edges of pipes shall be bevelled by grinding to make a 'butt' joint. Welding shall be done exactly to specifications (for example, accepted standard [D-3(45)]. Fabrication of supports shall be as per site requirements and specifications. Spacing between supports shall be as specified.

17.3.13.3 Overhead piping may be supported on walls/ columns with appropriately sized brackets or may be suspended from ceiling/roof truss with MS suspension rods or hangars. Anchor fasteners of adequate sizes shall be used for pipe supports. Vertical risers shall be supported by horizontal supports mounted at regular intervals.

17.3.13.4 Vertical pipes passing through floors shall be parallel to the wall and shall be straight (as verified with a plumb line).

17.3.13.5 Where pipes pass through floor slab or walls, gaps between the pipe and floor/wall opening should be sealed as detailed in Part F 'Fire and Life Safety' of this NBCS.

17.3.13.6 Chilled water piping shall be supported on PUF or wooden gutties. Single gutties shall be used for horizontal runs and double gutties for vertical risers. While installing insulated pipes, care shall be taken that insulation is not unduly compressed.

17.3.13.7 Condenser water piping shall be placed on serrated rubber pads at each support and held in place by U-clamps.

17.3.13.8. Piping shall have expansion joints as required. Fittings, valves and instruments shall be as per approved drawings. Drain points and air vents shall be provided as required. Auto purge valves shall be provided at the highest point in each pipe loop with discharge piped out to nearest drain or sump. While T connections shall be permitted on suction side of pump, shoe connections shall be provided on the discharge side. Elbows/bends shall be minimum and reducers shall be eccentric type. Welding scale shall be removed before applying red oxide primer. Readymade fittings to specifications are recommended. Welder shall be competent and hydraulic pressure testing with 1.5 times the working pressure shall be done for 24 h (*see* Fig. 33, Fig. 34 and Fig. 35).

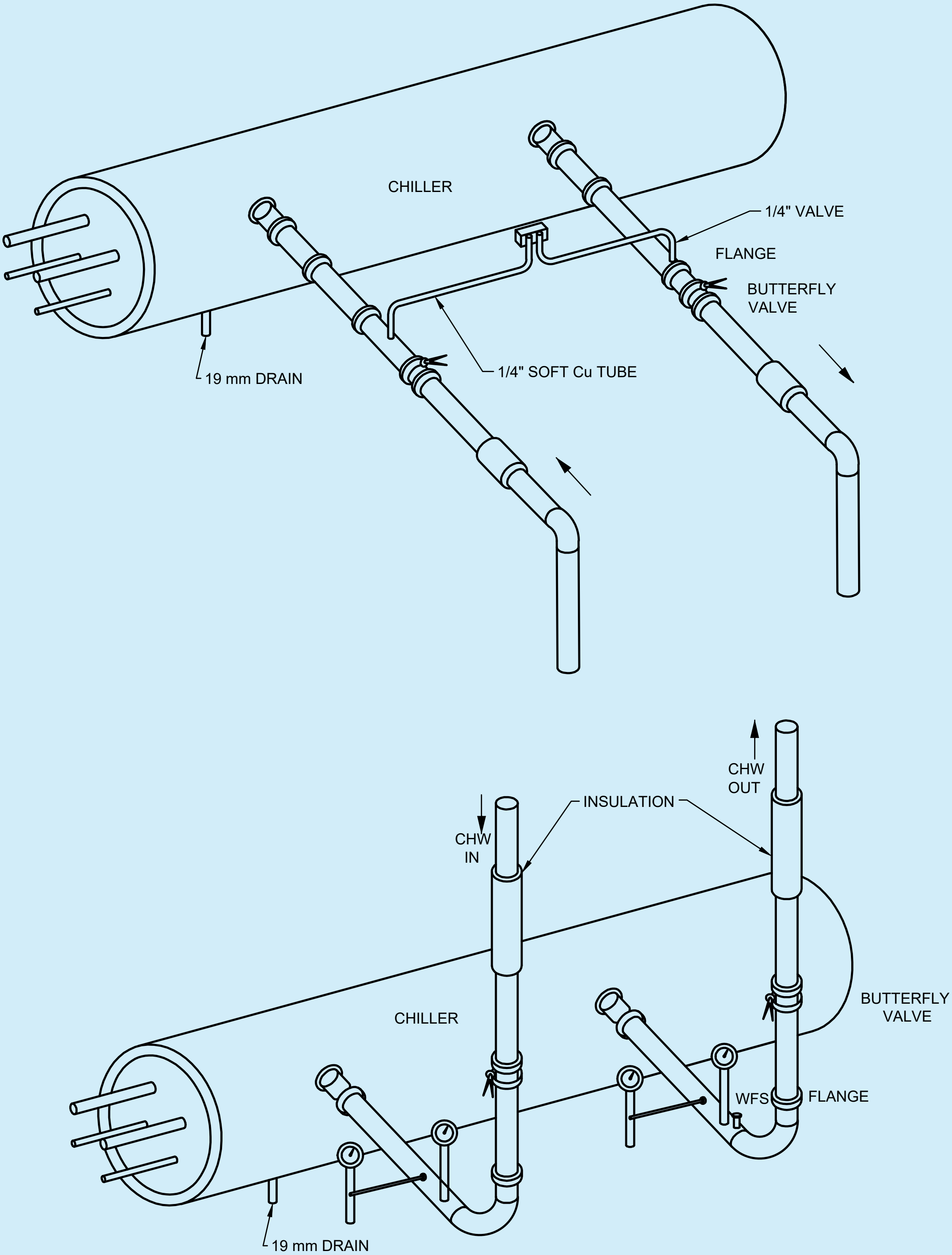


FIG. 33 TYPICAL CHILLER CONNECTIONS

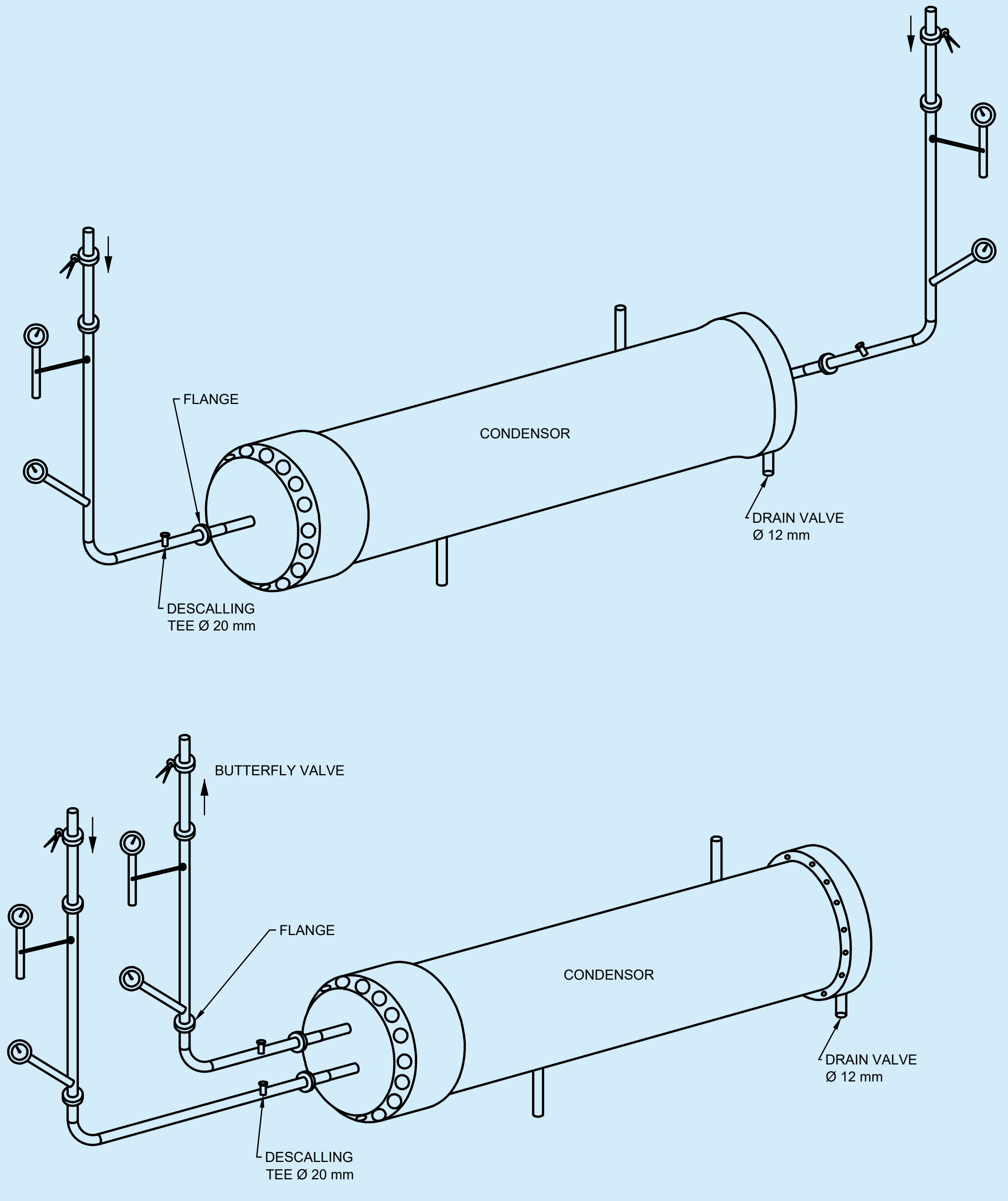


FIG. 34 TYPICAL WATER COOLED CONDENSER CONNECTIONS

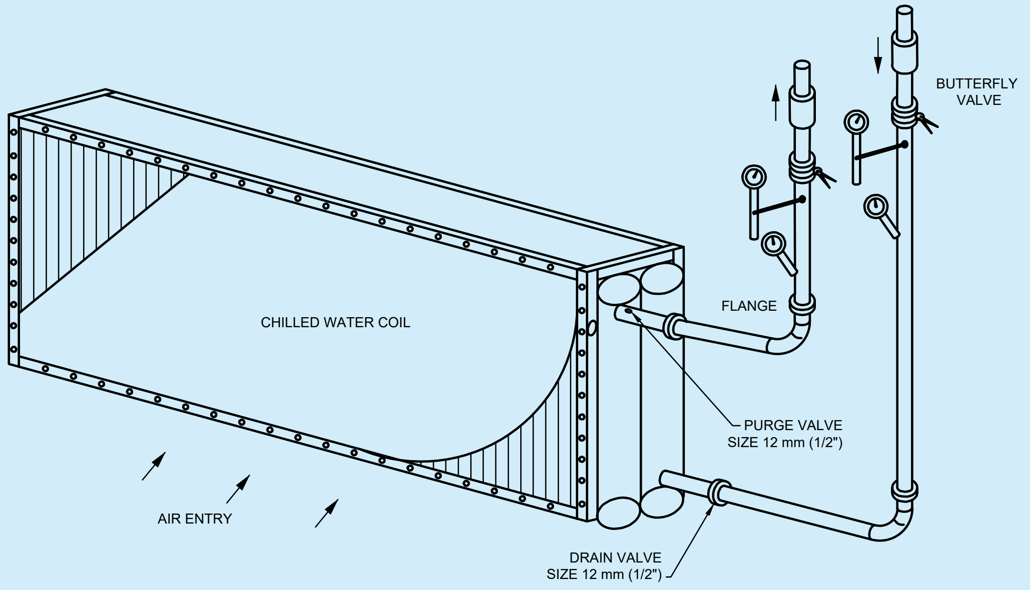


FIG. 35 TYPICAL COIL CONNECTIONS

17.3.13.9 Flanges shall be provided near each equipment connection and all flanges shall be with gaskets.

17.3.13.10 During installation process, open pipe ends shall be always kept closed.

17.3.14 *Insulation Works*

The following are key points for the proper installation of insulation works, ensuring correct material selection, proper fitting, and adherence to safety and performance.

17.3.14.1 *Thermal insulation of sheet metal ducts*

Duct surface shall be cleaned with wire brush to remove dust and grease. Two coats of cold setting fire resistant adhesive compound shall be applied over the duct surface. GI stick pins shall be used in case of glass wool/rock wool. Thermal insulation shall be wrapped around the duct with aluminium foil facing the outer side. All transverse and longitudinal joints to be sealed with aluminium tape. PVC strips shall be fixed at suitable intervals so that insulation remains in place. Insulation shall be covered with GI wire mesh (0.63 mm thick × 19 mm) netting on the outside where duct is open and exposed to weather and shall be additionally protected with tar felt or two coats of 10 mm sand cement plaster followed by two coats of water proofing chemical.

17.3.14.2 *Duct made of pre insulated boards*

Pre-insulated duct boards may be used for efficient air distribution with reduced energy loss and improved installation speed (*see* Fig. 36) providing both thermal insulation and structural stability.

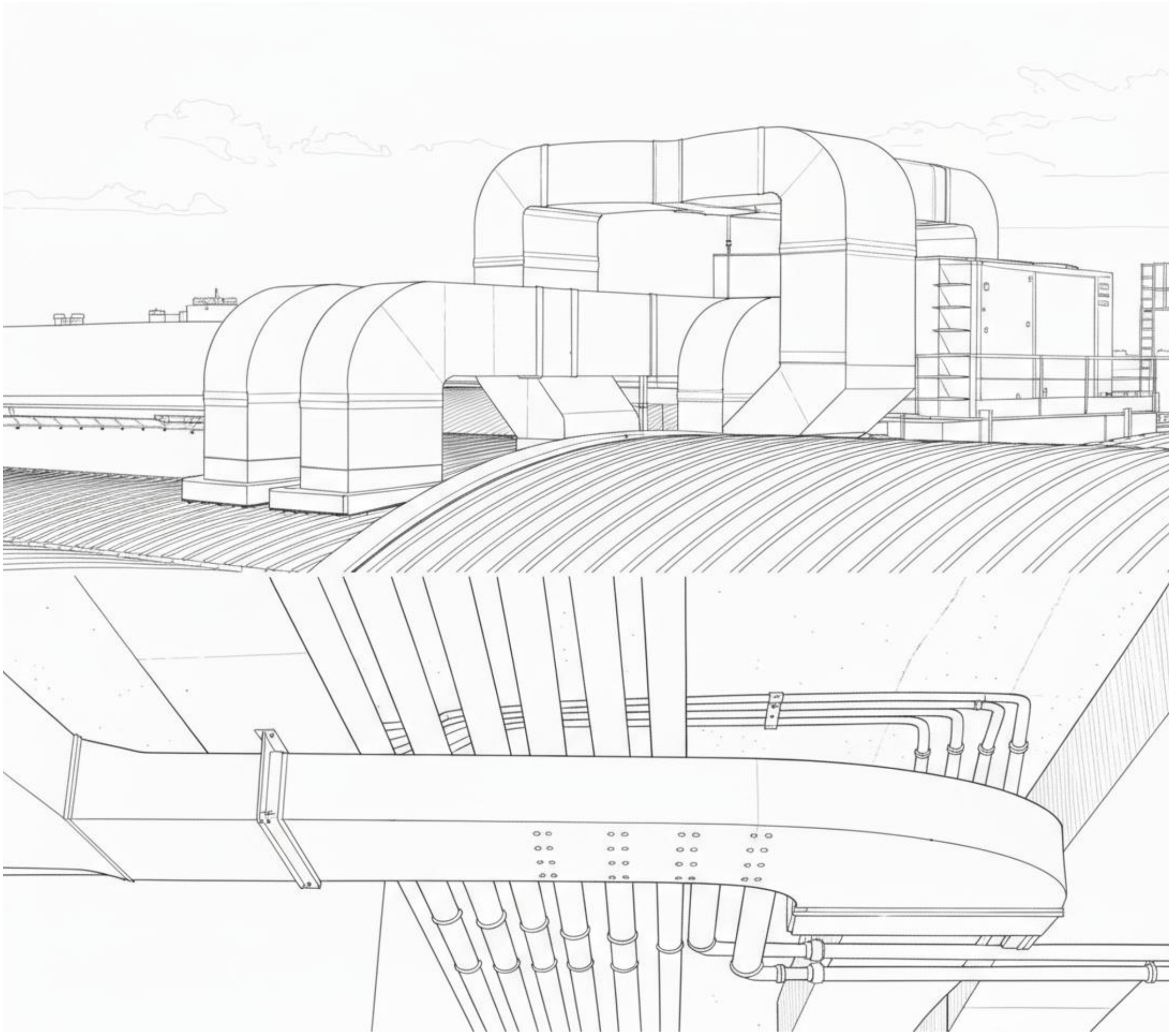


FIG. 36 PRE INSULATED DUCTS.

17.3.14.3 *Acoustic lining of duct*

Inside surface on which lining is to be provided shall be cleaned of dust and grease. Material used shall be 12 mm/25 mm thick resin bonded fibre glass/rock wool rigid board with 48 kg/m^3 density. Board shall be fixed inside duct using fire retardant adhesive and covered with tissue paper. Insulation shall be covered with 0.5 mm thick perforated Aluminium sheet having polysurlyn lamination on inner side with minimum 20 percent perforation. Insulation board and aluminium sheet shall be secured with Cadmium coated bolts, nuts, cup washers and steel screws. Ends shall be securely sealed so that no lining material is exposed. Length of lining shall be as per approved drawing.

As an alternative to the above, open cell elastomeric foams of fire-retardant type specially formulated for sound attenuation may be used. These shall be fixed inside the duct with fire retardant cold setting adhesive (*see* Fig. 37).

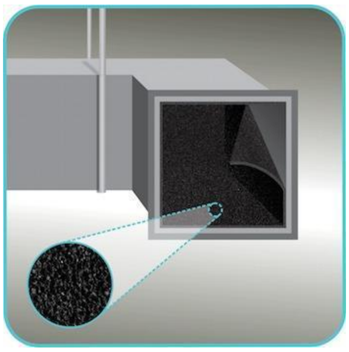


FIG. 37 ACOUSTIC INSULATION OF DUCT WITH OPEN CELL ELASTOMERIC FOAM.

17.3.14.4 *Acoustic lining of equipment rooms*

Where specified in drawings, inside walls and ceilings of equipment rooms may be lined with acoustic insulation. Surface to be cleaned with a wire brush. A frame work of U-shaped channel with height equal to proposed lining thickness made of 0.63 mm thick GI sheet shall be fixed to the walls by means of raw plugs. Similar frame work shall be fixed to ceiling by means of dash fasteners (*see* Fig. 38).

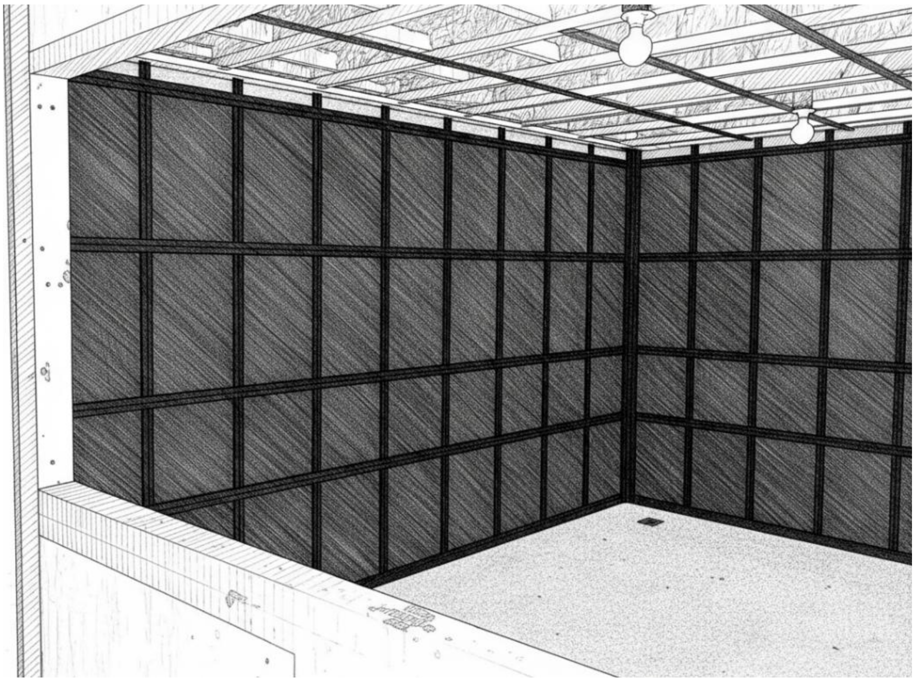


FIG. 38 PLANT/AHU ROOM ACOUSTIC INSULATION.

Resin bonded glass wool/rockwool cut to size shall be fitted in the framework and covered with tissue paper. Surface shall be covered with 0.5 mm thick perforated (minimum 20 percent perforations) aluminium sheet with polysurlyn lamination on the inside. These shall be fixed with brass screws.

17.3.14.5 *Pipe insulation*

Insulation of pipes shall be carried out with specified material. Pipe insulation material is available in pre-moulded half sections.

Pipe surface shall be cleaned of dust, rust and grease with wire brush. One coat of red oxide primer and two coats of cold setting, fire retardant adhesive compound shall be applied. After tightly fixing the insulation sections, joints shall be sealed with adhesive compound.

Piping within covered space shall be covered with aluminium cladding of specified thickness having polysurlyn lamination on the inside over a vapor barrier. Cladding shall have 50 mm overlap and shall be tied down with lacing wire.

Piping outside the building and exposed to weather shall be finished with vapor barrier and GI or Aluminium sheet cladding with polysurlyn lamination on inner side. Two layers each of 6 mm thick sand cement plaster each cured for minimum 48 h followed by two coats of water proofing chemicals or fixed with bitumen membranes with hot bitumen adhesive or cold quick setting fire resistant mastic compound.

Buried piping shall be covered with suitable gauge polythene faced hessian cloth (with polythene facing outward) with 50 mm overlap. Joints shall be sealed with bitumen. A layer of 0.5 mm × 20 mm thick sand cement plaster (1:4) shall be applied and water proofed by applying hot bitumen or cold quick setting fire resistant mastic compound with a layer of tar felt over the plaster. Final finishing shall again be with hot bitumen or cold mastic compound.

Alternatively, the insulation sections shall be protected with a 3 mm to 5 mm thick HDPE sheet cladding hot air welded and sealed at all joints.

In respect of pre insulated pipe sections, manufacturer’s instructions shall be followed. For pipes inside the building, the factory provided Aluminium foil lamination shall suffice both as vapor barrier and also as finishing material. Otherwise, GI or aluminium cladding shall be provided with 1.016 mm (40 mil) polysurlyn coating on inside. After application with adhesives, all joints shall be sealed with Aluminium tape.

17.3.14.6 *Insulation of pumps, valves, expansion tank, fittings and pipe supports*

Pumps, valves, fittings and strainers in the chilled water line shall be insulated to the same thickness and using the same as the pipeline. Care shall be taken to apply the insulation in such a way that it shall be possible to dismantle the pumps and operate the valves without damaging the insulation. Moulded PUF insulation in sections may be used where access is desired. These shall be easily reset/ reinsulated after dismantling. Option is available to use pre insulated valves and flanges.

Expansion tanks shall be insulated with 50 mm thick TF quality expanded polystyrene or 40 mm thick polyurethane or poly-isocyanurate foam as specified. Surface shall be cleaned and two layers of hot bitumen or fire resistant cold adhesive compound shall be applied. Insulation slabs shall be placed abutting one another and joints sealed with same compound. Slabs shall be covered with 0.63 mm thick × 20 mm GI wire mesh netting to be fixed with brass/GI screws. Final finishing shall be with 0.5 mm thick Aluminium cladding with polysurlyn coating on inner side.

Chilled water pipe supports shall be insulated with high density (120 kg/m^3) phenolic foam of thickness equal to pipe insulation. Supports shall have same shape as the outside surface of the pipe. Shape/ dimensions of pipe supports shall be as specified.

Pipe bends shall be insulated with sections cut into bevelled lags and applied over the bends with fire retardant cold adhesive compound and all joints sealed with cut insulation pieces or factory-made shaped bends in two halves fixed with fire retardant cold adhesive compound and tied with lacing wire. Vapor barrier similar to that used for pipes shall be applied and the whole assembly may be clad with Aluminium sheet having 1.016 mm (40 mil) polysurlyn lamination on inner side. Cladding to have same thickness as pipe cladding. Joints in the cladding shall be grooved joints fixed with screws and sealed with aluminium tape or mastic compound.

17.3.14.7 *Under deck and over deck insulation*

Underdeck roof insulation shall be with phenotherm 35 kg/m^3 or polyisocynurate foam 32 kg/m^3 or resin bonded glass wool or rock wool 48 kg/m^3. Thickness shall be as specified and method of application same as that of acoustic insulation for equipment rooms (*see* **17.3.14.4**).

Overdeck insulation shall be with polyurethane foam 40 kg/m^3 to 45 kg/m^3 of specified thickness sprayed in multiple layers to form a jointless homogeneous surface. Geo – textile cloth shall cover the insulation followed by plaster and suitable water proofing system. Alternatively, extruded polystyrene slabs or polyurethane/ poly-isocynurate foam slabs shall be fixed over fully dried roof surface with fire resistant mastic and joints sealed with adhesive compound.

17.3.15 *Thermal Storage System*

Storage tanks shall be fabricated out of carbon steel of adequate thickness, hydraulically tested for 1.5 times the design pressure and shall be positioned at the location ensuring that its operating load is safely transferred. Tanks may also be fabricated on-site. Red oxide primer as per the accepted standard [D-3(46)] shall be applied to all exposed welds and edges. Tank shall be able to withstand freezing of water inside through repeated cycles without damage. Tank shall be thoroughly cleaned and flushed before use. External surface shall be insulated similar to that of expansion tank.

Where multiple tanks are installed adjacent to one another, minimum clearance of 600 mm shall be kept. Adequate height to allow for piping (minimum 1 m) shall be available above the tank. Each pair of manifold inlet/outlet connections shall include a shutoff valve to isolate the unit. Bypass circuit shall be installed with a 3-way modulating valve. Thermometer/pressure gauge shall be installed at inlet and outlet. Before start up, air shall be completely purged out from the system for which air bleed valves shall be located at the uppermost points in the piping network.

When glycol is used as the storage medium, add inhibitors as recommended by the manufacturers.

When tanks are located in the open outside the building, insulation shall be covered with a layer of 26 gauge aluminium cladding.

17.3.16 *Smoke Management Systems*

Purpose of smoke management is:

a) Inhibit smoke from entering stairwells and other means of egress, refuge spaces, elevator lobbies, elevator shafts etc;

b) Maintain tenable environment in refuge areas and egress paths during evacuation;

c) Provide conditions to enable emergency response teams to fight fire and conduct rescue operations; and

d) Protect life and property.

Smoke control systems should be as per **19** as well as Part F ‘Fire and Life Safety’ of this NBCS.

Installation aspects of smoke and fire dampers is covered in **17.3.11.13** and installation of fans and ducting is covered in **17.3.10** and **17.3.11**.

17.3.17 *Installation of Electrical Systems*

These are the key provisions for the installation of electrical systems in electrical rooms and switchboards.

17.3.17.1 All the equipment used in the electrical room shall be of sufficient current rating for its normal duty as well as during the fault current. The equipment shall be of sufficient mechanical strength to withstand the fault condition to ensure human safety.

17.3.17.2 Every switch board of medium or high voltage shall comply the following provisions:

A clear space of not less than 1 meter in width shall be provided in front of the switch board. If there is any attachment at the back of the switchboard, the space shall be either less than 20 cm or more than 750 mm in width from the farthest outstanding part of the switchgear. If the space behind the switchboard exceeds 75 cm in width, there shall be passage-way from either end of the switchboard clear to height of 1.8 meters.

All disconnect switches are to be identified and labelled stating the source of the circuit and the equipment/circuit it is feeding.

17.3.17.3 Insulation mat as per the accepted standard [D-3(47)] of proper grade to be used in front of 415 V and above electrical switchgears. All panel doors shall be locked, made vermin proof by sealing entry holes specially from cable entry points.

17.3.17.4 Bus coupler panel and its cubicle shall be handled with utmost care. Red painting of bus coupler, adaptor panel and tie feeder panel from all sides (front, top and back) for easy identification and treat Adaptor panel as a part of bus coupler.

17.3.17.5 Incoming barrier of power supply and phase barrier inside the switchgear panel to be ensured.

17.3.17.6 There should not be any loose wirings, exposed joints, distribution box or junction box in cover open condition in the workplace. No switchgear which has been kept disconnected, for period of six months or more for any maintenance job shall be energized without testing.

17.3.17.7 Over current protection to be provided to disconnect the supply automatically if the rated current of the equipment, cable or supply line is exceeded for a time which the equipment, cable or supply line is not designed to withstand. Earth fault or earth leakage protection to disconnect the supply automatically if the earth fault current exceeds the limit of current for keeping the contact potential within the reasonable value.

17.3.17.8 The operation of all electrical and mechanical interlocks on the switchgear shall be checked to ensure that they operate in a positive manner. All electrical connections shall be proved for mechanical integrity that is terminal tightness, shrouding etc. The panels, relays and control modules shall be visually inspected to ensure freedom from debris and mechanical damage.

17.3.17.9 Proper illumination in work area shall be ensured and no lone working shall be allowed inside HV/LV switch gear room

17.3.17.10 Refer to Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS for installation of all electrical equipment including motors and switch/control gear.

17.3.18 *Installation of Electric Motors*

These are the key provisions for the installation of electric motors

17.3.18.1 Motors shall be installed at a location compatible with its enclosure type as well as the ambient conditions.

17.3.18.2 Before installation, the insulation resistance shall be checked by applying 500 V (d.c.). If insulation resistance is low, the windings shall be dried out till the resistance increases to 5 megaohms. While doing this, the frame shall be grounded and winding discharged against frame immediately after measurement.

17.3.18.3 Foundation details and bolting arrangements shall be as specified by the manufacturer.

17.3.18.4 For belt drives, the sheave pully shall be mounted close to motor housing and clearance may be provided for end-to-end movement of motor shaft. Drive shafts shall be parallel and belt tensions may be checked periodically during the first 24 hours of operation.

17.3.18.5 Lubrication shall be done with grease recommended by the manufacturer.

17.3.18.6 Terminal housing shall be of adequate size to allow minimum terminal spacing as per article 430 of accepted standard [D-3(39)]. Adequate earthing shall be provided. Minimum earth wire permitted is No. 14 standard wire gauge (SWG).

17.3.18.7 All motor connections and protections shall comply with accepted standard [D-3(39)].

17.3.19 *Installation of Instrumentation/Controls/BMS/IOT*

These are the key provisions for the installation of Instrumentation/Controls/BMS/IOT

17.3.19.1 Field devices such as temperature and RH sensors, pressure transmitters, flow meters etc shall be installed as per manufacturer's instructions. Components of the BMS System shall be housed in an enclosure suitable for the location where they are installed in respect of temperature, RH and IP rating. Where required, suitable transformers shall be installed in the enclosure and adequate earthing shall be provided. Uninterrupted power supply shall be provided to the BMS system.

17.3.19.2 Field cabling and containment shall be as per Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS. Kept at a minimum distance of 300 mm from power cables. Field cabling shall be carried out with screened cables in conduits work stations/servers shall be installed in clean and dust free rooms.

17.3.19.3 While installing IoT – Wireless field devices, availability of repeaters to form a mesh network shall be ensured.

17.3.19.4 Method statement for installation and commissioning shall be provided by the supplier/installer and pre-approved by the Engineer-in-Charge. If energy dashboard is part of the scope of supply, all screens shall be pre-approved by the Engineer-in-Charge. Where specified, IoT interface for specified equipment shall be provided and these shall be tested as part of the commissioning process.

17.3.19.5 All back up data for restoring the system functionally, in case of any failure, shall be provided and these shall be safely stored separately.

17.3.19.6 System shall be protected from cyber threats by ensuring that all remote access is password protected. Default user names/passwords shall not be retained.

17.4 Vibration Isolation and Seismic Protection

These are the key provisions to be considered for vibration isolation and seismic protection.

17.4.1 When machinery with moving parts rests on any support or foundation, vibration gets transmitted usually accompanied by noise. Objective of vibration isolation is to minimize the effect of the dynamic forces generated by the moving parts being transmitted to surrounding structure. When a resilient material is introduced between the machine and its foundation, it deflects due to the static load and by so doing establishes the natural frequency of the isolation system. Natural frequency f_n of the machine set on resilient material is a function of the static deflection of the resilient material under the imposed load.

This shall be expressed as:

$$f_n = (946.5/V) - d$$

where

d is the static deflection in mm; and
V is the velocity in m/s.

The disturbing frequency f_d of the machine is defined by the operating characteristics of the equipment. Efficiency of isolation (percentage of vibration isolated) may be calculated as:

$$E = 100.1 - 1/(f_d/f_n)^2\text{-}1$$

E represents the amount of reduction in the amplitude of the transmitted mechanical vibration. Based on the graph below (Fig. 39) the static deflection required to attain the desired isolation efficiency shall be ascertained.

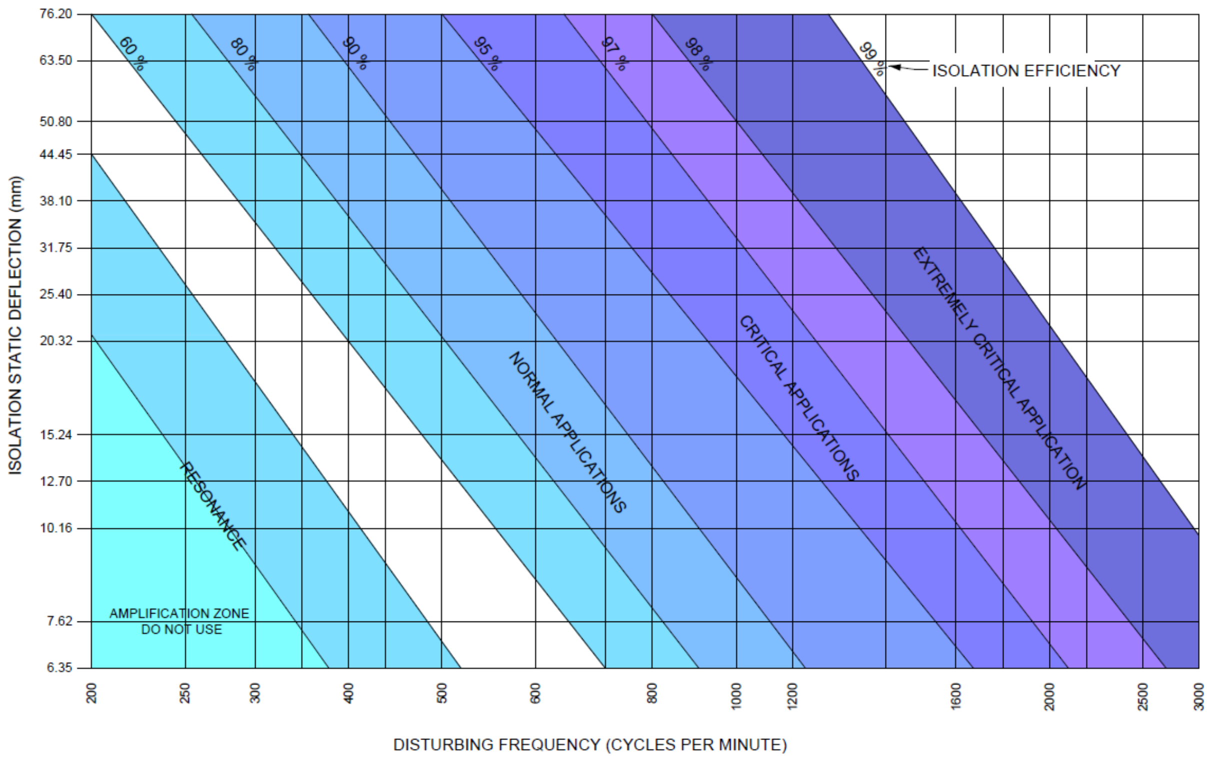


FIG. 39 STATIC DEFLECTION REQUIRED FOR DESIRED ISOLATION EFFICIENCY

17.4.2 Selection of isolator in a particular situation depends on:

a) Weight to be supported;

b) Disturbing frequency; and

c) Rigidity of supporting structure.

17.4.2.1 Anti vibration mounts (AVMs) are generally elastomer based, springs, or fluid based. These need to be selected and positioned carefully since incorrect use shall worsen the problem.

17.4.2.2 *Selection of suitable AVMs involves the following steps*:

a) Estimating the weight distribution at each support which depends on its position relative to the centre of gravity of the total mass. This is best done by a computer program. If the equipment is placed on suspended supports, same shall be factored into the selection program;

b) Selecting the appropriate AVM based on type of equipment being supported, its operating speed, nature of supporting structure, environmental conditions and space availability;

c) The load rating of the mount shall be determined based on isolation efficiency derived from charts available for that purpose; and

d) Usually, the AVM is selected for the next higher load rating than the one determined in (c) above.

17.4.3 *Seismic Isolation of Non – Structural Components*

These are the key provisions to be considered for Seismic Isolation of non- structural components

17.4.3.1 The non-structural components in the building which shall be affected by earthquake effects include:

a) Mechanical systems and components;

b) Electrical systems;

c) Fire protection systems; and

d) Plumbing systems.

17.4.3.2 Seismic protection of the above systems shall ensure

a) The components are securely attached to the supporting structure;

b) Life safety equipment such as emergency lighting and fire suppression systems continue to function; and

c) For critical applications, non-structural components continue to operate. Critical applications shall include:

   1) Health care facilities;

   2) Airports;

   3) Emergency response centres;

   4) Critical defence installations; and

   5) Data centres.

17.4.3.3 Performance objectives for seismic protection of non-structural components and how the objectives shall be met shall be as indicated below:

a) *Position retention* — Non-structural components shall retain their position under design level earthquake demands without consideration of frictional resistance produced by effects of gravity. This shall be validated by analysis of seismic calculations and such analysis shall also validate the non- structural anchorage, force resisting skeleton and attachments shall have position retention capacity equal to or greater than project specific design level demand for the installation location. Both, strength design and allowable stress design approaches shall be accepted in such calculations. This shall be done in accordance with good practice [D-3(48)].

   The use of earthquake experience data may be used to establish non-structural position retention capacity. Seismic simulation testing may be used to establish non – structural position retention capacity.

b) *System interactive avoidance* — Unwanted interaction, under design level earthquake demands, between the non-structural systems and anything else that might be located in the vicinity shall be taken into account. This is to ensure that failure of one system or contact between systems shall not cause consequential damage to system. This may also be validated through visual inspection.

c) *Active operation* — Following application of design level earthquake demands, active operational functionality of mechanical and electrical equipment and electrical distribution system shall be maintained.

17.5 Painting and Colour Codes

The following are guidelines for painting and colour coding in air conditioning installations:

17.5.1 All surfaces to be painted shall be thoroughly cleaned with a wire brush. One coat of zinc chromate primer shall be applied and after the prime coat is completely dry finish painting shall be done. Second coat of finish painting shall be done just before hand over with permission from Engineer-in-Charge. Galvanized steel ducts may be painted without a prime coat. Dry film thickness (DFT) of finished painting shall be 250 microns minimum.

17.5.2 Colour code shall be used for easy identification of various items in an air conditioning installation for correct interpretation and identification.

17.5.3 Colour band shall be 150 mm wide, superimposed on ground colour to distinguish type and condition of fluid. The spacing of band shall not exceed 4.0 m.

17.5.4 Further identification may also be carried out using lettering and marking direction of flow.

17.5.4.1 The scheme of colour code for painting of pipe work services for air conditioning installation shall be as indicated in Table 42.

17.5.4.2 In addition to the colour bands specified above, all pipe work shall be legibly marked with black or white letters to indicate the type of service and the direction of flow, as identified below:

Hot water	:	HW
Chilled water	:	CHW
Condenser water	:	CDW
Steam	:	ST
Condensate drain	:	CN

Table 42 Scheme of Colour Code of Pipe Work Services for Air Conditioning Installation

(*Clause* 17.5.4.1)

Sl No.	Description	Ground Colour	Lettering Colouring	First Colour Band
(1)	(2)	(3)	(4)	(5)
i)	Cooling water	Sea green	Black	French blue
ii)	Chilled water	Sky Blue	Black	Black
iii)	Central heating	Dark Blue	Black	Canary yellow
iv)	Condensate drain pipe	Black	White	
v)	Vents	White	Black	
vi)	Valves and pipe line fittings	White with black handles	Black	
vii)	Belt guard	Black yellow diagonal strips		
viii)	Machine bases, inertia bases and plinth	Charcoal grey		

17.6 Testing and Balancing

17.6.1 *Air System Balancing*

Air distribution system shall be balanced so that air quantities measured at fan discharge and at various outlets shall be within ±10 percent of the design airflows. Branch duct adjustments shall be permanently marked after the balancing is complete. Throttling losses in the system shall be minimum. Tests shall be carried out for each fan and AHU. All dampers after adjustment shall be set and locked in position. All air and static pressure measurements shall be done with probe type meters. After balancing the fan speed shall be suitably adjusted. Duct leakages shall be tested and validated as specified.

17.6.1.1 Airflow balancing shall be carried out by adjusting the flow at each diffuser to match the right percentage of total airflow in the ductwork. Balancing shall be done based on the ratio between measured and designed flow rate. Following steps may be followed for proportional balancing:

a) Ensure all VCD, VAV and fire dampers are in fully open position.

b) Adjust the damper of each diffuser proportionately to the same ratio in each branch duct.

c) Adjust each branch duct VCD in each main duct so that each branch duct has the same proportion of airflow.

d) Adjust each main duct VCD.

e) Adjust fan speed (through VFD or belt pully) or the main VCD after the AHU to ensure that each diffuser has the right airflow rate.

f) Measure and record airflow rate at each diffuser and compare it with design airflow (within $\pm$ 10 percent)

17.6.2 *Water System Balancing*

The following are guidelines for testing and balancing in air and water systems:

17.6.2.1 Water balancing (of chilled water/brine) shall be done as under:

a) Keep all balancing, control and isolating valves in fully open position;

b) Measure the water flow across each balancing valve and compare it with design flow;

c) Identify the index value (lowest flow ratio) in each branch;

d) Keep index valve fully open;

e) Throttle the farthest valve to the same ratio and after that every valve in the branch starting from the lowest ratio;

f) Monitor index valve percentage after throttling each valve gradually. By the time the last valve is completed, the flow is balanced at all valves including the index valve; and

g) Record the reading at all valves.

17.6.2.2 Where pressure independent balancing and control valves (PIBCV) are used, the proportional balancing described above is not required. Each valve may be set to correct water flow rate as per manufacturer's instructions.

17.7 Cleaning/Flushing of Duct/Pipe work

The following are guidelines for cleaning and flushing of duct and pipe work:

17.7.1 In duct work, access doors shall be provided next to dampers and these access doors shall be used to clean the duct work. Prior to commissioning of the system, entire duct shall be cleaned. Where specified, such cleaning shall be done by duct cleaning systems using robots. Otherwise, the fans shall be run with only pre-filters in place (to protect coils) and the entire dust shall be blown off. After the cleaning, pre filters shall be replaced.

17.7.2 Pipe work shall be cleaned and flushed before commissioning. Flushing of system shall be done bypassing sensitive parts such as chiller and heat exchanger tubes. Before filling up the system with clean water, all shut off and control valves shall be kept fully open. Continuous air venting shall be done. Pump shall be started only after filling up and air venting is completed. Additional flushing pumps may be used over and above system pumps and the primary pump strainer basket mesh shall not be less than 3 mm when flushing starts. Flushing velocity from 1.0 m/s to 1.25 m/s is recommended. Pressure drop across pumps shall be monitored during the process so that pump cavitation is avoided.

17.7.2.1 Dynamic flushing in several steps is recommended. First the primary ring main may be flushed, thereafter the main pipes and lastly the horizontal mains in each floor. After each round of flushing, the strainer basket shall be changed. Each loop needs to be flushed till the water is clean and the strainer basket does not collect sediments. Flushing of horizontal main pipes shall start at the topmost floor and finish at the lowest floor. After flushing in sections is completed, full system flushing shall be carried out. After flushing is complete, the water sample shall be analysed to determine successful flushing.

17.8 Commissioning, Testing and Balancing, Performance Validation, System Handover

The following are guidelines for effective commissioning, testing and balancing, performance validation, and system handover to ensure that all systems operate as per design intent and meet performance requirements:

17.8.1 Commissioning shall mean putting an equipment or a system into active service as intended by the owner.

17.8.1.1 In respect of unitary equipment such as room (window air conditioners), Non – ducted split units, package units or VRF, this process shall involve connecting the unit (after installation) to a suitably sized power outlet, switching on the unit and recording the temperature/RH/ air movement/air quality in different locations in the conditioned space. The process shall also involve whether the unit's contracted features are operational and whether the compressor switches off once the set temperature is achieved. Air flow from the unit as well as noise level shall be appropriately measured and recorded. Capacity/power consumption is verified with factory issued test certificates.

17.8.1.2 For a central air conditioning plant or a mechanical ventilation system, the process of commissioning shall follow a structured and documented process. This shall comply with special publication.

NOTE — Special publication may be ISHRAE Standard 10003-2020.

17.8.1.3 The commissioning flow chart given below in Fig. 40 sets out the stages of this process.

17.8.1.4 Initiating the process. Owner shall designate commissioning authority (CxA) at the start of the process. CxA shall either be a member of the owner's project team or an independent agency appointed for this specific purpose. CxA shall have adequate knowledge and experience to guide the project through various stages of the process. Engagement of CxA shall be for a minimum period extending to one year from date the building/facility is put to use. It shall be the responsibility of CxA that all non-conformances, warranty issues, seasonal and occupancy related performance issues are fully resolved during her tenure with the project.

Owner shall be responsible for defining scope of the commissioning plan, incorporating CxA into the project team and establishing the commissioning budget.

Role of CxA and contractual requirements for commissioning shall be a part of all sub contracts including contracts with architects/consultants.

17.8.1.5 Owner's project requirements (OPR) shall include all functional requirements of the project and owner's expectation regarding its use and operation.

17.8.1.6 Commissioning plan shall be a written document scheduling the commissioning activities. This shall include scope, resources required at each stage, agreed timelines and roles and responsibilities of each member of the project team.

17.8.1.7 Basis of design document details design team's approach to meet OPR.

17.8.1.8 Commissioning specifications shall be established for all equipment and systems.

17.8.1.9 Commissioning design review shall involve verifying that all equipment and systems meet OPR. This shall not be a peer review of the design.

17.8.1.10 Testing shall involve preparation of check lists, test procedures and report forms which shall be agreed by all concerned. Person responsible for executing test protocols shall be identified.

17.8.1.11 Functional performance testing shall involve recording the operation, then initiating a change in one or more parameters and recording the changes in performance. This shall be done in auto mode with functional BMS and controls.

17.8.1.12 All information required to operate, maintain and if required, modify the system in future. Systems manual shall be approved by owner and kept updated.

17.8.1.13 Operation and maintenance training shall be documented. Evaluation and feedback are important.

17.8.1.14 Final commissioning report shall summarize the entire process and shall have owner's acceptance.

17.8.1.15 Post occupancy commissioning shall include all delayed and seasonal tests, warranty actions/records and resolution of all non-conformances with owner's specific approval.

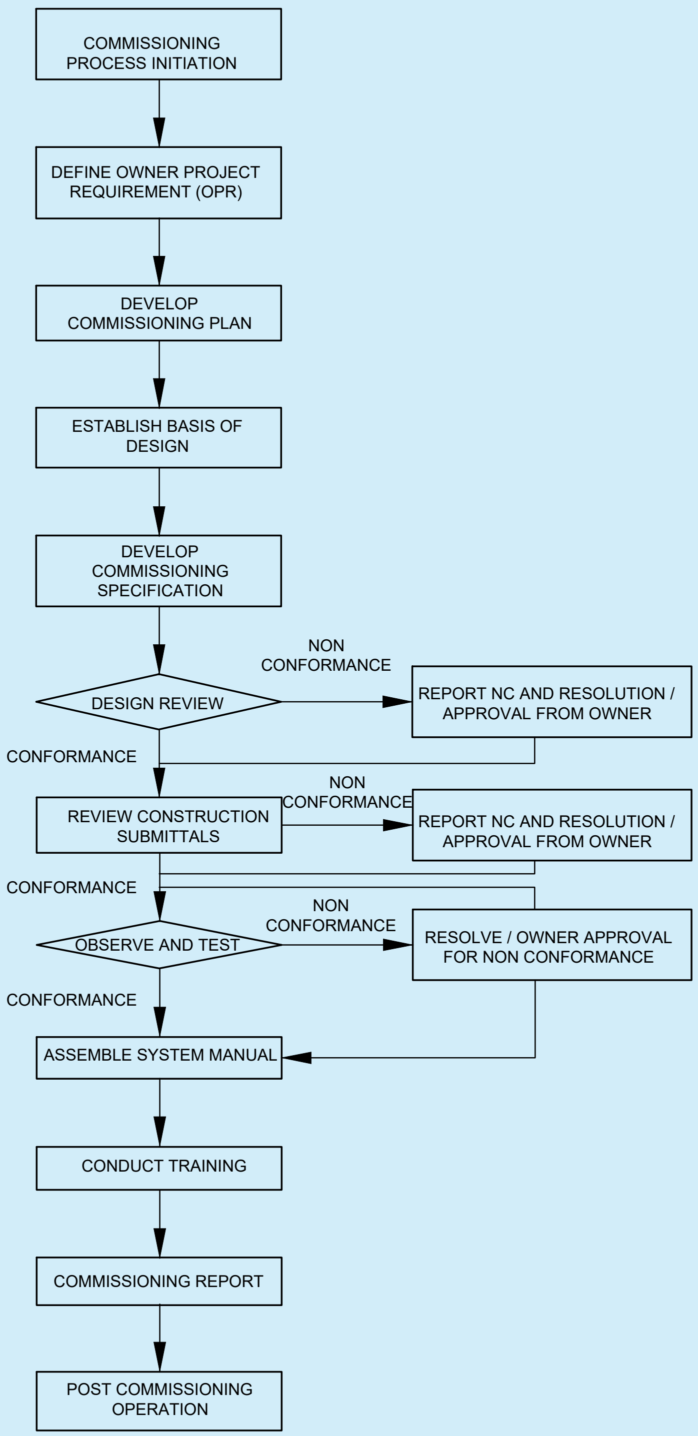


FIG. 40 FLOW CHART FOR HVAC COMMISSIONING

17.8.2 For testing of equipment, receipt and inspection at site, safe storage and handling at site, refer **17.2**. For each piece of equipment, sub system and system check lists shall be made use of for delivery, installation, start up and readiness of commissioning. Sample check lists available in special publication may be suitably adapted for use.

NOTE — Special publication maybe ISHRAE Standard 10003-2020.

17.8.3 Individual pieces of equipment, sub systems shall be considered ready for testing only when the complete installation is over, pressure testing/leak testing (as specified) are completed and required utility connections are made to the satisfaction of CxA. It shall be verified that all blank offs, shipping bolts etc have been removed. Where specified, manufacturer/supplier shall inspect and certify that the equipment/system is ready to start.

17.8.4 Testing of all equipment/ systems shall be in line with the documented protocol which is agreed between the manufacturer/vendor, CxA, design consultant, project management consultant and any other agency who may have jurisdiction over the project. Check lists, reporting formats and standards applicable shall all be agreed in writing beforehand.

17.8.5 Training and documentation for data collection leading to energy analysis shall form part of the process. It is recommended that all individual loads greater than 15 KW be monitored for energy management.

17.8.6 Static inspections that verify proper installation (for example, belt tensions, coupling alignments, oil levels, labelling, gauges and instrumentation in place, sensors calibrated etc) shall be submitted by the vendor to CxA with the request for startup.

17.8.7 Pre-commissioning cleaning of the installed ducting and leak testing of the ducts (where specified) shall be carried out prior to staring of AHU/Fans.

17.8.8 Activities where the equipment is initially energized, tested and operated shall be completed before functional performance testing (FPT) (*see* **17.8.1.11**). FPT shall involve testing under dynamic situations such as low load, full load, varying ambient, power outage, distress and emergency conditions etc. FPT shall establish that the equipment/system performs as designed under all these conditions. Systems shall be operated integrally through all specified operation sequences. Each step of FTP shall be planned in advance, agreed between vendor and CxA, performed under supervision and results tabulated in a pre-agreed format. In case FPT shall not be performed due to any reason (such as ramp up period) before the equipment /plant is put to beneficial use, same shall be recorded and owner's approval taken.

17.8.9 Operation of all controls and safety cut outs shall be demonstrated and recorded. All measuring instruments shall have valid calibration certificates which shall be part of the record.

17.8.10 Handover documentation shall include all information owner/user will need to operate, maintain and if required, modify the system at a later date. Hand over documents shall include the following as a minimum requirement:

a) Updated and approved (by owner) systems manual;

b) Final commissioning report;

c) Set of as built drawings;

d) Full set of design calculations;

e) Spare parts lists/ordering procedure;

f) Warranty records, escalation matrix;

g) List of non-conformance with resolution records;

h) List of accessories/tools/spare parts supplied with system;

j) Operation and maintenance manuals provided by equipment supplier; and

k) List of spares and escalation matrix.

17.9 Safety Points Specific to HVAC Systems

The following are safety guidelines specific to HVAC systems to ensure safe operation and maintenance:

17.9.1 Persons working on installation of HVAC Systems, large and small, are exposed to the same risks applicable to any construction activity such as bodily injuries, fall from height, electrical shocks etc. It is of prime importance to follow all safety requirements mandated by national as well as local codes, rules and regulations, including but not limited to, the safety provisions of Indian electricity act and rules framed thereunder and fire safety regulations prescribed by authority having jurisdiction (AHJ).

17.9.2 Following are the safety guidelines specific to HVAC Systems:

17.9.2.1 Nature and level of smoke hazard from the type of insulation material used in the system shall be carefully evaluated and proper ventilation and smoke management provisions shall be made dep, ending on the type and scalability of potential smoke generation.

17.9.2.2 Refrigerant gas being heavier than air, shall cause suffocation in enclosed spaces. Personnel shall use prescribed (by the refrigerant supplier) safety equipment while handling refrigerants and the plant room shall be well ventilated. Refrigerant cylinders shall be stored in upright position. Appropriate safeguard shall be in place based on manufacturer supplied information with respect to physical and chemical health hazards as well as storage and handling information.

17.9.2.3 While pressure testing systems, the following precautions shall be taken:

a) Obtain necessary work permit;

b) Barricade the area;

c) Use calibrated instruments and gauges;

d) Store gas cylinders in covered area and in upright position. Use colour codes to identify the gas inside the cylinder. Avoid exposing cylinders to sunlight;

e) Isolation/relief valves shall be provided appropriately;

f) Air shall be completely purged from water systems before pressure testing. identify drain valves and drainage routes/sumps; and

g) Have documented SOP approved by engineer-in-charge before work is taken up. Maintain appropriate records.

18 POST OCCUPANCY OPERATION AND MAINTENANCE

Proper maintenance schedules, regular inspections, and timely repairs are critical to optimizing system functionality and extending equipment lifespan. These practices also contribute to energy efficiency and occupant comfort in the building.

18.1 Operation and Maintenance Procedures

The following guidelines for post occupancy operation and maintenance to ensure long-term performance, safety, and efficiency of the HVAC systems after installation.

18.1.1 *Operating Plan*

The operating plan is given below:

a) Facility operating plan shall describe the intended use of this facility to ensure safety to the users, maintain thermal comfort, optimize resources efficiency, monitoring and data collection of the systems being operated in the facility;

b) Key focus of this plan to ensure safety, reliability, comfort and healthy environment for the users/occupants;

c) During the post occupancy phase of operations, the facility shall develop a systems operations manual;

d) This manual shall comprise the owner's project requirements (OPR) and also the current facility requirements (CFR);

e) These documents shall be maintained throughout the operations of the facilities and shall update the changes in the operational procedures and functions of the facility;

f) Maintaining the information for the facility/building operators to understand, operate and maintain the building systems for entire life cycle of the facility as per design;

g) Operations and maintenance staffs shall effectively utilize the procedures for documenting and tracking of information on any updates, modifications to systems that occurred during occupancy and post occupancy of the facility;

h) The document shall include the modifications performed and the reasoning for the modification to the building operations;

j) As part of the operating plan the facility/operations team shall ensure all the safe operational practices recommended by the original equipment manufacturers (OEM); and

k) The facility/operations team shall ensure that the design documents like basis of design (BOD), as built drawings/documents updated time to time, are kept intact to ensure the intended operations of the systems in the facility/building.

18.1.2 *Facility Layouts*

This includes simple line diagrams (schematics) capturing the facility site/building plan, floor layouts and locations of major equipment and systems, control centers (like, electrical power isolations points), utility shut off points (like, dampers, isolation valves) to assist orientation/training of operators.

18.1.3 *Data Logging*

Maintain daily log of each system operations, record values as per OEM checklist for daily readings and record any abnormalities like sound, vibration, leakages, etc.

18.1.4 *Scheduling of Operation*

Operating hours for the facility to be described and listed and this should be the basis to perform more detailed scheduled and operational sequences.

18.1.5 *Roles and Responsibilities*

The building owner/operator shall describe their facility team/operating staff on the requirements of the building operations.

18.1.6 *Minimum Qualifications for the Building HVAC Operators*

The minimum qualifications for the HVAC Operators should be:

a) The operator should be having a required education qualification of reading operating manuals, standard operating procedures in the language that written with;

b) *Engineering-in-Charge* — Degree or diploma in electrical/mechanical/HVAC with not less than 5 years of experience respectively in operating and maintaining the HVAC systems;

c) *Supervisor* — Diploma in electrical/ mechanical/ HVAC with 3 years of experience or ITI or equivalent in HVAC with 5 years of experience;

d) *Technician* — Diploma in electrical/mechanical/HVAC with minimum 1 year of experience or ITI or equivalent in HVAC with 3 years of experience;

e) *Operator* — Should have passed 10th Class/Intermediate from a state board of education [like CBSE or higher secondary education (HSC)]. Also, operator should have minimum technical qualification of certificate course or refrigeration and air-conditioning from an institute having reputation with state board of technical education (like ITI/NCVT/ AICTE); and

f) *Trainee* — Anybody not meeting the above qualifications may be inducted as a Trainee attached to any of the above positions. He/she may not be allowed to operate systems without supervision or support during their training period of minimum one year.

18.1.7 *Building and Equipment Operating Schedules, Set Points and Ranges*

The following guidelines for building and equipment operating schedules, set points, and ranges are designed to maintain optimal system performance and energy efficiency:

a) List and describe the planned/intended operating hours of the facility/building;

b) Building owner/operator shall define the initial operating schedules, anticipated occupancy, desired temperatures in the occupied zones;

c) All the set points of equipment with their safe operational requirements/adjustments shall be listed and provided to the operations team to perform their tasks;

d) The list shall comprise safe operating set point values, acceptable ranges and limitations; and

e) Any change to be implemented on the set points shall be recorded and implemented with reason for changing the set values to eliminate human errors.

18.1.8 *Sequence of Operations and Limitations*

The sequence of operations and limitations are listed as follows:

a) List out and provide operational sequences of all equipment to be operated in language and format that is understandable to the operating staffs;

b) The operators shall be briefed on the occupied and un occupied mode of operations of system and components; and

c) The operators shall be briefed on the safe shutdown of plant and equipment.

18.1.9 *Start-Up and Shut Down Procedures/Actions*

Start-up procedures shall be adopted as stated in **18.1.10** to **18.1.12**. The maintenance procedure shall be as laid down in **18.1.15**.

18.1.10 Before starting the plant, the operator shall ensure that:

a) All the valves in the refrigeration system, condenser water lines and chilled water lines are kept open, except those of the standby items of equipment;

b) Adequate water in the cooling tower basin and the make-up water system is working satisfactorily;

c) The make-up water system to the expansion tank of the chilled water circuit is working and there is a regular water supply. Water is lost from the chilled water system through pump gland drips/leaky mechanical seal. If the level in the expansion tank is not maintained, air shall enter the chilled water system, affecting the system performance substantially. It shall even lead to system/plant breakdown;

d) All the air filters and water strainers are clean. Any dirty filter and/or the blocked strainer shall be cleaned periodically;

e) All doors and windows of the air conditioned/refrigerated area are closed;

f) The crankcase of the compressor is warm (to the physical touch). If it is not warm, defects in the crankcase heater and/or circuit should be checked fully. The compressor shall not be started until the defect is rectified and the crankcase warms up, else it will result to poor lubrication, thereby substantially reducing the life of the bearings of the compressor; and

g) The supply voltage is within permissible limits. The windings of the motors shall get affected, if run on low voltage conditions.

18.1.11 The starting sequence shall be as follows:

a) Switch 'ON' the mains. Observe the voltage. If it is less than the permissible level, do not start any component of the system;

b) Start air handling unit motors (all dampers have to be checked based on their specific application);

c) Start condenser water pumps. Check that sufficient water pressure is obtained;

d) Start cooling tower fan;

e) Start chilled water pumps and check the pressure;

f) Switch 'ON' the compressor control switch; and

g) Start the compressor motor.

18.1.12 Following aspects require special attention:

a) In the compressor with oil pump, the compressor oil pressure should build up as the compressor is started. Check that corrects net oil pressure is built up;

b) Check the oil level in the oil sight glass of the compressor. The oil level should be about 40 to 50 percent of the sight glass. In operation, certain amount of oil gets entrained in the refrigerant vapor in the compressor and is carried to the system along with the refrigerant;

c) After the plant operation has stabilized, check all the pressures and temperatures and ensure that the system is working satisfactorily;

d) Record periodically the readings of temperature, pressure, current and other required data in the log sheet;

e) Check for any unusual noise/vibration in the plant. If something unusual is noticed, trace out the reason for it and rectify the cause; and

f) During the operation of the system, if any major component stops suddenly, trace out its reason, before starting the component/system again.

18.1.13 The stopping sequence of the system shall be as follows:

a) Switch off the compressor on the low-pressure switch, as the system gets pumped down;

b) Switch off the power supply to compressor;

c) Check that the crankcase electric heater comes on as soon as compressor stops and ensure that the heater is working;

d) Stop chilled water pumps;

e) Stop air handling units; and

f) Stop condenser water pumps and cooling tower fans.

18.1.14 Following are standard operating instructions which the operator should follow:

a) In case stand-by chiller, pumps, etc, are provided, systematically change over the stand-by components, periodically. This will ensure uniform wear and tear of the system. Further, this also helps in ensuring that all equipment is in good shape and the stand-by is in working condition;

b) Do not switch off the main switch on the main electrical board or switch off any component of the system, when the plant is in operation;

c) All the water valves in the system shall be kept open and need not be closed, unless specifically instructed by designer. But it shall be done to close and open each valve periodically to ensure that the valves work and are not stuck by scale formation or dirt accumulation; and

d) Keep the plant room clean. Do not use the plant room, particularly the air handling unit rooms for storage.

NOTE — An automatic sequence of operation shall be covered under Chapter 15 - Building Management System.

18.1.15 *Maintenance Procedures, Check Points and Records*

Procedure for establishing and maintaining effective maintenance practices, ensuring comprehensive inspection points, and maintaining accurate records for facility management.

18.1.15.1 *Preventive maintenance*

The preventive maintenance is:

a) The goal of preventive maintenance is to prevent equipment failure caused normal wear and necessary action by replacing worn out components prior to actual failure of an equipment;

b) Maintenance activities shall include partial or complete overhauls at specified periods and shall include oil changes, lubrication of components, minor adjustments and so on;

c) Operators shall record equipment deterioration by inspection to facilitate the replacement or repair of worn-out parts prior to any system failures;

d) Implement preventive maintenance of installations and equipment to maintain efficient working of building HVAC system and its components;

e) Implement periodic inspection schedule, testing and maintenance throughout their life cycle;

f) Conduct periodic maintenance and testing, servicing, checking, calibration, overhauls and certification, annual inspection, testing, witness regular test and certification;

g) Perform overhauling or replacement at a certain usage (as recommended by OEM) or due to specific causes. This shall be outlined specifically in a preventive maintenance program;

h) Implement preventive maintenance program (PMP) and shall include the methodology and record for all action that are necessary to maintain the efficient working order of the HVAC system;

j) The maintenance procedures shall be unique to each property and the installations and equipment within these facilities;

k) Conduct preventive maintenance in accordance with statutory requirements. (pressurization system, smoke control system, fire alarm interlock with air handling units are few example);

m) Conduct preventive maintenance and overhauls with clear maintenance schedule, instruction and procedures;

n) Ensure O&M personnel induction training on safety, statutory requirements and performance target and work manner;

p) Adopt condition based/reliability-based maintenance where appropriate;

q) Conduct a regular review of all procedures/standards;

r) Conduct a regular update of relevant procedures/standards against the latest statutory requirements, good industrial practices, maintenance records and fault history by a working/facility team comprising competent personnel; and

s) Adopt a web based/mobile application-based performance monitoring system for maintenance that is contracted out.

18.1.15.2 *Maintenance schedule*

18.1.15.2.1 *Day-to-day maintenance schedule*

a) *Clean the air filters* — Dirty filters cause poor air flow through air handlers and results in thermal discomfort. Unattended filters could result in damage to the filtering media and the unfiltered air shall pass through them and settle on the cooling coils. This results in poor heat transfer and impacts the efficiency of the air handler system and total efficiency of the HVAC system as a cascading effect;

b) *Leak testing for refrigerant leaks* — Check physically for oil traces, which is indicates source of refrigerant leaks. Perform leak testing using soap bubble or electronic leak detectors. Refrigerant leaks impact the system operation and leads to breakdown of system. Refrigerant leaks impact our environment due to global warming potential of the refrigerants being used in the HVAC system;

c) *Water pump (packed) glands* — Certain amount of water should drip through the gland which is necessary to keep the gland cool. However, if the drip develops into a flow of water, then, it is an indication of leak gland. This should be attended by tightening the gland nuts to reduce the water leaks or gland packing to be replace to prevent loss of water from the distribution system. Unattended gland water leaks shall result in loss of water and facilitates air entry into to the system which could result in abnormality in the chilled water distribution system; and

d) Check and monitor any abnormal noise or vibration.

18.1.15.2.2 *Weekly maintenance schedule*:

a) *Belt Tension of belt drives* — Check the belt tensions and tighten whenever necessary if found loose. A loose belt reduces the transmission efficiency, reduced air volume of the air handlers and thus affects the thermal comfort. Loose belts cause noisy air handlers and results disturbance to the building occupants;

b) *Check the cooling coil drain tray* — Check and ensure the drain tray is clean from dirt and debris to prevent mould growth or water leakage due to drain block;

c) *Check the lubrication* — Check the lubrication of bearings to ensure smooth operation and prevent breakdowns due to worn out bearings;

d) *Check the Cooling tower* — Check the cooling tower sumps to keep them clean from debris (tree leaves if the towers are located at ground level). The towers are kept open to ambient and dust from the air shall settle on the fills, sumps and requires cleaning to ensure proper operation of the cooling towers which are very critical for a water-cooled system efficiency and performance; and

e) Conduct water quality test for cooling towers periodically.

18.1.15.2.3 *Monthly Maintenance Schedule*:

a) Replace the air handling unit filters with a set of replacement filters kept under stock;

b) Check the starters and check loose connections of cables and ensure tightness;

c) Check the operation of automatic blow down system for cooling towers;

d) Check and clean the Y-strainers and Pot Strainers;

e) Check the preventive maintenance schedule of components and equipment and conduct maintenance activities; and

f) Check and tighten all base bolts to prevent breakdown due to vibration.

18.1.15.2.4 *Quarterly maintenance schedule*:

a) Perform pre scheduled preventive maintenance activities equipment and components;

b) Check the log book and conduct thorough study of equipment to ensure healthiness of its operation;

c) Check and lubricate all the moving parts;

d) Check and clean the cooling tower sumps;

e) Check and clean all the strainers and pot strainers;

f) Check and replace belts if necessary;

g) Check the safety interlock of all air handling units;

h) Check and replace defective instruments like pressure gauges, thermometers etc;

j) Check and rectify any refrigerant leaks; and

k) Check and replace any defective/malfunctioning components of the system.

18.1.15.2.5 *Half year maintenance schedule*:

a) Perform all the activities listed under quarterly maintenance schedule;

b) Check the conditions of the cooling tower fills and clean them;

c) Check and clean the air handling unit blowers from dust/dirt accumulation; and

d) Check the operation of cooling tower blow down system (automatic) and chemical dosing system.

18.1.15.2.6 *Yearly Maintenance Schedule*:

a) Perform all the activities listed under half yearly maintenance schedule;

b) Collect oil sample from the chiller system and send them for laboratory test to analyze presence of metal particles, viscosity etc;

c) Post lab results replace the oil to ensure safe operation of the compressors to enhance the life cycle;

d) Check and clean the condenser tubes of the water-cooled system to improve heat transfer. If, required only, conduct de-scaling of the tubes. Frequent de-scaling is not recommended and cause potential damage to the copper tubes;

e) Check the operation of auto tube cleaning system if it is in place and ensure its components are intact or replace if required;

f) Conduct vibration test of all major equipment and record the test readings for further comparisons till life cycle of the equipment;

g) Witness maintenance of all the electrical starter panels, switch gears, earthing conductors and perform interlock testing for safe operations of the total system; and

h) Conduct indoor air quality test and if required perform duct cleaning procedures to ensure good indoor air quality.

18.1.15.2.7 *Check points*:

a) Compare the previous maintenance log and record any abnormalities;

b) Conduct necessary actions to rectify or replace component/equipment as noticed on abnormalities;

c) Ensure O&M personnel induction training on safety, statutory requirements and performance targets; and

d) Conduct and analyze on operating costs due to lack of maintenance and ensure corrective actions.

18.1.15.2.8 *Records*:

a) Good maintenance records for ensuring that a piece of equipment is performing in line with OEM recommendations, warranties and thus help to determine an equipment preventive maintenance schedule;

b) Maintenance records to be maintained to assist the service matter experts, (SME's) service technicians for diagnosing any repeat problems with or within the plant or its equipment;

c) Maintain paper records of all maintenance related activities including testing and commissioning certificates, test records, as-built documents, statutory approved submission documents, water quality test reports, oil quality test report, indoor air quality reports, calibration certificates of instruments;

d) Maintain emergency call/fault attendance reports, fault history etc;

e) Maintain records of usage of refrigerants;

f) Maintain soft copies of all the documents listed under paper records;

g) Assign designated persons/s responsible to review and update routine maintenance inspection schedule, emergency call/fault attendance reports etc, on a monthly basis and the findings shall be documented for future reference;

h) Set up record system able to automatically provide alerts for outstanding shutdown notices and annual maintenance renewal etc; and

j) Digitize all documents and records with a standardized file naming system in a reliable database server for easy retrieval. (Facility management system/manager shall perform this activity).

18.1.15.2.9 *Training*:

a) Provide training to equip staff with knowledge to work safely and without risk to health;

b) To Ensure maintenance plans are followed correctly, facility managers and building owners should provide sufficient periodical training to all maintenance technicians. Good sources of trainings are:

   1) An experienced technician or facility staff member;

   2) OEM's training;

   3) Technical and community colleges; and

   4) Videos, seminars and short-term courses by professional associations like ISHRAE.

c) The minimum training content delivered should generally include below:

   1) Basis of design or design brief;

   2) Documentation requirements;

   3) Operating procedures such as startup and shutdown procedures;

   4) Basic diagnostic instructions; and

5) Fire, life safety and emergency procedures.

18.1.15.10 *Spare parts inventory management*:

a) Managing spare parts in an optimal way is an inherent and substantial part of O&M aimed at ensuring that spare parts are available in a timely manner for corrective maintenance in order to minimize the downtime of a system or equipment;

b) Maintain a spare parts list for plant and equipment;

c) Maintain an updated contact list of suppliers with lead time details of each spare;

d) Review constantly the quality and quantity of spare parts in stock and restock when necessary;

e) Maintain the log of spares utilized;

f) Assign designated person/s responsible for regular updates of any changes in spare parts inventory;

g) Monitor the condition of spare parts to ensure their quality is maintained;

h) Identify long lead items/components for early procurement; and

j) Identify discontinued items (obsolete) for sourcing an alternative parts or upgrade of system.

18.2 Predictive Maintenance

a) Predictive maintenance is a planned maintenance basis diagnosis of equipment condition;

b) Measurements that detect the onset of a degradation mechanism, thereby allowing causal stressors to be eliminated or controlled prior to any significant deterioration in the component physical state;

c) Predictive maintenance techniques may include vibration analysis, thermography, oil analysis, motor current analysis, Eddy current testing and pressure measurement etc; and

d) Performance trending.

18.2.1 *Emergency Operating Procedures*

a) Emergency event requires planning, preparation and practice;

b) Regardless of type of risk, every facility shall have a risk event plan or emergency management in place to minimize the damage and danger to following:

1) Building occupants;

2) Documents and assets within facility;

3) Building, IAQ and surrounding community, air and water; and

4) Establish and implement environmental management systems and safety management systems. For additional information special publications may be referred.

NOTE — Special publication maybe ISO 14001 for Environmental Management Systems and OHSAS 18001 or ISO45001 for Safety Management Systems.

18.2.2 *Performance Based Operation and Maintenance Contracts*

a) Performance based or output based operation and maintenance contracts gaining traction;

b) The advantages of performance-based O&M contracts are;

c) Payment for actual performance or usage;

d) Contractor responsibility for full package;

e) Continuous improvement/Innovation; and

f) Reduced risks due to lower capital costs.

18.2.3 User feedback mechanisms for optimizing maintenance schedules or identifying:

a) Metering and sub-metering of energy and resource use is a critical component of a comprehensive O&M program;

b) Metering provides the information that when analyzed allows the building operations staff to make informed decisions on how to best operate mechanical/electrical systems and equipment:

   1) Metering approaches;

   2) One time/spot measurement;

   3) Run time measurement;

   4) Short term monitoring;

   5) Long term monitoring;

   6) Measure, verify, and optimize equipment performance; and

   7) Identify improvement opportunities on environmental (especially energy, water efficiency) and safety aspects.

18.3 Computerized Maintenance Management System (CMMS)

CMMS is a computerized maintenance management system, which is a software application that helps controlling, monitoring and analysing operation of HVAC system. The centralized CMMS would help in optimization of cost, resources, and information at one place.

Benefits that the CMMS shall provide to the facility during operation and maintenance shall include

a) *Electronic records* — Use of electronic records reduces the amount of time spent by office personnel entering data, which improves the efficiency of the maintenance department;

b) *Reduced repair costs* — A CMMS shall store and provide easy access to equipment records. When historical records are easily accessible, they shall be used to determine the most economical decision for a piece of equipment: repair or replace;

c) *Personnel management* — Records for each employee shall be stored and easily accessed and updated. Electronic personnel records shall include, but are not limited to, training and certifications earned by each employee, pay scales, and hire dates;

d) *Asset management* — A CMMS shall be used to store electronic records for any type of asset, including equipment and supplies;

e) *Process automation* — When a CMMS is set up correctly, preventive and predictive maintenance work orders, parts and supply reordering, and other notifications shall be automated, saving time and increasing the efficiency of the maintenance team;

f) *Improved work control* — CMMSs provide increased ability to manage work orders and prioritize and schedule work;

g) *Increased budget accountability* — Accurate records of labour hours and inventory items increase the accuracy of expenses and budgets; and

h) *Increased ability to measure performance and service* — When data are available, performance shall be determined for a period of time, such as that required to complete a specific type of maintenance activity or respond to a service call.

CCMS Modules shall be considered for facility operations are as below:

1) *Asset management* — Its asset operations, monitoring and control, repair and PM history;
2) *Work order management* — Helpdesk ticket end-to-end user complaint to closure of calls;
3) *Procurement* — Inventory management, parts ordering; and
4) *Dashboard reporting* — This module allows you to generate reports on a variety of maintenance metrics, such as downtime, costs, and asset health.

18.3.1 Broad CMMS selection process that shall be considered as follows:

a) *Team involvement* — To include the facility managers, clients, property management team, consultants and technicians shall be involved during the planning phase to select the suitable CMMS software, scope of working of CMMS and integration requirement with other third-party systems (BMS, EMS and AI); and

b) When selecting the software, the following shall be part of the process:

   1) Continued focus and understanding of the true need for the CMMS; and
   2) Clear understanding of how the CMMS implementation will support maintenance best practices desired by the organization.

18.3.1.1 Process of selection given below shall be followed:

a) For planning:

   1) Establish a CMMS selection task force;
   2) Perform interviews with all members of the facilities team who will use the CMMS;
   3) Determine the needs, wants, reporting features, and goals for the CMMS;
   4) Determine what CMMS outputs are needed to meet the goals;
   5) Determine input data that shall be collected;
   6) Evaluate the value of data versus the cost of collecting, inputting, and maintaining the data within the CMMS;
   7) Integration of third-party software data to CMMS;
   8) Determine available budget;
   9) Determine CMMS modules needed;
   10) Write CMMS specification;
   11) Prepare for CMMS vendor meetings;
   12) Meet with vendors and participate in vendor demonstrations;
   13) Prepare bid package;
   14) Develop a continual implementation plan;
   15) Determine if a phased implementation plan is needed; and

16) Develop a phased implementation plan (if needed).

b) CMMS software/ CMMS module selection shall fulfil the operational needs of facility and shall address the following:

1) Areas for improvement in current maintenance practices (for example, reduced downtime, improved preventive maintenance);

2) Advanced features and functionalities (for example, mobile access, work order management, reporting);

3) Strong asset management, work order creation and tracking, preventive maintenance scheduling, and inventory control;

4) Consider advance features like AI-based functionalities for predictive maintenance, asset analysis from R and M, PM perspective, dash-board reporting, mobile capabilities for Equipment operative team;

5) Integration with any existing software you use;

6) Options for future scalability;

7) Requirements towards company where CMMS to be installed IT/data security and compliance with relevant regulations;

8) Implementation costs, licensing fees (one time and recurring), availability of software key, and ongoing software support; and

9) Strong after sale service and customer support.

18.3.2 *Implement a CMMS, Going-Live Phase*

a) Select a CMMS;

b) Collect input data;

c) Input data into CMMS;

d) Determine how users will be trained;

e) Implement training;

f) Go live and cut over:

1) Going live is the transition between setting up a system and using the CMMS for daily maintenance activities. This project phase reveals how successful the planning and implementation processes were. Before going live, the CMMS shall be tested and checked to ensure correct system configuration.

g) Implement continual implementation plan; and

h) Create standard operating procedures (SOPs) for using the CMMS.

18.3.3 *Integration of CMMS with Building Automation Systems (BAS) for Data Collection and Analysis*

a) Integrating of CMMS with a building automation system (BAS) will be an intense tool for property and facility management engineering team. The BAS regularly monitors and collects data on several characteristics of a building's management systems;

b) Integration complexity shall be looked during planning phase for example, specific software and desired functionality, shall ensure that both the systems communicate with each other properly;

c) To protect sensitive building information, a common IT security protocol, proper data security protocol shall be ensured;

d) It shall be ensured that periodic data export from the BAS and import into the CMMS be established or worked during RFP phase. Make certain consistent data formats and naming principles between the CMMS and BAS for seamless integration; and

e) Protocol for two-way communication between both the systems, allowing for real-time data exchange shall be established.

18.3.4 *Automation and Integration of Chiller plant and Pumps*

Following systems shall be considered for automation and integration in the BMS and CMMS. Refer to relevant guidelines/best practices ('Handbook on Asset and Facility Management in Buildings') for further details.

a) Building automation system (BAS);

b) Chiller plant manager; and

c) Sensors, VFD, communication system/PLCs .

18.4 Controls to Measure and Monitor Building Energy Performance

18.4.1 *System for Data Collection and Analysis*

Building controls such as CMMS and BMS systems shall consists of following metering systems, controls and monitoring systems for data collection and analysis:

a) Main meters for sources of energy used by facility;

b) Submetering for applications that represent 10 percent or more of the total building consumption such as chillers, pumps, cooling towers, AHU's, ventilation systems, hotel rooms (cumulative), lighting and UPS systems;

c) Data collection systems: collecting data from energy meters, sub-meters on pumps, cooling towers, AHU feeders;

d) Building Automation System (BAS);

e) Chiller plant manager; and

f) Sensors, VFD, communication system/PLCs.

NOTE — Refer to relevant guidelines/best practices ('Handbook on Asset and Facility Management in Buildings') for further details.

18.4.2 *Maintenance Strategy and Controls Including Integration of CMMS with Building Automation Systems* (*BAS*)

a) *Predictive maintenance* — Using CMMS/CPM/ BMS, BAS data for example, equipment performance metrics and sensor readings, helps to predict potential breakdowns and shall schedule maintenance before failures occur. CMMS shall be used to analyse this data and initiate work orders;

b) *Condition* — Based Maintenance: BAS/ CMMS shall use this data to prioritize maintenance tasks based on actual equipment condition, optimizing resource allocation;

c) *Preventive maintenance* — CMMS should be used to schedule PM tasks based on OEM recommendations, historical data, and BAS/CMMS/BAS alerts; and

d) A 52 week PPM planner shall be used aligned with checks prescribed in OEM manuals.

18.4.2.1 Critical systems identification and analysis of efficiency and healthiness checks:

a) AHUs;

b) Compressors;

c) Chiller;

d) Condenser;

e) Cooling towers;

f) CMMS, PLC and controls;

g) Electrical, power distribution network;

h) Water cycle – open and closed circuits;

j) Boilers;

k) PAHUs/PACs, domestic air conditioners; and

m) STPs in case STP water is used in cooling towers.

18.4.2.2 Analysis of efficiency and healthiness checks:

a) ikW and COP test shall be performed with the help of OEM and maintenance agency the analysis report should contain performance designed vs actual;

b) Check for the AHU performance; and

c) A regular health check-up should be conducted by a professional HVAC engineer, preferably on annual basis, the test should be considered during Peak load for example, summer.

Further an equipment wise check shall be conducted using calibrated equipment's.

18.4.2.3 *Selection of maintenance class and transitioning from reactive to proactive maintenance*

To transition from one particular maintenance approach to another, a detailed analysis shall be made to determine what will be the most effective. Table 43 below indicates the type of maintenance to be considered for the systems/facility.

Table 43 Type of Maintenance to be Considered for the Systems/Facility

(*Clause* 18.4.2.3)

Sl No.	Maintenance Class	Effect of Failure	Desired Result	Analysis Used to Determine Most Effective Maintenance Technique	Implementation of Maintenance Practices
(1)	(2)	(3)	(4)	(5)	(6)
i)	A: Mission Critical	Significant financial, safety and/or operational impact	Risk mitigation and maximize equipment and system availability	Reliability centered maintenance and risk assessment	Preventive and predictive approaches
ii)	B: Optimize life-cycle costs	Minor impact on core business activities	Minimize equipment LCC over time	Reliability centered maintenance and risk assessment	Preventive, predictive and reactive approaches
iii)	C: Minimize short-term costs	No impact on core business activities	Minimize short-term costs	Risk assessment	Risk mitigation measures, visual inspection, and preventive maintenance
iv)	D: Industry standard maintenance	No impact on core business activities	Minimize short-term costs	Risk assessment	RS Means or original equipment manufacturer job plans, preventive maintenance
v)	E: Out-of-Service	No impact oncore business	Ability to operate	Risk assessment	Minimum maintenance

To select the appropriate maintenance class, the facility mangers to develop an effective maintenance program and shall consider the following:

a) Focus on maintenance that results in the best return on investment (ROI);

b) Make sure results shall be measured;

c) Avoid intrusive maintenance; and

d) Employ an effective management system, such as a computerized maintenance management system (CMMS).

18.4.2.3.1 The Systems to consider for proactive maintenance shall include:

a) AHUs;

b) Compressors;

c) Chiller;

d) Condenser;

e) Cooling towers;

f) CMMS, PLC and controls;

g) Electrical, power distribution network;

h) Water cycle – open and closed circuits;

j) Boilers;

k) PAHUs/PACs, domestic air conditioners; and

m) STPs in case STP water is used in cooling towers.

18.4.2.3.2 Transition from reactive to proactive maintenance shall benefits the facility operation and maintenance and that shall include:

a) Fixing things only when they break is a reactive maintenance, which leads to a sudden failure to equipment, causing down-time, production loss, Impact equipment life span/age and requires more budget/cost to manage;

b) Making a switch towards proactive maintenance for example, prevention over repairs requires time and efforts but helpful enhance equipment life, up-time, reliability, maintenance cost, safety and increased efficiency; and

c) A proactive maintenance should have deep understanding of your assets, thorough study of OEM manuals, drawings, design brief/project reports, regular inspection, analysis of failure history/equipment down-time/break-down historical data, condition monitoring, a regular training to operators understanding various failures code as described in OEM manuals.

18.5 Benchmarking of Building Performance with KPI Index

18.5.1 *Strategies for Implementing the Measurement and Verification* (*M and V*)

M and V strategies/ techniques shall be used by facility owners or energy efficiency project investors for the following purposes:

a) Increase energy savings;

b) Enhance financing for efficiency projects;

c) Improve engineering design and facility operations and maintenance; and

d) Manage energy budgets.

M and V plan produce verifiable savings reports. An M and V Plan shall be developed for each facility by qualified professional. M and V applies to a wide variety of facilities including existing and new buildings and industrial processes. M and V strategies consist of some or all of the following:

1) Meter installation calibration and maintenance.

2) Submetering for applications such as HVAC (Chillers, Pumps, CT), AHU's, ventilation fans, internal lighting, external lighting and UPS.

3) Data gathering and screening,

4) Development of a computation method and acceptable estimates,

5) Computations with measured data, and

6) Reporting, quality assurance and third-party verification of reports.

M and V options to be followed shall include as below:

i) Option-1 – Retrofit Isolation: key parameter measurement

ii) Option-2 – Whole facility

18.5.1.1 *Option*-1 – *Retrofit isolation: key parameter measurement*

Retrofit isolation techniques are best applied where:

a) Only the performance of the systems affected by the ECM is of concern, either due to the responsibilities assigned to the parties in an energy performance contract, or due to the savings of the ECM being too small to be detected;

b) Interactive effects of the ECM on the energy use of other facility equipment shall be reasonably estimated or assumed to be insignificant;

c) Sub-meters already exist to isolate energy use of systems;

d) Meters added at the measurement boundary shall be used for other purposes such as operational feedback or tenant billing; and

e) Long term testing is not warranted.

18.5.1.2 *Option*-2 – *Monitoring performance of whole facility*

Monitoring of whole facility Involves use of utility meters, whole-facility meters, or sub-meters to assess the energy performance of a total facility. The measurement boundary encompasses either the whole facility or a major section. This option determines the collective savings of all ECMs applied to the part of the facility monitored by the energy meter.

Facility team shall select the M and V options based on the energy metering infrastructure availability and energy conservation measures (ECM) implementation and monitoring of ECM.

a) *Overall HVAC system efficiency for benchmarking*

The overall HVAC systems efficiency shall be measured with the help of submetering as suggested in the M and V section and overall HVAC systems efficiency shall be calculated as overall HVAC energy consumption in kWh and divided by conditioned area of the facility.

$$\text{Overall HVAC systems efficiency} = \frac{\text{overall HVAC consumption (kWh)}}{\text{conditioned area-sq.m}}$$

b) Scheduled intervals, equipment performance to be measured and monitored and compared with bench marking of efficiency till the end-of-life cycle of the equipment. The equipment performance to be measured on monthly basis in kWh and performance to be tabulated for comparing with benchmarking.

c) Corrective action to ensure performance of the system.

The corrective actions to ensure performance of systems shall be as per Table 44.

Table 44 Potential Corrective Measures List

(*Clause* 18.5.1.2)

Sl No. (1)	Possible Cause (2)	Recommended Corrective Action (3)
i)	General	
	Human error in data collection and analysis	Train staff in data collection and analysis
	Instrumentation error	Check error and calibrate to attain manufacturer recommended level
	Equipment operating schedules	Change equipment operating schedules to more closely follow actual building activities.
	Equipment load	Identify correct plug load and ensure their proper use
ii)	Lighting	a) Fine tune on hours more closely to reflect actual.
		b) Measure actual lux level and take corrective action to ensure task-based lux level
		c) Check voltage level to ensure there is no over voltage to avoid over consumption
		d) Ensure fixture and accessories cleaning to increase lighting efficiency
iii)	HVAC- Chilled Water System	a) Ensure preventive maintenance as per schedule.
		b) Check DPT function.
		c) Check pump operates at its best efficiency point.
		d) Check COP at full load condition.
iv)	Water	a) Check for leak in pipes, fittings.
		b) Ensure periodical cleaning of the filters provided at the roof for rainwater harvesting.
		c) Rainwater collection sumps to be have a maintenance plan in place.

d) Measure the indoor air quality, thermal comfort, water performance and energy performance as part of KPI.

Metrics and trends are important parts of maintenance organization. Metrics, also known as key performance indicators (KPIs), shall be used to drive improvement, minimize problems, help set priorities and goals, and help determine milestones and evaluate success.

To effectively use metrics and trends, reports shall be set up and generated using the CMMS. There are two basic types of reports

1) *KPIs and measures reports* — Reports that contain key performance indicators, trends, and measurements that identify systems and equipment, processes, other maintenance management topics that need the most attention.

2) *Functional reports* — Reports used Measure the indoor air quality, thermal comfort, water performance and energy performance.

18.6 HVAC System Life Expectancy Comparison

a) Comparison of HVAC systems to check the end of life/replacement of systems:

   1) The typical lifespan of an HVAC system is known. Nevertheless, after years of regular use, the efficiency and performance of an HVAC system naturally decreases overtime;

   2) There are few signs that signal this decrease are:

      i) Increase in breakdown frequency;

      ii) Increased energy use;

      iii) Performance degradation; and

      iv) Unusual noise.

   3) Risk based approach to plan and schedule replacement works in accordance with spare parts availability as well as any specific law and safety requirements, etc.

b) Performance based approach to analyse the HVAC systems life served including the factor affecting the HVAC systems lifespans:

   1) Output based approach to understand the life of HVAC system;

   2) Factors that affect the HVAC systems lifespan are:

      i) Quality of installation;

      ii) Utilization time of equipment;

      iii) Location; and

      iv) Operation and maintenance.

19 SMOKE MITIGATION – FIRE AND LIFE SAFETY

Smoke mitigation towards fire and life safety is towards providing human life safety with tenability to enable the occupants ideally to exit from event (fire incident) zone or in some occupancies to have occupants towards defend in place.

Compartmentation to ensure space by space integrity to prevent the spread of fire. Such spaces enable occupants to proceed to exits and to be provided with aspects of smoke mitigation though ideally pressurization or in some cases with smoke ventilation.

Smoke ventilation predominately post fire incident to enable reducing smoke concentration levels for firefighters and emergency responders shall more effectively check reoccurrence of fires and rescue occupants from the defend in place safe zone to exits. This also helps in reducing property damage by purge of smoke post fire incident in the fire incident area/ spaces.
The various strategies and spaces shall be identified to enable designers to adopt the required strategy of smoke mitigation and also have specific design, systems provided and its functions on respective occupancy of the project.

Technical, design, performance aspects with integration to fire alarm system and also importantly evacuation strategy are specified.

Aspects of pressurization (Pascals) for defend in place [One door open], phased evacuation [Three door open], assisted firefighting shaft [Three door open] shall be detailed in the smoke mitigation aspects towards life safety.

The power supplies, fire rating, interlocking of the smoke mitigation system on mechanical systems, equipment, elevators, lobby compartmentation, damper control shall be included for successful implementation of the smoke mitigation system.

19.1 Smoke Mitigation – Sub Structure

19.1.1 *Car Parking in Sub Structure*

19.1.1.1 *Basement car parking non-mechanical*

In case of smoke exhaust and pressurization of areas below ground, each basement shall be separately ventilated. Vents with cross-sectional area (aggregate) not less than 2.5 percent of the floor area spread evenly round the perimeter of the basement shall be provided in the form of grills, or breakable stall board lights or pavement lights or by way of shafts.

19.1.1.2 *Basement car parking stack/ mechanical*

In case of smoke exhaust and pressurization of areas below ground, mechanical ventilation system be provided with following requirements:

a) Mechanical ventilation system shall be designed to permit 12 ACPH in case of fire or distress call. However, for normal operation, air changes schedule shall be as given in 3;

b) In multi-level basements, independent air intake and smoke exhaust shafts (masonry or reinforced concrete) for respective basement levels and compartments therein shall be planned with its make-up air and exhaust air fans located on the respective level and in the respective compartment. Alternatively, in multi-level basements, common intake masonry (or reinforced cement concrete) shaft may serve respective compartments aligned at all basement levels. Similarly, common smoke exhaust/outlet masonry (or reinforced cement concrete) shafts may also be planned to serve such compartments at all basement levels. All supply air and exhaust air fans on respective levels shall be installed in fire resisting room of 120 min. Exhaust fans at the respective levels shall be provided with back draft damper connection to the common smoke exhaust shaft ensuring complete isolation and compartmentation of floor isolation to eliminate spread of fire and smoke to the other compartments/floors;

c) Due consideration shall be taken for ensuring proper drainage of such shafts to avoid insanitation condition. Inlets and extracts may be terminated at ground level with stall board or pavement lights as before. Stall board and pavement lights should be in positions easily accessible to the fire brigade and clearly marked 'AIR INLET' or 'SMOKE OUTLET' with an indication of area served at or near the opening;

d) Smoke from any fire in the basement shall not obstruct any exit serving the ground and upper floors of the building;

e) The smoke exhaust fans in the mechanical ventilation system shall be fire rated, that is, 250 °C for 120 min;

f) The smoke ventilation of the basement car parking areas shall be through provision of supply and exhaust air ducts duly installed with its supports and connected to supply air and exhaust fans. Alternatively, a system of impulse fans (jet fans) may be used for meeting the requirement of smoke ventilation complying with the following:

   1) Structural aspects of beams and other down stands/services shall be taken care of in the planning and provision of the jet fans;

   2) Fans shall be fire rated, that is, 250 °C for 120 min;

3) Fans shall be adequately supported to enable operations for the duration as above;

4) Power supply panels for the fans shall be located in fire safe zone to ensure continuity of power supply;

5) Power supply cabling shall meet circuit integrity requirement in accordance with accepted standard [D-3(39)];

6) The jet fan system may be validated through final computational fluid dynamics (CFD) analysis of the basement; and

7) The jet fan system after commissioning may be verified by Smoke Gun Test for at least one compartment.

The smoke extraction system shall operate on actuation of flow switch actuation of sprinkler system. In addition, a local and/or remote 'manual start-stop control/switch' shall be provided for operations by the fire fighters. Visual indication of the operation status of the fans shall also be provided with the remote control. Smoke exhaust system having make-up air and exhaust air system for areas other than car parking shall be required for common areas and exit access corridor in basements/underground structures and shall be completely separate and independent of car parking areas and other mechanical areas supply air shall not be less than 5 m from any exhaust discharge openings.

19.1.1.3 *EV Car Parking*

The aspect of smoke ventilation for EV car parking shall be as above in **19.1.1.2**.

19.1.2 *Habitat Spaces in Substructure*

Smoke exhaust system having make-up air and exhaust air system for areas other than car parking shall be required for common areas and exit access corridor in basements/underground structures and shall be completely separate and independent of car parking areas and other mechanical areas.

19.1.3 *Services Spaces in Substructure*

Smoke exhaust system having make-up air and exhaust air system for areas other than car parking shall be required for common areas and exit access corridor in basements/underground structures and shall be completely separate and independent of car parking areas and other mechanical areas.

19.2 Smoke Mitigation – Super Structure/Above Ground

19.2.1 *Group A* (*Residential Buildings*) *to Group G* (*Industrial Buildings*)

For the residential buildings (Group A), all kitchen exhaust fans, where provided shall be fixed to an outside wall or to a duct of non-combustible material, which leads directly to the outside. The ducts shall not pass through areas having combustible materials. However, in case of centralized ducting, the duct shall be provided with adequate protection to limit the spread of fire.

For the industrial buildings (Group G), occupancy requiring undivided floor areas so large that the distances from points within the area to the nearest outside walls where exit doors are 45 m and more, requirements for distance to exits should provide stairs leading to exit tunnels or to overhead passageways. In cases where such arrangements are not practicable, the authority may, by special ruling, permit other exit arrangements for single storeyed buildings with distances higher than distances specified in **4**, if completely automatic sprinkler protection is provided and if the heights of ceiling curtain boards and roof ventilation are such as to minimise the possibility that employees will be overtaken by the spread of fire or smoke within 1.8 m of the floor level before they have time to reach exits, provided, however, that in no case may the distance of travel to reach the nearest exit exceed 65 m where smoke venting is required as a condition for permitting distances of travel to exits in excess of the maximum otherwise allowed.

Additional precautions for smoke mitigation are as follows:

a) In any room in which volatile flammable substances are used or stored, no device generating a glow or flame capable of igniting flammable vapour shall be installed or used, such a room shall be provided with a suitably designed exhaust ventilation system; and

b) Storage buildings, such as for warehouses, natural draft smoke venting shall utilize roof vents or vents in walls at or near the ceiling level; such vents shall be normally open, or, if closed, shall be designed for automatic opening in case of fire, by release of smoke sensitive devices.

19.3 Smoke Mitigation of Exits - Natural/ Mechanical

19.3.1 *Staircases and Exits Passageways*

19.3.1.1 *Smoke control of exits*:

a) In building design, compartmentation plays a vital part in limiting the spread of fire and smoke. The design should ensure avoidance of spread of smoke to adjacent spaces through the various leakage openings in the compartment enclosure, such as cracks, openings annular gaps around pipes ducts, airflow grills and doors;

b) The pressurization of staircases and lift lobbies shall be adopted:

    1) The pressure difference for staircases shall be minimum 50 Pa; and

    2) Pressure differences for lobbies (or corridors) shall be between 25 Pa and 30 Pa. Further, the pressure differential for enclosed staircase adjacent to such lobby (or corridors) shall be 50 Pa. For enclosed staircases adjacent to non-pressurized lobby (or corridors), the pressure differential shall be 50 Pa.

c) Equipment and ductwork for staircase pressurization shall be in accordance with one of the following:

    1) Directly connected to the stairway by ductwork enclosed in non-combustible construction; and

    2) If ducts used to pressurize the system are passed through shafts and grills are provided at each level, it shall be ensured that hot gases and smoke from the building shall not ingress into the staircases under any circumstances.

d) The normal air conditioning system and the pressurization system shall be designed and interfaced to meet the requirements of emergency services. When the emergency pressurization is brought into action, the following changes in the normal air conditioning system shall be affected:

    1) Any re-circulation of air shall be stopped and all exhaust air vented to atmosphere;

    2) Any air supply to the spaces/areas other than exits shall be stopped;

    3) The exhaust system may be continued, provided:

        i) The positions of the extraction grills permit a general air flow away from the means of egress;

        ii) The construction of the ductwork and fans is such that, it will not be rendered inoperable by hot gases and smoke; and

        iii) There is no danger of spread of smoke to other floors by the path of the extraction system which shall be ensured by keeping the extraction fans running.

e) For pressurized stair enclosure systems, the activation of the systems shall be initiated by signalling from fire alarm panel;

f) Pressurization system shall be integrated and supervised with the automatic/manual fire alarm system for actuation;

g) Wherever pressurized staircase is to be connected to unpressurized area, the two areas shall be segregated by 120 min fire resistant wall. Such as, in high-rise office buildings, a pressurized staircase can discharge into an unpressurized ground-level lobby or covered parking floor, provided that the staircase itself remains protected as a smoke-free escape path up to the point of final exit; and

h) Fresh air intake for pressurization shall be away (at least 4 m) from any of the exhaust outlets/grille.

19.3.1.2 In case of smoke exhaust and pressurization of areas above ground, corridors in exit access (exit access corridor) are created for meeting the requirement of use, privacy and layout in various occupancies. These are most often noted in hospitality, health care occupancies and sleeping accommodations. Exit access corridors of guest rooms and indoor patient department/areas having patients lacking self-preservation and for sleeping accommodations such as apartments, custodial, penal and mental institutions, etc, shall be provided with 60 min fire resistant wall and 20 min self-closing fire doors along with all fire stop sealing of penetrations. The fire doors shall be as per the accepted standard [D-3(49)].

Smoke exhaust system having make-up air and exhaust air system or alternatively pressurization system with supply air system for these exit access corridors shall be required. Smoke exhaust system having make-up air and exhaust air system shall also be required for theatres/auditoria. Such smoke exhaust system shall also be required for large lobbies and which have exit through staircase leading to exit discharge. This would enable eased exit of people through smoke-controlled area to exit discharge. All exit passageway (from exit-to-exit discharge) shall be pressurized or naturally ventilated. The mechanical pressurization system shall be automatic in action with manual controls in addition. All such exit passageway shall be maintained with integrity for safe means of egress and evacuation. Doors provided in such exit passageway shall be fire rated doors of 120 min rating.

Smoke exhaust system where provided, for above areas and occupancies shall have a minimum of 12 air changes per hour smoke exhaust mechanism. Pressurization system where provided shall have a minimum pressure differential of 25 Pa to 30 Pa in relationship to other areas. The smoke exhaust fans in the mechanical ventilation system shall be fire rated, that is, 250 °C for 120 min. For naturally cross-ventilated corridors or corridors with operable windows, such smoke exhaust system or pressurization system will not be required.

19.3.2 *Lift Lobby*

Smoke control of exits – pressure differences for lobbies (or corridors) shall be between 25 Pa and 30 Pa.

Firefighting shaft (fire tower) is an enclosed shaft having protected area of 120 min fire resistance rating comprising protected lobby, staircase and fireman's lift, connected directly to exit discharge or through exit passageway with 120 min fire resistant wall at the level of exit discharge to exit discharge.

19.3.3 *Hoist Way*

If the lift lobby is not provided at any of the levels in air-conditioned buildings or in internal spaces where funnel/flue effect may be created, lift hoist way shall be pressurized at 50 Pa.

19.3.4 *Fire Tower/Firefighting Shaft*

Firefighting shaft (Fire tower) is an enclosed shaft having protected area of 120 min fire resistance rating comprising protected lobby, staircase and fireman's lift, connected directly to exit discharge or through exit passageway with 120 min fire resistant wall at the level of exit discharge to exit discharge.

If lift lobby shall not be provided at any of the levels in air-conditioned buildings or in internal spaces where funnel/flue effect may be created, lift hoist way shall be pressurized at 50 Pa.

19.3.5 *Egress and Evacuation Strategy*

Staircase and fire lift lobby of a firefighting shaft shall be smoke controlled. It is recommended that the pressurization requirement for staircase in firefighting shaft and for other fire exit staircases in buildings greater than 60 m in height be evaluated to limit the force required to operate the door assembly (in the direction of door opening) to not more than 133 N to set the door leaf in motion. The aspect of pressurization, door area/width and door closure shall be planned in consideration to the above.

19.4 Smoke Mitigation – Special Occupancy and Spaces

19.4.1 *Atrium*

Given below are the requirements for atrium concerning with the smoke prevention and management for occupancy and spaces:

a) Atrium in business occupancy shall be planned with 6 ACPH while atrium in hotels and assembly occupancy shall be planned with 8 ACPH smoke extraction system. Such air changes shall be planned in atrium for a height of 15 m from the top;

b) Smoke exhaust fans shall be capable of operating effectively at 250 °C for 120 minutes;

c) Makeup air supply points shall be located beneath the smoke layer and on the lower levels connected by the atrium. Makeup air shall be provided by fans, openings to outside to allow infiltration, or the combination thereof;

d) It is recommended that makeup air be designed at 85 percent to 95 percent of the exhaust flow rate, not including the leakage through these small paths. The makeup air shall not cause door-opening force to exceed allowable limits. The makeup air velocity shall not exceed 1.02 m/s where the makeup air could come in contact with the plume unless a higher makeup air velocity is supported by engineering analysis; and

e) Atrium smoke management system fans shall be provided with emergency power. If so, required by the authority, an engineering analysis should be performed which demonstrates that the smoke system for the atrium is designed to keep the smoke layer interface 1 800 mm above the highest occupied floor level of exit access, open to the atrium, for a period equal to 1.5 times the calculated egress time or 20 min, whichever is greater.

19.4.2 *Commercial Kitchens*

Given below are the requirements for fire separation for commercial kitchens:

a) Where the flue or exhaust duct passes through the compartment wall or floor, the flue or duct shall be encased by non-combustible construction and no damper shall be permitted to be installed in such flue or duct. Also, such flue or ductwork shall be clear from combustible materials; and

b) It is advisable to locate the kitchen/cooking operations on the external periphery of the building so that in the event of mechanical ventilation failure, it shall be naturally ventilated.

19.4.2.1 *Exhaust ventilation system*

Following guidelines shall be ensured for proper functioning of exhaust ventilation system for cooking equipment:

a) Provision of cleaning of the kitchen exhaust every six months to ensure that the carbon soot accumulated in the exhaust duct is cleaned to avoid the chances of outbreak of fire shall be made;

b) Independent exhaust ducts shall be provided for equipment using dry fuel like wood/ charcoal which produce spark and are likely to ignite the grease which might have accumulated in the common duct; and

c) Alternatively, approved spark arrestors may be provided before the duct from equipment using dry fuel meets the main duct. These spark arrestors shall be so provided that these are easily accessible and removable for cleaning.

19.4.2.2 *Cooking equipment*

Following procedures shall be adopted for handling of cooking equipment:

a) Any penetrations to the outside of a hood, be either welded or fit with a sealing device in accordance to the relevant Indian standard for not allowing cooking grease, oil to migrate to the outer portion of the hood. The fitment arrangements shall be of approved type. Gaskets for the panels shall be certified to withstand a temperature of 815.6 °C;

b) Grease strip shall be readily available for efficient and regular cleaning of concrete or paved floors of kitchen and restaurant and also the drainage areas;

c) The hood or that portion of a primary collection means designed for collecting cooking vapours and residues shall be constructed of and be supported by steel not less than 1.09 mm in thickness or stainless steel not less than 0.94 mm in thickness or other approved material of equivalent strength and fire and corrosion resistance;

d) All seams, joint, and penetrations of the hood enclosure that direct and capture grease-laden vapours and exhaust gases shall have a liquid tight continuous external weld to the hood's lower outermost perimeter;

e) Grease filters shall be of steel rigid construction that will not distort or crush under normal operation handling and cleaning conditions. They shall be so arranged that all exhaust air passes through the grease filters. Filters shall be easily accessible and removable for periodic cleaning;

f) Grease filters shall be installed at an angle not less than 45° from the horizontal;

g) Grease filters shall be equipped with a grease drip tray beneath their lower edges and shall have a suitable minimum depth needed to collect grease. The grease drip trays shall be pitched to drain into an enclosed metal container having a capacity not exceeding 3.8 1itre; and

h) The exhaust ducts shall be constructed of and supported by carbon steel not less than 1.37 mm in thickness or stainless steel not less than 1.09 mm in thickness.

19.4.2.3 *Rooftop terminations — Exhaust systems*

The exhaust system shall terminate either outside the building with a fan or duct or through the roof or to the roof from outside with minimum 3 m of horizontal clearance from the outlet to the adjacent buildings, property lines and air intakes. There shall be a minimum of 1.5 m of horizontal clearance from the outlet (fan housing) to any combustible structure. There shall be a vertical separation of 1.0 m below any exhaust outlets for air intakes within 3.0 m of the exhaust outlet.

19.4.3 *Auditoriums and Cinemas*

Smoke exhaust and pressurization of areas above ground – smoke exhaust system having make-up air and exhaust air system shall also be required for theatres/auditorium.

19.4.4 *Multilevel Car Parking* (*MLCP*) *Non-Mechanical*

Open parking structures (including multi-level parking and stilt parking) specifies the degree to which the structure's exterior walls shall have openings. Parking structures that meet the definition of the term open parking structure provide sufficient area in exterior walls to vent the products of combustion to a greater degree than an enclosed parking structure.

A parking structure having each parking level wall openings open to the atmosphere, for an area of not less than 0.4 m^2 for each linear metre of its exterior perimeter shall be construed as open parking structure. Such openings shall be distributed over 40 percent of the building perimeter or uniformly over two opposing sides. Interior wall lines shall be at least 20 percent open, with openings distributed to provide ventilation, else, the structure shall be deemed as enclosed parking structures.

NOTE — A car park located at the stilt level of a building (not open to sky) shall be considered an open or an unenclosed car park if any part of the car park is within 30 m of a permanent natural ventilation opening and any one of the following is complied with towards the permanent natural ventilation requirement: i) 50 percent of the car park perimeter shall be open to permanent natural ventilation. ii) At least 75 percent of the car park perimeter is having the 50 percent natural ventilation opening.

19.4.5 *Multilevel Car Parking Stack/ Mechanical*

For enclosed multilevel car parking structures utilizing stack or mechanical systems, provisions for smoke ventilation should be considered to enhance safety during fire incidents. Smoke ventilation systems may include natural or mechanical ventilation strategies to ensure the rapid removal of smoke and hot gases. Such systems should comply with the applicable fire safety standards and guidelines, as detailed in Part F 'Fire and Life Safety' of this NBCS.

For smoke ventilation requirement of car parking, *refer* **19.1.1.2**.

19.5 Aspects of Pressurization (Pascals) Smoke Mitigation

Smoke ventilation systems should be designed in line with the principles of pressurization to manage smoke spread effectively. These include:

a) *Defend in Place* (*One Door Open*) — Maintaining a pressure differential to protect occupants who remain within the structure;

b) *Phased Evacuation* (*Three Doors Open*) — Ensuring controlled evacuation by balancing pressurization for multiple door openings;

c) *Firefighting Shaft* (*Three Doors Open*) — Providing safe access for firefighters with maintained pressure differentials;

d) *Pressure Differential Measurement* — Ensuring pressure is appropriately maintained in staircases, lift lobbies, and lifts; and

e) *Door Opening Force Measurement* — Verifying door operation under pressurized conditions to allow safe egress.

19.6 Aspects of Ventilation for Smoke Mitigation

Ventilation systems play a crucial role in maintaining occupant safety during fire emergencies by controlling smoke spread and ensuring adequate air changes per hour (ACPH). Proper compartmentalization, isolation of air handling units, and adherence to fire resistance standards for ducts and dampers to prevent smoke infiltration and maintain safe evacuation routes. These units ensure effective fire containment and support emergency response operations.

19.6.1 *Air Handling Unit*

From fire safety point of view, separate air handling units (AHU) for each floor shall be provided so as to avoid the hazards arising from spread of fire and smoke through the air conditioning ducts. The air ducts shall be separate from each AHU to its floor and in no way shall interconnect with the duct of any other floor. Within a floor it would be desirable to have separate air handling unit provided for each compartment. Air handling unit shall be provided with effective means for preventing circulation of smoke through the system in the case of a fire in air filters or from other sources drawn into the system and shall have smoke sensitive devices for actuation in accordance with best practices.

Shafts or ducts, if penetrating multiple floors, shall be of masonry construction with fire damper in connecting ductwork or shall have fire rated ductwork with fire dampers at floor crossing. Alternatively, the duct and equipment should be installed in room having walls, doors and fire damper in duct exiting/entering the room of 120 min fire resistance rating. Such shafts and ducts shall have all passive fire control meeting 120 min fire resistance rating requirement to meet the objective of isolation of the floor from spread of fire to upper and lower floors through shaft/duct work.

NOTES

1 Zoned and compartmented HVAC systems are encouraged with an approach to avoid common exhaust shafts and fresh air intake shafts which will limit the requirement of such passive measure and fire rated duct work and dampers.

2 The air filters of the air handling units shall be made of non-combustible materials.

3 The air handling unit room shall not be used for storage of any combustible materials.

19.6.2 *Duct Work*

Air ducts serving main floor areas, corridors, etc, shall not pass through the exits/exit passageway/ exit enclosure. Exits and lift lobbies, etc, shall not be used as return air passage.

As far as possible, metallic ducts shall be used even for the return air instead of space above the false ceiling.

Wherever the ducts pass through fire walls or floors, the opening around the ducts shall be sealed with materials having fire resistance rating of the compartment. Such duct shall also be provided with fire dampers at all fire walls and floors unless such ducts are required to perform for fire safety operation; and in such case fire damper may be avoided at fire wall and floor while integrity of the duct shall be maintained with 120 min fire resistance rating to allow the emergency operations for fire safety requirements.

The ducting within compartment would require minimum fire resistance rating of 30 min. If such duct crosses adjacent compartment/floor and not having fire dampers in such compartment/floor, it shall require fire resistance duct work rating of 120 min. The requirements of support of the duct shall meet its functional time requirement as above.

The materials used for insulating the duct system (inside or outside) shall be of non-combustible type. Any such insulating material shall not be wrapped or secured by any material of combustible nature.

Inspection panels shall be provided in the ductwork to facilitate the cleaning accumulated dust in ducts and to obtain access for maintenance of fire dampers. For more information, special publications may be referred.

NOTE — Special publication may be UL 2079 'Tests for Fire Resistance of Building Joint Systems'.

19.6.3 *Fire or Fire/Smoke Dampers*

These dampers shall be evaluated to be located in supply air ducts, fresh air and return air ducts/passages at the following points:

a) At the fire separation wall;

b) Where ducts/passages enter the vertical shaft;

c) Where the ducts pass through floors; and

d) At the inlet of supply air duct and the return air duct of each compartment on every floor.

Damper shall be so installed to provide complete integrity of the compartment with all passive fire protection sealing. Damper should be accessible to maintain, test and also replace, if so required. Damper shall be integrated with fire alarm panel and shall be sequenced to operate as per requirement and have interlocking arrangement for fire safety of the building. Manual operation facilities for damper operation shall also be provided.

19.6.4 *Glass Facade*

Openable panels shall be provided on each floor and shall be spaced not more than 10 m apart measured along the external wall from centre-to-centre of the access openings. Such openings shall be operable at a height between 1.2 m and 1.5 m from the floor and shall be in the form of openable panels (fire access panels) of size not less than 1 000 mm × 1 000 mm opening outwards. The wordings, 'FIRE OPENABLE PANEL — OPEN IN CASE OF FIRE, DO NOT OBSTRUCT' of minimum 25 mm letter height shall be marked on the internal side. Such panels shall be suitably distributed on each floor based on occupant concentration. These shall not be limited to cubicle areas and shall be also located in common areas/corridors to facilitate access by the building occupants and fire personnel for smoke exhaust in times of distress.

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The standards listed may be as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section of this NBCS.

	IS No.	*Title*
(1)	IS 3615 : 2024	Glossary of terms used in refrigeration and air conditioning (*third revision*)
(2)	IS16656 : 2017/ ISO 817 : 2014	Refrigerants — Designation and safety classification
(3)	IS 7896 : 2023	Air conditioning outdoor design conditions data for Indian cities (*second revision*)
(4)	IS 1391 (Part 1) : 2023	Room Air Conditioners Specification Part 1 Unitary air conditioners (*fourth revision*)
(5)	IS 60335-2-40 : 2018/ IEC 60335-2-40	Household and similar electrical appliances — safety: Part 2-40 Particular requirements for electrical heat pumps air-conditioners and dehumidifiers
(6)	IS 1391 (Part 2) : 2023	Room air conditioners specification: Part 2 Split air conditioners (*fourth revision*)
(7)	IS 18689 : 2024	Household and similar electrical appliances — Safety — Particular requirements for commercial refrigerating appliances and ice-makers with an incorporated or remote refrigerant unit or motor-compressor (IEC 60335-2-89 : 2019, MOD)
(8)	IS 16678 (Part 1) : 2018/ ISO 5149-1 : 2014	Refrigerating systems and heat pumps safety and environmental requirements: Part 1 Definitions, classification and selection criteria
(9)	IS 16678 (Part 2) : 2018/ ISO 5149-2 : 2014	Refrigerating systems and heat pumps — Safety and environmental requirements: Part 2 Design, construction, testing, marking and documentation
(10)	IS 16678 (Part 3) : 2018/ ISO 5149-3 : 2014	Refrigerating systems and heat pumps — Safety and environmental requirements: Part 3 installation site
(11)	IS 16678 (Part 4) : 2024/ ISO 5149-4 : 2022	Refrigerating systems and heat pumps — Safety and environmental requirements: Part 4 Operation, maintenance, repair and recovery (*first revision*)
(12)	IS 5610 : 2025	Refrigerants — Specification (*third revision*)
(13)	IS 3315 : 2024	Direct evaporative air cooler — Specification (*fourth revision*)
(14)	IS 655 : 2006	Air ducts — Specification (*second revision*)
(15)	IS 17570 (Part 1) : 2021/ ISO 16890-1:2016	Air Filters for general ventilation: Part 1 Technical specifications, requirements and classification system based upon particulate matter efficiency (ePM)
(16)	17570 (Part 2) : 2025	Air Filters for general ventilation: Part 2 Measurement of fractional efficiency and air flow resistance (ISO 16890-2 : 2016, MOD)

	IS No.	*Title*
(17)	IS 17570 (Part 3) : 2021/ ISO 16890-3 : 2016	Air Filters for general ventilation: Part 3 Determination of the gravimetric efficiency and the air flow resistance versus the mass of test dust captured
(18)	IS 17570 (Part 4) : 2025/ ISO 16890-4 : 2016	Air Filters for general ventilation: Part 4 Conditioning method to determine the minimum fractional test efficiency
(19)	IS 16753 (Part 1) : 2022/ ISO 29463-1 : 2017	High efficiency filters and filter media for removing particles from air Part 1 Classification, performance, testing and marking (*first revision*)
(20)	IS 8188 : 1999	Treatment of water for cooling towers — Code of practice (*first revision*)
(21)	IS 18794 : 2024	Air handling units — Specification
(22)	IS 12615 : 2018	Line operated three phase a.c. motors (IE CODE) “efficiency classes and performance specification” (*third revision*)
(23)	IS 13204 : 2024	Rigid phenolic foam for thermal insulation — Specification (*first revision*)
(24)	IS 16590 : 2023	Liquid chilling package units — Specification (*first revision*)
(25)	IS 8148 : 2018	Specification for packaged air conditioners (*second revision*)
(26)	IS 18728 : 2024	Multiple split-system air conditioners and air-to-air heat pumps (VRF air conditioners) — Specification
(27)	IS 3601 : 2006	Steel tubes for mechanical and general engineering purposes — Specification (*second revision*)
(28)	IS 661 : 2019	Thermal insulation of cold storage — Code of practice (*fourth revision*)
(29)	IS IS 17773 : 2022	Closed-circuit ammonia refrigeration system — Code of practice for design and installation (ANSI/IIAR 2 : 2014, NEQ)
(30)	IS 1893 (Part 1) : 2025	Design for earthquake hazard and criteria for earthquake-resistant design of structures: Part 1 General provisions (*seventh revision*)
(31)	IS 3043 : 2018	Code of practice for earthing (*second revision*)
(32)	IS/IEC 62305-4 : 2010	Protection against lightning: Part 4 Electrical and electronic systems within structures
(33)	IS 732 : 2019	Code of practice for electrical wiring installations (*fourth revision*)
(34)	IS 1255 : 1983	Code of practice for installation and maintenance of power cables up to and including 33 kV rating (*second revision*)
(35)	IS 16753 (Part 2) : 2018/ ISO 29463-2 : 2011	High-Efficiency filters and filter media for removing particles in air: Part 2 aerosol production, measuring equipment and particle — Counting statistics
(36)	IS 16753 (Part 3) : 2018/ ISO 29463-3 : 2011	High-Efficiency filters and filter media for removing particles in air: Part 3 testing flat sheet filter media
(37)	IS 16753 (Part 4) : 2018/ ISO 29463-4 : 2011	High-Efficiency filters and filter media for removing particles in air: Part 4 testing method for determining leakage of filter elements — Scan method

	IS No.	*Title*
(38)	IS 16753 (Part 5) : 2024/ ISO 29463-5	High-efficiency filters and filter media for removing particles in air: Part 5 Test method for filter elements (*first revision*)
(39)	SP 30 : 2023	National Electrical Code of India 2023 (*second revision*)
(40)	IS 17025 : 2017/ ISO/IEC 17025 : 2017	General requirements for the competence of testing and calibration laboratories (*second revision*)
(41)	IS 13301 : 1992	Vibration isolation for machine foundations — Guidelines
(42)	IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP Code)
(43)	IS 4759 : 1996	Hot — Dip zinc coatings on structural steel and other allied products — Specification (*third revision*)
(44)	IS 737 : 2024	Wrought aluminium and aluminium alloy sheet and strip for general engineering purposes — Specification (*fourth revision*)
(45)	IS 10234 : 1982	Recommendations for general pipeline welding
(46)	IS 2075 : 2017	Ready mixed paint, stoving, red oxide zinc chrome, priming — Specification (*third revision*)
(47)	IS 15652 : 2006/ IEC 61111	Insulating mats for electrical purposes — Specification
(48)	IS 1946 (Part 2) : 2025	Fixing devices in walls, ceilings and floors of solid construction: Part 2 Design of post-installed anchorage to concrete — Code of practice (*first revision*)
(49)	IS 3614 : 2021	Fire doors and doorsets — Specification (*first revision*)
	IS 61439-1 : 2020/ IEC 61439-1:2020	Low-voltage switchgear and controlgear assemblies Part 1: General rules (*first revision*)
(50)	IS 2370: 2014	Walk-in cold rooms — Specification